

an *antisymmetric tensor*. So we say: the electric and magnetic fields are both part of an antisymmetric tensor of the second rank in four dimensions.

You've come a long way. Remember way back when we defined what a velocity meant? Now we are talking about "an antisymmetric tensor of the second rank in four dimensions."

Now we have to find the law of the transformation of $F_{\mu\nu}$. It isn't at all difficult to do; it's just laborious—the brains involved are nil, but the work is not. What we want is the Lorentz transformation of $\nabla_\mu A_\nu - \nabla_\nu A_\mu$. Since ∇_μ is just a special case of a vector, we will work with the general antisymmetric vector combination, which we can call $G_{\mu\nu}$:

$$G_{\mu\nu} = a_\mu b_\nu - a_\nu b_\mu. \quad (26.20)$$

(For our purposes, a_μ will eventually be replaced by ∇_μ and b_μ will be replaced by the potential A_μ .) The components of a_μ and b_μ transform by the Lorentz formulas, which are

$$\begin{aligned} a'_t &= \frac{a_t - va_x}{\sqrt{1-v^2}}, & b'_t &= \frac{b_t - vb_x}{\sqrt{1-v^2}}, \\ a'_x &= \frac{a_x - va_t}{\sqrt{1-v^2}}, & b'_x &= \frac{b_x - vb_t}{\sqrt{1-v^2}}, \\ a'_y &= a_y, & b'_y &= b_y, \\ a'_z &= a_z, & b'_z &= b_z. \end{aligned} \quad (26.21)$$

Now let's transform the components of $G_{\mu\nu}$. We start with G_{tx} :

$$\begin{aligned} G'_{tx} &= a'_t b'_x - a'_x b'_t \\ &= \left(\frac{a_t - va_x}{\sqrt{1-v^2}} \right) \left(\frac{b_x - vb_t}{\sqrt{1-v^2}} \right) - \left(\frac{a_x - va_t}{\sqrt{1-v^2}} \right) \left(\frac{b_t - vb_x}{\sqrt{1-v^2}} \right) \\ &= a_t b_x - a_x b_t. \end{aligned}$$

But that is just G_{tx} ; so we have the simple result

$$G'_{tx} = G_{tx}.$$

We will do one more

$$G'_{ty} = \frac{a_t - va_x}{\sqrt{1-v^2}} b_y - a_y \frac{b_t - vb_x}{\sqrt{1-v^2}} = \frac{(a_t b_y - a_y b_t) - v(a_x b_y - a_y b_x)}{\sqrt{1-v^2}}.$$

So we get that

$$G'_{ty} = \frac{G_{ty} - vG_{xy}}{\sqrt{1-v^2}}.$$

And, of course, in the same way,

$$G'_{tz} = \frac{G_{tz} - vG_{xz}}{\sqrt{1-v^2}}.$$

It is clear how the rest will go. Let's make a table of all six terms, only now we may as well write them for $F_{\mu\nu}$:

$$\begin{aligned} F'_{tx} &= F_{tx}, & F'_{xy} &= \frac{F_{xy} - vF_{ty}}{\sqrt{1-v^2}}, \\ F'_{ty} &= \frac{F_{ty} - vF_{xy}}{\sqrt{1-v^2}}, & F'_{yz} &= F_{yz}, \\ F'_{tz} &= \frac{F_{tz} - vF_{xz}}{\sqrt{1-v^2}}, & F'_{zx} &= \frac{F_{zx} - vF_{zt}}{\sqrt{1-v^2}}. \end{aligned} \quad (26.22)$$

Of course, we still have $F'_{\mu\nu} = -F'_{\nu\mu}$ and $F_{\mu\mu} = 0$.

So we have the transformation of the electric and magnetic fields. All we have to do is look at Table 26-1 to find out what our grand notation in terms of $F_{\mu\nu}$ means in terms of $\mathbf{E}$ and $\mathbf{B}$. It's just a matter of substitution. So that we can see how it looks in the ordinary symbols, we'll rewrite our transformation of the field components in Table 26-2.

Table 26-2

The Lorentz transformation of the electric and magnetic fields (Note: $c = 1$)

$E'_x = E_x$	$B'_x = B_x$
$E'_y = \frac{E_y - vB_z}{\sqrt{1 - v^2}}$	$B'_y = \frac{B_y + vE_z}{\sqrt{1 - v^2}}$
$E'_z = \frac{E_z + vB_y}{\sqrt{1 - v^2}}$	$B'_z = \frac{B_z - vE_y}{\sqrt{1 - v^2}}$

The equations in Table 26-2 tell us how $\mathbf{E}$ and $\mathbf{B}$ change if we go from one inertial frame to another. If we know $\mathbf{E}$ and $\mathbf{B}$ in one system, we can find what they are in another that moves by with the speed v .

We can write these equations in a form that is easier to remember if we notice that since v is in the x -direction, all the terms with v are components of the cross products $\mathbf{v} \times \mathbf{E}$ and $\mathbf{v} \times \mathbf{B}$. So we can rewrite the transformations as shown in Table 26-3

Table 26-3

An alternative form for the field transformations (Note: $c = 1$)

$E'_x = E_x$	$B'_x = B_x$
$E'_y = \frac{(E + \mathbf{v} \times \mathbf{B})_y}{\sqrt{1 - v^2}}$	$B'_y = \frac{(\mathbf{B} - \mathbf{v} \times \mathbf{E})_y}{\sqrt{1 - v^2}}$
$E'_z = \frac{(E + \mathbf{v} \times \mathbf{B})_z}{\sqrt{1 - v^2}}$	$B'_z = \frac{(\mathbf{B} - \mathbf{v} \times \mathbf{E})_z}{\sqrt{1 - v^2}}$

It is now easier to remember which components go where. In fact, the transformation can be written even more simply if we define the field components along x as the “parallel” components $E_{\parallel}$ and $B_{\parallel}$ (because they are parallel to the relative velocity of S and S'), and the total transverse components—the vector sums of the y - and z -components—as the “perpendicular” components $E_{\perp}$ and $B_{\perp}$. Then we get the equations in Table 26-4. (We have also put back the c 's, so it will be more convenient when we want to refer back later.)

Table 26-4

Still another form for the Lorentz transformation of $\mathbf{E}$ and $\mathbf{B}$

$E'_{\parallel} = E_{\parallel}$	$B'_{\parallel} = B_{\parallel}$
$E'_{\perp} = \frac{(\mathbf{E} + \mathbf{v} \times \mathbf{B})_{\perp}}{\sqrt{1 - v^2/c^2}}$	$B'_{\perp} = \frac{(\mathbf{B} - \frac{\mathbf{v} \times \mathbf{E}}{c^2})_{\perp}}{\sqrt{1 - v^2/c^2}}$

The field transformations give us another way of solving some problems we have done before—for instance, for finding the fields of a moving point charge. We have worked out the fields before by differentiating the potentials. But we could now do it by transforming the Coulomb field. If we have a point charge at rest in the S -frame, then there is only the simple radial $\mathbf{E}$ -field. In the S' -frame we will see a point charge moving with the velocity u , if the S' -frame moves by the

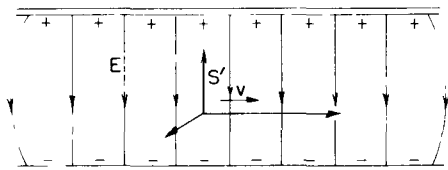


Fig. 26-7. The coordinate frame S' moving through a static electric field.

S -frame with the speed $v = -u$. We will let you show that the transformations of Tables 26-3 and 26-4 give the same electric and magnetic fields we got in Section 26-2.

The transformation of Table 26-2 gives us an interesting and simple answer for what we see if we move past *any* system of fixed charges. For example, suppose we want to know the fields in *our* frame S' if we are moving along between the plates of a condenser, as shown in Fig. 26-7. (It is, of course, the same thing if we say that a charged condenser is moving past *us*.) What do we see? The transformation is easy in this case because the $\mathbf{B}$ -field in the original system is zero. Suppose, first, that our motion is perpendicular to $\mathbf{E}$, then we will see an $\mathbf{E}' = \mathbf{E}/\sqrt{1 - v^2/c^2}$ which is still completely transverse. We will see, in addition, a magnetic field $\mathbf{B}' = -\mathbf{v} \times \mathbf{E}'/c^2$. (The $\sqrt{1 - v^2/c^2}$ doesn't appear in our formula for $\mathbf{B}'$ because we wrote it in terms of $\mathbf{E}'$ rather than $\mathbf{E}$; but it's the same thing.) So when we move along perpendicular to a static electric field, we see a reduced $\mathbf{E}$ and an added transverse $\mathbf{B}$. If our motion is not perpendicular to $\mathbf{E}$, we break $\mathbf{E}$ into $\mathbf{E}_{\parallel}$ and $\mathbf{E}_{\perp}$. The parallel part is unchanged, $E'_{\parallel} = E_{\parallel}$, and the perpendicular component does as just described.

Let's take the opposite case, and imagine we are moving through a pure static *magnetic* field. This time we would see an *electric* field $\mathbf{E}'$ equal to $\mathbf{v} \times \mathbf{B}'$, and the magnetic field changed by the factor $1/\sqrt{1 - v^2/c^2}$ (assuming it is transverse). So long as v is much less than c , we can neglect the change in the magnetic field, and the main effect is that an electric field appears. As one example of this effect, consider this once famous problem of determining the speed of an airplane. It's no longer famous, since radar can now be used to determine the air speed from ground reflections, but for many years it was very hard to find the speed of an airplane in bad weather. You could not see the ground and you didn't know which way was up, and so on. Yet it was important to know how fast you were moving relative to the earth. How can this be done without seeing the earth? Many who knew the transformation formulas thought of the idea of using the fact that the airplane moves in the magnetic field of the earth. Suppose that an airplane is flying where there is a magnetic field more or less known. Let's just take the simple case where the magnetic field is vertical. If we were flying through it with a horizontal velocity v , then, according to our formula, we should see an electric field which is $\mathbf{v} \times \mathbf{B}$, i.e., perpendicular to the line of motion. If we hang an insulated wire across the airplane, this electric field will induce charges on the ends of the wire. That is nothing new. From the point of view of someone on the ground, we are moving a wire through a field, and the $\mathbf{v} \times \mathbf{B}$ force causes charges to move to the ends of the wire. The transformation equations just say the same thing in a different way. (The fact that we can say the thing more than one way doesn't mean that one way is better than another. We are getting so many different methods and tools that we can usually get the same result in 65 different ways!)

So to measure v , all we have to do is measure the voltage between the ends of the wire. We can't do it with a voltmeter because the same fields will act on the wires in the voltmeter, but there are ways of measuring such fields. We talked about some of them when we discussed atmospheric electricity in Chapter 9. So it should be possible to measure the speed of the airplane.

This important problem was, however, never solved this way. The reason is that the electric field that is developed is of the order of millivolts per meter. It is possible to measure such fields, but the trouble is that these fields are, unfortunately, not any different from any other electric fields. The field that is produced by motion through the magnetic field can't be distinguished from some electric field that was already in the air from another cause, say from electrostatic charges in the air, or on the clouds. We described in Chapter 9 that there are, typically, electric fields above the surface of the earth with strengths of about 100 volts per meter. But they are quite irregular. So as the airplane flies through the air, it sees fluctuations of atmospheric electric fields which are enormous in comparison to the tiny fields produced by the $\mathbf{v} \times \mathbf{B}$ term, and it turns out for practical reasons to be impossible to measure speeds of an airplane by its motion through the earth's magnetic field.

26-4 The equations of motion in relativistic notation*

It doesn't do much good to find electric and magnetic fields from Maxwell's equations unless we know what the fields do when we have them. You may remember that the fields are required to find the forces on charges, and that those forces determine the motion of the charge. So, of course, part of the theory of electrodynamics is the relation between the motion of charges and the forces.

For a single charge in the fields $\mathbf{E}$ and $\mathbf{B}$, the force is

$$\mathbf{F} = q(\mathbf{E} + \mathbf{v} \times \mathbf{B}). \quad (26.23)$$

This force is equal to the mass times the acceleration for low velocities, but the correct law for any velocity is that the force is equal to $d\mathbf{p}/dt$. Writing $\mathbf{p} = m_0\mathbf{v}/\sqrt{1 - v^2/c^2}$, we find that the relativistically correct equation of motion is

$$\frac{d}{dt} \left(\frac{m_0\mathbf{v}}{\sqrt{1 - v^2/c^2}} \right) = \mathbf{F} = q(\mathbf{E} + \mathbf{v} \times \mathbf{B}). \quad (26.24)$$

We would like now to discuss this equation from the point of view of relativity. Since we have put our Maxwell equations in relativistic form, it would be interesting to see what the equations of motion would look like in relativistic form. Let's see whether we can rewrite the equation in a four-vector notation.

We know that the momentum is part of a four-vector p_μ whose time component is the energy $m_0c/\sqrt{1 - v^2/c^2}$. So we might think to replace the left-hand side of Eq. (26.24) by dp_μ/dt . Then we need only find a fourth component to go with $\mathbf{F}$. This fourth component must equal the rate-of-change of the energy, or the rate of doing work, which is $\mathbf{F} \cdot \mathbf{v}$. We would then like to write the right-hand side of Eq. (26.24) as a four-vector like $(\mathbf{F} \cdot \mathbf{v}, F_x, F_y, F_z)$. But this does not make a four-vector.

The *time* derivative of a four-vector is no longer a four-vector, because the d/dt requires the choice of some special frame for measuring t . We got into that trouble before when we tried to make $\mathbf{v}$ into a four-vector. Our first guess was that the time component would be $cdt/dt = c$. But the quantities

$$\left(c, \frac{dx}{dt}, \frac{dy}{dt}, \frac{dz}{dt} \right) = (c, \mathbf{v}) \quad (26.25)$$

are *not* the components of a four-vector. We found that they could be made into one by multiplying each component by $1/\sqrt{1 - v^2/c^2}$. The "four-velocity" u_μ is the four-vector

$$u_\mu = \left(\frac{c}{\sqrt{1 - v^2/c^2}}, \frac{\mathbf{v}}{\sqrt{1 - v^2/c^2}} \right). \quad (26.26)$$

So it appears that the trick is to multiply d/dt by $1/\sqrt{1 - v^2/c^2}$, if we want the derivatives to make a four-vector.

Our second guess then is that

$$\frac{1}{\sqrt{1 - v^2/c^2}} \frac{d}{dt} (p_\mu) \quad (26.27)$$

should be a four-vector. But what is $\mathbf{v}$? It is the velocity of the particle—not of a coordinate frame! Then the quantity f_μ defined by

$$f_\mu = \left(\frac{\mathbf{F} \cdot \mathbf{v}}{\sqrt{1 - v^2/c^2}}, \frac{\mathbf{F}}{\sqrt{1 - v^2/c^2}} \right) \quad (26.28)$$

is the extension into four dimensions of a force—we can call it the "four-force." It is indeed a four-vector, and its space components are not the components of $\mathbf{F}$ but of $\mathbf{F}/\sqrt{1 - v^2/c^2}$.

* In this section we will put back all of the c 's.

The question is—why is f_μ a four-vector? It would be nice to get a little understanding of that $1/\sqrt{1 - v^2/c^2}$ factor. Since it has come up twice now, it is time to see why the d/dt can always be fixed by the same factor. The answer is in the following: When we take the time derivative of some function x , we compute the increment Δx in a small interval Δt in the variable t . But in another frame, the interval Δt might correspond to a change in both t' and x' , so if we vary only t' , the change in x will be different. We have to find a variable for our differentiation that is a measure of an “interval” in *space-time*, which will then be the same in all coordinate systems. When we take Δx for that interval, it will be the same for all coordinate frames. When a particle “moves” in four-space, there are the changes $\Delta t, \Delta x, \Delta y, \Delta z$. Can we make an invariant interval out of them? Well, they are the components of the four-vector $x_\mu = (ct, x, y, z)$ so if we define a quantity Δs by

$$(\Delta s)^2 = \frac{1}{c^2} \Delta x_\mu \Delta x_\mu = \frac{1}{c^2} (c^2 \Delta t^2 - \Delta x^2 - \Delta y^2 - \Delta z^2) \quad (26.29)$$

—which is a four-dimensional dot product—we then have a good four-scalar to use as a measure of a four-dimensional interval. From Δs —or its limit ds —we can define a parameter $s = \int ds$. And a derivative with respect to s , d/ds , is a nice four-dimensional operation, because it is invariant with respect to a Lorentz transformation.

It is easy to relate ds to dt for a moving particle. For a moving point particle,

$$dx = v_x dt, \quad dy = v_y dt, \quad dz = v_z dt, \quad (26.30)$$

and

$$ds = \sqrt{(dt^2/c^2)(c^2 - v_x^2 - v_y^2 - v_z^2)} = dt\sqrt{1 - v^2/c^2}. \quad (26.31)$$

So the operator

$$\frac{1}{\sqrt{1 - v^2/c^2}} \frac{d}{dt}$$

is an *invariant operator*. If we operate on any four-vector with it, we get another four-vector. For instance, if we operate on (ct, x, y, z) , we get the four-velocity u_μ :

$$\frac{dx_\mu}{ds} = u_\mu.$$

We see now why the factor $\sqrt{1 - v^2/c^2}$ fixes things up.

The invariant variable s is a useful physical quantity. It is called the “proper time” along the path of a particle, because ds is always an interval of time in a frame that is moving with the particle at any particular instant. (Then, $\Delta x = \Delta y = \Delta z = 0$, and $\Delta s = \Delta t$.) If you can imagine some “clock” whose rate doesn’t depend on the acceleration, such a clock carried along with the particle would show the time s .

We can now go back and write Newton’s law (as corrected by Einstein) in the neat form

$$\frac{dp_\mu}{ds} = f_\mu, \quad (26.32)$$

where f_μ is given in Eq. (26.28). Also, the momentum p_μ can be written as

$$p_\mu = m_0 u_\mu = m_0 \frac{dx_\mu}{ds}, \quad (26.33)$$

where the coordinates $x_\mu = (ct, x, y, z)$ now describe the trajectory of the particle. Finally, the four-dimensional notation gives us this very simple form of the equations of motion:

$$f_\mu = m_0 \frac{d^2 x_\mu}{ds^2}, \quad (26.34)$$

which is reminiscent of $F = ma$. It is important to notice that Eq. (26.34) is *not* the same as $F = ma$, because the four-vector formula Eq. (26.34) has in it the

relativistic mechanics which are different from Newton's law for high velocities. It is unlike the case of Maxwell's equations, where we were able to rewrite the equations in the relativistic form *without any change in the meaning at all*—but with just a change of notation.

Now let's return to Eq. (26.24) and see how we can write the right-hand side in four-vector notation. The three components—when divided by $\sqrt{1 - v^2/c^2}$ —are the components of f_μ , so

$$f_x = \frac{q(\mathbf{E} + \mathbf{v} \times \mathbf{B})_x}{\sqrt{1 - v^2/c^2}} = q \left[\frac{E_x}{\sqrt{1 - v^2/c^2}} + \frac{v_y B_z}{\sqrt{1 - v^2/c^2}} - \frac{v_z B_y}{\sqrt{1 - v^2/c^2}} \right]. \quad (26.35)$$

Now we must put all quantities in their relativistic notation. First, $c/\sqrt{1 - v^2/c^2}$ and $v_y/\sqrt{1 - v^2/c^2}$ and $v_z/\sqrt{1 - v^2/c^2}$ are the t -, y -, and z -components of the four-velocity u_μ . And the components of $\mathbf{E}$ and $\mathbf{B}$ are components of the second-rank tensor of the fields $F_{\mu\nu}$. Looking back in Table 26-1 for the components of $F_{\mu\nu}$ that correspond to E_x , B_z , and B_y , we get

$$f_x = q(u_t F_{xt} - u_y F_{xy} - u_z F_{xz}),$$

which begins to look interesting. Every term has the subscript x , which is reasonable, since we're finding an x -component. Then all the others appear in pairs: tt , yy , zz —except that the xx -term is missing. So we just stick it in, and write

$$f_x = q(u_t F_{xt} - u_x F_{xx} - u_y F_{xy} - u_z F_{xz}). \quad (26.36)$$

We haven't changed anything because $F_{\mu\nu}$ is antisymmetric, and F_{xx} is zero. The reason for wanting to put in the xx -term is so that we can write Eq. (26.36) in the short-hand form

$$f_\mu = q u_\nu F_{\mu\nu}. \quad (26.37)$$

This equation is the same as Eq. (26.36) if we make the *rule* that whenever any subscript occurs *twice* (as ν does here), you automatically sum over terms in the same way as for the scalar product, *using the same convention for the signs*.

You can easily believe that (26.37) works equally well for $\mu = y$ or $\mu = z$, but what about $\mu = t$? Let's see, for fun, what it says:

$$f_t = q(u_t F_{tt} - u_x F_{tx} - u_y F_{ty} - u_z F_{tz})$$

Now we have to translate back to E 's and B 's. We get

$$f_t = q \left(0 + \frac{v_x}{\sqrt{1 - v^2/c^2}} E_x + \frac{v_y}{\sqrt{1 - v^2/c^2}} E_y + \frac{v_z}{\sqrt{1 - v^2/c^2}} E_z \right), \quad (26.38)$$

or

$$f_t = \frac{q \mathbf{v} \cdot \mathbf{E}}{\sqrt{1 - v^2/c^2}}$$

But from Eq. (26.28), f_t is supposed to be

$$\frac{\mathbf{F} \cdot \mathbf{v}}{\sqrt{1 - v^2/c^2}} = \frac{q(\mathbf{E} + \mathbf{v} \times \mathbf{B}) \cdot \mathbf{v}}{\sqrt{1 - v^2/c^2}}.$$

This is the same thing as Eq. (26.38), since $(\mathbf{v} \times \mathbf{B}) \cdot \mathbf{v}$ is zero. So everything comes out all right.

Summarizing, our equation of motion can be written in the elegant form

$$m_0 \frac{d^2 x_\mu}{ds^2} = f_\mu = q u_\nu F_{\mu\nu}. \quad (26.39)$$

Although it is nice to see that the equations can be written that way, this form is not particularly useful. It's usually more convenient to solve for particle motions by using the original equations (26.24), and that's what we will usually do.

Field Energy and Field Momentum

27-1 Local conservation

It is clear that the energy of matter is not conserved. When an object radiates light it loses energy. However, the energy lost is possibly describable in some other form, say in the light. Therefore the theory of the conservation of energy is incomplete without a consideration of the energy which is associated with the light or, in general, with the electromagnetic field. We take up now the law of conservation of energy and, also, of momentum for the fields. Certainly, we cannot treat one without the other, because in the relativity theory they are different aspects of the same four-vector.

Very early in Volume I, we discussed the conservation of energy; we said then merely that the total energy in the world is constant. Now we want to extend the idea of the energy conservation law in an important way—in a way that says something in *detail* about *how* energy is conserved. The new law will say that if energy goes away from a region, it is because it *flows* away through the boundaries of that region. It is a somewhat stronger law than the conservation of energy without such a restriction.

To see what the statement means, let's look at how the law of the conservation of charge works. We described the conservation of charge by saying that there is a current density $\mathbf{j}$ and a charge density ρ , and that when the charge decreases at some place there must be a flow of charge away from that place. We call that the conservation of charge. The mathematical form of the conservation law is

$$\nabla \cdot \mathbf{j} = -\frac{\partial \rho}{\partial t}. \quad (27.1)$$

This law has the consequence that the total charge in the world is always constant—there is never any net gain or loss of charge. However, the total charge in the world could be constant in another way. Suppose that there is some charge Q_1 near some point (1) while there is no charge near some point (2) some distance away (Fig. 27-1). Now suppose that, as time goes on, the charge Q_1 were to gradually fade away and that *simultaneously* with the decrease of Q_1 some charge Q_2 would appear near point (2), and in such a way that at every instant the sum of Q_1 and Q_2 was a constant. In other words, at any intermediate state the amount of charge lost by Q_1 would be added to Q_2 . Then the total amount of charge in the world would be conserved. That's a “world-wide” conservation, but not what we will call a “local” conservation, because in order for the charge to get from (1) to (2), it didn't have to appear anywhere in the space between point (1) and point (2). Locally, the charge was just “lost.”

There is a difficulty with such a “world-wide” conservation law in the theory of relativity. The concept of “simultaneous moments” at distant points is one which is not equivalent in different systems. Two events that are simultaneous in one system are not simultaneous for another system moving past. For “world-wide” conservation of the kind described, it is necessary that the charge lost from Q_1 should appear *simultaneously* in Q_2 . Otherwise there would be some moments when the charge was not conserved. There seems to be no way to make the law of charge conservation relativistically invariant without making it a “local” conservation law. As a matter of fact, the requirement of the Lorentz relativistic invariance seems to restrict the possible laws of nature in surprising ways. In modern quantum field theory, for example, people have often wanted to alter the theory by allowing what we call a “nonlocal” interaction—where something *here*

27-1 Local conservation

27-2 Energy conservation and electromagnetism

27-3 Energy density and energy flow in the electromagnetic field

27-4 The ambiguity of the field energy

27-5 Examples of energy flow

27-6 Field momentum

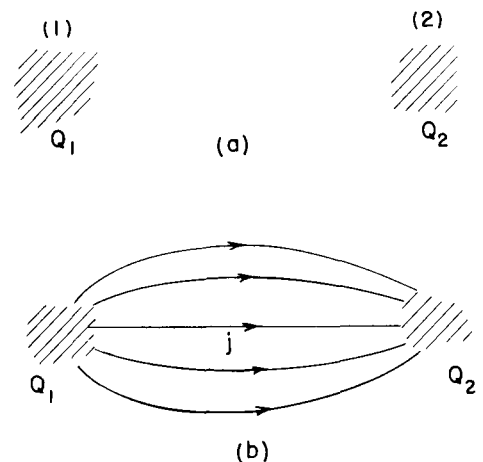


Fig. 27-1. Two ways to conserve charge: (a) $Q_1 + Q_2$ is constant; (b) $dQ_1/dt = \int \mathbf{j} \cdot \mathbf{n} da = -dQ_2/dt$.

has a direct effect on something *there*—but we get in trouble with the relativity principle.

“Local” conservation involves another idea. It says that a charge can get from one place to another only if there is something happening in the space between. To describe the law we need not only the density of charge, ρ , but also another kind of quantity, namely $\mathbf{j}$, a vector giving the rate of flow of charge across a surface. Then the flow is related to the rate of change of the density by Eq. (27.1). This is the more extreme kind of a conservation law. It says that charge is conserved in a special way—conserved “locally.”

It turns out that energy conservation is also a *local* process. There is not only an energy density in a given region of space but also a vector to represent the rate of flow of the energy through a surface. For example, when a light source radiates, we can find the light energy moving out from the source. If we imagine some mathematical surface surrounding the light source, the energy lost from inside the surface is equal to the energy that flows out through the surface.

27-2 Energy conservation and electromagnetism

We want now to write quantitatively the conservation of energy for electromagnetism. To do that, we have to describe how much energy there is in any volume element of space, and also the rate of energy flow. Suppose we think first only of the electromagnetic field energy. We will let u represent the *energy density* in the field (that is, the amount of energy per unit volume in space) and let the vector $\mathbf{S}$ represent the *energy flux* of the field (that is, the flow of energy per unit time across a unit area perpendicular to the flow). Then, in perfect analogy with the conservation of charge, Eq. (27.1), we can write the “local” law of energy conservation in the field as

$$\frac{\partial u}{\partial t} = -\nabla \cdot \mathbf{S}. \quad (27.2)$$

Of course, this law is not true in general; it is not true that the field energy is conserved. Suppose you are in a dark room and then turn on the light switch. All of a sudden the room is full of light, so there is energy in the field, although there wasn’t any energy there before. Equation (27.2) is not the complete conservation law, because the *field energy alone* is not conserved, only the total energy in the world—there is also the energy of matter. The field energy will change if there is some work being done by matter on the field or by the field on matter.

However, if there is matter inside the volume of interest, we know how much energy it has: Each particle has the energy $m_0 c^2 / \sqrt{1 - v^2/c^2}$. The total energy of the matter is just the sum of all the particle energies, and the flow of this energy through a surface is just the sum of the energy carried by each particle that crosses the surface. We want now to talk only about the energy of the electromagnetic field. So we must write an equation which says that the total *field* energy in a given volume decreases *either* because field energy flows out of the volume *or* because the field loses energy to matter (or gains energy, which is just a negative loss). The field energy inside a volume V is

$$\int_V u \, dV,$$

and its rate of decrease is minus the time derivative of this integral. The flow of field energy out of the volume V is the integral of the normal component of $\mathbf{S}$ over the surface Σ that encloses V ,

$$\int_{\Sigma} \mathbf{S} \cdot \mathbf{n} \, da.$$

So

$$-\frac{\partial}{\partial t} \int_V u \, dV = \int_{\Sigma} \mathbf{S} \cdot \mathbf{n} \, da + (\text{work done on matter inside } V). \quad (27.3)$$

We have seen before that the field does work on each unit volume of matter at the rate $\mathbf{E} \cdot \mathbf{j}$. [The force on a particle is $\mathbf{F} = q(\mathbf{E} + \mathbf{v} \times \mathbf{B})$, and the rate of doing work is $\mathbf{F} \cdot \mathbf{v} = q\mathbf{E} \cdot \mathbf{v}$. If there are N particles per unit volume, the rate of doing work per unit volume is $Nq\mathbf{E} \cdot \mathbf{v}$, but $Nq\mathbf{v} = \mathbf{j}$] So the quantity $\mathbf{E} \cdot \mathbf{j}$ must be equal to the loss of energy per unit time and per unit volume by the field. Equation (27.3) then becomes

$$-\frac{\partial}{\partial t} \int_V u dV = \int_S \mathbf{S} \cdot \mathbf{n} da + \int_V \mathbf{E} \cdot \mathbf{j} dV. \quad (27.4)$$

This is our conservation law for energy in the field. We can convert it into a differential equation like Eq. (27.2) if we can change the second term to a volume integral. That is easy to do with Gauss' theorem. The surface integral of the normal component of $\mathbf{S}$ is the integral of its divergence over the volume inside. So Eq. (27.3) is equivalent to

$$-\int_V \frac{du}{dt} dV = \int_V \nabla \cdot \mathbf{S} dV + \int_V \mathbf{E} \cdot \mathbf{j} dV,$$

where we have put the time derivative of the first term inside the integral. Since this equation is true for any volume, we can take away the integrals and we have the energy equation for the electromagnetic fields:

$$-\frac{\partial u}{\partial t} = \nabla \cdot \mathbf{S} + \mathbf{E} \cdot \mathbf{j}. \quad (27.5)$$

Now this equation doesn't do us a bit of good unless we know what u and $\mathbf{S}$ are. Perhaps we should just tell you what they are in terms of $\mathbf{E}$ and $\mathbf{B}$, because all we really want is the result. However, we would rather show you the kind of argument that was used by Poynting in 1884 to obtain formulas for $\mathbf{S}$ and u , so you can see where they come from. (You won't, however, need to learn this derivation for our later work.)

27-3 Energy density and energy flow in the electromagnetic field

The idea is to suppose that there is a field energy density u and a flux $\mathbf{S}$ that depend only upon the fields $\mathbf{E}$ and $\mathbf{B}$. (For example, we know that in electrostatics, at least, the energy density can be written $\frac{1}{2}\epsilon_0 \mathbf{E} \cdot \mathbf{E}$.) Of course, the u and $\mathbf{S}$ might depend on the potentials or something else, but let's see what we can work out. We can try to rewrite the quantity $\mathbf{E} \cdot \mathbf{j}$ in such a way that it becomes the sum of two terms, one that is the time derivative of one quantity and another that is the divergence of a second quantity. The first quantity would then be u and the second would be $\mathbf{S}$ (with suitable signs). Both quantities must be written in terms of the fields only; that is, we want to write our equality as

$$\mathbf{E} \cdot \mathbf{j} = -\frac{\partial u}{\partial t} - \nabla \cdot \mathbf{S}. \quad (27.6)$$

The left-hand side must first be expressed in terms of the fields only. How can we do that? By using Maxwell's equations, of course. From Maxwell's equation for the curl of $\mathbf{B}$,

$$\mathbf{j} = \epsilon_0 c^2 \nabla \times \mathbf{B} - \epsilon_0 \frac{\partial \mathbf{E}}{\partial t}.$$

Substituting this in (27.6) we will have only $\mathbf{E}$'s and $\mathbf{B}$'s:

$$\mathbf{E} \cdot \mathbf{j} = \epsilon_0 c^2 \mathbf{E} \cdot (\nabla \times \mathbf{B}) - \epsilon_0 \mathbf{E} \cdot \frac{\partial \mathbf{E}}{\partial t}. \quad (27.7)$$

We are already partly finished. The last term is a time derivative—it is $(\partial/\partial t)(\frac{1}{2}\epsilon_0 \mathbf{E} \cdot \mathbf{E})$. So $\frac{1}{2}\epsilon_0 \mathbf{E} \cdot \mathbf{E}$ is at least one part of u . It's the same thing we found in electrostatics. Now, all we have to do is to make the other term into the divergence of something.

Notice that the first term on the right-hand side of (27.7) is the same as

$$(\nabla \times \mathbf{B}) \cdot \mathbf{E}. \quad (27.8)$$

And, as you know from vector algebra, $(\mathbf{a} \times \mathbf{b}) \cdot \mathbf{c}$ is the same as $\mathbf{a} \cdot (\mathbf{b} \times \mathbf{c})$; so our term is also the same as

$$\nabla \cdot (\mathbf{B} \times \mathbf{E}) \quad (27.9)$$

and we have the divergence of “something,” just as we wanted. Only that’s wrong! We warned you before that ∇ is “like” a vector, but not “exactly” the same. The reason it is not is because there is an additional *convention* from calculus: when a derivative operator is in front of a product, it works on everything to the right. In Eq (27.7), the ∇ operates only on $\mathbf{B}$, not on $\mathbf{E}$. But in the form (27.9), the normal convention would say that ∇ operates on both $\mathbf{B}$ and $\mathbf{E}$. So it’s *not* the same thing. In fact, if we work out the components of $\nabla \cdot (\mathbf{B} \times \mathbf{E})$ we can see that it is equal to $\mathbf{E} \cdot (\nabla \times \mathbf{B})$ *plus* some other terms. It’s like what happens when we take a derivative of a product in algebra. For instance,

$$\frac{d}{dx}(fg) = \frac{df}{dx}g + f\frac{dg}{dx}.$$

Rather than working out all the components of $\nabla \cdot (\mathbf{B} \times \mathbf{E})$, we would like to show you a trick that is very useful for this kind of problem. It is a trick that allows you to use all the rules of vector algebra on expressions with the ∇ operator, without getting into trouble. The trick is to throw out—for a while at least—the rule of the calculus notation about what the derivative operator works on. You see, ordinarily, the order of terms is used for *two* separate purposes. One is for calculus: $f(d/dx)g$ is not the same as $g(d/dx)f$; and the other is for vectors: $\mathbf{a} \times \mathbf{b}$ is different from $\mathbf{b} \times \mathbf{a}$. We can, if we want, choose to abandon momentarily the calculus rule. Instead of saying that a derivative operates on everything to the right, we make a *new* rule that doesn’t depend on the order in which terms are written down. Then we can juggle terms around without worrying.

Here is our new convention. We show, by a subscript, what a differential operator works on; the *order* has no meaning. Suppose we let the operator D stand for $\partial/\partial x$. Then D_f means that only the derivative of the variable quantity f is taken. Then

$$D_f f = \frac{\partial f}{\partial x}.$$

But if we have $D_f fg$, it means

$$D_f fg = \left(\frac{\partial f}{\partial x}\right)g.$$

But notice now that according to our new rule, $f D_f g$ means the same thing. We can write the same thing any which way.

$$D_f fg = g D_f f = f D_f g = fg D_f.$$

You see, the D_f can even come *after* everything. (It’s surprising that such a handy notation is never taught in books on mathematics or physics.)

You may wonder: What if I *want* to write the derivative of fg ? I *want* the derivative of *both* terms. That’s easy, you just say so; you write $D_f(fg) + D_g(fg)$. That is just $g(\partial f/\partial x) + f(\partial g/\partial x)$, which is what you mean in the old notation by $\partial(fg)/\partial x$.

You will see that it is now going to be very easy to work out a new expression for $\nabla \cdot (\mathbf{B} \times \mathbf{E})$. We start by changing to the new notation; we write

$$\nabla \cdot (\mathbf{B} \times \mathbf{E}) = \nabla_B \cdot (\mathbf{B} \times \mathbf{E}) + \nabla_E \cdot (\mathbf{B} \times \mathbf{E}). \quad (27.10)$$

The moment we do that we don’t have to keep the order straight any more. We always know that ∇_E operates on $\mathbf{E}$ only, and ∇_B operates on $\mathbf{B}$ only. In these circumstances, we can use ∇ as though it were an ordinary vector. (Of course,

when we are finished, we will want to return to the “standard” notation that everybody usually uses) So now we can do the various things like interchanging dots and crosses and making other kinds of rearrangements of the terms. For instance, the middle term of Eq. (27.10) can be rewritten as $\mathbf{E} \cdot \nabla_B \times \mathbf{B}$. (You remember that $\mathbf{a} \cdot \mathbf{b} \times \mathbf{c} = \mathbf{b} \cdot \mathbf{c} \times \mathbf{a}$.) And the last term is the same as $\mathbf{B} \cdot \mathbf{E} \times \nabla_E$. It looks freakish, but it is all right. Now if we try to go back to the ordinary convention, we have to arrange that the ∇ operates only on its “own” variable. The first one is already that way, so we can just leave off the subscript. The second one needs some rearranging to put the ∇ in front of the $\mathbf{E}$, which we can do by reversing the cross product and changing sign:

$$\mathbf{B} \cdot (\mathbf{E} \times \nabla_E) = -\mathbf{B} \cdot (\nabla_E \times \mathbf{E}).$$

Now it is in a conventional order, so we can return to the usual notation. Equation (27.10) is equivalent to

$$\nabla \cdot (\mathbf{B} \times \mathbf{E}) = \mathbf{E} \cdot (\nabla \times \mathbf{B}) - \mathbf{B} \cdot (\nabla \times \mathbf{E}). \quad (27.11)$$

(A quicker way would have been to use components in this special case, but it was worth taking the time to show you the mathematical trick. You probably won’t see it anywhere else, and it is very good for unlocking vector algebra from the rules about the order of terms with derivatives.)

We now return to our energy conservation discussion and use our new result, Eq. (27.11), to transform the $\nabla \times \mathbf{B}$ term of Eq. (27.7). That energy equation becomes

$$\mathbf{E} \cdot \mathbf{j} = \epsilon_0 c^2 \nabla \cdot (\mathbf{B} \times \mathbf{E}) + \epsilon_0 c^2 \mathbf{B} \cdot (\nabla \times \mathbf{E}) - \frac{\partial}{\partial t} \left(\frac{1}{2} \epsilon_0 \mathbf{E} \cdot \mathbf{E} \right) \quad (27.12)$$

Now you see we’re almost finished. We have one term which is a nice derivative with respect to t to use for u and another that is a beautiful divergence to represent $\mathbf{S}$. Unfortunately, there is the center term left over, which is neither a divergence nor a derivative with respect to t . So we almost made it, but not quite. After some thought, we look back at the differential equations of Maxwell and discover that $\nabla \times \mathbf{E}$ is, fortunately, equal to $-\partial \mathbf{B} / \partial t$, which means that we can turn the extra term into something that is a pure time derivative:

$$\mathbf{B} \cdot (\nabla \times \mathbf{E}) = \mathbf{B} \cdot \left(-\frac{\partial \mathbf{B}}{\partial t} \right) = -\frac{\partial}{\partial t} \left(\frac{\mathbf{B} \cdot \mathbf{B}}{2} \right).$$

Now we have exactly what we want. Our energy equation reads

$$\mathbf{E} \cdot \mathbf{j} = \nabla \cdot (\epsilon_0 c^2 \mathbf{B} \times \mathbf{E}) - \frac{\partial}{\partial t} \left(\frac{\epsilon_0 c^2}{2} \mathbf{B} \cdot \mathbf{B} + \frac{\epsilon_0}{2} \mathbf{E} \cdot \mathbf{E} \right), \quad (27.13)$$

which is exactly like Eq. (27.6), if we make the *definitions*

$$u = \frac{\epsilon_0}{2} \mathbf{E} \cdot \mathbf{E} + \frac{\epsilon_0 c^2}{2} \mathbf{B} \cdot \mathbf{B} \quad (27.14)$$

and

$$\mathbf{S} = \epsilon_0 c^2 \mathbf{E} \times \mathbf{B}. \quad (27.15)$$

(Reversing the cross product makes the signs come out right.)

Our program was successful. We have an expression for the energy density that is the sum of an “electric” energy density and a “magnetic” energy density, whose forms are just like the ones we found in statics *when we worked out the energy in terms of the fields*. Also, we have found a formula for the energy flow vector of the electromagnetic field. This new vector, $\mathbf{S} = \epsilon_0 c^2 \mathbf{E} \times \mathbf{B}$, is called “Poynting’s vector,” after its discoverer. It tells us the rate at which the field energy moves around in space. The energy which flows through a small area da per second is $\mathbf{S} \cdot \mathbf{n} da$, where $\mathbf{n}$ is the unit vector perpendicular to da . (Now that we have our formulas for u and $\mathbf{S}$, you can forget the derivations if you want.)

27-4 The ambiguity of the field energy

Before we take up some applications of the Poynting formulas [Eqs. (27.14) and (27.15)], we would like to say that we have not really “proved” them. All we did was to find a *possible* “ u ” and a *possible* “ S .” How do we know that by juggling the terms around some more we couldn’t find another formula for “ u ” and another formula for “ S ”? The new S and the new u would be different, but they would still satisfy Eq. (27.6). It’s possible. It can be done, but the forms that have been found always involve various *derivatives* of the field (and always with second-order terms like a second derivative or the square of a first derivative). There are, in fact, an infinite number of different possibilities for u and S , and so far no one has thought of an experimental way to tell which one is right! People have guessed that the simplest one is probably the correct one, but we must say that we do not know for certain what is the actual location in space of the electromagnetic field energy. So we too will take the easy way out and say that the field energy is given by Eq. (27.14). Then the flow vector S must be given by Eq. (27.15).

It is interesting that there seems to be no unique way to resolve the indefiniteness in the location of the field energy. It is sometimes claimed that this problem can be resolved by using the theory of gravitation in the following argument. In the theory of gravity, all energy is the source of gravitational attraction. Therefore the energy density of electricity must be located properly if we are to know in which direction the gravity force acts. As yet, however, no one has done such a delicate experiment that the precise location of the gravitational influence of electromagnetic fields could be determined. That electromagnetic fields alone can be the source of gravitational force is an idea it is hard to do without. It has, in fact, been observed that light is deflected as it passes near the sun—we could say that the sun pulls the light down toward it. Do you not want to allow that the light pulls equally on the sun? Anyway, everyone always accepts the simple expressions we have found for the location of electromagnetic energy and its flow. And although sometimes the results obtained from using them seem strange, nobody has ever found anything wrong with them—that is, no disagreement with experiment. So we will follow the rest of the world—besides, we believe that it is probably perfectly right.

We should make one further remark about the energy formula. In the first place, the energy per unit volume in the field is very simple: It is the electrostatic energy plus the magnetic energy, *if* we write the electrostatic energy in terms of E^2 and the magnetic energy as B^2 . We found two such expressions as *possible* expressions for the energy when we were doing static problems. We also found a number of other formulas for the energy in the electrostatic field, such as $\rho\phi$, which is *equal* to the integral of $\mathbf{E} \cdot \mathbf{E}$ in the electrostatic case. However, in an electrodynamic field the equality failed, and there was no obvious choice as to which was the right one. Now we know which is the right one. Similarly, we have found the formula for the magnetic energy that is correct in general. The right formula for the energy density of *dynamic* fields is Eq. (27.14)

27-5 Examples of energy flow

Our formula for the energy flow vector S is something quite new. We want now to see how it works in some special cases and also to see whether it checks out with anything that we knew before. The first example we will take is light. In a light wave we have an $\mathbf{E}$ vector and a $\mathbf{B}$ vector at right angles to each other and to the direction of the wave propagation. (See Fig. 27-2.) In an electromagnetic wave, the magnitude of $\mathbf{B}$ is equal to $1/c$ times the magnitude of $\mathbf{E}$, and since they are at right angles,

$$|\mathbf{E} \times \mathbf{B}| = \frac{E^2}{c}$$

Therefore, for light, the flow of energy per unit area per second is

$$S = \epsilon_0 c E^2. \quad (27.16)$$

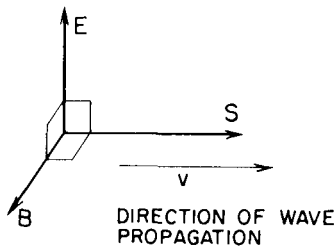


Fig. 27-2. The vectors $\mathbf{E}$, $\mathbf{B}$, and $\mathbf{S}$ for a light wave.

For a light wave in which $E = E_0 \cos \omega(t - x/c)$, the average rate of energy flow per unit area, $\langle S \rangle_{av}$ —which is called the “intensity” of the light—is the mean value of the square of the electric field times $\epsilon_0 c$:

$$\text{Intensity} = \langle S \rangle_{av} = \epsilon_0 c \langle E^2 \rangle_{av}. \quad (27.17)$$

Believe it or not, we have already derived this result in Section 31-3 of Vol. I, when we were studying light. We can believe that it is right because it also checks against something else. When we have a light beam, there is an energy density in space given by Eq. (27.14). Using $cB = E$ for a light wave, we get that

$$u = \frac{\epsilon_0}{2} E^2 + \frac{\epsilon_0 c^2}{2} \left(\frac{E^2}{c^2} \right) = \epsilon_0 E^2.$$

But E varies in space, so the average energy density is

$$\langle u \rangle_{av} = \epsilon_0 \langle E^2 \rangle_{av}. \quad (27.18)$$

Now the wave travels at the speed c , so we should think that the energy that goes through a square meter in a second is c times the amount of energy in one cubic meter. So we would say that

$$\langle S \rangle_{av} = \epsilon_0 c \langle E^2 \rangle_{av}.$$

And it's right; it is the same as Eq. (27.17).

Now we take another example. Here is a rather curious one. We look at the energy flow in a capacitor that we are charging slowly. (We don't want frequencies so high that the capacitor is beginning to look like a resonant cavity, but we don't want DC either.) Suppose we use a circular parallel plate capacitor of our usual kind, as shown in Fig. 27-3. There is a nearly uniform electric field inside which is changing with time. At any instant the total electromagnetic energy inside is u times the volume. If the plates have a radius a and a separation h , the total energy between the plates is

$$U = \left(\frac{\epsilon_0}{2} E^2 \right) (\pi a^2 h). \quad (27.19)$$

This energy changes when E changes. When the capacitor is being charged, the volume between the plates is receiving energy at the rate

$$\frac{dU}{dt} = \epsilon_0 \pi a^2 h E \dot{E}. \quad (27.20)$$

So there must be a flow of energy into that volume from somewhere. Of course you know that it must come in on the charging wires—not at all! It can't enter the space between the plates from that direction, because E is perpendicular to the plates; $E \times B$ must be *parallel* to the plates.

You remember, of course, that there is a magnetic field that circles around the axis when the capacitor is charging. We discussed that in Chapter 23. Using the last of Maxwell's equations, we found that the magnetic field at the edge of the capacitor is given by

$$2\pi a c^2 B = \dot{E} \cdot \pi a^2,$$

or

$$B = \frac{a}{2c^2} \dot{E}.$$

Its direction is shown in Fig. 27-3. So there is an energy flow proportional to $E \times B$ that comes in all around the edges, as shown in the figure. The energy isn't actually coming down the wires, but from the space surrounding the capacitor.

Let's check whether or not the total amount of flow through the whole surface between the edges of the plates checks with the rate of change of the energy inside—it had better; we went through all that work proving Eq. (27.15) to make sure,

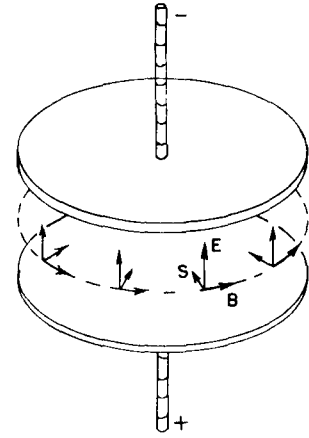


Fig. 27-3. Near a charging capacitor, the Poynting vector S points inward toward the axis.

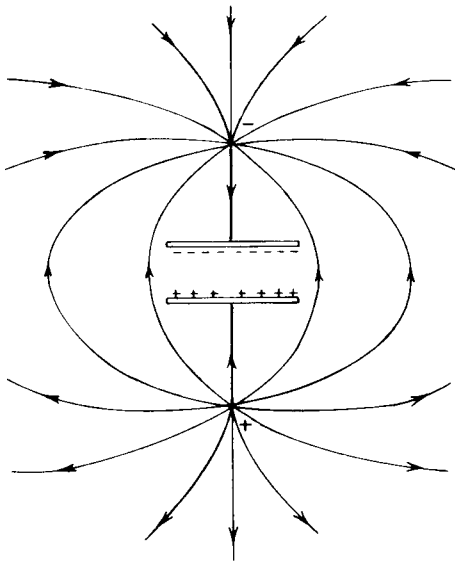


Fig. 27-4. The fields outside a capacitor when it is being charged by bringing two charges from a large distance.

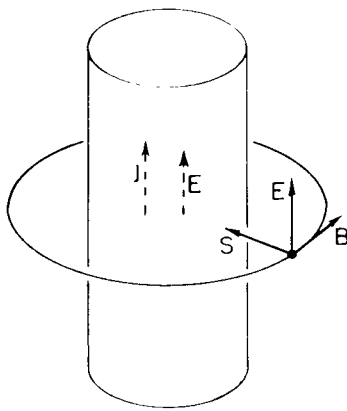


Fig. 27-5 The Poynting vector $\mathbf{S}$ near a wire carrying a current.

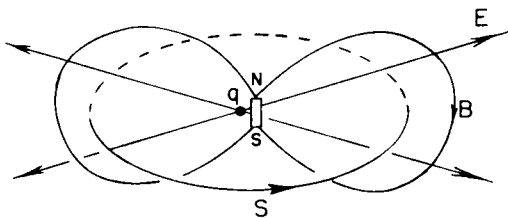


Fig. 27-6. A charge and a magnet produce a Poynting vector that circulates in closed loops.

but let's see. The area of the surface is $2\pi ah$, and $\mathbf{S} = \epsilon_0 c^2 \mathbf{E} \times \mathbf{B}$ is in magnitude

$$\epsilon_0 c^2 E \left(\frac{a}{2c^2} \dot{E} \right),$$

so the total flux of energy is

$$\pi a^2 h \epsilon_0 E \dot{E}.$$

It does check with Eq. (27.20). But it tells us a peculiar thing: that when we are charging a capacitor, the energy is not coming down the wires; it is coming in through the edges of the gap. That's what this theory says!

How can that be? That's *not* an easy question, but here is one way of thinking about it. Suppose that we had some charges above and below the capacitor and far away. When the charges are far away, there is a weak but enormously spread-out field that surrounds the capacitor. (See Fig. 27-4.) Then, as the charges come together, the field gets stronger nearer to the capacitor. So the field energy which is way out moves toward the capacitor and eventually ends up between the plates.

As another example, we ask what happens in a piece of resistance wire when it is carrying a current. Since the wire has resistance, there is an electric field along it, driving the current. Because there is a potential drop along the wire, there is also an electric field just outside the wire, parallel to the surface. (See Fig. 27-5.) There is, in addition, a magnetic field which goes around the wire because of the current. The $\mathbf{E}$ and $\mathbf{B}$ are at right angles; therefore there is a Poynting vector directed radially inward, as shown in the figure. There is a flow of energy into the wire all around. It is, of course, equal to the energy being lost in the wire in the form of heat. So our "crazy" theory says that the electrons are getting their energy to generate heat because of the energy flowing into the wire from the field outside. Intuition would seem to tell us that the electrons get their energy from being pushed along the wire, so the energy should be flowing down (or up) along the wire. But the theory says that the electrons are really being pushed by an electric field, which has come from some charges very far away, and that the electrons get their energy for generating heat from these fields. The energy somehow flows from the distant charges into a wide area of space and then inward to the wire.

Finally, in order to really convince you that this theory is obviously nuts, we will take one more example—an example in which an electric charge and a magnet are *at rest* near each other—both sitting quite still. Suppose we take the example of a point charge sitting near the center of a bar magnet, as shown in Fig. 27-6. Everything is at rest, so the energy is not changing with time. Also, $\mathbf{E}$ and $\mathbf{B}$ are quite static. But the Poynting vector says that there is a flow of energy, because there is an $\mathbf{E} \times \mathbf{B}$ that is not zero. If you look at the energy flow, you find that it just circulates around and around. There isn't any change in the energy anywhere—everything which flows into one volume flows out again. It is like incompressible water flowing around. So there is a circulation of energy in this so-called static condition. How absurd it gets!

Perhaps it isn't so terribly puzzling, though, when you remember that what we called a "static" magnet is really a circulating permanent current. In a permanent magnet the electrons are spinning permanently inside. So maybe a circulation of the energy outside isn't so queer after all.

You no doubt begin to get the impression that the Poynting theory at least partially violates your intuition as to where energy is located in an electromagnetic field. You might believe that you must revamp all your intuitions, and, therefore have a lot of things to study here. But it seems really not necessary. You don't need to feel that you will be in great trouble if you forget once in a while that the energy in a wire is flowing into the wire from the outside, rather than along the wire. It seems to be only rarely of value, when using the idea of energy conservation, to notice in detail what path the energy is taking. The circulation of energy around a magnet and a charge seems, in most circumstances, to be quite unimportant. It is not a vital detail, but it is clear that our ordinary intuitions are quite wrong.

27-6 Field momentum

Next we would like to talk about the *momentum* in the electromagnetic field. Just as the field has energy, it will have a certain momentum per unit volume. Let us call that momentum density $\mathbf{g}$. Of course, momentum has various possible directions, so that $\mathbf{g}$ must be a vector. Let's talk about one component at a time; first, we take the x -component. Since each component of momentum is conserved we should be able to write down a law that looks something like this:

$$-\frac{\partial}{\partial t} \left(\text{momentum of matter} \right)_x = \frac{\partial g_x}{\partial t} + \left(\text{momentum outflow} \right)_x.$$

The left side is easy. The rate-of-change of the momentum of matter is just the force on it. For a particle, it is $\mathbf{F} = q(\mathbf{E} + \mathbf{v} \times \mathbf{B})$; for a distribution of charges, the force per unit volume is $(\rho\mathbf{E} + \mathbf{j} \times \mathbf{B})$. The "momentum outflow" term, however, is strange. It cannot be the divergence of a vector because it is not a scalar; it is, rather, an x -component of some vector. Anyway, it should probably look something like

$$\frac{\partial a}{\partial x} + \frac{\partial b}{\partial y} + \frac{\partial c}{\partial z},$$

because the x -momentum could be flowing in any one of the three directions. In any case, whatever a , b , and c are, the combination is supposed to equal the outflow of the x -momentum.

Now the game would be to write $\rho\mathbf{E} + \mathbf{j} \times \mathbf{B}$ in terms only of $\mathbf{E}$ and $\mathbf{B}$ —eliminating ρ and $\mathbf{j}$ by using Maxwell's equations—and then to juggle terms and make substitutions to get it into a form that looks like

$$\frac{\partial g_x}{\partial t} + \frac{\partial a}{\partial x} + \frac{\partial b}{\partial y} + \frac{\partial c}{\partial z}.$$

Then, by identifying terms, we would have expressions for g_x , a , b , and c . It's a lot of work, and we are not going to do it. Instead, we are only going to find an expression for $\mathbf{g}$, the momentum density—and by a different route.

There is an important theorem in mechanics which is this: whenever there is a flow of energy in any circumstance at all (field energy or any other kind of energy), the energy flowing through a unit area per unit time, when multiplied by $1/c^2$, is equal to the momentum per unit volume in the space. In the special case of electrodynamics, this theorem gives the result that $\mathbf{g}$ is $1/c^2$ times the Poynting vector

$$\mathbf{g} = \frac{1}{c^2} \mathbf{S}. \quad (27.21)$$

So the Poynting vector gives not only energy flow but, if you divide by c^2 , also the momentum density. The same result would come out of the other analysis we suggested, but it is more interesting to notice this more general result. We will now give a number of interesting examples and arguments to convince you that the general theorem is true.

First example: Suppose that we have a lot of particles in a box—let's say N per cubic meter—and that they are moving along with some velocity $\mathbf{v}$. Now let's consider an imaginary plane surface perpendicular to $\mathbf{v}$. The energy flow through a unit area of this surface per second is equal to Nv , the number which flow through the surface per second, times the energy carried by each one. The energy in each particle is $m_0 c^2 / \sqrt{1 - v^2/c^2}$. So the energy flow per second is

$$Nv \frac{m_0 c^2}{\sqrt{1 - v^2/c^2}}.$$

But the momentum of each particle is $m_0 v / \sqrt{1 - v^2/c^2}$, so the *density* of momentum is

$$N \frac{m_0 v}{\sqrt{1 - v^2/c^2}},$$

which is just $1/c^2$ times the energy flow—as the theorem says. So the theorem is true for a bunch of particles.

It is also true for light. When we studied light in Volume I, we saw that when the energy is absorbed from a light beam, a certain amount of momentum is delivered to the absorber. We have, in fact, shown in Chapter 36 of Vol. I that the momentum is $1/c$ times the energy absorbed [Eq. (36.24) of Vol. I]. If we let U_0 be the energy arriving at a unit area per second, then the momentum arriving at a unit area per second is U_0/c . But the momentum is travelling at the speed c , so its *density* in front of the absorber must be U_0/c^2 . So again the theorem is right.

Finally we will give an argument due to Einstein which demonstrates the same thing once more. Suppose that we have a railroad car on wheels (assumed frictionless) with a certain big mass M . At one end there is a device which will shoot out some particles or light (or anything, it doesn't make any difference what it is), which are then stopped at the opposite end of the car. There was some energy originally at one end—say the energy U indicated in Fig. 27-7(a)—and then later it is at the opposite end, as shown in Fig. 27-7(c). The energy U has been displaced the distance L , the length of the car. Now the energy U has the mass U/c^2 , so if the car stayed still, the center of gravity of the car would be moved. Einstein didn't like the idea that the center of gravity of an object could be moved by fooling around only on the inside, so he assumed that it is impossible to move the center of gravity by doing anything inside. But if that is the case, when we moved the energy U from one end to the other, the whole car must have recoiled some distance x , as shown in part (c) of the figure. You can see, in fact, that the total mass of the car, times x , must equal the mass of the energy moved, U/c^2 times L (assuming that U/c^2 is much less than M):

$$Mx = \frac{U}{c^2} L. \quad (27.22)$$

Let's now look at the special case of the energy being carried by a light flash. (The argument would work as well for particles, but we will follow Einstein, who was interested in the problem of light.) What causes the car to be moved? Einstein argued as follows: When the light is emitted there must be a recoil, some unknown recoil with momentum p . It is this recoil which makes the car roll backward. The recoil velocity v of the car will be this momentum divided by the mass of the car:

$$v = \frac{p}{M}.$$

The car moves with this velocity until the light energy U gets to the opposite end. Then, when it hits, it gives back its momentum and stops the car. If x is small, then the time the car moves is nearly equal to L/c ; so we have that

$$x = vt = v \frac{L}{c} = \frac{p}{M} \frac{L}{c}.$$

Putting this x in Eq. (27.22), we get that

$$p = \frac{U}{c}.$$

Again we have the relation of energy and momentum for light. Dividing by c to get the momentum density $g = p/c$, we get once more that

$$g = \frac{U}{c^2}. \quad (27.23)$$

You may well wonder: What is so important about the center-of-gravity theorem? Maybe it is wrong. Perhaps, but then we would also lose the conservation of angular momentum. Suppose that our boxcar is moving along a track at some speed v and that we shoot some light energy from the *top* to the *bottom* of the car—say, from A to B in Fig. 27-8. Now we look at the angular momentum of the system about the point P . Before the energy U leaves A , it has the mass

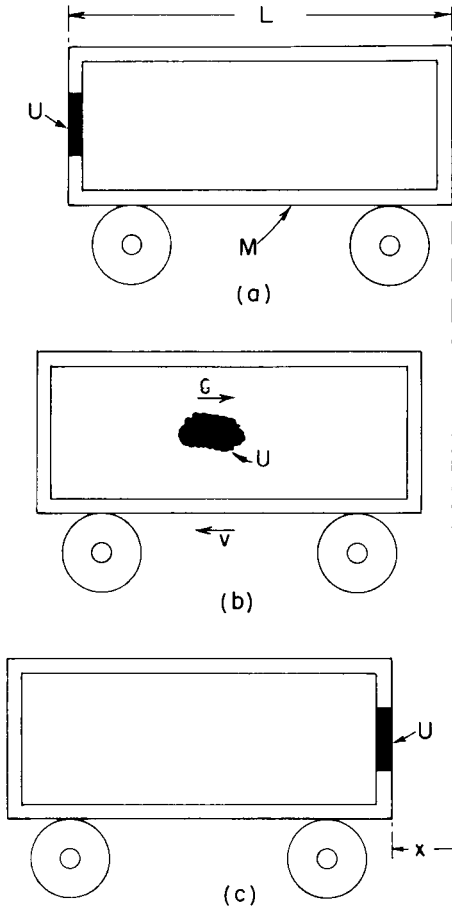


Fig. 27-7. The energy U in motion at the speed c carries the momentum U/c .

$m = U^2/c$ and the velocity v , so it has the angular momentum mvr_a . When it arrives at B , it has the same mass and, if the *linear* momentum of the whole boxcar is not to change, it must still have the velocity v . Its angular momentum about P is then mvr_B . The angular momentum will be changed *unless* the right recoil momentum was given to the car when the light was emitted—that is, unless the light carries the momentum U/c . It turns out that the angular momentum conservation and the theorem of center-of-gravity are closely related in the relativity theory. So the conservation of angular momentum would also be destroyed if our theorem were not true. At any rate, it does turn out to be a true general law, and in the case of electrodynamics we can use it to get the momentum in the field.

We will mention two further examples of momentum in the electromagnetic field. We pointed out in Section 26-2 the failure of the law of action and reaction when two charged particles were moving on orthogonal trajectories. The forces on the two particles don't balance out, so the action and reaction are not equal, therefore the net momentum of the matter must be changing. It is not conserved. But the momentum in the field is also changing in such a situation. If you work out the amount of momentum given by the Poynting vector, it is not constant. However, the change of the particle momenta is just made up by the field momentum, so the total momentum of particles plus field is conserved.

Finally, another example is the situation with the magnet and the charge, shown in Fig. 27-6. We were unhappy to find that energy was flowing around in circles, but now, since we know that energy flow and momentum are proportional, we know also that there is momentum circulating in the space. But a *circulating* momentum means that there is *angular* momentum. So there is *angular* momentum in the field. Do you remember the paradox we described in Section 17-4 about a solenoid and some charges mounted on a disc? It seemed that when the current turned off, the whole disc should start to turn. The puzzle was: Where did the angular momentum come from? The answer is that if you have a magnetic field and some charges, there will be some angular momentum in the field. It must have been put there when the field was built up. When the field is turned off, the angular momentum is given back. So the disc in the paradox *would* start rotating. This mystic circulating flow of energy, which at first seemed so ridiculous, is absolutely necessary. There is really a momentum flow. It is needed to maintain the conservation of angular momentum in the whole world.

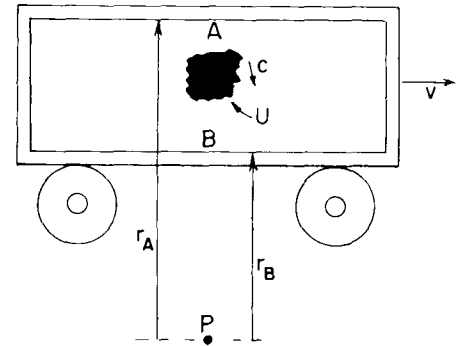


Fig. 27-8. The energy U must carry the momentum U/c if the angular momentum about P is to be conserved.

Electromagnetic Mass

28-1 The field energy of a point charge

In bringing together relativity and Maxwell's equations, we have finished our main work on the theory of electromagnetism. There are, of course, some details we have skipped over and one large area that we will be concerned with in the future—the interaction of electromagnetic fields with matter. But we want to stop for a moment to show you that this tremendous edifice, which is such a beautiful success in explaining so many phenomena, ultimately falls on its face. When you follow any of our physics too far, you find that it always gets into some kind of trouble. Now we want to discuss a serious trouble—the failure of the classical electromagnetic theory. You can appreciate that there is a failure of all classical physics because of the quantum-mechanical effects. Classical mechanics is a mathematically consistent theory; it just doesn't agree with experience. It is interesting, though, that the classical theory of electromagnetism is an unsatisfactory theory all by itself. There are difficulties associated with the *ideas* of Maxwell's theory which are not solved by and not directly associated with quantum mechanics. You may say, "Perhaps there's no use worrying about these difficulties. Since the quantum mechanics is going to change the laws of electrodynamics, we should wait to see what difficulties there are after the modification." However, when electromagnetism is joined to quantum mechanics, the difficulties remain. So it will not be a waste of our time now to look at what these difficulties are. Also, they are of great historical importance. Furthermore, you may get some feeling of accomplishment from being able to go far enough with the theory to see everything—including all of its troubles.

The difficulty we speak of is associated with the concepts of electromagnetic momentum and energy, when applied to the electron or any charged particle. The concepts of simple charged particles and the electromagnetic field are in some way inconsistent. To describe the difficulty, we begin by doing some exercises with our energy and momentum concepts.

First, we compute the energy of a charged particle. Suppose we take a simple model of an electron in which all of its charge q is uniformly distributed on the surface of a sphere of radius a , which we may take to be zero for the special case of a point charge. Now let's calculate the energy in the electromagnetic field. If the charge is standing still, there is no magnetic field, and the energy per unit volume is proportional to the square of the electric field. The magnitude of the electric field is $q/4\pi\epsilon_0 r^2$, and the energy density is

$$u = \frac{\epsilon_0}{2} E^2 = \frac{q^2}{32\pi^2\epsilon_0 r^4}.$$

To get the total energy, we must integrate this density over all space. Using the volume element $4\pi r^2 dr$, the total energy, which we will call U_{elec} , is

$$U_{\text{elec}} = \int \frac{q^2}{8\pi\epsilon_0 r^2} dr.$$

This is readily integrated. The lower limit is a , and the upper limit is ∞ , so

$$U_{\text{elec}} = \frac{1}{2} \frac{q^2}{4\pi\epsilon_0} \frac{1}{a}. \quad (28.1)$$

28-1 The field energy of a point charge

28-2 The field momentum of a moving charge

28-3 Electromagnetic mass

28-4 The force of an electron on itself

28-5 Attempts to modify the Maxwell theory

28-6 The nuclear force field

If we use the electronic charge q_e for q and the symbol e^2 for $q_e^2/4\pi\epsilon_0$, then

$$U_{\text{elec}} = \frac{1}{2} \frac{e^2}{a}. \quad (28.2)$$

It is all fine until we set a equal to zero for a point charge—there's the great difficulty. Because the energy of the field varies inversely as the fourth power of the distance from the center, its volume integral is infinite. There is an infinite amount of energy in the field surrounding a point charge.

What's wrong with an infinite energy? If the energy can't get out, but must stay there forever, is there any real difficulty with an infinite energy? Of course, a quantity that comes out infinite may be annoying, but what really matters is only whether there are any *observable* physical effects. To answer that question, we must turn to something else besides the energy. Suppose we ask how the energy *changes* when we *move* the charge. Then, if the *changes* are infinite, we will be in trouble.

28-2 The field momentum of a moving charge

Suppose an electron is moving at a uniform velocity through space, assuming for a moment that the velocity is low compared with the speed of light. Associated with this moving electron there is a momentum—even if the electron had no mass before it was charged—because of the momentum in the electromagnetic field. We can show that the field momentum is in the direction of the velocity v of the charge and is, for small velocities, proportional to v . For a point P at the distance r from the center of the charge and at the angle θ with respect to the line of motion (see Fig. 28-1) the electric field is radial and, as we have seen, the magnetic field is $v \times E/c^2$. The momentum density, Eq. (27.21), is

$$g = \epsilon_0 E \times B.$$

It is directed obliquely toward the line of motion, as shown in the figure, and has the magnitude

$$g = \frac{\epsilon_0 v}{c^2} E^2 \sin \theta.$$

The fields are symmetric about the line of motion, so when we integrate over space, the transverse components will sum to zero, giving a resultant momentum parallel to v . The component of g in this direction is $g \sin \theta$, which we must integrate over all space. We take as our volume element a ring with its plane perpendicular to v , as shown in Fig. 28-2. Its volume is $2\pi r^2 \sin \theta d\theta dr$. The total momentum is then

$$p = \int \frac{\epsilon_0 v}{c^2} E^2 \sin^2 \theta 2\pi r^2 \sin \theta d\theta dr.$$

Since E is independent of θ (for $v \ll c$), we can immediately integrate over θ ; the integral is

$$\int \sin^3 \theta d\theta = - \int (1 - \cos^2 \theta) d(\cos \theta) = -\cos \theta + \frac{\cos^3 \theta}{3}.$$

The limits of θ are 0 and π , so the θ -integral gives merely a factor of $4/3$, and

$$p = \frac{8\pi}{3} \frac{\epsilon_0 v}{c^2} \int E^2 r^2 dr.$$

The integral (for $v \ll c$) is the one we have just evaluated to find the energy; it is $q^2/16\pi^2\epsilon_0 a$, and

$$p = \frac{2}{3} \frac{q^2}{4\pi\epsilon_0} \frac{v}{ac^2},$$

or

$$p = \frac{2}{3} \frac{e^2}{ac^2} v. \quad (28.3)$$

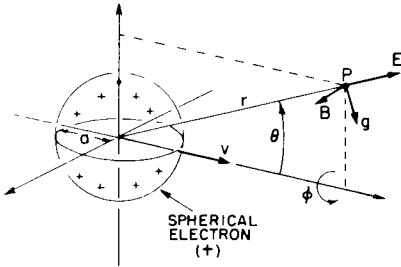


Fig. 28-1. The fields E and B and the momentum density g for a positive electron. For a negative electron, E and B are reversed but g is not.

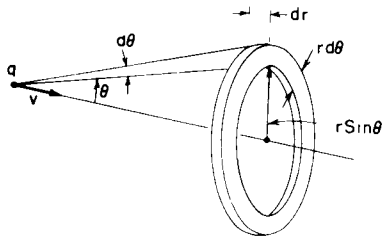


Fig. 28-2. The volume element $2\pi r^2 \sin \theta d\theta dr$ used for calculating the field momentum.

The momentum in the field—the electromagnetic momentum—is proportional to v . It is just what we should have for a particle with the mass equal to the coefficient of v . We can, therefore, call this coefficient the *electromagnetic mass*, m_{elec} , and write it as

$$m_{\text{elec}} = \frac{2}{3} \frac{e^2}{ac^2}. \quad (28.4)$$

28-3 Electromagnetic mass

Where does the mass come from? In our laws of mechanics we have supposed that every object “carries” a thing we call the mass—which also means that it “carries” a momentum proportional to its velocity. Now we discover that it is understandable that a charged particle carries a momentum proportional to its velocity. It might, in fact, be that the mass is just the effect of electrodynamics. The origin of mass has until now been unexplained. We have at last in the theory of electrodynamics a grand opportunity to understand something that we never understood before. It comes out of the blue—or rather, from Maxwell and Poynting—that any charged particle will have a momentum proportional to its velocity just from electromagnetic influences.

Let's be conservative and say, for a moment, that there are two kinds of mass—that the total momentum of an object could be the sum of a mechanical momentum and the electromagnetic momentum. The mechanical momentum is the “mechanical” mass, m_{mech} , times v . In experiments where we measure the mass of a particle by seeing how much momentum it has, or how it swings around in an orbit, we are measuring the total mass. We say generally that the momentum is the total mass ($m_{\text{mech}} + m_{\text{elec}}$) times the velocity. So the observed mass can consist of two pieces (or possibly more if we include other fields): a mechanical piece plus an electromagnetic piece. We know that there is definitely an electromagnetic piece, and we have a formula for it. And there is the thrilling possibility that the mechanical piece is not there at all—that the mass is all electromagnetic.

Let's see what size the electron must have if there is to be no mechanical mass. We can find out by setting the electromagnetic mass of Eq. (28.4) equal to the observed mass m_e of an electron. We find

$$a = \frac{2}{3} \frac{e^2}{m_e c^2}. \quad (28.5)$$

The quantity

$$r_0 = \frac{e^2}{m_e c^2} \quad (28.6)$$

is called the “classical electron radius”; it has the numerical value 2.82×10^{-13} cm, about one one-hundred-thousandth of the diameter of an atom.

Why is r_0 called the electron radius, rather than our a ? Because we could equally well do the same calculation with other assumed distributions of charges—the charge might be spread uniformly through the volume of a sphere or it might be smeared out like a fuzzy ball. For any particular assumption the factor $2/3$ would change to some other fraction. For instance, for a charge uniformly distributed throughout the volume of a sphere, the $2/3$ gets replaced by $4/5$. Rather than to argue over which distribution is correct, it was decided to define r_0 as the “nominal” radius. Then different theories could supply their pet coefficients.

Let's pursue our electromagnetic theory of mass. Our calculation was for $v \ll c$; what happens if we go to high velocities? Early attempts led to a certain amount of confusion, but Lorentz realized that the charged sphere would contract into an ellipsoid at high velocities and that the fields would change in accordance with the formulas (26.6) and (26.7) we derived for the relativistic case in Chapter 26. If you carry through the integrals for $\mathbf{p}$ in that case, you find that for an arbitrary velocity v , the momentum is altered by the factor $1/\sqrt{1 - v^2/c^2}$:

$$\mathbf{p} = \frac{2}{3} \frac{e^2}{ac^2} \frac{\mathbf{v}}{\sqrt{1 - v^2/c^2}}. \quad (28.7)$$

In other words, the electromagnetic mass rises with velocity inversely as $\sqrt{1 - v^2/c^2}$ —a discovery that was made before the theory of relativity.

Early experiments were proposed to measure the changes with velocity in the observed mass of a particle in order to determine how much of the mass was mechanical and how much was electrical. It was believed at the time that the electrical part *would* vary with velocity, whereas the mechanical part would *not*. But while the experiments were being done, the theorists were also at work. Soon the theory of relativity was developed, which proposed that no matter what the origin of the mass, it *all* should vary as $m_0/\sqrt{1 - v^2/c^2}$. Equation (28.7) was the beginning of the theory that mass depended on velocity.

Let's now go back to our calculation of the energy in the field, which led to Eq. (28.2). According to the theory of relativity, the energy U will have the mass U/c^2 ; Eq. (28.2) then says that the field of the electron should have the mass

$$m'_{\text{elec}} = \frac{U_{\text{elec}}}{c^2} = \frac{1}{2} \frac{e^2}{ac^2}, \quad (28.8)$$

which is not the same as the electromagnetic mass, m_{elec} , of Eq. (28.4). In fact, if we just combine Eqs. (28.2) and (28.4), we would write

$$U_{\text{elec}} = \frac{3}{4} m_{\text{elec}} c^2.$$

This formula was discovered before relativity, and when Einstein and others began to realize that it must always be that $U = mc^2$, there was great confusion.

28-4 The force of an electron on itself

The discrepancy between the two formulas for the electromagnetic mass is especially annoying, because we have carefully proved that the theory of electrodynamics is consistent with the principle of relativity. Yet the theory of relativity implies without question that the momentum must be the same as the energy times v/c^2 . So we are in some kind of trouble; we must have made a mistake. We did not make an algebraic mistake in our calculations, but we have left something out.

In deriving our equations for energy and momentum, we assumed the conservation laws. We assumed that *all* forces were taken into account and that any work done and any momentum carried by other “nonelectrical” machinery was included. Now if we have a sphere of charge, the electrical forces are all repulsive and an electron would tend to fly apart. Because the system has unbalanced forces, we can get all kinds of errors in the laws relating energy and momentum. To get a *consistent* picture, we must imagine that something holds the electron together. The charges must be *held* to the sphere by some kind of rubber bands—something that keeps the charges from flying off. It was first pointed out by Poincaré that the rubber bands—or whatever it is that holds the electron together—must be included in the energy and momentum calculations. For this reason the extra nonelectrical forces are also known by the more elegant name “the Poincaré stresses.” If the extra forces are included in the calculations, the masses obtained in two ways are changed (in a way that depends on the detailed assumptions). And the results are consistent with relativity; i.e., the mass that comes out from the momentum calculation is the same as the one that comes from the energy calculation. However, both of them contain *two* contributions: an electromagnetic mass and contribution from the Poincaré stresses. Only when the two are added together do we get a consistent theory.

It is therefore impossible to get all the mass to be electromagnetic in the way we hoped. It is not a legal theory if we have nothing but electrodynamics. Something else has to be added. Whatever you call them—“rubber bands,” or “Poincaré stresses,” or something else—there have to be other forces in nature to make a consistent theory of this kind.

Clearly, as soon as we have to put forces on the inside of the electron, the beauty of the whole idea begins to disappear. Things get very complicated. You would want to ask: How strong are the stresses? How does the electron shake? Does it oscillate? What are all its internal properties? And so on. It might be possible that an electron does have some complicated internal properties. If we made a theory of the electron along these lines, it would predict odd properties, like modes of oscillation, which haven't apparently been observed. We say "apparently" because we observe a lot of things in nature that still do not make sense. We may someday find out that one of the things we don't understand today (for example, the muon) can, in fact, be explained as an oscillation of the Poincaré stresses. It doesn't seem likely, but no one can say for sure. There are so many things about fundamental particles that we still don't understand. Anyway, the complex structure implied by this theory is undesirable, and the attempt to explain all mass in terms of electromagnetism—at least in the way we have described—has led to a blind alley.

We would like to think a little more about why we say we have a mass when the momentum in the field is proportional to the velocity. Easy! The mass is the coefficient between momentum and velocity. But we can look at the mass in another way: a particle has mass if you have to exert a force in order to accelerate it. So it may help our understanding if we look a little more closely at where the forces come from. How do we know that there has to be a force? Because we have proved the law of the conservation of momentum for the fields. If we have a charged particle and push on it for awhile, there will be some momentum in the electromagnetic field. Momentum must have been poured into the field somehow. Therefore there must have been a force pushing on the electron in order to get it going—a force in addition to that required by its mechanical inertia, a force due to its electromagnetic interaction. And there must be a corresponding force back on the "pusher." But where does that force come from?

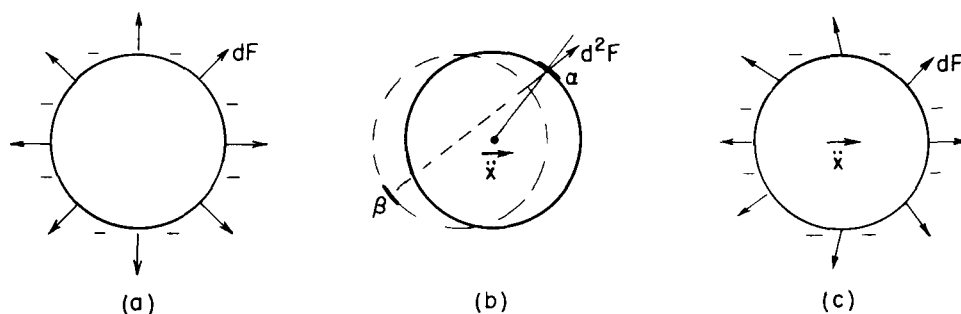


Fig. 28-3. The self-force on an accelerating electron is not zero because of the retardation. (By dF we mean the force on a surface element da ; by d^2F we mean the force on the surface element da_α from the charge on the surface element da_β .)

The picture is something like this. We can think of the electron as a charged sphere. When it is at rest, each piece of charge repels electrically each other piece, but the forces all balance in pairs, so that there is no *net* force. [See Fig. 28-3(a).] However, when the electron is being accelerated, the forces will no longer be in balance because of the fact that the electromagnetic influences take time to go from one piece to another. For instance, the force on the piece α in Fig. 28-3(b) from a piece β on the opposite side depends on the position of β at an earlier time, as shown. Both the magnitude and direction of the force depend on the motion of the charge. If the charge is accelerating, the forces on various parts of the electron might be as shown in Fig. 28-3(c). When all these forces are added up, they don't cancel out. They would cancel for a uniform velocity, even though it looks at first glance as though the retardation would give an unbalanced force even for a uniform velocity. But it turns out that there is no net force unless the electron is being accelerated. With acceleration, if we look at the forces between

the various parts of the electron, action and reaction are not exactly equal, and the electron exerts a force *on itself* that tries to hold back the acceleration. It holds itself back by its own bootstraps.

It is possible, but difficult, to calculate this self-reaction force; however, we don't want to go into such an elaborate calculation here. We will tell you what the result is for the special case of relatively uncomplicated motion in one dimension, say x . Then, the self-force can be written in a series. The first term in the series depends on the acceleration $\ddot{x}$, the next term is proportional to $\dot{x}$, and so on.* The result is

$$F = \alpha \frac{e^2}{ac^2} \dot{x} - \frac{2}{3} \frac{e^2}{c^3} \ddot{x} + \gamma \frac{e^2 a}{c^4} \ddot{\ddot{x}} + \cdots \quad (28.9)$$

where α and γ are numerical coefficients of the order of 1. The coefficient α of the $\ddot{x}$ term depends on what charge distribution is assumed; if the charge is distributed uniformly on a sphere, then $\alpha = 2/3$. So there is a term, proportional to the acceleration, which varies inversely as the radius a of the electron and agrees exactly with the value we got in Eq. (28.4) for m_{elec} . If the charge distribution is chosen to be different, so that α is changed, the fraction $2/3$ in Eq. (28.4) would be changed in the same way. The term in $\ddot{x}$ is *independent* of the assumed radius a , and also of the assumed distribution of the charge; its coefficient is *always* $2/3$. The next term is proportional to the radius a , and its coefficient γ depends on the charge distribution. You will notice that if we let the electron radius a go to zero, the last term (and all higher terms) will go to zero; the second term remains constant, but the first term—the electromagnetic mass—goes to infinity. And we can see that the infinity arises because of the force of one part of the electron on another—because we have allowed what is perhaps a silly thing, the possibility of the “point” electron acting on itself.

28-5 Attempts to modify the Maxwell theory

We would like now to discuss how it might be possible to modify Maxwell's theory of electrodynamics so that the idea of an electron as a simple point charge could be maintained. Many attempts have been made, and some of the theories were even able to arrange things so that all the electron mass was electromagnetic. But all of these theories have died. It is still interesting to discuss some of the possibilities that have been suggested—to see the struggles of the human mind.

We started out our theory of electricity by talking about the interaction of one charge with another. Then we made up a theory of these interacting charges and ended up with a field theory. We believe it so much that we allow it to tell us about the force of one part of an electron on another. Perhaps the entire difficulty is that electrons do not act on themselves; perhaps we are making too great an extrapolation from the interaction of separate electrons to the idea that an electron interacts with itself. Therefore some theories have been proposed in which the possibility that an electron acts on itself is ruled out. Then there is no longer the infinity due to the self-action. Also, there is no longer any electromagnetic mass associated with the particle; all the mass is back to being mechanical, but there are new difficulties in the theory.

We must say immediately that such theories require a modification of the idea of the electromagnetic field. You remember we said at the start that the force on a particle at any point was determined by just two quantities— $\mathbf{E}$ and $\mathbf{B}$. If we abandon the “self-force” this can no longer be true, because if there is an electron in a certain place, the force isn't given by the total $\mathbf{E}$ and $\mathbf{B}$, but by only those parts due to *other* charges. So we have to keep track always of how much of $\mathbf{E}$ and $\mathbf{B}$ is due to the charge on which you are calculating the force and how much is due to the other charges. This makes the theory much more elaborate, but it gets rid of the difficulty of the infinity.

* We are using the notation: $x = dx/dt$, $\dot{x} = d^2x/dt^2$, $\ddot{x} = d^3x/dt^3$, etc.

So we can, *if we want to*, say that there is no such thing as the electron acting upon itself, and throw away the whole set of forces in Eq. (28.9). However, we have then thrown away the baby with the bath! Because the second term in Eq. (28.9), the term in $\ddot{x}$, is needed. That force does something very definite. If you throw it away, you're in trouble again. When we accelerate a charge, it radiates electromagnetic waves, so it loses energy. Therefore, to accelerate a charge, we must require more force than is required to accelerate a neutral object of the same mass; otherwise energy wouldn't be conserved. The rate at which we do work on an accelerating charge must be equal to the rate of loss of energy per second by radiation. We have talked about this effect before—it is called the radiation resistance. We still have to answer the question: Where does the extra force, against which we must do this work, come from? When a big antenna is radiating, the forces come from the influence of one part of the antenna current on another. For a single accelerating electron radiating into otherwise empty space, there would seem to be only one place the force could come from—the action of one part of the electron on another part.

We found back in Chapter 32 of Vol. I that an oscillating charge radiates energy at the rate

$$\frac{dW}{dt} = \frac{2}{3} \frac{e^2 (\ddot{x})^2}{c^3}. \quad (28.10)$$

Let's see what we get for the rate of doing work *on* an electron against the bootstrap force of Eq. (28.9). The rate of work is the force times the velocity, or $F \dot{x}$:

$$\frac{dW}{dt} = \alpha \frac{e^2}{ac^2} \ddot{x} \dot{x} - \frac{2}{3} \frac{e^2}{c^3} \ddot{x} \dot{x} + \dots \quad (28.11)$$

The first term is proportional to $d\dot{x}^2/dt$, and therefore just corresponds to the rate of change of the kinetic energy $\frac{1}{2} m v^2$ associated with the electromagnetic mass. The second term should correspond to the radiated power in Eq. (28.10). But it is different. The discrepancy comes from the fact that the term in Eq. (28.11) is generally true, whereas Eq. (28.10) is right only for an *oscillating* charge. We can show that the two are equivalent if the motion of the charge is periodic. To do that, we rewrite the second term of Eq. (28.11) as

$$- \frac{2}{3} \frac{e^2}{c^3} \frac{d}{dt} (\dot{x} \ddot{x}) + \frac{2}{3} \frac{e^2}{c^3} (\ddot{x})^2,$$

which is just an algebraic transformation. If the motion of the electron is periodic, the quantity $\dot{x} \ddot{x}$ returns periodically to the same value, so that if we take the *average* of its time derivative, we get zero. The second term, however, is always positive (it's a square), so its average is also positive. This term gives the net work done and is just equal to Eq. (28.10).

The term in $\ddot{x}$ of the bootstrap force is required in order to have energy conservation in radiating systems, and we can't throw it away. It was, in fact, one of the triumphs of Lorentz to show that there is such a force and that it comes from the action of the electron on itself. We must believe in the idea of the action of the electron on itself, and we *need* the term in $\ddot{x}$. The problem is how we can get that term without getting the first term in Eq. (28.9), which gives all the trouble. We don't know how. You see that the classical electron theory has pushed itself into a tight corner.

There have been several other attempts to modify the laws in order to straighten the thing out. One way, proposed by Born and Infeld, is to change the Maxwell equations in a complicated way so that they are no longer linear. Then the electromagnetic energy and momentum can be made to come out finite. But the laws they suggest predict phenomena which have never been observed. Their theory also suffers from another difficulty we will come to later, which is common to all the attempts to avoid the troubles we have described.

The following peculiar possibility was suggested by Dirac. He said: Let's admit that an electron acts on itself through the *second* term in Eq. (28.9) but not through the first. He then had an ingenious idea for getting rid of one but not the

other. Look, he said, we made a special assumption when we took only the *retarded* wave solutions of Maxwell's equations; if we were to take the *advanced* waves instead, we would get something different. The formula for the self-force would be

$$F = \alpha \frac{e^2}{ac^2} \ddot{x} + \frac{2}{3} \frac{e^2}{c^3} \ddot{x} + \gamma \frac{e^2 a}{c^4} \dot{x} \quad (28.12)$$

This equation is just like Eq. (28.9) except for the sign of the second term—and some higher terms—of the series [Changing from retarded to advanced waves is just changing the *sign* of the delay which, it is not hard to see, is equivalent to changing the sign of t everywhere. The only effect on Eq. (28.9) is to change the sign of all the odd time derivatives.] So, Dirac said, let's make the new rule that an electron acts on itself by one-half the *difference* of the retarded and advanced fields which it produces. The difference of Eqs. (28.9) and (28.12), divided by two, is then

$$F = -\frac{2}{3} \frac{e^2}{c^3} \ddot{x} + \text{higher terms.}$$

In all the higher terms, the radius a appears to some positive power in the numerator. Therefore, when we go to the limit of a point charge, we get only the one term—just what is needed. In this way, Dirac got the radiation resistance force and none of the inertial forces. There is no electromagnetic mass, and the classical theory is saved—but at the expense of an arbitrary assumption about the self-force.

The arbitrariness of the extra assumption of Dirac was removed, to some extent at least, by Wheeler and Feynman, who proposed a still stranger theory. They suggest that point charges interact *only* with other charges, but that the interaction is half through the advanced and half through the retarded waves. It turns out, most surprisingly, that in most situations you won't see any effects of the advanced waves, but they do have the effect of producing just the radiation reaction force. The radiation resistance is *not* due to the electron acting on itself, but from the following peculiar effect. When an electron is accelerated at the time t , it shakes all the other charges in the world at a *later* time $t' = t + r/c$ (where r is the distance to the other charge), because of the *retarded* waves. But then these other charges react back on the original electron through their *advanced* waves, which will arrive at the time t'' , equal to t' minus r/c , which is, of course, just t . (They also react back with their retarded waves too, but that just corresponds to the normal “reflected” waves.) The combination of the advanced and retarded waves means that at the instant it is accelerated an oscillating charge feels a force from all the charges that are “going to” absorb its radiated waves. You see what tight knots people have gotten into in trying to get a theory of the electron!

We'll describe now still another kind of theory, to show the kind of things that people think of when they are stuck. This is another modification of the laws of electrodynamics, proposed by Bopp. You realize that once you decide to change the equations of electromagnetism you can start anywhere you want. You can change the force law for an electron, or you can change the Maxwell equations (as we saw in the examples we have described), or you can make a change somewhere else. One possibility is to change the formulas that give the potentials in terms of the charges and currents. One of our formulas has been that the potentials at some point are given by the current density (or charge) at each other point at an earlier time. Using our four-vector notation for the potentials, we write

$$A_\mu(1, t) = \frac{1}{4\pi\epsilon_0 c^2} \int \frac{j_\mu(2, t - r_{12}/c)}{r_{12}} dV_2 \quad (28.13)$$

Bopp's beautifully simple idea is that: Maybe the trouble is in the $1/r$ factor in the integral. Suppose we were to start out by assuming only that the potential at one point depends on the charge density at any other point as *some* function of the distance between the points, say as $f(r_{12})$. The total potential at point (1)

will then be given by the integral of j_μ times this function over all space:

$$A_\mu(1) = \int j_\mu(2) f(r_{12}) dV_2.$$

That's all. No differential equation, nothing else. Well, one more thing. We also ask that the result should be relativistically invariant. So by "distance" we should take the invariant "distance" between two points in space-time. This distance squared (within a sign which doesn't matter) is

$$\begin{aligned} s_{12}^2 &= c^2(t_1 - t_2)^2 - r_{12}^2 \\ &= c^2(t_1 - t_2)^2 - (x_1 - x_2)^2 - (y_1 - y_2)^2 - (z_1 - z_2)^2. \end{aligned} \quad (28.14)$$

So, for a relativistically invariant theory, we should take some function of the magnitude of s_{12} , or what is the same thing, some function of s_{12}^2 . So Bopp's theory is that

$$A_\mu(1, t_1) = \int j_\mu(2, t_2) F(s_{12}^2) dV_2 dt_2. \quad (28.15)$$

(The integral must, of course, be over the four-dimensional volume $dt_2 dx_2 dy_2 dz_2$)

All that remains is to choose a suitable function for F . We assume only one thing about F —that it is very small except when its argument is near zero—so that a graph of F would be a curve like the one in Fig. 28-4. It is a narrow spike with a finite area centered at $s^2 = 0$, and with a width which we can say is roughly a^2 . We can say, crudely, that when we calculate the potential at point (1), only those points (2) produce any appreciable effect if $s_{12}^2 = c^2(t_2 - t_1)^2 - r_{12}^2$ is within $\pm a^2$ of zero. We can indicate this by saying that F is important only for

$$s_{12}^2 = c^2(t_1 - t_2)^2 - r_{12}^2 \approx \pm a^2. \quad (28.16)$$

You can make it more mathematical if you want to, but that's the idea.

Now suppose that a is very small in comparison with the size of ordinary objects like motors, generators, and the like so that for normal problems $r_{12} \gg a$. Then Eq. (28.16) says that charges contribute to the integral of Eq. (28.15) only when $t_1 - t_2$ is in the small range

$$c(t_1 - t_2) \approx \sqrt{r_{12}^2 \pm a^2} \approx r_{12} \sqrt{1 \pm \frac{a^2}{r_{12}^2}}.$$

Since $a^2/r_{12}^2 \ll 1$, the square root can be approximated by $1 \pm a^2/2r_{12}^2$, so

$$t_1 - t_2 = \frac{r_{12}}{c} \left(1 \pm \frac{a^2}{2r_{12}^2} \right) = \frac{r_{12}}{c} \pm \frac{a^2}{2r_{12}c}.$$

What is the significance? This result says that the only *times* t_2 that are important in the integral of A_μ are those which differ from the time t_1 , at which we want the potential, by the delay r_{12}/c —with a negligible correction so long as $r_{12} \gg a$. In other words, this theory of Bopp approaches the Maxwell theory—so long as we are far away from any particular charge—in the sense that it gives the retarded wave effects.

We can, in fact, see approximately what the integral of Eq. (28.15) is going to give. If we integrate first over t_2 from $-\infty$ to $+\infty$ —keeping r_{12} fixed—then s_{12}^2 is also going to go from $-\infty$ to $+\infty$. The integral will all come from t_2 's in a small interval of width $\Delta t_2 = 2 \times a^2/2r_{12}c$, centered at $t_1 - r_{12}/c$. Say that the function $F(s^2)$ has the value K at $s^2 = 0$; then the integral over t_2 gives approximately $K j_\mu \Delta t_2$, or

$$\frac{Ka^2}{c} \frac{j_\mu}{r_{12}}.$$

We should, of course, take the value of j_μ at $t_2 = t_1 - r_{12}/c$, so that Eq. (28.15) becomes

$$A_\mu(1, t_1) = \frac{Ka^2}{c} \int \frac{j_\mu(2, t_1 - r_{12}/c)}{r_{12}} dV_2.$$

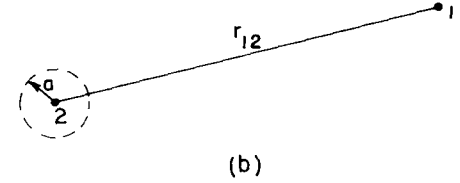
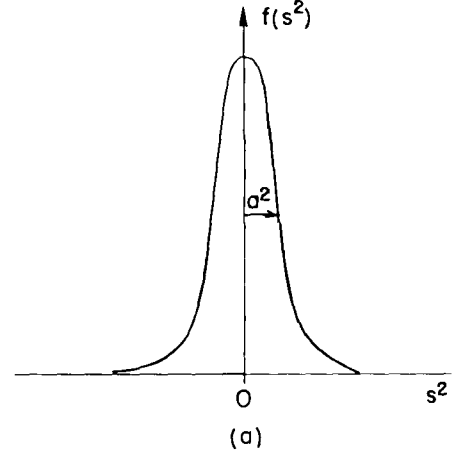


Fig. 28-4. The function $F(s^2)$ used in the nonlocal theory of Bopp.

If we pick $K = q^2 c / 4\pi\epsilon_0 a^2$, we are right back to the retarded potential solution of Maxwell's equations—including automatically the $1/r$ dependence! And it all came out of the simple proposition that the potential at one point in space-time depends on the current density at all other points in space-time, but with a weighting factor that is some narrow function of the four-dimensional distance between the two points. This theory again predicts a finite electromagnetic mass for the electron, and the energy and mass have the right relation for the relativity theory. They must, because the theory is relativistically invariant from the start, and everything seems to be all right.

There is, however, one fundamental objection to this theory and to all the other theories we have described. All particles we know obey the laws of quantum mechanics, so a quantum-mechanical modification of electrodynamics has to be made. Light behaves like photons. It isn't 100 percent like the Maxwell theory. So the electrodynamic theory has to be changed. We have already mentioned that it might be a waste of time to work so hard to straighten out the classical theory, because it could turn out that in quantum electrodynamics the difficulties will disappear or may be resolved in some other fashion. But the difficulties do not disappear in quantum electrodynamics. That is one of the reasons that people have spent so much effort trying to straighten out the classical difficulties, hoping that if they *could* straighten out the classical difficulty and *then* make the quantum modifications, everything would be straightened out. The Maxwell theory still has the difficulties after the quantum mechanics modifications are made.

The quantum effects do make some changes—the formula for the mass is modified, and Planck's constant $\hbar$ appears—but the answer still comes out infinite unless you cut off an integration somehow—just as we had to stop the classical integrals at $r = a$. And the answers depend on how you stop the integrals. We cannot, unfortunately, demonstrate for you here that the difficulties are really basically the same, because we have developed so little of the theory of quantum mechanics and even less of quantum electrodynamics. So you must just take our word that the quantized theory of Maxwell's electrodynamics gives an infinite mass for a point electron.

It turns out, however, that nobody has ever succeeded in making a *self-consistent* quantum theory out of *any* of the modified theories. Born and Infeld's ideas have never been satisfactorily made into a quantum theory. The theories with the advanced and retarded waves of Dirac, or of Wheeler and Feynman, have never been made into a satisfactory quantum theory. The theory of Bopp has never been made into a satisfactory quantum theory. So today, there is no known solution to this problem. We do not know how to make a consistent theory—including the quantum mechanics—which does not produce an infinity for the self-energy of an electron, or any point charge. And at the same time, there is no satisfactory theory that describes a non-point charge. It's an unsolved problem.

In case you are deciding to rush off to make a theory in which the action of an electron on itself is completely removed, so that electromagnetic mass is no longer meaningful, and then to make a quantum theory of it, you should be warned that you are certain to be in trouble. There is definite experimental evidence of the existence of electromagnetic inertia—there is evidence that some of the mass of charged particles is electromagnetic in origin.

It used to be said in the older books that since Nature will obviously not present us with two particles—one neutral and the other charged, but otherwise the same—we will never be able to tell how much of the mass is electromagnetic and how much is mechanical. But it turns out that Nature *has* been kind enough to present us with just such objects, so that by comparing the observed mass of the charged one with the observed mass of the neutral one, we can tell whether there is any electromagnetic mass. For example, there are the neutrons and protons. They interact with tremendous forces—the nuclear forces—whose origin is unknown. However, as we have already described, the nuclear forces have one remarkable property. So far as they are concerned, the neutron and proton are exactly the same. The *nuclear* forces between neutron and neutron, neutron and proton, and proton and proton are all identical as far as we can tell. Only the little

electromagnetic forces are different; electrically the proton and neutron are as different as night and day. This is just what we wanted. There are two particles, identical from the point of view of the strong interactions, but different electrically. And they have a small difference in mass. The mass difference between the proton and the neutron—expressed as the difference in the rest-energy mc^2 in units of Mev—is about 1.3 Mev, which is about 2.6 times the electron mass. The classical theory would then predict a radius of about $\frac{1}{3}$ to $\frac{1}{2}$ the classical electron radius, or about 10^{-13} cm. Of course, one should really use the quantum theory, but by some strange accident, all the constants— 2π 's and $\hbar$'s, etc.—come out so that the quantum theory gives roughly the same radius as the classical theory. The only trouble is that the *sign* is wrong! The neutron is *heavier* than the proton.

Table 28-1
Particle Masses

Particle	Charge (electronic)	Mass (Mev)	Δm^* (Mev)
n (neutron)	0	939.5	
p (proton)	+1	938.2	-1.3
π (π -meson)	0	135.0	
	± 1	139.6	+4.6
K (K-meson)	0	497.8	
	± 1	493.9	-3.9
Σ (sigma)	0	1191.5	
	+1	1189.4	-2.1
	-1	1196.0	+4.5

* Δm = (mass of charged) - (mass of neutral).

Nature has also given us several other pairs—or triplets—of particles which appear to be exactly the same except for their electrical charge. They interact with protons and neutrons, through the so-called “strong” interactions of the nuclear forces. In such interactions, the particles of a given kind—say the π -mesons—behave in every way like one object *except* for their electrical charge. In Table 28-1 we give a list of such particles, together with their measured masses. The charged π -mesons—positive or negative—have a mass of 139.6 Mev, but the neutral π -meson is 4.6 Mev lighter. We believe that this mass difference is electromagnetic; it would correspond to a particle radius of 3 to 4×10^{-14} cm. You will see from the table that the mass differences of the other particles are usually of the same general size.

Now the size of these particles can be determined by other methods, for instance by the diameters they appear to have in high-energy collisions. So the electromagnetic mass seems to be in general agreement with electromagnetic theory, if we stop our integrals of the field energy at the same radius obtained by these other methods. That’s why we believe that the differences do represent electromagnetic mass.

You are no doubt worried about the different signs of the mass differences in the table. It is easy to see why the charged ones should be heavier than the neutral ones. But what about those pairs like the proton and the neutron, where the measured mass comes out the other way? Well, it turns out that these particles are complicated, and the computation of the electromagnetic mass must be more elaborate for them. For instance, although the neutron has no *net* charge, it *does* have a charge distribution inside it—it is only the *net* charge that is zero. In fact, we believe that the neutron looks—at least sometimes—like a proton with a negative π -meson in a “cloud” around it, as shown in Fig. 28-5. Although the neutron is “neutral,” because its total charge is zero, there are still electromagnetic energies

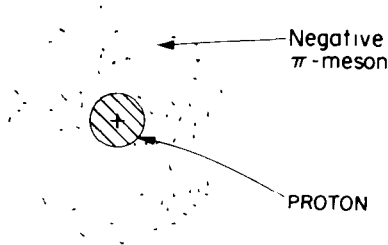


Fig. 28-5. A neutron may exist, at times, as a proton surrounded by a negative π -meson.

(for example, it has a magnetic moment), so it's not easy to tell the sign of the electromagnetic mass difference without a detailed theory of the internal structure.

We only wish to emphasize here the following points: (1) the electromagnetic theory predicts the existence of an electromagnetic mass, but it also falls on its face in doing so, because it does not produce a consistent theory—and the same is true with the quantum modifications; (2) there is experimental evidence for the existence of electromagnetic mass; and (3) all these masses are roughly the same as the mass of an electron. So we come back again to the original idea of Lorentz—maybe all the mass of an electron is purely electromagnetic, maybe the whole 0.511 Mev is due to electrodynamics. Is it or isn't it? We haven't got a theory, so we cannot say.

We must mention one more piece of information, which is the most annoying. There is another particle in the world called a *muon*—or μ -meson—which, so far as we can tell, differs in no way whatsoever from an electron except for its mass. It acts in every way like an electron: it interacts with neutrinos and with the electromagnetic field, and it has no nuclear forces. It does nothing different from what an electron does—at least, nothing which cannot be understood as merely a consequence of its higher mass (206.77 times the electron mass). Therefore, whenever someone finally gets the explanation of the mass of an electron, he will then have the puzzle of where a muon gets its mass. Why? Because whatever the electron does, the muon does the same—so the mass ought to come out the same. There are those who believe faithfully in the idea that the muon and the electron are the same particle and that, in the final theory of the mass, the formula for the mass will be a quadratic equation with two roots—one for each particle. There are also those who propose it will be a transcendental equation with an infinite number of roots, and who are engaged in guessing what the masses of the other particles in the series must be, and why these particles haven't been discovered yet.

28-6 The nuclear force field

We would like to make some further remarks about the part of the mass of nuclear particles that is not electromagnetic. Where does this other large fraction come from? There are other forces besides electrodynamics—like nuclear forces—that have their own field theories, although no one knows whether the current theories are right. These theories also predict a field energy which gives the nuclear particles a mass term analogous to electromagnetic mass; we could call it the “ π -mesic-field-mass.” It is presumably very large, because the forces are great, and it is the possible origin of the mass of the heavy particles. But the meson field theories are still in a most rudimentary state. Even with the well-developed theory of electromagnetism, we found it impossible to get beyond first base in explaining the electron mass. With the theory of the mesons, we strike out.

We may take a moment to outline the theory of the mesons, because of its interesting connection with electrodynamics. In electrodynamics, the field can be described in terms of a four-potential that satisfies the equation

$$\square^2 A_\mu = \text{sources.}$$

Now we have seen that pieces of the field can be radiated away so that they exist separated from the sources. These are the photons of light, and they are described by a differential equation without sources:

$$\square^2 A_\mu = 0.$$

People have argued that the field of nuclear forces ought also to have its own “photons”—they would presumably be the π -mesons—and that they should be described by an analogous differential equation. (Because of the weakness of the human brain, we can't think of something really new, so we argue by analogy with what we know.) So the meson equation might be

$$\square^2 \phi = 0,$$

where ϕ could be a different four-vector or perhaps a scalar. It turns out that the pion has no polarization, so ϕ should be a scalar. With the simple equation $\square^2 \phi = 0$, the meson field would vary with distance from a source as $1/r^2$, just as the electric field does. But we know that nuclear forces have much shorter distances of action, so the simple equation won't work. There is one way we can change things without disrupting the relativistic invariance: we can add or subtract from the D'Alembertian a constant, times ϕ . So Yukawa suggested that the free quanta of the nuclear force field might obey the equation

$$\square^2 \phi - \mu^2 \phi = 0, \quad (28.17)$$

where μ^2 is a constant—that is, an invariant scalar. (Since $\square^2$ is a scalar differential operator in four dimensions, its invariance is unchanged if we add another scalar to it.)

Let's see what Eq. (28.17) gives for the nuclear force when things are not changing with time. We want a spherically symmetric solution of

$$\nabla^2 \phi - \mu^2 \phi = 0$$

around some point source at, say, the origin. If ϕ depends only on r , we know that

$$\nabla^2 \phi = \frac{1}{r} \frac{\partial^2}{\partial r^2} (r\phi).$$

So we have the equation

$$\frac{1}{r} \frac{\partial^2}{\partial r^2} (r\phi) - \mu^2 \phi = 0$$

or

$$\frac{\partial^2}{\partial r^2} (r\phi) = \mu^2 (r\phi).$$

Thinking of $(r\phi)$ as our dependent variable, this is an equation we have seen many times. It's solution is

$$r\phi = Ke^{\pm\mu r}.$$

Clearly, ϕ cannot become infinite for large r , so the $+$ sign in the exponent is ruled out. The solution is

$$\phi = K \frac{e^{-\mu r}}{r}. \quad (28.18)$$

This function is called the *Yukawa potential*. For an attractive force, K is a negative number whose magnitude must be adjusted to fit the experimentally observed strength of the forces.

The Yukawa potential of the nuclear forces dies off more rapidly than $1/r$ by the exponential factor. The potential—and therefore the force—falls to zero much more rapidly than $1/r$ for distances beyond $1/\mu$, as shown in Fig. 28-6. The “range” of nuclear forces is much less than the “range” of electrostatic forces. It is found experimentally that the nuclear forces do not extend beyond about 10^{-13} cm, so $\mu \approx 10^{15} \text{ m}^{-1}$.

Finally, let's look at the free-wave solution of Eq. (28.17). If we substitute

$$\phi = \phi_0 e^{i(\omega t - kz)}$$

into Eq. (28.17), we get that

$$\frac{\omega^2}{c^2} - k^2 - \mu^2 = 0.$$

Relating frequency to energy and wave number to momentum, as we did at the end of Chapter 36 of Vol. I, we get that

$$\frac{E^2}{c^2} - p^2 = \mu^2 \hbar^2,$$

which says that the Yukawa “photon” has a mass equal to $\mu\hbar/c$. If we use for μ

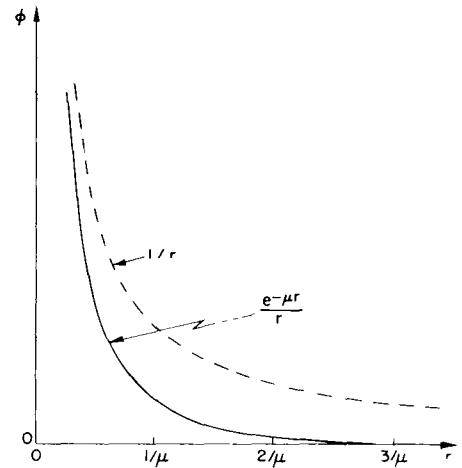


Fig. 28-6. The Yukawa potential $e^{-\mu r}/r$, compared with the Coulomb potential $1/r$.

the estimate 10^{15} m^{-1} , which gives the observed range of the nuclear forces, the mass comes out to $3 \times 10^{-25} \text{ gm}$, or 170 Mev, which is roughly the observed mass of the π -meson. So, by an analogy with electrodynamics, we would say that the π -meson is the “photon” of the nuclear force field. But now we have pushed the ideas of electrodynamics into regions where they may not really be valid—we have gone beyond electrodynamics to the problem of the nuclear forces.

The Motion of Charges in Electric and Magnetic Fields

29-1 Motion in a uniform electric or magnetic field

We want now to describe—mainly in a qualitative way—the motions of charges in various circumstances. Most of the interesting phenomena in which charges are moving in fields occur in very complicated situations, with many, many charges all interacting with each other. For instance, when an electromagnetic wave goes through a block of material or a plasma, billions and billions of charges are interacting with the wave and with each other. We will come to such problems later, but now we just want to discuss the much simpler problem of the motions of a single charge in a *given* field. We can then disregard all other charges—except, of course, those charges and currents which exist somewhere to produce the fields we will assume.

We should probably ask first about the motion of a particle in a uniform electric field. At low velocities, the motion is not particularly interesting—it is just a uniform acceleration in the direction of the field. However, if the particle picks up enough energy to become relativistic, then the motion gets more complicated. But we will leave the solution for that case for you to play with.

Next, we consider the motion in a uniform magnetic field with zero electric field. We have already solved this problem—one solution is that the particle goes in a circle. The magnetic force $q\mathbf{v} \times \mathbf{B}$ is always at right angles to the motion, so $d\mathbf{p}/dt$ is perpendicular to $\mathbf{p}$ and has the magnitude $v p/R$, where R is the radius of the circle.

$$F = qvB = \frac{vp}{R}.$$

The radius of the circular orbit is then

$$R = \frac{p}{qB}. \quad (29.1)$$

That is only one possibility. If the particle has a component of its motion along the field direction, that motion is constant, since there can be no component of the magnetic force in the direction of the field. The general motion of a particle in a uniform magnetic field is a constant velocity parallel to $\mathbf{B}$ and a circular motion at right angles to $\mathbf{B}$ —the trajectory is a cylindrical helix (Fig. 29-1). The radius of the helix is given by Eq. (29.1) if we replace p by $p_{\perp}$, the component of momentum at right angles to the field.

29-2 Momentum analysis

A uniform magnetic field is often used in making a “momentum analyzer,” or “momentum spectrometer,” for high-energy charged particles. Suppose that charged particles are shot into a uniform magnetic field at the point *A* in Fig. 29-2(a), the magnetic field being perpendicular to the plane of the drawing. Each particle will go into an orbit which is a circle whose radius is proportional to its momentum. If all the particles enter perpendicular to the edge of the field, they will leave the field at a distance x (from *A*) which is proportional to their momentum p . A counter placed at some point such as *C* will detect only those particles whose momentum is in an interval Δp near the momentum $p = qBx/2$.

It is, of course, not necessary that the particles go through 180° before they are counted, but the so-called “ 180° spectrometer” has a special property. It is not

29-1 Motion in a uniform electric or magnetic field

29-2 Momentum analysis

29-3 An electrostatic lens

29-4 A magnetic lens

29-5 The electron microscope

29-6 Accelerator guide fields

29-7 Alternating-gradient focusing

29-8 Motion in crossed electric and magnetic fields

Review Chapter 30, Vol. I, *Diffraction*

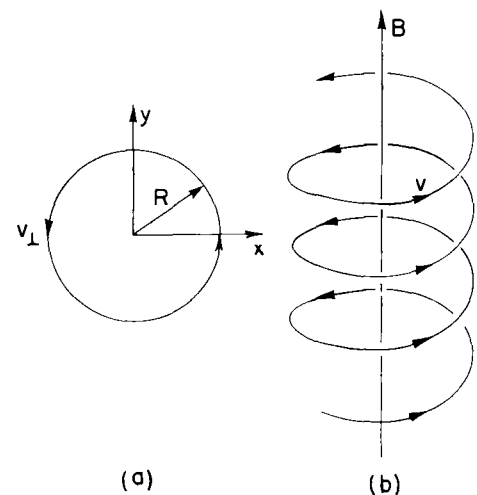


Fig. 29-1. Motion of a particle in a uniform magnetic field.

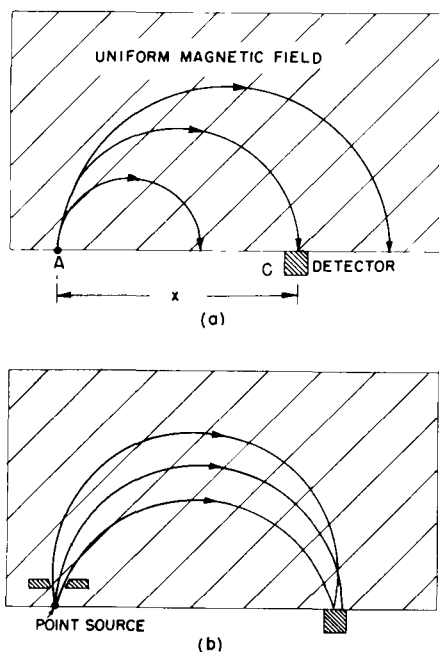


Fig. 29-2. A uniform-field, momentum spectrometer with 180° focusing: (a) different momenta; (b) different angles. (The magnetic field is directed perpendicular to the plane of the figure.)

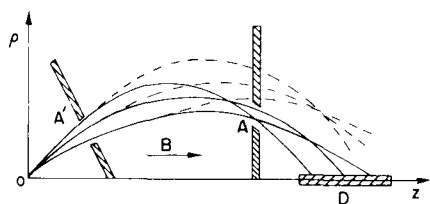


Fig. 29-3. An axial-field spectrometer.

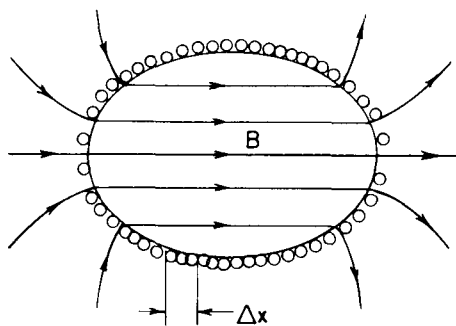


Fig. 29-4. An ellipsoidal coil with equal currents in each axial interval Δx produces a uniform magnetic field inside.

necessary that all the particles enter at right angles to the field edge. Figure 29-2(b) shows the trajectories of three particles, all with the *same* momentum but entering the field at different angles. You see that they take different trajectories, but all leave the field very close to the point C. We say that there is a “focus.” Such a focusing property has the advantage that larger angles can be accepted at A—although some limit is usually imposed, as shown in the figure. A larger angular acceptance usually means that more particles are counted in a given time, decreasing the time required for a given measurement.

By varying the magnetic field, or moving the counter along in x , or by using many counters to cover a range of x , the “spectrum” of momenta in the incoming beam can be measured. [By the “momentum spectrum” $f(p)$, we mean that the number of particles with momenta between p and $(p + dp)$ is $f(p) dp$.] Such measurements have been made, for example, to determine the distribution of energies in the β -decay of various nuclei.

There are many other forms of momentum spectrometers, but we will describe just one more, which has an especially large *solid* angle of acceptance. It is based on the helical orbits in a uniform field, like the one shown in Fig. 29-1. Let’s think of a cylindrical coordinate system— ρ, θ, z —set up with the z -axis along the direction of the field. If a particle is emitted from the origin at some angle α with respect to the z -axis, it will move along a spiral whose equation is

$$\rho = a \sin kz, \quad \theta = bz,$$

where a, b , and k are parameters you can easily work out in terms of p, α , and the magnetic field B . If we plot the distance ρ from the axis as a function of z for a given momentum, but for several starting angles, we will get curves like the solid ones drawn in Fig. 29-3. (Remember that this is just a kind of projection of a helical trajectory.) When the angle between the axis and the starting direction is larger, the peak value of ρ is large but the longitudinal velocity is less, so the trajectories for different angles tend to come to a kind of “focus” near the point A in the figure. If we put a narrow aperture of A , particles with a range of initial angles can still get through and pass on to the axis, where they can be counted by the long detector D .

Particles which leave the source at the origin with a higher momentum but at the same angles, follow the paths shown by the broken lines and do not get through the aperture at A . So the apparatus selects a small interval of momenta. The advantage over the first spectrometer described is that the aperture A —and the aperture A' —can be an annulus, so that particles which leave the source in a rather large solid angle are accepted. A large fraction of the particles from the source are used—an important advantage for weak sources or for very precise measurements.

One pays a price for this advantage, however, because a large volume of uniform magnetic field is required, and this is usually only practical for low-energy particles. One way of making a uniform field, you remember, is to wind a coil on a sphere, with a surface current density proportional to the sine of the angle. You can also show that the same thing is true for an ellipsoid of rotation. So such spectrometers are often made by winding an elliptical coil on a wooden (or aluminum) frame. All that is required is that the current in each interval of axial distance Δx be the same, as shown in Fig. 29-4

29-3 An electrostatic lens

Particle focusing has many applications. For instance, the electrons that leave the cathode in a TV picture tube are brought to a focus at the screen—to make a fine spot. In this case, one wants to take electrons all of the same energy but with different initial angles and bring them together in a small spot. The problem is like focusing light with a lens, and devices which do the corresponding job for particles are also called lenses.

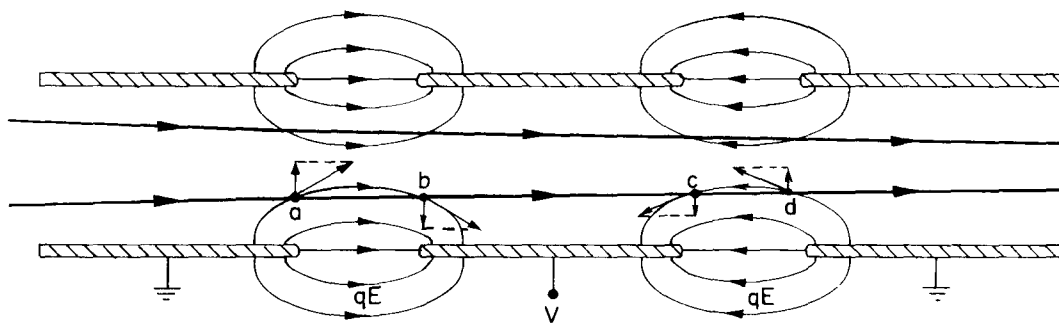


Fig. 29-5. An electrostatic lens. The field lines shown are "lines of force," that is, of qE .

One example of an electron lens is sketched in Fig. 29-5. It is an "electrostatic" lens whose operation depends on the electric field between two adjacent electrodes. Its operation can be understood by considering what happens to a parallel beam that enters from the left. When the electrons arrive at the region *a*, they feel a force with a sidewise component and get a certain impulse that bends them toward the axis. You might think that they would get an equal and opposite impulse in the region *b*, but that is not so. By the time the electrons reach *b* they have gained energy and so *spend less time* in the region *b*. The forces are the same, but the time is shorter, so the impulse is less. In going through the regions *a* and *b*, there is a net axial impulse, and the electrons are bent toward a common point. In leaving the high-voltage region, the particles get another kick toward the axis. The force is outward in region *c* and inward in region *d*, but the particles stay longer in the latter region, so there is again a net impulse. For distances not too far from the axis (Can you see why?), and this is just the condition necessary for lens-type focusing.

You can use the same arguments to show that there is focusing if the potential of the middle electrode is either positive or negative with respect to the other two. Electrostatic lenses of this type are commonly used in cathode-ray tubes and in some electron microscopes.

29-4 A magnetic lens

Another kind of lens—often found in electron microscopes—is the magnetic lens sketched schematically in Fig. 29-6. A cylindrically symmetric electromagnet has very sharp circular pole tips which produce a strong, nonuniform field in a small region. Electrons which travel vertically through this region are focused. You can understand the mechanism by looking at the magnified view of the pole-tip region drawn in Fig. 29-7. Consider two electrons *a* and *b* that leave the source *S* at some angle with respect to the axis. As electron *a* reaches the beginning of the field, it is deflected *away from you* by the horizontal component of the field. But then it will have a lateral velocity, so that when it passes through the strong vertical field, it will get an impulse toward the axis. Its lateral motion is taken out by the magnetic force as it leaves the field, so the net effect is an impulse toward the axis, plus a "rotation" about the axis. All the forces on particle *b* are opposite, so it also is deflected toward the axis. In the figure, the divergent electrons are brought into parallel paths. The action is like a lens with an object at the focal point. Another similar lens upstream can be used to focus the electrons back to a single point, making an image of the source *S*.

29-5 The electron microscope

You know that electron microscopes can "see" objects too small to be seen by optical microscopes. We discussed in Chapter 30 of Vol. I the basic limitations of any optical system due to diffraction of the lens opening. If a lens opening sub-

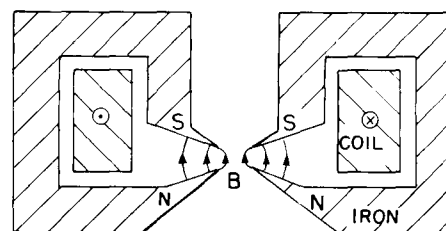


Fig. 29-6. A magnetic lens.

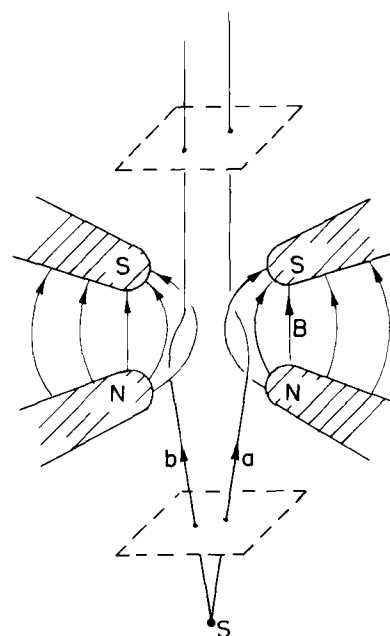


Fig. 29-7. Electron motion in the magnetic lens.

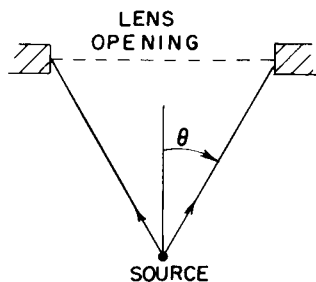


Fig. 29-8. The resolution of a microscope is limited by the angle subtended from the source.

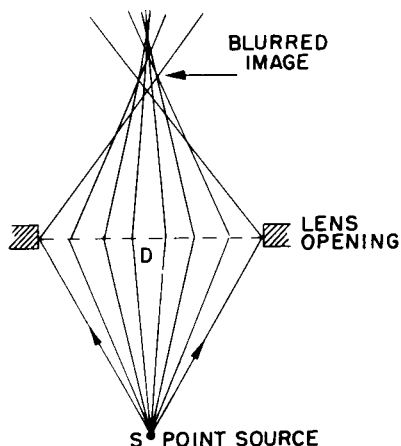


Fig. 29-9. Spherical aberration of a lens.

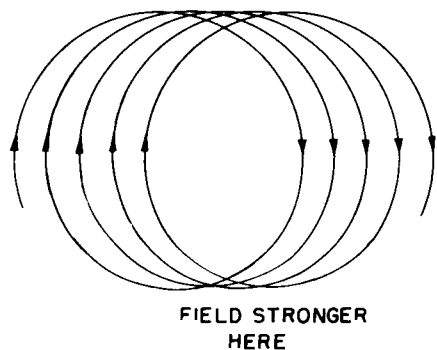


Fig. 29-10. Particle motion in a slightly nonuniform field.

tends the angle 2θ from a source (see Fig. 29-8), two neighboring spots at the source cannot be seen as separate if they are closer than about

$$\delta \approx \frac{\lambda}{\sin \theta},$$

where λ is the wavelength of the light. With the best optical microscope, θ approaches the theoretical limit of 90° , so δ is about equal to λ , or approximately 5000 angstroms.

The same limitation would also apply to an electron microscope, but there the wavelength is—for 50-kilovolt electrons—about 0.05 angstrom. If one could use a lens opening of near 30° , it would be possible to see objects only $\frac{1}{5}$ of an angstrom apart. Since the atoms in molecules are typically 1 or 2 angstroms apart, we could get photographs of molecules. Biology would be easy; we would have a photograph of the DNA structure. What a tremendous thing that would be! Most of present-day research in molecular biology is an attempt to figure out the shapes of complex organic molecules. If we could only see them!

Unfortunately, the best resolving power that has been achieved in an electron microscope is more like 20 angstroms. The reason is that no one has yet designed a lens with a large opening. All lenses have “spherical aberration,” which means that rays at large angles from the axis have a different point of focus than the rays nearer the axis, as shown in Fig. 29-9. By special techniques, optical microscope lenses can be made with a negligible spherical aberration, but no one has yet been able to make an electron lens which avoids spherical aberration.

In fact, one can show that any electrostatic or magnetic lens of the types we have described must have an irreducible amount of spherical aberration. This aberration—together with diffraction—limits the resolving power of electron microscopes to their present value.

The limitation we have mentioned does not apply to electric and magnetic fields which are not axially symmetric or which are not constant in time. Perhaps some day someone will think of a new kind of electron lens that will overcome the inherent aberration of the simple electron lens. Then we will be able to photograph atoms directly. Perhaps one day chemical compounds will be analyzed by looking at the positions of the atoms rather than by looking at the color of some precipitate!

29-6 Accelerator guide fields

Magnetic fields are also used to produce special particle trajectories in high-energy particle accelerators. Machines like the cyclotron and synchrotron bring particles to high energies by passing the particles repeatedly through a strong electric field. The particles are held in their cyclic orbits by a magnetic field.

We have seen that a particle in a uniform magnetic field will go in a circular orbit. This, however, is true only for a perfectly uniform field. Imagine a field B which is nearly uniform over a large area but which is slightly stronger in one region than in another. If we put a particle of momentum p in this field, it will go in a nearly circular orbit with the radius $R = p/qB$. The radius of curvature will, however, be slightly smaller in the region where the field is stronger. The orbit is not a closed circle but will “walk” through the field, as shown in Fig. 29-10. We can, if we wish, consider that the slight “error” in the field produces an extra angular kick which sends the particle off on a new track. If the particles are to make millions of revolutions in an accelerator, some kind of “radial focusing” is needed which will tend to keep the trajectories close to some design orbit.

Another difficulty with a uniform field is that the particles do not remain in a plane. If they start out with the slightest angle—or are given a slight angle by any small error in the field—they will go in a helical path that will eventually take them into the magnet pole or the ceiling or floor of the vacuum tank. Some arrangement must be made to inhibit such vertical drifts; the field must provide “vertical focusing” as well as radial focusing.

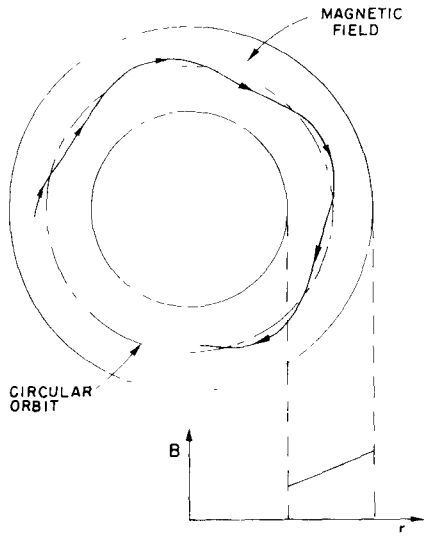


Fig. 29-11. Radial motion of a particle in a magnetic field with a large positive slope.

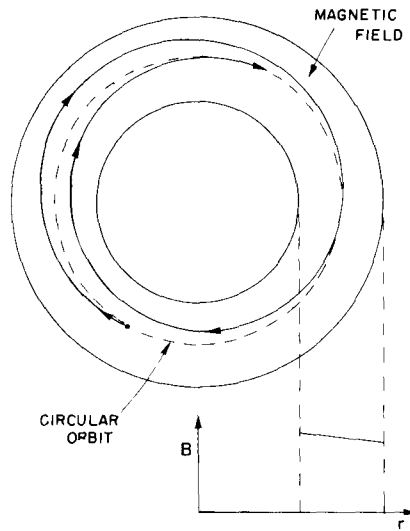


Fig. 29-12. Radial motion of a particle in a magnetic field with a small negative slope.

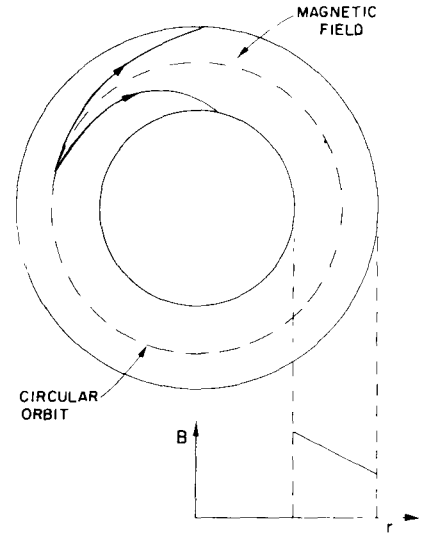


Fig. 29-13. Radial motion of a particle in a magnetic field with a large negative slope.

One would, at first, guess that radial focusing could be provided by making a magnetic field which increases with increasing distance from the center of the design path. Then if a particle goes out to a large radius, it will be in a stronger field which will bend it back toward the correct radius. If it goes to too small a radius, the bending will be less, and it will be returned toward the design radius. If a particle is once started at some angle with respect to the ideal circle, it will oscillate about the ideal circular orbit, as shown in Fig. 29-11. The radial focusing would keep the particles near the circular path.

Actually there is still some radial focusing even with the *opposite* field slope. This can happen if the radius of curvature of the trajectory does not increase more rapidly than the increase in the distance of the particle from the center of the field. The particle orbits will be as drawn in Fig. 29-12. If the gradient of the field is too large, however, the orbits will not return to the design radius but will spiral inward or outward, as shown in Fig. 29-13.

We usually describe the slope of the field in terms of the "relative gradient" or *field index*, n :

$$n = \frac{dB/B}{dr/r}. \quad (29.2)$$

A guide field gives radial focusing if this relative gradient is greater than -1 .

A radial field gradient will also produce *vertical* forces on the particles. Suppose we have a field that is stronger nearer to the center of the orbit and weaker at the outside. A vertical cross section of the magnet at right angles to the orbit might be as shown in Fig. 29-14. (For protons the orbits would be coming out of the page.) If the field is to be stronger to the left and weaker to the right, the lines of the magnetic field must be curved as shown. We can see that this must be so by using the law that the circulation of $\mathbf{B}$ is zero in free space. If we take coordinates as shown in the figure, then

$$(\nabla \times \mathbf{B})_y = \frac{\partial B_x}{\partial z} - \frac{\partial B_z}{\partial x} = 0,$$

or

$$\frac{\partial B_x}{\partial z} = \frac{\partial B_z}{\partial x}. \quad (29.3)$$

Since we assume that $\partial B_z / \partial x$ is negative, there must be an equal negative $\partial B_x / \partial z$.

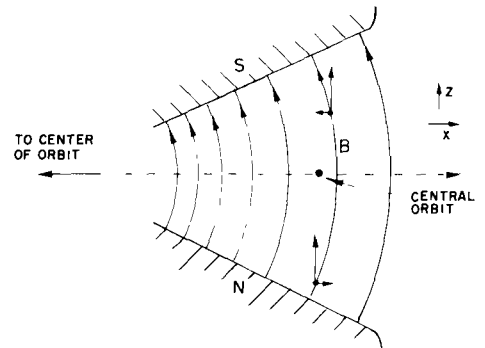


Fig. 29-14. A vertical guide field as seen in a cross section perpendicular to the orbits.

If the “nominal” plane of the orbit is a plane of symmetry where $B_r = 0$, then the radial component B_r will be negative above the plane and positive below. The lines must be curved as shown.

Such a field will have vertical focusing properties. Imagine a proton that is travelling more or less parallel to the central orbit but above it. The horizontal component of $\mathbf{B}$ will exert a downward force on it. If the proton is below the central orbit, the force is reversed. So there is an effective “restoring force” toward the central orbit. From our arguments there will be vertical focusing, provided that the *vertical* field decreases with increasing radius; but if the field gradient is positive, there will be “vertical defocusing.” So for vertical focusing, the field index n must be less than zero. We found above that for radial focusing n had to be greater than -1 . The two conditions together give the condition that

$$-1 < n < 0$$

if the particles are to be kept in stable orbits. In cyclotrons, values very near zero are used; in betatrons and synchrotrons, the value $n = -0.6$ is typically used.

29-7 Alternating-gradient focusing

Such small values of n give rather “weak” focusing. It is clear that much more effective radial focusing would be given by a large positive gradient ($n \gg 1$), but then the vertical forces would be strongly defocusing. Similarly, large negative slopes ($n \ll -1$) would give stronger vertical forces but would cause radial defocusing. It was realized about 10 years ago, however, that a force that alternates between strong focusing and strong defocusing can still have a *net* focusing force.

To explain how *alternating-gradient focusing* works, we will first describe the operation of a quadrupole lens, which is based on the same principle. Imagine that a uniform negative magnetic field is added to the field of Fig. 29-14, with the strength adjusted to make zero field at the orbit. The resulting field—for small displacements from the neutral point—would be like the field shown in Fig. 29-15. Such a four-pole magnet is called a “quadrupole lens.” A positive particle that enters (from the reader) to the right or left of the center is pushed back toward the center. If the particle enters above or below, it is pushed *away* from the center. This is a horizontal focusing lens. If the horizontal gradient is reversed—as can be done by reversing all the polarities—the signs of all the forces are reversed and we have a vertical focusing lens, as in Fig. 29-16. For such lenses, the field strength—and therefore the focusing forces—increase linearly with the distance of the lens from the axis.

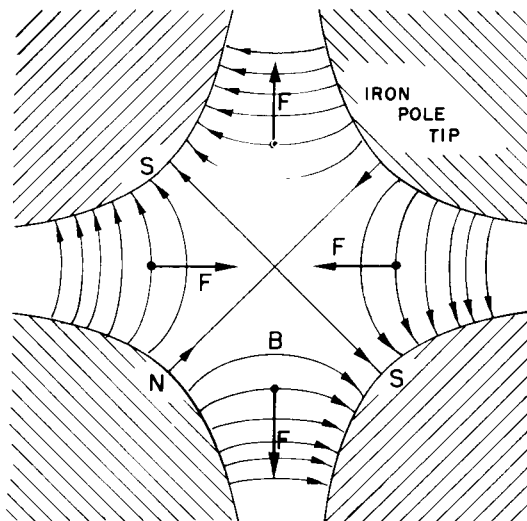


Fig. 29-15. A horizontal focusing quadrupole lens.

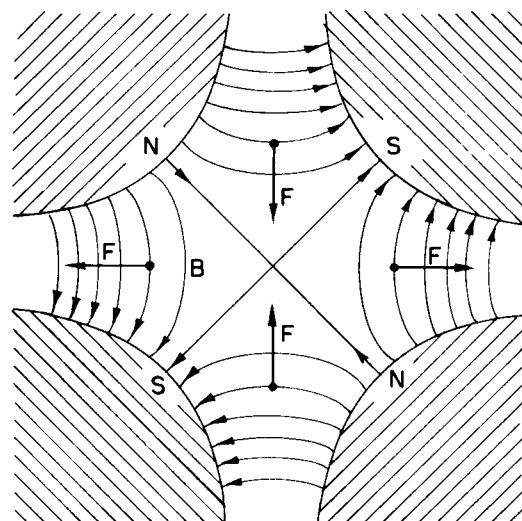


Fig. 29-16. A vertical focusing quadrupole lens.

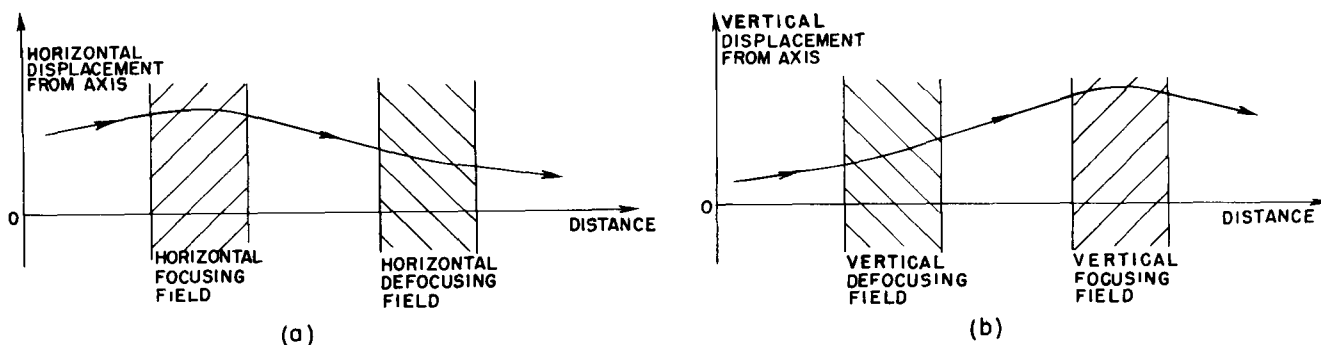


Fig. 29-17. Horizontal and vertical focusing with a pair of quadrupole lenses.

Now imagine that two such lenses are placed in series. If a particle enters with some horizontal displacement from the axis, as shown in Fig. 29-17(a), it will be deflected toward the axis in the first lens. When it arrives at the second lens it is closer to the axis, so the force outward is less and the outward deflection is less. There is a net bending toward the axis; the *average* effect is horizontally focusing. On the other hand, if we look at a particle which enters off the axis in the vertical direction, the path will be as shown in Fig. 29-17(b). The particle is first deflected *away* from the axis, but then it arrives at the second lens with a larger displacement, feels a stronger force, and so is bent toward the axis. Again the net effect is focusing. Thus a pair of quadrupole lenses acts independently for horizontal and vertical motion—very much like an optical lens. Quadrupole lenses are used to form and control beams of particles in much the same way that optical lenses are used for light beams.

We should point out that an alternating-gradient system does not *always* produce focusing. If the gradients are too large (in relation to the particle momentum or to the spacing between the lenses), the net effect can be a defocusing one. You can see how that could happen if you imagine that the spacing between the two lenses of Fig. 29-17 were increased, say, by a factor of three or four.

Let's return now to the synchrotron guide magnet. We can consider that it consists of an alternating sequence of "positive" and "negative" lenses with a superimposed uniform field. The uniform field serves to bend the particles, on the average, in a horizontal circle (with no effect on the vertical motion), and the alternating lenses act on any particles that might tend to go astray—pushing them always toward the central orbit (on the average).

There is a nice mechanical analog which demonstrates that a force which alternates between a "focusing" force and a "defocusing" force can have a net "focusing" effect. Imagine a mechanical "pendulum" which consists of a *solid* rod with a weight on the end, suspended from a pivot which is arranged to be moved rapidly up and down by a motor driven crank. Such a pendulum has *two* equilibrium positions. Besides the normal, downward-hanging position, the pendulum is also in equilibrium "hanging upward"—with its "bob" *above* the pivot! Such a pendulum is drawn in Fig. 29-18.

By the following argument you can see that the vertical pivot motion is equivalent to an alternating focusing force. When the pivot is accelerated downward, the "bob" tends to move inward, as indicated in Fig. 29-19. When the pivot is accelerated upward, the effect is reversed. The force restoring the "bob" toward the axis alternates, but the average effect is a force toward the axis. So the pendulum will swing back and forth about a neutral position which is just opposite the normal one.

There is, of course, a much easier way of keeping a pendulum upside down, and that is by *balancing* it on your finger! But try to balance *two independent* sticks on the *same finger*! Or one stick with your eyes closed! Balancing involves making a correction for what is going wrong. And this is not possible, in general, if there are several things going wrong at once. In a synchrotron there are billions of particles going around together, each one of which may start out with a different "error." The kind of focusing we have been describing works on them all.

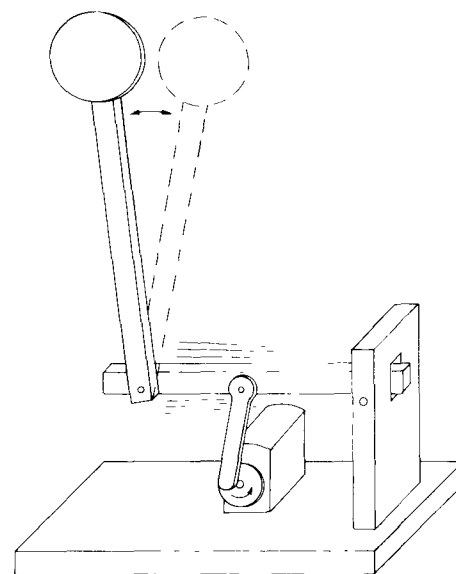


Fig. 29-18. A pendulum with an oscillating pivot can have a stable position with the bob above the pivot.

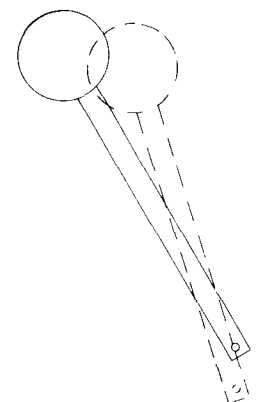


Fig. 29-19. A downward acceleration of the pivot causes the pendulum to move toward the vertical.

29-8 Motion in crossed electric and magnetic fields

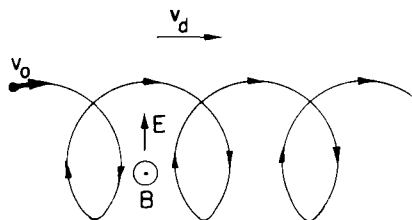


Fig. 29-20. Path of a particle in crossed electric and magnetic fields.

So far we have talked about particles in electric fields only or in magnetic fields only. There are some interesting effects when there are both kinds of fields at the same time. Suppose we have a uniform magnetic field $\mathbf{B}$ and an electric field $\mathbf{E}$ at right angles. Particles that start out perpendicular to $\mathbf{B}$ will move in a curve like the one in Fig 29-20 (The figure is a *plane* curve, *not* a helix!) We can understand this motion qualitatively. When the particle (assumed positive) moves in the direction of $\mathbf{E}$, it picks up speed, and so it is bent less by the magnetic field. When it is going against the $\mathbf{E}$ -field, it loses speed and is continually bent more by the magnetic field. The net effect is that it has an average “drift” in the direction of $\mathbf{E} \times \mathbf{B}$.

We can, in fact, show that the motion is a uniform circular motion superimposed on a uniform sidewise motion at the speed $v_d = E/B$ —the trajectory in Fig. 29-20 is a cycloid. Imagine an observer who is moving to the right at a constant speed. In his frame our magnetic field gets transformed to a new magnetic field *plus* an electric field in the *downward* direction. If he has just the right speed, his total electric field will be zero, and he will see the electron going in a circle. So the motion *we* see is a circular motion, plus a translation at the drift speed $v_d = E/B$. The motion of electrons in crossed electric and magnetic fields is the basis of the *magnetron* tubes, i.e., oscillators used for generating microwave energy.

There are many other interesting examples of particle motions in electric and magnetic fields—such as the orbits of the electrons and protons trapped in the Van Allen belts—but we do not, unfortunately, have the time to deal with them here

The Internal Geometry of Crystals

30-1 The internal geometry of crystals

We have finished the study of the basic laws of electricity and magnetism, and we are now going to study the electromagnetic properties of matter. We begin by describing solids—that is, crystals. When the atoms of matter are not moving around very much, they get stuck together and arrange themselves in a configuration with as low an energy as possible. If the atoms in a certain place have found a pattern which seems to be of low energy, then the atoms somewhere else will probably make the same arrangement. For these reasons, we have in a solid material a repetitive pattern of atoms

In other words, the conditions in a crystal are this way: The environment of a particular atom in a crystal has a certain arrangement, and if you look at the same kind of an atom at another place farther along, you will find one whose surroundings are exactly the same. If you pick an atom farther along by the same distance, you will find the conditions exactly the same once more. The pattern is repeated over and over again—and, of course, in three dimensions.

Imagine the problem of designing a wallpaper—or a cloth, or some geometric design for a plane area—in which you are supposed to have a design element which repeats and repeats and repeats, so that you can make the area as large as you want. This is the two-dimensional analog of a problem which a crystal solves in three dimensions. For example, Fig. 30-1(a) shows a common kind of wallpaper design. There is a single element repeated in a pattern that can go on forever. The geometric characteristics of this wallpaper design, considering only its repetition properties and not worrying about the geometry of the flower itself or its artistic merit, are contained in Fig. 30-1(b). If you start at any point, you can find the *corresponding* point by moving the distance a along the direction of arrow 1. You can also get to a corresponding point if you move the distance b in the direction of the other arrow. There are, of course, many other directions. You can go, for example, from point α to point β and reach a corresponding position, but such a step can be considered as a combination of a step along direction 1, followed by a step along direction 2. One of the basic properties of the pattern can be described by the two shortest steps to nearby equal positions. By “equal” positions we mean that if you were to stand in any one of them and look around you, you would see exactly the same thing as if you were to stand in another one. That’s the fundamental property of a crystal. The only difference is that a crystal is a three-dimensional arrangement instead of a two-dimensional arrangement; and naturally, instead of flowers, each element of the lattice is some kind of an arrangement of atoms—perhaps six hydrogen atoms and two carbon atoms—in some kind of pattern. The pattern of atoms in a crystal can be found out experimentally by x-ray diffraction. We have mentioned this method briefly before, and won’t say any more now except that the precise arrangement of the atoms in space has been worked out for most simple crystals and also for some fairly complex ones.

The internal pattern of a crystal shows up in several ways. First, the binding strength of the atoms in certain directions is usually stronger than in other directions. This means that there are certain planes through the crystal where it is more easily broken than others. They are called the *cleavage planes*. If you crack a crystal with a knife blade it will often split apart along such a plane. Second, the internal structure often appears at the surface because of the way the crystal was formed. Imagine a crystal being deposited out of a solution. There are the atoms floating around in the solution and finally settling down when they find a position

30-1 The internal geometry of crystals

30-2 Chemical bonds in crystals

30-3 The growth of crystals

30-4 Crystal lattices

30-5 Symmetries in two dimensions

30-6 Symmetries in three dimensions

30-7 The strength of metals

30-8 Dislocations and crystal growth

30-9 The Bragg-Nye crystal model

Reference: C. Kittel, *Introduction to Solid State Physics*, John Wiley and Sons, Inc., New York, 2nd ed., 1956.

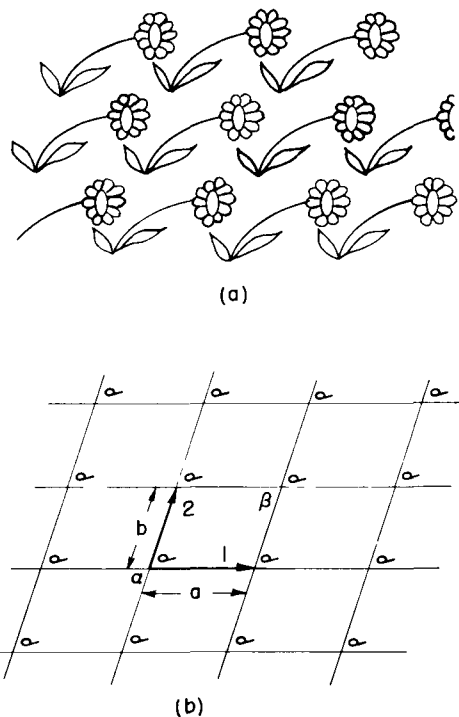


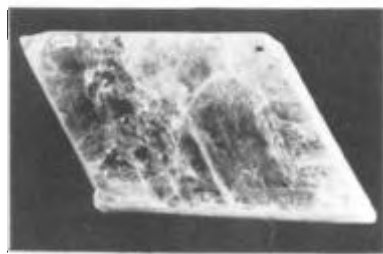
Fig. 30-1. A repeating pattern in two dimensions.



(a)



(b)



(c)

Fig. 30-2. Natural crystals: (a) quartz, (b) sodium chloride, (c) mica.

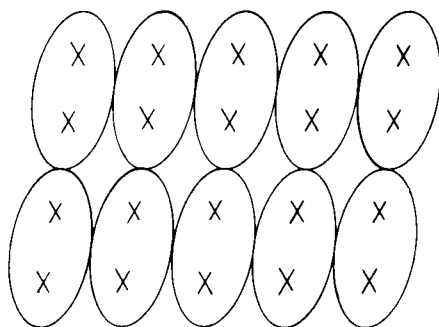


Fig. 30-3. The lattice of a molecular crystal.

of lowest energy. (It's as if the wallpaper got made by flowers drifting around until one drifted accidentally into place and got stuck, and then the next, and the next so that the pattern gradually grows.) You can appreciate that there will be certain directions in which it will grow at a different speed than in other directions, thereby growing into some kind of geometrical shape. Because of such effects, the outside surfaces of many crystals show some of the character of the internal arrangement of the atoms

For example, Fig. 30-2(a) shows the shape of a typical quartz crystal whose internal pattern is hexagonal. If you look closely at such a crystal, you will notice that the outside does not make a very good hexagon because the sides are not all of equal length—they are, in fact, often very unequal. But in one respect it is a very good hexagon: the *angles* between the faces are exactly 120° . Clearly, the size of any particular face is an accident of the growth, but the *angles* are a representation of the internal geometry. So every crystal of quartz has a different shape, even though the angles between corresponding faces are always the same.

The internal geometry of a crystal of sodium chloride is also evident from its external shape. Figure 30-2(b) shows the shape of a typical grain of salt. Again the crystal is not a perfect cube, but the faces *are* exactly at right angles to one another.

A more complicated crystal is mica, which has the shape shown in Fig. 30-2(c). It is a highly anisotropic crystal, as is easily seen from the fact that it is very tough if you try to pull it apart in one direction (horizontally in the figure), but very easy to split by pulling apart in the other direction (vertically). It has commonly been used to obtain very tough, thin sheets. Mica and quartz are two examples of natural minerals containing silica. A third example of a mineral with silica is asbestos, which has the interesting property that it is easily pulled apart in two directions but not in the third. It appears to be made of very strong, *linear* fibers.

30-2 Chemical bonds in crystals

The mechanical properties of crystals clearly depend on the kind of chemical bindings between the atoms. The strikingly different strength of mica along different directions depends on the kinds of interatomic binding in the different directions. You have already learned in chemistry, no doubt, about the different kinds of chemical bonds. First, there are ionic bonds, as we have already discussed for sodium chloride. Roughly speaking, the sodium atoms have lost an electron and become positive ions, the chlorine atoms have gained an electron and become negative ions. The positive and negative ions are arranged in a three-dimensional checkerboard and are held together by electrical forces.

The covalent bond—in which electrons are shared between two atoms—is more common and is usually very strong. In a diamond, for example, the carbon atoms have covalent bonds in all four directions to the nearest neighbors, so the crystal is very hard indeed. There is also covalent bonding between silicon and oxygen in a quartz crystal, but there the bond is really only partially covalent. Because there is not complete sharing of the electrons, the atoms are partly charged, and the crystal is somewhat ionic. Nature is not as simple as we try to make it; there are really all possible gradations between covalent and ionic bonding.

A sugar crystal has still another kind of binding. In it there are large molecules in which the atoms are held strongly together by covalent bonds, so that the molecule is a tough structure. But since the strong bonds are completely satisfied, there are only relatively weak attractions between the separate, individual molecules. In such *molecular* crystals the molecules keep their individual identity, so to speak, and the internal arrangement might be as shown in Fig. 30-3. Since the molecules are not held strongly to each other, the crystals are easy to break. They are quite different from something like diamond, which is really one giant molecule that cannot be broken anywhere without disrupting strong covalent bonds. Paraffin is another example of a molecular crystal.

An extreme example of a molecular crystal occurs in a substance like solid argon. There is very little attraction between the atoms—each atom is a completely

saturated monatomic molecule. But at very low temperatures, the thermal motion is very small, so the slight interatomic forces can cause the atoms to settle down into a regular array like a pile of closely packed spheres.

The metals form a completely different class of substances. The bonding is of an entirely different kind. In a metal the bonding is not between adjacent atoms but is a property of the whole crystal. The valence electrons are not attached to one atom or to a pair of atoms but are shared throughout the crystal. Each atom contributes an electron to a universal pool of electrons, and the atomic positive ions reside in the sea of negative electrons. The electron sea holds the ions together like some kind of glue.

In the metals, since there are no special bonds in any particular direction, there is no strong directionality in the binding. They are still crystalline, however, because the total energy is lowest when the atomic ions are arranged in some definite array—although the energy of the preferred arrangement is not usually much lower than other possible ones. To a first approximation, the atoms of many metals are like small spheres packed in as tightly as possible.

30-3 The growth of crystals

Try to imagine the natural formation of crystals in the earth. In the earth's surface there is a big mixture of all kinds of atoms. They are being continually churned about by volcanic action, by wind, and by water—continually being moved about and mixed. Yet, by some trick, silicon atoms gradually begin to find each other, and to find oxygen atoms, to make silica. One atom at a time is added to the others to build up a crystal—the mixture gets unmixed. And somewhere nearby, sodium and chlorine atoms are finding each other and building up a crystal of salt.

How does it happen that once a crystal is started, it permits only a particular kind of atom to join on? It happens because the whole system is working toward the lowest possible energy. A growing crystal will accept a new atom if it is going to make the energy as low as possible. But how does it *know* that a silicon—or an oxygen—atom at some particular spot is going to result in the lowest possible energy? It does it by trial and error. In the liquid, all of the atoms are in perpetual motion. Each atom bounces against its neighbors about 10^{13} times every second. If it hits against the right spot of growing crystal, it has a somewhat smaller chance of jumping off again if the energy is low. By continually testing over periods of millions of years at a rate of 10^{13} tests per second, the atoms gradually build up at the places where they find their lowest energy. Eventually they grow into big crystals.

30-4 Crystal lattices

The arrangement of the atoms in a crystal—the crystal *lattice*—can take on many geometric forms. We would like to describe first the simplest lattices, which are characteristic of most of the metals and of the solid form of the inert gases. They are the cubic lattices which can occur in two forms: the body-centered cubic, shown in Fig. 30-4(a), and the face-centered cubic shown in Fig. 30-4(b). The drawings show, of course, only one cube of the lattice; you are to imagine that the pattern is repeated indefinitely in three dimensions. Also, to make the drawing clearer, only the “centers” of the atoms are shown. In an actual crystal, the atoms are more like spheres in contact with each other. The dark and light spheres in the drawings may, in general, stand for different kinds of atoms or may be the same kind. For instance, iron has a body-centered cubic lattice at low temperatures, but a face-centered cubic lattice at higher temperatures. The physical properties are quite different in the two crystalline forms.

How do such forms come about? Imagine that you have the problem of packing spherical atoms together as tightly as possible. One way would be to start by making a layer in a “hexagonal close-packed array,” as shown in Fig. 30-5(a). Then you could build up a second layer like the first, but displaced horizontally,

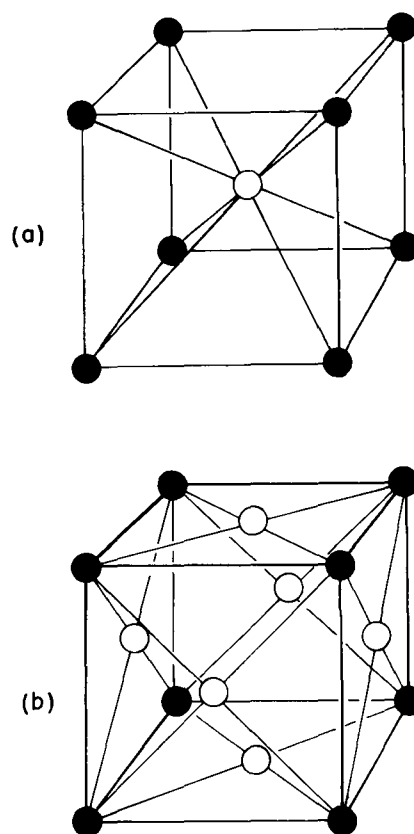


Fig. 30-4. The unit cell of cubic crystals: (a) body-centered, (b) face-centered.

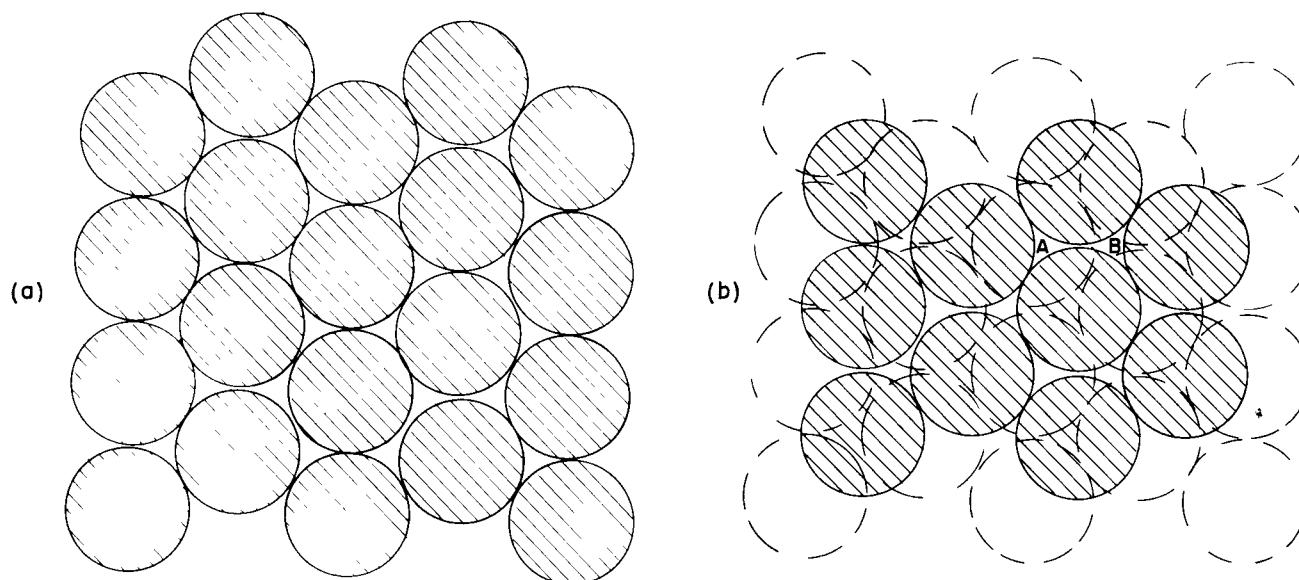


Fig. 30-5. Building up a hexagonal close-packed lattice.

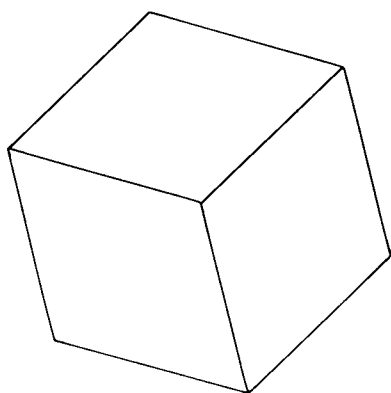


Fig. 30-6. Is this a hexagon or a cube seen from one corner?

as shown in Fig. 30-5(b). Next, you can put on the third layer. But notice! There are *two* distinct ways of placing the *third* layer. If you start the third layer by placing an atom at *A* in Fig. 30-5(b), each atom in the third layer is directly above an atom of the bottom layer. On the other hand, if you start the third layer by putting an atom at the position *B*, the atoms of the third layer will be centered at points exactly in the middle of a triangle formed by three atoms of the bottom layer. Any other starting place is equivalent to *A* or *B*, so there are only two ways of placing the third layer.

If the third layer has an atom at point *B*, the crystal lattice is a face-centered cubic—but seen at an angle. It seems funny that starting with hexagons you can end up with cubes. But notice that a cube looked at from a corner has a hexagonal outline. For instance, Fig. 30-6 could represent a plane hexagon or a cube seen in perspective!

If a third layer is added to Fig. 30-5(b) by starting with an atom at *A*, there is no cubical structure, and the lattice has instead only a hexagonal symmetry. It is clear that both possibilities we have described are equally close-packed.

Some metals—for example, copper and silver—choose the first alternative, the face-centered cubic. Others—for example, beryllium and magnesium—choose the other alternatives; they form hexagonal crystals. Clearly, which crystal lattice appears cannot depend only on the packing of little spheres, but must also be determined in part by other factors. In particular, it depends on the slight remaining angular dependence of the interatomic forces (or, in the case of the metals, on the energy of the electron pool). You will, no doubt, learn all about such things in your chemistry courses.

30-5 Symmetries in two dimensions

We would now like to discuss some of the properties of crystals from the point of view of their internal symmetries. The main feature of a crystal is that if you start at one atom and move to a corresponding atom one lattice unit away, you are again in the same kind of an environment. That's the fundamental proposition. But if you were an atom, there would be another kind of change that could take you again to the same environment—that is, another possible "symmetry." Figure 30-7(a) shows another possible "wallpaper-type" design (though one you have probably never seen). Suppose we compare the environments for points *A* and *B*. You might, at first, think that they are the same—but not quite. Points *C* and *D* are equivalent to *A*, but the environment of *B* is like that of *A* only if the surroundings are reversed, as in a mirror reflection.

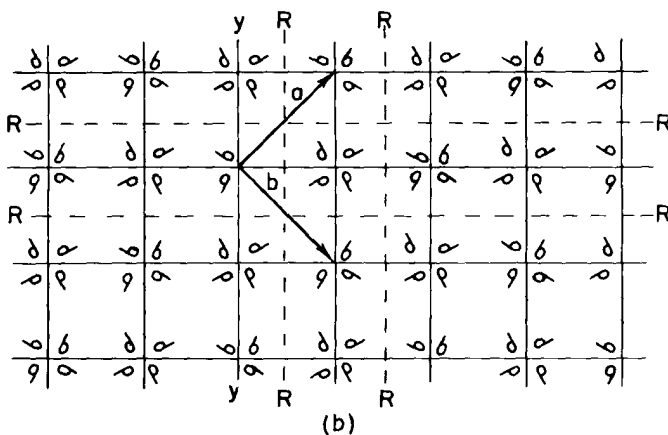
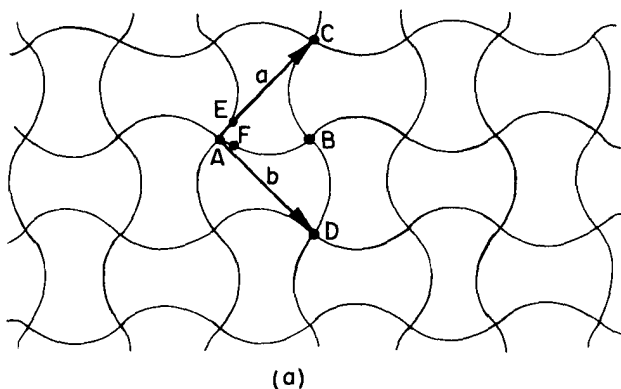


Fig. 30-7. A pattern of high symmetry.

There are other kinds of “equivalent” points in the pattern. For instance, the points *E* and *F* have the “same” environments except that one is rotated 90° with respect to the other. The pattern is quite special. A rotation of 90° —or any multiple of it—about a vertex such as *A* gives the same pattern all over again. A crystal with such a structure would have square corners on the outside, but inside it is more complicated than a simple cube.

Now that we have described some special examples, let’s try to figure out all the possible symmetries a crystal can have. First, we consider what happens in a plane. A *plane* lattice can be defined by the two so-called *primitive* vectors that go from one point of the lattice to the two *nearest* equivalent points. The two vectors **1** and **2** are the primitive vectors of the lattice of Fig. 30-1. The two vectors *a* and *b* of Fig. 30-7(a) are the primitive vectors of the pattern there. We could, of course, equally well replace *a* by $-a$, or *b* by $-b$. Since *a* and *b* are equal in magnitude and at right angles, a rotation of 90° turns *a* into *b*, and *b* into $-a$, giving the same lattice once again.

We see that there are lattices which have a “four-sided” symmetry. And we have described earlier a close-packed array based on a hexagon which could have a six-sided symmetry. A rotation of the array of circles in Fig. 30-5(a) by an angle of 60° about the center of any circle brings the pattern back to itself.

What other kinds of rotational symmetry are there? Can we have, for example, a fivefold or an eightfold rotational symmetry? It is easy to see that they are impossible. *The only symmetry with more sides than four is a six-sided symmetry.* First, let’s show that more than sixfold symmetry is impossible. Suppose we try to imagine a lattice with two equal primitive vectors with an enclosed angle less than 60° , as in Fig. 30-8(a). We are to suppose that points *B* and *C* are equivalent to *A*, and that *a* and *b* are the two *shortest* vectors from *A* to its equivalent neighbors. But that is clearly wrong, because the distance between *B* and *C* is shorter than from either one to *A*. There must be a neighbor at *D* equivalent to *A* which is closer than *B* or *C*. We should have chosen *b'* as one of our primitive vectors. So the angle between the two primitive vectors must be 60° or larger. Octagonal symmetry is not possible.

What about fivefold symmetry? If we assume that the primitive vectors *a* and *b* have equal lengths and make an angle of $2\pi/5 = 72^\circ$, as in Fig. 30-8(b), then there should also be an equivalent lattice point at *D*, at 72° from *C*. But the vector *b'* from *E* to *D* is then less than *b*, so *b* is not a primitive vector. There can be no fivefold symmetry. The only possibilities that do not get us into this kind of difficulty are $\theta = 60^\circ, 90^\circ$, or 120° . Zero or 180° are also clearly possible. One way of stating our result is that the pattern can be left unchanged by a rotation of one full turn (no change at all), one-half of a turn, one-third, one-fourth, or one-sixth of a turn. And those are all the possible rotational symmetries in a plane—a total of five. If $\theta = 2\pi/n$, we speak of an “*n*-fold” symmetry. We say

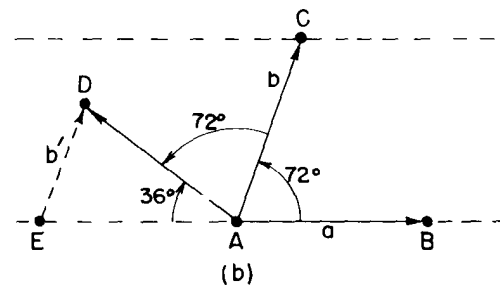
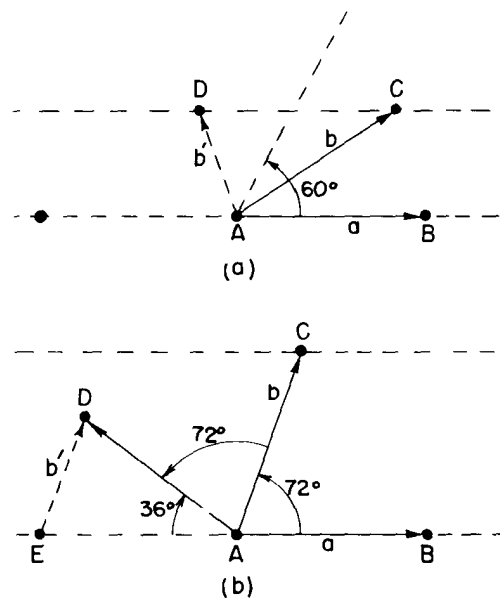


Fig. 30-8. (a) Rotational symmetries greater than sixfold are not possible. (b) Fivefold rotational symmetry is not possible.

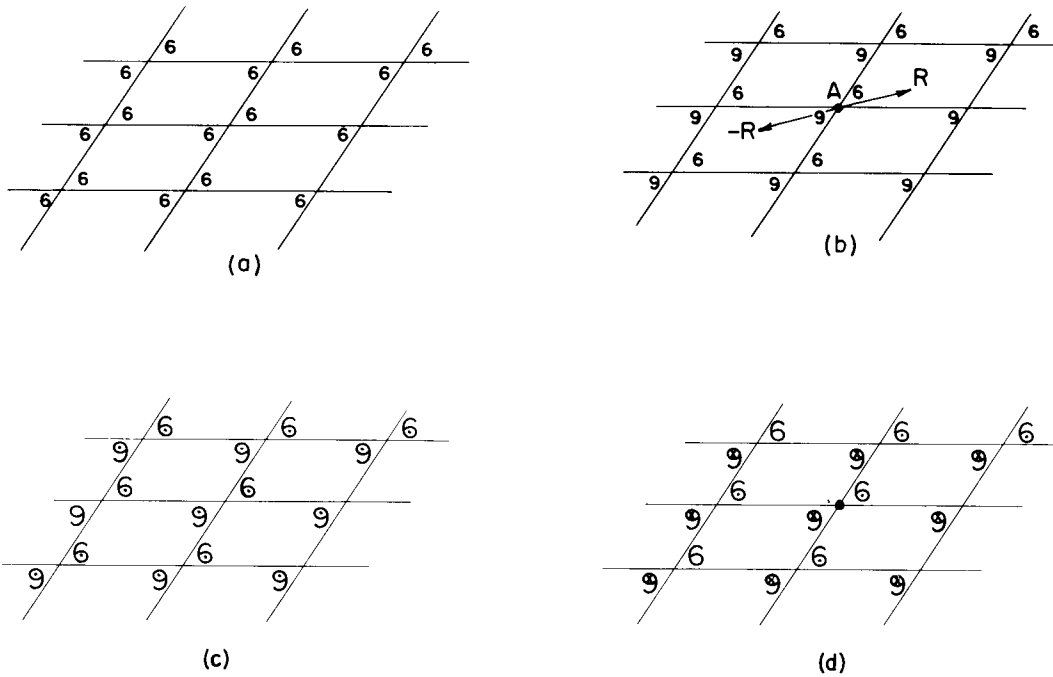


Fig. 30-9. Symmetry under inversion. Pattern (b) is unchanged if $R \rightarrow -R$, but pattern (a) is changed. In three dimensions pattern (d) is symmetric under an inversion but (c) is not.

that a pattern with n equal to 4 or to 6 has a “higher symmetry” than one with n equal to 1 or to 2.

Returning to Fig. 30-7(a), we see that the pattern has a fourfold rotational symmetry. We have drawn in Fig. 30-7(b) another design which has the same symmetry properties as part (a). The little comma-like figures are asymmetric objects which serve to define the symmetry of the design inside of each square. Notice that the commas are reversed in alternate squares, so that the unit cell is larger than one of the small squares. If there were no commas, the pattern would still have fourfold symmetry, but the unit cell would be smaller. The patterns of Fig. 30-7 also have other symmetry properties. For instance, a reflection about any of the broken lines $R-R$ reproduces the same pattern.

The patterns of Fig. 30-7 have still another kind of symmetry. If the pattern is reflected about the line $Y-Y$ and shifted one square to the right (or left), we get back the original pattern. The line $Y-Y$ is called a “glide” line.

These are all the possible symmetries in two dimensions. There is one more spatial symmetry operation which is equivalent *in two dimensions* to a 180° rotation, but which is a quite distinct operation in three dimensions. It is *inversion*. By an inversion we mean that any point at the vector displacement R from some origin [for instance, the point A in Fig. 30-9(b)] is moved to the point at $-R$.

An inversion of pattern (a) of Fig. 30-9 produces a new pattern, but an inversion of pattern (b) reproduces the same pattern. For a two-dimensional pattern (as you can see from the figure), an inversion of the pattern (b) through the point A is equivalent to a rotation of 180° about the same point. Suppose, however, we make the pattern in Fig. 30-9(b) three dimensional by imagining that the little 6's and 9's each have an “arrow” *pointing out of the page*. After an inversion in three dimensions all the arrows will be reversed, so the pattern is *not* reproduced. If we indicate the heads and tails of the arrows by dots and crosses, respectively, we can make a *three-dimensional* pattern, as in Fig. 30-9(c), which is *not* symmetric under an inversion, or we can make a pattern like the one shown in (d), which *does* have such a symmetry. Notice that it is *not* possible to imitate a three-dimensional inversion by any combination of rotations.

If we characterize the “symmetry” of a pattern—or lattice—by the kinds of symmetry operations we have been describing, it turns out that for two dimensions 17 distinct patterns are possible. We have drawn one pattern of the lowest possible

symmetry in Fig. 30-1, and one of high symmetry in Fig. 30-7. We will leave you with the game of trying to figure out all of the 17 possible patterns.

It is peculiar how few of the 17 possible patterns are used in making wallpaper and fabrics. One always sees the same three or four basic patterns. Is this because of a lack of imagination of designers, or because many of the possible patterns are not pleasing to the eye?

30-6 Symmetries in three dimensions

So far we have talked only about patterns in two dimensions. What we are really interested in, however, are patterns of atoms in three dimensions. First, it is clear that a three-dimensional crystal will have *three* primitive vectors. If we then ask about the possible symmetry operations in three dimensions, we find that there are 230 different possible symmetries! For some purposes, these 230 types can be grouped into seven classes, which are drawn in Fig. 30-10. The lattice with the least symmetry is called the *triclinic*. Its unit cell is a parallelepiped. The primitive vectors are of different lengths, and no two of the angles between them are equal. There is no possibility of any rotational or reflection symmetry. There are, however, still two possible symmetries—the unit cell is, or is not, changed by an inversion through the vertex. (By an inversion in three dimensions, we again mean that spatial displacements $\mathbf{R}$ are replaced by $-\mathbf{R}$ —in other words, that (x, y, z) goes into $(-x, -y, -z)$). So the triclinic lattice has only two possible symmetries, unless there is some special relation among the primitive vectors. For example, if all the vectors are equal and are separated by equal angles, one has the *trigonal* lattice shown in the figure. This figure can have an additional symmetry, it may be unchanged by a rotation about the long, body diagonal.

If one of the primitive vectors, say c , is at right angles to the other two, we get a *monoclinic* unit cell. A new symmetry is possible—a rotation by 180° about c . The *hexagonal* cell is a special case in which the vectors a and b are equal and the angle between them is 60° , so that a rotation of 60° , or 120° , or 180° about the vector c repeats the same lattice (for certain internal symmetries).

If all three primitive vectors are at right angles, but of different lengths, we get the *orthorhombic* cell. The figure is symmetric for rotations of 180° about the three axes. Higher-order symmetries are possible with the *tetragonal* cell, which has all right angles and two equal primitive vectors. Finally, there is the *cubic* cell, which is the most symmetric of all.

The point of all this discussion about symmetries is that the internal symmetries of the crystals show up—sometimes in subtle ways—in the macroscopic physical properties of the crystal. For instance, a crystal will, in general, have a tensor electric polarizability. If we describe the tensor in terms of the ellipsoid of polarization, we should expect that some of the crystal symmetries should show up also in the ellipsoid. For example, a cubic crystal is symmetric with respect to a rotation of 90° about any one of three orthogonal directions. Clearly, the only ellipsoid with this property is a sphere. *A cubic crystal must be an isotropic dielectric.*

On the other hand, a tetragonal crystal has a fourfold rotational symmetry. Its ellipsoid must have two of its principal axes equal, and the third must be parallel to the axis of the crystal. Similarly, since the orthorhombic crystal has twofold rotational symmetry about three orthogonal axes, its axes must coincide with the axes of the polarization ellipsoid. In a like manner, *one* of the axes of a monoclinic crystal must be parallel to *one* of the principal axes of the ellipsoid, though we can't say anything about the other axes. Since a triclinic crystal has no rotational symmetry, the ellipsoid can have any orientation at all.

As you can see, we can make a big game of figuring out the possible symmetries and relating them to the possible physical tensors. We have considered only the polarization tensor, but things get more complicated for others—for instance, for the tensor of elasticity. There is a branch of mathematics called “group theory” that deals with such subjects, but usually you can figure out what you want with common sense.

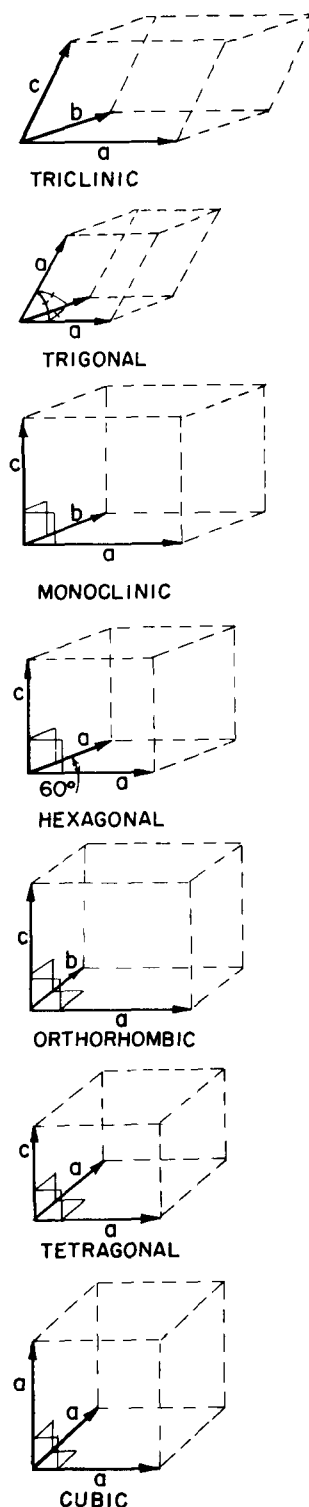


Fig. 30-10. The seven classes of crystal lattices.

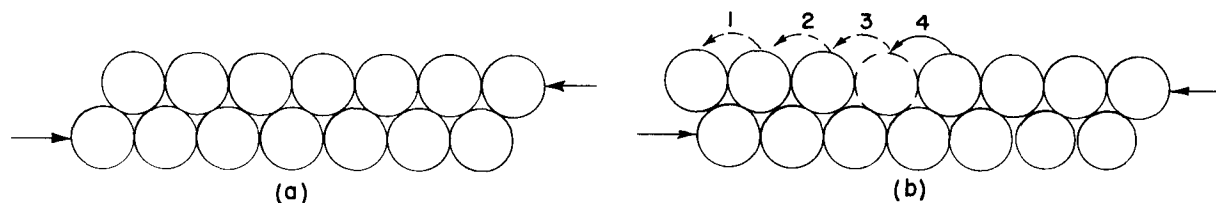


Fig. 30-11. Slippage of crystal planes.

30-7 The strength of metals

We have said that metals usually have a simple cubic crystal structure; we want now to discuss their mechanical properties—which depend on this structure. Metals are, generally speaking, very “soft,” because it is easy to slide one layer of the crystal over the next. You may think: “That’s ridiculous; metals are strong.” Not so, a *single crystal* of a metal can be distorted very easily.

Suppose we look at two layers of a crystal subjected to a shear force, as shown in the diagram of Fig. 30-11(a). You might at first think the whole layer would resist motion until the force was big enough to push the whole layer “over the hump,” so that it shifted one notch to the left. Although slipping does occur along a plane, it doesn’t happen that way (If it did, you would calculate that the metal is much stronger than it really is.) What happens is more like one atom going at a time; first the atom on the left makes its jump, then the next, and so on, as indicated in Fig. 30-11(b). In effect it is the vacant space between two atoms that quickly travels to the right, with the net result that the whole second layer has moved over one atomic spacing. The slipping goes this way because it takes much less energy to lift one atom at a time over the hump than to lift a whole row. Once the force is enough to start the process, it goes the rest of the way very fast.

It turns out that in a real crystal, slipping will occur repeatedly at one plane, then will stop there and start at some other plane. The details of why it starts and stops are quite mysterious. It is, in fact, quite strange that successive regions of slip are often fairly evenly spaced. Figure 30-12 shows a photograph of a tiny, thin copper crystal that has been stretched. You can see the various planes where slipping has occurred.

Fig. 30-12. A photograph of a small crystal of copper after stretching. [Courtesy of S. S. Brenner, Senior Scientist, United States Steel Research Center, Monroeville, Pa.]

The sudden slipping of individual crystal planes is quite apparent if you take a piece of tin wire that has large crystals in it and stretch it while holding it next to your ear. You can hear a rush of “ticks” as the planes snap to their new positions, one after the other.

The problem of having a “missing” atom in one row is somewhat more difficult than it might appear from Fig. 30-11. When there are more layers, the situation must be something like that shown in Fig. 30-13. Such an imperfection in a crystal is called a *dislocation*. It is presumed that such dislocations are either present when the crystal was formed or are generated at some notch or crack at the surface. Once they are produced, they can move relatively freely through the crystal. The gross distortions result from the motions of many of such dislocations.

Dislocations can move freely—that is, they require little extra energy—so long as the rest of the crystal has a perfect lattice. But they may get “stuck” if they encounter some other kind of imperfection in the crystal. If it takes a lot of energy for them to pass the imperfection, they will be stopped. This is precisely the mechanism that gives strength to *imperfect* metal crystals. Pure iron crystals are quite soft, but a small concentration of impurity atoms may cause enough imperfections to effectively immobilize the dislocations. As you know, steel, which is primarily iron, is very hard. To make steel, a small amount of carbon is dissolved in the iron melt; if the melt is cooled rapidly, the carbon precipitates out in little grains, making many microscopic distortions in the lattice. The dislocations can no longer move about, and the metal is hard.

Pure copper is very soft, but can be “work-hardened.” This is done by hammering on it or bending it back and forth. In this case, many new dislocations of various kinds are made which interfere with one another, cutting down their

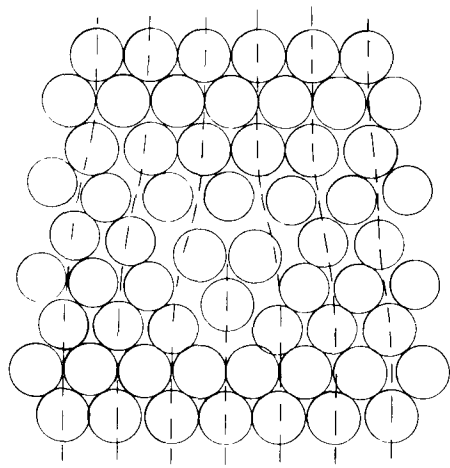


Fig. 30-13. A dislocation in a crystal.

mobility. Perhaps you've seen the trick of taking a bar of "dead soft" copper and gently bending it around someone's wrist as a bracelet. In the process, it becomes work-hardened and cannot easily be unbent again! A work-hardened metal like copper can be made soft again by annealing at a high temperature. The thermal motion of the atoms "irons out" the dislocations and makes large single crystals again. We have, so far, described only the so-called *slip* dislocation. There are many other kinds, one of which is the *screw* dislocation shown in Fig. 30-14. Such dislocations often play an important part in crystal growth.

30-8 Dislocations and crystal growth

One of the great puzzles for a long time was how crystals can possibly grow. We have described how it is that each atom might, by repeated testing, determine whether it was better to be in the crystal or not. But that means that each atom must find a place of low energy. However, an atom put on a new surface is only bound by one or two bonds from below, and doesn't have the same energy it would have if it were placed in a corner, where it would have atoms on three sides. Suppose we imagine a growing crystal as a stack of blocks, as shown in Fig. 30-15. If we try a new block at, say, position *A*, it will have only one of the six neighbors it should ultimately get. With so many bonds lacking, its energy is not very low. It would be better off at position *B*, where it already has one-half of its quota of bonds. Crystals do indeed grow by attaching new atoms at places like *B*.

What happens, though, when that line is finished? To start a new line, an atom must come to rest with only two sides attached, and that is again not very likely. Even if it did, what would happen when the layer was finished? How could a new layer get started? One answer is that the crystal prefers to grow at a dislocation, for instance around a screw dislocation like the one shown in Fig. 30-14. As blocks are added to this crystal, there is always some place where there are three available bonds. The crystal prefers, therefore, to grow with a dislocation built in. Such a spiral pattern of growth is shown in Fig. 30-16, which is a photograph of a single crystal of paraffin.

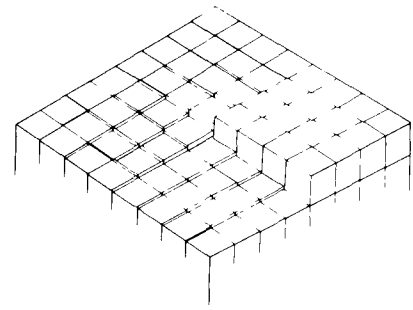


Fig. 30-14. A screw dislocation. [From Charles Kittel, *Introduction to Solid State Physics*, John Wiley and Sons, Inc., New York, 2nd ed., 1956.]

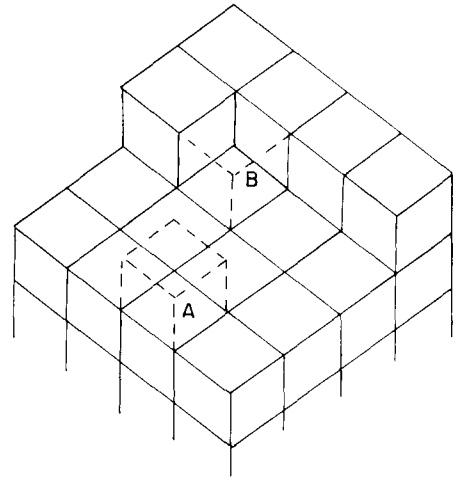


Fig. 30-15. Crystal growth.

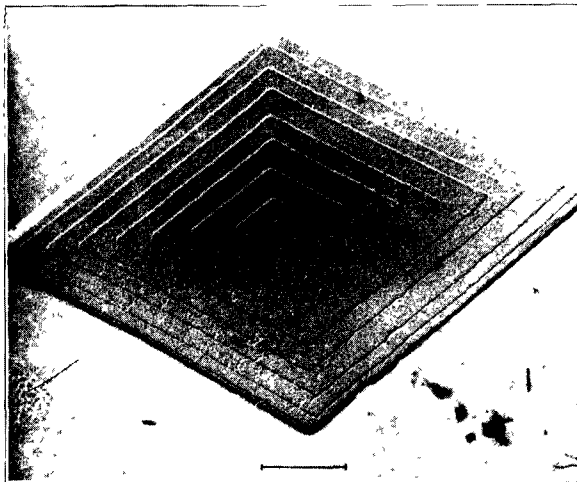


Fig. 30-16. A paraffin crystal which has grown around a screw dislocation. [From Charles Kittel, *Introduction to Solid State Physics*, John Wiley and Sons, Inc., New York, 2nd ed., 1956.]

30-9 The Bragg-Nye crystal model

We cannot, of course, see what goes on with the individual atoms in a crystal. Also, as you realize by now, there are many complicated phenomena that are not easy to treat quantitatively. Sir Lawrence Bragg and J. F. Nye have devised a scheme for making a model of a metallic crystal which shows in a striking way many of the phenomena that are believed to occur in a real metal. In the following pages we have reproduced their original article, which describes their method and shows some of the results they obtained with it. (The article is reprinted from the *Proceedings of the Royal Society of London*, Vol. 190, September 1947, pp. 474-481—with the permission of the authors and of the Royal Society.)

A dynamical model of a crystal structure

BY SIR LAWRENCE BRAGG, F.R.S. AND J. F. NYE

Cavendish Laboratory, University of Cambridge

(Received 9 January 1947—Read 19 June 1947)

[Plates 8 to 21]

The crystal structure of a metal is represented by an assemblage of bubbles, a millimetre or less in diameter, floating on the surface of a soap solution. The bubbles are blown from a fine pipette beneath the surface with a constant air pressure, and are remarkably uniform in size. They are held together by surface tension, either in a single layer on the surface or in a three-dimensional mass. An assemblage may contain hundreds of thousands of bubbles and persists for an hour or more. The assemblages show structures which have been supposed to exist in metals, and simulate effects which have been observed, such as grain boundaries, dislocations and other types of fault, slip, recrystallization, annealing, and strains due to 'foreign' atoms.

1 THE BUBBLE MODEL

Models of crystal structure have been described from time to time in which the atoms are represented by small floating or suspended magnets, or by circular disks floating on a water surface and held together by the forces of capillary attraction. These models have certain disadvantages, for instance, in the case of floating objects in contact, frictional forces impede their free relative movement. A more serious disadvantage is that the number of components is limited for a large number of components is required in order to approach the state of affairs in a real crystal. The present paper describes the behaviour of a model in which the atoms are represented by small bubbles from 2.0 to 0.1 mm in diameter floating on the surface of a soap solution. These small bubbles are sufficiently persistent for experiments lasting an hour or more, they slide past each other without friction, and they can be produced in large numbers. Some of the illustrations in this paper were taken from assemblages of bubbles numbering 100,000 or more. The model most nearly represents the behaviour of a metal structure, because the bubbles are of one type only and are held together by a general capillary attraction which represents the binding force of the free electrons in the metal. A brief description of the model has been given in the *Journal of Scientific Instruments* (Bragg 1942b).

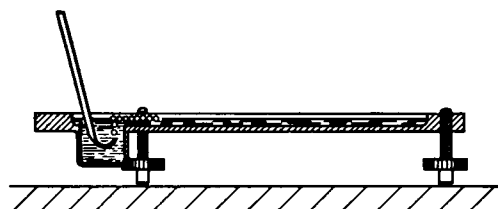


FIGURE 1. Apparatus for producing rafts of bubbles.

2 METHOD OF FORMATION

The bubbles are blown from a fine orifice, beneath the surface of a soap solution. We have had the best results with a solution the formula of which was given to us by Mr Green of the Royal Institution. 15.2 c.c. of oleic acid (pure redistilled) is well shaken in 50 c.c. of distilled water. This is mixed thoroughly with 73 c.c. of 10% solution of tri-ethanolamine and the mixture made up to 200 c.c. To this is added 164 c.c. of pure glycerine. It is left to stand and the clear liquid is drawn off from below. In some experiments this was diluted in three times its volume of water to reduce viscosity. The orifice of the jet is about 5 mm. below the surface. A constant air pressure of 50 to 200 cm. of water is supplied by means of two Winchester flasks. Normally the bubbles are remarkably uniform in size. Occasionally they issue in an irregular manner, but this can be corrected by a change of jet or of pressure. Unwanted bubbles can easily be destroyed by playing a small flame over the surface. Figure 1 shows the apparatus. We have found it of advantage to blacken the bottom of the vessel, because details of structure, such as grain boundaries and dislocations, then show up more clearly.

Figure 2, plate 8, shows a portion of a 'raft' or two-dimensional crystal of bubbles. Its regularity can be judged by looking at the figure in a glancing direction. The size of the bubbles varies with the aperture, but does not appear to vary to any marked degree with the pressure or the depth of the orifice beneath the surface. The main effect of increasing the pressure is to increase the rate of issue of the bubbles. As an example, a thick-walled jet of 49 μ bore with a pressure of 100 cm. produced bubbles of 1.2 mm. in diameter. A thin-walled jet of 27 μ diameter and a pressure of 180 cm. produced bubbles of 0.6 mm. diameter. It is convenient to refer to bubbles of 2.0 to 1.0 mm. diameter as 'large' bubbles, those from 0.8 to 0.6 mm. diameter as 'medium' bubbles, and those from 0.3 to 0.1 mm. diameter as 'small' bubbles, since their behaviour varies with their size.

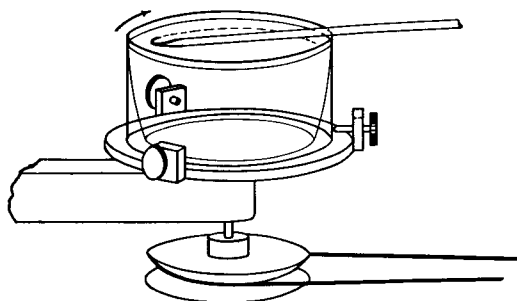


FIGURE 3. Apparatus for producing bubbles of small size.

With this apparatus we have not found it possible to reduce the size of the jet and so produce bubbles of smaller diameter than 0.6 mm. As it was desired to experiment with very small bubbles, we had recourse to placing the soap solution in a rotating vessel and introducing a fine jet as nearly as possible parallel to a stream line. The bubbles are swept away as they form, and under steady conditions are reasonably uniform. They issue at a rate of one thousand or more per second, giving a high-pitched note. The soap solution mounts up in a steep wall around the perimeter of the vessel while it is rotating, but carries back most of the bubbles with it when rotation ceases. With this device, illustrated in figure 3, bubbles down to 0.12 mm. in diameter can be obtained. As an example, an orifice 38 μ across in a thin-walled jet, with a pressure of 190 cm. of water, and a speed of the fluid of 180 cm./sec. past the orifice, produced bubbles of 0.14 mm. diameter. In this case a dish of diameter 9.5 cm. and speed of 6 rev./sec. was used. Figure 4, plate 8, is an enlarged picture of these 'small' bubbles and shows their degree of regularity; the pattern is not as perfect with a rotating as with a stationary vessel, the rows being seen to be slightly irregular when viewed in a glancing direction.

These two-dimensional crystals show structures which have been supposed to exist in metals, and simulate effects which have been observed, such as grain boundaries, dislocations and other types of fault, slip, recrystallization, annealing, and strains due to 'foreign' atoms.

3 GRAIN BOUNDARIES

Figures 5a, 5b and 5c, plates 9 and 10, show typical grain boundaries for bubbles of 1.87, 0.76 and 0.30 mm. diameter respectively. The width of the disturbed area at the boundary, where the bubbles have an irregular distribution, is in general greater the smaller the bubbles. In figure 5a, which shows portions of several adjacent grains, bubbles at a boundary between two grains adhere definitely to one crystalline arrangement or the other. In figure 5c there is a marked 'Beilby layer' between the two grains. The small bubbles, as will be seen, have a greater rigidity than the large ones, and this appears to give rise to more irregularity at the interface.

Separate grains show up distinctly when photographs of polycrystalline rafts such as figures 5a to 5c, plates 9 and 10, and figures 12a to 12e, plates 14 to 16, are viewed obliquely. With suitable lighting, the floating raft of bubbles itself when viewed obliquely resembles a polished and etched metal in a remarkable way.

It often happens that some 'impurity atoms', or bubbles which are markedly larger or smaller than the average, are found in a polycrystalline raft, and when this is so a large proportion of them are situated at the grain boundaries. It would be incorrect to say that the irregular bubbles make their way to the boundaries; it is a defect of the model that no diffusion of bubbles through the structure can take place, mutual adjustments of neighbours alone being possible. It appears that the boundaries tend to readjust themselves by the growth of one crystal at the expense of another till they pass through the irregular atoms.

4 DISLOCATIONS

When a single crystal or polycrystalline raft is compressed, extended, or otherwise deformed it exhibits a behaviour very similar to that which has been pictured for metals subjected to strain. Up to a certain limit the model is within its elastic range. Beyond that point it yields by slip along one of the three equally inclined directions of closely packed rows. Slip takes place by the bubbles in one row moving forward over those in the next row by an amount equal to the distance between neighbours. It is very interesting to watch this process taking place. The movement is not simultaneous along the whole row but begins at one end with the appearance of a 'dislocation', where there is locally one more bubble in the rows on one side of the slip line as compared with those on the other. This dislocation then runs along the slip line from one side of the crystal to the other, the final result being a slip by one 'inter-atomic' distance. Such a process has been invoked by Orowan, by Polanyi and by Taylor to explain the small forces required to produce plastic gliding in metal structures. The theory put forward by Taylor (1934) to explain the mechanism of plastic deformation of crystals considers the mutual action and equilibrium of such dislocations. The bubbles afford a very striking picture of what has been supposed to take place in the metal. Sometimes

the dislocations run along quite slowly, taking a matter of seconds to cross a crystal, stationary dislocations also are to be seen in crystals which are not homogeneously strained. They appear as short black lines, and can be seen in the series of photographs, figures 12a to 12e, plates 14 to 16. When a polycrystalline raft is compressed, these dark lines are seen to be dashing about in all directions across the crystals.

Figures 6a, 6b and 6c, plates 10 and 11, show examples of dislocations. In figure 6a, where the diameter of the bubbles is 1.9 mm, the dislocation is very local, extending over about six bubbles. In figure 6b (diameter 0.76 mm) it extends over twelve bubbles, and in figure 6c (diameter 0.30 mm) its influence can be traced for a length of about fifty bubbles. The greater rigidity of the small bubbles leads to longer dislocations. The study of any mass of bubbles shows, however, that there is not a standard length of dislocation for each size. The length depends upon the nature of the strain in the crystal. A boundary between two crystals with corresponding axes at approximately 30° (the maximum angle which can occur) may be regarded as a series of dislocations in alternate rows, and in this case the dislocations are very short. As the angle between the neighbouring crystals decreases, the dislocations occur at wider intervals and at the same time become longer, till one finally has single dislocations in a large body of perfect structure as shown in figures 6a, 6b and 6c.

Figure 7, plate 11, shows three parallel dislocations. If we call them positive and negative (following Taylor) they are positive, negative, positive, reading from left to right. The strip between the last two has three bubbles in excess, as can be seen by looking along the rows in a horizontal direction. Figure 8, plate 12, shows a dislocation projecting from a grain boundary, an effect often observed.

Figure 9, plate 12, shows a place where two bubbles take the place of one. This may be regarded as a limiting case of positive and negative dislocations on neighbouring rows, with the compressive sides of the dislocations facing each other. The contrary case would lead to a hole in the structure, one bubble being missing at the point where the dislocations met.

5. OTHER TYPES OF FAULT

Figure 10, plate 12, shows a narrow strip between two crystals of parallel orientation, the strip being crossed by a number of fault lines where the bubbles are not in close packing. It is in such places as these that recrystallization may be expected. The boundaries approach and the strip is absorbed into a wider area of perfect crystal.

Figures 11a to 11g, plates 13 and 14 are examples of arrangements which frequently appear in places where there is a local deficiency of bubbles. While a dislocation is seen as a dark stripe in a general view, these structures show up in the shape of the letter V or as triangles. A typical V structure is seen in figure 11a. When the model is being distorted, a V structure is formed by two dislocations meeting at an inclination of 60°, it is destroyed by the dislocations continuing along their paths. Figure 11b shows a small triangle, which also embodies a dislocation, for it will be noticed that the rows below the fault have one more bubble than those below. If a mild amount of 'thermal movement' is imposed by gentle agitation of one side of the crystal, such faulty places disappear and a perfect structure is formed.

Here and there in the crystals there is a blank space where a bubble is missing, showing as a black dot in a general view. Examples occur in figure 11g. Such a gap cannot be closed by a local readjustment, since filling the hole causes another to appear. Such holes both appear and disappear when the crystal is 'cold-worked'.

These structures in the model suggest that similar local faults may exist in an actual metal. They may play a part in processes such as diffusion or the order-disorder change by reducing energy barriers in their neighbourhood, and act as nuclei for crystallization in an allotropic change.

6. RECRYSTALLIZATION AND ANNEALING

Figures 12a to 12e, plates 14 to 16, show the same raft of bubbles at successive times. A raft covering the surface of the solution was given a vigorous stirring with a glass rake, and then left to adjust itself. Figure 12a shows its aspect about 1 sec. after stirring has ceased. The raft is broken into a number of small 'crystallites', these are in a high state of non-homogeneous strain as is shown by the numerous dislocations and other faults. The following photograph (figure 12b) shows the same raft 32 sec. later. The small grains have coalesced to form larger grains, and much of the strain has disappeared in the process. Recrystallization takes place right through the series, the last three photographs of which show the appearance of the raft 2, 14 and 25 min. after the initial stirring. It is not possible to follow the rearrangement for much longer times, because the bubbles shrink after long standing, apparently due to the diffusion of air through their walls, and they also become thin and tend to burst. No agitation was given to the model during this process. An ever slower process of rearrangement goes on, the movement of the bubbles in one part of the raft setting up strains which activate a rearrangement in a neighbouring part, and that in its turn still another.

A number of interesting points are to be seen in this series. Note the three small grains at the points indicated by the co-ordinates AA, BB, CC. A persists, though

changed in form, throughout the whole series. B is still present after 14 min., but has disappeared in 25 min., leaving behind it four dislocations marking internal strain in the grain. Grain C shrinks and finally disappears in figure 12d, leaving a hole and a V which has disappeared in figure 12e. At the same time the ill-defined boundary in figure 12d at DD has become a definite one in figure 12e. Note also the straightening out of the grain boundary in the neighbourhood of EE in figures 12b to 12e. Dislocations of various lengths can be seen, marking all stages between a slight warping of the structure and a definite boundary. Holes where bubbles are missing show up as black dots. Some of these holes are formed or filled up by movements of dislocations, but others represent places where a bubble has burst. Many examples of V's and some of triangles can be seen. Other interesting points will be apparent from a study of this series of photographs.

Figures 13a, 13b and 13c, plate 17, show a portion of a raft 1 sec., 4 sec. and 4 min. after the stirring process, and is interesting as showing two successive stages in the relaxation towards a more perfect arrangement. The changes show up well when one looks in a glancing direction across the page. The arrangement is very broken in figure 13a. In figure 13b the bubbles have grouped themselves in rows, but the curvature of these rows indicates a high degree of internal strain. In figure 13c this strain has been relieved by the formation of a new boundary at A-A, the rows on either side now being straight. It would appear that the energy of this strained crystal is greater than that of the intercrystalline boundary. We are indebted to Messrs Kodak for the photographs of figure 13, which were taken when the cinematograph film referred to below was produced.

7. EFFECT OF IMPURITY ATOM

Figure 14, plate 18, shows the widespread effect of a bubble which is of the wrong size. If this figure is compared with the perfect rafts shown in figures 2 and 4, plate 8, it will be seen that three bubbles, one larger and two smaller than normal, disturb the regularity of the rows over the whole of the figure. As has been mentioned above, bubbles of the wrong size are generally found in the grain boundaries, where holes of irregular size occur which can accommodate them.

8. MECHANICAL PROPERTIES OF THE TWO-DIMENSIONAL MODEL

The mechanical properties of a two-dimensional perfect raft have been described in the paper referred to above (Bragg 1942b). The raft lies between two parallel springs dipping horizontally in the surface of the soap solution. The pitch of the springs is adjusted to fit the spacing of the rows of bubbles, which then adhere firmly to them. One spring can be translated parallel to itself by a micrometer screw, and the other is supported by two thin vertical glass fibres. The shearing stress can be measured by noting the deflexion of the glass fibres. When subjected to a shearing strain, the raft obeys Hooke's law of elasticity up to the point where the elastic limit is reached. It then slips along some intermediate row by an amount equal to the width of one bubble. The elastic shear and slip can be repeated several times. The elastic limit is approximately reached when one side of the raft has been sheared by an amount equal to a bubble width past the other side. This feature supports the basic assumption made by one of us in the calculation of the elastic limit of a metal (Bragg 1942a), in which it is supposed that each crystallite in a cold-worked metal only yields when the strain in it has reached such a value that energy is released by the slip.

A calculation has been made by M. M. Nicolson of the forces between the bubbles, and will be published shortly. It shows two interesting points. The curve for the variation of potential energy with distance between centres is very similar to those which have been plotted for atoms. It has a minimum for a distance between centres slightly less than a free bubble diameter, and rises sharply for smaller distances. Further, the rise is extremely sharp for bubbles of 0.1 mm. diameter but much less so for bubbles of 1 mm. diameter, thus confirming the impression given by the model that the small bubbles behave as if they were much more rigid than the large ones.

9. THREE-DIMENSIONAL ASSEMBLAGES

If the bubbles are allowed to accumulate in multiple layers on the surface, they form a mass of three-dimensional 'crystals' with one of the arrangements of closest packing. Figure 15, plate 18, shows an oblique view of such a mass, its resemblance to a polished and etched metal surface is noticeable. In figure 16, plate 20, a similar mass is seen viewed normally. Parts of the structure are definitely in cubic closest packing, the outer surface being the (111) face or (100) face. Figure 17a, plate 19, shows a (111) face. The outlines of the three bubbles on which each upper bubble rests can be clearly seen, and the next layer of these bubbles is faintly visible in a position not beneath the uppermost layer, showing that the packing of the (111) planes has the well-known cubic succession. Figure 17b, plate 19, shows a (100) face with each bubble resting on four others. The cubic axes are of course inclined at 45° to the close-packed rows of the surface layer. Figure 17c, plate 19, shows a twin in the cubic structure across the face (111). The uppermost faces are (111) and (100), and they make a small angle with each other, though this is not apparent in the figure, it shows up in an oblique view. Figure 17d, plate 19, appears to show both the cubic and hexagonal succession of closely packed planes, but it is difficult to verify whether the left-hand side follows the true hexagonal close-packed struc-

ture because it is not certain that the assemblage had a depth of more than two layers at this point. Many instances of twins, and of intercrystalline boundaries, can be seen in figure 16, plate 20.

Figure 18, plate 21, shows several dislocations in a three-dimensional structure subjected to a bending strain.

10 DEMONSTRATION OF THE MODEL

With the co-operation of Messrs Kodak, a 16 mm cinematograph film has been made of the movements of the dislocations and grain boundaries when single crystal and polycrystalline rafts are sheared, compressed, or extended. Moreover, if the soap solution is placed in a glass vessel with a flat bottom, the model lends itself to projection on a large scale by transmitted light. Since a certain depth is required for producing the bubbles, and the solution is rather opaque, it is desirable to make the projection through a glass block resting on the bottom of the vessel and just submerged beneath the surface.

In conclusion, we wish to express our thanks to Mr C. E. Harrold, of King's College, Cambridge, who made for us some of the pipettes which were used to produce the bubbles.

REFERENCES

- Bragg, W. L. 1942a *Nature*, **149**, 511.
Bragg, W. L. 1942b *J. Sci. Instrum.* **19**, 148.
Taylor, G. I. 1934 *Proc. Roy. Soc. A*, **145**, 362.

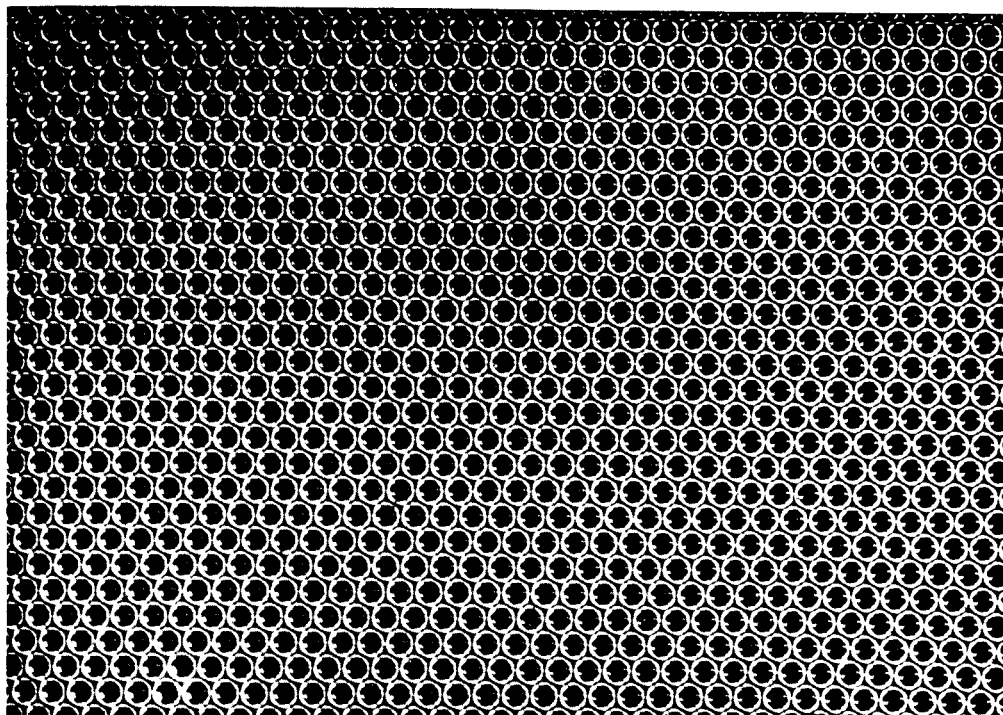


FIGURE 2. Perfect crystalline raft of bubbles. Diameter 1.41 mm.

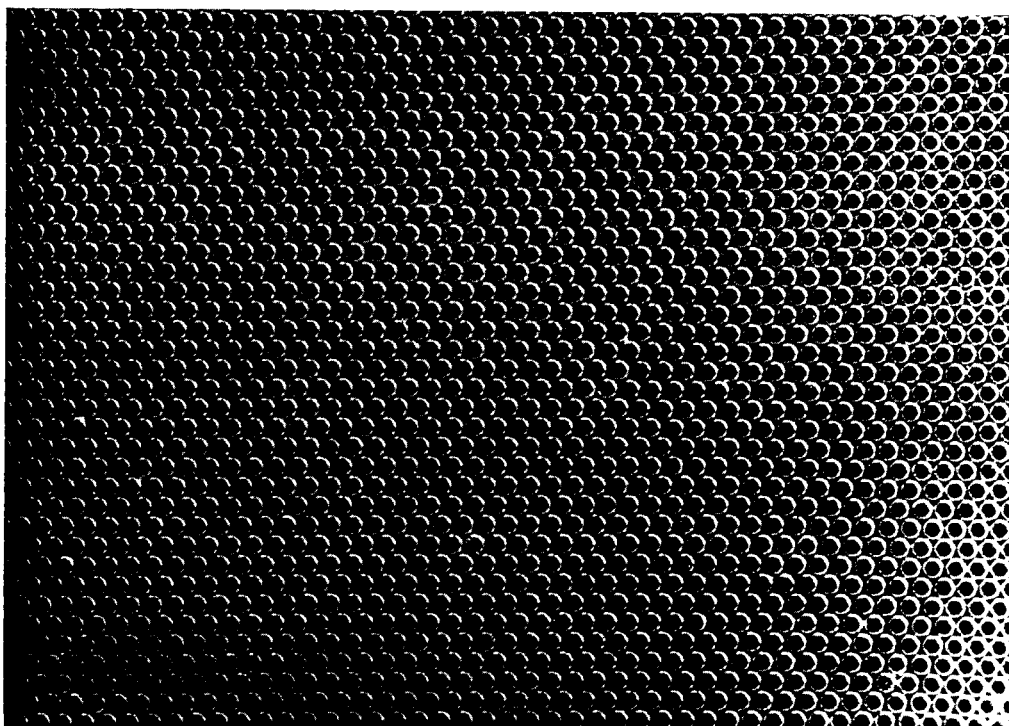


FIGURE 4. Perfect crystalline raft of bubbles. Diameter 0.30 mm.

Grain boundaries

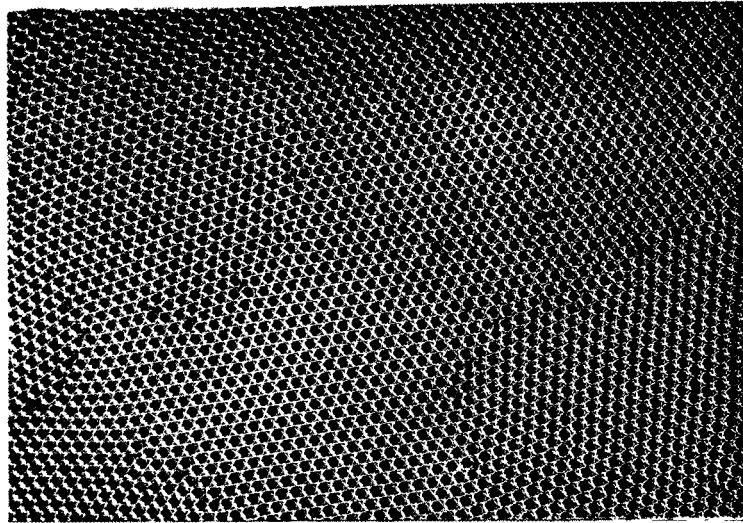


FIGURE 5a. Diameter 1.87 mm.

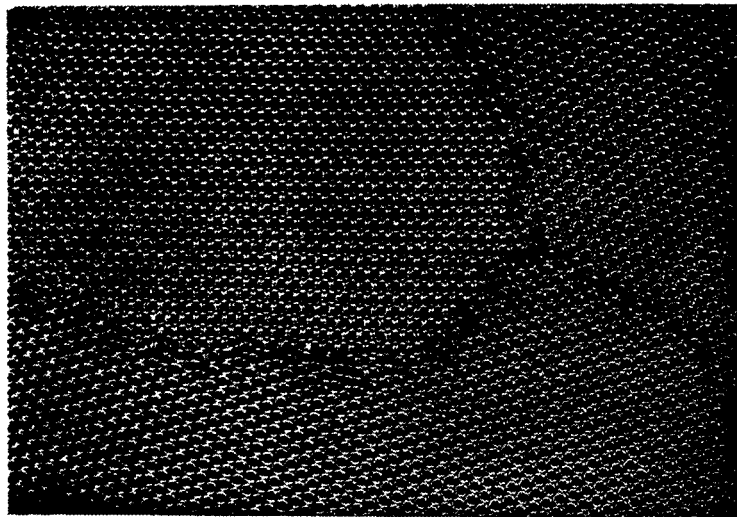


FIGURE 5b. Diameter 0.76 mm.

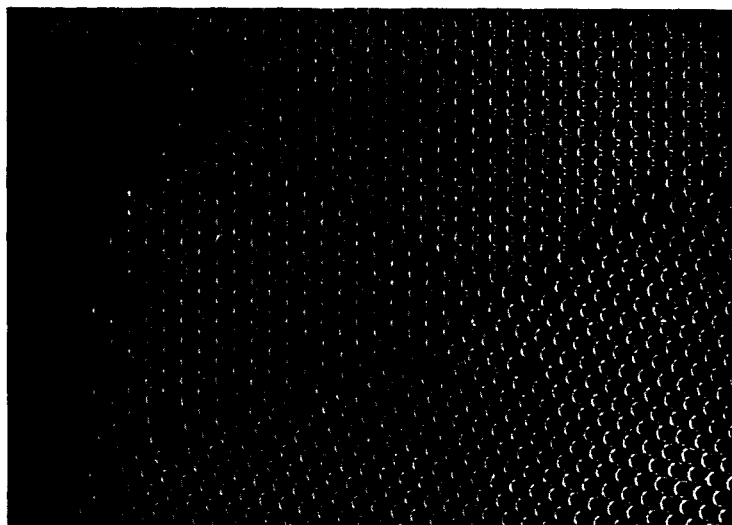


FIGURE 5*c*. A grain boundary. Diameter 0.30 mm.

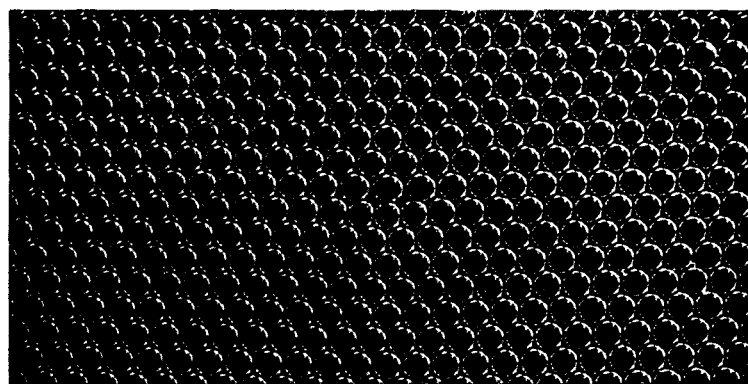


FIGURE 6*a*. A dislocation. Diameter 1.9 mm.

Dislocations

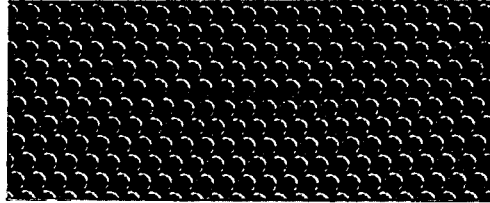


FIGURE 6*b*. Diameter 0.76 mm.

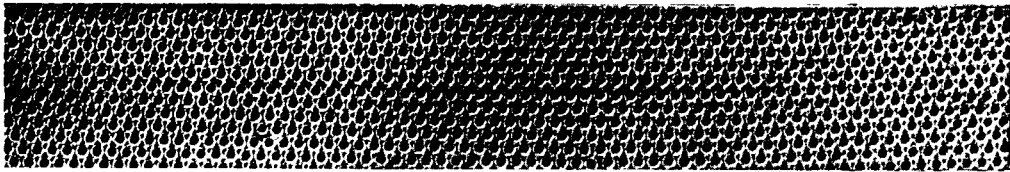


FIGURE 6*c*. Diameter 0.30 mm.

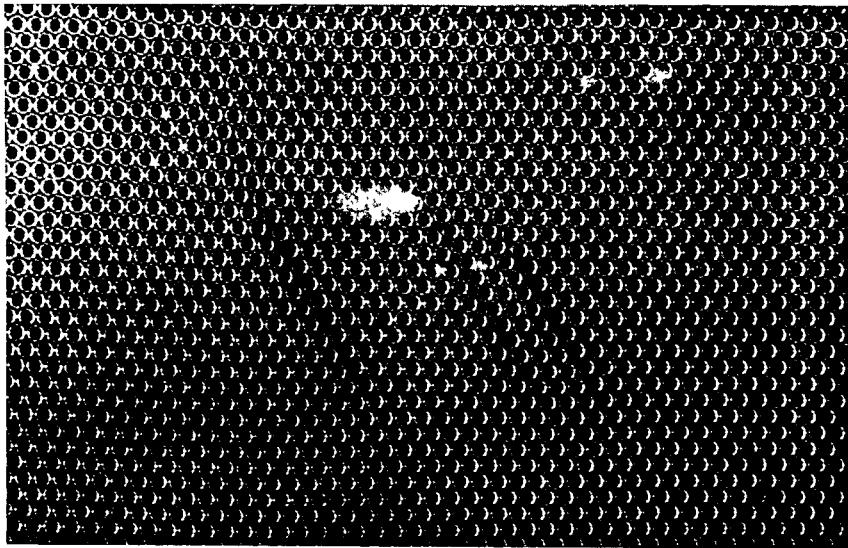


FIGURE 7. Parallel dislocations. Diameter 0.76 mm.

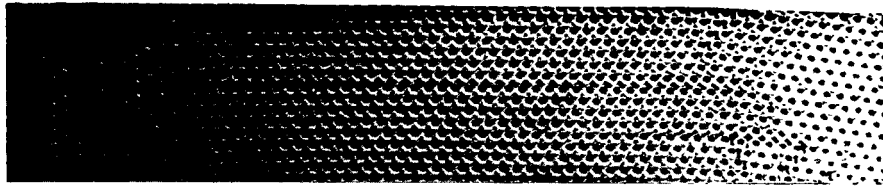


FIGURE 8. Dislocation projecting from a grain boundary. Diameter 0.30 mm.

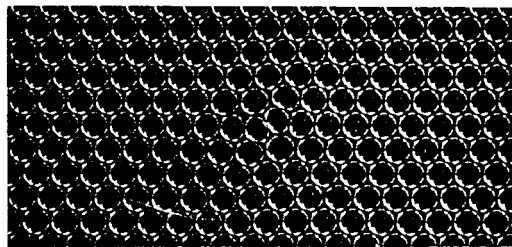


FIGURE 9. Dislocations in adjacent rows. Diameter 1.9 mm.

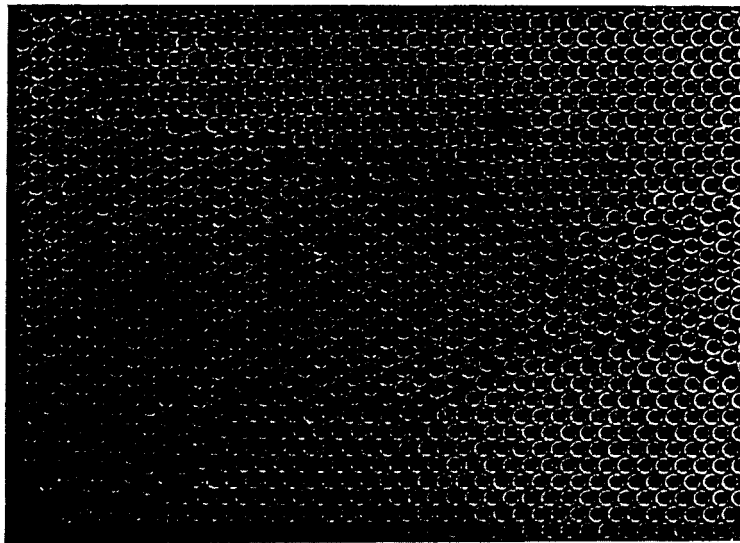
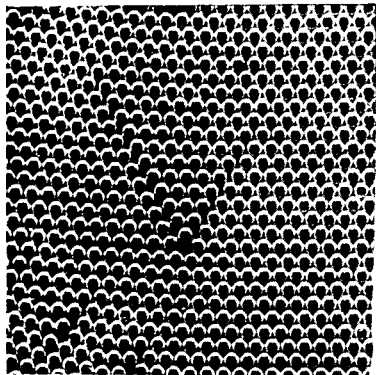
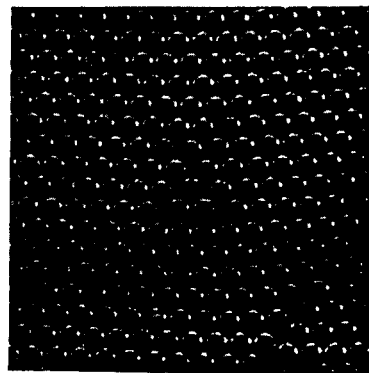


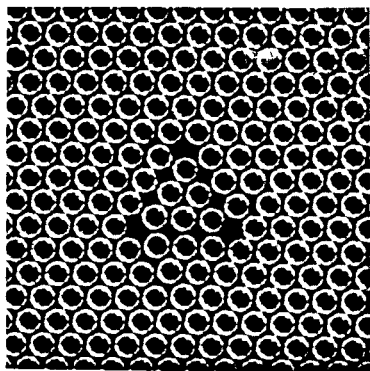
FIGURE 10. Series of fault lines between two areas of parallel orientation. Diameter 0.30 mm.



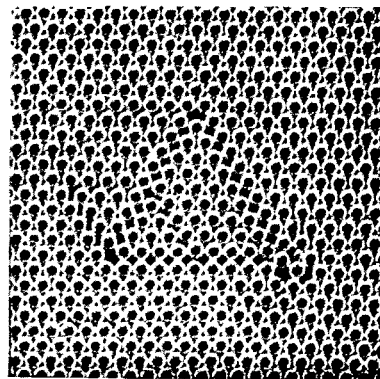
Diameter 0.68 mm.
a



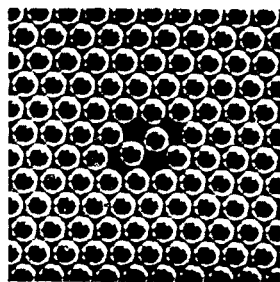
Diameter 0.68 mm.
b



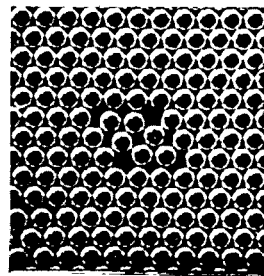
Diameter 0.6 mm.
c



Diameter 0.30 mm.
d

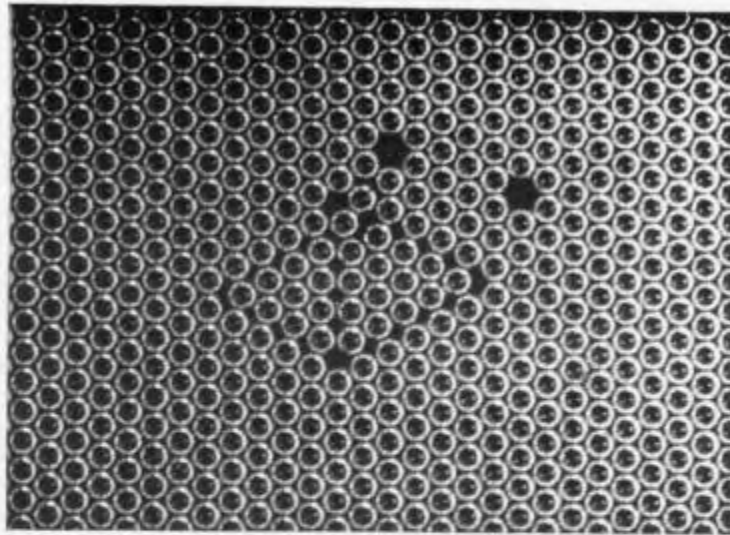


Diameter 0.6 mm
e

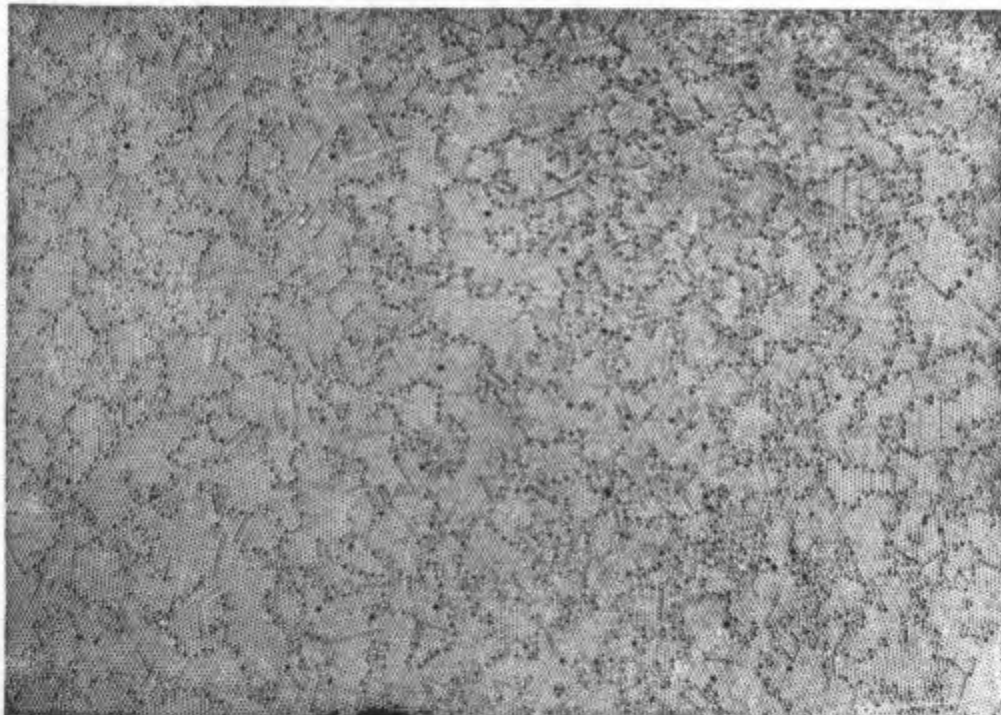


Diameter 0.6 mm
f

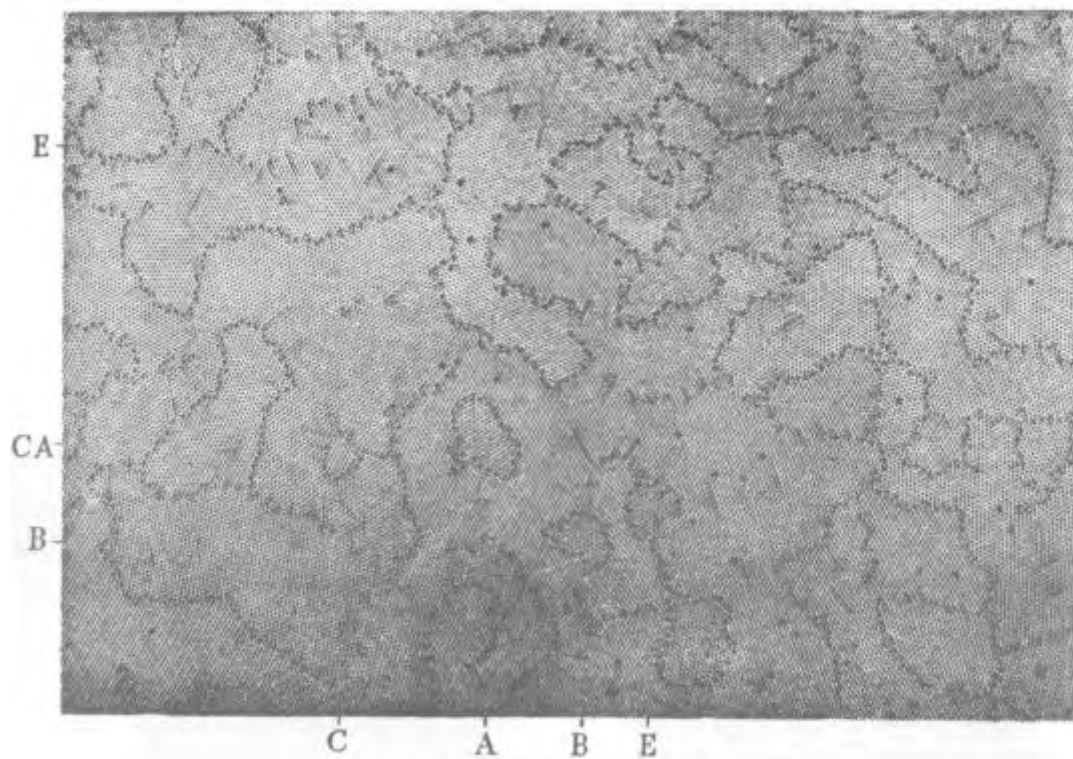
FIGURE 11. Types of fault.



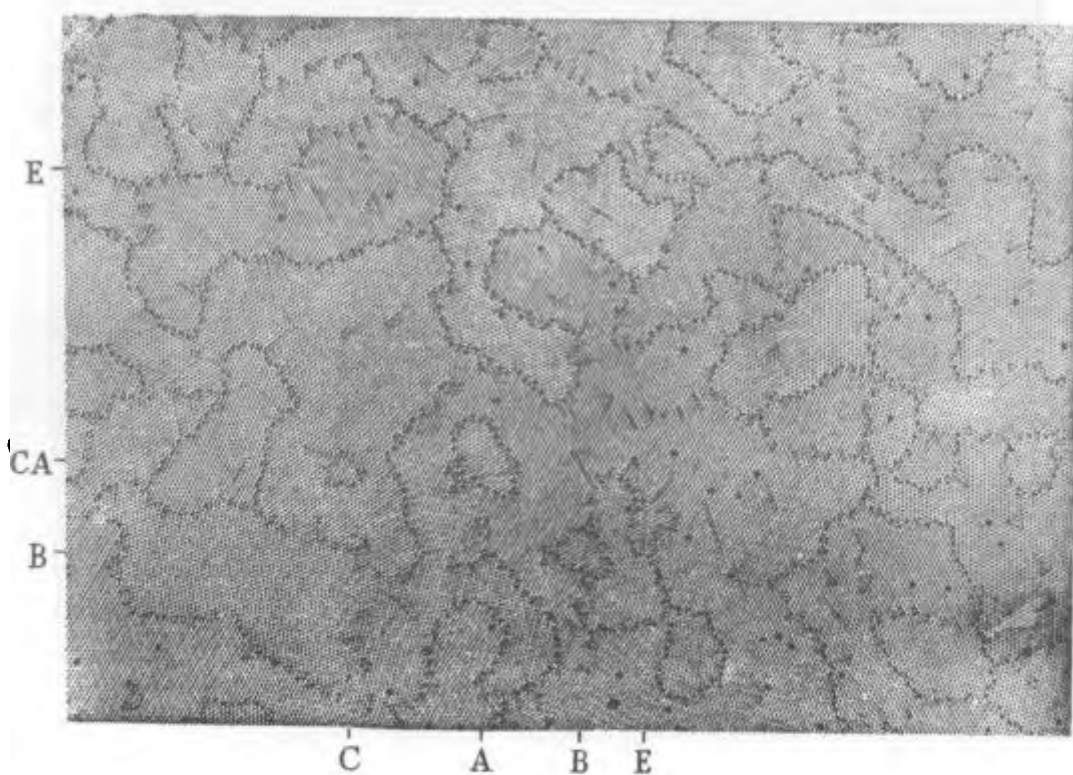
g
FIGURE 11. Types of fault. Diameter 0.68 mm.



a. Immediately after stirring;
FIGURE 12. Recrystallization. Diameter 0.60 mm.

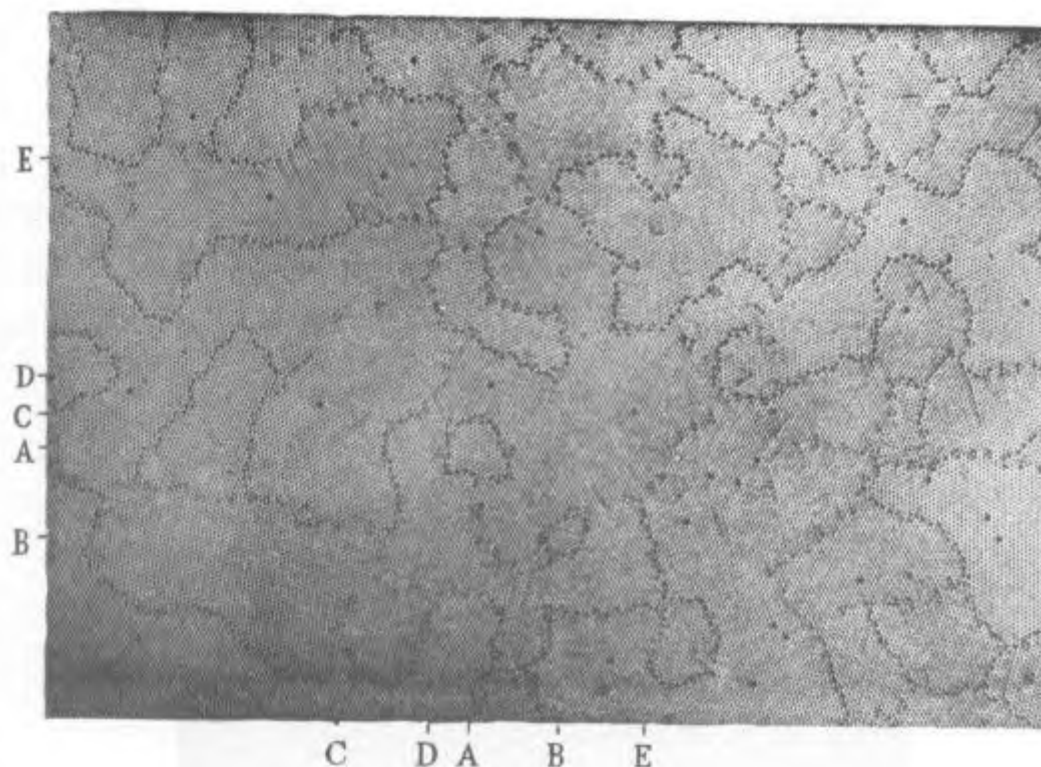


b. After 33 sec.

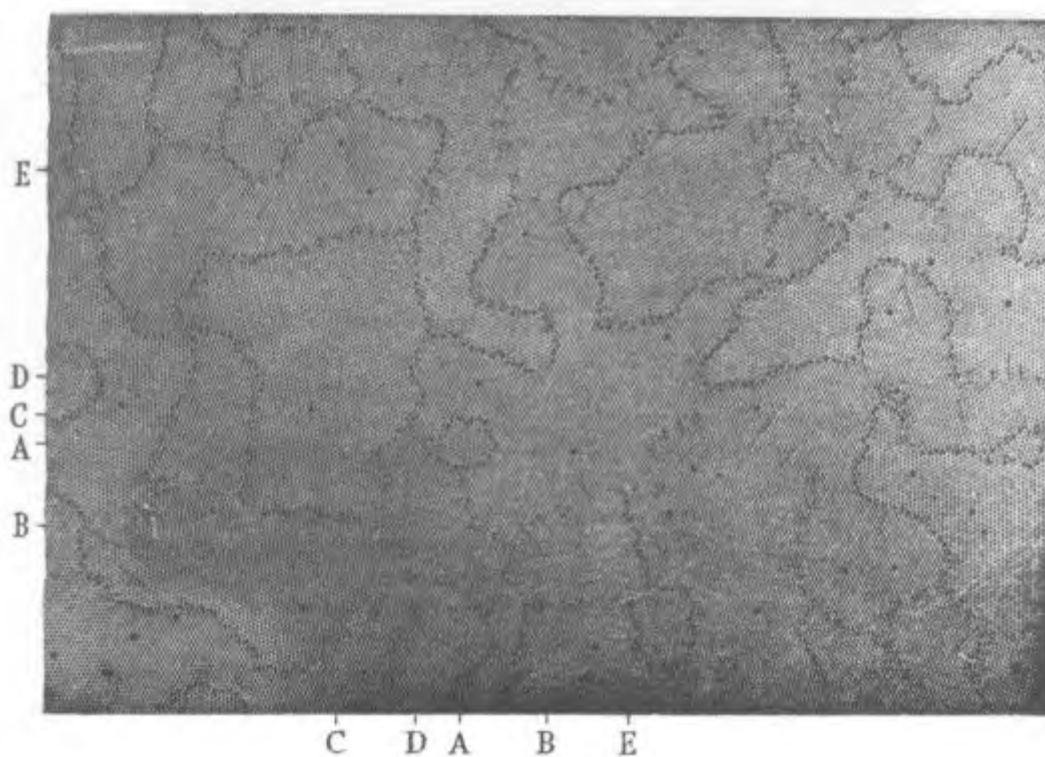


c. After 2 min.

FIGURE 12

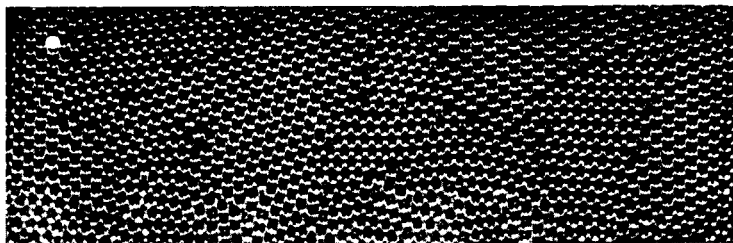


d. After 14 min.

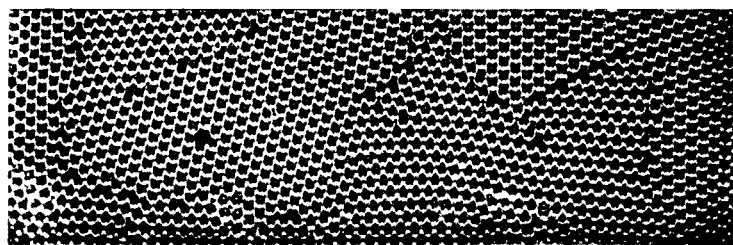


e. After 25 min.

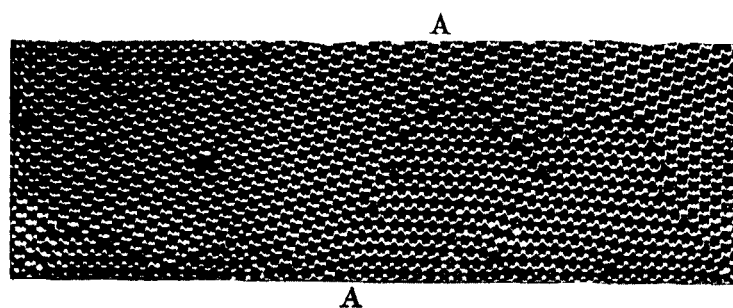
FIGURE 12



a After 1 sec.



b After 4 sec.



c After 4 min.

FIGURE 13. Two stages of recrystallization. Diameter 1.64 mm.

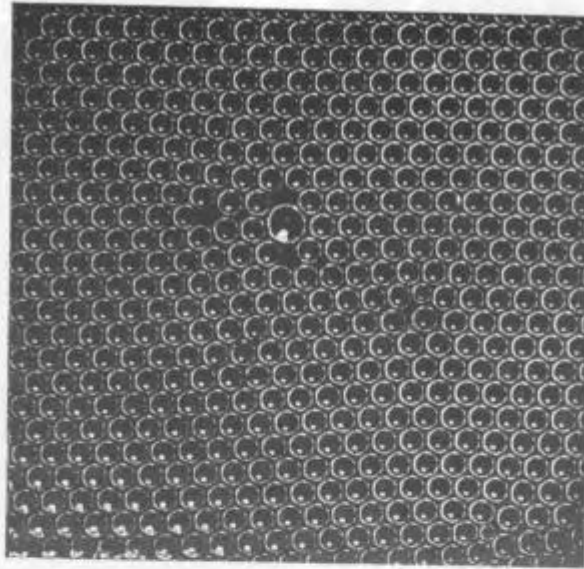
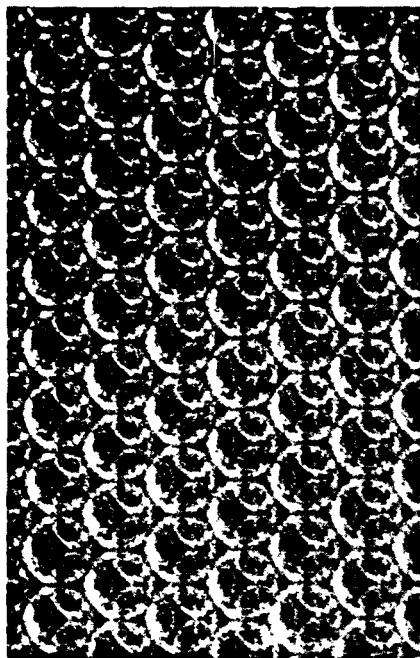


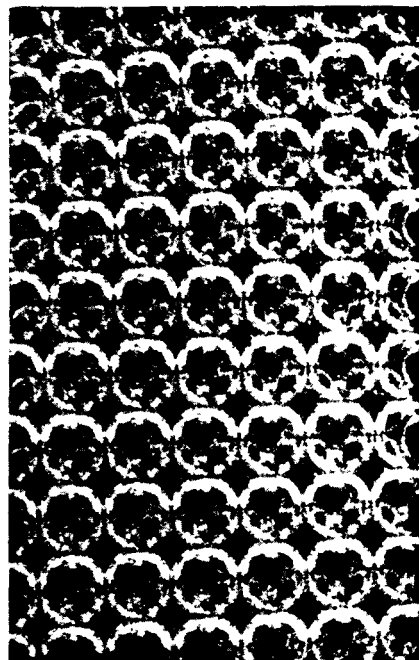
FIGURE 14. Effect of atoms of impurity. Diameter of uniform bubbles about 1.3 μ m.



FIGURE 15. Oblique view of three-dimensional raft.

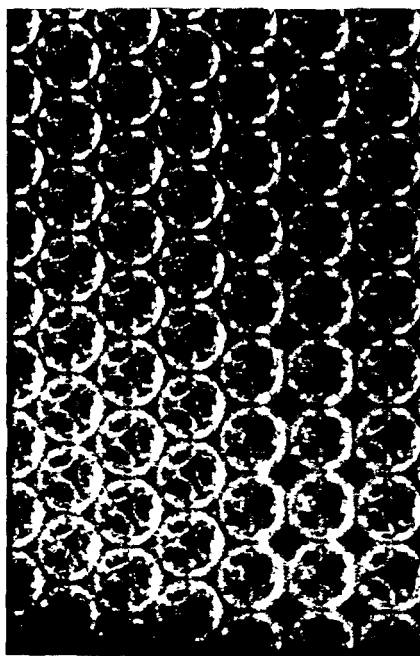


a. (111) face

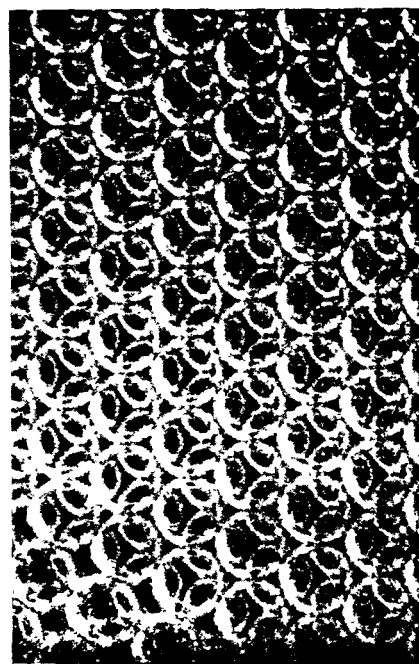


b. (100) face.

Face-centred cubic structure



c. Twin across (111), cubic structure.



d. Possible example of hexagonal close-packing

Diameter 0.70 mm.

FIGURE 17

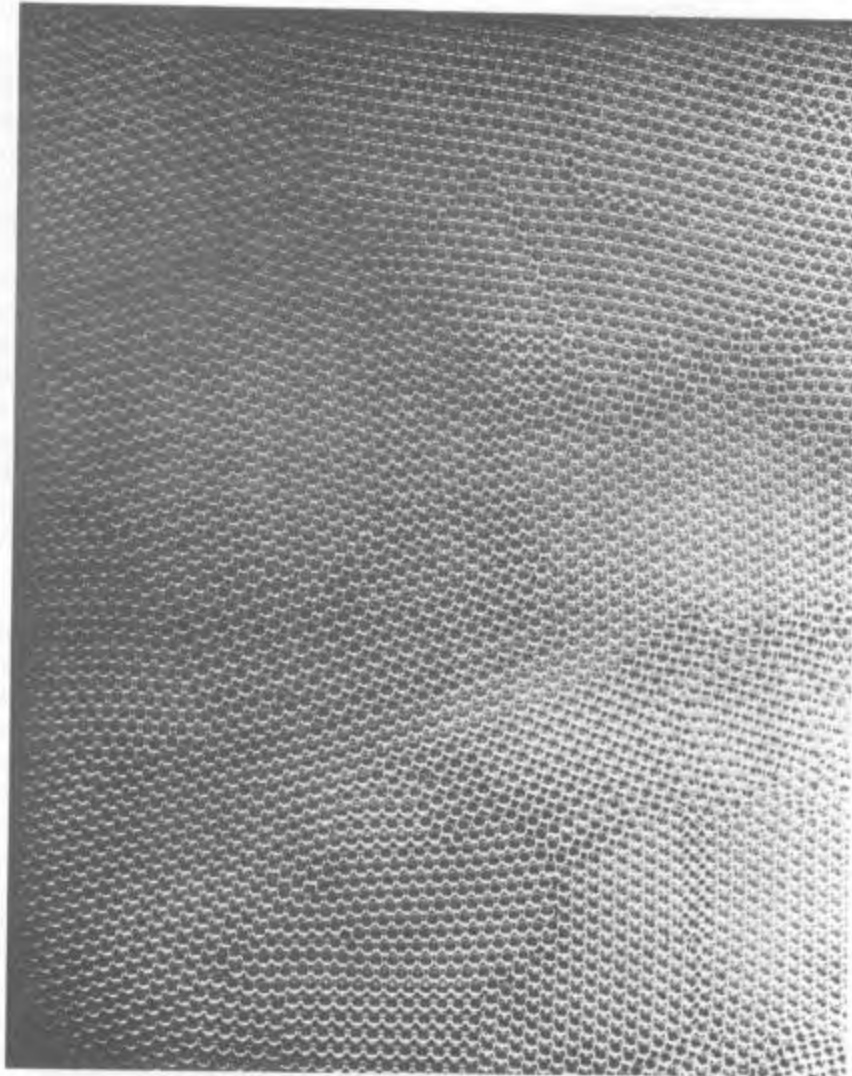


FIGURE 16. A three-dimensional raft viewed normally. Diameter 0.70 mm.

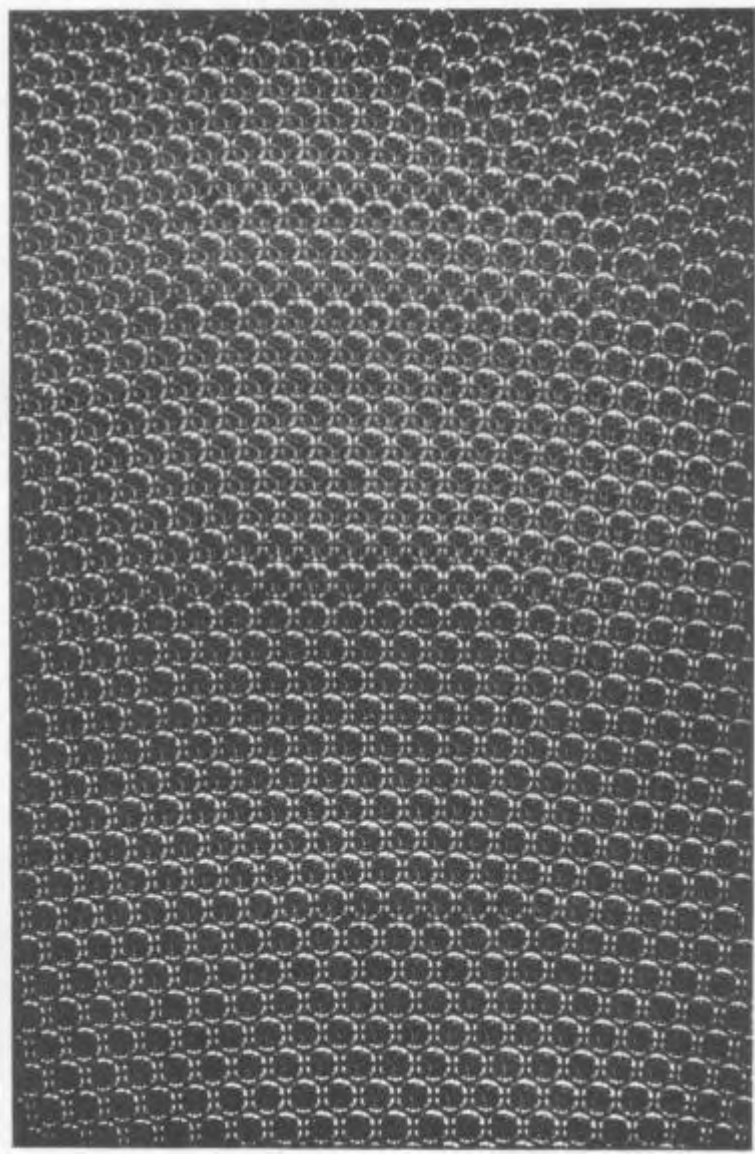


FIGURE 18. Dislocations in three-dimensional structure. Diameter 0.70 mm.

Tensors

31-1 The tensor of polarizability

Physicists always have a habit of taking the simplest example of any phenomenon and calling it “physics,” leaving the more complicated examples to become the concern of other fields—say of applied mathematics, electrical engineering, chemistry, or crystallography. Even solid-state physics is almost only half physics because it worries too much about special substances. So in these lectures we will be leaving out many interesting things. For instance, one of the important properties of crystals—or of most substances—is that their electric polarizability is different in different directions. If you apply a field in any direction, the atomic charges shift a little and produce a dipole moment, but the magnitude of the moment depends very much on the direction of the field. That is, of course, quite a complication. But in physics we usually start out by talking about the special case in which the polarizability is the same in all directions, to make life easier. We leave the other cases to some other field. Therefore, for our later work, we will not need at all what we are going to talk about in this chapter.

The mathematics of tensors is particularly useful for describing properties of substances which vary in direction—although that’s only one example of their use. Since most of you are not going to become physicists, but are going to go into the *real* world, where things depend severely upon direction, sooner or later you will need to use tensors. In order not to leave anything out, we are going to describe tensors, although not in great detail. We want the feeling that our treatment of physics is complete. For example, our electrodynamics is complete—as complete as any electricity and magnetism course, even a graduate course. Our mechanics is not complete, because we studied mechanics when you didn’t have a high level of mathematical sophistication, and we were not able to discuss subjects like the principle of least action, or Lagrangians, or Hamiltonians, and so on, which are *more elegant ways* of describing mechanics. Except for general relativity, however, we do have the complete *laws* of mechanics. Our electricity and magnetism is complete, and a lot of other things are quite complete. The quantum mechanics, naturally, will not be—we have to leave something for the future. But you should at least know what a tensor is.

We emphasized in Chapter 30 that the properties of crystalline substances are different in different directions—we say they are *anisotropic*. The variation of the induced dipole moment with the direction of the applied electric field is only one example, the one we will use for our example of a tensor. Let’s say that for a given direction of the electric field the induced dipole moment per unit volume $\mathbf{P}$ is proportional to the strength of the applied field $\mathbf{E}$. (This is a good approximation for many substances if $\mathbf{E}$ is not too large.) We will call the proportionality constant α .* We want now to consider substances in which α depends on the direction of the applied field, as, for example, in crystals like calcite, which make double images when you look through them.

Suppose, in a particular crystal, we find that an electric field $\mathbf{E}_1$ in the x -direction produces the polarization $\mathbf{P}_1$ in the x -direction. Then we find that an electric field $\mathbf{E}_2$ in the y -direction, with the same *strength*, as $\mathbf{E}_1$ produces a different polar-

31-1 The tensor of polarizability

31-2 Transforming the tensor components

31-3 The energy ellipsoid

31-4 Other tensors; the tensor of inertia

31-5 The cross product

31-6 The tensor of stress

31-7 Tensors of higher rank

31-8 The four-tensor of electromagnetic momentum

Review Chapter 11, Vol. I, *Vectors*
Chapter 20, Vol. I, *Rotation in Space*

* In Chapter 10 we followed the usual convention and wrote $\mathbf{P} = \epsilon_0 \chi \mathbf{E}$ and called χ (“khi”) the “susceptibility.” Here, it will be more convenient to use a single letter, so we write α for $\epsilon_0 \chi$. For isotropic dielectrics, $\alpha = (\kappa - 1)\epsilon_0$, where κ is the dielectric constant (see Section 10-4).

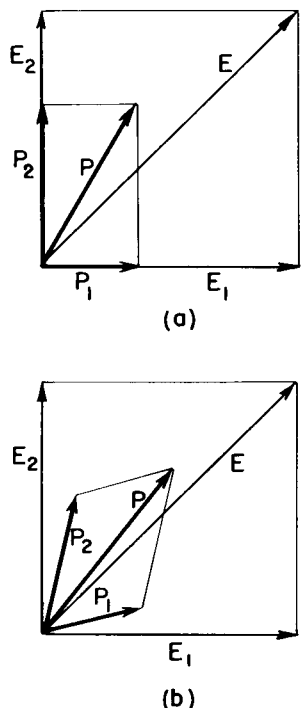


Fig. 31-1. The vector addition of polarizations in an anisotropic crystal.

ization P_2 in the y -direction. What would happen if we put an electric field at 45° ? Well, that's a superposition of two fields along x and y , so the polarization P will be the vector sum of P_1 and P_2 , as shown in Fig. 31-1(a). The polarization is no longer in the same direction as the electric field. You can see how that might come about. There may be charges which can move easily up and down, but which are rather stiff for sidewise motions. When a force is applied at 45° , the charges move farther up than they do toward the side. The displacements are not in the direction of the external force, because there are asymmetric internal elastic forces.

There is, of course, nothing special about 45° . It is *generally* true that the induced polarization of a crystal is *not* in the direction of the electric field. In our example above, we happened to make a "lucky" choice of our x - and y -axes, for which P was along E for both the x - and y -directions. If the crystal were rotated with respect to the coordinate axes, the electric field E_2 in the y -direction would have produced a polarization P with both an x - and a y -component. Similarly, the polarization due to an electric field in the x -direction would have produced a polarization with an x -component and a y -component. Then the polarizations would be as shown in Fig. 31-1(b), instead of as in part (a). Things get more complicated—but for any field E , the *magnitude* of P is still proportional to the magnitude of E .

We want now to treat the general case of an arbitrary orientation of a crystal with respect to the coordinate axes. An electric field in the x -direction will produce a polarization P with x -, y -, and z -components; we can write

$$P_x = \alpha_{xx}E_x, \quad P_y = \alpha_{yx}E_x, \quad P_z = \alpha_{zx}E_x. \quad (31.1)$$

All we are saying here is that if the electric field is in the x -direction, the polarization does not have to be in that same direction, but rather has an x -, a y -, and a z -component—each proportional to E_x . We are calling the constants of proportionality α_{xx} , α_{yx} , and α_{zx} , respectively (the first letter to tell us which component of P is involved, the last to refer to the direction of the electric field).

Similarly, for a field in the y -direction, we can write

$$P_x = \alpha_{xy}E_y, \quad P_y = \alpha_{yy}E_y, \quad P_z = \alpha_{zy}E_y; \quad (31.2)$$

and for a field in the z -direction,

$$P_x = \alpha_{xz}E_z, \quad P_y = \alpha_{yz}E_z, \quad P_z = \alpha_{zz}E_z. \quad (31.3)$$

Now we have said that polarization depends linearly on the fields, so if there is an electric field E that has both an x - and a y -component, the resulting x -component of P will be the sum of the two P_x 's of Eqs. (31.1) and (31.2). If E has components along x , y , and z , the resulting components of P will be the sum of the three contributions in Eqs. (31.1), (31.2), and (31.3). In other words, P will be given by

$$\begin{aligned} P_x &= \alpha_{xx}E_x + \alpha_{xy}E_y + \alpha_{xz}E_z, \\ P_y &= \alpha_{yx}E_x + \alpha_{yy}E_y + \alpha_{yz}E_z, \\ P_z &= \alpha_{zx}E_x + \alpha_{zy}E_y + \alpha_{zz}E_z. \end{aligned} \quad (31.4)$$

The dielectric behavior of the crystal is then completely described by the nine quantities (α_{xx} , α_{xy} , α_{xz} , α_{yx} , $\dots$), which we can represent by the symbol α_{ij} . (The subscripts i and j each stand for any one of the three possible letters x , y , and z .) Any arbitrary electric field E can be resolved with the components E_x , E_y , and E_z ; from these we can use the α_{ij} to find P_x , P_y , and P_z , which together give the total polarization P . The set of nine coefficients α_{ij} is called a *tensor*—in this instance, the *tensor of polarizability*. Just as we say that the three numbers (E_x , E_y , E_z) "form the vector E ," we say that the nine numbers (α_{xx} , α_{xy} , $\dots$) "form the tensor α_{ij} ."

31-2 Transforming the tensor components

You know that when we change to a different coordinate system x' , y' , and z' , the components $E_{x'}$, $E_{y'}$, and $E_{z'}$ of the vector will be quite different—as will also the components of $\mathbf{P}$. So all the coefficients α_{ij} will be different for a different set of coordinates. You can, in fact, see how the α 's must be changed by changing the components of $\mathbf{E}$ and $\mathbf{P}$ in the proper way, because if we describe the *same physical* electric field in the new coordinate system we should get the same polarization. For any new set of coordinates, $P_{x'}$ is a linear combination of P_x , P_y , and P_z :

$$P_{x'} = aP_x + bP_y + cP_z,$$

and similarly for the other components. If you substitute for P_x , P_y , and P_z in terms of the E 's, using Eq. (31.4), you get

$$\begin{aligned} P_{x'} &= a(\alpha_{xx}E_x + \alpha_{xy}E_y + \alpha_{xz}E_z) \\ &+ b(\alpha_{yx}E_x + \alpha_{yy}E_y + \cdots) \\ &+ c(\alpha_{zx}E_x + \cdots + \cdots). \end{aligned}$$

Then you write E_x , E_y , and E_z in terms of $E_{x'}$, $E_{y'}$, and $E_{z'}$; for instance,

$$E_x = a'E_{x'} + b'E_{y'} + c'E_{z'},$$

where a' , b' , c' are related to, but not equal to, a , b , c . So you have $P_{x'}$, expressed in terms of the components $E_{x'}$, $E_{y'}$, and $E_{z'}$; that is, you have the new α_{ij} . It is fairly messy, but quite straightforward.

When we talk about changing the axes we are assuming that the crystal stays put *in space*. If the crystal were rotated *with* the axes, the α 's would not change. Conversely, if the orientation of the crystal were changed with respect to the axes, we would have a new set of α 's. But if they are known for *any* one orientation of the crystal, they can be found for any other orientation by the transformation we have just described. In other words, the dielectric property of a crystal is described *completely* by giving the components of the polarization tensor α_{ij} , with respect to any arbitrarily chosen set of axes. Just as we can associate a vector velocity $\mathbf{v} = (v_x, v_y, v_z)$ with a particle, knowing that the three components will change in a certain definite way if we change our coordinate axes, so with a crystal we associate its polarization tensor α_{ij} , whose nine components will transform in a certain definite way if the coordinate system is changed.

The relation between $\mathbf{P}$ and $\mathbf{E}$ written in Eq. (31.4) can be put in the more compact notation:

$$P_i = \sum_j \alpha_{ij} E_j, \quad (31.5)$$

where it is understood that i represents either x , y , or z and that the sum is taken on $j = x, y$, and z . Many special notations have been invented for dealing with tensors, but each of them is convenient only for a limited class of problems. One common convention is to omit the sum sign ($\sum$) in Eq. (31.5), leaving it *understood* that whenever the same subscript occurs twice (here j), a sum is to be taken over that index. Since we will be using tensors so little, we will not bother to adopt any such special notations or conventions.

31-3 The energy ellipsoid

We want now to get some experience with tensors. Suppose we ask the interesting question: What energy is required to polarize the crystal (in addition to the energy in the electric field which we know is $\epsilon_0 E^2/2$ per unit volume)? Consider for a moment the atomic charges that are being displaced. The work done in displacing the charge the distance dx is $qE_x dx$, and if there are N charges per unit volume, the work done is $qE_x N dx$. But $qN dx$ is the change dP_x in the dipole

moment per unit volume. So the energy required *per unit volume* is

$$E_x dP_x.$$

Combining the work for the three components of the field, the work per unit volume is found to be

$$\mathbf{E} \cdot d\mathbf{P}.$$

Since the magnitude of $\mathbf{P}$ is proportional to $\mathbf{E}$, the work done per unit volume in bringing the polarization from 0 to $\mathbf{P}$ is the integral of $\mathbf{E} \cdot d\mathbf{P}$. Calling this work u_P ,* we write

$$u_P = \frac{1}{2} \mathbf{E} \cdot \mathbf{P} = \frac{1}{2} \sum_i E_i P_i. \quad (31.6)$$

Now we can express $\mathbf{P}$ in terms of $\mathbf{E}$ by Eq. (31.5), and we have that

$$u_P = \frac{1}{2} \sum_i \sum_j \alpha_{ij} E_i E_j. \quad (31.7)$$

The energy density u_P is a number independent of the choice of axes, so it is a scalar. A tensor has then the property that when it is summed over one index (with a vector), it gives a new vector; and when it is summed over *both* indexes (with *two* vectors), it gives a scalar.

The tensor α_{ij} should really be called a “tensor of second rank,” because it has two indexes. A vector—with *one* index—is a tensor of the first rank, and a scalar—with no index—is a tensor of zero rank. So we say that the electric field $\mathbf{E}$ is a tensor of the first rank and that the energy density u_P is a tensor of zero rank. It is possible to extend the ideas of a tensor to three or more indexes, and so to make tensors of ranks higher than two.

The subscripts of the polarization tensor range over three possible values—they are tensors in three dimensions. The mathematicians consider tensors in four, five, or more dimensions. We have already used a four-dimensional tensor $F_{\mu\nu}$ in our relativistic description of the electromagnetic field (Chapter 26).

The polarization tensor α_{ij} has the interesting property that it is *symmetric*, that is, that $\alpha_{xy} = \alpha_{yx}$, and so on for any pair of indexes. (This is a *physical* property of a real crystal and not necessary for all tensors.) You can prove for yourself that this must be true by computing the change in energy of a crystal through the following cycle: (1) Turn on a field in the x -direction; (2) turn on a field in the y -direction; (3) turn *off* the x -field; (4) turn off the y -field. The crystal is now back where it started, and the net work done on the polarization must be back to zero. You can show, however, that for this to be true, α_{xy} must be equal to α_{yx} . The same kind of argument can, of course, be given for α_{xz} , etc. So the polarization tensor is symmetric.

This also means that the polarization tensor can be measured by just measuring the energy required to polarize the crystal in various directions. Suppose we apply an $\mathbf{E}$ -field with only an x - and a y -component; then according to Eq. (31.7),

$$u_P = \frac{1}{2} [\alpha_{xx} E_x^2 + (\alpha_{xy} + \alpha_{yx}) E_x E_y + \alpha_{yy} E_y^2]. \quad (31.8)$$

With an E_x alone, we can determine α_{xx} ; with an E_y alone, we can determine α_{yy} ; with both E_x and E_y , we get an extra energy due to the term with $(\alpha_{xy} + \alpha_{yx})$. Since the α_{xy} and α_{yx} are equal, this term is $2\alpha_{xy}$ and can be related to the energy.

The energy expression, Eq. (31.8), has a nice geometric interpretation. Suppose we ask what fields E_x and E_y correspond to some *given* energy density—say u_0 . That is just the mathematical problem of solving the equation

$$\alpha_{xx} E_x^2 + 2\alpha_{xy} E_x E_y + \alpha_{yy} E_y^2 = 2u_0.$$

This is a quadratic equation, so if we plot E_x and E_y , the solutions of this equation

* This work done in *producing* the polarization by an electric field is not to be confused with the potential energy $-\mathbf{p}_0 \cdot \mathbf{E}$ of a permanent dipole moment $\mathbf{p}_0$.

are all the points on an ellipse (Fig. 31-2). (It must be an ellipse, rather than a parabola or a hyperbola, because the energy for any field is always positive and finite.) The vector E with components E_x and E_y can be drawn from the origin to the ellipse. So such an “energy ellipse” is a nice way of “visualizing” the polarization tensor.

If we now generalize to include all three components, the electric vector E in any direction required to give a unit energy density gives a point which will be on the surface of an ellipsoid, as shown in Fig. 31-3. The shape of this ellipsoid of constant energy uniquely characterizes the tensor polarizability.

Now an ellipsoid has the nice property that it can always be described simply by giving the directions of three “principal axes” and the diameters of the ellipse along these axes. The “principal axes” are the directions of the longest and shortest diameters and the direction at right angles to both. They are indicated by the axes a , b , and c in Fig. 31-3. With respect to these axes, the ellipsoid has the particularly simple equation

$$\alpha_{aa}E_a^2 + \alpha_{bb}E_b^2 + \alpha_{cc}E_c^2 = 2u_0.$$

So with respect to these axes, the dielectric tensor has only three components that are not zero: α_{aa} , α_{bb} , and α_{cc} . That is to say, no matter how complicated a crystal is, it is always possible to choose a set of axes (not necessarily the crystal axes) for which the polarization tensor has only three components. With such a set of axes, Eq. (31.4) becomes simply

$$P_a = \alpha_{aa}E_a, \quad P_b = \alpha_{bb}E_b, \quad P_c = \alpha_{cc}E_c. \quad (31.9)$$

An electric field along any one of the principal axes produces a polarization along the same axis, but the coefficients for the three axes may, of course, be different.

Often, a tensor is described by listing the nine coefficients in a table inside of a pair of brackets:

$$\begin{bmatrix} \alpha_{xx} & \alpha_{xy} & \alpha_{xz} \\ \alpha_{yx} & \alpha_{yy} & \alpha_{yz} \\ \alpha_{zx} & \alpha_{zy} & \alpha_{zz} \end{bmatrix}. \quad (31.10)$$

For the principal axes a , b , and c , only the diagonal terms are not zero; we say then that “the tensor is diagonal.” The complete tensor is

$$\begin{bmatrix} \alpha_{aa} & 0 & 0 \\ 0 & \alpha_{bb} & 0 \\ 0 & 0 & \alpha_{cc} \end{bmatrix}. \quad (31.11)$$

The important point is that any polarization tensor (in fact, any symmetric tensor of rank two in any number of dimensions) can be put in this form by choosing a suitable set of coordinate axes.

If the three elements of the polarization tensor in diagonal form are all equal, that is, if

$$\alpha_{aa} = \alpha_{bb} = \alpha_{cc} = \alpha, \quad (31.12)$$

the energy ellipsoid becomes a sphere, and the polarizability is the same in all directions. The material is isotropic. In the tensor notation,

$$\alpha_{ij} = \alpha \delta_{ij}, \quad (31.13)$$

where δ_{ij} is the *unit tensor*

$$\delta_{ij} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}. \quad (31.14)$$

That means, of course,

$$\begin{aligned} \delta_{ij} &= 1, & \text{if } i &= j; \\ \delta_{ij} &= 0, & \text{if } i &\neq j. \end{aligned} \quad (31.15)$$

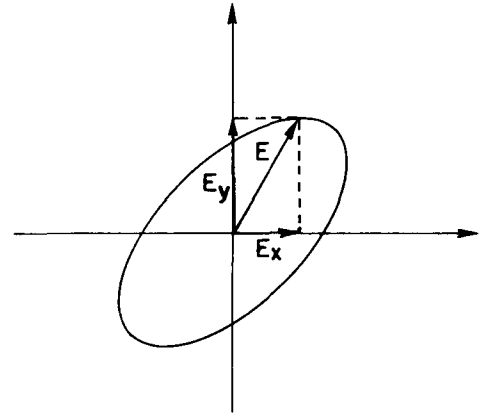


Fig. 31-2. Locus of the vector $E = (E_x, E_y)$ that gives a constant energy of polarization.

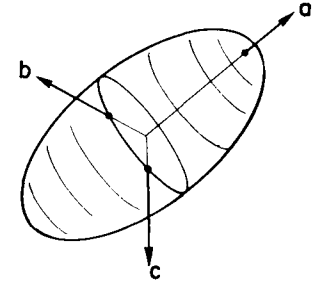


Fig. 31-3. The energy ellipsoid of the polarization tensor.

The tensor δ_{ij} is often called the “Kronecker delta.” You may amuse yourself by proving that the tensor (31.14) has exactly the same form if you change the coordinate system to any other rectangular one. The polarization tensor of Eq. (31.13) gives

$$P_i = \alpha \sum_j \delta_{ij} E_j = \alpha E_i,$$

which means the same as our old result for isotropic dielectrics:

$$\mathbf{P} = \alpha \mathbf{E}.$$

The shape and orientation of the polarization ellipsoid can sometimes be related to the symmetry properties of the crystal. We have said in Chapter 30 that there are 230 different possible internal symmetries of a three-dimensional lattice and that they can, for many purposes, be conveniently grouped into seven classes, according to the shape of the unit cell. Now the ellipsoid of polarizability must share the internal geometric symmetries of the crystal. For example, a triclinic crystal has low symmetry—the ellipsoid of polarizability will have unequal axes, and its orientation will not, in general, be aligned with the crystal axes. On the other hand, a monoclinic crystal has the property that its properties are unchanged if the crystal is rotated 180° about one axis. So the polarization tensor must be the same after such a rotation. It follows that the ellipsoid of the polarizability must return to itself after a 180° rotation. That can happen only if one of the axes of the ellipsoid is in the same direction as the symmetry axis of the crystal. Otherwise, the orientation and dimensions of the ellipsoid are unrestricted.

For an orthorhombic crystal, however, the axes of the ellipsoid must correspond to the crystal axes, because a 180° rotation about any one of the three axes repeats the same lattice. If we go to a tetragonal crystal, the ellipse must have the same symmetry, so it must have two equal diameters. Finally, for a cubic crystal, all three diameters of the ellipsoid must be equal, it becomes a sphere, and the polarizability of the crystal is the same in all directions.

There is a big game of figuring out the possible kinds of tensors for all the possible symmetries of a crystal. It is called a “group-theoretical” analysis. But for the simple case of the polarizability tensor, it is relatively easy to see what the relations must be.

31-4 Other tensors; the tensor of inertia

There are many other examples of tensors appearing in physics. For example, in a metal, or in any conductor, one often finds that the current density $\mathbf{j}$ is approximately proportional to the electric field $\mathbf{E}$; the proportionality constant is called the conductivity σ :

$$\mathbf{j} = \sigma \mathbf{E}.$$

For crystals, however, the relation between $\mathbf{j}$ and $\mathbf{E}$ is more complicated; the conductivity is not the same in all directions. The conductivity is a tensor, and we write

$$j_i = \sum_j \sigma_{ij} E_j.$$

Another example of a physical tensor is the moment of inertia. In Chapter 18 of Volume I we saw that a solid object rotating about a fixed axis has an angular momentum $\mathbf{L}$ proportional to the angular velocity $\boldsymbol{\omega}$, and we called the proportionality factor I , the moment of inertia:

$$\mathbf{L} = I \boldsymbol{\omega}.$$

For an arbitrarily shaped object, the moment of inertia depends on its orientation with respect to the axis of rotation. For instance, a rectangular block will have different moments about each of its three orthogonal axes. Now angular velocity $\boldsymbol{\omega}$ and angular momentum $\mathbf{L}$ are both vectors. For rotations about one of the axes of symmetry, they are parallel. But if the moment of inertia is different for the

three principal axes, then ω and L are, in general, not in the same direction (see Fig. 31-4). They are related in a way analogous to the relation between E and P . In general, we must write

$$\begin{aligned} L_x &= I_{xx}\omega_x + I_{xy}\omega_y + I_{xz}\omega_z, \\ L_y &= I_{xy}\omega_x + I_{yy}\omega_y + I_{yz}\omega_z, \\ L_z &= I_{zx}\omega_x + I_{zy}\omega_y + I_{zz}\omega_z. \end{aligned} \quad (31.16)$$

The nine coefficients I_{ij} are called the tensor of inertia. Following the analogy with the polarization, the kinetic energy for any angular momentum must be some quadratic form in the components ω_x , ω_y , and ω_z :

$$\text{KE} = \frac{1}{2} \sum_{ij} I_{ij} \omega_i \omega_j. \quad (31.17)$$

We can use the energy to define the ellipsoid of inertia. Also, energy arguments can be used to show that the tensor is symmetric—that $I_{ij} = I_{ji}$.

The tensor of inertia for a rigid body can be worked out if the shape of the object is known. We need only to write down the total kinetic energy of all the particles in the body. A particle of mass m and velocity v has the kinetic energy $\frac{1}{2}mv^2$, and the total kinetic energy is just the sum

$$\sum \frac{1}{2}mv^2$$

over all of the particles of the body. The velocity v of each particle is related to the angular velocity ω of the solid body. Let's assume that the body is rotating about its center of mass, which we take to be at rest. Then if r is the displacement of a particle from the center of mass, its velocity v is given by $\omega \times r$. So the total kinetic energy is

$$\text{KE} = \sum \frac{1}{2}m(\omega \times r)^2. \quad (31.18)$$

Now all we have to do is write $\omega \times r$ out in terms of the components ω_x , ω_y , ω_z , and x , y , z , and compare the result with Eq. (31.17); we find I_{ij} by identifying terms. Carrying out the algebra, we write

$$\begin{aligned} (\omega \times r)^2 &= (\omega \times r)_x^2 + (\omega \times r)_y^2 + (\omega \times r)_z^2 \\ &= (\omega_y z - \omega_z y)^2 + (\omega_z x - \omega_x z)^2 + (\omega_x y - \omega_y x)^2 \\ &= +\omega_y^2 z^2 - 2\omega_y \omega_z zy + \omega_z^2 y^2 \\ &\quad + \omega_z^2 x^2 - 2\omega_z \omega_x xz + \omega_x^2 z^2 \\ &\quad + \omega_x^2 y^2 - 2\omega_x \omega_y yx + \omega_y^2 x^2. \end{aligned}$$

Multiplying this equation by $m/2$, summing over all particles, and comparing with Eq. (31.17), we see that I_{xx} , for instance, is given by

$$I_{xx} = \sum m(y^2 + z^2).$$

This is the formula we have had before (Chapter 19, Vol. I) for the moment of inertia of a body about the x -axis. Since $r^2 = x^2 + y^2 + z^2$, we can also write this term as

$$I_{xx} = \sum m(r^2 - x^2).$$

Working out all of the other terms, the tensor of inertia can be written as

$$I_{ij} = \begin{bmatrix} \sum m(r^2 - x^2) & -\sum mxy & -\sum mxz \\ -\sum myx & \sum m(r^2 - y^2) & -\sum myz \\ -\sum mzx & -\sum mzy & \sum m(r^2 - z^2) \end{bmatrix}. \quad (31.19)$$

If you wish, this may be written in "tensor notation" as

$$I_{ij} = \sum m(r^2 \delta_{ij} - r_i r_j). \quad (31.20)$$

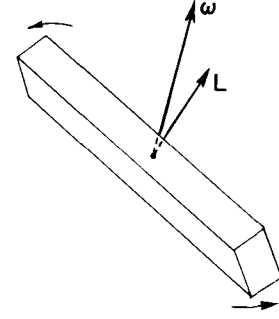


Fig. 31-4. The angular momentum L of a solid object is not, in general, parallel to its angular velocity ω .

where the r_i are the components (x, y, z) of the position vector of a particle and the $\sum$ means to sum over all the particles. The moment of inertia, then, is a tensor of the second rank whose terms are a property of the body and relate $\mathbf{L}$ to $\boldsymbol{\omega}$ by

$$L_i = \sum_j I_{ij} \omega_j \quad (31.21)$$

For a body of any shape whatever, we can find the ellipsoid of inertia and, therefore, the three principal axes. Referred to these axes, the tensor will be diagonal, so for any object there are always three orthogonal axes for which the angular velocity and angular momentum are parallel. They are called the principal axes of inertia.

31-5 The cross product

We should point out that we have been using tensors of the second rank since Chapter 20 of Volume I. There, we defined a “torque in a plane,” such as τ_{xy} , by

$$\tau_{xy} = xF_y - yF_x.$$

Generalized to three dimensions, we could write

$$\tau_{ij} = r_i F_j - r_j F_i. \quad (31.22)$$

The quantity τ_{ij} is a tensor of the second rank. One way to see that this is so is by combining τ_{ij} with some vector, say the unit vector $\mathbf{e}$, according to

$$\sum_j \tau_{ij} e_j.$$

If this quantity is a *vector*, then τ_{ij} must transform as a tensor—this is our definition of a tensor. Substituting for τ_{ij} , we have

$$\begin{aligned} \sum_j \tau_{ij} e_j &= \sum_j r_i F_j e_j - \sum_j r_j e_j F_i \\ &= r_i (\mathbf{F} \cdot \mathbf{e}) - (\mathbf{r} \cdot \mathbf{e}) F_i. \end{aligned}$$

Since the dot products are scalars, the two terms on the right-hand side are vectors, and likewise their difference. So τ_{ij} is a tensor.

But τ_{ij} is a special kind of tensor; it is *antisymmetric*, that is,

$$\tau_{ij} = -\tau_{ji},$$

so it has only three nonzero terms— τ_{xy} , τ_{yz} , and τ_{zx} . We were able to show in Chapter 20 of Volume I that these three terms, almost “by accident,” transform like the three components of a vector, so that we could *define*

$$\boldsymbol{\tau} = (\tau_x, \tau_y, \tau_z) = (\tau_{yz}, \tau_{zx}, \tau_{xy})$$

We say “by accident,” because it happens only in three dimensions. In four dimensions, for instance, an antisymmetric tensor of the second rank has *six* nonzero terms and certainly cannot be replaced by a vector with *four* components.

Just as the axial vector $\boldsymbol{\tau} = \mathbf{r} \times \mathbf{F}$ is a tensor, so also is every cross product of two polar vectors—all the same arguments apply. By luck, however, they are also representable by vectors (really pseudovectors), so our mathematics has been made easier for us.

Mathematically, if $\mathbf{a}$ and $\mathbf{b}$ are any two vectors, the nine quantities $a_i b_j$ form a tensor (although it may have no useful physical purpose). Thus, for the position vector r_i , $r_i r_j$ is a tensor, and since δ_{ij} is also, we see that the right side of Eq. (31.20) is indeed a tensor. Likewise Eq. (31.22) is a tensor, since the two terms on the right-hand side are tensors.

31-6 The tensor of stress

The symmetric tensors we have described so far arose as coefficients in relating one vector to another. We would like to look now at a tensor which has a different physical significance—the tensor of *stress*. Suppose we have a solid object with various forces on it. We say that there are various “stresses” inside, by which we mean that there are internal forces between neighboring parts of the material. We have talked a little about such stresses in a two-dimensional case when we considered the surface tension in a stretched diaphragm in Section 12-3. We will now see that the internal forces in the material of a three-dimensional body can be described in terms of a tensor.

Consider a body of some elastic material—say a block of jello. If we make a cut through the block, the material on each side of the cut will, in general, get displaced by the internal forces. Before the cut was made, there must have been forces between the two parts of the block that kept the material in place; we can define the stresses in terms of these forces. Suppose we look at an imaginary plane perpendicular to the x -axis—like the plane σ in Fig. 31-5—and ask about the force across a small area $\Delta y \Delta z$ in this plane. The material on the left of the area exerts the force $\Delta \mathbf{F}_1$ on the material to the right, as shown in part (b) of the figure. There is, of course, the opposite reaction force $-\Delta \mathbf{F}_1$ exerted on the material to the left of the surface. If the area is small enough, we expect that $\Delta \mathbf{F}_1$ is proportional to the area $\Delta y \Delta z$.

You are already familiar with one kind of stress—the pressure in a static liquid. There the force is equal to the pressure times the area and is at right angles to the surface element. For solids—also for viscous liquids in motion—the force need not be normal to the surface; there are *shear* forces in addition to pressures (positive or negative). (By a “shear” force we mean the *tangential* components of the force across a surface.) All three components of the force must be taken into account. Notice also that if we make our cut on a plane with some other orientation, the forces will be different. A complete description of the internal stress requires a tensor.

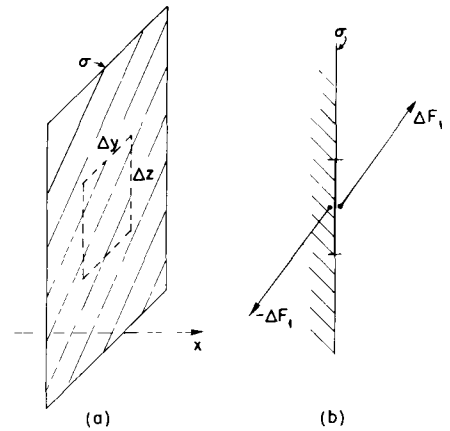


Fig. 31-5. The material to the left of the plane σ exerts across the area $\Delta y \Delta z$ the force $\Delta \mathbf{F}_1$ on the material to the right of the plane.

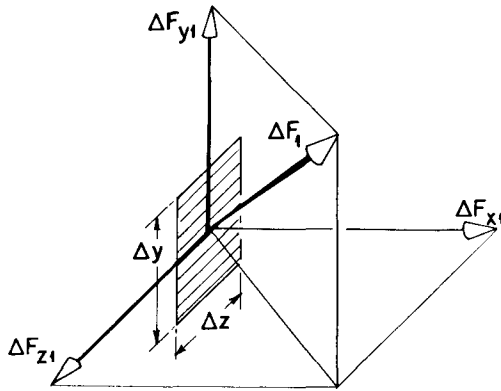


Fig. 31-6. The force $\Delta \mathbf{F}_1$ across an element of area $\Delta y \Delta z$ perpendicular to the x -axis is resolved into the three components ΔF_{x1} , ΔF_{y1} , and ΔF_{z1} .

We define the stress tensor in the following way: First, we imagine a cut perpendicular to the x -axis and resolve the force $\Delta \mathbf{F}_1$ across the cut into its components ΔF_{x1} , ΔF_{y1} , ΔF_{z1} , as in Fig. 31-6. The ratio of these forces to the area $\Delta y \Delta z$, we call S_{xx} , S_{yx} , and S_{zx} . For example,

$$S_{yx} = \frac{\Delta F_{y1}}{\Delta y \Delta z}.$$

The first index y refers to the direction force component; the second index x is normal to the area. If you wish, you can write the area $\Delta y \Delta z$ as Δa_x , meaning an element of area perpendicular to x . Then

$$S_{yx} = \frac{\Delta F_{y1}}{\Delta a_x}.$$

Next, we think of an imaginary cut perpendicular to the y -axis. Across a small

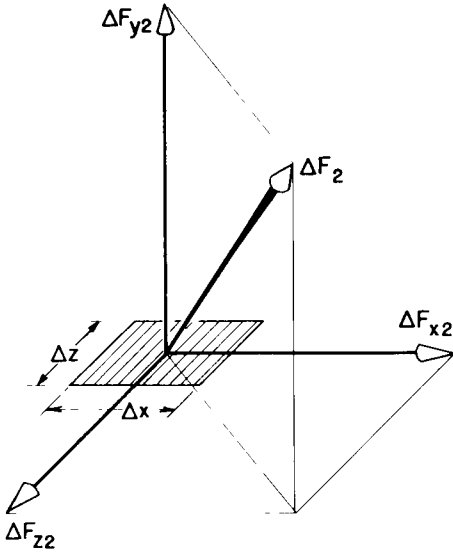


Fig. 31-7. The force across an element of area perpendicular to y is resolved into three rectangular components

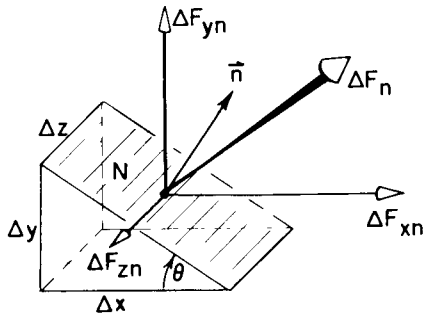


Fig. 31-8. The force F_n across the face N (whose unit normal is $\mathbf{n}$) is resolved into components.

area $\Delta x \Delta z$ there will be a force ΔF_2 . Again we resolve this force into three components, as shown in Fig. 31-7, and define the three components of the stress, S_{xy} , S_{yy} , S_{zy} , as the force per unit area in the three directions. Finally, we make an imaginary cut perpendicular to z and define the three components S_{xz} , S_{yz} , and S_{zz} . So we have the nine numbers

$$S_{ij} = \begin{bmatrix} S_{xx} & S_{xy} & S_{xz} \\ S_{yx} & S_{yy} & S_{yz} \\ S_{zx} & S_{zy} & S_{zz} \end{bmatrix}. \quad (31.23)$$

We want to show now that these nine numbers are sufficient to describe completely the internal state of stress, and that S_{ij} is indeed a tensor. Suppose we want to know the force across a surface oriented at some arbitrary angle. Can we find it from S_{ij} ? Yes, in the following way: We imagine a little solid figure which has one face N in the new surface, and the other faces parallel to the coordinate axes. If the face N happened to be parallel to the z -axis, we would have the triangular piece shown in Fig. 31-8. (This is a somewhat special case, but will illustrate well enough the general method.) Now the stress forces on the little solid triangle in Fig. 31-8 are in equilibrium (at least in the limit of infinitesimal dimensions), so the total force on it must be zero. We know the forces on the faces parallel to the coordinate axes directly from S_{ij} . Their vector sum must equal the force on the face N , so we can express this force in terms of S_{ij} .

Our assumption that the surface forces on the small triangular volume are in equilibrium neglects any other body forces that might be present, such as gravity or pseudo forces if our coordinate system is not an inertial frame. Notice, however, that such body forces will be proportional to the volume of the little triangle and, therefore, to Δx , Δy , Δz , whereas all the surface forces are proportional to the areas such as $\Delta x \Delta y$, $\Delta y \Delta z$, etc. So if we take the scale of the little wedge small enough, the body forces can always be neglected in comparison with the surface forces.

Let's now add up the forces on the little wedge. We take first the x -component, which is the sum of five parts—one from each face. However, if Δz is small enough, the forces on the triangular faces (perpendicular to the z -axis) will be equal and opposite, so we can forget them. The x -component of the force on the bottom rectangle is

$$\Delta F_{x2} = S_{xy} \Delta x \Delta z.$$

The x -component of the force on the vertical rectangle is

$$\Delta F_{x1} = S_{xx} \Delta y \Delta z.$$

These two must be equal to the x -component of the force outward across the face N . Let's call $\mathbf{n}$ the unit vector normal to the face N , and the force on it $\mathbf{F}_n$, then we have

$$\Delta F_{xn} = S_{xx} \Delta y \Delta z + S_{xy} \Delta x \Delta z.$$

The x -component S_{xn} of the stress across this plane is equal to ΔF_{xn} divided by the area, which is $\Delta \sqrt{\Delta x^2 + \Delta y^2}$, or

$$S_{xn} = S_{xx} \frac{\Delta y}{\sqrt{\Delta x^2 + \Delta y^2}} + S_{xy} \frac{\Delta x}{\sqrt{\Delta x^2 + \Delta y^2}}.$$

Now $\Delta x / \sqrt{\Delta x^2 + \Delta y^2}$ is the cosine of the angle θ between $\mathbf{n}$ and the y -axis, as shown in Fig. 31-8, so it can also be written as n_y , the y -component of $\mathbf{n}$. Similarly, $\Delta y / \sqrt{\Delta x^2 + \Delta y^2}$ is $\sin \theta = n_x$. We can write

$$S_{xn} = S_{xx} n_x + S_{xy} n_y.$$

If we now generalize to an arbitrary surface element, we would get that

$$S_{xn} = S_{xx} n_x + S_{xy} n_y + S_{xz} n_z$$

or, in general,

$$S_{in} = \sum_j S_{ij} n_j. \quad (31.24)$$

We can find the force across any surface element in terms of the S_{ij} , so it does describe completely the state of internal stress of the material.

Equation (31.24) says that the tensor S_{ij} relates the force S_n to the unit vector $\mathbf{n}$, just as α_{ij} relates $\mathbf{P}$ to $\mathbf{E}$. Since $\mathbf{n}$ and $\mathbf{S}_n$ are vectors, the components of S_{ij} must transform as a tensor with changes in coordinate axes. So S_{ij} is indeed a tensor.

We can also show that S_{ij} is a *symmetric* tensor by looking at the forces on a little cube of material. Suppose we take a little cube, oriented with its faces parallel to our coordinate axes, and look at it in cross section, as shown in Fig. 31-9. If we let the edge of the cube be one unit, the x - and y -components of the forces on the faces normal to the x - and y -axes might be as shown in the figure. If the cube is small, the stresses do not change appreciably from one side of the cube to the opposite side, so the force components are equal and opposite as shown. Now there must be no torque on the cube, or it would start spinning. The total torque about the center is $(S_{yx} - S_{xy})$ (times the unit edge of the cube), and since the total is zero, S_{yx} is equal to S_{xy} , and the stress tensor is symmetric.

Since S_{ij} is a symmetric tensor, it can be described by an ellipsoid which will have three principal axes. For surfaces normal to these axes, the stresses are particularly simple—they correspond to pushes or pulls perpendicular to the surfaces. There are no shear forces along these faces. For *any* stress, we can always choose our axes so that the shear components are zero. If the ellipsoid is a sphere, there are only normal forces in *any* direction. This corresponds to a hydrostatic pressure (positive or negative). So for a hydrostatic pressure, the tensor is diagonal and all three components are equal; they are, in fact, just equal to the pressure p . We can write

$$S_{ij} = p \delta_{ij}. \quad (31.25)$$

The stress tensor—and also its ellipsoid—will, in general, vary from point to point in a block of material; to describe the whole block we need to give the value of each component of S_{ij} as a function of position. So the stress tensor is a *field*. We have had *scalar fields*, like the temperature $T(x, y, z)$, which give one number for each point in space, and *vector fields* like $\mathbf{E}(x, y, z)$, which give three numbers for each point. Now we have a *tensor field* which gives nine numbers for each point in space—or really six for the symmetric tensor S_{ij} . A complete description of the internal forces in an arbitrarily distorted solid requires six functions of x , y , and z .

31-7 Tensors of higher rank

The stress tensor S_{ij} describes the internal *forces* of matter. If the material is elastic, it is convenient to describe the internal *distortion* in terms of another tensor T_{ij} —called the *strain* tensor. For a simple object like a bar of metal, you know that the change in length, ΔL , is approximately proportional to the force, so we say it obeys Hooke's law:

$$\Delta L = \gamma F.$$

For a solid elastic body with arbitrary distortions, the strain T_{ij} is related to the stress S_{ij} by a set of linear equations:

$$T_{ij} = \sum_{k,l} \gamma_{ijkl} S_{kl}. \quad (31.26)$$

Also, you know that the potential energy of a spring (or bar) is

$$\frac{1}{2} F \Delta L = \frac{1}{2} \gamma F^2.$$

The generalization for the elastic energy *density* in a solid body is

$$U_{\text{elastic}} = \sum_{ijkl} \frac{1}{2} \gamma_{ijkl} S_{ij} S_{kl}. \quad (31.27)$$

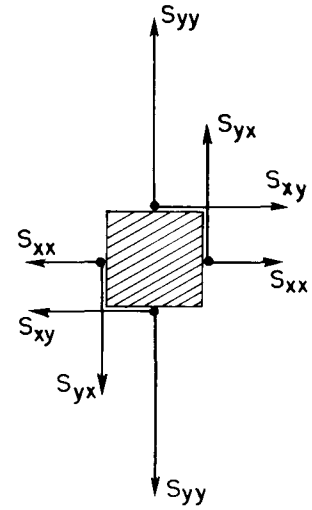


Fig. 31-9. The x - and y -forces on four faces of a small unit cube.

The complete description of the elastic properties of a crystal must be given in terms of the coefficients γ_{ijkl} . This introduces us to a new beast. It is a tensor of the *fourth* rank. Since each index can take on any one of three values, x, y , or z , there are $3^4 = 81$ coefficients. But there are really only 21 *different* numbers. First, since S_{ij} is symmetric, it has only six different values, and only 36 *different* coefficients are needed in Eq. (31.27). But also, S_{ij} can be interchanged with S_{kl} without changing the energy, so γ_{ijkl} must be symmetric if we interchange ij and kl . This reduces the number of different coefficients to 21. So to describe the elastic properties of a crystal of the lowest possible symmetry requires 21 elastic constants! This number is, of course, reduced for crystals of higher symmetry. For example, a cubic crystal has only three elastic constants, and an isotropic substance has only two.

That the latter is true can be seen as follows. How can the components of γ_{ijkl} be independent of the direction of the axes, as they must be if the material is isotropic? *Answer:* They can be independent *only* if they are expressible in terms of the tensor δ_{ij} . There are two possible expressions, $\delta_{ij}\delta_{kl}$ and $\delta_{ik}\delta_{jl} + \delta_{il}\delta_{jk}$, which have the required symmetry, so γ_{ijkl} must be a linear combination of them. Therefore, for isotropic materials,

$$\gamma_{ijkl} = a(\delta_{ij}\delta_{kl}) + b(\delta_{ik}\delta_{jl} + \delta_{il}\delta_{jk}),$$

and the material requires two constants, a and b , to describe its elastic properties. We will leave it for you to show that a cubic crystal needs only three

As a final example, this time of a third-rank tensor, we have the piezoelectric effect. Under stress, a crystal generates an electric field proportional to the stress; hence, in general, the law is

$$E_i = \sum_{j,k} P_{ijk} S_{jk},$$

where E_i is the electric field, and the P_{ijk} are the piezoelectric coefficients—or the piezoelectric tensor. Can you show that if the crystal has a center of inversion (invariant under $x, y, z \rightarrow -x, -y, -z$) the piezoelectric coefficients are all zero?

31-8 The four-tensor of electromagnetic momentum

All the tensors we have looked at so far in this chapter relate to the three dimensions of space; they are defined to have a certain transformation property under spatial rotations. In Chapter 26 we had occasion to use a tensor in the four dimensions of relativistic space-time—the electromagnetic field tensor $F_{\mu\nu}$. The components of such a four-tensor transform under a Lorentz transformation of the coordinates in a special way that we worked out. (Although we did not do it that way, we could have considered the Lorentz transformation as a “rotation” in a four-dimensional “space” called Minkowski space; then the analogy with what we are doing here would have been clearer.)

As our last example, we want to consider another tensor in the four dimensions (t, x, y, z) of relativity theory. When we wrote the stress tensor, we defined S_{ij} as a component of a force across a unit area. But a force is equal to the time rate of change of a momentum. Therefore, instead of saying “ S_{xy} is the x -component of the force across a unit area perpendicular to y ,” we could equally well say, “ S_{xy} is the rate of flow of the x -component of momentum through a unit area perpendicular to y .” In other words, each term of S_{ij} also represents the flow of the i -component of momentum through a unit area perpendicular to the j -direction. These are pure space components, but they are parts of a “larger” tensor $S_{\mu\nu}$ in four dimensions (μ and $\nu = t, x, y, z$) containing additional components like S_{tx} , S_{yt} , S_{tt} , etc. We will now try to find the physical meaning of these extra components.

We know that the space components represent flow of momentum. We can get a clue on how to extend this to the time dimension by studying another kind of “flow”—the flow of electric charge. For the *scalar* quantity, charge, the rate of flow (per unit area perpendicular to the flow) is a space *vector*—the current density

vector $\mathbf{j}$. We have seen that the time component of this flow vector is the density of the stuff that is flowing. For instance, $\mathbf{j}$ can be combined with a time component, $j_t = \rho$, the charge density, to make the four-vector $j_\mu = (\rho, \mathbf{j})$; that is, the μ in j_μ takes on the values t, x, y, z to mean “density, rate of flow in the x -direction, rate of flow in y , rate of flow in z ” of the scalar charge.

Now by analogy with our statement about the time component of the flow of a scalar quantity, we might expect that with S_{xx} , S_{xy} , and S_{xz} , describing the flow of the x -component of momentum, there should be a time component S_{xt} which would be the density of whatever is flowing; that is, S_{xt} should be the density of x -momentum. So we can extend our tensor horizontally to include a t -component. We have

$$\begin{aligned} S_{xt} &= \text{density of } x\text{-momentum,} \\ S_{xx} &= x\text{-flow of } x\text{-momentum,} \\ S_{xy} &= y\text{-flow of } x\text{-momentum,} \\ S_{xz} &= z\text{-flow of } x\text{-momentum.} \end{aligned}$$

Similarly, for the y -component of momentum we have the three components of flow— S_{yx} , S_{yy} , S_{yz} —to which we should add a fourth term:

$$S_{yt} = \text{density of } y\text{-momentum.}$$

And, of course, to S_{zx} , S_{zy} , S_{zz} we would add

$$S_{zt} = \text{density of } z\text{-momentum.}$$

In four dimensions there is also a t -component of momentum, which is, we know, energy. So the tensor S_{ij} should be extended vertically with S_{tx} , S_{ty} , and S_{tz} , where

$$\begin{aligned} S_{tx} &= x\text{-flow of energy,} \\ S_{ty} &= y\text{-flow of energy,} \\ S_{tz} &= z\text{-flow of energy;} \end{aligned} \tag{31.28}$$

that is, S_{tx} is the flow of energy per unit area and per unit time across a surface perpendicular to the x -axis, and so on. Finally, to complete our tensor we need S_{tt} , which would be the *density of energy*. We have extended our stress tensor S_{ij} of three dimensions to the four-dimensional *stress-energy tensor* $S_{\mu\nu}$. The index μ can take on the four values t, x, y , and z , meaning, respectively, “density,” “flow per unit area in the x -direction,” “flow per unit area in the y -direction,” and “flow per unit area in the z -direction.” In the same way, ν takes on the four values t, x, y, z to tell us *what* flows, namely, “energy,” “momentum in the x -direction,” “momentum in the y -direction,” and “momentum in the z -direction.”

As an example, we will discuss this tensor not in matter, but in a region of free space in which there is an electromagnetic field. We know that the flow of energy is the Poynting vector $\mathbf{S} = \epsilon_0 c^2 \mathbf{E} \times \mathbf{B}$. So the x -, y -, and z -components of $\mathbf{S}$ are, from the relativistic point of view, the components S_{tx} , S_{ty} , and S_{tz} of our four-dimensional stress-energy tensor. The symmetry of the tensor S_{ij} carries over into the time components as well, so the four-dimensional tensor $S_{\mu\nu}$ is symmetric:

$$S_{\mu\nu} = S_{\nu\mu}. \tag{31.29}$$

In other words, the components S_{xt} , S_{yt} , S_{zt} , which are the *densities* of x , y , and z momentum, are also equal to the x -, y -, and z -components of the Poynting vector $\mathbf{S}$, the *energy flow*—as we have already shown in an earlier chapter by a different kind of argument.

The remaining components of the electromagnetic stress tensor $S_{\mu\nu}$ can also be expressed in terms of the electric and magnetic fields $\mathbf{E}$ and $\mathbf{B}$. That is to say, we must admit stress or, to put it less mysteriously, flow of momentum in the electromagnetic field. We discussed this in Chapter 27 in connection with Eq (27.21), but did not work out the details

Those who want to exercise their prowess in tensors in four dimensions might like to see the formula for $S_{\mu\nu}$ in terms of the fields:

$$S_{\mu\nu} = \frac{\epsilon_0}{2} \left(\sum_{\alpha} F_{\mu\alpha} F_{\nu\alpha} - \frac{1}{4} \delta_{\mu\nu} \sum_{\alpha,\beta} F_{\beta\alpha} F_{\beta\alpha} \right),$$

where sums on α, β are on t, x, y, z but (as usual in relativity) we adopt a special meaning for the sum sign $\sum$ and for the symbol δ . In the sums the x, y, z terms are to be *subtracted* and $\delta_{tt} = +1$, while $\delta_{xx} = \delta_{yy} = \delta_{zz} = -1$ and $\delta_{\mu\nu} = 0$ for $\mu \neq \nu$ ($c = 1$). Can you verify that it gives the energy density $S_{tt} = (\epsilon_0/2) (E^2 + B^2)$ and the Poynting vector $\epsilon_0 \mathbf{E} \times \mathbf{B}$? Can you show that in an electrostatic field with $\mathbf{B} = 0$ the principal axes of stress are in the direction of the electric field, that there is a *tension* $(\epsilon_0/2)E^2$ along the direction of the field, and that there is an equal *pressure* in directions perpendicular to the field direction?

Refractive Index of Dense Materials

32-1 Polarization of matter

We want now to discuss the phenomenon of the refraction of light—and also, therefore, the absorption of light—by dense materials. In Chapter 31 of Volume I we discussed the theory of the index of refraction, but because of our limited mathematical abilities at that time, we had to restrict ourselves to finding the index only for materials of low density, like gases. The physical principles that produced the index were, however, made clear. The electric field of the light wave polarizes the molecules of the gas, producing oscillating dipole moments. The acceleration of the oscillating charges radiates new waves of the field. This new field, interfering with the old field, produces a changed field which is equivalent to a phase shift of the original wave. Because this phase shift is proportional to the thickness of the material, the effect is equivalent to having a different phase velocity in the material. When we looked at the subject before, we neglected the complications that arise from such effects as the new wave changing the fields at the oscillating dipoles. We assumed that the forces on the charges in the atoms came just from the *incoming* wave, whereas, in fact, their oscillations are driven not only by the incoming wave but also by the radiated waves of all the other atoms. It would have been difficult for us at that time to include this effect, so we studied only the rarefied gas, where such effects are not important.

Now, however, we will find that it is very easy to treat the problem by the use of differential equations. This method obscures the physical origin of the index (as coming from the re-radiated waves interfering with the original waves), but it makes the theory for dense materials much simpler. This chapter will bring together a large number of pieces from our earlier work. We've taken up practically everything we will need, so there are relatively few really new ideas to be introduced. Since you may need to refresh your memory about what we are going to need, we give in Table 32-1 a list of the equations we are going to use, together with a reference to the place where each can be found. In most instances, we will not take the time to give the physical arguments again, but will just use the equations.

Table 32-1

**Our work in this chapter will be based on the following material,
already covered in earlier chapters**

Subject	Reference	Equation
Damped oscillations	Vol. I, Chap. 23	$m(\ddot{x} + \gamma\dot{x} + \omega_0^2x) = F$
Index of gases	Vol. I, Chap. 31	$n = 1 + \frac{1}{2} \frac{Nq_e^2}{\epsilon_0(\omega_0^2 - \omega^2)}$ $n = n' - in''$
Mobility	Vol. I, Chap. 41	$m\dot{x} + \mu x = F$
Electrical conductivity	Vol. I, Chap. 43	$\mu = \frac{\tau}{m}; \sigma = \frac{Nq_e^2\tau}{m}$
Polarizability	Vol. II, Chap. 10	$\rho_{\text{pol}} = -\nabla \cdot \mathbf{P}$
Inside dielectrics	Vol. II, Chap. 11	$\mathbf{E}_{\text{local}} = \mathbf{E} + \frac{1}{3\epsilon_0} \mathbf{P}$

32-1 Polarization of matter

32-2 Maxwell's equations in a dielectric

32-3 Waves in a dielectric

32-4 The complex index of refraction

32-5 The index of a mixture

32-6 Waves in metals

32-7 Low-frequency and high-frequency approximations; the skin depth and the plasma frequency

Review: See Table 32-1.

We begin by recalling the machinery of the index of refraction for a gas. We suppose that there are N particles per unit volume and that each particle behaves as a harmonic oscillator. We use a model of an atom or molecule in which the electron is bound with a force proportional to its displacement (as though the electron were held in place by a spring). We emphasized that this was not a legitimate *classical* model of an atom, but we will show later that the correct quantum mechanical theory gives results equivalent to this model (in simple cases). In our earlier treatment, we did not include the possibility of a damping force in the atomic oscillators, but we will do so now. Such a force corresponds to a resistance to the motion, that is, to a force proportional to the velocity of the electron. Then the equation of motion is

$$F = q_e E = m(\ddot{x} + \gamma \dot{x} + \omega_0^2 x), \quad (32.1)$$

where x is the displacement parallel to the direction of E . (We are assuming an *isotropic* oscillator whose restoring force is the same in all directions. Also, we are taking, for the moment, a linearly polarized wave, so that E doesn't change direction.) If the electric field acting on the atom varies sinusoidally with time, we write

$$E = E_0 e^{i\omega t}. \quad (32.2)$$

The displacement will then oscillate with the same frequency, and we can let

$$x = x_0 e^{i\omega t}$$

Substituting $\dot{x} = i\omega x$ and $\ddot{x} = -\omega^2 x$, we can solve for x in terms of E :

$$x = \frac{q_e/m}{-\omega^2 + i\gamma\omega + \omega_0^2} E \quad (32.3)$$

Knowing the displacement, we can calculate the acceleration $\ddot{x}$ and find the radiated wave responsible for the index. This was the way we computed the index in Chapter 31 of Volume I.

Now, however, we want to take a different approach. The induced dipole moment p of an atom is $q_e x$ or, using Eq. (32.3),

$$p = \frac{q_e^2/m}{-\omega^2 + i\gamma\omega + \omega_0^2} E. \quad (32.4)$$

Since p is proportional to E , we write

$$p = \epsilon_0 \alpha(\omega) E, \quad (32.5)$$

where α is called the *atomic polarizability*.* With this definition, we have

$$\alpha = \frac{q_e^2/m\epsilon_0}{-\omega^2 + i\gamma\omega + \omega_0^2}. \quad (32.6)$$

The quantum mechanical solution for the motions of electrons in atoms gives a similar answer except with the following modifications. The atoms have several natural frequencies, each frequency with its own dissipation constant γ . Also the effective "strength" of each mode is different, which we can represent by multiplying the polarizability for each frequency by a strength factor f , which is a number we expect to be of the order of 1. Representing the three parameters ω , γ , and f by ω_k , γ_k , and f_k for each mode of oscillation, and summing over the

* Throughout this chapter we follow the notation of Chapter 31 of Volume I, and let α represent the *atomic* polarizability as defined here. In the last chapter, we used α to represent the *volume* polarizability—the ratio of P to E . In the notation of *this* chapter $P = N\alpha\epsilon_0 E$ (see Eq. 32.8)

various modes, we modify Eq. (32.6) to read

$$\alpha(\omega) = \frac{q_e^2}{\epsilon_0 m} \sum_k \frac{f_k}{-\omega^2 + i\gamma_k \omega + \omega_{0k}^2}. \quad (32.7)$$

If N is the number of atoms per unit volume in the material, the polarization P is just $Np = \epsilon_0 N \alpha E$, and is proportional to E :

$$\mathbf{P} = \epsilon_0 N \alpha(\omega) \mathbf{E}. \quad (32.8)$$

In other words, when there is a sinusoidal electric field acting in a material, there is an induced dipole moment per unit volume which is proportional to the electric field—with a proportionality constant α that, we emphasize, depends upon the frequency. At very high frequencies, α is small; there is not much response. However, at low frequencies there can be a strong response. Also, the proportionality constant is a complex number, which means that the polarization does not exactly follow the electric field, but may be shifted in phase to some extent. At any rate, there is a polarization per unit volume whose magnitude is proportional to the strength of the electric field.

32-2 Maxwell's equations in a dielectric

The existence of polarization in matter means that there are polarization charges and currents inside of the material, and these must be put into the complete Maxwell equations in order to find the fields. We are going to solve Maxwell's equations this time in a situation in which the charges and currents are not zero, as in a vacuum, but are given implicitly by the polarization vector. Our first step is to find explicitly the charge density ρ and current density $\mathbf{j}$, averaged over a small volume of the same size we had in mind when we defined $\mathbf{P}$. Then the ρ and $\mathbf{j}$ we need can be obtained from the polarization.

We have seen in Chapter 10 that when the polarization $\mathbf{P}$ varies from place to place, there is a charge density given by

$$\rho_{\text{pol}} = -\nabla \cdot \mathbf{P}. \quad (32.9)$$

At that time, we were dealing with static fields, but the same formula is valid also for time-varying fields. However, when $\mathbf{P}$ varies with time, there are charges in motion, so there is also a polarization *current*. Each of the oscillating charges contributes a current equal to its charge q_e , times its velocity v . With N such charges per unit volume, the current density $\mathbf{j}$ is

$$\mathbf{j} = N q_e \mathbf{v}.$$

Since we know that $v = dx/dt$, then $j = N q_e (dx/dt)$, which is just dP/dt . Therefore the current density from the varying polarization is

$$\mathbf{j}_{\text{pol}} = \frac{d\mathbf{P}}{dt} \quad (32.10)$$

Our problem is now direct and simple. We write Maxwell's equations with the charge density and current density expressed in terms of $\mathbf{P}$, using Eqs. (32.9) and (32.10). (We assume that there are no other currents and charges in the material.) We then relate $\mathbf{P}$ to $\mathbf{E}$ with Eq. (32.5), and we solve the equation for $\mathbf{E}$ and $\mathbf{B}$ —looking for the wave solutions.

Before we do this, we would like to make an historical note. Maxwell originally wrote his equations in a form which was different from the one we have been using. Because the equations were written in this different form for many years—and are still written that way by many people—we will explain the difference. In the early days, the mechanism of the dielectric constant was not fully and clearly appreciated. The nature of atoms was not understood, nor that there was a polarization of the material. So people did not appreciate that there was a contribution

to the charge density ρ from $\nabla \cdot \mathbf{P}$. They thought only in terms of charges that were not bound to atoms (such as the charges that flow in wires or are rubbed off surfaces).

Today, we prefer to let ρ represent the *total* charge density, including the part from the bound atomic charges. If we call that part ρ_{pol} , we can write

$$\rho = \rho_{\text{pol}} + \rho_{\text{other}},$$

where ρ_{other} is the charge density considered by Maxwell and refers to the charges not bound to individual atoms. We would then write

$$\nabla \cdot \mathbf{E} = \frac{\rho_{\text{pol}} + \rho_{\text{other}}}{\epsilon_0}.$$

Substituting ρ_{pol} from Eq. (32.9),

$$\nabla \cdot \mathbf{E} = \frac{\rho_{\text{other}}}{\epsilon_0} - \frac{1}{\epsilon_0} \nabla \cdot \mathbf{P}$$

or

$$\nabla \cdot (\epsilon_0 \mathbf{E} + \mathbf{P}) = \rho_{\text{other}}. \quad (32.11)$$

The current density in the Maxwell equations for $\nabla \times \mathbf{B}$ also has, in general, contributions from bound atomic currents. We can therefore write

$$\mathbf{j} = \mathbf{j}_{\text{pol}} + \mathbf{j}_{\text{other}},$$

and the Maxwell equation becomes

$$c^2 \nabla \times \mathbf{B} = \frac{\mathbf{j}_{\text{other}}}{\epsilon_0} + \frac{\mathbf{j}_{\text{pol}}}{\epsilon_0} + \frac{\partial \mathbf{E}}{\partial t}. \quad (32.12)$$

Using Eq. (32.10), we get

$$\epsilon_0 c^2 \nabla \times \mathbf{B} = \mathbf{j}_{\text{other}} + \frac{\partial}{\partial t} (\epsilon_0 \mathbf{E} + \mathbf{P}). \quad (32.13)$$

Now you can see that if we were to *define* a new vector $\mathbf{D}$ by

$$\mathbf{D} = \epsilon_0 \mathbf{E} + \mathbf{P}, \quad (32.14)$$

the two field equations would become

$$\nabla \cdot \mathbf{D} = \rho_{\text{other}} \quad (32.15)$$

and

$$\epsilon_0 c^2 \nabla \times \mathbf{B} = \mathbf{j}_{\text{other}} + \frac{\partial \mathbf{D}}{\partial t}. \quad (32.16)$$

These are actually the forms that Maxwell used for dielectrics. His two remaining equations were

$$\nabla \times \mathbf{E} = - \frac{\partial \mathbf{B}}{\partial t},$$

and

$$\nabla \cdot \mathbf{B} = 0,$$

which are the same as we have been using.

Maxwell and the other early workers also had a problem with magnetic materials (which we will take up soon). Because they did not know about the circulating currents responsible for atomic magnetism, they used a current density that was missing still another part. Instead of Eq. (32.16), they actually wrote

$$\nabla \times \mathbf{H} = \mathbf{j}' + \frac{\partial \mathbf{D}}{\partial t}, \quad (32.17)$$

where $\mathbf{H}$ differs from $\epsilon_0 c^2 \mathbf{B}$ because it includes the effects of atomic currents. (Then $\mathbf{j}'$ represents what is left of the currents.) So Maxwell had *four* field vectors— $\mathbf{E}$, $\mathbf{D}$, $\mathbf{B}$, and $\mathbf{H}$ —the $\mathbf{D}$ and $\mathbf{H}$ were hidden ways of not paying attention to what

was going on inside the material. You will find the equations written this way in many places.

To solve the equations, it is necessary to relate $\mathbf{D}$ and $\mathbf{H}$ to the other fields, and people used to write

$$\mathbf{D} = \epsilon \mathbf{E} \quad \text{and} \quad \mathbf{B} = \mu \mathbf{H}. \quad (32.18)$$

However, these relations are only approximately true for some materials and even then only if the fields are not changing rapidly with time. (For sinusoidally varying fields one often *can* write the equations this way by making ϵ and μ complex functions of the frequency, but not for an arbitrary time variation of the fields.) So there used to be all kinds of cheating in solving the equations. We think the right way is to keep the equations in terms of the fundamental quantities as we now understand them—and that's how we have done it.

32-3 Waves in a dielectric

We want now to find out what kind of electromagnetic waves can exist in a dielectric material in which there are no extra charges other than those bound in atoms. So we take $\rho = -\nabla \cdot \mathbf{P}$ and $\mathbf{j} = \partial \mathbf{P} / \partial t$. Maxwell's equations then become

$$\begin{aligned} \text{(a)} \quad \nabla \cdot \mathbf{E} &= -\frac{\nabla \cdot \mathbf{P}}{\epsilon_0} & \text{(b)} \quad c^2 \nabla \times \mathbf{B} &= \frac{\partial}{\partial t} \left(\frac{\mathbf{P}}{\epsilon_0} + \mathbf{E} \right) \\ \text{(c)} \quad \nabla \times \mathbf{E} &= -\frac{\partial \mathbf{B}}{\partial t} & \text{(d)} \quad \nabla \cdot \mathbf{B} &= 0 \end{aligned} \quad (32.19)$$

We can solve these equations as we have done before. We start by taking the curl of Eq. (32.19c):

$$\nabla \times (\nabla \times \mathbf{E}) = -\frac{\partial}{\partial t} \nabla \times \mathbf{B}.$$

Next, we make use of the vector identity

$$\nabla \times (\nabla \times \mathbf{E}) = \nabla(\nabla \cdot \mathbf{E}) - \nabla^2 \mathbf{E},$$

and also substitute for $\nabla \times \mathbf{B}$, using Eq. (32.19b); we get

$$\nabla(\nabla \cdot \mathbf{E}) - \nabla^2 \mathbf{E} = -\frac{1}{\epsilon_0 c^2} \frac{\partial^2 \mathbf{P}}{\partial t^2} - \frac{1}{c^2} \frac{\partial^2 \mathbf{E}}{\partial t^2}.$$

Using Eq. (32.19a) for $\nabla \cdot \mathbf{E}$, we get

$$\nabla^2 \mathbf{E} - \frac{1}{c^2} \frac{\partial^2 \mathbf{E}}{\partial t^2} = -\frac{1}{\epsilon_0 c^2} \nabla(\nabla \cdot \mathbf{P}) + \frac{1}{\epsilon_0 c^2} \frac{\partial^2 \mathbf{P}}{\partial t^2}. \quad (32.20)$$

So instead of the wave equation, we now get that the D'Alembertian of $\mathbf{E}$ is equal to two terms involving the polarization $\mathbf{P}$.

Since $\mathbf{P}$ depends on $\mathbf{E}$, however, Eq. (32.20) can still have wave solutions. We will now limit ourselves to *isotropic* dielectrics, so that $\mathbf{P}$ is always in the same direction as $\mathbf{E}$. Let's try to find a solution for a wave going in the z -direction. Then, the electric field might vary as $e^{i(\omega t - kz)}$. We will also suppose that the wave is polarized in the x -direction—that the electric field has only an x -component. We write

$$E_x = E_0 e^{i(\omega t - kz)}. \quad (32.21)$$

You know that any function of $(z - vt)$ represents a wave that travels with the speed v . The exponent of Eq. (32.21) can be written as

$$-ik \left(z - \frac{\omega}{k} t \right),$$

so, Eq. (32.21) represents a wave with the phase velocity

$$v_{\text{ph}} = \omega/k.$$

The index of refraction n is defined (see Chapter 31, Vol. I) by letting

$$v_{\text{ph}} = \frac{c}{n}.$$

Thus Eq. (32.21) becomes

$$E_x = E_0 e^{i\omega(t - nz/c)}.$$

So we can find n by finding what value of k is required if Eq. (32.21) is to satisfy the proper field equations, and then using

$$n = \frac{kc}{\omega}. \quad (32.22)$$

In an isotropic material, there will be only an x -component of the polarization; then $\mathbf{P}$ has no variation with the x -coordinate, so $\nabla \cdot \mathbf{P} = 0$, and we get rid of the first term on the right-hand side of Eq. (32.20). Also, since we are assuming a linear dielectric, P_x will vary as $e^{i\omega t}$, and $\partial^2 P_x / \partial t^2 = -\omega^2 P_x$. The Laplacian in Eq. (32.20) becomes simply $\partial^2 E_x / \partial z^2 = -k^2 E_x$, so we get

$$-k^2 E_x + \frac{\omega^2}{c^2} E_x = -\frac{\omega^2}{\epsilon_0 c^2} P_x. \quad (32.23)$$

Now let us assume for the moment that since $\mathbf{E}$ is varying sinusoidally, we can set $\mathbf{P}$ proportional to $\mathbf{E}$, as in Eq. (32.5). (We'll come back to discuss this assumption later.) We write

$$P_x = \epsilon_0 N \alpha E_x.$$

Then E_x drops out of Eq. (32.23), and we find

$$k^2 = \frac{\omega^2}{c^2} (1 + N\alpha). \quad (32.24)$$

We have found that a wave like Eq. (32.21), with the wave number k given by Eq. (32.24), will satisfy the field equations. Using Eq. (32.22), the index n is given by

$$n^2 = 1 + N\alpha. \quad (32.25)$$

Let's compare this formula with what we obtained in our theory of the index of a gas (Chapter 31, Vol. I). There, we got Eq. (31.29), which is

$$n = 1 + \frac{1}{2} \frac{Nq_e^2}{m\epsilon_0} \frac{1}{-\omega^2 + \omega_0^2}. \quad (32.26)$$

Taking α from Eq. (32.6), Eq. (32.25) would give us

$$n^2 = 1 + \frac{Nq_e^2}{m\epsilon_0} \frac{1}{-\omega^2 + i\gamma\omega + \omega_0^2}. \quad (32.27)$$

First, we have the new term in $i\gamma\omega$, because we are including the dissipation of the oscillators. Second, the left-hand side is n instead of n^2 , and there is an extra factor of $1/2$. But notice that if N is small enough so that n is close to one (as it is for a gas), then Eq. (32.27) says that n^2 is one plus a small number: $n^2 = 1 + \epsilon$. We can then write $n = \sqrt{1 + \epsilon} \approx 1 + \epsilon/2$, and the two expressions are equivalent. Thus our new method gives for a gas the same result we found earlier.

Now you might think that Eq. (32.27) should give the index of refraction for dense materials also. It needs to be modified, however, for several reasons. First, the derivation of this equation assumes that the polarizing field on each atom is the field E_x . That assumption is *not* right, however, because in dense materials there is also the field produced by other atoms in the vicinity, which may be comparable to E_x . We considered a similar problem when we studied the static fields in dielectrics. (See Chapter 11.) You will remember that we estimated the field at a single atom by imagining that it sat in a spherical hole in the surrounding dielectric. The field in such a hole—which we called the *local* field—is increased

over the average field E by the amount $P/3\epsilon_0$. (Remember, however, that this result is only strictly true in isotropic materials—including the special case of a cubic crystal.)

The same arguments will hold for the electric field in a wave, so long as the wavelength of the wave is much longer than the spacing between atoms. Limiting ourselves to such cases, we write

$$E_{\text{local}} = E + \frac{P}{3\epsilon_0}. \quad (32.28)$$

This local field is the one that should be used for E in Eq. (32.3); that is, Eq.(32.8) should be rewritten:

$$P = \epsilon_0 N \alpha E_{\text{local}}. \quad (32.29)$$

Using E_{local} from Eq. (32.28), we find

$$P = \epsilon_0 N \alpha \left(E + \frac{P}{3\epsilon_0} \right)$$

or

$$P = \frac{N\alpha}{1 - (N\alpha/3)} \epsilon_0 E. \quad (32.30)$$

In other words, for dense materials P is still proportional to E (for sinusoidal fields). However, the constant of proportionality is not $\epsilon_0 N \alpha$, as we wrote below Eq. (32.23), but should be $\epsilon_0 N \alpha / [1 - (N\alpha/3)]$. So we should correct Eq. (32.25) to read

$$n^2 = 1 + \frac{N\alpha}{1 - (N\alpha/3)}. \quad (32.31)$$

It will be more convenient if we rewrite this equation as

$$3 \frac{n^2 - 1}{n^2 + 2} = N\alpha, \quad (32.32)$$

which is algebraically equivalent. This is known as the Clausius-Mosotti equation.

There is another complication in dense materials. Because neighboring atoms are so close, there are strong interactions between them. The internal modes of oscillation are, therefore, modified. The natural frequencies of the atomic oscillations are spread out by the interactions, and they are usually quite heavily damped—the resistance coefficient becomes quite large. So the ω_0 's and γ 's of the solid will be quite different from those of the free atoms. With these reservations, we can still represent α , at least approximately, by Eq. (32.7). We have then that

$$3 \frac{n^2 - 1}{n^2 + 2} = \frac{Nq_e^2}{m\epsilon_0} \sum_k \frac{f_k}{-\omega^2 + i\gamma_k\omega + \omega_{0k}^2}. \quad (32.33)$$

One final complication. If the dense material is a mixture of several components, each will contribute to the polarization. The total α will be the sum of the contributions from each component of the mixture [except for the inaccuracy of the local field approximation, Eq. (32.28), in ordered crystals—effects we discussed when analyzing ferroelectrics]. Writing N_j as the number of atoms of each component per unit volume, we should replace Eq. (32.32) by

$$3 \left(\frac{n^2 - 1}{n^2 + 2} \right) = \sum_j N_j \alpha_j, \quad (32.34)$$

where each α_j will be given by an expression like Eq. (32.7). Equation (32.34) completes our theory of the index of refraction. The quantity $3(n^2 - 1)/(n^2 + 2)$ is given by some complex function of frequency, which is the mean atomic polarizability $\alpha(\omega)$. The precise evaluation of $\alpha(\omega)$ (that is, finding f_k , γ_k and ω_{0k}) in dense substances is a difficult problem of quantum mechanics. It has been done from first principles only for a few especially simple substances.

32-4 The complex index of refraction

We want to look now at the consequences of our result, Eq. (32.33). First, we notice that α is complex, so the index n is going to be a complex number. What does that mean? Let's say that we write n as the sum of a real and an imaginary part:

$$n = n_R - in_I, \quad (32.35)$$

where n_R and n_I are real functions of ω . We write in_I with a minus sign, so that n_I will be a positive quantity in all ordinary optical materials. (In ordinary inactive materials—that are not, like lasers, light sources themselves— γ is a positive number, and that makes the imaginary part of n negative.) Our plane wave of Eq. (32.21) is written in terms of n as

$$E_x = E_0 e^{-i\omega(t - nz/c)}$$

Writing n as in Eq. (32.35), we would have

$$E_x = E_0 e^{-\omega n_I z/c} e^{i\omega(t - n_R z/c)}. \quad (32.36)$$

The term $e^{i\omega(t - n_R z/c)}$ represents a wave travelling with the speed c/n_R , so n_R represents what we normally think of as the index of refraction. But the *amplitude* of this wave is

$$E_0 e^{-\omega n_I z/c},$$

which decreases exponentially with z . A graph of the strength of the electric field at some instant as a function of z is shown in Fig. 32-1, for $n_I \approx n_R/2\pi$. The imaginary part of the index represents the attenuation of the wave due to the energy losses in the atomic oscillators. The *intensity* of the wave is proportional to the square of the amplitude, so

$$\text{Intensity} \propto e^{-2\omega n_I z/c}.$$

This is often written as

$$\text{Intensity} \propto e^{-\beta z},$$

where $\beta = 2\omega n_I/c$ is called the *absorption coefficient*. Thus we have in Eq. (32.33) not only the theory of the index of refraction of materials, but the theory of their absorption of light as well.

In what we usually consider to be transparent material, the quantity $c/\omega n_I$ —which has the dimensions of a length—is quite large in comparison with the thickness of the material.

32-5 The index of a mixture

There is another prediction of our theory of the index of refraction that we can check against experiment. Suppose we consider a mixture of two materials. The index of the mixture is not the average of the two indexes, but should be given in terms of the sum of the two polarizabilities, as in Eq. (32.34). If we ask about the index of, say, a sugar solution, the total polarizability is the sum of the polarizability of the water and that of the sugar. Each must, of course, be calculated using for N the number per unit volume of the molecules of the particular kind. In other words, if a given solution has N_1 molecules of water, whose polarizability is α_1 , and N_2 molecules of sucrose ($\text{C}_{12}\text{H}_{22}\text{O}_{11}$), whose polarizability is α_2 , we should have that

$$3 \left(\frac{n^2 - 1}{n^2 + 2} \right) = N_1 \alpha_1 + N_2 \alpha_2. \quad (32.37)$$

We can use this formula to test our theory against experiment by measuring the index for various concentrations of sucrose in water. We are making several assumptions here, however. Our formula assumes that there is no chemical action when the sucrose is dissolved and that the disturbances to the individual atomic

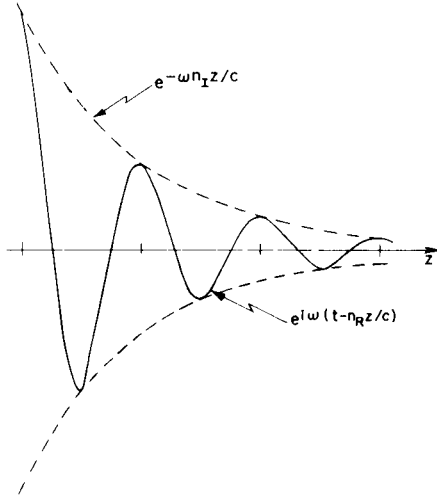


Fig. 32-1. A graph of E_x for some instant t , if $n_I \approx n_R/2\pi$.

Table 32-2

Refractive index of sucrose solutions, and comparison with predictions of Eq. (32.37).

Data from Handbook

A	B	C	D	E	F	G	H	J
Fraction of sucrose by weight	density (gm/cm ³)	n at 20°C	Moles of sucrose ^d per liter, N_2/N_0	Moles of water ^e per liter, N_1/N_0	$3 \left(\frac{n^2 - 1}{n^2 + 2} \right)$	$N_1\alpha_1$	$N_2\alpha_2$	$N_0\alpha_2$ (gm./liter)
0 ^a	0.9982	1.333	0	55.5	0.617	0.617	0	—
0.30	1.1270	1.3811	0.970	43.8	0.698	0.487	0.211	0.213
0.50	1.2296	1.4200	1.798	34.15	0.759	0.379	0.380	0.211
0.85	1.4454	1.5033	3.59	12.02	0.886	0.1335	0.752	0.210
1.00 ^b	1.588	1.5577 ^c	4.64	0	0.960	0	0.960	0.207

^a pure water^c average (see text)^e molecular weight of water = 18^b sugar crystals^d molecular weight of sucrose = 342

oscillators are not too different for various concentrations. So our result is certainly only approximate. Anyway, let's see how good it is.

We have picked the example of a sugar solution because there is a good table of measurements of the index of refraction in the *Handbook of Chemistry and Physics* and also because sugar is a molecular crystal that goes into solution without ionizing or otherwise changing its chemical state.

We give in the first three columns of Table 32-2 the data from the handbook. Column A is the percent of sucrose by weight, column B is the measured density (gm/cm³), and column C is the measured index of refraction for light whose wavelength is 589.3 millimicrons. For pure sugar we have taken the measured index of sugar crystals. The crystals are not isotropic, so the measured index is different along different directions. The handbook gives three values:

$$n_1 = 1.5376, \quad n_2 = 1.5651, \quad n_3 = 1.5705.$$

We have taken the average.

Now we could try to compute n for each concentration, but we don't know what value to take for α_1 or α_2 . Let's test the theory this way: We will assume that the polarizability of water (α_1) is the same at all concentrations and compute the polarizability of sucrose by using the experiment of values for n and solving Eq. (38.27) for α_2 . If the theory is correct, we should get the same α_2 for all concentrations.

First, we need to know N_1 and N_2 : let's express them in terms of Avogadro's number, N_0 . Let's take one liter (1000 cm³) for our unit of volume. Then N_i/N_0 is the weight per liter divided by the gram-molecular weight. And the weight per liter is the density (multiplied by 1000 to get grams per liter) times the fractional weight of either the sucrose or the water. In this way, we get N_2/N_0 and N_1/N_0 as in columns D and E of the table.

In column F we have computed $3(n^2 - 1)/(n^2 + 2)$ from the experimental values of n in column C. For pure water, $3(n^2 - 1)/(n^2 + 2)$ is 0.617, which is equal to just $N_1\alpha_1$. We can then fill in the rest of Column G, since for each row row G/E may be in the same ratio—namely, 0.617:55.5. Subtracting column G from column F, we get the contribution $N_2\alpha_2$ of the sucrose, shown in column H. Dividing these entries by the values of N_2/N_0 in column D, we get the value of $N_0\alpha_2$ shown in column J.

From our theory we would expect all the values of $N_0\alpha_2$ to be the same. They are not exactly equal, but pretty close. We can conclude that our ideas are fairly correct. Even more, we find that the polarizability of the sugar molecule doesn't seem to depend much on its surroundings—its polarizability is nearly the same in a dilute solution as it is in the crystal.

32-6 Waves in metals

The theory we have worked out in this chapter for solid materials can also be applied to good conductors, like metals, with very little modification. In metals some of the electrons have no binding force holding them to any particular atom; it is these “free” electrons which are responsible for the conductivity. There are other electrons which are bound, and the theory above is directly applicable to them. Their influence, however, is usually swamped by the effects of the conduction electrons. We will consider now only the effects of the free electrons.

If there is no restoring force on an electron—but still some resistance to its motion—its equation of motion differs from Eq. (32.1) only because the term in $\omega_0^2 x$ is lacking. So all we have to do is set $\omega_0^2 = 0$ in the rest of our derivations—except that there is one more difference. The reason that we had to distinguish between the average field and the local field in a dielectric is that in an insulator each of the dipoles is fixed in position, so that it has a definite relationship to the position of the others. But because the conduction electrons in a metal move around all over the place, the field on them *on the average* is just the average field E . So the correction we made to Eq. (32.5) by using Eq. (32.28) should *not* be made for conduction electrons. Therefore the formula for the index of refraction for metals should look like Eq. (32.27), except with ω_0 set equal to zero, namely,

$$n^2 = 1 + \frac{Nq_e^2}{m\epsilon_0} \frac{1}{-\omega^2 + i\gamma\omega}. \quad (32.38)$$

This is only the contribution from the conduction electrons, which we will assume is the major term for metals.

Now we even know how to find what value to use for γ , because it is related to the conductivity of the metal. In Chapter 43 of Volume I we discussed how the conductivity of a metal comes from the diffusion of the free electrons through the crystal. The electrons go on a jagged path from one scattering to the next, and between scatterings they move freely except for an acceleration due to any average electric field (as shown in Fig. 32-2). We found in Chapter 43 of Volume I that the average drift velocity is just the acceleration times the average time τ between collisions. The acceleration is $q_e E/m$, so

$$v_{\text{drift}} = \frac{q_e E}{m} \tau. \quad (32.39)$$

This formula assumed that E was constant, so that v_{drift} was a steady velocity. Since there is no average acceleration, the drag force is equal to the applied force. We have defined γ by saying that $\gamma m v$ is the drag force [see Eq. (32.1)], which is $q_e E$; therefore we have that

$$\gamma = \frac{1}{\tau}. \quad (32.40)$$

Although we cannot easily measure τ directly, we can determine it by measuring the conductivity of the metal. It is found experimentally that an electric field E in a metal produces a current with the density j proportional to E (for isotropic materials):

$$j = \sigma E.$$

The proportionality constant σ is called the *conductivity*. This is just what we expect from Eq. (32.39) if we set

$$j = Nq_e v_{\text{drift}}.$$

Then

$$\sigma = \frac{Nq_e^2}{m} \tau. \quad (32.41)$$

So τ —and therefore γ —can be related to the observed electrical conductivity. Using Eqs. (32.40) and (32.41), we can rewrite our formula for the index, Eq.

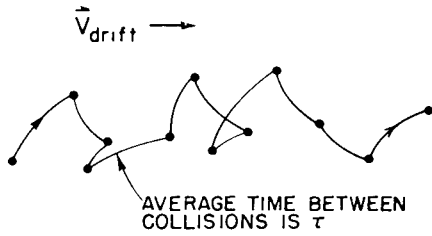


Fig. 32-2. The motion of a free electron.

(32.38), in the following form:

$$n^2 = 1 + \frac{\sigma/\epsilon_0}{i\omega(1 + i\omega\tau)}, \quad (32.42)$$

where

$$\tau = \frac{1}{\gamma} = \frac{m\sigma}{Nq_e^2}. \quad (32.43)$$

This is a convenient formula for the index of refraction of metals.

32-7 Low-frequency and high-frequency approximations; the skin depth and the plasma frequency

Our result, Eq. (32.42), for the index of refraction for metals predicts quite different characteristics for wave propagation at different frequencies. Let's first see what happens at very *low* frequencies. If ω is small enough, we can approximate Eq. (32.42) by

$$n^2 = -i \frac{\sigma}{\epsilon_0\omega}. \quad (32.44)$$

Now, as you can check by taking the square,*

$$\sqrt{-i} = \frac{1-i}{\sqrt{2}};$$

so for low frequencies,

$$n = \sqrt{\sigma/2\epsilon_0\omega} (1 - i). \quad (32.45)$$

The real and imaginary parts of n have the same magnitude. With such a large imaginary part to n , the wave is rapidly attenuated in the metal. Referring to Eq. (32.36), the amplitude of a wave going in the z -direction decreases as

$$\exp[-\sqrt{\sigma\omega/2\epsilon_0c^2} z]. \quad (32.46)$$

Let's write this as

$$e^{-z/\delta}, \quad (32.47)$$

where δ is then the distance in which the wave amplitude decreases by the factor $e^{-1} = 1/2.72$ —or roughly one-third. The amplitude of such a wave as a function of z is shown in Fig. 32-3. Since electromagnetic waves will penetrate into a metal only this distance, δ is called the *skin depth*. It is given by

$$\delta = \sqrt{2\epsilon_0c^2/\sigma\omega}. \quad (32.48)$$

Now what do we mean by "low" frequencies? Looking at Eq. (32.42), we see that it can be approximated by Eq. (32.44) only if $\omega\tau$ is much less than one and if $\omega\epsilon_0/\sigma$ is also much less than one—that is, our low-frequency approximation applies when

$$\omega \ll \frac{1}{\tau}$$

and

$$\omega \ll \frac{\sigma}{\epsilon_0}. \quad (32.49)$$

Let's see what frequencies these correspond to for a typical metal like copper. We compute τ by using Eq. (32.43), and σ/ϵ_0 , by using the measured conductivity. We take the following data from a handbook:

$$\begin{aligned} \sigma &= 5.76 \times 10^7 \text{ (ohm-meter)}^{-1}, \\ \text{atomic weight} &= 63.5 \text{ grams}, \\ \text{density} &= 8.9 \text{ grams/cm}^3, \\ \text{Avogadro's number} &= 6.02 \times 10^{23} \text{ (gram atomic weight)}^{-1}. \end{aligned}$$

* Or writing $-i = e^{-i\pi/2}$; $\sqrt{-i} = e^{-i\pi/4} = \cos \pi/4 - i \sin \pi/4$, which gives the same result.

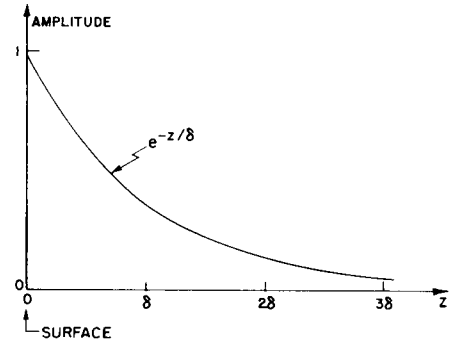


Fig. 32-3. The amplitude of a transverse electromagnetic wave as a function of distance into a metal.

If we assume that there is one free electron per atom, then the number of electrons per cubic meter is

$$N = 8.5 \times 10^{28} \text{ meter}^{-3}.$$

Using

$$q_e = 1.6 \times 10^{-19} \text{ coulomb},$$

$$\epsilon_0 = 8.85 \times 10^{-12} \text{ farad-meter}^{-1},$$

$$m = 9.11 \times 10^{-31} \text{ kgm},$$

we get

$$\tau = 2.4 \times 10^{-14} \text{ sec},$$

$$\frac{1}{\tau} = 4.1 \times 10^{13} \text{ sec}^{-1},$$

$$\frac{\sigma}{\epsilon_0} = 6.5 \times 10^{18} \text{ sec}^{-1}.$$

So for frequencies less than about 10^{12} cycles per second, copper will have the “low-frequency” behavior we describe (that means for waves whose free-space wavelength is longer than 0.3 millimeters—*very* short radio waves!).

For these waves, the skin depth in copper is

$$\delta = \sqrt{\frac{0.028 \text{ m}^2\text{-sec}^{-1}}{\omega}}.$$

For microwaves of 10,000 megacycles per second (3-cm waves)

$$\delta = 6.7 \times 10^{-4} \text{ cm}.$$

The wave penetrates a very small distance.

We can see from this why in studying cavities (or waveguides) we needed to worry only about the fields inside the cavity, and not in the metal or outside the cavity. Also, we see why the losses in a cavity are reduced by a thin plating of silver or gold. The losses come from the current, which are appreciable only in a thin layer equal to the skin depth.

Suppose we look now at the index of a metal like copper at high frequencies. For very high frequencies $\omega\tau$ is much greater than one, and Eq. (32.42) is well approximated by

$$n^2 = 1 - \frac{\sigma}{\epsilon_0\omega^2\tau}. \quad (32.50)$$

For waves of high frequencies the index of a metal becomes real—and less than one! This is also evident from Eq. (32.38) if the dissipation term with γ is neglected, as can be done for very large ω . Equation (32.38) gives

$$n^2 = 1 - \frac{Nq_e^2}{m\epsilon_0\omega^2}, \quad (32.51)$$

which is, of course, the same as Eq. (32.50). We have seen before the quantity $Nq_e^2/m\epsilon_0$, which we called the square of the plasma frequency (Section 7-3):

$$\omega_p^2 = \frac{Nq_e^2}{\epsilon_0 m},$$

so we can write Eq. (32.50) or Eq. (32.51) as

$$n^2 = 1 - \left(\frac{\omega_p}{\omega}\right)^2.$$

The plasma frequency is a kind of “critical” frequency.

For $\omega < \omega_p$ the index of a metal has an imaginary part, and waves are attenuated; but for $\omega \gg \omega_p$ the index is real, and the metal becomes transparent. You know, of course, that metals are reasonably transparent to x-rays. But some metals are even transparent in the ultraviolet. In Table 32-3 we give for

several metals the experimental observed wavelength at which they begin to become transparent. In the second column we give the calculated critical wavelength $\lambda_p = 2\pi c/\omega_p$. Considering that the experimental wavelength is not too well defined, the fit of the theory is fairly good.

You may wonder why the plasma frequency ω_p should have anything to do with the propagation of electromagnetic waves in metals. The plasma frequency came up in Chapter 7 as the natural frequency of *density* oscillations of the free electrons. (A clump of electrons is repelled by electric forces, and the inertia of the electrons leads to an oscillation of density.) So *longitudinal* plasma waves are resonant at ω_p . But we are now talking about *transverse* electromagnetic waves, and we have found that transverse waves are absorbed for frequencies below ω_p . (It's an interesting and *not* accidental coincidence.)

Although we have been talking about wave propagation in metals, you appreciate by this time the universality of the phenomena of physics—that it doesn't make any difference whether the free electrons are in a metal or whether they are in the plasma of the ionosphere of the earth, or in the atmosphere of a star. To understand radio propagation in the ionosphere, we can use the same expressions—using, of course, the proper values for N and τ . We can see now why long radio waves are absorbed or reflected by the ionosphere, whereas short waves go right through. (Short waves must be used for communication with satellites.)

We have talked about the high- and low-frequency extremes for wave propagation in metals. For the in-between frequencies the full-blown formula of Eq. (32.42) must be used. In general, the index will have real and imaginary parts; the wave is attenuated as it propagates into the metal. For very thin layers, metals are somewhat transparent even at optical frequencies. As an example, special goggles for people who work around high-temperature furnaces are made by evaporating a thin layer of gold on glass. The visible light is transmitted fairly well—with a strong green tinge—but the infrared is strongly absorbed.

Finally, it cannot have escaped the reader that many of these formulas resemble in some ways those for the dielectric constant κ discussed in Chapter 10. The dielectric constant κ measures the response of the material to a constant field, that is, for $\omega = 0$. If you look carefully at the definition of n and κ you see that κ is simply the limit of n^2 as $\omega \rightarrow 0$. Indeed, placing $\omega = 0$ and $n^2 = \kappa$ in equations of this chapter will reproduce the equations of the theory of the dielectric constant of Chapter 11.

Table 32-3*

Wavelengths below which the metal becomes transparent

Metal	λ (experimental)	$\lambda_p = 2\pi c/\omega_p$
Li	1550 Å	1550 Å
Na	2100	2090
K	3150	2870
Rb	3400	3220

* From: C. Kittel, *Introduction to Solid State Physics*, John Wiley and Sons, Inc., New York, 2nd ed., 1956, p. 266.

Reflection from Surfaces

33-1 Reflection and refraction of light

The subject of this chapter is the reflection and refraction of light—or electromagnetic waves in general—at surfaces. We have already discussed the laws of reflection and refraction in Chapter 35 of Volume I. Here's what we found out there:

1. The angle of reflection is equal to the angle of incidence. With the angles defined as shown in Fig. 33-1,

$$\theta_r = \theta_i. \quad (33.1)$$

2. The product $n \sin \theta$ is the same for the incident and transmitted beams (Snell's law).

$$n_1 \sin \theta_i = n_2 \sin \theta_t. \quad (33.2)$$

3. The intensity of the reflected light depends on the angle of incidence and also on the direction of polarization. For E perpendicular to the plane of incidence, the reflection coefficient $R_\perp$ is

$$R_\perp = \frac{I_r}{I_i} = \frac{\sin^2 (\theta_i - \theta_t)}{\sin^2 (\theta_i + \theta_t)} \quad (33.3)$$

For E parallel to the plane of incidence, the reflection coefficient $R_\parallel$ is

$$R_\parallel = \frac{I_r}{I_i} = \frac{\tan^2 (\theta_i - \theta_t)}{\tan^2 (\theta_i + \theta_t)}. \quad (33.4)$$

4. For normal incidence (any polarization, of course!),

$$\frac{I_r}{I_i} = \left(\frac{n_2 - n_1}{n_2 + n_1} \right)^2. \quad (33.5)$$

(Earlier, we used i for the incident angle and r for the refracted angle. Since we can't use r for both "refracted" and "reflected" angles, we are now using θ_i = incident angle, θ_r = reflected angle, and θ_t = transmitted angle.)

Our earlier discussion is really about as far as anyone would normally need to go with the subject, but we are going to do it all over again a different way. Why? One reason is that we assumed before that the indexes were real (no absorption in the materials). But another reason is that you should know how to deal with what happens to waves at surfaces from the point of view of Maxwell's equations. We'll get the same answers as before, but now from a straightforward solution of the wave problem, rather than by some clever arguments.

We want to emphasize that the amplitude of a surface reflection is not a property of the *material*, as is the index of refraction. It is a "surface property," one that depends precisely on how the surface is made. A thin layer of extraneous junk on the surface between two materials of indices n_1 and n_2 will usually change the reflection. (There are all kinds of possibilities of interference here—like the colors of oil films. Suitable thickness can even reduce the reflected amplitude to zero for a given frequency; that's how coated lenses are made.) The formulas we will derive are correct only if the change of index is sudden—within a distance very small compared with one wavelength. For light, the wavelength is about 5000 Å, so by a "smooth" surface we mean one in which the conditions change in

33-1 Reflection and refraction of light

33-2 Waves in dense materials

33-3 The boundary conditions

33-4 The reflected and transmitted waves

33-5 Reflection from metals

33-6 Total internal reflection

Review. Chapter 35, Vol. I, Polarization

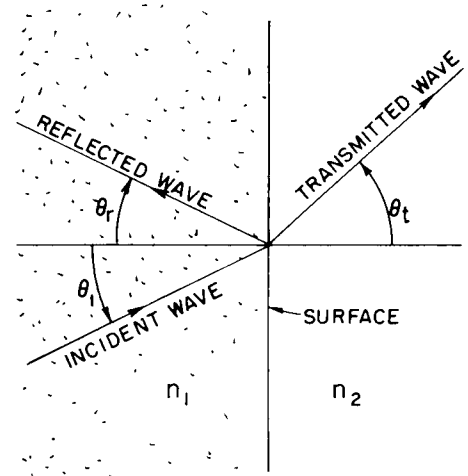


Fig. 33-1. Reflection and refraction of light waves at a surface. (The wave directions are normal to the wave crests.)

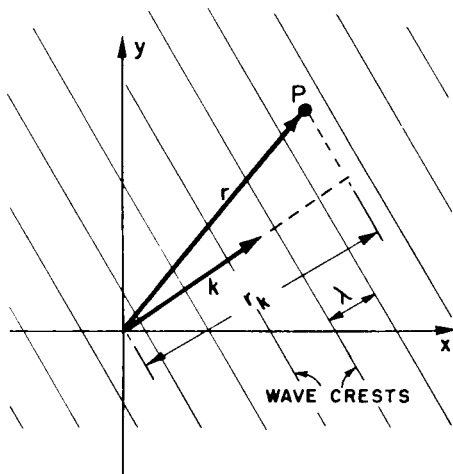


Fig. 33-2. For a wave moving in the direction $\mathbf{k}$, the phase at any point P is $(\omega t - \mathbf{k} \cdot \mathbf{r})$.

going a distance of only a few atoms (or a few angstroms). Our equations will work for light for highly polished surfaces. In general, if the index changes gradually over a distance of several wavelengths, there is very little reflection at all.

33-2 Waves in dense materials

First, we remind you about the convenient way of describing a sinusoidal plane wave we used in Chapter 36 of Volume I. Any field *component* in the wave (we use E as an example) can be written in the form

$$E = E_0 e^{i(\omega t - \mathbf{k} \cdot \mathbf{r})}, \quad (33.6)$$

where E represents the amplitude at the point $\mathbf{r}$ (from the origin) at the time t . The vector $\mathbf{k}$ points in the direction the wave is travelling, and its magnitude $|\mathbf{k}| = k = 2\pi/\lambda$ is the wave number. The phase velocity of the wave is $v_{ph} = \omega/k$, for a light wave in a material of index n , $v_{ph} = c/n$, so

$$k = \frac{\omega n}{c}. \quad (33.7)$$

Suppose $\mathbf{k}$ is in the z -direction, then $\mathbf{k} \cdot \mathbf{r}$ is just kz , as we have often used it. For $\mathbf{k}$ in any other direction, we should replace z by r_k , the distance from the origin in the $\mathbf{k}$ -direction; that is, we should replace kz by kr_k , which is just $\mathbf{k} \cdot \mathbf{r}$. (See Fig. 33-2.) So Eq. (33.6) is a convenient representation of a wave in any direction.

We must remember, of course, that

$$\mathbf{k} \cdot \mathbf{r} = k_x x + k_y y + k_z z,$$

where k_x , k_y , and k_z are the components of $\mathbf{k}$ along the three axes. In fact, we pointed out once that (ω, k_x, k_y, k_z) is a four-vector, and that its scalar product with (t, x, y, z) is an invariant. So the *phase* of a wave is an invariant, and Eq. (33.6) could be written

$$E = E_0 e^{ik_\mu x_\mu}.$$

But we don't need to be that fancy now.

For a sinusoidal E , as in Eq. (33.6), $\partial E / \partial t$ is the same as $i\omega E$, and $\partial E / \partial x$ is $-ik_x E$, and so on for the other components. You can see why it is very convenient to use the form in Eq. (33.6) when working with differential equations—differentiations are replaced by multiplications. One further useful point: The operation $\nabla = (\partial/\partial x, \partial/\partial y, \partial/\partial z)$ gets replaced by the three multiplications $(-ik_x, -ik_y, -ik_z)$. But these three factors transform as the components of the vector $\mathbf{k}$, so the operator ∇ gets replaced by multiplication with $-i\mathbf{k}$:

$$\begin{aligned} \frac{\partial}{\partial t} &\rightarrow i\omega, \\ \nabla &\rightarrow -i\mathbf{k}. \end{aligned} \quad (33.8)$$

This remains true for any ∇ operation—whether it is the gradient, or the divergence, or the curl. For instance, the z -component of $\nabla \times \mathbf{E}$ is

$$\frac{\partial E_y}{\partial x} - \frac{\partial E_x}{\partial y}.$$

If both E_y and E_x vary as $e^{-i\mathbf{k} \cdot \mathbf{r}}$, then we get

$$-ik_x E_y + ik_y E_x,$$

which is, you see, the z -component of $-i\mathbf{k} \times \mathbf{E}$.

So we have the very useful general fact that whenever you have to take the gradient of a vector that varies as a wave in three dimensions (they are an important part of physics), you can always take the derivations quickly and almost without thinking by remembering that the operation ∇ is equivalent to multiplication by $-i\mathbf{k}$.

For instance, the Faraday equation

$$\nabla \times \mathbf{E} = -\frac{\partial \mathbf{B}}{\partial t}$$

becomes for a wave

$$-ik \times \mathbf{E} = -i\omega \mathbf{B}.$$

This tells us that

$$\mathbf{B} = \frac{\mathbf{k} \times \mathbf{E}}{\omega}, \quad (33.9)$$

which corresponds to the result we found earlier for waves in free space—that $\mathbf{B}$, in a wave, is at right angles to $\mathbf{E}$ and to the wave direction. (In free space, $\omega/k = c$.) You can remember the sign in Eq. (33.9) from the fact that $\mathbf{k}$ is in the direction of Poynting's vector $\mathbf{S} = \epsilon_0 c^2 \mathbf{E} \times \mathbf{B}$.

If you use the same rule with the other Maxwell equations, you get again the results of the last chapter and, in particular, that

$$\mathbf{k} \cdot \mathbf{k} = k^2 = \frac{\omega^2 n^2}{c^2}. \quad (33.10)$$

But since we know that, we won't do it again.

If you want to entertain yourself, you can try the following terrifying problem that was the ultimate test for graduate students back in 1890: solve Maxwell's equations for plane waves in an *anisotropic* crystal, that is, when the polarization $\mathbf{P}$ is related to the electric field $\mathbf{E}$ by a tensor of polarizability. You should, of course, choose your axes along the principal axes of the tensor, so that the relations are simplest (then $P_x = \alpha_a E_x$, $P_y = \alpha_b E_y$, and $P_z = \alpha_c E_z$), but let the waves have an arbitrary direction and polarization. You should be able to find the relations between $\mathbf{E}$ and $\mathbf{B}$, and how $\mathbf{k}$ varies with direction and wave polarization. Then you will understand the optics of an anisotropic crystal. It would be best to start with the simpler case of a birefringent crystal—like calcite—for which two of the polarizabilities are equal (say, $\alpha_b = \alpha_c$), and see if you can understand why you see double when you look through such a crystal. If you can do that, then try the hardest case, in which all three α 's are different. Then you will know whether you are up to the level of a graduate student of 1890. In this chapter, however, we will consider only isotropic substances

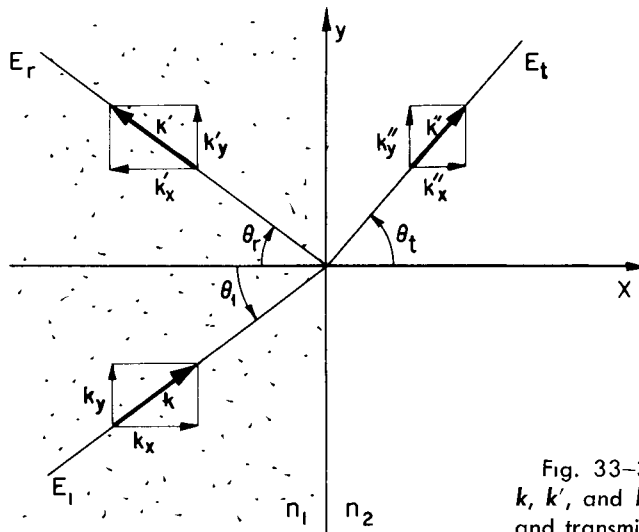


Fig. 33-3. The propagation vectors $\mathbf{k}$, $\mathbf{k}'$, and $\mathbf{k}''$ for the incident, reflected, and transmitted waves.

We know from experience that when a plane wave arrives at the boundary between two different materials—say, air and glass, or water and oil—there is a wave reflected and a wave transmitted. Suppose we assume no more than that and see what we can work out. We choose our axes with the yz -plane in the surface and the xy -plane perpendicular to the incident wave surfaces, as shown in Fig. 33-3.

The electric vector of the incident wave can then be written as

$$\mathbf{E}_i = \mathbf{E}_0 e^{i(\omega t - \mathbf{k} \cdot \mathbf{r})}. \quad (33.11)$$

Since $\mathbf{k}$ is perpendicular to the z -axis,

$$\mathbf{k} \cdot \mathbf{r} = k_x x + k_y y. \quad (33.12)$$

We write the reflected wave as

$$\mathbf{E}_r = \mathbf{E}'_0 e^{i(\omega' t - \mathbf{k}' \cdot \mathbf{r})}, \quad (33.13)$$

so that its frequency is ω' , its wave number is $\mathbf{k}'$, and its amplitude is $\mathbf{E}'_0$. (We know, of course, that the frequency is the same and the magnitude of $\mathbf{k}$ is the same as for the incident wave, but we are not going to assume even that. We will let it come out of the mathematical machinery.) Finally, we write for the transmitted wave,

$$\mathbf{E}_t = \mathbf{E}''_0 e^{i(\omega'' t - \mathbf{k}'' \cdot \mathbf{r})}. \quad (33.14)$$

We know that one of Maxwell's equations gives Eq (33.9), so for each of the waves we have

$$\mathbf{B}_i = \frac{\mathbf{k} \times \mathbf{E}_i}{\omega}, \quad \mathbf{B}_r = \frac{\mathbf{k}' \times \mathbf{E}_i}{\omega'}, \quad \mathbf{B}_t = \frac{\mathbf{k}'' \times \mathbf{E}_t}{\omega''}. \quad (33.15)$$

Also, if we call the indexes of the two media n_1 and n_2 , we have from Eq. (33.10)

$$k^2 = k_x^2 + k_y^2 = \frac{\omega^2 n_1^2}{c^2}. \quad (33.16)$$

Since the reflected wave is in the same material, then

$$k'^2 = \frac{\omega'^2 n_1^2}{c^2}, \quad (33.17)$$

whereas for the transmitted wave,

$$k''^2 = \frac{\omega''^2 n_2^2}{c^2}. \quad (33.18)$$

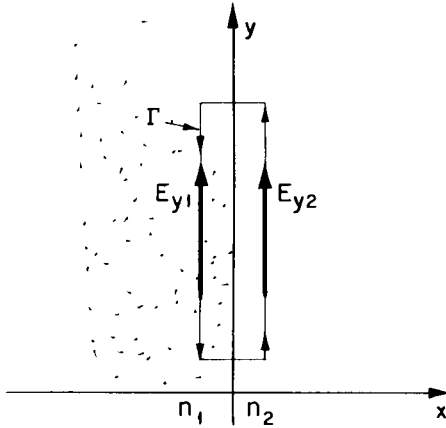


Fig. 33-4. A boundary condition $E_{y2} = E_{y1}$ is obtained from $\oint_{\Gamma} \mathbf{E} \cdot d\mathbf{s} = 0$.

33-3 The boundary conditions

All we have done so far is to describe the three waves; our problem now is to work out the parameters of the reflected and transmitted waves in terms of those of the incident wave. How can we do that? The three waves we have described satisfy Maxwell's equations in the uniform material, but Maxwell's equations must also be satisfied *at* the boundary between the two different materials. So we must now look at what happens right *at* the boundary. We will find that Maxwell's equations demand that the three waves fit together in a certain way.

As an example of what we mean, the y -component of the electric field $\mathbf{E}$ must be the *same* on both sides of the boundary. This is required by Faraday's law,

$$\nabla \times \mathbf{E} = - \frac{\partial \mathbf{B}}{\partial t}, \quad (33.19)$$

as we can see in the following way. Consider a little rectangular loop Γ which straddles the boundary, as shown in Fig 33-4. Equation (33.19) says that the line integral of $\mathbf{E}$ around Γ is equal to the rate of change of the flux of $\mathbf{B}$ through the loop:

$$\oint_{\Gamma} \mathbf{E} \cdot d\mathbf{s} = - \frac{\partial}{\partial t} \int \mathbf{B} \cdot \mathbf{n} \, da.$$

Now imagine that the rectangle is very narrow, so that the loop encloses an infinitesimal area. If $\mathbf{B}$ remains finite (and there's no reason it should be infinite at the boundary!) the flux through the area is zero. So the line integral of $\mathbf{E}$ must

be zero. If E_{y1} and E_{y2} are the components of the field on the two sides of the boundary and if the length of the rectangle is l , we have

$$E_{y1}l - E_{y2}l = 0$$

or

$$E_{y1} = E_{y2}, \quad (33.20)$$

as we have said. This gives us one relation among the fields of the three waves.

The procedure of working out the consequences of Maxwell's equations at the boundary is called "determining the boundary conditions." Ordinarily, it is done by finding as many equations like Eq. (33.20) as one can, by making arguments about little rectangles like Γ in Fig. 33-4, or by using little gaussian surfaces that straddle the boundary. Although that is a perfectly good way of proceeding, it gives the impression that the problem of dealing with a boundary is different for every different physical problem.

For example, in a problem of heat flow across a boundary, how are the temperatures on the two sides related? Well, you could argue, for one thing, that the heat flow *to* the boundary from one side would have to equal the flow *away* from the other side. It is usually possible, and generally quite useful, to work out the boundary conditions by making such physical arguments. There may be times, however, when in working on some problem you have only some equations, and you may not see right away what physical arguments to use. So although we are at the moment interested only in an electromagnetic problem, where we *can* make the physical arguments, we want to show you a method that can be used for any problem—a *general* way of finding what happens at a boundary directly from the differential equations.

We begin by writing all the Maxwell equations for a dielectric—and this time we are very specific and write out explicitly all the components:

$$\begin{aligned} \nabla \cdot \mathbf{E} &= -\frac{\nabla \cdot \mathbf{P}}{\epsilon_0} \\ \epsilon_0 \left(\frac{\partial E_x}{\partial x} + \frac{\partial E_y}{\partial y} + \frac{\partial E_z}{\partial z} \right) &= - \left(\frac{\partial P_x}{\partial x} + \frac{\partial P_y}{\partial y} + \frac{\partial P_z}{\partial z} \right) \end{aligned} \quad (33.21)$$

$$\begin{aligned} \nabla \times \mathbf{E} &= -\frac{\partial \mathbf{B}}{\partial t} \\ \frac{\partial E_z}{\partial y} - \frac{\partial E_y}{\partial z} &= -\frac{\partial B_x}{\partial t} \end{aligned} \quad (33.22a)$$

$$\frac{\partial E_x}{\partial z} - \frac{\partial E_z}{\partial x} = -\frac{\partial B_y}{\partial t} \quad (33.22b)$$

$$\frac{\partial E_y}{\partial x} - \frac{\partial E_x}{\partial y} = -\frac{\partial B_z}{\partial t} \quad (33.22c)$$

$$\nabla \cdot \mathbf{B} = 0$$

$$\frac{\partial B_x}{\partial x} + \frac{\partial B_y}{\partial y} + \frac{\partial B_z}{\partial z} = 0 \quad (33.23)$$

$$\begin{aligned} c^2 \nabla \times \mathbf{B} &= \frac{1}{\epsilon_0} \frac{\partial \mathbf{P}}{\partial t} + \frac{\partial \mathbf{E}}{\partial t} \\ c^2 \left(\frac{\partial B_z}{\partial y} - \frac{\partial B_y}{\partial z} \right) &= \frac{1}{\epsilon_0} \frac{\partial P_x}{\partial t} + \frac{\partial E_x}{\partial t} \end{aligned} \quad (33.24a)$$

$$c^2 \left(\frac{\partial B_x}{\partial z} - \frac{\partial B_z}{\partial x} \right) = \frac{1}{\epsilon_0} \frac{\partial P_y}{\partial t} + \frac{\partial E_y}{\partial t} \quad (33.24b)$$

$$c^2 \left(\frac{\partial B_y}{\partial x} - \frac{\partial B_x}{\partial y} \right) = \frac{1}{\epsilon_0} \frac{\partial P_z}{\partial t} + \frac{\partial E_z}{\partial t} \quad (33.24c)$$

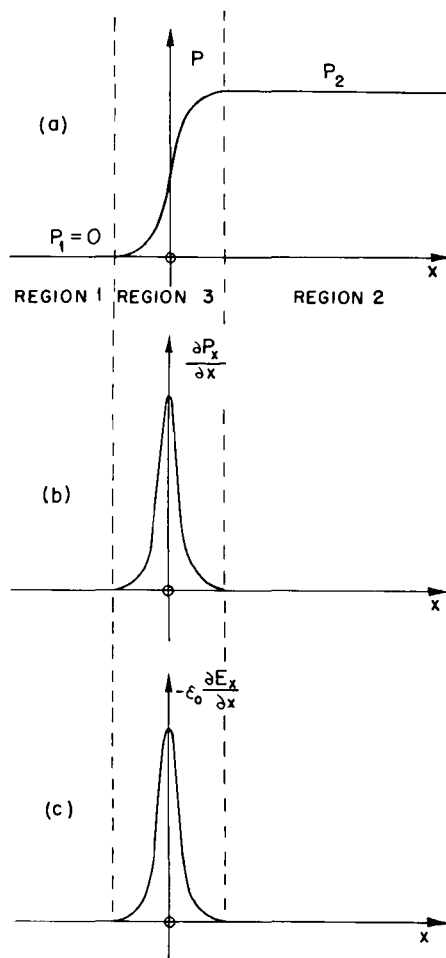


Fig. 33-5. The fields in the transition region (3) between two different materials in regions (1) and (2).

Now these equations must all hold in region 1 (to the left of the boundary) and in region 2 (to the right of the boundary). We have already written the solutions in regions 1 and 2. Finally, they must also be satisfied *in* the boundary, which we can call region 3. Although we usually think of the boundary as being sharply discontinuous, in reality it is not. The physical properties change very rapidly but not infinitely fast. In any case, we can imagine that there is a very rapid, but *continuous*, transition of the index between region 1 and 2, in a short distance we can call region 3. Also, any field quantity like P_x , or E_y , etc., will make a similar kind of transition in region 3. In this region, the differential equations must still be satisfied, and it is by following the differential equations in this region that we can arrive at the needed “boundary conditions.”

For instance, suppose that we have a boundary between vacuum (region 1) and glass (region 2). There is nothing to polarize in the vacuum, so $P_1 = 0$. Let's say there is some polarization P_2 in the glass. Between the vacuum and the glass there is a smooth, but rapid, transition. If we look at any component of P , say P_x , it might vary as drawn in Fig. 33-5(a). Suppose now we take the first of our equations, Eq. (33.21). It involves derivatives of the components of P with respect to x , y , and z . The y - and z -derivatives are not interesting; nothing spectacular is happening in those directions. But the x -derivative of P_x will have some very large values in region 3, because of the tremendous slope of P_x . The derivative $\partial P_x / \partial x$ will have a sharp spike at the boundary, as shown in Fig. 33-5(b). If we imagine squashing the boundary to an even thinner layer, the spike would get much higher. If the boundary is really sharp for the waves we are interested in, the magnitude of $\partial P_x / \partial x$ in region 3 will be much, much greater than any contributions we might have from the variation of P in the wave away from the boundary—so we ignore any variations other than those due to the boundary.

Now how can Eq. (33.21) be satisfied if there is a whopping big spike on the right-hand side? Only if there is an equally whopping big spike on the other side. Something on the left-hand side must also be big. The only candidate is $\partial E_x / \partial x$, because the variations with y and z are only those small effects in the wave we just mentioned. So $-\epsilon_0(\partial E / \partial x)$ must be as drawn in Fig. 33-5(c)—just a copy of $\partial P_x / \partial x$. We have that

$$\epsilon_0 \frac{\partial E_x}{\partial x} = - \frac{\partial P_x}{\partial x}.$$

If we integrate this equation with respect to x across region 3, we conclude that

$$\epsilon_0(E_{x2} - E_{x1}) = -(P_{x2} - P_{x1}). \quad (33.25)$$

In other words, the jump in $\epsilon_0 E_x$ in going from region 1 to region 2 must be equal to the jump in $-P_x$.

We can rewrite Eq. (33.25) as

$$\epsilon_0 E_{x2} + P_{x2} = \epsilon_0 E_{x1} + P_{x1}, \quad (33.26)$$

which says that the quantity $(\epsilon_0 E_x + P_x)$ has equal values in region 2 and region 1. People say: the quantity $(\epsilon_0 E_x + P_x)$ is *continuous* across the boundary. We have, in this way, one of our boundary conditions.

Although we took as an illustration the case in which P_1 was zero because region 1 was a vacuum, it is clear that the same argument applies for any two materials in the two regions, so Eq. (33.26) is true in general.

Let's now go through the rest of Maxwell's equations and see what each of them tells us. We take next Eq. (33.22a). There are no x -derivatives, so it doesn't tell us anything. (Remember that the fields *themselves* do not get especially large at the boundary; only the derivatives with respect to x can become so huge that they dominate the equation.) Next, we look at Eq. (33.22b). Ah! There is an x -derivative! We have $\partial E_z / \partial x$ on the left-hand side. Suppose it has a huge derivative. But wait a moment! There is nothing on the right-hand side to match it with; therefore E_z *cannot* have any jump in going from region 1 to region 2. [If it did, there would be a spike on the left of Eq. (33.22a) but none on the right,

and the equation would be false.] So we have a new condition:

$$E_{z2} = E_{z1}. \quad (33.27)$$

By the same argument, Eq. (33.22c) gives

$$E_{y2} = E_{y1}. \quad (33.28)$$

This last result is just what we got in Eq. (33.20) by a line integral argument.

We go on to Eq. (33.23). The only term that could have a spike is $\partial B_x / \partial x$. But there's nothing on the right to match it, so we conclude that

$$B_{x2} = B_{x1}. \quad (33.29)$$

On to the last of Maxwell's equations! Equation (33.24a) gives nothing, because there are no λ -derivatives. Equation (33.23b) has one, $-c^2 \partial B_z / \partial x$, but again, there is nothing to match it with. We get

$$B_{z2} = B_{z1}. \quad (33.30)$$

The last equation is quite similar, and gives

$$B_{y2} = B_{y1}. \quad (33.31)$$

The last three equations give us that $\mathbf{B}_2 = \mathbf{B}_1$. We want to emphasize, however, that we get this result only when the materials on both sides of the boundary are nonmagnetic—or rather, when we can neglect any magnetic effects of the materials. This can usually be done for most materials, except ferromagnetic ones. (We will treat the magnetic properties of materials in some later chapters.)

Our program has netted us the six relations between the fields in region 1 and those in region 2. We have put them all together in Table 33-1. We can now use them to match the waves in the two regions. We want to emphasize, however, that the idea we have just used will work in *any* physical situation in which you have differential equations and you want a solution that crosses a sharp boundary between two regions where some property changes. For our present purposes, we could have easily derived the same equations by using arguments about the fluxes and circulations at the boundary. (You might see whether you can get the same result that way.) But now you have seen a method that will work in case you ever get stuck and don't see any easy argument about the physics of what is happening at the boundary—you can just work with the equations.

Table 33-1

Boundary conditions at the surface of a dielectric

$$(\epsilon_0 \mathbf{E}_1 + \mathbf{P}_1)_x = (\epsilon_0 \mathbf{E}_2 + \mathbf{P}_2)_x$$

$$(\mathbf{E}_1)_y = (\mathbf{E}_2)_y$$

$$(\mathbf{E}_1)_z = (\mathbf{E}_2)_z$$

$$\mathbf{B}_1 = \mathbf{B}_2$$

(The surface is in the yz -plane)

33-4 The reflected and transmitted waves

Now we are ready to apply our boundary conditions to the waves we wrote down in Section 33-2. We had:

$$\mathbf{E}_i = E_0 e^{i(\omega t - k_x x - k_y y)}, \quad (33.32)$$

$$\mathbf{E}_r = E'_0 e^{i(\omega' t - k'_x x - k'_y y)}, \quad (33.33)$$

$$\mathbf{E}_t = E''_0 e^{i(\omega'' t - k''_x x - k''_y y)}, \quad (33.34)$$

$$\mathbf{B}_i = \frac{\mathbf{k} \times \mathbf{E}_i}{\omega}, \quad (33.35)$$

$$\mathbf{B}_r = \frac{\mathbf{k}' \times \mathbf{E}_r}{\omega'}, \quad (33.36)$$

$$\mathbf{B}_t = \frac{\mathbf{k}'' \times \mathbf{E}_t}{\omega''}. \quad (33.37)$$

We have one further bit of knowledge: $\mathbf{E}$ is perpendicular to its propagation vector $\mathbf{k}$ for each wave.

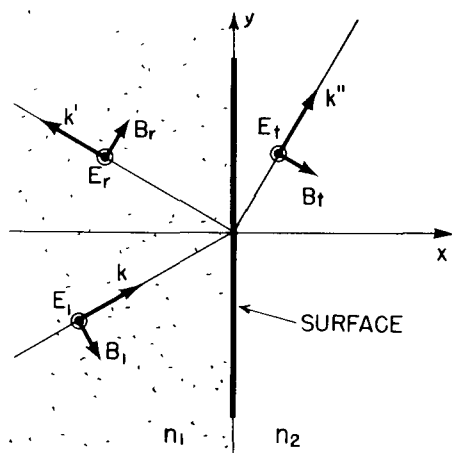


Fig. 33-6. Polarization of the reflected and transmitted waves when the E -field of the incident wave is perpendicular to the plane of incidence.

The results will depend on the direction of the E -vector (the “polarization”) of the incoming wave. The analysis is much simplified if we treat separately the case of an incident wave with its E -vector *parallel* to the “plane of incidence” (that is, the xy -plane) and the case of an incident wave with the E -vector *perpendicular* to the plane of incidence. A wave of any other polarization is just a linear combination of two such waves. In other words, the reflected and transmitted intensities are different for different polarizations, and it is easiest to pick the two simplest cases and treat them separately.

We will carry through the analysis for an incoming wave polarized perpendicular to the plane of incidence and then just give you the result for the other. We are cheating a little by taking the simplest case, but the principle is the same for both. So we take that E_i has only a z -component, and since all the E -vectors are in the same direction we can leave off the vector signs.

So long as both materials are isotropic, the induced oscillations of charges in the material will also be in the z -direction, and the E -field of the transmitted and radiated waves will have only z -components. So for all the waves, E_x and E_y and P_x and P_y are zero. The waves will have their E - and B -vectors as drawn in Fig. 33-6 (We are cutting a corner here on our original plan of getting everything from the equations. This result would also come out of the boundary conditions, but we can save a lot of algebra by using the physical argument. When you have some spare time, see if you can get the same result from the equations. It is clear that what we have said agrees with the equations; it is just that we have not shown that there are no *other* possibilities.)

Now our boundary conditions, Eqs. (33.26) through (33.31), give relations between the components of E and B in regions 1 and 2. For region 2 we have only the transmitted wave, but in region 1 we have *two* waves. Which one do we use? The fields in region 1 are, of course, the superposition of the fields of the incident and reflected waves. (Since each satisfies Maxwell’s equations, so does the sum.) So when we use the boundary conditions, we must use that

$$E_1 = E_i + E_r, \quad E_2 = E_t,$$

and similarly for the B ’s.

For the polarization we are considering, Eqs. (33.26) and (33.28) give us no new information; only Eq. (33.27) is useful. It says that

$$E_i + E_r = E_t$$

at the boundary, that is, for $x = 0$. So we have that

$$E_0 e^{i(\omega t - k_y y)} + E'_0 e^{i(\omega' t - k'_y y)} = E''_0 e^{i(\omega'' t - k''_y y)}, \quad (33.38)$$

which must be true for *all* t and for *all* y . Suppose we look first at $y = 0$. Then we have

$$E_0 e^{i\omega t} + E'_0 e^{i\omega' t} = E''_0 e^{i\omega'' t}$$

This equation says that two oscillating terms are equal to a third oscillation. That can happen only if all the oscillations have the same frequency. (It is impossible for three—or any number—of such terms with different frequencies to add to zero for all times.) So

$$\omega'' = \omega' = \omega. \quad (33.39)$$

As we knew all along, the frequencies of the reflected and transmitted waves are the same as that of the incident wave.

We should really have saved ourselves some trouble by putting that in at the beginning, but we wanted to show you that it can also be got out of the equations. When you are doing a real problem, it is usually the best thing to put everything you know into the works right at the start and save yourself a lot of trouble.

By definition, the *magnitude* of k is given by $k^2 = n^2 \omega^2 / c^2$, so we have also that

$$\frac{k''^2}{n_2^2} = \frac{k'^2}{n_1^2} = \frac{k^2}{n_1^2} \quad (33.40)$$

Now look at Eq. (33.38) for $t = 0$. Using again the same kind of argument we have just made, but this time based on the fact that the equation must hold for all values of y , we get that

$$k''_y = k'_y = k_y. \quad (33.41)$$

From Eq. (33.40), $k'^2 = k^2$, so

$$k'^2_x + k'^2_y = k^2_x + k^2_y.$$

Combining this with Eq. (33.41), we have that

$$k'^2_x = k^2_x,$$

or that $k'_x = \pm k_x$. The positive sign makes no sense; that would not give a *reflected* wave, but another *incident* wave, and we said at the start that we were solving the problem of only one incident wave. So we have

$$k'_x = -k_x. \quad (33.42)$$

The two equations (33.41) and (33.42) give us that the angle of reflection is equal to the angle of incidence, as we expected. (See Fig. 33-3) The reflected wave is

$$E_r = E'_0 e^{i(\omega t - k_x x + k_y y)}. \quad (33.43)$$

For the transmitted wave we already have that

$$k''_y = k_y,$$

and

$$\frac{k''^2}{n_2^2} = \frac{k^2}{n_1^2}; \quad (33.44)$$

so we can solve these to find k''_x . We get

$$k''^2_x = k''^2 - k''^2_y = \frac{n_2^2}{n_1^2} k^2 - k^2_y. \quad (33.45)$$

Suppose for a moment that n_1 and n_2 are real numbers (that the imaginary parts of the indexes are very small). Then all the k 's are also real numbers, and from Fig. 33-3 we find that

$$\frac{k_y}{k} = \sin \theta_i, \quad \frac{k''_y}{k''} = \sin \theta_t. \quad (33.46)$$

From (33.44) we get that

$$n_2 \sin \theta_t = n_1 \sin \theta_i, \quad (33.47)$$

which is Snell's law of refraction—again, something we already knew. If the indexes are not real, the wave numbers are complex, and we have to use Eq. (33.45). [We could still *define* the angles θ_i and θ_t by Eq. (33.46), and Snell's law, Eq. (33.47), would be true in general. But then the “angles” also are complex numbers, thereby losing their simple geometrical interpretation as angles. It is best then to describe the behavior of the waves by their complex k_x or k'_x values]

So far, we haven't found anything new. We have just had the simple-minded delight of getting some obvious answers from a complicated mathematical machinery. Now we are ready to find the amplitudes of the waves which we have not yet known. Using our results for the ω 's and k 's, the exponential factors in Eq. (33.38) can be cancelled, and we get

$$E_0 + E'_0 = E''_0. \quad (33.48)$$

Since both E'_0 and E''_0 are unknown, we need one more relationship. We must use another of the boundary conditions. The equations for E_x and E_y are no help, because all the E 's have only a z -component. So we must use the conditions on B . Let's try Eq. (33.29):

$$B_{x2} = B_{x1}.$$

From Eqs. (33.35) through (33.37),

$$B_{ri} = \frac{k_y E_i}{\omega}, \quad B_{ri} = \frac{k'_y E_i}{\omega'}, \quad B_{xt} = \frac{k''_y E_t}{\omega''}.$$

Recalling that $\omega'' = \omega' = \omega$ and $k''_y = k'_y = k_y$, we get that

$$E_0 + E'_0 = E''_0.$$

But this is just Eq. (33.48) all over again! We've just wasted time getting something we already knew.

We could try Eq. (33.30), $B_{z2} = B_{z1}$, but there are no z -components of $\mathbf{B}'$. So there's only one equation left: Eq. (33.31), $B_{y2} = B_{y1}$. For the three waves.

$$B_{yi} = -\frac{k_x E_i}{\omega}, \quad B_{yr} = -\frac{k'_x E_r}{\omega'}, \quad B_{yt} = -\frac{k''_x E_t}{\omega''}. \quad (33.49)$$

Putting for E_i , E_r , and E_t the wave expression for $x = 0$ (to be at the boundary), the boundary condition is

$$\frac{k_x}{\omega} E_0 e^{i(\omega t - k_y y)} + \frac{k'_x}{\omega'} E'_0 e^{i(\omega' t - k'_y y)} = \frac{k''_x}{\omega''} E''_0 e^{i(\omega'' t - k''_y y)}.$$

Again all ω 's and k_y 's are equal, so this reduces to

$$k_x E_0 + k'_x E'_0 = k''_x E''_0. \quad (33.50)$$

This gives us an equation for the E 's that is different from Eq. (33.48). With the two, we can solve for E'_0 and E''_0 . Remembering that $k'_x = -k_x$, we get

$$E'_0 = \frac{k_x - k''_x}{k_x + k''_x} E_0, \quad (33.51)$$

$$E''_0 = \frac{2k_x}{k_x + k''_x} E_0. \quad (33.52)$$

These, together with Eq. (33.45) or Eq. (33.46) for k''_x , give us what we wanted to know. We will discuss the consequences of this result in the next section.

If we begin with a wave polarized with its $\mathbf{E}$ -vector *parallel* to the plane of incidence, $\mathbf{E}$ will have both x - and y -components, as shown in Fig. 33-7. The algebra is straightforward but more complicated. (The work can be somewhat reduced by expressing things in this case in terms of the *magnetic* fields, which are all in the z -direction.) One finds that

$$|E'_0| = \frac{n_2^2 k_x - n_1^2 k''_x}{n_2^2 k_x + n_1^2 k''_x} |E_0| \quad (33.53)$$

and

$$|E''_0| = \frac{2n_1 n_2 k_x}{n_2^2 k_x + n_1^2 k''_x} |E_0|. \quad (33.54)$$

Let's see whether our results agree with those we got earlier. Equation (33.3) is the result we worked out in Chapter 35 of Volume I for the ratio of the intensity of the reflected wave to the intensity of the incident wave. Then, however, we were considering only *real* indexes. For real indexes (and k 's), we can write

$$k_x = k \cos \theta_i = \frac{\omega n_1}{c} \cos \theta_i,$$

$$k''_x = k'' \cos \theta_t = \frac{\omega n_2}{c} \cos \theta_t$$

Substituting in Eq. (33.51), we have

$$\frac{E'_0}{E_0} = \frac{n_1 \cos \theta_i - n_2 \cos \theta_t}{n_1 \cos \theta_i + n_2 \cos \theta_t}, \quad (33.55)$$

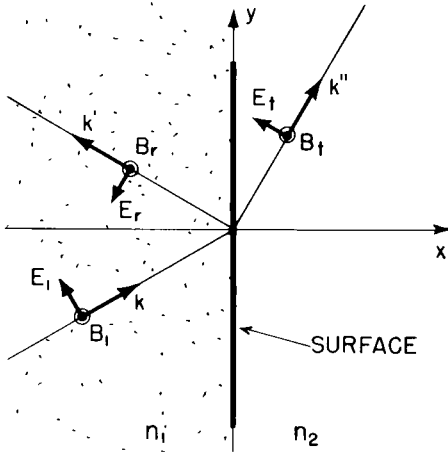


Fig. 33-7. Polarization of the waves when the $\mathbf{E}$ -field of the incident wave is parallel to the plane of incidence.

which does not look the same as Eq. (33.3). It will, however, if we use Snell's law to get rid of the n 's. Setting $n_2 = n_1 \sin \theta_i / \sin \theta_t$, and multiplying the numerator and denominator by $\sin \theta_t$, we get

$$\frac{E'_0}{E_0} = \frac{\cos \theta_i \sin \theta_t - \sin \theta_i \cos \theta_t}{\cos \theta_i \sin \theta_t + \sin \theta_i \cos \theta_t}.$$

The numerator and denominator are just the sines of $(\theta_i - \theta_t)$ and $(\theta_i + \theta_t)$; we get

$$\frac{E'_0}{E_0} = \frac{\sin(\theta_i - \theta_t)}{\sin(\theta_i + \theta_t)}. \quad (33.56)$$

Since E'_0 and E_0 are in the same material, the intensities are proportional to the squares of the electric fields, and we get the same result as before. Similarly, Eq. (33.53) is the same as Eq. (33.4).

For waves which arrive at normal incidence, $\theta_i = 0$ and $\theta_t = 0$. Equation (33.56) gives $0/0$, which is not very useful. We can, however, go back to Eq. (33.55), which gives

$$\frac{I_r}{I_i} = \left(\frac{E'_0}{E_0} \right)^2 = \left(\frac{n_1 - n_2}{n_1 + n_2} \right)^2. \quad (33.57)$$

This result, naturally, applies for "either" polarization, since for normal incidence there is no special "plane of incidence."

33-5 Reflection from metals

We can now use our results to understand the interesting phenomenon of reflection from metals. Why is it that metals are shiny? We saw in the last chapter that metals have an index of refraction which, for some frequencies, has a large imaginary part. Let's see what we would get for the reflected intensity when light shines from air (with $n = 1$) onto a material with $n = -in_I$. Then Eq. (33.55) gives (for normal incidence)

$$\frac{E'_0}{E_0} = \frac{1 + in_I}{1 - in_I}.$$

For the *intensity* of the reflected wave, we want the square of the absolute values of E'_0 and E_0 :

$$\frac{I_r}{I_i} = \frac{|E'_0|^2}{|E_0|^2} = \frac{|1 + in_I|^2}{|1 - in_I|^2},$$

or

$$\frac{I_r}{I_i} = \frac{1 + n_I^2}{1 + n_I^2} = 1. \quad (33.58)$$

For a material with an index which is a pure imaginary number, there is 100 percent reflection!

Metals do not reflect 100 percent, but many do reflect visible light very well. In other words, the imaginary part of their indexes is very large. But we have seen that a large imaginary part of the index means a strong absorption. So there is a general rule that if *any* material gets to be a *very* good absorber at any frequency, the waves are strongly reflected at the surface and very little gets inside to be absorbed. You can see this effect with strong dyes. Pure crystals of the strongest dyes have a "metallic" shine. Probably you have noticed that at the edge of a bottle of purple ink the dried dye will give a golden metallic reflection, or that dried red ink will sometimes give a greenish metallic reflection. Red ink absorbs out the greens of *transmitted* light, so if the ink is very concentrated, it will exhibit a strong surface *reflection* for the frequencies of green light.

You can easily show this effect by coating a glass plate with red ink and letting it dry. If you direct a beam of white light at the back of the plate, as shown in Fig. 33-8, there will be a transmitted beam of red light and a reflected beam of green light.

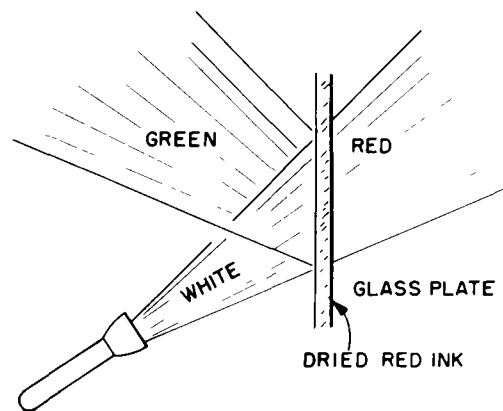


Fig. 33-8. A material which absorbs light strongly at the frequency ω also reflects light of that frequency.

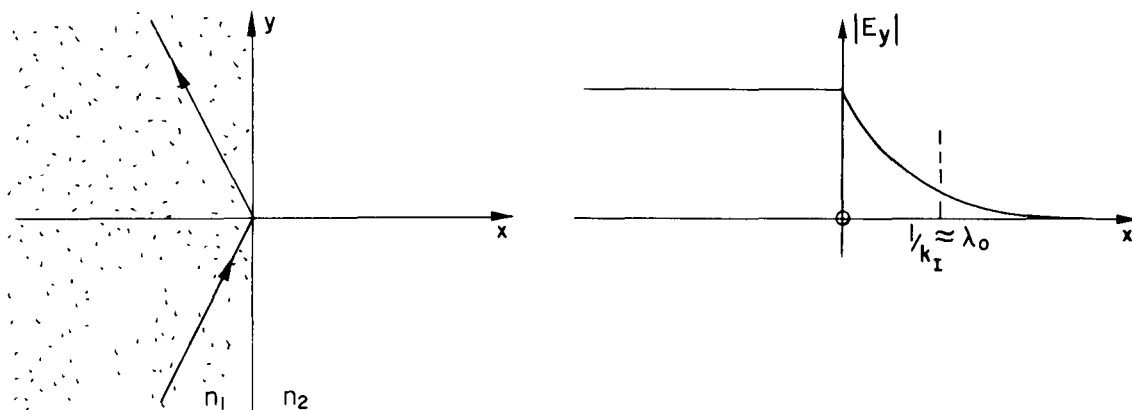


Fig. 33-9. Total internal reflection.

33-6 Total internal reflection

If light goes from a material like glass, with a real index n greater than 1, toward, say, air, with an index n_2 equal to 1, Snell's law says that

$$\sin \theta_t = n \sin \theta_i.$$

The angle θ_t of the transmitted wave becomes 90° when the incident angle θ_i is equal to the "critical angle" θ_c given by

$$n \sin \theta_c = 1. \quad (33.59)$$

What happens for θ_i greater than the critical angle? You know that there is total internal reflection. But how does that come about?

Let's go back to Eq (33.45) which gives the wave number k'' for the transmitted wave. We would have

$$k_x'^2 = \frac{k^2}{n^2} - k_y^2.$$

Now $k_y = k \sin \theta_i$ and $k = \omega n/c$, so

$$k_x'^2 = \frac{\omega^2}{c^2} (1 - n^2 \sin^2 \theta_i).$$

If $n \sin \theta_i$ is greater than one, $k_x'^2$ is *negative* and k_x' is a pure imaginary, say $\pm i k_I$. You know by now what that means! The "transmitted" wave (Eq. 33.34) will have the form

$$E_t = E_0'' e^{\pm k_I x} e^{i(\omega t - k_y y)}.$$

The wave amplitude either grows or drops off exponentially with increasing x . Clearly, what we want here is the negative sign. Then the *amplitude* of the wave to the right of the boundary will go as shown in Fig. 33-9. Notice that k_I is of the order ω/c —which is λ_0 , the free-space wavelength of the light. When light is totally reflected from the inside of a glass-air surface, there are fields in the air, but they extend beyond the surface only a distance of the order of the wavelength of the light

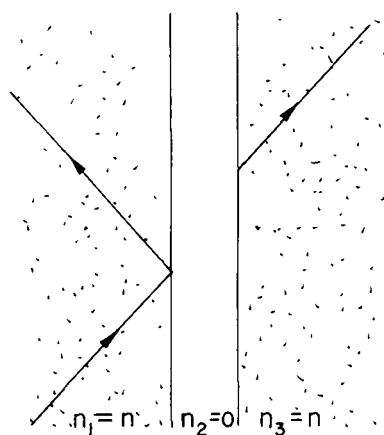


Fig. 33-10. If there is a small gap, internal reflection is not "total"; a transmitted wave appears beyond the gap.

We can now see how to answer the following question: If a light wave in glass arrives at the surface at a large enough angle, it is reflected, if another piece of glass is brought up to the surface (so that the "surface" in effect disappears) the light is transmitted. Exactly when does this happen? Surely there must be continuous change from total reflection to no reflection! The answer, of course, is that if the air gap is so small that the exponential tail of the wave in the air has an appreciable strength at the second piece of glass, it will shake the electrons there and generate a new wave, as shown in Fig. 33-10. Some light will be transmitted. (Clearly, our solution is incomplete, we should solve all the equations again for a thin layer of air between two regions of glass.)

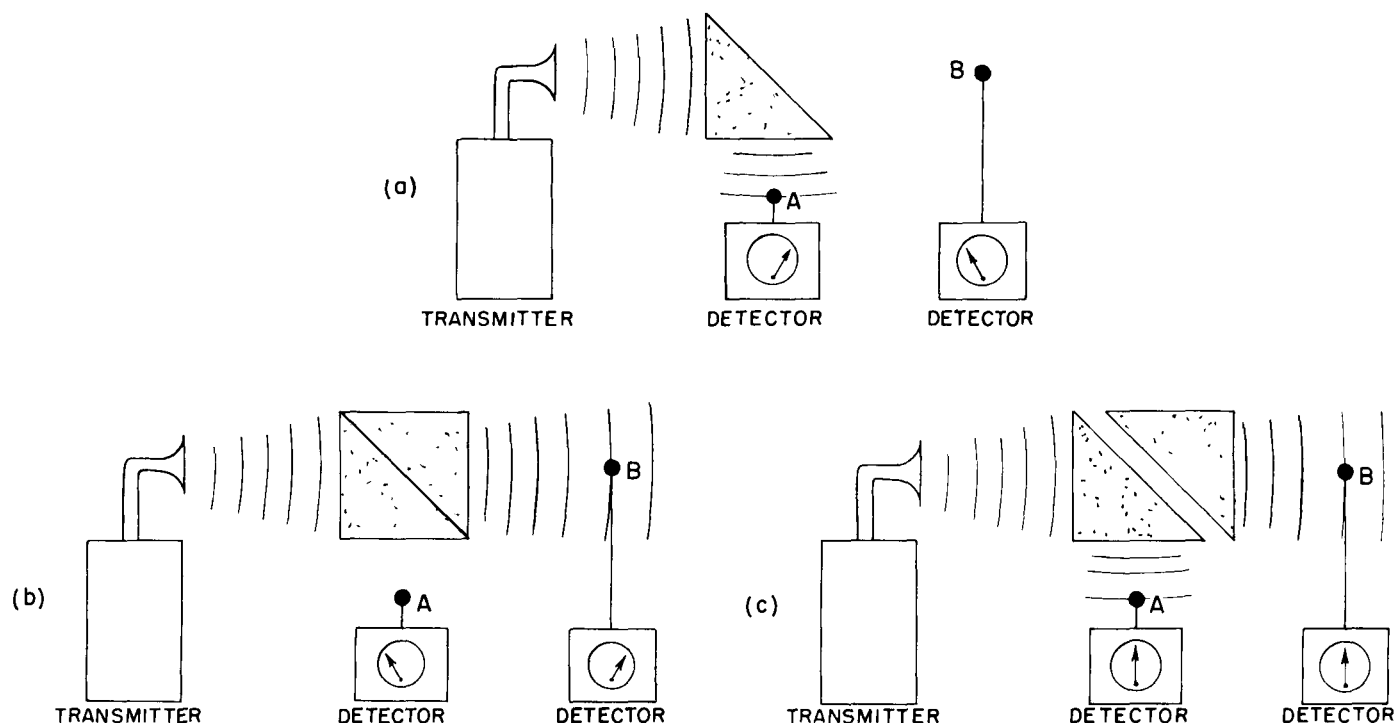


Fig. 33-11. A demonstration of the penetration of internally reflected waves.

This transmission effect can be observed with ordinary light only if the air gap is very small (of the order of the wavelength of light, like 10^{-5} cm), but it is easily demonstrated with three-centimeter waves. Then the exponentially decreasing field extends several centimeters. A microwave apparatus that shows the effect is drawn in Fig. 33-11. Waves from a small three-centimeter transmitter are directed at a 45° prism of paraffin. The index of refraction of paraffin for these frequencies is 1.50, and therefore the critical angle is 41.5° . So the wave is totally reflected from the 45° face and is picked up by detector *A*, as indicated in Fig. 33-11(a). If a second paraffin prism is placed in contact with the first, as shown in part (b) of the figure, the wave passes straight through and is picked up at detector *B*. If a gap of a few centimeters is left between the two prisms, as in part (c), there are both transmitted and reflected waves. The electric field outside the 45° face of the prism in Fig. 33-11(a) can also be shown by bringing detector *B* to within a few centimeters of the surface.

The Magnetism of Matter

34-1 Diamagnetism and paramagnetism

In this chapter we are going to talk about the magnetic properties of materials. The material which has the most striking magnetic properties is, of course, iron. Similar magnetic properties are shared also by the elements nickel, cobalt, and—at sufficiently low temperatures (below 16°C)—by gadolinium, as well as by a number of peculiar alloys. That kind of magnetism, called *ferromagnetism*, is sufficiently striking and complicated that we will discuss it in a special chapter. However, all ordinary substances do show some magnetic effects, although very small ones—a thousand to a million times less than the effects in ferromagnetic materials. Here we are going to describe ordinary magnetism, that is to say, the magnetism of substances other than the ferromagnetic ones.

This small magnetism is of two kinds. Some materials are *attracted* toward magnetic fields; others are *repelled*. Unlike the electrical effect in matter, which always causes dielectrics to be attracted, there are two signs to the magnetic effect. These two signs can be easily shown with the help of a strong electromagnet which has one sharply pointed pole piece and one flat pole piece, as drawn in Fig. 34-1. The magnetic field is much stronger near the pointed pole than near the flat pole. If a small piece of material is fastened to a long string and suspended between the poles, there will, in general, be a small force on it. This small force can be seen by the slight displacement of the hanging material when the magnet is turned on. The few ferromagnetic materials are attracted very strongly toward the pointed pole; all other materials feel only a very weak force. Some are weakly attracted to the pointed pole; and some are weakly repelled.

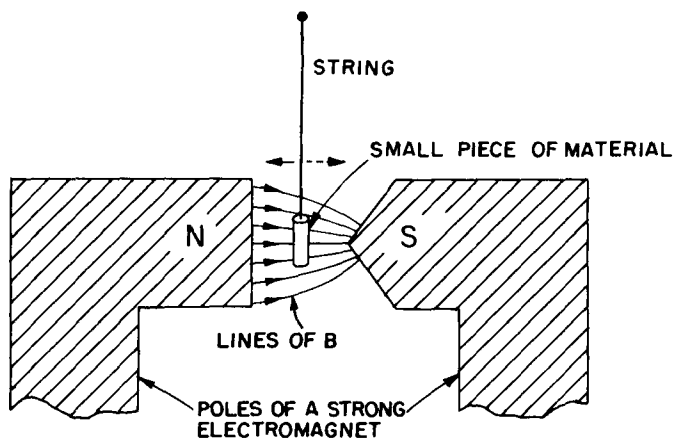


Fig. 34-1. A small cylinder of bismuth is weakly repelled by the sharp pole; a piece of aluminum is attracted.

The effect is most easily seen with a small cylinder of bismuth, which is *repelled* from the high-field region. Substances which are repelled in this way are called *diamagnetic*. Bismuth is one of the strongest diamagnetic materials, but even with it, the effect is still quite weak. Diamagnetism is always very weak. If a small piece of aluminum is suspended between the poles, there is also a weak force, but *toward* the pointed pole. Substances like aluminum are called *paramagnetic*. (In such an experiment, eddy-current forces arise when the magnet is turned on and off, and these can give off strong impulses. You must be careful to look for the net displacement after the hanging object settles down.)

34-1 Diamagnetism and paramagnetism

34-2 Magnetic moments and angular momentum

34-3 The precession of atomic magnets

34-4 Diamagnetism

34-5 Larmor's theorem

34-6 Classical physics gives neither diamagnetism nor paramagnetism

34-7 Angular momentum in quantum mechanics

34-8 The magnetic energy of atoms

Review: Section 15-1, "The forces on a current loop; energy of a dipole."

We want now to describe briefly the mechanisms of these two effects. First, in many substances the atoms have no permanent magnetic moments, or rather, all the magnets within each atom balance out so that the *net* moment of the atom is zero. The electron spins and orbital motions all exactly balance out, so that any particular atom has no average magnetic moment. In these circumstances, when you turn on a magnetic field little extra currents are generated inside the atom by induction. According to Lenz's law, these currents are in such a direction as to oppose the increasing field. So the induced magnetic moments of the atoms are directed *opposite* to the magnetic field. This is the mechanism of diamagnetism.

Then there are some substances for which the atoms do have a permanent magnetic moment—in which the electron spins and orbits have a net circulating current that is not zero. So besides the diamagnetic effect (which is always present), there is also the possibility of lining up the individual atomic magnetic moments. In this case, the moments try to line up *with* the magnetic field (in the way the permanent dipoles of a dielectric are lined up by the electric field), and the induced magnetism tends to enhance the magnetic field. These are the paramagnetic substances. Paramagnetism is generally fairly weak because the lining-up forces are relatively small compared with the forces from the thermal motions which try to derange the order. It also follows that paramagnetism is usually sensitive to the temperature. (The paramagnetism arising from the spins of the electrons responsible for conduction in a metal constitutes an exception. We will not be discussing this phenomenon here.) For ordinary paramagnetism, the lower the temperature, the stronger the effect. There is more lining-up at low temperatures when the deranging effects of the collisions are less. Diamagnetism, on the other hand, is more or less independent of the temperature. In any substance with built-in magnetic moments there is a diamagnetic as well as a paramagnetic effect, but the paramagnetic effect usually dominates.

In Chapter 11 we described a *ferroelectric* material, in which all the electric dipoles get lined up by their own mutual electric fields. It is also possible to imagine the magnetic analog of ferroelectricity, in which all the atomic moments would line up and lock together. If you make calculations of how this should happen, you will find that because the magnetic forces are so much smaller than the electric forces, thermal motions should knock out this alignment even at temperatures as low as a few tenths of a degree Kelvin. So it would be impossible at room temperature to have any permanent lining up of the magnets.

On the other hand, this is exactly what does happen in iron—it does get lined up. There is an effective force between the magnetic moments of the different atoms of iron which is much, much greater than the *direct magnetic* interaction. It is an indirect effect which can be explained only by quantum mechanics. It is about ten thousand times stronger than the direct magnetic interaction, and is what lines up the moments in ferromagnetic materials. We discuss this special interaction in a later chapter.

Now that we have tried to give you a qualitative explanation of diamagnetism and paramagnetism, we must correct ourselves and say that *it is not possible* to understand the magnetic effects of materials in any honest way from the point of view of classical physics. Such magnetic effects are a *completely quantum-mechanical phenomenon*. It is, however, possible to make some phoney classical arguments and to get some idea of what is going on. We might put it this way. You can make some classical arguments and get guesses as to the behavior of the material, but these arguments are not “legal” in any sense because it is absolutely essential that quantum mechanics be involved in every one of these magnetic phenomena. On the other hand, there are situations, such as in a plasma or a region of space with many free electrons, where the electrons do obey the laws of classical mechanics. And in those circumstances, some of the theorems from classical magnetism are worth while. Also, the classical arguments are of some value for historical reasons. The first few times that people were able to guess at the meaning and behavior of magnetic materials, they used classical arguments. Finally, as we have already illustrated, classical mechanics can give us some useful guesses

as to what might happen—even though the really honest way to study this subject would be to learn quantum mechanics first and then to understand the magnetism in terms of quantum mechanics.

On the other hand, we don't want to wait until we learn quantum mechanics inside out to understand a simple thing like diamagnetism. We will have to lean on the classical mechanics as kind of half showing what happens, realizing, however, that the arguments are really not correct. We therefore make a series of theorems about classical magnetism that will confuse you because they will prove different things. Except for the last theorem, every one of them will be wrong. Furthermore, they will all be wrong as a description of the physical world, because quantum mechanics is left out.

34-2 Magnetic moments and angular momentum

The first theorem we want to prove from classical mechanics is the following: If an electron is moving in a circular orbit (for example, revolving around a nucleus under the influence of a central force), there is a definite ratio between the magnetic moment and the angular momentum. Let's call $\mathbf{J}$ the angular momentum and $\boldsymbol{\mu}$ the magnetic moment of the electron in the orbit. The magnitude of the angular momentum is the mass of the electron times the velocity times the radius (See Fig. 34-2.) It is directed perpendicular to the plane of the orbit.

$$\mathbf{J} = m\mathbf{v}r. \quad (34.1)$$

(This is, of course, a nonrelativistic formula, but it is a good approximation for atoms, because for the electrons involved v/c is generally of the order of $e^2/\hbar c = 1/137$, or about 1 percent.)

The magnetic moment of the same orbit is the current times the area. (See Section 14-5.) The current is the charge per unit time which passes any point on the orbit, namely, the charge q times the frequency of rotation. The frequency is the velocity divided by the circumference of the orbit; so

$$I = q \frac{v}{2\pi r}.$$

The area is πr^2 , so the magnetic moment is

$$\boldsymbol{\mu} = \frac{qvr}{2}. \quad (34.2)$$

It is also directed perpendicular to the plane of the orbit. So $\mathbf{J}$ and $\boldsymbol{\mu}$ are in the same direction:

$$\boldsymbol{\mu} = \frac{q}{2m} \mathbf{J} \text{ (orbit)}. \quad (34.3)$$

Their ratio depends neither on the velocity nor on the radius. For any particle moving in a circular orbit the magnetic moment is equal to $q/2m$ times the angular momentum. For an electron, the charge is negative—we can call it $-q_e$; so for an electron

$$\boldsymbol{\mu} = -\frac{q_e}{2m} \mathbf{J} \text{ (electron orbit)}. \quad (34.4)$$

That's what we would expect classically and, miraculously enough, it is also true quantum-mechanically. It's one of those things. However, if you keep going with the classical physics, you find other places where it gives the wrong answers, and it is a great game to try to remember which things are right and which things are wrong. We might as well give you immediately what is true *in general* in quantum mechanics. First, Eq. (34.4) is true for *orbital motion*, but that's not the only magnetism that exists. The electron also has a spin rotation about its own axis (something like the earth rotating on its axis), and as a result of that spin it has both an angular momentum and a magnetic moment. But for reasons that are purely quantum-mechanical—there is no classical explanation—the ratio of $\boldsymbol{\mu}$

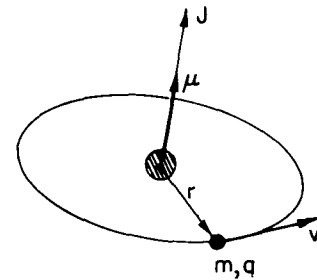


Fig. 34-2. For any circular orbit the magnetic moment $\boldsymbol{\mu}$ is $q/2m$ times the angular momentum $\mathbf{J}$.

to $\mathbf{J}$ for the electron spin is twice as large as it is for orbital motion of the spinning electron:

$$\boldsymbol{\mu} = -\frac{q_e}{m} \mathbf{J} \text{ (electron spin).} \quad (34.5)$$

In any atom there are, generally speaking, several electrons and some combination of spin and orbit rotations which builds up a total angular momentum and a total magnetic moment. Although there is no classical reason why it should be so, it is *always true* in quantum mechanics that (for an isolated atom) the direction of the magnetic moment is exactly opposite to the direction of the angular momentum. The ratio of the two is not necessarily either $-q_e/m$ or $-q_e/2m$, but somewhere in between, because there is a mixture of the contributions from the orbits and the spins. We can write

$$\boldsymbol{\mu} = -g \left(\frac{q_e}{2m} \right) \mathbf{J}, \quad (34.6)$$

where g is a factor which is characteristic of the state of the atom. It would be 1 for a pure orbital moment, or 2 for a pure spin moment, or some other number in between for a complicated system like an atom. This formula does not, of course, tell us very much. It says that the magnetic moment is *parallel* to the angular momentum, but can have any magnitude. The form of Eq. (34.6) is convenient, however, because g —called the “Landé g -factor”—is a dimensionless constant whose magnitude is of the order of one. It is one of the jobs of quantum mechanics to predict the g -factor for any particular atomic state.

You might also be interested in what happens in nuclei. In nuclei there are protons and neutrons which may move around in some kind of orbit and at the same time, like an electron, have an intrinsic spin. Again the magnetic moment is parallel to the angular momentum. Only now the order of magnitude of the ratio of the two is what you would expect for a *proton* going around in a circle, with m in Eq. (34.3) equal to the *proton* mass. Therefore it is usual to write for nuclei

$$\boldsymbol{\mu} = g \left(\frac{q_e}{2m_p} \right) \mathbf{J}, \quad (34.7)$$

where m_p is the mass of the proton, and g —called the *nuclear* g -factor—is a number near one, to be determined for each nucleus.

Another important difference for a nucleus is that the *spin* magnetic moment of the proton does *not* have a g -factor of 2, as the electron does. For a proton, $g = 2(2.79)$. Surprisingly enough, the *neutron* also has a spin magnetic moment, and its magnetic moment relative to its angular momentum is $2(-1.93)$. The neutron, in other words, is not exactly “neutral” in the magnetic sense. It is like a little magnet, and it has the kind of magnetic moment that a rotating *negative* charge would have.

34-3 The precession of atomic magnets

One of the consequences of having the magnetic moment proportional to the angular momentum is that an atomic magnet placed in a magnetic field will *precess*. First we will argue classically. Suppose that we have the magnetic moment $\boldsymbol{\mu}$ suspended freely in a uniform magnetic field. It will feel a torque $\boldsymbol{\tau}$, equal to $\boldsymbol{\mu} \times \mathbf{B}$, which tries to bring it in line with the field direction. But the atomic magnet is a gyroscope—it has the angular momentum $\mathbf{J}$. Therefore the torque due to the magnetic field will not cause the magnet to line up. Instead, the magnet will *precess*, as we saw when we analyzed a gyroscope in Chapter 20 of Volume I. The angular momentum—and with it the magnetic moment—precesses about an axis parallel to the magnetic field. We can find the rate of precession by the same method we used in Chapter 20 of the first volume.

Suppose that in a small time Δt the angular momentum changes from $\mathbf{J}$ to $\mathbf{J}'$, as drawn in Fig. 34-3, staying always at the same angle θ with respect to the direction of the magnetic field $\mathbf{B}$. Let's call ω_p the angular velocity of the precession, so that in the time Δt the angle of *precession* is $\omega_p \Delta t$. From the geometry of the

figure, we see that the change of angular momentum in the time Δt is

$$\Delta J = (J \sin \theta)(\omega_p \Delta t).$$

So the rate of change of the angular momentum is

$$\frac{dJ}{dt} = \omega_p J \sin \theta, \quad (34.8)$$

which must be equal to the torque:

$$\tau = \mu B \sin \theta. \quad (34.9)$$

The angular velocity of precession is then

$$\omega_p = \frac{\mu}{J} B. \quad (34.10)$$

Substituting μ/J from Eq. (34.6), we see that for an atomic system

$$\omega_p = g \frac{q_e B}{2m}, \quad (34.11)$$

the precession frequency is proportional to B . It is handy to remember that for an atom (or electron)

$$f_p = \frac{\omega_p}{2\pi} = (1.4 \text{ megacycles/gauss})gB, \quad (34.12)$$

and that for a nucleus

$$f_p = \frac{\omega_p}{2\pi} = (0.76 \text{ kilocycles/gauss})gB. \quad (34.13)$$

(The formulas for atoms and nuclei are different only because of the different conventions for g for the two cases.)

According to the *classical* theory, then, the electron orbits—and spins—in an atom should precess in a magnetic field. Is it also true quantum-mechanically? It is essentially true, but the meaning of the “precession” is different. In quantum mechanics one cannot talk about the *direction* of the angular momentum in the same sense as one does classically, nevertheless, there is a very close analogy—so close that we continue to call it “precession.” We will discuss it later when we talk about the quantum-mechanical point of view.

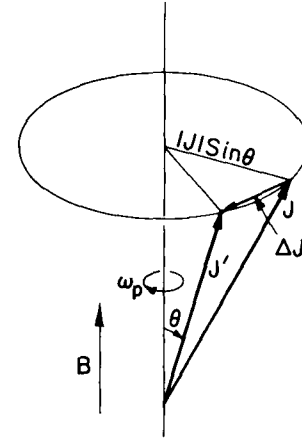


Fig. 34-3. An object with angular momentum J and a parallel magnetic moment μ placed in a magnetic field B precesses with the angular velocity ω_p .

34-4 Diamagnetism

Next we want to look at *diamagnetism* from the classical point of view. It can be worked out in several ways, but one of the nice ways is the following. Suppose that we slowly turn on a magnetic field in the vicinity of an atom. As the magnetic field changes an *electric* field is generated by magnetic induction. From Faraday's law, the line integral of E around any closed path is the rate of change of the magnetic flux through the path. Suppose we pick a path Γ which is a circle of radius r concentric with the center of the atom, as shown in Fig. 34-4. The average tangential electric field E around this path is given by

$$E 2\pi r = - \frac{d}{dt} (B \pi r^2),$$

and there is a circulating electric field whose strength is

$$E = - \frac{r}{2} \frac{dB}{dt}.$$

The induced electric field acting on an electron in the atom produces a torque equal to $-q_e E r$, which must equal the rate of change of the angular momentum dJ/dt :

$$\frac{dJ}{dt} = \frac{q_e r^2}{2} \frac{dB}{dt}. \quad (34.14)$$

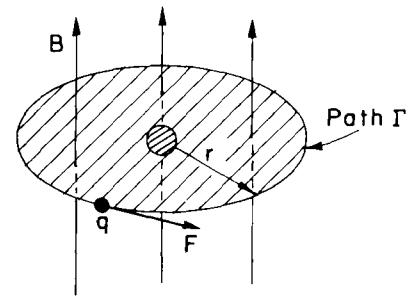


Fig. 34-4. The induced electric forces on the electrons in an atom.

Integrating with respect to time from zero field, we find that the change in angular momentum due to turning on the field is

$$\Delta J = \frac{q_e r^2}{2} B. \quad (34.15)$$

This is the extra angular momentum from the twist given to the electrons as the field is turned on.

This added angular momentum makes an extra magnetic moment which, because it is an *orbital* motion, is just $-q_e/2m$ times the angular momentum. The induced diamagnetic moment is

$$\Delta\mu = -\frac{q_e}{2m} \Delta J = -\frac{q_e^2 r^2}{4m} B. \quad (34.16)$$

The minus sign (as you can see is right by using Lenz's law) means that the added moment is opposite to the magnetic field.

We would like to write Eq. (34.16) a little differently. The r^2 which appears is the radius from an axis through the atom parallel to B , so if B is along the z -direction, it is $x^2 + y^2$. If we consider spherically symmetric atoms (or average over atoms with their natural axes in all directions) the average of $x^2 + y^2$ is $2/3$ of the average of the square of the true radial distance from the center *point* of the atom. It is therefore usually more convenient to write Eq. (34.16) as

$$\Delta\mu = -\frac{q_e^2}{6m} \langle r^2 \rangle_{av} B. \quad (34.17)$$

In any case, we have found an induced atomic moment proportional to the magnetic field B and opposing it. This is diamagnetism of matter. It is this magnetic effect that is responsible for the small force on a piece of bismuth in a nonuniform magnetic field. (You could compute the force by working out the energy of the induced moments in the field and seeing how the energy changes as the material is moved into or out of the high-field region.)

We are still left with the problem: What is the mean square radius, $\langle r^2 \rangle_{av}$? Classical mechanics cannot supply an answer. We must go back and start over with quantum mechanics. In an atom we cannot really say where an electron is, but only know the probability that it will be at some place. If we interpret $\langle r^2 \rangle_{av}$ to mean the average of the square of the distance from the center for the probability distribution, the diamagnetic moment given by quantum mechanics is just the same as formula (34.17). This equation, of course, is the moment for one electron. The total moment is given by the sum over all the electrons in the atom. The surprising thing is that the classical argument and quantum mechanics give the same answer, although, as we shall see, the classical argument that gives Eq. (34.17) is not really valid in classical mechanics.

The same diamagnetic effect occurs even when an atom already has a permanent moment. Then the system will precess in the magnetic field. As the whole atom precesses, it takes up an additional small angular velocity, and that slow turning gives a small current which represents a correction to the magnetic moment. This is just the diamagnetic effect represented in another way. But we don't really have to worry about that when we talk about paramagnetism. If the diamagnetic effect is first computed, as we have done here, we don't have to worry about the fact that there is an extra little current from the precession. That has already been included in the diamagnetic term.

34-5 Larmor's theorem

We can already conclude something from our results so far. First of all, in the classical theory the moment μ was always proportional to J , with a given constant of proportionality for a particular atom. There wasn't any spin of the electrons, and the constant of proportionality was always $-q_e/2m$; that is to say, in Eq. (34.6) we should set $g = 1$. The ratio of μ to J was independent of the internal motion of the electrons. Thus, according to the classical theory, all systems

of electrons would precess with *the same* angular velocity. (This is *not* true in quantum mechanics.) This result is related to a theorem in classical mechanics that we would now like to prove. Suppose we have a group of electrons which are all held together by attraction toward a central point—as the electrons are attracted by a nucleus. The electrons will also be interacting with each other, and can, in general, have complicated motions. Suppose you have solved for the motions with *no* magnetic field and then want to know what the motions would be *with* a weak magnetic field. The theorem says that the motion with a weak magnetic field is always one of the no-field solutions with an added rotation, about the axis of the field, with the angular velocity $\omega_L = q_e B / 2m$. (This is the same as ω_p , if $g = 1$.) There are, of course, many possible motions. The point is that for every motion without the magnetic field there is a corresponding motion in the field, which is the original motion plus a uniform rotation. This is called Larmor's theorem, and ω_L is called the *Larmor frequency*.

We would like to show how the theorem can be proved, but we will let you work out the details. Take, first, one electron in a central force field. The force on it is just $F(r)$, directed toward the center. If we now turn on a uniform magnetic field, there is an additional force, $qv \times B$; so the total force is

$$F(r) + qv \times B. \quad (34.18)$$

Now let's look at the same system from a coordinate system rotating with angular velocity ω about an axis through the center of force and parallel to B . This is no longer an inertial system, so we have to put in the proper pseudoforces—the centrifugal and Coriolis forces we talked about in Chapter 19 of Volume I. We found there that in a frame rotating with angular velocity ω , there is an apparent *tangential* force proportional to v_r , the radial component of velocity:

$$F_t = -2m\omega v_r, \quad (34.19)$$

And there is an apparent radial force which is given by

$$F_r = m\omega^2 r + 2m\omega v_t, \quad (34.20)$$

where v_t is the tangential component of the velocity, measured *in* the rotating frame. (The radial component v_r for rotating and inertial frames is the same.)

Now for small enough angular velocities (that is, if $\omega r \ll v_t$), we can neglect the first term (centrifugal) in Eq. (34.20) in comparison with the second (Coriolis). Then Eqs. (34.19) and (34.20) can be written together as

$$F = -2m\omega \times v. \quad (34.21)$$

If we now *combine* a rotation and a magnetic field, we must add the force in Eq. (34.21) to that in Eq. (34.18). The total force is

$$F(r) + qv \times B + 2mv \times \omega \quad (34.22)$$

[we reverse the cross product and the sign of Eq. (34.21) to get the last term]. Looking at our result, we see that if

$$2m\omega = -qB$$

the two terms on the right cancel, and in the moving frame the only force is $F(r)$. The motion of the electron is just the same as with no magnetic field—and, of course, no rotation. We have proved Larmor's theorem for one electron. Since the proof assumes a small ω , it also means that the theorem is true only for weak magnetic fields. The only thing we could ask you to improve on is to take the case of many electrons mutually interacting with each other, but all in the same central field, and prove the same theorem. So no matter how complex an atom is, if it has a central field the theorem is true. But that's the end of the classical mechanics, because it isn't true in fact that the motions precess in that way. The precession frequency ω_p of Eq. (34.11) is only equal to ω_L if g happens to be equal to 1.

34-6 Classical physics gives neither diamagnetism nor paramagnetism

Now we would like to demonstrate that according to classical mechanics there can be no diamagnetism and no paramagnetism at all. It sounds crazy—first, we have proved that there are paramagnetism, diamagnetism, precessing orbits, and so on, and now we are going to prove that it is all wrong. Yes!—We are going to prove that *if you follow the classical mechanics far enough*, there are no such magnetic effects—they all cancel out. If you start a classical argument in a certain place and don't go far enough, you can get any answer you want. But the only legitimate and correct proof shows that there is no magnetic effect whatever.

It is a consequence of classical mechanics that if you have any kind of system—a gas with electrons, protons, and whatever—kept in a box so that the whole thing can't turn, there will be no magnetic effect. It is possible to have a magnetic effect if you have an isolated system, like a star held together by itself, which can start rotating when you put on the magnetic field. But if you have a piece of material that is held in place so that it can't start spinning, then there will be no magnetic effects. What we mean by holding down the spin is summarized this way: At a given temperature we suppose that there is *only one state* of thermal equilibrium. The theorem then says that if you turn on a magnetic field and wait for the system to get into thermal equilibrium, there will be no paramagnetism or diamagnetism—there will be no induced magnetic moment. Proof: According to statistical mechanics, the probability that a system will have any given state of motion is proportional to $e^{-U/kT}$, where U is the energy of that motion. Now what is the energy of motion? For a particle moving in a constant magnetic field, the energy is the ordinary potential energy plus $mv^2/2$, with nothing additional for the magnetic field. [You know that the forces from electromagnetic fields are $q(\mathbf{E} + \mathbf{v} \times \mathbf{B})$, and that the rate of work $\mathbf{F} \cdot \mathbf{v}$ is just $q\mathbf{E} \cdot \mathbf{v}$, which is not affected by the magnetic field.] So the energy of a system, whether it is in a magnetic field or not, is always given by the kinetic energy plus the potential energy. Since the probability of any motion depends only on the energy—that is, on the velocity and position—it is the same whether or not there is a magnetic field. For *thermal* equilibrium, therefore, the magnetic field has no effect. If we have one system in a box, and then have another system in a second box, this time with a magnetic field, the probability of any particular velocity at any point in the first box is the same as in the second. If the first box has no average circulating current (which it will not have if it is in equilibrium with the stationary walls), there is no average magnetic moment. Since in the second box all the motions are the same, there is no average magnetic moment there either. Hence, if the temperature is kept constant and thermal equilibrium is re-established after the field is turned on, there can be no magnetic moment induced by the field—according to classical mechanics. We can only get a satisfactory understanding of magnetic phenomena from quantum mechanics.

Unfortunately, we cannot assume that you have a thorough understanding of quantum mechanics, so this is hardly the place to discuss the matter. On the other hand, we don't always have to learn something first by learning the exact rules and then by learning how they are applied in different cases. Almost every subject that we have taken up in this course has been treated in a different way. In the case of electricity, we wrote the Maxwell equations on "Page One" and then deduced all the consequences. That's one way. But we will *not* now try to begin a new "Page One," writing the equations of quantum mechanics and deducing everything from them. We will just have to tell you some of the consequences of quantum mechanics, before you learn where they come from. So here we go.

34-7 Angular momentum in quantum mechanics

We have already given you a relation between the magnetic moment and the angular momentum. That's pleasant. But what do the magnetic moment and the angular momentum *mean* in quantum mechanics? In quantum mechanics it turns out to be best to define things like magnetic moments in terms of the other concepts such as energy, in order to make sure that one knows what it means. Now,

it is easy to define a magnetic moment in terms of energy, because the energy of a moment in a magnetic field is, in the classical theory, $\boldsymbol{\mu} \cdot \mathbf{B}$. Therefore, the following definition has been taken in quantum mechanics: If we calculate the energy of a system in a magnetic field and we find that it is proportional to the field strength (for small field), the coefficient is called the component of magnetic moment in the direction of the field. (We don't have to get so elegant for our work now, we can still think of the magnetic moment in the ordinary, to some extent classical, sense.)

Now we would like to discuss the idea of angular momentum in quantum mechanics—or rather, the characteristics of what, in quantum mechanics, is called angular momentum. You see, when you go to new kinds of laws, you can't just assume that each word is going to mean exactly the same thing. You may think, say, “Oh, I know what angular momentum is. It's that thing that is changed by a torque.” But what's a torque? In quantum mechanics we have to have new definitions of old quantities. It would, therefore, be legally best to call it by some other name such as “quantangular momentum,” or something like that, because it is the angular momentum as defined in quantum mechanics. But if we can find a quantity in quantum mechanics which is identical to our old idea of angular momentum when the system becomes large enough, there is no use in inventing an extra word. We might as well just call it angular momentum. With that understanding, this odd thing that we are about to describe *is* angular momentum. It is the thing which in a large system we recognize as angular momentum in classical mechanics.

First, we take a system in which angular momentum is conserved, such as an atom all by itself in empty space. Now such a thing (like the earth spinning on its axis) could, in the ordinary sense, be spinning around any axis one wished to choose. And for a given spin, there could be many different “states,” all of the same energy, each “state” corresponding to a particular direction of the axis of the angular momentum. So in the classical theory, with a given angular momentum, there is an infinite number of possible states, all of the same energy.

It turns out in quantum mechanics, however, that several strange things happen. First, the number of states in which such a system *can exist* is limited—there is only a finite number. If the system is small, the finite number is very small, and if the system is large, the finite number gets very, very large. Second, we *cannot* describe a “state” by giving the *direction* of its angular momentum, but only by giving the *component* of the angular momentum along some direction—say in the *z*-direction. Classically, an object with a given total angular momentum J could have, for its *z*-component, any value from $+J$ to $-J$. But quantum-mechanically, the *z*-component of angular momentum can have only certain discrete values. Any given system—a particular atom, or a nucleus, or anything—with a given energy, has a characteristic number j , and its *z*-component of angular momentum can only be one of the following set of values:

$$\begin{array}{l} j\hbar \\ (j-1)\hbar \\ (j-2)\hbar \\ \vdots \\ -(j-2)\hbar \\ -(j-1)\hbar \\ -j\hbar \end{array} \quad (34.23)$$

The largest *z*-component is j times $\hbar$; the next smaller is one unit of $\hbar$ less, and so on down to $-j\hbar$. The number j is called “the spin of the system.” (Some people call it the “total angular momentum quantum number”; but we'll call it the “spin.”)

You may be worried that what we are saying can only be true for some “special” *z*-axis. But that is not so. For a system whose spin is j , the component of angular momentum along *any* axis can have only one of the values in (34.23). Although it is quite mysterious, we ask you just to accept it for the moment. We

will come back and discuss the point later. You may at least be pleased to hear that the z -component goes from some number to minus the *same* number, so that we at least don't have to decide which is the plus direction of the z -axis. (Certainly, if we said that it went from $+j$ to minus a different amount, that would be infinitely mysterious, because we wouldn't have been able to define the z -axis, pointing the other way.)

Now if the z -component of angular momentum must go down by integers from $+j$ to $-j$, then j must be an integer. No! Not quite; twice j must be an integer. It is only the *difference* between $+j$ and $-j$ that must be an integer. So, in general, the spin j is either an integer or a half-integer, depending on whether $2j$ is even or odd. Take, for instance, a nucleus like lithium, which has a spin of three-halves, $j = 3/2$. Then the angular momentum around the z -axis, in units of $\hbar$, is one of the following:

$$\begin{aligned} &+3/2 \\ &+1/2 \\ &-1/2 \\ &-3/2. \end{aligned}$$

There are four possible states, each of the same energy, if the nucleus is in empty space with no external fields. If we have a system whose spin is two, then the z -component of angular momentum has only the values, in units of $\hbar$,

$$\begin{aligned} &2 \\ &1 \\ &0 \\ &-1 \\ &-2. \end{aligned}$$

If you count how many states there are for a given j , there are $(2j + 1)$ possibilities. In other words, if you tell me the energy and also the spin j , it turns out that there are exactly $(2j + 1)$ states with that energy, each state corresponding to one of the different possible values of the z -component of the angular momentum.

We would like to add one other fact. If you pick out any atom of known j at random and measure the z -component of the angular momentum, then you may get any one of the possible values, and each of the values is *equally* likely. All of the states are in fact single states, and each is just as good as any other. Each one has the same "weight" in the world. (We are assuming that nothing has been done to sort out a special sample.) This fact has, incidentally, a simple classical analog. If you ask the same question classically: What is the likelihood of a particular z -component of angular momentum if you take a random sample of systems, all with the same total angular momentum?—the answer is that all values from the maximum to the minimum are equally likely. (You can easily work that out.) The classical result corresponds to the equal probability of the $(2j + 1)$ possibilities in quantum mechanics.

From what we have so far, we can get another interesting and somewhat surprising conclusion. In certain classical calculations the quantity that appears in the final result is the *square* of the magnitude of the angular momentum $\mathbf{J}$ —in other words, $\mathbf{J} \cdot \mathbf{J}$. It turns out that it is often possible to *guess* at the correct quantum-mechanical formula by using the classical calculation and the following simple rule: Replace $J^2 = \mathbf{J} \cdot \mathbf{J}$ by $j(j + 1)\hbar^2$. This rule is commonly used, and usually gives the correct result, but *not* always. We can give the following argument to show why you might expect this rule to work.

The scalar product $\mathbf{J} \cdot \mathbf{J}$ can be written as

$$\mathbf{J} \cdot \mathbf{J} = J_x^2 + J_y^2 + J_z^2.$$

Since it is a scalar, it should be the same for any orientation of the spin. Suppose we pick samples of any given atomic system at random and make measurements of J_x^2 , or J_y^2 , or J_z^2 , the *average value* should be the same for each. (There is no special distinction for any one of the directions.) Therefore, the average of $\mathbf{J} \cdot \mathbf{J}$ is just

equal to three times the average of any component squared, say of J_z^2 ;

$$\langle \mathbf{J} \cdot \mathbf{J} \rangle_{\text{av}} = 3 \langle J_z^2 \rangle.$$

But since $\mathbf{J} \cdot \mathbf{J}$ is the same for all orientations, its average is, of course, just its constant value; we have

$$\mathbf{J} \cdot \mathbf{J} = 3 \langle J_z^2 \rangle_{\text{av}}. \quad (34.24)$$

If we now say that we will use the same equation for quantum mechanics, we can easily find $\langle J_z^2 \rangle_{\text{av}}$. We just have to take the sum of the $(2j + 1)$ possible values of J_z^2 , and divide by the total number;

$$\langle J_z^2 \rangle_{\text{av}} = \frac{j^2 + (j-1)^2 + \cdots + (-j+1)^2 + (-j)^2}{2j+1} \hbar^2. \quad (34.25)$$

For a system with a spin of $3/2$, it goes like this:

$$\langle J_z^2 \rangle_{\text{av}} = \frac{(3/2)^2 + (1/2)^2 + (-1/2)^2 + (-3/2)^2}{4} \hbar^2 = \frac{5}{4} \hbar^2.$$

We conclude that

$$\mathbf{J} \cdot \mathbf{J} = 3 \langle J_z^2 \rangle_{\text{av}} = 3 \frac{5}{4} \hbar^2 = \frac{3}{2} \left(\frac{3}{2} + 1 \right) \hbar^2.$$

We will leave it for you to show that Eq. (34.25), together with Eq. (34.24), gives the general result

$$\mathbf{J} \cdot \mathbf{J} = j(j+1) \hbar^2. \quad (34.26)$$

Although we would think classically that the largest possible value of the z -component of $\mathbf{J}$ is just the magnitude of $\mathbf{J}$ —namely, $\sqrt{\mathbf{J} \cdot \mathbf{J}}$ —quantum mechanically the maximum of J_z is always a little less than that, because $j\hbar$ is always less than $\sqrt{j(j+1)} \hbar$. The angular momentum is never “completely along the z -direction.”

34-8 The magnetic energy of atoms

Now we want to talk again about the magnetic moment. We have said that in quantum mechanics the magnetic moment of a particular atomic system can be written in terms of the angular momentum by Eq. (34.6);

$$\boldsymbol{\mu} = -g \left(\frac{q_e}{2m} \right) \mathbf{J}, \quad (34.27)$$

where $-q_e$ and m are the charge and mass of the electron.

An atomic magnet placed in an external magnetic field will have an extra magnetic energy which depends on the component of its magnetic moment along the field direction. We know that

$$U_{\text{mag}} = -\boldsymbol{\mu} \cdot \mathbf{B}. \quad (34.28)$$

Choosing our z -axis along the direction of $\mathbf{B}$,

$$U_{\text{mag}} = -\mu_z B. \quad (34.29)$$

Using Eq. (34.27), we have that

$$U_{\text{mag}} = g \left(\frac{q_e}{2m} \right) J_z B.$$

Quantum mechanics says that J_z can have only certain values: $j\hbar, (j-1)\hbar, \dots, -j\hbar$. Therefore, the magnetic energy of an atomic system is not arbitrary; it can have only certain values. Its maximum value, for instance, is

$$g \left(\frac{q_e}{2m} \right) \hbar j B.$$

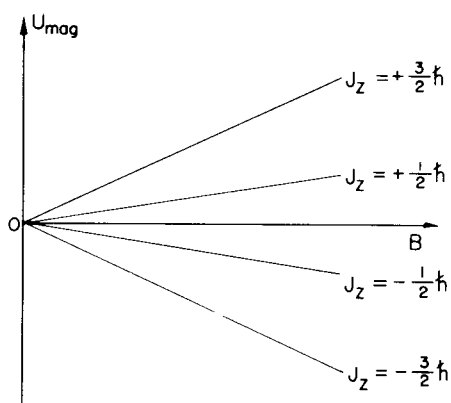


Fig. 34-5. The possible magnetic energies of an atomic system with a spin of $3/2$ in a magnetic field B .

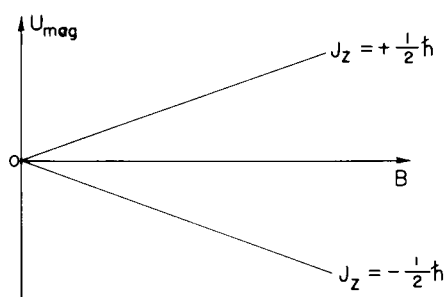


Fig. 34-6. The two possible energy states of an electron in a magnetic field B .

The quantity $q_e\hbar/2m$ is usually given the name “the Bohr magneton” and written μ_B :

$$\mu_B = \frac{q_e\hbar}{2m}.$$

The possible values of the magnetic energy are

$$U_{\text{mag}} = g\mu_B B \frac{J_z}{\hbar},$$

where $J_z/\hbar$ takes on the possible values $j, (j-1), (j-2), \dots, (-j+1), -j$.

In other words, the energy of an atomic system is changed when it is put in a magnetic field by an amount that is proportional to the field, and proportional to J_z . We say that the energy of an atomic system is “split into $2j+1$ levels” by a magnetic field. For instance, an atom whose energy is U_0 outside a magnetic field and whose j is $3/2$, will have four possible energies when placed in a field. We can show these energies by an energy-level diagram like that drawn in Fig 34-5. Any particular atom can have only one of the four possible energies in any given field B . That is what quantum mechanics says about the behavior of an atomic system in a magnetic field.

The simplest “atomic” system is a single electron. The spin of an electron is $1/2$, so there are two possible states. $J_z = \hbar/2$ and $J_z = -\hbar/2$. For an electron at rest (no orbital motion), the spin magnetic moment has a g -value of 2, so the magnetic energy can be either $\pm\mu_B B$. The possible energies in a magnetic field are shown in Fig. 34-6. Speaking loosely we say that the electron either has its spin “up” (along the field) or “down” (opposite the field).

For systems with higher spins, there are more states. We can think that the spin is “up” or “down” or cocked at some “angle” in between, depending on the value of J_z .

We will use these quantum mechanical results to discuss the magnetic properties of materials in the next chapter.

Paramagnetism and Magnetic Resonance

35-1 Quantized magnetic states

In the last chapter we described how in quantum mechanics the angular momentum of a thing does not have an arbitrary direction, but its component along a given axis can take on only certain equally spaced, discrete values. It is a shocking and peculiar thing. You may think that perhaps we should not go into such things until your minds are more advanced and ready to accept this kind of an idea. Actually, your minds will never become more advanced—in the sense of being able to accept such a thing easily. There isn't any descriptive way of making it intelligible that isn't so subtle and advanced in its own form that it is more complicated than the thing you were trying to explain. The behavior of matter on a small scale—as we have remarked many times—is different from anything that you are used to and is very strange indeed. As we proceed with classical physics, it is a good idea to try to get a growing acquaintance with the behavior of things on a small scale, at first as a kind of experience without any deep understanding. Understanding of these matters comes very slowly, if at all. Of course, one does get better able to know what is going to happen in a quantum-mechanical situation—if that is what understanding means—but one never gets a comfortable feeling that these quantum-mechanical rules are “natural.” Of course they *are*, but they are not natural to our own experience at an ordinary level. We should explain that the attitude that we are going to take with regard to this rule about angular momentum is quite different from many of the other things we have talked about. We are not going to try to “explain” it, but we must at least *tell* you what happens; it would be dishonest to describe the magnetic properties of materials without mentioning the fact that the classical description of magnetism—of angular momentum and magnetic moments—is incorrect.

One of the most shocking and disturbing features about quantum mechanics is that if you take the angular momentum along any particular axis you find that it is always an integer or half-integer times $\hbar$. This is so no matter which axis you take. The subtleties involved in that curious fact—that you can take any other axis and find that the component for it is also locked to the same set of values—we will leave to a later chapter, when you will experience the delight of seeing how this apparent paradox is ultimately resolved.

We will now just accept the fact that for every atomic system there is a number j , called the *spin* of the system—which must be an integer or a half-integer—and that the component of the angular momentum along any particular axis will always have one of the following values between $+j\hbar$ and $-j\hbar$:

$$J_z = \text{one of } \left\{ \begin{array}{c} j \\ j-1 \\ j-2 \\ \vdots \\ -j+2 \\ -j+1 \\ -j \end{array} \right\} \cdot \hbar. \quad (35.1)$$

We have also mentioned that every simple atomic system has a magnetic moment which has the same direction as the angular momentum. This is true not only for atoms and nuclei but also for the fundamental particles. Each fundamental particle has its own characteristic value of j and its magnetic moment.

35-1 Quantized magnetic states

35-2 The Stern-Gerlach experiment

35-3 The Rabi molecular-beam method

35-4 The paramagnetism of bulk materials

35-5 Cooling by adiabatic demagnetization

35-6 Nuclear magnetic resonance

Review. Chapter 11, *Inside Dielectrics*

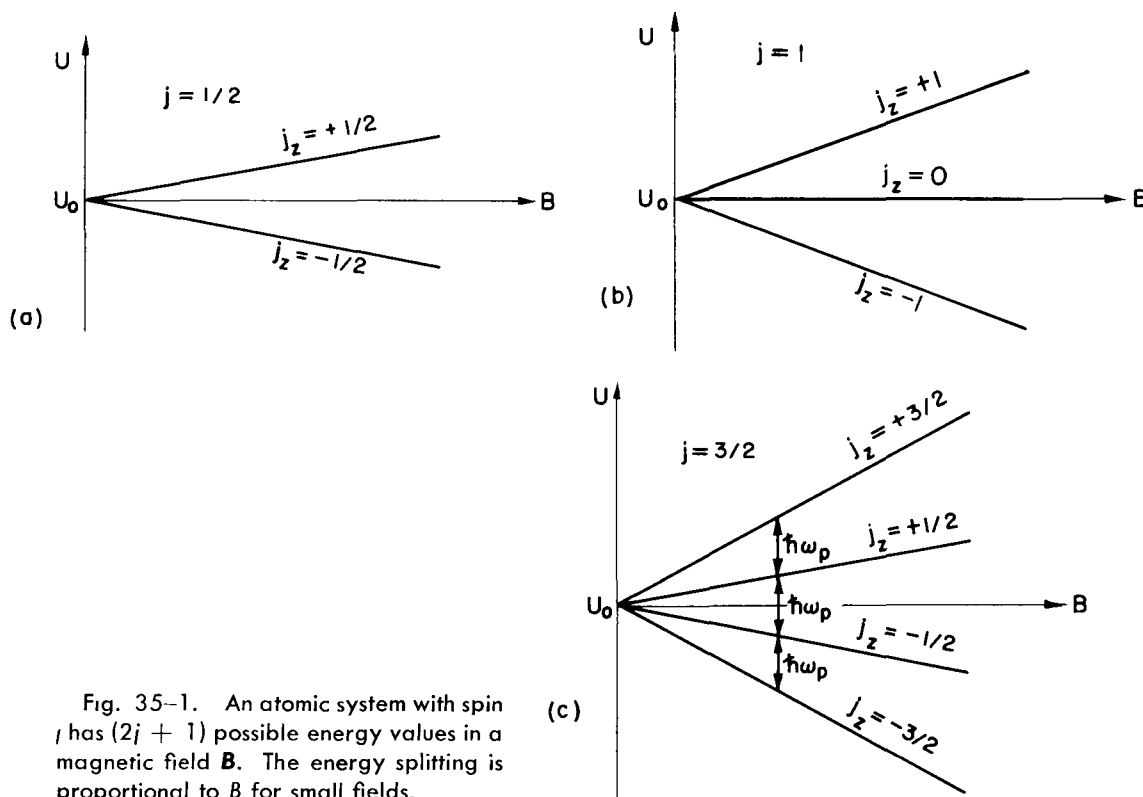


Fig. 35-1. An atomic system with spin j has $(2j + 1)$ possible energy values in a magnetic field B . The energy splitting is proportional to B for small fields.

(For some particles, both are zero.) What we mean by “the magnetic moment” in this statement is that the energy of the system in a magnetic field, say in the z -direction, can be written as $-\mu_z B$ for small magnetic fields. We must have the condition that the field should not be too great, otherwise it could disturb the internal motions of the system and the energy would not be a measure of the magnetic moment that was there before the field was turned on. But if the field is sufficiently weak, the field changes the energy by the amount

$$\Delta U = -\mu_z B, \quad (35.2)$$

with the understanding that in this equation we are to replace μ_z by

$$\mu_z = g \left(\frac{q}{2m} \right) J_z, \quad (35.3)$$

where J_z has one of the values in Eq. (35.1).

Suppose we take a system with a spin $j = 3/2$. Without a magnetic field, the system has four different possible states corresponding to the different values of J_z , all of which have exactly the same energy. But the moment we turn on the magnetic field, there is an additional energy of interaction which separates these states into four slightly different energy levels. The energies of these levels are given by a certain energy proportional to B , multiplied by $\hbar$ times $3/2, 1/2, -1/2$, and $-3/2$ —the values of J_z . The splitting of the energy levels for atomic systems with spins of $1/2, 1$, and $3/2$ are shown in the diagrams of Fig. 35-1. (Remember that for any arrangement of electrons the magnetic moment is always directed opposite to the angular momentum.)

You will notice from the diagrams that the “center of gravity” of the energy levels is the same with and without a magnetic field. Also notice that the spacings from one level to the next are always equal for a given particle in a given magnetic field. We are going to write the energy spacing, for a given magnetic field B , as $\hbar\omega_p$ —which is just a definition of ω_p . Using Eqs. (35.2) and (35.3), we have

$$\begin{aligned} \hbar\omega_p &= g \frac{q}{2m} \hbar B \\ \text{or} \quad \omega_p &= g \frac{q}{2m} B. \end{aligned} \quad (35.4)$$

The quantity $g(q/2m)$ is just the ratio of the magnetic moment to the angular momentum—it is a property of the particle. Equation (35.4) is the same formula that we got in Chapter 34 for the angular velocity of precession in a magnetic field, for a gyroscope whose angular momentum is $\mathbf{J}$ and whose magnetic moment is $\boldsymbol{\mu}$.

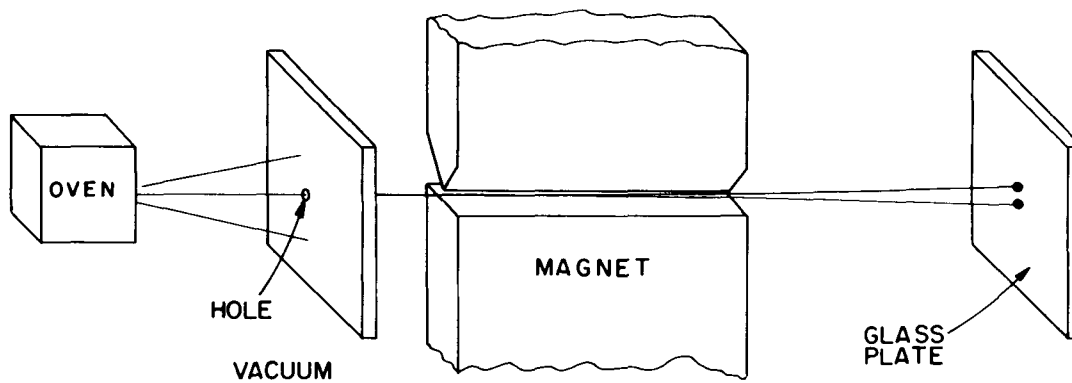


Fig. 35-2. The experiment of Stern and Gerlach.

35-2 The Stern-Gerlach experiment

The fact that the angular momentum is quantized is such a surprising thing that we will talk a little bit about it historically. It was a shock from the moment it was discovered (although it was expected theoretically). It was first observed in an experiment done in 1922 by Stern and Gerlach. If you wish, you can consider the experiment of Stern-Gerlach as a direct justification for a belief in the quantization of angular momentum. Stern and Gerlach devised an experiment for measuring the magnetic moment of individual silver atoms. They produced a beam of silver atoms by evaporating silver in a hot oven and letting some of them come out through a series of small holes. This beam was directed between the pole tips of a special magnet, as shown in Fig. 35-2. Their idea was the following. If the silver atom has a magnetic moment $\boldsymbol{\mu}$, then in a magnetic field $\mathbf{B}$ it has an energy $-\boldsymbol{\mu}_z B$, where z is the direction of the magnetic field. In the classical theory, μ_z would be equal to the magnetic moment times the cosine of the angle between the moment and the magnetic field, so the extra energy in the field would be

$$\Delta U = -\boldsymbol{\mu} B \cos \theta. \quad (35.5)$$

Of course, as the atoms come out of the oven, their magnetic moments would point in every possible direction, so there would be all values of θ . Now if the magnetic field varies very rapidly with z —if there is a strong field gradient—then the magnetic energy will also vary with position, and there will be a force on the magnetic moments whose direction will depend on whether cosine θ is positive or negative. The atoms will be pulled up or down by a force proportional to the derivative of the magnetic energy; from the principle of virtual work,

$$F_z = -\frac{\partial U}{\partial z} = \mu \cos \theta \frac{\partial B}{\partial z}. \quad (35.6)$$

Stern and Gerlach made their magnet with a very sharp edge on one of the pole tips in order to produce a very rapid variation of the magnetic field. The beam of silver atoms was directed right along this sharp edge, so that the atoms would feel a vertical force in the inhomogeneous field. A silver atom with its magnetic moment directed horizontally would have no force on it and would go straight past the magnet. An atom whose magnetic moment was exactly vertical would have a force pulling it up toward the sharp edge of the magnet. An atom whose magnetic moment was pointed downward would feel a downward push. Thus,

as they left the magnet, the atoms would be spread out according to their vertical components of magnetic moment. In the classical theory all angles are possible, so that when the silver atoms are collected by deposition on a glass plate, one should expect a smear of silver along a vertical line. The height of the line would be proportional to the magnitude of the magnetic moment. The abject failure of classical ideas was completely revealed when Stern and Gerlach saw what actually happened. They found on the glass plate two distinct spots. The silver atoms had formed two beams.

That a beam of atoms whose spins would apparently be randomly oriented gets split up into two separate beams is most miraculous. How does the magnetic moment *know* that it is only allowed to take on certain components in the direction of the magnetic field? Well, that was really the beginning of the discovery of the quantization of angular momentum, and instead of trying to give you a theoretical explanation, we will just say that you are stuck with the result of this experiment just as the physicists of that day had to accept the result when the experiment was done. It is an *experimental fact* that the energy of an atom in a magnetic field takes on a series of individual values. For each of these values the energy is proportional to the field strength. So in a region where the field varies, the principle of virtual work tells us that the possible magnetic force on the atoms will have a set of separate values, the force is different for each state, so the beam of atoms is split into a small number of separate beams. From a measurement of the deflection of the beams, one can find the strength of the magnetic moment.

35-3 The Rabi molecular-beam method

We would now like to describe an improved apparatus for the measurement of magnetic moments which was developed by I. I. Rabi and his collaborators. In the Stern-Gerlach experiment the deflection of atoms is very small, and the measurement of the magnetic moment is not very precise. Rabi's technique permits a fantastic precision in the measurement of the magnetic moments. The method is based on the fact that the original energy of the atoms in a magnetic field is split up into a finite number of energy levels. That the energy of an atom in the magnetic field can have only certain discrete energies is really not more surprising than the fact that atoms *in general* have only certain discrete energy levels—something we mentioned often in Volume I. Why should the same thing *not* hold for atoms in a magnetic field? It does. But it is the attempt to correlate this with the idea of an *oriented magnetic moment* that brings out some of the strange implications of quantum mechanics.

When an atom has two levels which differ in energy by the amount ΔU , it can make a transition from the upper level to the lower level by emitting a light quantum of frequency ω , where

$$\hbar\omega = \Delta U \quad (35.7)$$

The same thing can happen with atoms in a magnetic field. Only then, the energy differences are so small that the frequency does not correspond to light, but to microwaves or to radiofrequencies. The transitions from the lower energy level to an upper energy level of an atom can also take place with the absorption of light or, in the case of atoms in a magnetic field, by the absorption of microwave energy. Thus if we have an atom in a magnetic field, we can cause transitions from one state to another by applying an additional electromagnetic field of the proper frequency. In other words, if we have an atom in a strong magnetic field and we “tickle” the atom with a weak varying electromagnetic field, there will be a certain probability of knocking it to another level if the frequency is near to the ω in Eq. (35.7). For an atom in a magnetic field, this frequency is just what we have earlier called ω_p , and it is given in terms of the magnetic field by Eq. (35.4). If the atom is tickled with the wrong frequency, the chance of causing a transition is very small. Thus there is a sharp *resonance* at ω_p in the probability of causing a transition. By measuring the frequency of this resonance in a known magnetic field B , we can measure the quantity $g(q/2m)$ —and hence the g -factor—with great precision.

It is interesting that one comes to the same conclusion from a classical point of view. According to the classical picture, when we place a small gyroscope with a magnetic moment μ and an angular momentum J in an external magnetic field, the gyroscope will precess about an axis parallel to the magnetic field. (See Fig 35-3.) Suppose we ask: How can we change the angle of the classical gyroscope with respect to the field—namely, with respect to the z -axis? The magnetic field produces a torque around a *horizontal* axis. Such a torque you would think is *trying* to line up the magnet with the field, but it only causes the precession. If we want to change the angle of the gyroscope with respect to the z -axis, we must exert a torque on it *about the z -axis*. If we apply a torque which goes in the same direction as the precession, the angle of the gyroscope will change to give a smaller component of J in the z -direction. In Fig. 35-3, the angle between J and the z -axis would increase. If we try to hinder the precession, J moves toward the vertical.

For our precessing atom in a uniform magnetic field, how can we apply the kind of torque we want? The answer is: with a weak magnetic field from the side. You might at first think that the direction of this magnetic field would have to rotate with the precession of the magnetic moment, so that it was always at right angles to the moment, as indicated by the field B' in Fig. 35-4(a). Such a field works very well, but an *alternating* horizontal field is almost as good. If we have a small horizontal field B' , which is always in the x -direction (plus or minus) and which oscillates with the frequency ω_p , then on each one-half cycle the torque on the magnetic moment reverses, so that it has a cumulative effect which is almost as effective as a rotating magnetic field. Classically, then, we would expect the component of the magnetic moment along the z -direction to change if we have a very weak oscillating magnetic field at a frequency which is exactly ω_p . Classically, of course, μ_z would change continuously, but in quantum mechanics the z -component of the magnetic moment cannot adjust continuously. It must jump suddenly from one value to another. We have made the comparison between the consequences of classical mechanics and quantum mechanics to give you some clue as to what might happen classically and how it is related to what actually happens in quantum mechanics. You will notice, incidentally, that the expected resonant frequency is the same in both cases.

One additional remark: From what we have said about quantum mechanics, there is no apparent reason why there couldn't also be transitions at the frequency $2\omega_p$. It happens that there isn't any analog of this in the classical case, and also it doesn't happen in the quantum theory either—at least not for the particular method of inducing the transitions that we have described. With an oscillating horizontal magnetic field, the probability that a frequency $2\omega_p$ would cause a jump of two steps at once is zero. It is only at the frequency ω_p that transitions, either upward or downward, are likely to occur.

Now we are ready to describe Rabi's method for measuring magnetic moments. We will consider here only the operation for atoms with a spin of $1/2$. A diagram of the apparatus is shown in Fig. 35-5. There is an oven which gives out a stream of neutral atoms which passes down a line of three magnets. Magnet 1

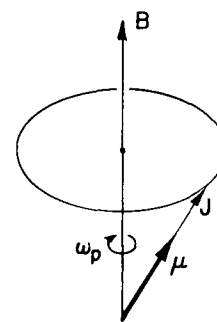


Fig. 35-3. The classical precession of an atom with the magnetic moment μ and the angular momentum J .

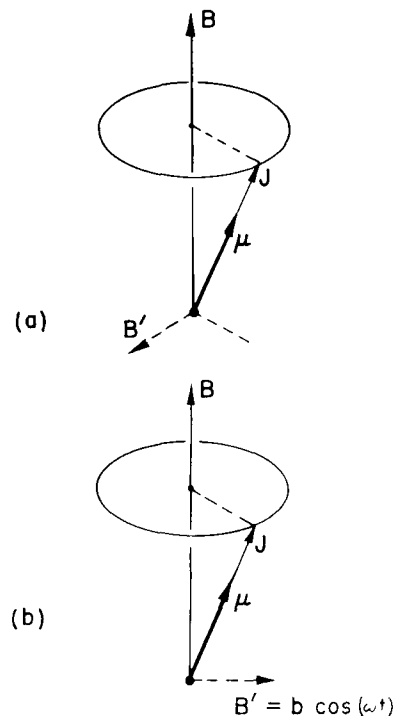


Fig. 35-4. The angle of precession of an atomic magnet can be changed by a horizontal magnetic field always at right angles to μ , as in (a), or by an oscillating field, as in (b).

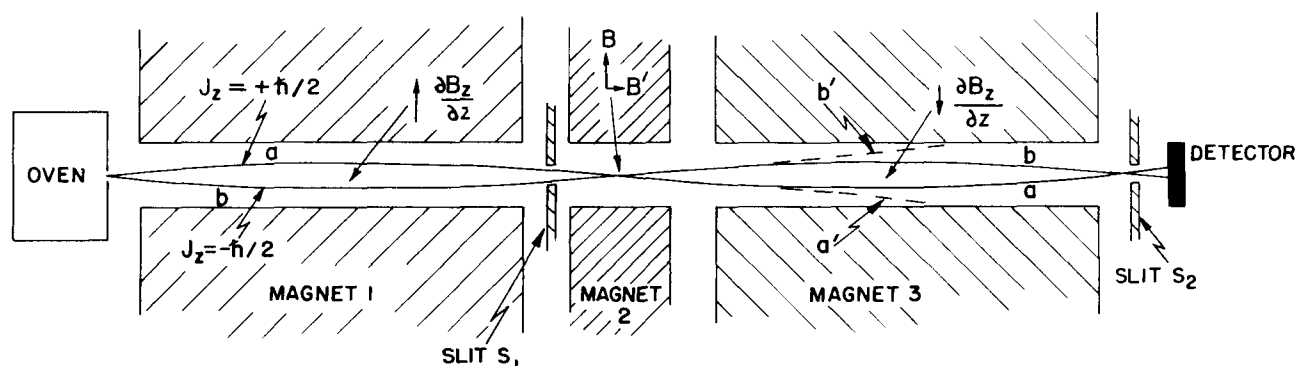


Fig. 35-5. The Rabi molecular-beam apparatus.

is just like the one in Fig. 35-2, and has a field with a strong field gradient—say, with $\partial B_z/\partial z$ positive. If the atoms have a magnetic moment, they will be deflected downward if $J_z = +\hbar/2$, or upward if $J_z = -\hbar/2$ (since for electrons μ is directed opposite to $\mathbf{J}$). If we consider only those atoms which can get through the slit S_1 , there are two possible trajectories, as shown. Atoms with $J_z = +\hbar/2$ must go along curve a to get through the slit, and those with $J_z = -\hbar/2$ must go along curve b . Atoms which start out from the oven along other paths will not get through the slit.

Magnet 2 has a uniform field. There are no forces on the atoms in this region, so they go straight through and enter magnet 3. Magnet 3 is just like magnet 1 but with the field *inverted*, so that $\partial B_z/\partial z$ has the opposite sign. The atoms with $J_z = +\hbar/2$ (we say “with spin up”), that felt a downward push in magnet 1, get an *upward* push in magnet 3; they continue on the path a and go through slit S_2 to a detector. The atoms with $J_z = -\hbar/2$ (“with spin down”) also have opposite forces in magnets 1 and 3 and go along the path b , which also takes them through slit S_2 to the detector.

The detector may be made in various ways, depending on the atom being measured. For example, for atoms of an alkali metal like sodium, the detector can be a thin, hot tungsten wire connected to a sensitive current meter. When sodium atoms land on the wire, they are evaporated off as Na^+ ions, leaving an electron behind. There is a current from the wire proportional to the number of sodium atoms arriving per second.

In the gap of magnet 2 there is a set of coils that produces a small horizontal magnetic field $\mathbf{B}'$. The coils are driven with a current which oscillates at a variable frequency ω . So between the poles of magnet 2 there is a strong, constant, vertical field $\mathbf{B}_0$ and a weak, oscillating, horizontal field $\mathbf{B}'$.

Suppose now that the frequency ω of the oscillating field is set at ω_p —the “precession” frequency of the atoms in the field $\mathbf{B}$. The alternating field will cause some of the atoms passing by to make transitions from one J_z to the other. An atom whose spin was initially “up” ($J_z = +\hbar/2$) may be flipped “down” ($J_z = -\hbar/2$). Now this atom has the direction of its magnetic moment reversed, so it will feel a *downward* force in magnet 3 and will move along the path a' , shown in Fig. 35-5. It will no longer get through the slit S_2 to the detector. Similarly, some of the atoms whose spins were initially down ($J_z = -\hbar/2$) will have their spins flipped up ($J_z = +\hbar/2$) as they pass through magnet 2. They will then go along the path b' and will not get to the detector.

If the oscillating field $\mathbf{B}'$ has a frequency appreciably different from ω_p , it will not cause any spin flips, and the atoms will follow their undisturbed paths to the detector. So you can see that the “precession” frequency ω_p of the atoms in the field $\mathbf{B}_0$ can be found by varying the frequency ω of the field $\mathbf{B}'$ until a decrease is observed in the current of atoms arriving at the detector. A decrease in the current will occur when ω is “in resonance” with ω_p . A plot of the detector current as a function of ω might look like the one shown in Fig. 35-6. Knowing ω_p , we can obtain the g -value of the atom.

Such atomic-beam or, as they are usually called, “molecular” beam resonance experiments are a beautiful and delicate way of measuring the magnetic properties of atomic objects. The resonance frequency ω_p can be determined with great precision—in fact, with a greater precision than we can measure the magnetic field $\mathbf{B}_0$, which we must know to find g .

35-4 The paramagnetism of bulk materials

We would like now to describe the phenomenon of the paramagnetism of bulk materials. Suppose we have a substance whose atoms have permanent magnetic moments, for example a crystal like copper sulfate. In the crystal there are copper ions whose inner electron shells have a net angular momentum and a net magnetic moment. So the copper ion is an object which has a permanent magnetic moment. Let's say just a word about which atoms have magnetic moments and which ones don't. Any atom, like sodium for instance, which has an *odd* number

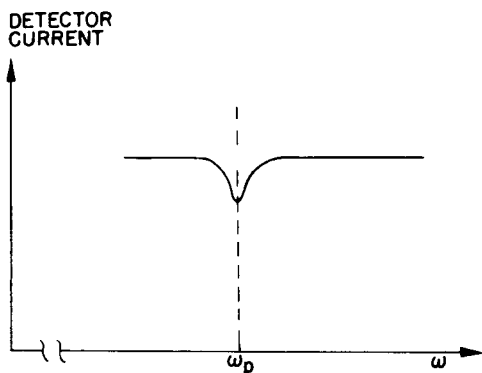


Fig. 35-6. The current of atoms in the beam decreases when $\omega = \omega_p$.

of electrons, will have a magnetic moment. Sodium has one electron in its unfilled shell. This electron gives the atom a spin and a magnetic moment. Ordinarily, however, when compounds are formed the extra electrons in the outside shell are coupled together with other electrons whose spin directions are exactly opposite, so that all the angular momenta and magnetic moments of the valence electrons usually cancel out. That's why, in general, molecules do not have a magnetic moment. Of course if you have a gas of sodium atoms, there is no such cancellation.* Also, if you have what is called in chemistry a "free radical"—an object with an odd number of valence electrons—then the bonds are not completely satisfied, and there is a net angular momentum.

In most bulk materials there is a net magnetic moment only if there are atoms present whose *inner* electron shell is not filled. Then there can be a net angular momentum and a magnetic moment. Such atoms are found in the "transition element" part of the periodic table—for instance, chromium, manganese, iron, nickel, cobalt, palladium, and platinum are elements of this kind. Also, all of the rare earth elements have unfilled inner shells and permanent magnetic moments. There are a couple of other strange things that also happen to have magnetic moments, such as liquid oxygen, but we will leave it to the chemistry department to explain the reason.

Now suppose that we have a box full of atoms or molecules with permanent moments—say a gas, or a liquid, or a crystal. We would like to know what happens if we apply an external magnetic field. With *no* magnetic field, the atoms are kicked around by the thermal motions, and the moments wind up pointing in all directions. But when there is a magnetic field, it acts to line up the little magnets; then there are more moments lying toward the field than away from it. The material is "magnetized."

We define the *magnetization* $\mathbf{M}$ of a material as the net magnetic moment per unit volume, by which we mean the vector sum of all the atomic magnetic moments in a unit volume. If there are N atoms per unit volume and their *average* moment is $\langle \boldsymbol{\mu} \rangle_{av}$ then $\mathbf{M}$ can be written as N times the average atomic moment:

$$\mathbf{M} = N \langle \boldsymbol{\mu} \rangle_{av} \quad (35.8)$$

The definition of $\mathbf{M}$ corresponds to the definition of the electric polarization $\mathbf{P}$ of Chapter 10.

The classical theory of paramagnetism is just like the theory of the dielectric constant we showed you in Chapter 11. One assumes that each of the atoms has a magnetic moment $\boldsymbol{\mu}$, which always has the same magnitude but which can point in any direction. In a field $\mathbf{B}$, the magnetic energy is $-\boldsymbol{\mu} \cdot \mathbf{B} = -\mu B \cos \theta$, where θ is the angle between the moment and the field. From statistical mechanics, the relative probability of having any angle is $e^{-\text{energy}/kT}$, so angles near zero are more likely than angles near π . Proceeding exactly as we did in Section 11-3, we find that for small magnetic fields $\mathbf{M}$ is directed parallel to $\mathbf{B}$ and has the magnitude

$$M = \frac{N\mu^2 B}{3kT}. \quad (35.9)$$

[See Eq. (11.20).] This approximate formula is correct only for $\mu B/kT$ much less than one.

We find that the induced magnetization—the magnetic moment per unit volume—is proportional to the magnetic field. This is the phenomenon of paramagnetism. You will see that the effect is stronger at lower temperatures and weaker at higher temperatures. When we put a field on a substance, it develops, for small fields, a magnetic moment proportional to the field. The ratio of M to B (for small fields) is called the magnetic *susceptibility*.

Now we want to look at paramagnetism from the point of view of quantum mechanics. We take first the case of an atom with a spin of $1/2$. In the absence of

* Ordinary Na vapor is mostly monatomic, although there are also some molecules of Na_2 .

a magnetic field the atoms have a certain energy, but in a magnetic field there are two possible energies, one for each value of J_z . For $J_z = +\hbar/2$, the energy is changed by the magnetic field by the amount

$$\Delta U_1 = +g \left(\frac{q_e \hbar}{2m} \right) \cdot \frac{1}{2} \cdot B. \quad (35.10)$$

(The energy shift ΔU is positive for an atom because the electron charge is negative.) For $J_z = -\hbar/2$, the energy is changed by the amount

$$\Delta U_2 = -g \left(\frac{q_e \hbar}{2m} \right) \cdot \frac{1}{2} \cdot B \quad (35.11)$$

To save writing, let's set

$$\mu_0 = g \left(\frac{q_e \hbar}{2m} \right) \cdot \frac{1}{2}; \quad (35.12)$$

then

$$\Delta U = \pm \mu_0 B. \quad (35.13)$$

The meaning of μ_0 is clear: $-\mu_0$ is the z-component of the magnetic moment in the up-spin case, and $+\mu_0$ is the z-component of the magnetic moment in the down-spin case.

Now statistical mechanics tells us that the probability that an atom is in one state or another is proportional to

$$e^{-(\text{Energy of state})/kT}.$$

With no magnetic field the two states have the same energy; so when there is equilibrium in a magnetic field, the probabilities are proportional to

$$e^{-\Delta U/kT}. \quad (35.14)$$

The number of atoms per unit volume with spin up is

$$N_{\text{up}} = a e^{-\mu_0 B/kT}, \quad (35.15)$$

and the number with spin down is

$$N_{\text{down}} = a e^{+\mu_0 B/kT}. \quad (35.16)$$

The constant a is to be determined so that

$$N_{\text{up}} + N_{\text{down}} = N, \quad (35.17)$$

the total number of atoms per unit volume. So we get that

$$a = \frac{N}{e^{+\mu_0 B/kT} + e^{-\mu_0 B/kT}}. \quad (35.18)$$

What we are interested in is the *average* magnetic moment along the z-axis. The atoms with spin up will contribute a moment of $-\mu_0$, and those with spin down will have a moment of $+\mu_0$, so the average moment is

$$\langle \mu \rangle_{\text{av}} = \frac{N_{\text{up}}(-\mu_0) + N_{\text{down}}(+\mu_0)}{N}. \quad (35.19)$$

The magnetic moment per unit volume M is then $N \langle \mu \rangle_{\text{av}}$. Using Eqs. (35.15), (35.16), and (35.17), we get that

$$M = N \mu_0 \frac{e^{+\mu_0 B/kT} - e^{-\mu_0 B/kT}}{e^{+\mu_0 B/kT} + e^{-\mu_0 B/kT}}. \quad (35.20)$$

This is the quantum-mechanical formula for M for atoms with $j = 1/2$. Incidentally, this formula can also be written somewhat more concisely in terms of the

hyperbolic tangent function:

$$M = N\mu_0 \tanh \frac{\mu_0 B}{kT}. \quad (35.21)$$

A plot of M as a function of B is given in Fig. 35.7. When B gets very large, the hyperbolic tangent approaches 1, and M approaches the limiting value $N\mu_0$. So at high fields, the magnetization *saturates*. We can see why that is; at high enough fields the moments are all lined up in the same direction. In other words, they are all in the spin-down state, and each atom contributes the moment μ_0 .

In most normal cases—say, for typical moments, room temperatures, and the fields one can normally get (like 10,000 gauss)—the ratio $\mu_0 B/kT$ is about 0.02. One must go to very low temperatures to see the saturation. For normal temperatures, we can usually replace $\tanh x$ by x , and write

$$M = \frac{N\mu_0^2 B}{kT}. \quad (35.22)$$

Just as we saw in the classical theory, M is proportional to B . In fact, the formula is almost exactly the same, except that there seems to be a factor of $1/3$ missing. But we still need to relate the μ_0 in our quantum formula to the μ that appears in the classical result, Eq. (35.9).

In the classical formula, what appears is $\mu^2 = \mu \cdot \mu$, the square of the vector magnetic moment, or

$$\mu \cdot \mu = \left(g \frac{q_e}{2m} \right)^2 J \cdot J. \quad (35.23)$$

We pointed out in the last chapter that you can very likely get the right answer from a classical calculation by replacing $J \cdot J$ by $j(j+1)\hbar^2$. In our particular example, we have $j = 1/2$, so

$$j(j+1)\hbar^2 = \frac{3}{4}\hbar^2.$$

Substituting this for $J \cdot J$ in Eq. (35.23), we get

$$\mu \cdot \mu = \left(g \frac{q_e}{2m} \right)^2 \frac{3\hbar^2}{4},$$

or in terms of μ_0 , defined in Eq. (35.12), we get

$$\mu \cdot \mu = 3\mu_0^2.$$

Substituting this for μ^2 in the classical formula, Eq. (35.9), does indeed reproduce the correct quantum formula, Eq. (35.22).

The quantum theory of paramagnetism is easily extended to atoms of any spin j . The low-field magnetization is

$$M = Ng^2 \frac{j(j+1)}{3} \frac{\mu_B^2 B}{kT}, \quad (35.24)$$

where

$$\mu_B = \frac{q_e \hbar}{2m} \quad (35.25)$$

is a combination of constants with the dimensions of a magnetic moment. Most atoms have moments of roughly this size. It is called the *Bohr magneton*. The spin magnetic moment of the electron is almost exactly one Bohr magneton.

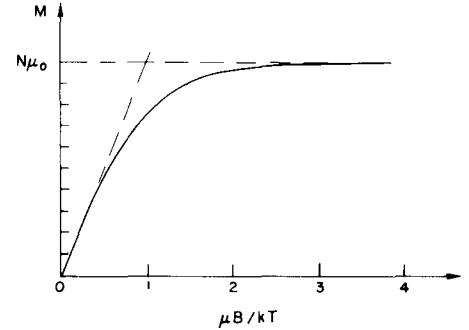


Fig. 35-7. The variation of the paramagnetic magnetization with the magnetic field strength B .

35-5 Cooling by adiabatic demagnetization

There is a very interesting special application of paramagnetism. At very low temperatures it is possible to line up the atomic magnets in a strong field. It is then possible to get down to *extremely* low temperatures by a process called *adiabatic demagnetization*. We can take a paramagnetic salt (for example, one

containing a number of rare-earth atoms like praseodymium-ammonium-nitrate), and start by cooling it down with liquid helium to one or two degrees absolute in a strong magnetic field. Then the factor $\mu B/kT$ is larger than 1—say more like 2 or 3. Most of the spins are lined up, and the magnetization is nearly saturated. Let's say, to make it easy, that the field is very powerful and the temperature is very low, so that nearly all the atoms are lined up. Then you isolate the salt thermally (say, by removing the liquid helium and leaving a good vacuum) and turn off the magnetic field. The temperature of the salt goes way down.

Now if you were to turn off the field *suddenly*, the jiggling and shaking of the atoms in the crystal lattice would gradually knock all the spins out of alignment. Some of them would be up and some down. But if there is no field (and disregarding the interactions between the atomic magnets, which will make only a slight error), it takes no energy to turn over the atomic magnets. They could randomize their spins without any energy change and, therefore, without any temperature change.

Suppose, however, that while the atomic magnets are being flipped over by the thermal motion there is still some magnetic field present. Then it requires some work to flip them over opposite to the field—they *must do work against the field*. This takes energy from the thermal motions and lowers the temperature. So if the strong magnetic field is not removed too rapidly, the temperature of the salt will decrease—it is cooled by the demagnetization. From the quantum-mechanical view, when the field is strong all the atoms are in the lowest state, because the odds against any being in the upper state are impossibly big. But as the field is lowered, it gets more and more likely that thermal fluctuations will knock an atom into the upper state. When that happens, the atom absorbs the energy $\Delta U = \mu_0 B$. So if the field is turned off slowly, the magnetic transitions can take energy out of the thermal vibrations of the crystal, cooling it off. It is possible in this way to go from a temperature of a few degrees absolute down to a temperature of a few thousandths of a degree.

Would you like to make something even colder than that? It turns out that Nature has provided a way. We have already mentioned that there are also magnetic moments for the atomic nuclei. Our formulas for paramagnetism work just as well for nuclei, except that the moments of nuclei are roughly a *thousand times smaller*. [They are of the order of magnitude of $qh/2m_p$, where m_p is the *proton* mass, so they are smaller by the ratio of the masses of the electron and proton.] With such magnetic moments, even at a temperature of 2°K, the factor $\mu B/kT$ is only a few parts in a thousand. But if we use the paramagnetic demagnetization process to get down to a temperature of a few thousandths of a degree, $\mu B/kT$ becomes a number near 1—at these low temperatures we can begin to saturate the nuclear moments. That is good luck, because we can then use the adiabatic demagnetization of the *nuclear* magnetism to reach still lower temperatures. Thus it is possible to do two stages of magnetic cooling. First we use adiabatic demagnetization of paramagnetic ions to reach a few thousandths of a degree. Then we use the cold paramagnetic salt to cool some material which has a strong nuclear magnetism. Finally, when we remove the magnetic field from this material, its temperature will go down to within a *millionth* of a degree of absolute zero—if we have done everything very carefully.

35-6 Nuclear magnetic resonance

We have said that atomic paramagnetism is very small and that nuclear magnetism is even a thousand times smaller. Yet it is relatively easy to observe the nuclear magnetism by the phenomenon of “nuclear magnetic resonance.” Suppose we take a substance like water, in which all of the electron spins are exactly balanced so that their net magnetic moment is zero. The molecules will still have a very, very tiny magnetic moment due to the nuclear magnetic moment of the hydrogen nuclei. Suppose we put a small sample of water in a magnetic field B . Since the protons (of the hydrogen) have a spin of $1/2$, they will have two possible energy states. If the water is in thermal equilibrium, there will be slightly more

protons in the lower energy states—with their moments directed parallel to the field. There is a small net magnetic moment per unit volume. Since the proton moment is only about one-thousandth of an atomic moment, the magnetization which goes as μ^2 —using Eq. (35.22)—is only about one-millionth as strong as typical atomic paramagnetism. (That's why we have to pick a material with no atomic magnetism.) If you work it out, the difference between the number of protons with spin up and with spin down is only one part in 10^8 , so the effect is indeed very small! It can still be observed, however, in the following way.

Suppose we surround the water sample with a small coil that produces a small horizontal oscillating magnetic field. If this field oscillates at the frequency ω_p , it will induce transitions between the two energy states—just as we described for the Rabi experiment in Section 35-3. When a proton flips from an upper energy state to a lower one, it will give up the energy $\mu_z B$ which, as we have seen, is equal to $\hbar\omega_p$. If it flips from the lower energy state to the upper one, it will *absorb* the energy $\hbar\omega_p$ from the coil. Since there are slightly more protons in the lower state than in the upper one, there will be a net *absorption* of energy from the coil. Although the effect is very small, the slight energy absorption can be seen with a sensitive electronic amplifier.

Just as in the Rabi molecular-beam experiment, the energy absorption will be seen only when the oscillating field is in resonance, that is, when

$$\omega = \omega_p = g \left(\frac{q_e}{2m_p} \right) B$$

It is often more convenient to search for the resonance by varying B while keeping ω fixed. The energy absorption will evidently appear when

$$B = \frac{2m_p}{gq_e} \omega$$

A typical nuclear magnetic resonance apparatus is shown in Fig. 35-8. A high-frequency oscillator drives a small coil placed between the poles of a large electromagnet. Two small auxiliary coils around the pole tips are driven with a 60-cycle current so that the magnetic field is “wobbled” about its average value by a very small amount. As an example, say that the main current of the magnet is set to give a field of 5000 gauss, and the auxiliary coils produce a variation of ± 1 gauss about this value. If the oscillator is set at 21.2 megacycles per second, it will then be at the proton resonance each time the field sweeps through 5000 gauss [using Eq. (34.13) with $g = 5.58$ for the proton].

The circuit of the oscillator is arranged to give an additional output signal proportional to any *change* in the power being absorbed from the oscillator. This signal is fed to the vertical deflection amplifier of an oscilloscope. The horizontal sweep of the oscilloscope is triggered once during each cycle of the field-wobbling frequency. (More usually, the horizontal deflection is made to follow in proportion to the wobbling field.)

Before the water sample is placed inside the high-frequency coil, the power drawn from the oscillator is some value. (It doesn't change with the magnetic field.) When a small bottle of water is placed in the coil, however, a signal appears on the oscilloscope, as shown in the figure. We see a picture of the power being absorbed by the flipping over of the protons!

In practice, it is difficult to know how to set the main magnet to exactly 5000 gauss. What one does is to adjust the main magnet current until the resonance signal appears on the oscilloscope. It turns out that this is now the most convenient way to make an accurate measurement of the strength of a magnetic field. Of course, at some time *someone* had to measure accurately the magnetic field and frequency to determine the g -value of the proton. But now that this has been done, a proton resonance apparatus like that of the figure can be used as a “proton resonance magnetometer.”

We should say a word about the shape of the signal. If we were to wobble the magnetic field very slowly, we would expect to see a normal resonance curve. The energy absorption would read a maximum when ω_p arrived exactly at the

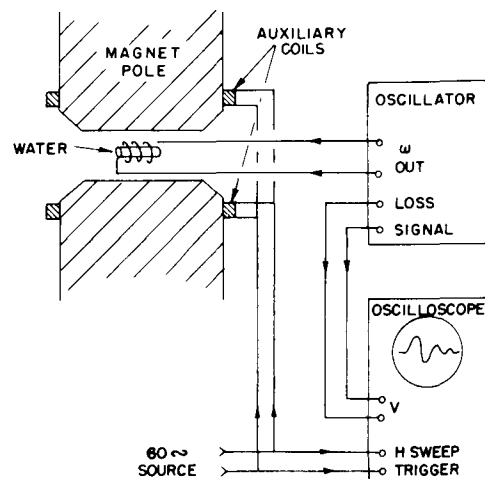


Fig 35-8. A nuclear magnetic resonance apparatus.

oscillator frequency. There would be some absorption at nearby frequencies because all the protons are not in exactly the same field—and different fields mean slightly different resonant frequencies.

One might wonder, incidentally, whether at the resonance frequency we should see any signal at all. Shouldn't we expect the high-frequency field to equalize the populations of the two states—so that there should be no signal except when the water is first put in? Not exactly, because although we are *trying* to equalize the two populations, the thermal motions on their part are trying to keep the proper ratios for the temperature T . If we sit at the resonance, the power being absorbed by the nuclei is just what is being lost to the thermal motions. There is, however, relatively little “thermal contact” between the proton magnetic moments and the atomic motions. The protons are relatively isolated down in the center of the electron distributions. So in pure water, the resonance signal is, in fact, usually too small to be seen. To increase the absorption, it is necessary to increase the “thermal contact.” This is usually done by adding a little iron oxide to the water. The iron atoms are like small magnets; as they jiggle around in their thermal dance, they make tiny jiggling magnetic fields at the protons. These varying fields “couple” the proton magnets to the atomic vibrations and tend to establish thermal equilibrium. It is through this “coupling” that protons in the higher energy states can lose their energy so that they are again capable of absorbing energy from the oscillator.

In practice the output signal of a nuclear resonance apparatus does not look like a normal resonance curve. It is usually a more complicated signal with oscillations—like the one drawn in the figure. Such signal shapes appear because of the changing fields. The explanation should be given in terms of quantum mechanics, but it can be shown that in such experiments the classical ideas of precessing moments always give the correct answer. Classically, we would say that when we arrive at resonance we start driving a lot of the precessing nuclear magnets synchronously. In so doing, we make them precess *together*. These nuclear magnets, all rotating together, will set up an induced emf in the oscillator coil at the frequency ω_p . But because the magnetic field is increasing with time, the precession frequency is increasing also, and the induced voltage is soon at a frequency a little higher than the oscillator frequency. As the induced emf goes alternately in phase and out of phase with the oscillator, the “absorbed” power goes alternately positive and negative. So on the oscilloscope we see the beat note between the proton frequency and the oscillator frequency. Because the proton frequencies are not all identical (different protons are in slightly different fields) and also possibly because of the disturbance from the iron oxide in the water, the freely precessing moments soon get out of phase, and the beat signal disappears.

These phenomena of magnetic resonance have been put to use in many ways as tools for finding out new things about matter—especially in chemistry and nuclear physics. It goes without saying that the numerical values of the magnetic moments of nuclei tell us something about their structure. In chemistry, much has been learned from the structure (or shape) of the resonances. Because of magnetic fields produced by nearby nuclei, the exact position of a nuclear resonance is shifted somewhat, depending on the environment in which any particular nucleus finds itself. Measuring these shifts helps determine which atoms are near which other ones and helps to elucidate the details of the structure of molecules. Equally important is the electron spin resonance of free radicals. Although not present to any very large extent in equilibrium, such radicals are often intermediate states of chemical reactions. A measurement of an electron spin resonance is a delicate test for the presence of free radicals and is often the key to understanding the mechanism of certain chemical reactions.

Ferromagnetism

36-1 Magnetization currents

In this chapter we will discuss some materials in which the net effect of the magnetic moments in the material is much greater than in the case of paramagnetism or diamagnetism. The phenomenon is called *ferromagnetism*. In paramagnetic and diamagnetic materials the induced magnetic moments are usually so weak that we don't have to worry about the additional fields produced by the magnetic moments. For *ferromagnetic* materials, however, the magnetic moments induced by applied magnetic fields are quite enormous and have a great effect on the fields themselves. In fact, the induced moments are so strong that they are often the dominant effect in producing the observed fields. So one of the things we will have to worry about is the mathematical theory of large induced magnetic moments. That is, of course, just a technical question. The real problem is, why are the magnetic moments so strong—how does it all work? We will come to that question in a little while.

Finding the magnetic fields of ferromagnetic materials is something like the problem of finding the electrostatic field in the presence of dielectrics. You will remember that we first described the internal properties of a dielectric in terms of a vector field $\mathbf{P}$, the dipole moment per unit volume. Then we figured out that the effects of this polarization are equivalent to a charge density ρ_{pol} given by the divergence of $\mathbf{P}$:

$$\rho_{\text{pol}} = -\nabla \cdot \mathbf{P}. \quad (36.1)$$

The total charge in any situation can be written as the sum of this polarization charge plus all other charges, whose density we write* ρ_{other} . Then the Maxwell equation which relates the divergence of $\mathbf{E}$ to the charge density becomes

$$\nabla \cdot \mathbf{E} = \frac{\rho}{\epsilon_0} = \frac{\rho_{\text{pol}} + \rho_{\text{other}}}{\epsilon_0},$$

or

$$\nabla \cdot \mathbf{E} = -\frac{\nabla \cdot \mathbf{P}}{\epsilon_0} + \frac{\rho_{\text{other}}}{\epsilon_0}.$$

We can then pull out the polarization part of the charge and put it on the other side of the equation, to get the new law

$$\nabla \cdot (\epsilon_0 \mathbf{E} + \mathbf{P}) = \rho_{\text{other}}. \quad (36.2)$$

The new law says the divergence of the quantity $(\epsilon_0 \mathbf{E} + \mathbf{P})$ is equal to the density of the other charges.

Pulling $\mathbf{E}$ and $\mathbf{P}$ together as in Eq. (36.2), of course, is useful only if we know some relation between them. We have seen that the theory which relates the induced electric dipole moment to the field was a relatively complicated business and can really only be applied to certain simple situations, and even then as an approximation. We would like to remind you of one of the approximate ideas we used. To find the induced dipole moment of an atom inside a dielectric, it is necessary to know the electric field that acts on an individual atom. We made the approximation—which is not too bad in many cases—that the field on the atom

36-1 Magnetization currents

36-2 The field H

36-3 The magnetization curve

36-4 Iron-core inductances

36-5 Electromagnets

36-6 Spontaneous magnetization

Review: Chapter 10, *Dielectrics*
Chapter 17, *The Law of Induction*

* If all of the "other" charges were on conductors, ρ_{other} would be the same as our ρ_{free} of Chapter 10.

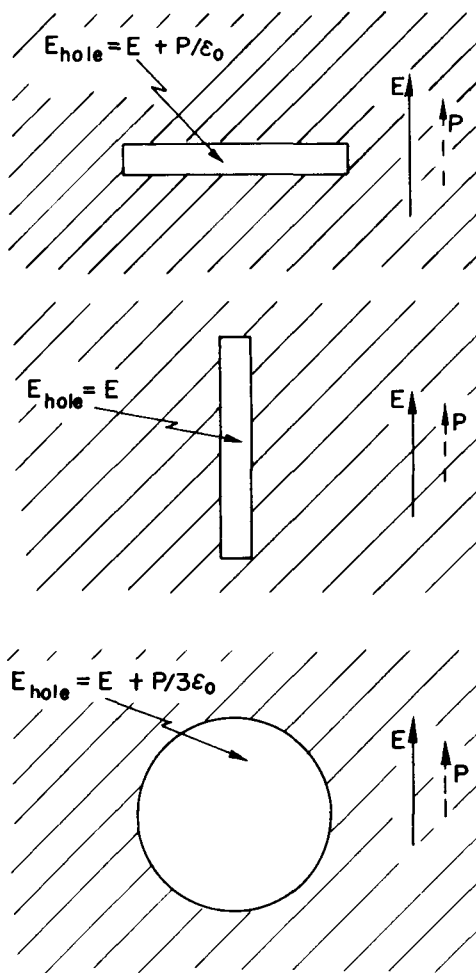


Fig. 36-1. The electric field in a cavity in a dielectric depends on the shape of the cavity.

is the same as it would be at the center of the small hole which would be left if we took out the atom (keeping the dipole moments of all the neighboring atoms the same). You will also remember that the electric field in a hole in a polarized dielectric depends on the shape of the hole. We summarize our earlier results in Fig. 36-1. For a thin, disc-shaped hole perpendicular to the polarization, the electric field in the hole is given by

$$E_{\text{hole}} = E_{\text{dielectric}} + \frac{P}{\epsilon_0},$$

which we showed by using Gauss' law. On the other hand, in a needle-shaped slot parallel to the polarization, we showed—by using the fact that the curl of $\mathbf{E}$ is zero—that the electric fields inside and outside of the slot are the same. Finally, we found that for a spherical hole the electric field was one-third of the way between the field of the slot and the field of the disc:

$$E_{\text{hole}} = E_{\text{dielectric}} + \frac{1}{3} \frac{P}{\epsilon_0} \text{ (spherical hole)}. \quad (36.3)$$

This was the field we used in thinking about what happens to an atom inside a polarized dielectric.

Now we have to discuss the analog of all this for the case of magnetism. One simple, short-cut way of doing this is to say the $\mathbf{M}$, the magnetic moment per unit volume, is just like $\mathbf{P}$, the electric dipole moment per unit volume, and that, therefore, the negative of the divergence of $\mathbf{M}$ is equivalent to a “magnetic charge density” ρ_m —whatever that may mean. The trouble is, of course, that there isn't any such thing as a “magnetic charge” in the physical world. As we know, the divergence of $\mathbf{B}$ is always zero. But that does not stop us from making an artificial *analog* and writing

$$\nabla \cdot \mathbf{M} = -\rho_m, \quad (36.4)$$

where it is to be understood that ρ_m is purely mathematical. Then we could make a complete analogy with the electrostatic case and use all our old equations from electrostatics. People have often done something like that. In fact, historically, people even believed that the analogy was right. They believed that the quantity ρ_m represented the density of “magnetic poles.” These days, however, we know that the magnetization of materials comes from circulating currents within the atoms—either from the spinning electrons or from the motion of the electrons in the atom. It is therefore nicer from a physical point of view to describe things realistically in terms of the atomic currents, rather than in terms of a density of some mythical “magnetic poles.” Incidentally, these currents are sometimes called “Amperian” currents, because Ampere first suggested that the magnetism of matter came from circulating atomic currents.

The actual microscopic current density in magnetized matter is, of course, very complicated. Its value depends on where you look in the atom—it's large in some places and small in others; it goes one way in one part of the atom and the opposite way in another part (just as the microscopic electric field varies enormously inside a dielectric). In many practical problems, however, we are interested only in the fields outside of the matter or in the *average* magnetic field inside of the matter—where we mean an average taken over many, many atoms. It is only for such *macroscopic* problems that it is convenient to describe the magnetic state of the matter in terms of $\mathbf{M}$, the average dipole moment per unit volume. What we want to show now is that the atomic currents of magnetized matter can give rise to certain large-scale currents which are related to $\mathbf{M}$.

What we are going to do, then, is to separate the current density $\mathbf{j}$ —which is the real source of the magnetic fields—into various parts: one part to describe the circulating currents of the atomic magnets, and the other parts to describe what other currents there may be. It is usually most convenient to separate the currents into three parts. In Chapter 32 we made a distinction between the currents which flow freely on conductors and the ones which are due to the back and forth motions

of the bound charges in dielectrics. In Section 32-2 we wrote

$$\mathbf{j} = \mathbf{j}_{\text{pol}} + \mathbf{j}_{\text{other}},$$

where $\mathbf{j}_{\text{pol}}$ represented the currents from the motion of the bound charges in dielectrics and $\mathbf{j}_{\text{other}}$ took care of all other currents. Now we want to go further. We want to separate $\mathbf{j}_{\text{other}}$ into one part, $\mathbf{j}_{\text{mag}}$, which describes the average currents inside of magnetized materials, and an additional term which we can call $\mathbf{j}_{\text{cond}}$ for whatever is left over. The last term will generally refer to currents in conductors, but it may also include other currents—for example the currents from charges moving freely through empty space. So we will write for the total current density:

$$\mathbf{j} = \mathbf{j}_{\text{pol}} + \mathbf{j}_{\text{mag}} + \mathbf{j}_{\text{cond}}. \quad (36.5)$$

Of course it is this total current which belongs in the Maxwell equation for the curl of $\mathbf{B}$:

$$c^2 \nabla \times \mathbf{B} = \frac{\mathbf{j}}{\epsilon_0} + \frac{\partial \mathbf{E}}{\partial t}. \quad (36.6)$$

Now we have to relate the current $\mathbf{j}_{\text{mag}}$ to the magnetization vector $\mathbf{M}$. So that you can see where we are going, we will tell you that the result is going to be that

$$\mathbf{j}_{\text{mag}} = \nabla \times \mathbf{M}. \quad (36.7)$$

If we are given the magnetization vector $\mathbf{M}$ everywhere in a magnetic material, the circulation current density is given by the curl of $\mathbf{M}$. Let's see if we can understand why this is so.

First, let's take the case of a cylindrical rod which has a uniform magnetization parallel to its axis. Physically, we know that such a uniform magnetization really means a uniform density of atomic circulating currents everywhere inside the material. Suppose we try to imagine what the actual currents would look like in a cross section of the material. We would expect to see currents something like those shown in Fig. 36-2. Each atomic current goes around and around in a little circle, with all the circulating currents going around in the same direction. Now what is the effective current of such a thing? Well, in most of the bar there is no effect at all, because right next to each current there is another current going in the opposite direction. If we imagine a small surface—but one still quite a bit larger than a single atom—such as is indicated in Fig. 36-2 by the line $\overline{AB}$, the net current through such a surface is zero. There is no net current anywhere inside the material. Note, however, that at the surface of the material there are atomic currents which are not cancelled by neighboring currents going the other way. At the surface there is a net current always going in the same direction around the rod. Now you see why we said earlier that a uniformly magnetized rod is equivalent to a long solenoid carrying an electric current.

How does this view fit with Eq. (36.7)? First, inside the material the magnetization $\mathbf{M}$ is constant, so all its derivatives are zero. This agrees with our geometric picture. At the surface, however, $\mathbf{M}$ is not really constant—it is constant up to the edge and then suddenly collapses to zero. So, right at the surface there are terrific gradients which, according to (36.7), will give a high current density. Suppose we look at what happens near the point C in Fig. 36-2. Taking the x - and y -directions as in the figure, the magnetization $\mathbf{M}$ is in the z -direction. Writing out the components of Eq. (36.7), we have

$$\begin{aligned} \frac{\partial M_z}{\partial y} &= (j_{\text{mag}})_x, \\ -\frac{\partial M_z}{\partial x} &= (j_{\text{mag}})_y. \end{aligned} \quad (36.8)$$

At the point C , the derivative $\partial M_z / \partial y$ is zero, but $\partial M_z / \partial x$ is large and positive. Equation (36.7) says that there is a large current density in the minus y -direction. This agrees with our picture of a surface current going around the bar.

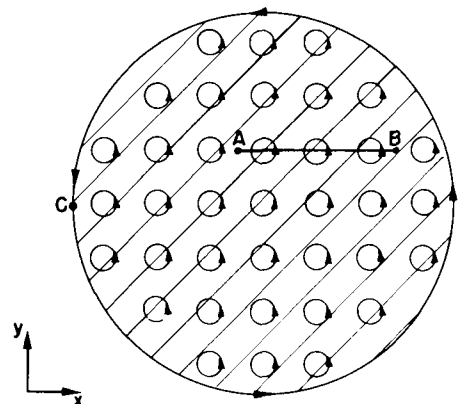


Fig. 36-2. Schematic diagram of the circulating atomic currents as seen in a cross section of an iron rod magnetized in the z -direction.

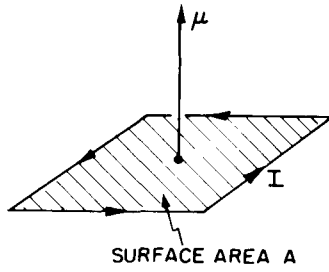


Fig. 36-3. The dipole moment μ of a current loop is IA .

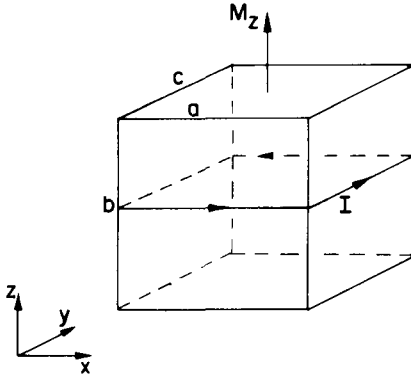
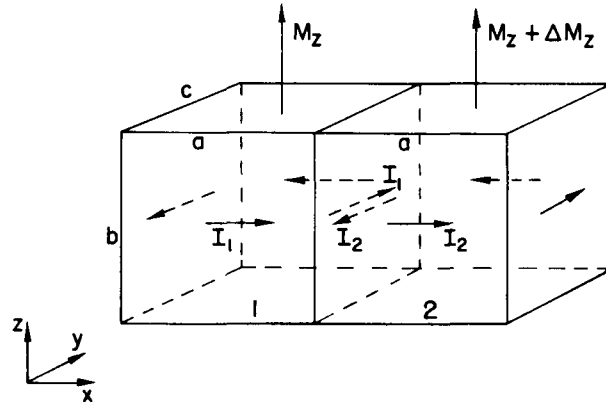


Fig. 36-4. A small magnetized block is equivalent to a circulating surface current.

Fig. 36-5. If the magnetization of two neighboring blocks is not the same, there is a net surface current in between.



Now we want to find the current density for a more complicated case in which the magnetization varies from point to point in a material. It is easy to see qualitatively that if the magnetization is different in two neighboring regions, there will not be a perfect cancellation of the circulating currents so that there will be a net current in the volume of the material. It is this effect that we want to work out quantitatively.

First, we need to recall the results of Section 14-5 that a circulating current I has a magnetic moment μ given by

$$\mu = IA, \quad (36.9)$$

where A is the area of the current loop (see Fig. 36-3). Now let's consider a small rectangular block inside of a magnetized material, as sketched in Fig. 36-4. We take the block so small that we can consider that the magnetization is uniform inside it. If this block has a magnetization M_z in the z -direction, the net effect will be the same as a surface current going around on the vertical faces, as shown. We can find the magnitude of these currents from Eq. (36.9). The total magnetic moment of the block is equal to the magnetization times the volume:

$$\mu = M_z(abc),$$

from which we get (remembering that the area of the loop is ac)

$$I = M_z b.$$

In other words, the current per unit length (vertically) on each of the vertical surfaces is equal to M_z .

Now suppose that we imagine two such little blocks next to each other, as shown in Fig. 36-5. Because block 2 is slightly displaced from block 1, it will have a slightly different vertical component of magnetization, which we call $M_z + \Delta M_z$. Now on the surface between the two blocks there will be two contributions to the total current. Block 1 will produce a current I_1 flowing in the positive y -direction, and block 2 will produce a surface current I_2 flowing in the negative y -direction. The total surface current in the positive y -direction is the sum:

$$\begin{aligned} I &= I_1 - I_2 = M_z b - (M_z + \Delta M_z)b \\ &= -\Delta M_z b. \end{aligned}$$

We can write ΔM_z as the derivative of M_z in the x -direction times the displacement from block 1 to block 2, which is just a :

$$\Delta M_z = \frac{\partial M_z}{\partial x} a.$$

The current flowing between the two blocks is then

$$I = -\frac{\partial M_z}{\partial x} ab.$$

To relate the current I to an average volume current density $\mathbf{j}$, we must realize that this current I is really spread over a certain cross-sectional area. If we imagine the whole volume of the material to be filled with such little blocks, one such side face (perpendicular to the x -axis) can be associated with each block.* Then we see that the area to be associated with the current I is just the area ab of one of the front faces. We get the result

$$j_y = \frac{I}{ab} = -\frac{\partial M_z}{\partial x}.$$

We have at least the beginning of the curl of $\mathbf{M}$.

There should be another term in j_y from the variation of the x -component of the magnetization with z . This contribution to $\mathbf{j}$ will come from the surface between two little blocks stacked one on top of the other, as shown in Fig. 36-6. Using the same arguments we have just made, you can show that this surface will contribute to j_y the amount $\partial M_x / \partial z$. These are the only surfaces which can contribute to the y -component of the current so we have that the total current density in the y -direction is

$$j_y = \frac{\partial M_x}{\partial z} - \frac{\partial M_z}{\partial x}.$$

Working out the currents on the remaining faces of a cube—or using the fact that our z -direction is completely arbitrary—we can conclude that the vector current density is indeed given by the equation

$$\mathbf{j} = \nabla \times \mathbf{M}.$$

So if we choose to describe the magnetic situation in matter in terms of the average magnetic moment per unit volume $\mathbf{M}$, we find that the circulating atomic currents are equivalent to an average current density in matter given by Eq. (36.7). If the material is also a dielectric, there may be, in addition, a polarization current $\mathbf{j}_{\text{pol}} = \partial \mathbf{P} / \partial t$. And if the material is also a conductor, we may have a conduction current $\mathbf{j}_{\text{cond}}$ as well. We can write the total current as

$$\mathbf{j} = \mathbf{j}_{\text{cond}} + \nabla \times \mathbf{M} + \frac{\partial \mathbf{P}}{\partial t}. \quad (36.10)$$

36-2 The field $\mathbf{H}$

Next, we want to insert the current as written in Eq. (36.10) into Maxwell's equations. We get

$$c^2 \nabla \times \mathbf{B} = \frac{\mathbf{j}}{\epsilon_0} + \frac{\partial \mathbf{E}}{\partial t} = \frac{1}{\epsilon_0} \left(\mathbf{j}_{\text{cond}} + \nabla \times \mathbf{M} + \frac{\partial \mathbf{P}}{\partial t} \right) + \frac{\partial \mathbf{E}}{\partial t}.$$

We can move the term in $\mathbf{M}$ to the left-hand side:

$$c^2 \nabla \times \left(\mathbf{B} - \frac{\mathbf{M}}{\epsilon_0 c^2} \right) = \frac{\mathbf{j}_{\text{cond}}}{\epsilon_0} + \frac{\partial}{\partial t} \left(\mathbf{E} + \frac{\mathbf{P}}{\epsilon_0} \right). \quad (36.11)$$

As we remarked in Chapter 32, many people like to write $(\mathbf{E} + \mathbf{P}/\epsilon_0)$ as a new vector field $\mathbf{D}/\epsilon_0$. Similarly, it is often convenient to write $(\mathbf{B} - \mathbf{M}/\epsilon_0 c^2)$ as a single vector field. We choose to define a new vector field $\mathbf{H}$ by

$$\mathbf{H} = \mathbf{B} - \frac{\mathbf{M}}{\epsilon_0 c^2}. \quad (36.12)$$

Then Eq. (36.11) becomes

$$\epsilon_0 c^2 \nabla \times \mathbf{H} = \mathbf{j}_{\text{cond}} + \frac{\partial \mathbf{D}}{\partial t}. \quad (36.13)$$

It looks simple, but all the complexity is just hidden in the letters $\mathbf{D}$ and $\mathbf{H}$.

* Or, if you prefer, the current I in each face should be split 50-50 with the blocks on the two sides.

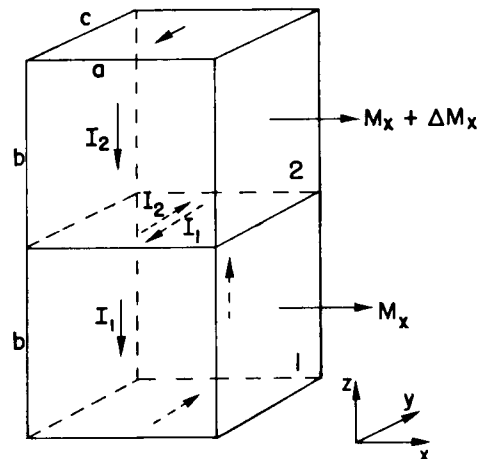


Fig. 36-6. Two blocks, one above the other, may also contribute to j_y .

Now we have to give you a warning. Most people who use the mks units have chosen to use a different definition of H . Calling *their* field H' (of course, they still call it H without the prime), it is defined by

$$H' = \epsilon_0 c^2 B - M. \quad (36.14)$$

(Also, they usually write $\epsilon_0 c^2$ as a new number $1/\mu_0$; then they have one more constant to keep track of!) With this definition, Eq. (36.13) looks even simpler:

$$\nabla \times H' = j_{\text{cond}} + \frac{\partial D}{\partial t}. \quad (36.15)$$

But the difficulties with this definition of H' are, first, that it doesn't agree with the definition of people who don't use the mks units, and second, that it makes H' and B have different units. We think it is more convenient for H to have the same units as B —rather than the units of M , as H' does. But if you are going to be an engineer and work on the design of transformers, magnets, and such, you will have to watch out. You will find many books which use for H the definition of Eq. (36.14) rather than our definition of Eq. (36.12), and many other books—especially handbooks about magnetic materials—that relate B and H the way we have done. You'll have to be careful to figure out which convention they are using.

One way to tell is by the units they use. Remember that in the mks system, B —and therefore *our* H —are measured with the unit: one weber per square meter, equal to 10,000 gauss. In the mks system, a magnetic moment (a current times an area) has the unit: one ampere-meter². The magnetization M , then, has the unit: one ampere *per* meter. For H' the units are the same as for M . You can see that this also agrees with Eq. (36.15), since ∇ has the dimensions of one over a length. People who are working with electromagnets also get in the habit of calling the unit of H (with the H' definition) “one ampere *turn* per meter”—thinking of the turns of wire on a winding. But a “turn” is really a dimensionless number, so that doesn't need to confuse you. Since our H is equal to $H'/\epsilon_0 c^2$, if you are using the mks system, H (in webers/meter²) is equal to $4\pi \times 10^{-7}$ times H' (in amperes per meter). It is perhaps more convenient to remember that H (in gauss) = 0.0126 H' (in amp/meter).

There is one more horrible thing. Many people who use *our* definition of H have decided to call the units of H and B by *different names*! Even though they have the same dimensions, they call the unit of B one *gauss*, and the unit of H one *oersted* (after Gauss and Oersted, of course). So, in many books you will find graphs with B plotted in gauss and H in oersteds. They are really the same unit— 10^{-4} of the mks unit. We have summarized the confusion about magnetic units in Table 36-1.

Table 36-1

Units of magnetic quantities

$[B]$	= weber/meter ² = 10^4 gauss
$[H]$	= weber/meter ² = 10^4 gauss or 10^4 oersted
$[M]$	= ampere/meter
$[H']$	= ampere/meter

Convenient conversions

B (gauss)	= $10^4 B$ (weber/meter ²)
H (gauss)	= H (oersted)
	= 0.0126 H' (amp/meter)

36-3 The magnetization curve

Now we will look at some simple situations in which the magnetic field is constant, or in which the fields change slowly enough that we can neglect $\partial D/\partial t$ in comparison with j_{cond} . Then the fields obey the equations

$$\nabla \cdot B = 0, \quad (36.16)$$

$$\nabla \times H = j_{\text{cond}}/\epsilon_0 c^2, \quad (36.17)$$

$$H = B - M/\epsilon_0 c^2. \quad (36.18)$$

Suppose we have a torus (a donut) of iron wrapped with a coil of copper wire, as shown in Fig. 36-7(a). A current I flows in the wire. What is the magnetic field? The magnetic field will be mainly inside the iron; there, the lines of B will be circles, as drawn in Fig. 36-7(b). Since the flux of B is continuous, its divergence is zero, and Eq. (36.16) is satisfied. Next, we write Eq. (36.17) in another form by

integrating around the closed loop Γ drawn in Fig. 36-7(b). From Stokes's theorem, we have that

$$\oint_{\Gamma} \mathbf{H} \cdot d\mathbf{s} = \frac{1}{\epsilon_0 c^2} \int_S \mathbf{j}_{\text{cond}} \cdot \mathbf{n} da, \quad (36.19)$$

where the integral of $\mathbf{j}$ is to be carried out over any surface S bounded by Γ . This surface is cut once by each turn of the winding. Each turn contributes the current I to the integral, and, if there are N turns in all, the integral is NI . From the symmetry of our problem, $\mathbf{B}$ is the same all around the curve Γ ; if we assume that the magnetization, and therefore, the field $\mathbf{H}$ is also constant along Γ , Eq. (36.19) becomes

$$Hl = \frac{NI}{\epsilon_0 c^2},$$

where l is the length of the curve Γ . So,

$$H = \frac{1}{\epsilon_0 c^2} \frac{NI}{l}. \quad (36.20)$$

It is because $\mathbf{H}$ is directly proportional to the magnetizing current in cases like this one that $\mathbf{H}$ is sometimes called the *magnetizing field*.

Now all we need is an equation which relates $\mathbf{H}$ to $\mathbf{B}$. But there isn't any such equation! There is, of course, Eq. (36.18), but it is no help because there is no direct relation between $\mathbf{M}$ and $\mathbf{B}$ for a ferromagnetic material like iron. The magnetization $\mathbf{M}$ depends on the whole past history of the iron, and not only on what $\mathbf{B}$ is at the moment.

All is not lost, though. We can get solutions in certain simple cases. If we start out with unmagnetized iron—let's say with iron that has been annealed at high temperatures—then in the simple geometry of the torus, all the iron will have the same magnetic history. Then we can say something about $\mathbf{M}$ —and therefore about the relation between $\mathbf{B}$ and $\mathbf{H}$ —from experimental measurements. The field $\mathbf{B}$ in the torus is, from Eq. (36.20), given as a constant times the current I in the winding. The field $\mathbf{B}$ can be measured by integrating over time the emf in the coil (or in an extra coil wound over the magnetizing coil shown in the figure). This emf is equal to the rate of change of the flux of $\mathbf{B}$, so the integral of the emf with time is equal to $\mathbf{B}$ times the cross-sectional area of the torus.

Figure 36-8 shows the relation between $\mathbf{B}$ and $\mathbf{H}$, observed with a torus of soft iron. When the current is first turned on, $\mathbf{B}$ increases with increasing $\mathbf{H}$ along the curve a . Note the different scales on $\mathbf{B}$ and $\mathbf{H}$; initially, it takes only a relatively small $\mathbf{H}$ to make a large $\mathbf{B}$. Why is $\mathbf{B}$ so much larger with the iron than it would be with air? Because there is a large magnetization $\mathbf{M}$ which is equivalent to a large surface current on the iron—the field $\mathbf{B}$ comes from the *sum* of this current and the conduction current in the winding. Why $\mathbf{M}$ should be so large, we will discuss later.

At higher values of $\mathbf{H}$, the magnetization curve levels off. We say that the iron *saturates*. With the scales of our figure, the curve appears to become horizontal. Actually, it continues to rise slightly—for large fields, $\mathbf{B}$ becomes proportional to $\mathbf{H}$, and with a unit slope. There is no further increase of $\mathbf{M}$. Incidentally, we should point out that if the torus were made of some nonmagnetic material, $\mathbf{M}$ would be zero and $\mathbf{B}$ would equal $\mathbf{H}$ for all fields.

The first thing we notice is that curve a in Fig. 36-8—which is the so-called *magnetization curve*—is highly nonlinear. But it's worse than that. If, after reaching saturation, we decrease the current in the coil to bring $\mathbf{H}$ back to zero, the magnetic field $\mathbf{B}$ falls along curve b . When $\mathbf{H}$ reaches zero, there is still some $\mathbf{B}$ left. Even with no magnetizing current there is a magnetic field in the iron—it has become permanently magnetized. If we now turn on a *negative* current in the coil, the $\mathbf{B}$ - $\mathbf{H}$ curve continues along b until the iron is saturated in the negative direction. If we then bring the current back to zero again, $\mathbf{B}$ goes along curve c . If we alternate the current between large positive and negative values, the $\mathbf{B}$ - $\mathbf{H}$ curve goes back and forth along very nearly the curves b and c . If we vary $\mathbf{H}$ in some arbitrary

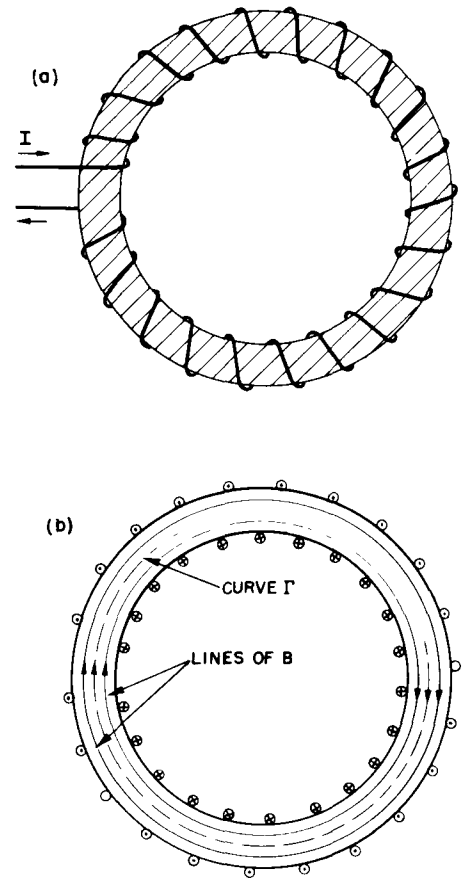


Fig. 36-7. (a) A torus of iron wound with a coil of insulated wire. (b) Cross section of torus showing field lines.

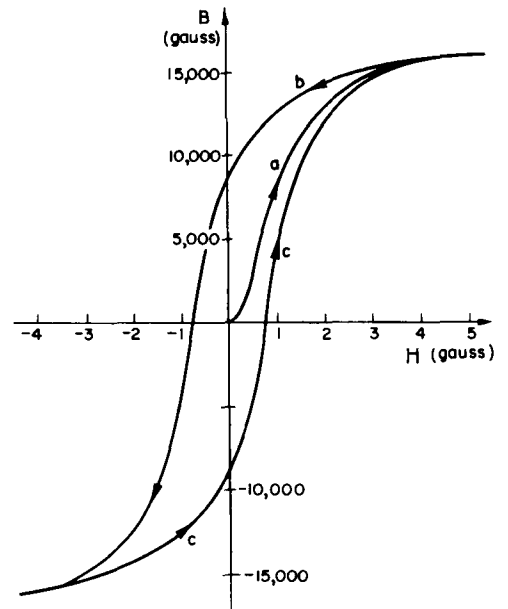


Fig. 36-8. Typical magnetization and hysteresis curves for soft iron.

way, however, we can get more complicated curves which will, in general, lie somewhere between the curves b and c . The loop made by repeated oscillation of the fields is called a *hysteresis* loop of the iron.

We see then that we cannot write a functional relationship like $B = f(H)$, because the value of B at any instant depends not only on what H is at that time, but on its whole past history. Naturally, the magnetization and hysteresis curves are different for different substances. The shape of the curves depends critically on the chemical composition of the material, and also on the details of its preparation and subsequent physical treatment. We will discuss some of the physical explanations for these complications in the next chapter.

36-4 Iron-core inductances

One of the most important applications of magnetic materials is in electrical circuits—for example, in transformers, electric motors, and so on. One reason is that with iron we can control where the magnetic fields go, and also get much larger fields for a given electric current. For example, the typical “toroidal” inductance is made very much like the object shown in Fig. 36-7. For a given inductance, it can be much smaller in volume and use much less copper than an equivalent “air-core” inductance. For a given inductance, we get a much smaller resistance in the winding, so the inductance is more nearly “ideal”—particularly for low frequencies. It is very easy to understand, qualitatively, how such an inductance works. If I is the current in the winding, then the field H which is produced in the inside is proportional to I —as given by Eq. (36.20). The voltage $\mathcal{V}$ across the terminals is related to the magnetic field B . Neglecting the resistance of the winding, the voltage $\mathcal{V}$ is proportional to $\partial B/\partial t$. The inductance $\mathcal{L}$, which is the ratio of $\mathcal{V}$ to dI/dt (see Section 17-7), thus involves the relation between B and H in the iron. Since the B is so much bigger than the H , we get a large factor in the inductance. Physically, what happens is that a small current in the coil, which would ordinarily produce a small magnetic field, causes the little “slave” magnets in the iron to line up and produce a tremendously greater “magnetic” current than the external current in the winding. It is as if we had a lot more current going through the coil than we really have. When we reverse the current, all the little magnets flip over—all those internal currents reverse—and we get a much higher induced emf than we would get without the iron. If we want to calculate the inductance, we can do so through the energy—as described in Section 17-8. The *rate* at which energy is delivered from the current source is $I\mathcal{V}$. The voltage $\mathcal{V}$ is the cross-sectional area A of the core, times N , times dB/dt . From Eq. (36.20), $I = (\epsilon_0 c^2 l/N)H$. So we have

$$\frac{dU}{dt} = \mathcal{V}I = (\epsilon_0 c^2 l A) H \frac{dB}{dt}.$$

Integrating over time, we have

$$U = (\epsilon_0 c^2 l A) \int H dB. \quad (36.21)$$

Notice that lA is the volume of the torus, so we have shown that the energy density $u = U/\text{vol}$ in a magnetic material is given by

$$u = \epsilon_0 c^2 \int H dB. \quad (36.22)$$

An interesting feature is involved here. When we use alternating currents, the iron is driven around a hysteresis loop. Since B is not a single-valued function of H , the integral of $\int H dB$ around one complete cycle is *not* equal to zero. It is the area enclosed inside the hysteresis curve. Thus, the driving source delivers a certain net energy each cycle—an energy proportional to the area inside the hysteresis loop. And that energy is “lost.” It is lost from the electromagnetic goings on, but turns up as heat in the iron. It is called the *hysteresis loss*. To keep such energy losses small, we would like the hysteresis loop to be as narrow as

possible. One way to decrease the area of the loop is to reduce the maximum field that is reached during each cycle. For smaller maximum fields, we get a hysteresis curve like the one shown in Fig. 36-9. Also, special materials are designed to have a very narrow loop. The so-called *transformer irons*—which are iron alloys with a small amount of silicon—have been developed to have this property.

When an inductance is run over a small hysteresis loop, the relationship between B and H can be approximated by a linear equation. People usually write

$$B = \mu H. \quad (36.23)$$

The constant μ is *not* the magnetic moment we have used before. It is called the *permeability* of the iron. (It is also sometimes called the “*relative permeability*.”) The permeability of ordinary irons is typically several thousand. There are special alloys alike “supermalloy” which can have permeabilities as high as a million.

If we use the approximation that $B = \mu H$ in Eq. (36.21), we can write the energy in a toroidal inductance as

$$U = (\epsilon_0 c^2 l A) \mu \int H dH = (\epsilon_0 c^2 l A) \frac{\mu H^2}{2}. \quad (36.24)$$

So the energy density is approximately

$$u \approx \frac{\epsilon_0 c^2}{2} \mu H^2.$$

We can now set the energy of Eq. (36.24) equal to the energy $\mathcal{L}I^2/2$ of an inductance, and solve for $\mathcal{L}$. We get

$$\mathcal{L} = (\epsilon_0 c^2 l A) \mu \left(\frac{H}{I} \right)^2.$$

Using H/I from Eq. (36.20), we have

$$\mathcal{L} = \frac{\mu N^2 A}{\epsilon_0 c^2 l}. \quad (36.25)$$

The inductance is proportional to μ . If you want inductances for such things as audio amplifiers, you will try to operate them on a hysteresis loop where the B - H relationship is as linear as possible. (You will remember that we spoke in Chapter 50, Vol. I, about the generation of harmonics in nonlinear systems.) For such purposes, Eq. (36.23) is a useful approximation. On the other hand, if you *want* to generate harmonics, you may use an inductance which is intentionally operated in a highly nonlinear way. Then you will have to use the complete B - H curves, and analyze what happens by graphical or numerical methods.

A “transformer” is often made by putting two coils on the same torus—or *core*—of a magnetic material. (For the larger transformers, the core is made with rectangular proportions for convenience.) Then a varying current in the “primary” winding causes the magnetic field in the core to change, which induces an emf in the “secondary” winding. Since the flux through *each turn* of both windings is the same, the emf’s in the two windings are in the same ratio as the number of turns on each. A voltage applied to the primary is transformed to a different voltage at the secondary. Since a certain *net* current around the core is needed to produce the required change in the magnetic field, the *algebraic* sum of the currents in the two windings will be fixed and equal to the required “magnetizing” current. If the current drawn from the secondary increases, the primary current must increase in proportion—there is a “transformation” of currents as well as voltage.

36-5 Electromagnets

Now let’s discuss a practical situation which is a little more complicated. Suppose we have an electromagnet of the rather standard form shown in Fig. 36-10—there is a “C-shaped” yoke of iron, with a coil of many turns of wire wrapped around the yoke. What is the magnetic field B in the gap?

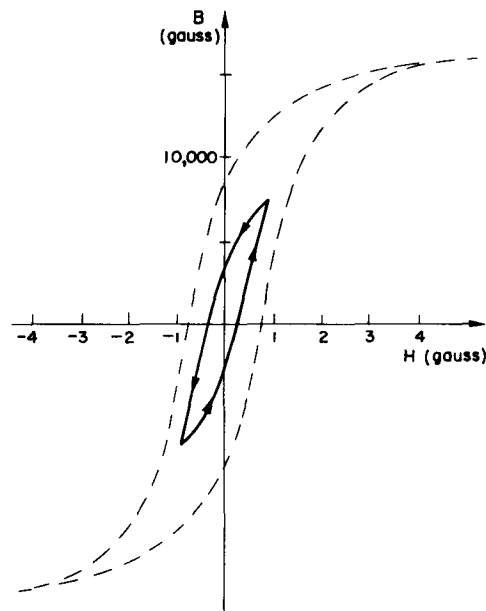


Fig. 36-9. A hysteresis loop that doesn't reach saturation.

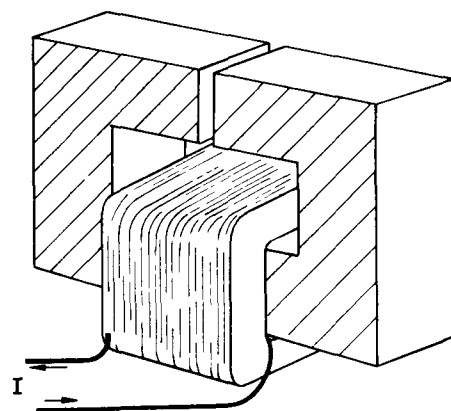


Fig. 36-10. An electromagnet.

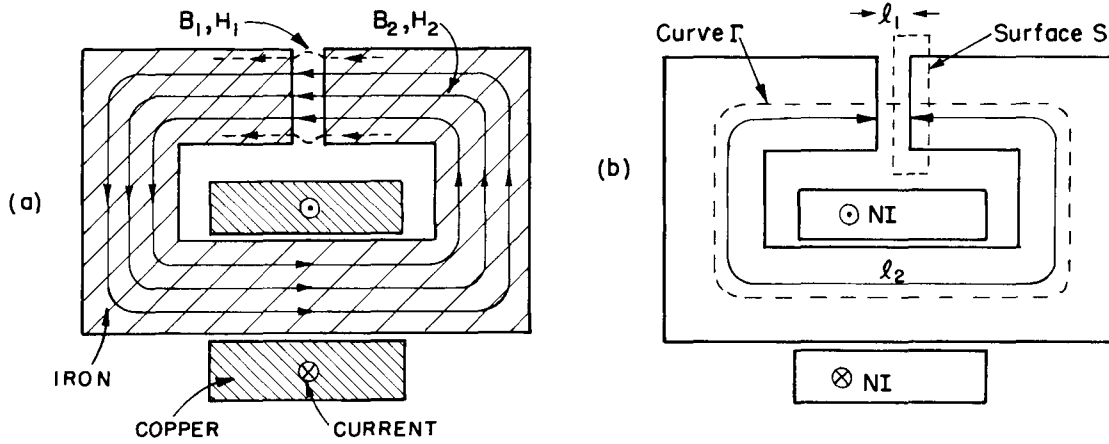


Fig. 36-11. Cross section of an electromagnet.

If the gap thickness is small compared with all the other dimensions, we can, as a first approximation, assume that the lines of $\mathbf{B}$ will go around through the loop, just as they did in the torus. They will look more or less as shown in Fig. 36-11(a). They tend to spread out somewhat in the gap, but if the gap is narrow, this will be a small effect. It is a fair approximation to assume that the flux of $\mathbf{B}$ through any cross section of the yoke is a constant. If the yoke has a uniform cross-sectional area—and if we neglect any edge effects at the gaps or at the corners—we can say that $\mathbf{B}$ is uniform around the yoke.

Also, $\mathbf{B}$ will have the same value in the gap. This follows from Eq. (36.16). Imagine the closed surface S , shown in Fig. 36-11(b), which has one face in the gap and the other in the iron. The total flux of $\mathbf{B}$ out of this surface must be zero. Calling B_1 the field in the gap and B_2 the field in the iron, we have that

$$B_1 A_1 - B_2 A_2 = 0.$$

Since $A_1 = A_2$ (to our approximation), it follows that $B_1 = B_2$.

Now let's look at H . We can again use Eq. (36.19), taking the line integral around the curve Γ in Fig. 36-11(b). As before, the right-hand side is NI , the number of turns times the current. Now, however, H will be different in the iron and in the air. Calling H_2 the field in the iron and l_2 the path length around the yoke, this part of the curve will contribute the amount $H_2 l_2$ to the integral. Calling H_1 the field in the gap and l_1 the gap thickness, we get the contribution $H_1 l_1$ from the gap. We have that

$$H_1 l_1 + H_2 l_2 = \frac{NI}{\epsilon_0 c^2}. \quad (36.26)$$

Now we know something else: that in the air gap, the magnetization is negligible, so that $B_1 = H_1$. Since $B_1 = B_2$, Eq. (36.26) becomes

$$B_2 l_1 + H_2 l_2 = \frac{NI}{\epsilon_0 c^2}. \quad (36.27)$$

We still have two unknowns. To find B_2 and H_2 , we need another relationship—namely, the one which relates B to H in the iron.

If we can make the approximation that $B_2 = \mu H_2$, we can solve the equation algebraically. However, let's do the general case, in which the magnetization curve of the iron is one like that shown in Fig. 36-8. What we want is the simultaneous solution of this functional relationship together with Eq. (36.27). We can find it by plotting a graph of Eq. (36.27) on the same graph with the magnetization curve, as is done in Fig. 36-12. Where the two curves intersect, we have our solution.

For a given current I , the function (36.27) is the straight line marked $I > 0$ in Fig. 36-12. The line intersects the H -axis ($B_2 = 0$) at $H_2 = NI/\epsilon_0 c^2 l_2$, and the slope is $-l_2/l_1$. Different currents just shift the line horizontally. From Fig.

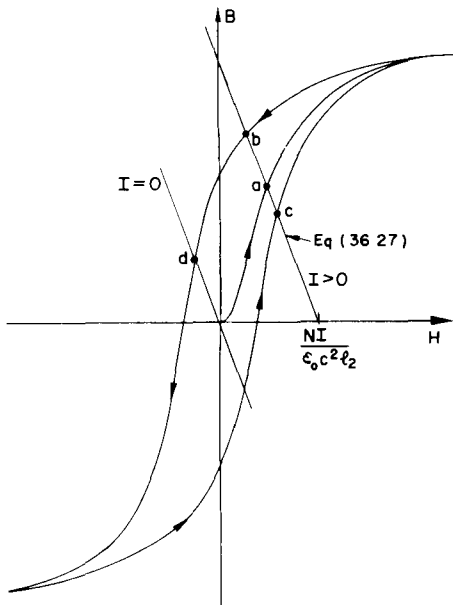


Fig. 36-12. Solving for the field in an electromagnet.

36-12, we see that for a given current there are several different solutions, depending on how you got there. If you have just built the magnet and turned the current up to I , the field B_2 (which is also B_1) will have the value given by point a . If you have run the current to some very high value and come down to I , the field will be given by point b . Or, if you have just had a high negative current in the magnet and then come up to I , the field is the one at point c . The field in the gap will depend on what you have done in the past.

When the current in the magnet is zero, the relation between B_2 and H_2 in Eq. (36.27) is shown by the line marked $I = 0$ in the figure. There are still various possible solutions. If you have first saturated the iron, there may be a considerable residual field in the magnet as given by point d . You can take the coil off, and you have a permanent magnet. You can see that for a good permanent magnet, you would want a material with a *wide* hysteresis loop. Special alloys, such as Alnico V, have very wide loops.

36-6 Spontaneous magnetization

We now turn to the question of why it is that in ferromagnetic materials a small magnetic field produces such a large magnetization. The magnetization of ferromagnetic materials like iron and nickel comes from the magnetic moment of the electrons in the inner shell of the atom. Each electron has a magnetic moment μ equal to $q/2m$ times its g -factor, times its angular momentum $\mathbf{J}$. For a single electron with no net orbital motion, $g = 2$, and the component of $\mathbf{J}$ in any direction—say the z -direction—is $\pm \hbar/2$, so the component of μ along the z -axis is

$$\mu_z = \frac{q\hbar}{2m} = 0.928 \times 10^{-23} \text{ amp}\cdot\text{m}^2. \quad (36.28)$$

In an iron atom, there are actually two electrons that contribute to the ferromagnetism, so to keep the discussion simpler we will talk about nickel, which is ferromagnetic like iron but which has only one electron in the inner shell. (It is easy to extend the arguments to iron.)

Now the point is that in the presence of an external field $\mathbf{B}$, the atomic magnets tend to line up with the field, but are knocked about by thermal motions just as we described for paramagnetic materials. In the last chapter we found out that the balance between a magnetic field trying to line up the atomic magnets and the thermal motions trying to derange them produced the result that the mean magnetic moment per unit volume will end up as

$$M = N\mu \tanh \frac{\mu B_a}{kT}. \quad (36.29)$$

By B_a we mean the field acting at the atom, and kT is the Boltzmann energy. In the theory of paramagnetism we used for B_a just B itself, neglecting the part of the field at any given atom contributed by the atoms nearby. In the ferromagnetic case, there is a complication. We shouldn't use the average field in the iron for the B_a acting on an individual atom. Instead, we must do as we did in the case of dielectrics—we have to find the *local* field acting at a single atom. For an exact calculation we should add up the fields at the atom in question contributed by all of the other atoms in the crystal lattice. But as we did for dielectrics, we will make the approximation that the field at an atom is the same as we would find in a small spherical hole in the material—assuming that the moments of the atoms in the neighborhood are not changed by the presence of the hole.

Following the arguments we made in Chapter 11, we might think that we could write

$$B_{\text{hole}} = B + \frac{1}{3} \frac{M}{\epsilon_0 c^2} \quad (\text{wrong!}).$$

But that is not right. We *can*, however, make use of the results of Chapter 11 if we make a careful comparison of the equations of Chapter 11 with the equations

for ferromagnetism in this chapter. Let's put together the corresponding equations. For regions where there are no conduction currents or charges we have:

Electrostatics	Static ferromagnetism
$\nabla \cdot \left(\mathbf{E} + \frac{\mathbf{P}}{\epsilon_0} \right) = 0$	$\nabla \cdot \mathbf{B} = 0$
$\nabla \times \mathbf{E} = 0$	$\nabla \times \left(\mathbf{B} - \frac{\mathbf{M}}{\epsilon_0 c^2} \right) = 0$

(36.30)

These two sets of equations can be thought of as analogous if we make the following *purely mathematical* correspondences:

$$\mathbf{E} \rightarrow \mathbf{B} - \frac{\mathbf{M}}{\epsilon_0 c^2}, \quad \mathbf{E} + \frac{\mathbf{P}}{\epsilon_0} \rightarrow \mathbf{B}.$$

This is the same as making the analogy

$$\mathbf{E} \rightarrow \mathbf{H}, \quad \mathbf{P} \rightarrow \mathbf{M}/c^2. \quad (36.31)$$

In other words, if we write the equations of ferromagnetism as

$$\begin{aligned} \nabla \cdot \left(\mathbf{H} + \frac{\mathbf{M}}{\epsilon_0 c^2} \right) &= 0, \\ \nabla \times \mathbf{H} &= 0, \end{aligned} \quad (36.32)$$

they *look like* the equations of electrostatics.

This purely algebraic correspondence has led to some confusion in the past. People tended to think that $\mathbf{H}$ was “*the* magnetic field.” But, as we have seen, $\mathbf{B}$ and $\mathbf{E}$ are physically the fundamental fields, and $\mathbf{H}$ is a derived idea. So although the *equations* are analogous, the *physics* is not analogous. However, that doesn't need to stop us from using the principle that the same equations have the same solutions.

We can use our earlier results for the electric field inside of holes of various shapes in dielectrics—summarized in Fig. 36-1—to find the field $\mathbf{H}$ inside of corresponding holes. Knowing $\mathbf{H}$, we can determine $\mathbf{B}$. For instance (using the results we summarized in Section 1), the field $\mathbf{H}$ in a needle-shaped hole parallel to $\mathbf{M}$ is the same as the $\mathbf{H}$ in the material,

$$\mathbf{H}_{\text{hole}} = \mathbf{H}_{\text{material}}.$$

But since $\mathbf{M}$ in the hole is zero, we have

$$\mathbf{B}_{\text{hole}} = \mathbf{B}_{\text{material}} - \frac{\mathbf{M}}{\epsilon_0 c^2}. \quad (36.33)$$

On the other hand, for a disc-shaped hole, perpendicular to $\mathbf{M}$, we have

$$\mathbf{E}_{\text{hole}} = \mathbf{E}_{\text{dielectric}} + \frac{\mathbf{P}}{\epsilon_0},$$

which translates into

$$\mathbf{H}_{\text{hole}} = \mathbf{H}_{\text{material}} + \frac{\mathbf{M}}{\epsilon_0 c^2}.$$

Or, in terms of $\mathbf{B}$,

$$\mathbf{B}_{\text{hole}} = \mathbf{B}_{\text{material}}. \quad (36.34)$$

Finally, for a spherical hole, by making our analogy with Eq. (36.3) we would have

$$\mathbf{H}_{\text{hole}} = \mathbf{H}_{\text{material}} + \frac{\mathbf{M}}{3\epsilon_0 c^2}$$

or

$$\mathbf{B}_{\text{hole}} = \mathbf{B}_{\text{material}} - \frac{2}{3} \frac{\mathbf{M}}{\epsilon_0 c^2}. \quad (36.35)$$

This result is quite different from what we got for $\mathbf{E}$.

It is, of course, possible to get these results in a more physical way, by using the Maxwell equations directly. For example, Eq. (36.34) follows directly from $\nabla \cdot \mathbf{B} = 0$. (You use a gaussian surface that is half in the material and half out.) Similarly, you can get Eq. (36.33) by using a line integral along a curve that goes up inside the hole and returns through the material. Physically, the field in the hole is reduced because of the surface currents—which are given by $\nabla \times \mathbf{M}$. We will leave it for you to show that Eq. (36.35) can also be obtained by considering the effects of the surface currents on the boundary of the spherical cavity.

In finding the equilibrium magnetization from Eq. (36.29), it turns out to be most convenient to deal with $\mathbf{H}$; so write

$$\mathbf{B}_a = \mathbf{H} + \lambda \frac{\mathbf{M}}{\epsilon_0 c^2}. \quad (36.36)$$

In the spherical hole approximation, we would have $\lambda = \frac{1}{3}$, but, as you will see, we will want later to use some other value, so we leave it as an adjustable parameter. Also, we will take all the fields in the same direction so that we won't need to worry about the vector directions. If we were now to substitute Eq. (36.36) into Eq. (36.29), we would have one equation that relates the magnetization M to the magnetizing field H :

$$M = N\mu \tanh \left(\frac{H + \lambda M / \epsilon_0 c^2}{kT} \right).$$

It is, however, an equation that cannot be solved explicitly, so we will do it graphically.

Let's put the problem in a generalized form by writing Eq. (36.29) as

$$\frac{M}{M_{\text{sat}}} = \tanh x, \quad (36.37)$$

where M_{sat} is the saturation value of the magnetization, namely, $N\mu$, and x represents $\mu B_a / kT$. The dependence of M/M_{sat} on x is shown by curve a in Fig. 36-13. We can also write x as a function of M —using Eq. (36.36) for B_a —as

$$x = \frac{\mu B_a}{kT} = \frac{\mu H}{kT} + \left(\frac{\mu \lambda M_{\text{sat}}}{\epsilon_0 c^2 kT} \right) \frac{M}{M_{\text{sat}}}. \quad (36.38)$$

For any given value of H , this is a straight-line relationship between M/M_{sat} and x . The x intercept is at $x = \mu H / kT$, and the slope is $\epsilon_0 c^2 kT / \mu \lambda M_{\text{sat}}$. For any particular H , we would have a line like the one marked b in Fig. 36-13. The intersection of curves a and b gives us the solution for M/M_{sat} . We have solved the problem.

Let's look at how the solutions will go for various circumstances. We start with $H = 0$. There are two possible situations, shown by the lines b_1 and b_2 in Fig. 36-14. You will notice from Eq. (36.38) that the slope of the line is proportional to the absolute temperature T . So, at *high temperatures* we would have a line like b_1 . The solution is $M/M_{\text{sat}} = 0$. When the magnetizing field H is zero, the magnetization is also zero. But at *low temperatures*, we would have a line like b_2 , and there are *two solutions* for M/M_{sat} —one with $M/M_{\text{sat}} = 0$ and one with M/M_{sat} near one. It turns out that only the upper solution is stable—as you can see by considering small variations about these solutions.

According to these ideas, then, a magnetic material should magnetize itself *spontaneously* at sufficiently low temperatures. In short, when the thermal motions are small enough, the coupling between the atomic magnets causes them all to line up parallel to each other—we have a permanently magnetized material analogous to the ferroelectrics we discussed in Chapter 11.

If we start at high temperatures and come down, there is a critical temperature, called the Curie temperature T_c , where the ferromagnetic behavior suddenly sets in. This temperature corresponds to the line b_3 of Fig. 36-14, which is tangent to the curve a , and has, therefore, a slope of 1. The Curie temperature is given by

$$\frac{\epsilon_0 c^2 k T_c}{\mu \lambda M_{\text{sat}}} = 1. \quad (36.39)$$

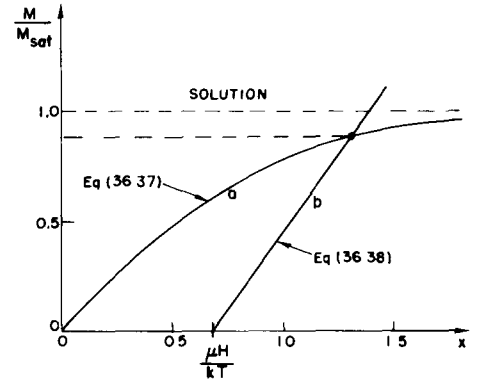


Fig. 36-13. A graphical solution of Eqs. (36.37) and (36.38).

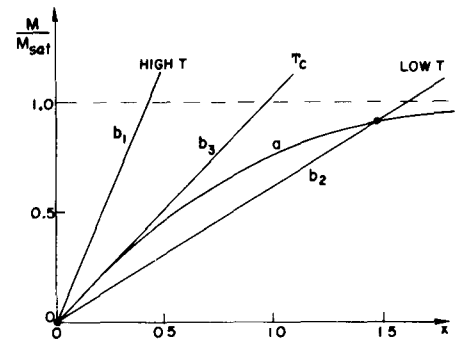


Fig. 36-14. Finding the magnetization when $H = 0$.

We can, if we wish, write Eq. (36.38) more simply in terms of T_c as

$$x = \frac{\mu H}{kT} + \frac{T_c}{T} \left(\frac{M}{M_{\text{sat}}} \right). \quad (36.40)$$

Now we want to see what happens for small magnetizing fields H . We can see from Fig. 36-14 how things will go if we shift our straight lines a little to the right. For the low-temperature case, the intersection point will move out a little bit along the low-slope part of curve a , and M will change relatively little. For the high-temperature case, however, the intersection point runs up the steep part of curve a , and M will change relatively rapidly. In fact, we can approximate this part of curve a by a straight line of unit slope, and write:

$$\frac{M}{M_{\text{sat}}} = x = \frac{\mu H}{kT} + \frac{T_c}{T} \left(\frac{M}{M_{\text{sat}}} \right).$$

Now we can solve for M/M_{sat} :

$$\frac{M}{M_{\text{sat}}} = \frac{\mu H}{k(T - T_c)}. \quad (36.41)$$

We have a law that is something like the one we had for paramagnetism. For paramagnetism, we had

$$\frac{M}{M_{\text{sat}}} = \frac{\mu B}{kT}. \quad (36.42)$$

One difference now is that we have the magnetization in terms of H , which includes some of the effects of the interaction of the atomic magnets, but the main difference is that the magnetization is inversely proportional to the *difference* between T and T_c , instead of to the absolute temperature T , alone. Neglecting the interactions between neighboring atoms corresponds to taking $\lambda = 0$, which from Eq. (36.39) means taking $T_c = 0$. Then the results are just what we had in Chapter 35.

We can check our theoretical picture with the experimental data for nickel. It is observed experimentally that the ferromagnetic behavior of nickel disappears when its temperature is raised above 631°K . We can compare this with T_c calculated from Eq. (36.39). Remembering that $M_{\text{sat}} = \mu N$, we have

$$T_c = \lambda \frac{N_\mu^2}{k\epsilon_0 c^2}.$$

From the density and atomic weight of nickel, we get

$$N = 9.1 \times 10^{28} \text{ m}^{-3}.$$

Taking μ from Eq. (36.28), and setting $\lambda = \frac{1}{3}$, we get

$$T_c = 0.24^\circ\text{K}.$$

There is a discrepancy of a factor of about 2600! Our theory of ferromagnetism fails completely.

We can try to “patch up” the theory as Weiss did by saying that for some unknown reason λ is not one-third, but $(2600) \times \frac{1}{3}$ —or about 900. It turns out that one gets similar values for other ferromagnetic materials like iron. To see what this means, let’s go back to Eq. (36.36). We see that a large λ means that B_a , the local field on the atom, appears to be much, much larger than we would think. In fact, writing $H = B - M/\epsilon_0 c^2$, we have

$$B_a = B + \frac{(\lambda - 1)M}{\epsilon_0 c^2}.$$

According to our original idea—with $\lambda = \frac{1}{3}$ —the local magnetization M *reduces* the effective field B_a by the amount $-\frac{2}{3}M/\epsilon_0$. Even if our model of a spherical hole were not very good, we would still expect *some* reduction. Instead, to explain

the phenomenon of ferromagnetism, we have to imagine that the magnetization of the field *enhances* the local field by some large factor—like one thousand or more. There doesn't seem to be any reasonable way to manufacture such tremendous fields at an atom—nor even fields of the proper sign! Clearly, our “magnetic” theory of ferromagnetism is a dismal failure. We must conclude, then, that ferromagnetism has to do with some *nonmagnetic* interaction between the spinning electrons in neighboring atoms. This interaction must generate a strong tendency for all of the nearby spins to line up in one direction. We will see later that it has to do with quantum mechanics and the Pauli exclusion principle.

Finally, we look at what happens at low temperatures—for $T < T_c$. We have seen that there will then be a spontaneous magnetization—even with $H = 0$ —given by the intersection of the curves a and b_2 of Fig. 36-14. If we solve for M for various temperatures—by varying the slope of the line b_2 —we get the theoretical curve shown in Fig. 36-15. This curve should be the same for all ferromagnetic materials for which the atomic moment comes from a single electron. The curves for other materials are only slightly different.

In the limit, as T goes to absolute zero, M goes to M_{sat} . As the temperature is increased, the magnetization decreases, falling to zero at the Curie temperature. The points in Fig. 36-15 are the experimental observations for nickel. They fit the theoretical curve fairly well. Even though we don't understand the basic mechanism, the general features of the theory seem to be correct.

Finally, there is one more disturbing discrepancy in our attempt to understand ferromagnetism. We have found that above some temperature the material should behave like a paramagnetic substance with a magnetization M proportional to H (or B), and that below that temperature it should become spontaneously magnetized. But that's not what we found when we measured the magnetization curve for iron. It only became permanently magnetized *after* we had “magnetized” it. According to the ideas just discussed, it would magnetize itself! What is wrong? Well, it turns out that if you look at a *small enough crystal* of iron or nickel, it is indeed completely magnetized! But in large pieces of iron, there are many small regions or “domains” that are magnetized in different directions, so that on a large scale the *average* magnetization appears to be zero. In each small domain, however, the iron has a locked-in magnetization with M nearly equal to M_{sat} . The consequences of this domain structure are that gross properties of large pieces of material are quite different from the microscopic properties that we have really been treating. We will take up in the next lecture the story of the practical behavior of bulk magnetic materials.

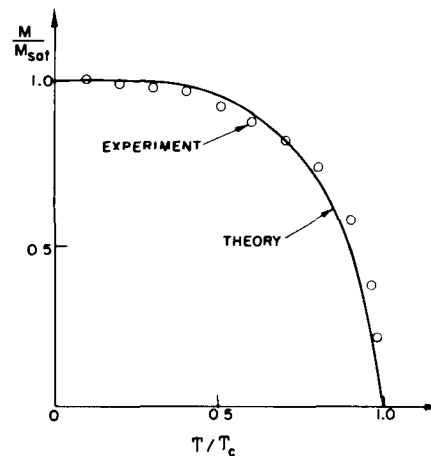


Fig. 36-15. Spontaneous magnetization as a function of temperature for nickel.

Magnetic Materials

37-1 Understanding ferromagnetism

In this chapter we will discuss the behavior and peculiarities of ferromagnetic materials and of other strange magnetic materials. Before proceeding to study magnetic materials, however, we will review very quickly some of the things about the general theory of magnets that we learned in the last chapter.

First, we imagine the atomic currents inside the material that are responsible for the magnetism, and then describe them in terms of a volume current density $\mathbf{j}_{\text{mag}} = \nabla \times \mathbf{M}$. We emphasize that this is not supposed to represent the *actual* currents. When the magnetization is uniform the currents do not *really* cancel out precisely; that is, the whirling currents of one electron in one atom and the whirling currents of an electron in another atom do not overlap in such a way that the sum is exactly zero. Even within a single atom the distribution of magnetism is *not* smooth. For instance, in an iron atom the magnetization is distributed in a more or less spherical shell, not too close to the nucleus and not too far away. Thus, magnetism in matter is quite a complicated thing in its details; it is very irregular. However, we are obliged now to ignore this detailed complexity and discuss phenomena from a gross, average point of view. Then it is true that the *average* current in the interior region, over any finite area that is big compared with an atom, is zero when $\mathbf{M} = 0$. So, what we mean by magnetization per unit volume and $\mathbf{j}_{\text{mag}}$ and so on, at the level we are now considering, is an average over regions that are large compared with the space occupied by a single atom.

In the last chapter, we also discovered that a ferromagnetic material has the following interesting property: above a certain temperature it is not strongly magnetic, whereas below this temperature it becomes magnetic. This fact is easily demonstrated. A piece of nickel wire at room temperature is attracted by a magnet. However, if we heat it above its Curie temperature with a gas flame, it becomes nonmagnetic and is not attracted toward the magnet—even when brought quite close to the magnet. If we let it lie near the magnet while it cools off, at the instant its temperature falls below the critical temperature it is suddenly attracted again by the magnet!

The general theory of ferromagnetism that we will use supposes that the spin of the electron is responsible for the magnetization. The electron has spin one-half and carries one Bohr magneton of magnetic moment $\mu = \mu_B = q_e \hbar / 2m$. The electron spin can be pointed either “up” or “down.” Because the electron has a negative charge, when its spin is “up” it has a *negative* moment, and when its spin is “down” it has a *positive* moment. With our usual conventions, the moment μ of the electron is opposite its spin. We have found that the energy of orientation of a magnetic dipole in a given applied field $\mathbf{B}$ is $-\mu \cdot \mathbf{B}$, but the energy of the spinning electrons depends on the neighboring spin alignments as well. In iron, if the moment of a nearby atom is “up,” there is a very strong tendency that the moment of the one next to it will also be “up.” That is what makes iron, cobalt, and nickel so strongly magnetic—the moments all want to be parallel. The first question we have to discuss is *why*.

Soon after the development of quantum mechanics, it was noticed that there is a very strong *apparent* force—not a magnetic force or any other kind of actual force, but only an apparent force—trying to line the spins of nearby electrons *opposite* to one another. These forces are closely related to chemical valence forces. There is a principle in quantum mechanics—called the *exclusion principle*—that

37-1 Understanding ferromagnetism

37-2 Thermodynamic properties

37-3 The hysteresis curve

37-4 Ferromagnetic materials

37-5 Extraordinary magnetic materials

References: Bozorth, R. M., “Magnetism,” *Encyclopaedia Britannica*, Vol. 14, 1957, pp. 636–667.

Kittel, C., *Introduction to Solid State Physics*, John Wiley and Sons, Inc., New York, 2nd ed., 1956.

two electrons cannot occupy exactly the same state, that they cannot be in exactly the same condition as to location and spin orientation.* For example, if they are at the same point, the only alternative is to have their spins opposite. So, if there is a region of space between atoms where electrons like to congregate (as in a chemical bond) and we want to put another electron on top of one already there, the only way to do it is to have the spin of the second one pointed opposite to the spin of the first one. To have the spins parallel is against the law, unless the electrons stay away from each other. This has the effect that a pair of parallel-spin electrons near to each other have much more energy than a pair of opposite-spin electrons; the net effect is as though there were a force trying to turn the spin over. Sometimes this spin-turning force is called the *exchange force*, but that only makes it more mysterious—it is not a very good term. It is just because of the exclusion principle that electrons have a tendency to make their spins opposite. In fact, that is the explanation of the *lack* of magnetism in almost all substances! The spins of the free electrons on the outside of the atoms have tremendous tendency to balance in opposite directions. The problem is to explain why for materials like iron it is just the reverse of what we should expect.

We have summarized the supposed alignment effect by adding a suitable term in the energy equation, by saying that if the electron magnets in the neighborhood have a mean magnetization M , then the moment of an electron has a strong tendency to be in the same direction as the average magnetization of the atoms in the neighborhood. Thus, we may write for the two possible spin orientations,†

$$\begin{aligned}\text{Spin "up" energy} &= +\mu \left(H + \frac{\lambda M}{\epsilon_0 c^2} \right), \\ \text{Spin "down" energy} &= -\mu \left(H + \frac{\lambda M}{\epsilon_0 c^2} \right).\end{aligned}\tag{37.1}$$

When it was clear that quantum mechanics could supply a tremendous spin-orientating force—even if, apparently, of the wrong sign—it was suggested that ferromagnetism might have its origin in this same force, that due to the complexities of iron and the large number of electrons involved, the sign of the interaction energy would come out the other way around. Since the time this was thought of—in about 1927 when quantum mechanics was first being understood—many people have been making various estimates and semicalculations, trying to get a theoretical prediction for λ . The most recent calculations of the energy between the two electron spins in iron—assuming that the interaction is a direct one between the two electrons in neighboring atoms—still give the *wrong sign*. The present understanding of this is again to assume that the complexity of the situation is somehow responsible and to hope that the next man who makes the calculation with a more complicated situation will get the right answer!

It is believed that the up-spin of one of the electrons in the inside shell, which is making the magnetism, tends to make the conduction electrons which fly around the outside have the opposite spin. One might expect this to happen because the conduction electrons come into the same region as the “magnetic” electrons. Since they move around, they can carry their prejudice for being upside down over to the next atom; that is, one “magnetic” electron tries to force the conduction electrons to be opposite, and the conduction electron then makes the next “magnetic” electron opposite to *it*. The double interaction is equivalent to an interaction which tries to line up the two “magnetic” electrons. In other words, the tendency to make parallel spins is the result of an intermediary that tends to some extent to be opposite to both. This mechanism does not require that the conduction electrons be completely “upside down.” They could just have a slight prejudice to be down, just enough to load the “magnetic” odds the other way. This is the mechanism that

* See Chapter 43.

† We write these equations with $H = B - M/\epsilon_0 c^2$ instead of B to agree with the work of the last chapter. You might prefer to write $U = \pm \mu B_a = \pm \mu (B + \lambda' M/\epsilon_0 c^2)$, where $\lambda' = \lambda - 1$. It's the same thing.

the people who have calculated such things now believe is responsible for ferromagnetism. But we must emphasize that to this day nobody can calculate the magnitude of λ simply by knowing that the material is number 26 in the periodic table. In short, we don't thoroughly understand it.

Now let us continue with the theory, and then come back later to discuss a certain error involved in the way we have set it up. If the magnetic moment of a certain electron is "up," energy comes both from the external field and also from the tendency of the spins to be parallel. Since the energy is lower when the spins are parallel, the effect is sometimes thought of as due to an "effective internal field." But remember, it is *not* due to a true *magnetic* force; it is an interaction that is more complicated. In any case, we take Eqs. (37.1) as the formulas for the energies of the two spin states of a "magnetic" electron. At a temperature T , the relative probability of these two states is proportional to $e^{-\text{energy}/kT}$, which we can write as e^{+x} , with $x = \mu(H + \lambda M/\epsilon_0 c^2)/kT$. Then, if we calculate the mean value of the magnetic moment, we find (as in the last chapter) that it is

$$M = N\mu \tanh x. \quad (37.2)$$

Now we would like to calculate the internal energy of the material. We note that the energy of an electron is exactly proportional to the magnetic moment, so that the calculation of the mean moment and the calculation of the mean energy are the same—except that in place of μ in Eq. (37.2) we would write $-\mu B$, which is $-\mu(H + \lambda M/\epsilon_0 c^2)$. The mean energy is then

$$\langle U \rangle_{av} = -N\mu \left(H + \frac{\lambda M}{\epsilon_0 c^2} \right) \tanh x.$$

Now this is not quite correct. The term $\lambda M/\epsilon_0 c^2$ represents interactions of all possible *pairs* of atoms, and we must remember to count each pair only *once*. (When we consider the energy of one electron in the field of the rest and then the energy of a second electron in the field of the rest, we have counted part of the first energy once more.) Thus, we must divide the *mutual interaction term* by two, and our formula for the energy then turns out to be

$$\langle U \rangle_{av} = -N\mu \left(H + \frac{\lambda M}{2\epsilon_0 c^2} \right) \tanh x. \quad (37.3)$$

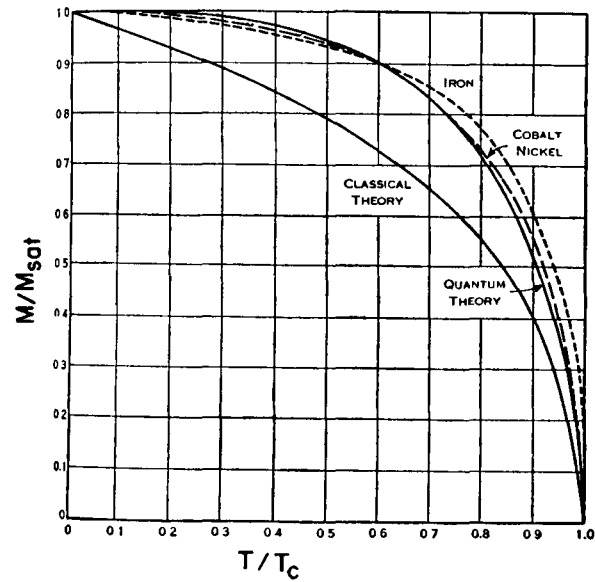
In the last chapter we discovered an interesting thing—that below a certain temperature the material finds a solution to the equations in which the magnetic moment *is not zero*, even with no external magnetizing field. When we set $H = 0$ in Eq. (37.2), we found that

$$\frac{M}{M_{sat}} = \tanh \left(\frac{T_c}{T} \frac{M}{M_{sat}} \right), \quad (37.4)$$

where $M_{sat} = N\mu$, and $T_c = \mu\lambda M_{sat}/k\epsilon_0 c^2$. When we solve this equation (graphically or otherwise), we find that the ratio M/M_{sat} as a function of T/T_c is a curve like that labeled "quantum theory" in Fig. 37-1. The dashed curve marked "cobalt, nickel" shows the experimental results for crystals of these elements. The theory and experiment are in reasonably good agreement. The figure also shows the result of the classical theory in which the calculation is carried out assuming that the atomic magnets can have all possible orientations in space. You can see that this assumption gives a prediction that is not even close to the experimental facts.

Even the quantum theory deviates from the observed behavior at both high and low temperatures. The reason for the deviations is that we have made a rather sloppy approximation in the theory: We have assumed that the energy of an atom depends upon the *mean* magnetization of its neighboring atoms. In other words, for each one that is "up" in the neighborhood of a given atom, there will be a contribution of energy due to that quantum mechanical alignment effect. But how many are there pointed "up"? On the average, that is measured by the

Fig. 37-1. The spontaneous magnetization ($H = 0$) of ferromagnetic crystals as a function of temperature. [Permission from Encyclopaedia Britannica.]



magnetization M —but only *on the average*. A particular atom somewhere might find *all* its neighbors “up.” Then its energy will be larger than the average. Another one might find some up and some down, perhaps averaging to zero, and it would have *no* energy from that term, and so on. What we ought to do is to use some more complicated kind of average, because the atoms in different places have different environments, and the numbers up and down are different for different ones. Instead of just taking one atom subjected to the average influence, we should take each one in its actual situation, compute its energy, and find the *average energy*. But how do we find out how many are “up” and how many are “down” in the neighborhood? That is, of course, just what we are trying to calculate—the number “up” and “down”—so we have a very complicated interconnected problem of correlations, a problem which has never been solved. It is an intriguing and exciting one which has existed for years and on which some of the greatest names in physics have written papers, but even they have not completely solved it.

It turns out that at low temperatures, when almost all the atomic magnets are “up” and only a few are “down,” it is easy to solve; and at high temperatures, far above the Curie temperature T_c when they are almost all random, it is again easy. It is often easy to calculate small departures from some simple, idealized situation, so it is fairly well understood why there are deviations from the simple theory at low temperature. It is also understood physically that for statistical reasons the magnetization *should* deviate at high temperatures. But the exact behavior near the Curie point has never been thoroughly figured out. That’s an interesting problem to work out some day if you want a problem that has never been solved.

37-2 Thermodynamic properties

In the last chapter we laid the groundwork necessary for calculating the thermodynamic properties of ferromagnetic materials. These are, naturally, related to the internal energy of the crystal, which includes interactions of the various spins, given by Eq. (37.3). For the energy of the spontaneous magnetization below the Curie point, we can set $H = 0$ in Eq. (37.3), and—noticing that $\tanh x = M/M_{\text{sat}}$ —we find a mean energy proportional to M^2 :

$$\langle U \rangle_{\text{av}} = - \frac{N\mu\lambda M^2}{2\epsilon_0 c^2 M_{\text{sat}}} \quad (37.5)$$

If we now plot the energy due to the magnetism as a function of temperature, we get a curve which is the negative of the square of the curve of Fig. 37-1, as drawn in Fig. 37-2(a). If we were to measure then the *specific heat* of such a material we would obtain a curve which is the derivative of 37-2(a). It is shown in Fig.

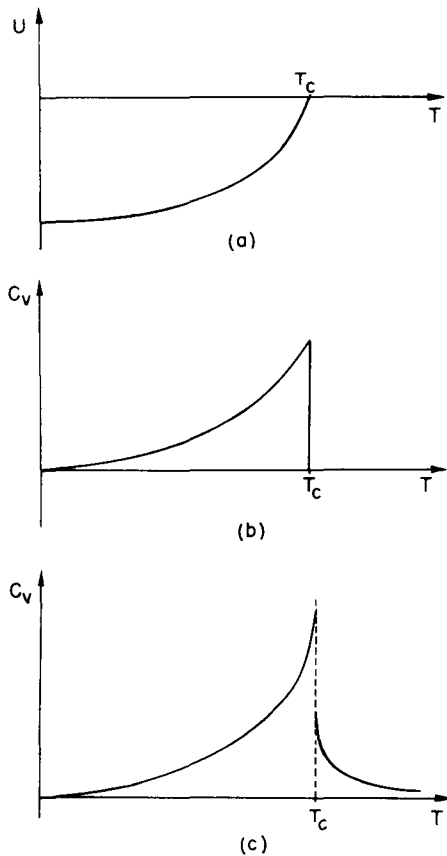


Fig. 37-2. The energy per unit volume and specific heat of a ferromagnetic crystal.

37-2(b). It rises slowly with increasing temperature, but falls suddenly to zero at $T = T_c$. The sharp drop is due to the change in slope of the magnetic energy and is reached right at the Curie point. So without any magnetic measurements at all we could have discovered that something was going on inside of iron or nickel by measuring this thermodynamic property. However, both experiment and improved theory (with fluctuations included) suggest that this simple curve is wrong and that the true situation is really more complicated. The curve goes higher at the peak and falls to zero somewhat slowly. Even if the temperature is high enough to randomize the spins *on the average*, there are still local regions where there is a certain amount of polarization, and in these regions the spins still have a little extra energy of interaction—which only dies out slowly as things get more and more random with further increases in temperature. So the actual curve looks like Fig. 37-2(c). One of the challenges of theoretical physics today is to find an exact theoretical description of the character of the specific heat near the Curie transition—an intriguing problem which has not yet been solved. Naturally, this problem is very closely related to the shape of the magnetization curve in the same region.

Now we want to describe some experiments, other than thermodynamic ones, which show that there is something *right* about our interpretation of magnetism. When the material is magnetized to saturation at low enough temperatures, M is very nearly equal to M_{sat} —nearly all the spins are parallel, as well as their magnetic moments. We can check this by an experiment. Suppose we suspend a bar magnet by a thin fiber and then surround it by a coil so that we can reverse the magnetic field without touching the magnet or putting any torque on it. This is a very difficult experiment because the magnetic forces are so enormous that any irregularities, any lopsidedness, or any lack of perfection in the iron will produce accidental torques. However, the experiment has been done under careful conditions in which such accidental torques are minimized. By means of the magnetic field from a coil that surrounds the bar, we turn all the atomic magnets over at once. When we do this we also change the angular momenta of all the spins from “up” to “down” (see Fig. 37-3). If angular momentum is to be conserved when the spins all turn over, the rest of the bar must have an opposite change in angular momentum. The whole magnet will start to spin. And sure enough, when we do the experiment, we find a slight turning of the magnet. We can measure the total angular momentum given to the whole magnet, and this is simply N times $\hbar$, the change in the angular momentum of each spin. The ratio of angular momentum to magnetic moment measured this way comes out to within about 10 percent of what we calculate. Actually, our calculations assume that the atomic magnets are due purely to the electron spin, but there is, in addition, some orbital motion also in most materials. The orbital motion is not completely free of the lattice and does not contribute much more than a few percent to the magnetism. As a matter of fact, the saturation magnetic field that one gets taking $M_{sat} = N\mu$ and using the density of iron of 7.9 and the moment μ of the spinning electron is about 20,000 gauss. But according to experiment, it is actually in the neighborhood of 21,500 gauss. This is a typical magnitude of error—5 or 10 percent—due to neglecting the contributions of the orbital moments that have not been included in making the analysis. Thus, a slight discrepancy with the gyromagnetic measurements is quite understandable.

37-3 The hysteresis curve

We have concluded from our theoretical analysis that a ferromagnetic material should spontaneously become magnetized below a certain temperature so that all the magnetism would be in the same direction. But we know that this is not true for an ordinary piece of *unmagnetized* iron. Why isn't all iron magnetized? We can explain it with the help of Fig. 37-4. Suppose the iron were all a big single crystal of the shape shown in Fig. 37-4(a) and spontaneously magnetized all in one direction. Then there would be a considerable external magnetic field, which would have a lot of energy. We can reduce that field energy if we arrange that one side of

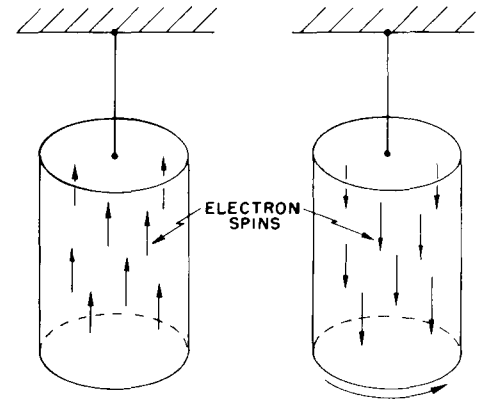


Fig. 37-3. When the magnetization of a bar of iron is reversed, the bar is given some angular velocity.

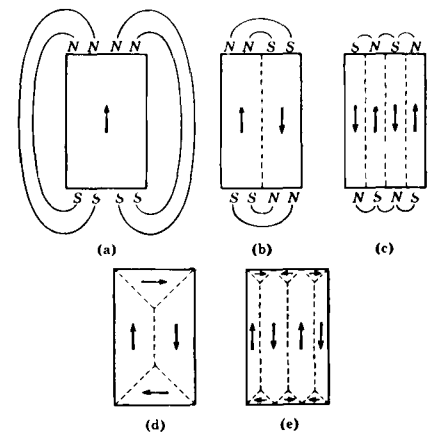


Fig. 37-4. The formation of domains in a single crystal of iron. [From Charles Kittel, *Introduction to Solid State Physics*, John Wiley and Sons, Inc., New York, 2nd ed., 1956.]

the block is magnetized “up” and the other side magnetized “down,” as in Fig. 37-4(b). Then, of course, the fields outside the iron would extend over less volume, so there would be less energy there.

Ah, but wait! In the layer between the two regions we have up-spinning electrons adjacent to down-spinning electrons. But ferromagnetism appears only in those materials for which the energy is *reduced* if the electrons are *parallel* rather than opposite. So, we have added some extra energy along the dotted line in Fig. 37-4(b); this energy is sometimes called *wall energy*. A region having only one direction of magnetization is called a *domain*. At the interface—the “wall”—between two domains, where we have atoms on opposite sides which are spinning in different directions, there is an energy per unit area of the wall. We have described it as though two adjacent atoms were spinning exactly opposite, but it turns out that nature adjusts things so that the transition is more gradual. But we don’t need to worry about such fine details at this point.

Now the question is: When is it better or worse to make a wall? The answer is that it depends on the *size* of the domains. Suppose that we were to scale up a block so that the whole thing was twice as big. The volume in the space outside filled with a given magnetic field strength would be *eight* times bigger, and the energy in the magnetic field, which is proportional to the volume, would also be eight times greater. But the *surface* area between two domains, which will give the wall energy, would be only *four* times as big. Therefore, if the piece of iron is big enough, it will pay to split it into more domains. This is why only the very tiny crystals can have but a single domain. Any large object—one more than about a hundredth of a millimeter in size—will have at least one domain wall; and any ordinary, “centimeter-size” object will be split into many domains, as shown in the figure. Splitting into domains goes on *until the energy needed to put in one extra wall is as large as the energy decrease in the magnetic field outside the crystal*.

Actually nature has discovered still another way to lower the energy: It is not necessary to have the field go outside at all, if a little triangular region is magnetized *sideways*, as in Fig. 37-4(d).^{*} Then with the arrangement of Fig. 37-4(d) we see that there is *no* external field, but instead only a little more domain wall.

But that introduces a new kind of problem. It turns out that when a single crystal of iron is magnetized, it changes its length in the direction of magnetization, so an “ideal” cube with its magnetization, say, “up,” is no longer a perfect cube. The “vertical” dimension will be different from the “horizontal” dimension. This effect is called *magnetostriction*. Because of such geometric changes, the little triangular pieces of Fig. 37-4(d) do not, so to speak, “fit” into the available space anymore—the crystal has got too long one way and too short the other way. Of course, it *does* fit, really, but only by being squashed in; and this involves some mechanical stresses. So, this arrangement *also* introduces an extra energy. It is the balance of all these various energies which determines how the domains finally arrange themselves in their complicated fashion in a piece of unmagnetized iron.

Now, what happens when we put on an external magnetic field? To take a simple case, consider a crystal whose domains are as shown in Fig. 37-4(d). If we apply an external magnetic field in the upward direction, in what manner does the crystal become magnetized? First, the middle domain wall can *move over sideways* (to the right) and reduce the energy. It moves over so that the region which is “up” becomes bigger than the region which is “down”. There are more elementary magnets lined up with the field, and this gives a lower energy. So, for a piece of iron in weak fields—at the very beginning of magnetization—the domain walls begin to move and eat into the regions which are magnetized opposite to the field. As the field continues to increase, a whole crystal shifts gradually into a single

^{*} You may be wondering how spins that have to be either “up” or “down” can also be “sideways”! That’s a good question, but we won’t worry about it right now. We’ll simply adopt the classical point of view, thinking of the atomic magnets as classical dipoles which can be polarized sideways. Quantum mechanics requires considerable expertness to understand how things can be quantized both “up-and-down,” and “right-and-left,” all at the same time.

large domain which the external field helps to keep lined up. In a strong field the crystal “likes” to be all one way *just because* its energy in the applied field is reduced—it is no longer merely the crystal’s own external field which matters.

What if the geometry is not so simple? What if the axes of the crystal and its spontaneous magnetization are in one direction, but we apply the magnetic field in *some other direction*—say at 45° ? We might think that domains would reform themselves with their magnetization parallel to the field, and then as before, they could all grow into one domain. But this is not easy for the iron to do, *for the energy needed to magnetize a crystal depends on the direction of magnetization relative to the crystal axis*. It is relatively easy to magnetize iron in a direction parallel to the crystal axes, but it takes *more* energy to magnetize it in some other direction—like 45° with respect to one of the axes. Therefore, if we apply a magnetic field in such a direction, what happens first is that the domains which point along one of the preferred directions which is *near* to the applied field grow until the magnetization is all along one of these directions. Then *with much stronger fields*, the magnetization is gradually pulled around parallel to the field, as sketched in Fig. 37-5.

In Fig. 37-6 are shown some observations of the magnetization curves of single crystals of iron. To understand them, we must first explain something about the notation that is used in describing directions in a crystal. There are many ways in which a crystal can be sliced so as to produce a face which is a plane of atoms. Everyone who has driven past an orchard or vineyard knows this—it is fascinating to watch. If you look one way, you see lines of trees—if you look another way, you see different lines of trees, and so on. In a similar way, a crystal has definite families of planes that hold many atoms, and the planes have this important characteristic (we consider a cubic crystal to make it easier): If we observe where the planes intersect the three coordinate axes—we find that the *reciprocals* of the three distances from the origin are in the ratio of simple whole numbers. These three whole numbers are taken as the definition of the planes. For example, in Fig. 37-7(a), a plane parallel to the yz -plane is shown. This is called a $[100]$ plane; the reciprocals of its intersection of the y - and z -axes are both zero. The direction perpendicular to such a plane (in a cubic crystal) is given the same set of numbers. It is easy to understand the idea in a cubic crystal, for then the indices $[100]$ mean a vector which has a unit component in the x -direction and none in the y - or z -directions. The $[110]$ direction is in a direction 45° from the x - and y -axes, as in Fig. 37-7(b); and the $[111]$ direction is in the direction of the cube diagonal, as in Fig. 37-7(c).

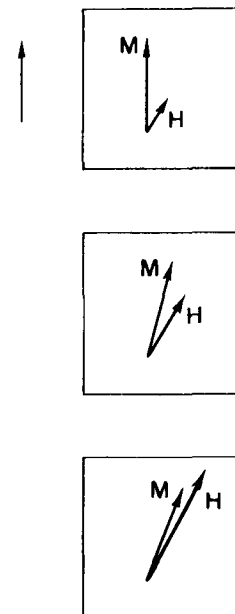


Fig. 37-5. A magnetizing field H at an angle with respect to the crystal axis will gradually change the direction of the magnetization without changing its magnitude.

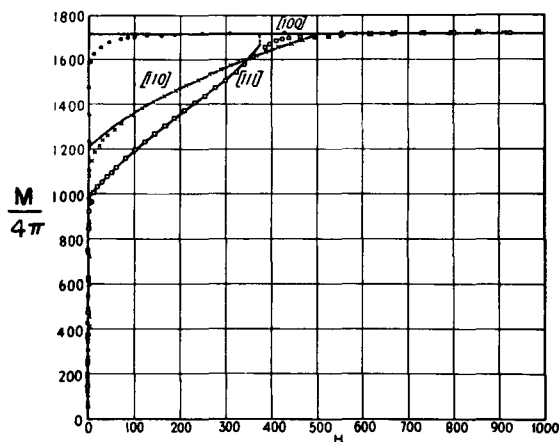


Fig. 37-6. The component of M parallel to H , for different directions of H (with respect to the crystal axes). [From F. Bitter, *Introduction to Ferromagnetism*, McGraw-Hill Book Co., Inc., 1937.]

Returning now to Fig. 37-6, we see the magnetization curves of a single crystal of iron for various directions. First, note that for very tiny fields—so weak that it is hard to see them on the scale at all—the magnetization increases extremely rapidly to quite large values. If the field is in the $[100]$ direction—namely along one of those nice, easy directions of magnetization—the curve goes up to a high value, curves around a little, and then is saturated. What happened is that the

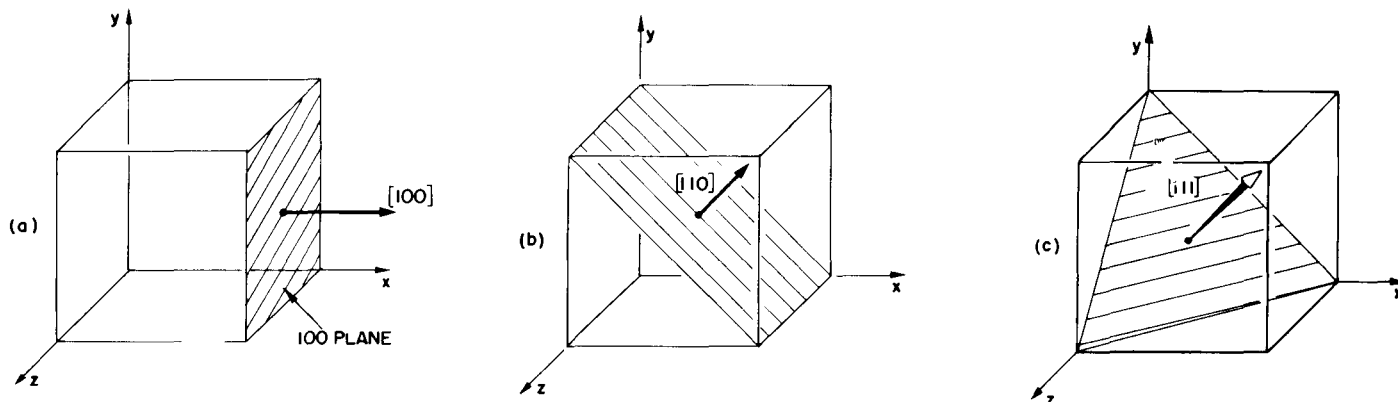


Fig. 37-7. The way the crystal planes are labeled.

domains which were already there are very easily removed. Only a small field is required to make the domain walls move and eat up all of the “wrong-way” domains. Single crystals of iron are enormously permeable (magnetic sense), much more so than ordinary polycrystalline iron. A perfect crystal magnetizes extremely easily. Why is it curved at all? Why doesn’t it just go right up to saturation? We are not sure. You might study that some day. We do understand why it is flat for high fields. When the whole block is a single domain, the extra magnetic field cannot make any more magnetization—it is already at M_{sat} , with all the electrons lines up.

Now, if we try to do the same thing in the $[110]$ direction—which is at 45° to the crystal axes—what will happen? We turn on a little bit of field and the magnetization leaps up as the domains grow. Then as we increase the field some more, we find that it takes quite a lot of field to get up to saturation, because now the magnetization is *turning away* from an “easy” direction. If this explanation is correct, the point at which the $[110]$ curve extrapolates back to the vertical axis should be at $1/\sqrt{2}$ of the saturation value. It turns out, in fact, to be very, very close to $1/\sqrt{2}$. Similarly, in the $[111]$ direction—which is along the cube diagonal—we find, as we would expect, that the curve extrapolates back to nearly $1/\sqrt{3}$ of saturation.

Figure 37-8 shows the corresponding situation for two other materials, nickel and cobalt. Nickel is different from iron. In nickel, it turns out that the $[111]$ direction is the easy direction of magnetization. Cobalt has a hexagonal crystal form, and people have botched up the system of nomenclature for this case. They want to have three axes on the bottom of the hexagon and one perpendicular to these, so they have used four indices. The $[0001]$ direction is the direction of the axis of the hexagon, and $[10\bar{1}0]$ is perpendicular to that axis. We see that crystals of different metals behave in different ways.

Now we must discuss a polycrystalline material, such as an ordinary piece of iron. Inside such materials there are many, many little crystals with their crystal-line axes pointing every which way. *These are not the same as domains.* Remember that the domains were all part of a *single crystal*, but in a piece of iron there are

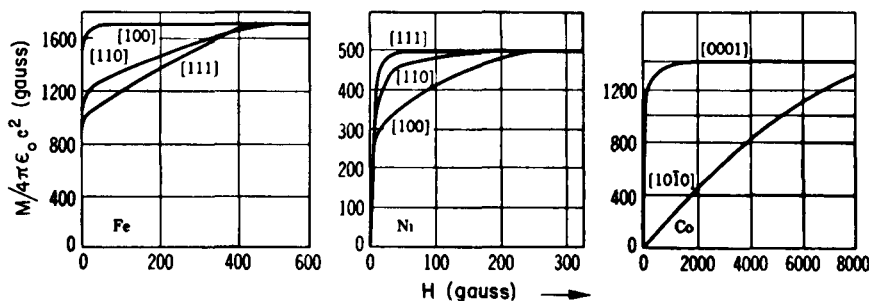


Fig. 37-8. Magnetization curves for single crystals of iron, nickel, and cobalt. [From Charles Kittel, *Introduction to Solid State Physics*, John Wiley and Sons, Inc., New York, 2nd ed., 1956.]

many *different crystals* with axes at different orientations, as shown in Fig. 37-9. Within each of these crystals, there will also generally be some domains. When we apply a *small* magnetic field to a piece of polycrystalline material, what happens is that the domain walls begin to move, and the domains which have a favorable direction of easy magnetization grow larger. This growth is reversible so long as the field stays very small—if we turn the field off, the magnetization will return to zero. This part of the magnetization curve is marked *a* in Fig. 37-10.

For larger fields—in the region *b* of the magnetization curve shown—things get much more complicated. In every small crystal of the material, there are strains and dislocations; there are impurities, dirt, and imperfections. And at all but the smallest fields, the domain wall, in moving, gets stuck on these. There is an interaction energy between the domain wall and a dislocation, or a grain boundary, or an impurity. So when the wall gets to one of them, it gets stuck; it sticks there at a certain field. But then if the field is raised some more, the wall suddenly snaps past. So the motion of the domain wall is not smooth the way it is in a perfect crystal—it gets hung up every once in a while and moves in jerks. If we were to look at the magnetization on a microscopic scale, we would see something like the insert of Fig. 37-10.

Now the important thing is that these jerks in the magnetization can cause an energy loss. In the first place, when a boundary finally slips past an impediment, it moves very quickly to the next one, since the field is already above what would be required for the unimpeded motion. The rapid motion means that there are rapidly changing magnetic fields which produce eddy currents in the crystal. These currents lose energy in heating the metal. A second effect is that when a domain suddenly changes, part of the crystal changes its dimensions from the magnetostriction. Each sudden shift of a domain wall sets up a little sound wave that carries away energy. Because of such effects, the second part of magnetization curve is *irreversible*, and *there is energy being lost*. This is the origin of the hysteresis effect, because to move a boundary wall forward—snap—and then to move it backward—snap—produces a different result. It's like "jerky" friction, and it takes energy.

Eventually, for high enough fields, when we have moved all the domain walls and magnetized each crystal in its best direction, there are still some crystallites which happen to have their easy directions of magnetization not in the direction of our external magnetic field. Then it takes a lot of extra field to turn those magnetic moments around. So the magnetization increases slowly, but smoothly, for high fields—namely in the region marked *c* in the figure. The magnetization does not come sharply to its saturation value, because in the last part of the curve the atomic magnets are *turning* in the strong field. So we see why the magnetization curve of an ordinary polycrystalline materials, such as the one shown in Fig. 37-10, rises a little bit and *reversibly* at first, then rises irreversibly, and then curves over slowly. Of course, there is no sharp break-point between the three regions—they blend smoothly, one into the other.

It is not hard to show that the magnetization process in the middle part of the magnetization curve is jerky—that the domain walls jerk and snap as they shift. All you need is a coil of wire—with many thousands of turns—connected to an amplifier and a loudspeaker, as shown in Fig. 37-11. If you put a few silicon steel sheets (of the type used in transformers) at the center of the coil and bring a bar magnet slowly near the stack, the sudden changes in magnetization will produce impulses of emf in the coil, which are heard as distinct clicks in the loudspeaker. As you move the magnet nearer to the iron you will hear a whole rush of clicks that sound something like the noise of sand grains falling over each other as a can of sand is tilted. The domain walls are jumping, snapping, and jiggling as the field is increased. This phenomenon is called the *Barkhausen effect*.

As you move the magnet even closer to the iron sheets, the noise grows louder and louder for a while but then there is relatively little noise when the magnet gets very close. Why? Because nearly all the domain walls have moved as far as they can go. Any greater field is merely *turning* the magnetization in each domain, which is a smooth process.

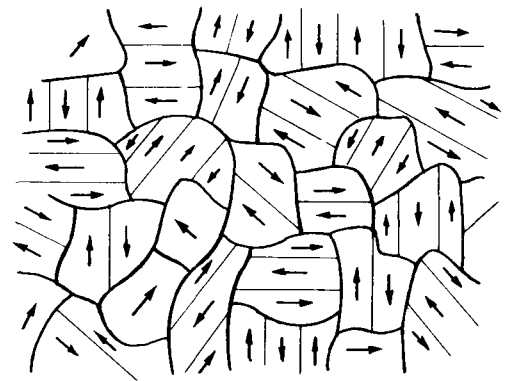


Fig. 37-9. The microscopic structure of an unmagnetized ferromagnetic material. Each crystal grain has an easy direction of magnetization and is broken up into domains which are spontaneously magnetized (usually) parallel to this direction.

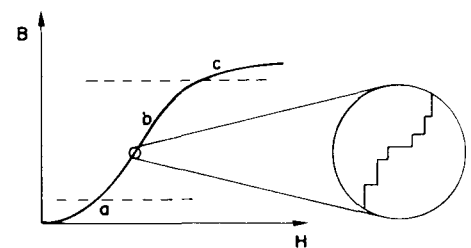


Fig. 37-10. The magnetization curve for polycrystalline iron.

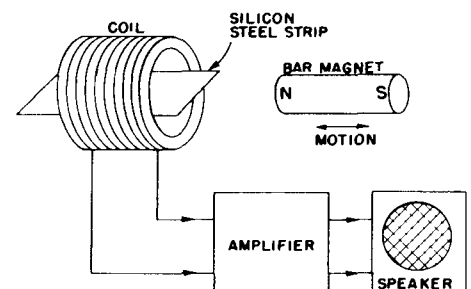


Fig. 37-11. The sudden changes in the magnetization of the steel strip are heard as clicks in the loudspeaker.

If you now withdraw the magnet, so as to come back on the downward branch of the hysteresis loop, the domains all try to get back to low energy again, and you hear another rush of backward-going jerks. You can also note that if you bring the magnet to a given place and move it back and forth a little bit, there is relatively little noise. It is again like tilting a can of sand—once the grains shift into place, small movements of the can don't disturb them. In the iron the small variations in the magnetic field aren't enough to move any boundaries over any of the "humps."

37-4 Ferromagnetic materials

Now we would like to talk about the various kinds of magnetic materials that there are in the technical world and to consider some of the problems involved in designing magnetic materials for different purposes. First, the term "the magnetic properties of iron," which one often hears, is a misnomer—there is no such thing. "Iron" is not a well-defined material—the properties of iron depend critically on the amount of impurities and also on *how* the iron is formed. You can appreciate that the magnetic properties will depend on how easily the domain walls move and that this is a *gross* property, not a property of the individual atoms. So practical ferromagnetism is not really a property of an iron *atom*—it is a property of *solid iron* in a *certain form*. For example, iron can take on two different crystalline forms. The common form has a body-centered cubic lattice, but it can also have a face-centered cubic lattice, which is, however, stable only at temperatures above 1100°C. Of course, at that temperature the body-centered cubic structure is already past the Curie point. However, by alloying chromium and nickel with the iron (one possible mixture is 18 percent chromium and 8 percent nickel) we can get what is called stainless steel, which, although it is mainly iron, retains the face-centered lattice even at low temperatures. Because its crystal structure is different, it has completely different magnetic properties. Most kinds of stainless steel are not magnetic to any appreciable degree, although there are some kinds which are somewhat magnetic—it depends on the composition of the alloy. Even when such an alloy is magnetic, it is not *ferromagnetic* like ordinary iron—even though it is mostly just iron.

We would like now to describe a few of the special materials which have been developed for their particular magnetic properties. First, if we want to make a *permanent* magnet, we would like material with an enormously *wide* hysteresis loop so that, when we turn the current off and come down to zero magnetizing field, the magnetization will remain large. For such materials the domain boundaries should be "frozen" in place as much as possible. One such material is the remarkable alloy "Alnico V" (51% Fe, 8% Al, 14% Ni, 24% Co, 3% Cu). (The rather complex composition of this alloy is indicative of the kind of detailed effort that has gone into making good magnets. What patience it takes to mix five things together and test them until you find the most ideal substance!) When Alnico solidifies, there is a "second phase" which precipitates out, making many tiny grains and very high internal strains. In this material, the domain boundaries have a hard time moving at all. In addition to having a precise composition, Alnico is mechanically "worked" in a way that makes the crystals appear in the form of long grains along the direction in which the magnetization is going to be. Then the magnetization will have a natural tendency to be lined up in these directions and will be held there from the anisotropic effects. Furthermore, the material is even cooled in an external magnetic field when it is manufactured, so that the grains will grow with the right crystal orientation. The hysteresis loop of Alnico V is shown in Fig 37-12. You see that it is about 500 times wider than the hysteresis curve for soft iron that we showed in the last chapter in Fig. 36-8.

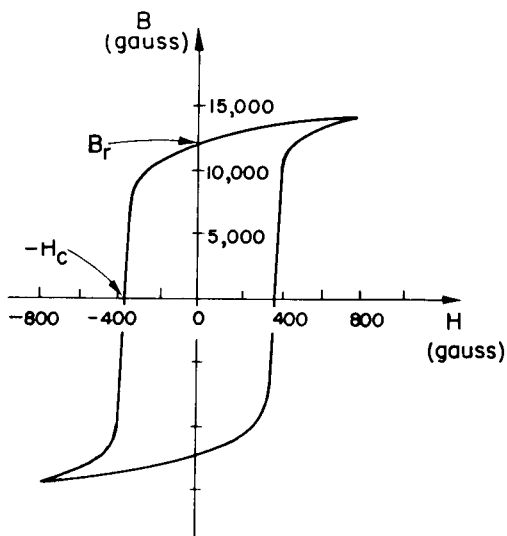


Fig 37-12. The hysteresis curve of Alnico V.

Let's turn now to a different kind of material. For building transformers and motors, we want a material which is magnetically "soft"—one in which the magnetism is easily changed so that an enormous amount of magnetization results from a very small applied field. To arrange this, we need pure, well-annealed material which will have very few dislocations and impurities so that the domain

walls can move easily. It would also be nice if we could make the anisotropy small. Then, even if a grain of the material sits at the wrong angle with respect to the field, it will still magnetize easily. Now we have said that iron prefers to magnetize along the [100] direction, whereas nickel prefers the [111] direction; so if we mix iron and nickel in various proportions, we might hope to find that with just the right proportions the alloy wouldn't prefer *any* direction—the [100] and [111] directions would be equivalent. It turns out that this happens with a mixture of 70 percent nickel and 30 percent iron. In addition—possibly by luck or maybe because of some physical relationship between the anisotropy and the magnetostriction effects—it turns out that the *magnetostriction* of iron and nickel has the opposite sign. And in an alloy of the two metals, this property goes through zero at about 80 percent nickel. So somewhere between 70 and 80 percent nickel we get very “soft” magnetic materials—alloys that are very easy to magnetize. They are called the *permalloys*. Permalloys are useful for high-quality transformers (at low signal levels), but they would be no good at all for permanent magnets. Permalloys must be very carefully made and handled. The magnetic properties of a piece of permalloy are drastically changed if it is stressed beyond its elastic limit—it mustn't be bent. Then, its permeability is reduced because of the dislocations, slip bands, and so on, which are produced by the mechanical deformations. The domain boundaries are no longer easy to move. The high permeability can, however, be restored by annealing at high temperatures.

It is often convenient to have some numbers to characterize the various magnetic materials. Two useful numbers are the intercepts of the hysteresis loop with the B - and H -axes, as indicated in Fig. 37-12. These intercepts are called the *remanent magnetic field* B_r and the *coercive force* H_c . In Table 37-1 we list these numbers for a few magnetic materials.

Table 37-1
Properties of some ferromagnetic materials

Material	B_r Residual magnetic field (gauss)	H_c Coercive force (gauss)
Supermalloy	(≈ 5000)	0.004
Silicon steel (transformer)	12,000	0.05
Armco iron	4000	0.6
Alnico V	13,000	550.

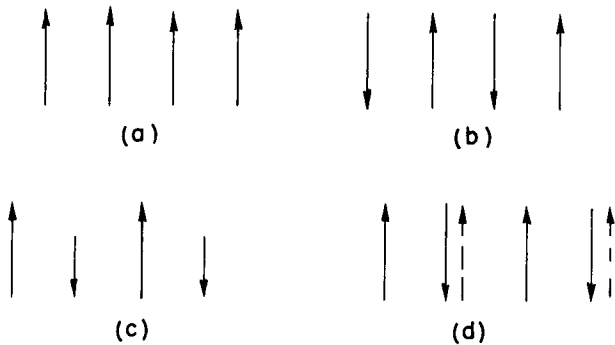


Fig. 37-13. Relative orientation of electron spins in various materials: (a) ferromagnetic, (b) antiferromagnetic, (c) ferrite, (d) yttrium-iron alloy. (Broken arrows show direction of total angular momentum, including orbital motion.)

37-5 Extraordinary magnetic materials

We would now like to discuss some of the more exotic magnetic materials. There are many elements in the periodic table which have incomplete inner electron shells and hence have atomic magnetic moments. For instance, right next to the ferromagnetic elements iron, nickel, and cobalt you will find chromium and manganese. Why aren't *they* ferromagnetic? The answer is that the λ term in Eq. (37.1) has the *opposite sign* for these elements. In the chromium lattice, for example, the spins of the chromium atoms alternate *atom by atom*, as shown in Fig. 37-13(b). So chromium *is* “magnetic” from its own point of view, but it is not technically interesting because there are no *external* magnetic effects. Chromium, then, is an example of a material in which quantum mechanical effects make the spins alternate. Such a material is called *antiferromagnetic*. The alignment in antiferromagnetic materials is also temperature dependent. Below a critical temperature, all the spins are lined up in the alternating array, but when the material is heated above a certain temperature—which is again called the Curie temperature—the spins suddenly become random. There is, internally, a sudden transition. This transition can be seen in the specific heat curve. Also it shows up in some special “magnetic” effects. For instance, the existence of the alternating spins can be verified by scattering neutrons from a crystal of chromium. Because a neutron itself has a spin

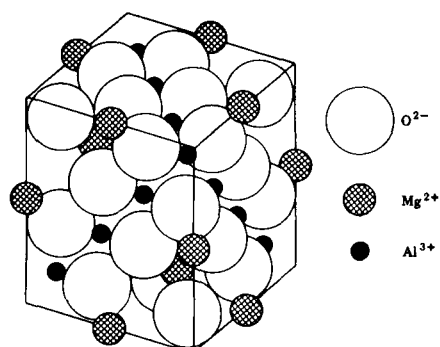


Fig. 37-14. Crystal structure of the mineral spinel (MgAl_2O_4); the Mg^{+2} ions occupy tetrahedral sites, each surrounded by four oxygen ions; the Al^{+3} ions occupy octahedral sites, each surrounded by six oxygen ions. [From Charles Kittel, *Introduction to Solid State Physics*, John Wiley and Sons, Inc., New York, 2nd ed., 1956]

(and a magnetic moment), it has a different amplitude to be scattered, depending on whether its spin is parallel or opposite to the spin of the scatterer. Thus, we get a different interference pattern when the spins in a crystal are alternating than we do when they have a random distribution

There is another kind of substance in which quantum mechanical effects make the electron spins alternate, but which is nevertheless *ferromagnetic*—that is, the crystal has a permanent net magnetization. The idea behind such materials is shown in Fig. 37-14. The figure shows the crystal structure of *spinel*, a magnesium-aluminum oxide, which—as it is shown—is *not* magnetic. The oxide has two kinds of metal atoms: magnesium and aluminum. Now if we replace the magnesium and the aluminum by two magnetic elements like iron and zinc, or by zinc and manganese—in other words, if we put in *magnetic* atoms instead of the nonmagnetic ones—an interesting thing happens. Let's call one kind of metal atom *a* and the other kind of metal atom *b*; then the following combination of forces must be considered. There is an *a-b* interaction which tries to make the *a* atoms and the *b* atoms have opposite spins—because quantum mechanics always gives the opposite sign (except for the mysterious crystals of iron, nickel, and cobalt). Then, there is a direct *a-a* interaction which tries to make the *a*'s opposite, and also a *b-b* interaction which tries to make the *b*'s opposite. Now, of course we cannot have everything opposite everything else—*a* opposite *b*, *a* opposite *a*, and *b* opposite *b*. Presumably because of the distances between the *a*'s and the presence of the oxygen (although we really don't know why), it turns out that the *a-b* interaction is stronger than the *a-a* or the *b-b*. So the solution that nature uses in this case is to make all the *a*'s *parallel* to each other, and all the *b*'s *parallel* to each other, but the two systems *opposite*. That gives the lowest energy because of the stronger *a-b* interaction. The result: all the *a*'s are spinning up and all the *b*'s are spinning down—or vice versa, of course. But if the *magnetic moments* of the *a*-type atom and the *b*-type atom are *not* equal, we can get the situation shown in Fig. 37-13(c), and there can be a net magnetization in the material. The material will then be ferromagnetic—although somewhat weak. Such materials are called *ferrites*. They do not have as high a saturation magnetization as iron—for obvious reasons—so they are only useful for smaller fields. But they have a very important difference—they are insulators; the ferrites are *ferromagnetic insulators*. In high-frequency fields, they will have very small eddy currents and so can be used, for example, in microwave systems. The microwave fields will be able to get inside such an insulating material, whereas they would be kept out by the eddy currents in a conductor like iron.

There is another class of magnetic materials which has only recently been discovered—members of the family of the orthosilicates called *garnets*. They are again crystals in which the lattice contains two kinds of metallic atoms, and we have again a situation in which two kinds of atoms can be substituted almost at will. Among the many compounds of interest there is one which is completely ferromagnetic. It has yttrium and iron in the garnet structure, and the reason it is ferromagnetic is very curious. Here again quantum mechanics is making the neighboring spins opposite, so that there is a locked-in system of spins with the electron spins of the iron one way and the electron spins of the yttrium the opposite way. But the yttrium atom is complicated. It is a rare-earth element and gets a large contribution to its magnetic moment from *orbital* motion of the electrons. For yttrium, the orbital motion contribution is *opposite* that of the spin and also is bigger. Thus, although quantum mechanics, working through the exclusion principle, makes the *spins* of the yttrium opposite those of the iron, it makes the *total* magnetic moment of the yttrium atom *parallel* to the iron because of the orbital effect—as sketched in Fig. 37-13(d). The compound is therefore a regular ferromagnet.

Another interesting example of ferromagnetism occurs in some of the rare-earth elements. It has to do with a still more peculiar arrangement of the spins. The material is not ferromagnetic in the sense that the spins are all parallel, nor is it antiferromagnetic in the sense that every atom is opposite. In these crystals all of the spins *in one layer* are parallel and lie in the plane of the layer. In the next

layer all spins are again parallel to each other, but point in a somewhat different direction. In the following layer they are in still another direction, and so on. The result is that the local magnetization vector varies in the form of a spiral—the magnetic moments of the successive layers rotate as we proceed along a line perpendicular to the layers. It is interesting to try to analyze what happens when a field is applied to such a spiral—all the twistings and turnings that must go on in all those atomic magnets. (Some people *like* to amuse themselves with the theory of these things!) Not only are there cases of “flat” spirals, but there are also cases in which the directions the magnetic moments of successive layers map out a cone, so that it has a spiral component and also a uniform ferromagnetic component in one direction!

The magnetic properties of materials, worked out on a more advanced level than we have been able to do here, have fascinated physicists of all kinds. In the first place, there are those practical people who love to work out ways of making things in a better way—they love to design better and more interesting magnetic materials. The discovery of things like ferrites, or their application, immediately delights people who like to see clever new ways of doing things. Besides this, there are those who find a fascination in the terrible complexity that nature can produce using a few basic laws. Starting with one and the same general idea, nature goes from the ferromagnetism of iron and its domains, to the antiferromagnetism of chromium, to the magnetism of ferrites and garnets, to the spiral structure of the rare earth elements, and on, and on. It is fascinating to discover experimentally all the strange things that go on in these special substances. Then, to the theoretical physicists, ferromagnetism presents a number of very interesting, unsolved, and beautiful challenges. One challenge is to understand why it exists at all. Another is to predict the statistics of the interacting spins in an ideal lattice. Even neglecting any possible extraneous complications, this problem has, so far, defied full understanding. The reason that it is so interesting is that it is such an easily stated problem: Given a lot of electron spins in a regular lattice, interacting with such-and-such a law, what do they do? It is simply stated, but it has defied complete analysis for years. Although it has been analyzed rather carefully for temperatures not too close to the Curie point, the theory of the sudden transition ~~at~~ the Curie point still needs to be completed.

Finally, the whole subject of the system of spinning atomic magnets—in ferromagnetic, or in paramagnetic materials and in nuclear magnetism, has also been a fascinating thing to advanced students in physics. The system of spins can be pushed on and pulled on with external magnetic fields, so one can do many tricks with resonances, with relaxation effects, with spin-echoes, and with other effects. It serves as a prototype of many complicated thermodynamic systems. But in paramagnetic materials the situation is often fairly simple, and people have been delighted both to do experiments and to explain the phenomena theoretically.

We now close our study of electricity and magnetism. In the first chapter, we spoke of the great strides that have been made since the early Greek observation of the strange behaviors of amber and of lodestone. Yet in all our long and involved discussion we have never explained *why it is that when we rub a piece of amber we get a charge on it*, nor have we explained *why a lodestone is magnetized*! You may say, “Oh, we just didn’t get the right sign.” No, it is worse than that. Even if we *did* get the right sign, we would still have the question: Why is the piece of lodestone in the ground magnetized? There is the earth’s magnetic field, of course, but *where does the earth’s field come from*? Nobody really knows—there have only been some good guesses. So you see, this physics of ours is a lot of fakery—we start out with the phenomena of lodestone and amber, and we end up not understanding either of them very well. But we *have* learned a tremendous amount of very exciting and very practical information in the process!

Elasticity

38-1 Hooke's law

The subject of elasticity deals with the behavior of those substances which have the property of recovering their size and shape when the forces producing deformations are removed. We find this elastic property to some extent in all solid bodies. If we had the time to deal with the subject at length, we would want to look into many things: the behavior of materials, the general laws of elasticity, the general theory of elasticity, the atomic machinery that determine the elastic properties, and finally the limitations of elastic laws when the forces become so great that plastic flow and fracture occur. It would take more time than we have to cover all these subjects in detail, so we will have to leave out some things. For example, we will not discuss plasticity or the limitations of the elastic laws. (We touched on these subjects briefly when we were talking about dislocations in metals.) Also, we will not be able to discuss the internal mechanisms of elasticity—so our treatment will not have the completeness we have tried to achieve in the earlier chapters. Our aim is mainly to give you an acquaintance with some of the ways of dealing with such practical problems as the bending of beams.

When you push on a piece of material, it “gives”—the material is deformed. If the force is small enough, the relative displacements of the various points in the material are proportional to the force—we say the behavior is *elastic*. We will discuss only the elastic behavior. First, we will write down the fundamental laws of elasticity, and then we will apply them to a number of different situations.

Suppose we take a rectangular block of material of length l , width w , and height h , as shown in Fig. 38-1. If we pull on the ends with a force F , then the length increases by an amount Δl . We will suppose in all cases that the change in length is a small fraction of the original length. As a matter of fact, for materials like wood and steel, the material will break if the change in length is more than a few percent of the original length. For a large number of materials, experiments show that for sufficiently small extensions the force is proportional to the extension

$$F \propto \Delta l. \quad (38.1)$$

This relation is known as *Hooke's law*.

The lengthening Δl of the bar will also depend on its length. We can figure out how by the following argument. If we cement two identical blocks together, end to end, the same forces act on each block, each will stretch by Δl . Thus, the stretch of a block of length $2l$ would be twice as big as a block of the same cross section, but of length l . In order to get a number more characteristic of the material, and less of any particular shape, we choose to deal with the ratio $\Delta l/l$ of the extension to the original length. This ratio is proportional to the force but independent of l :

$$F \propto \frac{\Delta l}{l}. \quad (38.2)$$

The force F will also depend on the area of the block. Suppose that we put two blocks side by side. Then for a given stretch Δl we would have the force F on each block, or twice as much on the combination of the two blocks. The force, for a given amount of stretch, must be proportional to the cross-sectional area A of the block. To obtain a law in which the coefficient of proportionality is independent of the dimensions of the body, we write Hooke's law for a rectangular

38-1 Hooke's law

38-2 Uniform strains

38-3 The torsion bar; shear waves

38-4 The bent beam

38-5 Buckling

Review: Chapter 47, Vol. I, *Sound; the Wave Equation*.

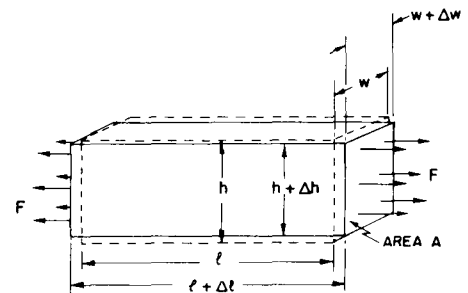


Fig. 38-1. The stretching of a bar under uniform tension.

block in the form

$$F = YA \frac{\Delta l}{l}. \quad (38.3)$$

The constant Y is a property only of the nature of the material; it is known as *Young's modulus*. (Usually you will see Young's modulus called E . But we've used E for electric fields, energy, and emf's, so we prefer to use a different letter.)

The *force per unit area* is called the *stress*, and the stretch per unit length—the *fractional stretch*—is called the *strain*. Equation (38.3) can therefore be rewritten in the following way:

$$\frac{F}{A} = Y \times \frac{\Delta l}{l}, \quad (38.4)$$

$$\text{Stress} = (\text{Young's modulus}) \times (\text{Strain}).$$

There is another part to Hooke's law: When you *stretch* a block of material in one direction it *contracts* at right angles to the stretch. The contraction in width is proportional to the width w and also to $\Delta l/l$. The sideways contraction is in the same proportion for both width and height, and is usually written

$$\frac{\Delta w}{w} = \frac{\Delta h}{h} = -\sigma \frac{\Delta l}{l}, \quad (38.5)$$

where the constant σ is another property of the material called *Poisson's ratio*. It is always positive in sign and is a number less than $1/2$. (It is "reasonable" that σ should be generally positive, but it is not quite clear that it *must* be so.)

The two constants Y and σ specify completely the elastic properties of a *homogeneous isotropic* (that is, noncrystalline) material. In crystalline materials the stretches and contractions can be different in different directions, so there can be many more elastic constants. We will restrict our discussion temporarily to homogeneous isotropic materials whose properties can be described by Y and σ . As usual there are different ways of describing things—some people like to describe the elastic properties of materials by different constants. It always takes two, and they can be related to σ and Y .

The last general law we need is the principle of superposition. Since the two laws (38.4) and (38.5) are linear in the forces and in the displacements, superposition will work. If you have one set of forces and get some displacements, and then you add a new set of forces and get some additional displacements, the resulting displacements will be the sum of the ones you would get with the two sets of forces acting independently.

Now we have all the general principles—the superposition principle and Eqs. (38.4) and (38.5)—and that's all there is to elasticity. But that is like saying that once you have Newton's laws that's all there is to mechanics. Or, given Maxwell's equations, that's all there is to electricity. It is, of course, true that with these principles you have a great deal, because with your present mathematical ability you could go a long way. We will, however, work out a few special applications.

38-2 Uniform strains

As our first example let's find out what happens to a rectangular block under uniform hydrostatic pressure. Let's put a block under water in a pressure tank. Then there will be a force acting inward on every face of the block proportional to the area (see Fig. 38-2). Since the hydrostatic pressure is uniform, the *stress* (force per unit area) on each face of the block is the same. We will work out first the change in the length. The change in length of the block can be thought of as the sum of changes in length that would occur in the three independent problems which are sketched in Fig. 38-3.

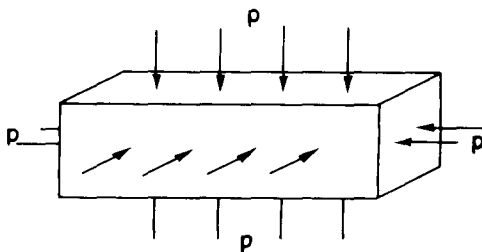


Fig. 38-2. A bar under uniform hydrostatic pressure.

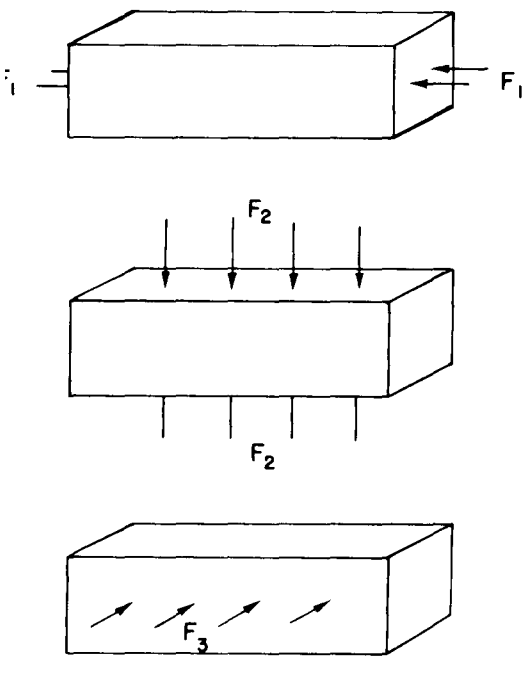


Fig. 38-3. Hydrostatic pressure is the superposition of three longitudinal compressions.

Problem 1. If we push on the ends of the block with a pressure p , the compressional strain is p/Y , and it is negative,

$$\frac{\Delta l_1}{l} = -\frac{p}{Y}.$$

Problem 2. If we push on the two sides of the block with pressure p , the compressional strain is again p/Y , but now we want the lengthwise strain. We can get that from the sideways strain multiplied by $-\sigma$. The sideways strain is

$$\frac{\Delta w}{w} = -\frac{p}{Y};$$

so

$$\frac{\Delta l_2}{l} = +\sigma \frac{p}{Y}.$$

Problem 3. If we push on the top of the block, the compressional strain is once more p/Y , and the corresponding strain in the sideways direction is again $-\sigma p/Y$. We get

$$\frac{\Delta l_3}{l} = +\sigma \frac{p}{Y}.$$

Combining the results of the three problems—that is, taking $\Delta l = \Delta l_1 + \Delta l_2 + \Delta l_3$ —we get

$$\frac{\Delta l}{l} = -\frac{p}{Y} (1 - 2\sigma). \quad (38.6)$$

The problem is, of course, symmetrical in all three directions; it follows that

$$\frac{\Delta w}{w} = \frac{\Delta h}{h} = -\frac{p}{Y} (1 - 2\sigma). \quad (38.7)$$

The change in the *volume* under hydrostatic pressure is also of some interest. Since $V = lwh$, we can write, for small displacements,

$$\frac{\Delta V}{V} = \frac{\Delta l}{l} + \frac{\Delta w}{w} + \frac{\Delta h}{h}.$$

Using (38.6) and (38.7), we have

$$\frac{\Delta V}{V} = -3 \frac{p}{Y} (1 - 2\sigma). \quad (38.8)$$

People like to call $\Delta V/V$ the *volume strain* and write

$$p = -K \frac{\Delta V}{V}.$$

The *volume stress* p is proportional to the volume strain—Hooke's law once more. The coefficient K is called the *bulk modulus*; it is related to the other constants by

$$K = \frac{Y}{3(1 - 2\sigma)}. \quad (38.9)$$

Since K is of some practical interest, many handbooks give Y and K instead of Y and σ . If you want σ you can always get it from Eq. (38.9). We can also see from Eq. (38.9) that Poisson's ratio, σ , must be less than one-half. If it were not, the bulk modulus K would be negative, and the material would expand under increasing pressure. That would allow us to get mechanical energy *out of* any old block—it would mean that the block was in unstable equilibrium. If it started to expand it would continue by itself with a release of energy.

Now we want to consider what happens when you put a “shear” strain on something. By shear strain we mean the kind of distortion shown in Fig. 38-4. As a preliminary to this, let us look at the strains in a *cube* of material subjected to the forces shown in Fig. 38-5. Again we can break it up into two problems: the vertical

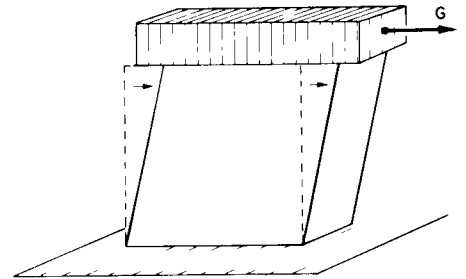


Fig. 38-4 A cube in uniform shear.

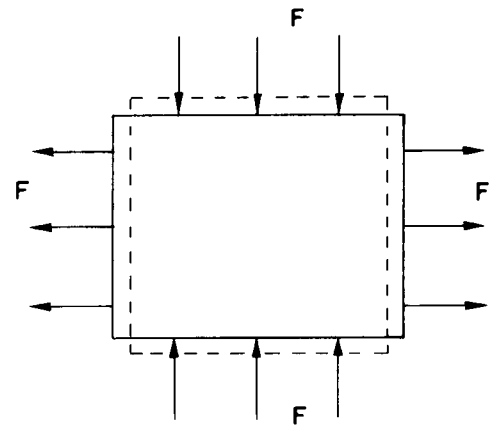


Fig. 38-5. A cube with compressing forces on top and bottom and equal stretching forces on two sides.

pushes, and the horizontal pulls. Calling A the area of the cube face, we have for the change in horizontal length

$$\frac{\Delta l}{l} = \frac{1}{Y} \frac{F}{A} + \sigma \frac{1}{Y} \frac{F}{A} = \frac{1 + \sigma}{Y} \frac{F}{A}. \quad (38.10)$$

The change in the vertical height is just the negative of this.

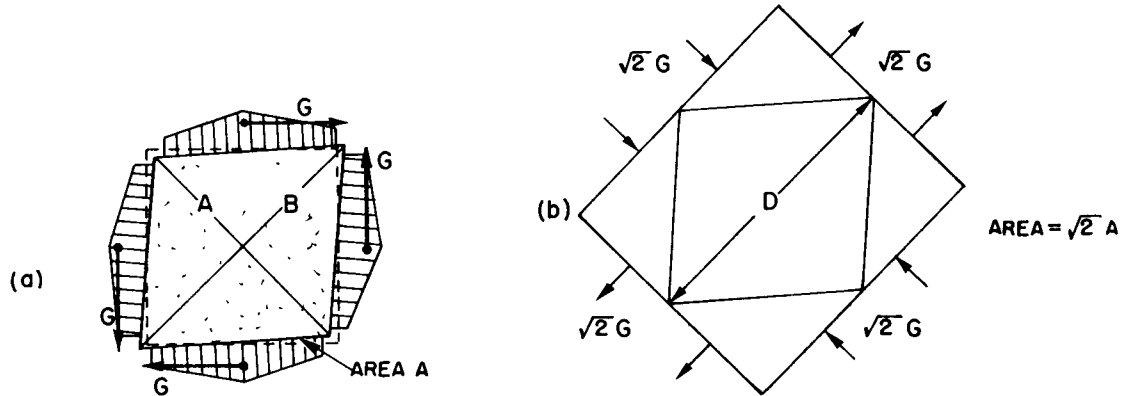


Fig. 38-6. The two pairs of shear forces in (a) produce the same stress as the compressing and stretching forces of (b).

Now suppose we have the same cube and subject it to the shearing forces shown in Fig. 38-6(a). Note that all the forces have to be equal if there are to be no net torques and the cube is to be in equilibrium. (Similar forces must also exist in Fig. 38-4, since the block is in equilibrium. They are provided through the “glue” that holds the block to the table.) The cube is then said to be in a state of pure shear. But note that if we cut the cube by a plane at 45° —say along the diagonal A in the figure—the total force acting across the plane is *normal* to plane and is equal to $\sqrt{2}G$. The area over which this force acts is $\sqrt{2}A$; therefore, the tensile stress normal to this plane is simply G/A . Similarly, if we examine a plane at an angle of 45° the other way—the diagonal B in the figure—we see that there is a compressional stress normal to this plane of $-G/A$. From this, we see that the *stress* in a “pure shear” is equivalent to a combination of tension and compression stresses of equal strength and at right angles to each other, and at 45° to the original faces of the cube. The internal stresses and strains are the same as we would find in the larger block of material with the forces shown in Fig. 38-6(b). But this is the problem we have already solved. The change in length of the diagonal is given by Eq. (38.10),

$$\frac{\Delta D}{D} = \frac{1 + \sigma}{Y} \frac{G}{A}. \quad (38.11)$$

(One diagonal is shortened; the other is elongated.)

It is often convenient to express a shear strain in terms of the angle by which the cube is twisted—the angle θ in Fig. 38-7. From the geometry of the figure you can see that the horizontal shift δ of the top edge is equal to $\sqrt{2} \Delta D$. So

$$\theta = \frac{\delta}{l} = \frac{\sqrt{2} \Delta D}{l} = 2 \frac{\Delta D}{D}. \quad (38.12)$$

The shear stress g is defined as the tangential force on one face divided by the area, $g = G/A$. Using Eq. (38.11) in (38.12), we get

$$\theta = 2 \frac{1 + \sigma}{Y} g.$$

Or, writing this in the form “stress = constant times strain,”

$$g = \mu \theta. \quad (38.13)$$

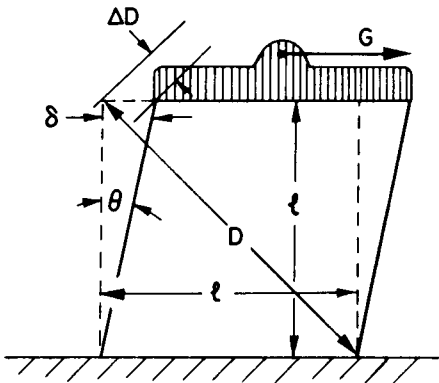


Fig. 38-7. The shear strain θ is $2 \Delta D/D$.

The proportionality coefficient μ is called the *shear modulus* (or, sometimes, the coefficient of rigidity). It is given in terms of Y and σ by

$$\mu = \frac{Y}{2(1 + \sigma)}. \quad (38.14)$$

Incidentally, the shear modulus must be positive—otherwise you could get work out of a self-shearing block. From Eq. (38.14), σ must be greater than -1 . We know, then, that σ must be between -1 and $+\frac{1}{2}$; in practice, however, it is always greater than zero.

As a last example of the type of situation where the stresses are uniform through the material, let's consider the problem of a block which is stretched, while it is at the same time *constrained* so that no lateral contraction can take place. (Technically, it's a little easier to compress it while keeping the sides from bulging out—but it's the same problem.) What happens? Well, there must be sideways forces which keep it from changing its thickness—forces we don't know off-hand but will have to calculate. It's the same kind of problem we have already done, only with a little different algebra. We imagine forces on all three sides, as shown in Fig. 38-8; we calculate the changes in dimensions, and we choose the transverse forces to make the width and height remain constant. Following the usual arguments, we get for the three strains:

$$\frac{\Delta l_x}{l_x} = \frac{1}{Y} \frac{F_x}{A_x} - \frac{\sigma}{Y} \frac{F_y}{A_y} - \frac{\sigma}{Y} \frac{F_z}{A_z} = \frac{1}{Y} \left[\frac{F_x}{A_x} - \sigma \left(\frac{F_y}{A_y} + \frac{F_z}{A_z} \right) \right], \quad (38.15)$$

$$\frac{\Delta l_y}{l_y} = \frac{1}{Y} \left[\frac{F_y}{A_y} - \sigma \left(\frac{F_x}{A_x} + \frac{F_z}{A_z} \right) \right], \quad (38.16)$$

$$\frac{\Delta l_z}{l_z} = \frac{1}{Y} \left[\frac{F_z}{A_z} - \sigma \left(\frac{F_x}{A_x} + \frac{F_y}{A_y} \right) \right]. \quad (38.17)$$

Now since Δl_y and Δl_z are supposed to be zero, Eqs. (38.16) and (38.17) give two equations relating F_y and F_z to F_x . Solving them together, we get that

$$\frac{F_y}{A_y} = \frac{F_z}{A_z} = \frac{\sigma}{1 - \sigma} \frac{F_x}{A_x}. \quad (38.18)$$

Substituting in (38.15), we have

$$\frac{\Delta l_x}{l_x} = \frac{1}{Y} \left(1 - \frac{2\sigma^2}{1 - \sigma} \right) \frac{F_x}{A_x} = \frac{1}{Y} \left(\frac{1 - \sigma - 2\sigma^2}{1 - \sigma} \right) \frac{F_x}{A_x}. \quad (38.19)$$

Often, you will see this turned around, and with the quadratic in σ factored out, it is then written

$$\frac{F}{A} = \frac{1 - \sigma}{(1 + \sigma)(1 - 2\sigma)} Y \frac{\Delta l}{l}. \quad (38.20)$$

When we constrain the sides, Young's modulus gets multiplied by a complicated function of σ . As you can most easily see from Eq. (38.19), the factor in front of Y is always greater than 1. It is harder to stretch the block when the sides are held—which also means that a block is *stronger* when the sides are held than when they are not.

38-3 The torsion bar; shear waves

Let's now turn our attention to an example which is more complicated because different parts of the material are stressed by different amounts. We consider a twisted rod such as you would find in a drive shaft of some machinery, or in a quartz fiber suspension used in a delicate instrument. As you probably know from experiments with the torsion pendulum, the *torque* on a twisted rod is proportional to the *angle*—the constant of proportionality obviously depending upon the length of the rod, on the radius of the rod, and on the properties of the material. The question is: In what way? We are now in a position to answer this question; it's just a matter of working out some geometry.

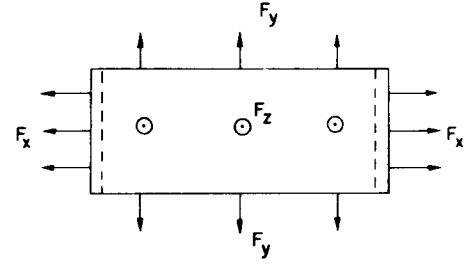


Fig. 38-8. Stretching without lateral contraction.

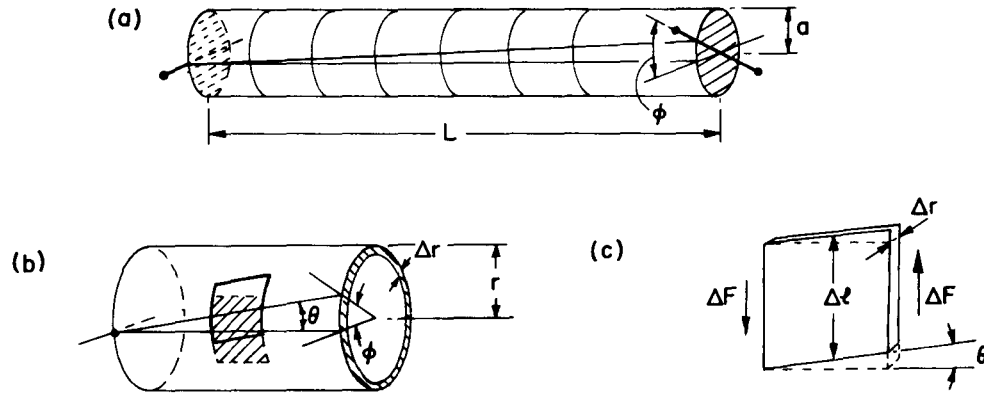


Fig. 38-9. (a) A cylindrical bar in torsion. (b) A cylindrical shell in torsion. (c) Each small piece of the shell is in shear.

Fig. 38-9(a) shows a cylindrical rod of length L , and radius a , with one end twisted by the angle ϕ with respect to the other. If we want to relate the strains to what we already know, we can think of the rod as being made up of many cylindrical shells and work out separately what happens to each shell. We start by looking at a thin, short cylinder of radius r (less than a) and thickness Δr —as drawn in Fig. 38-9(b). Now if we look at a piece of this cylinder that was originally a small square, we see that it has been distorted into a parallelogram. Each such element of the cylinder is in shear, and the shear angle θ is

$$\theta = \frac{r\phi}{L}.$$

The shear stress g in the material is, therefore [from Eq. (38.13)],

$$g = \mu\theta = \mu \frac{r\phi}{L}. \quad (38.21)$$

The shear stress is the tangential force ΔF on the end of the square divided by the area $\Delta l \Delta r$ of the end [see Fig. 38-9(c)]

$$g = \frac{\Delta F}{\Delta l \Delta r}.$$

The force ΔF on the end of such a square contributes a torque $\Delta\tau$ around the axis of the rod equal to

$$\Delta\tau = r \Delta F = rg \Delta l \Delta r. \quad (38.22)$$

The total torque τ is the sum of such torques around a complete circumference of the cylinder. So putting together enough pieces so that the Δl 's add up to $2\pi r$, we find that the total torque, for a *hollow tube*, is

$$rg(2\pi r) \Delta r. \quad (38.23)$$

Or, using (38.21),

$$\tau = 2\pi\mu \frac{r^3 \Delta r \phi}{L}. \quad (38.24)$$

We get that the rotational stiffness, τ/ϕ , of a hollow tube is proportional to the cube of the radius r and to the thickness Δr , and inversely proportional to the length L .

We can now imagine a solid rod to be made up of a series of concentric tubes, each twisted by the same angle ϕ (although the internal *stresses* are different for each tube). The total torque is the sum of the torques required to rotate each shell; for the *solid* rod

$$\tau = 2\pi\mu \frac{\phi}{L} \int r^3 dr,$$

where the integral goes from $r = 0$ to $r = a$, the radius of the rod. Integrating, we have

$$\tau = \mu \frac{\pi a^4}{2L} \phi. \quad (38.25)$$

For a rod in torsion, the torque is proportional to the angle and is proportional to the *fourth power* of the diameter—a rod twice as thick is sixteen times as stiff for torsion.

Before leaving the subject of torsion, let us apply what we have just learned to an interesting problem: torsional waves. If you take a long rod and suddenly twist one end, a wave of twist works its way along the rod, as sketched in Fig. 38-10(a). That's a little more exciting than a steady twist—let's see whether we can work out what happens.

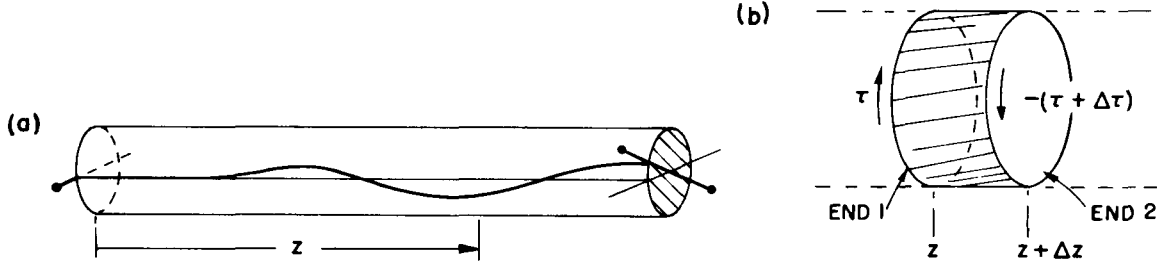


Fig. 38-10. (a) A torsional wave on a rod. (b) A volume element of the rod.

Let z be the distance to some point down the rod. For a static torsion the torque is the same everywhere along the rod, and is proportional to ϕ/L , the total torsion angle over the total length. What matters to the material is the local torsional strain, which is, you will appreciate, $\partial\phi/\partial z$. When the torsion along the rod is not uniform, we should replace Eq. (38.25) by

$$\tau(z) = \mu \frac{\pi a^4}{2} \frac{\partial\phi}{\partial z}. \quad (38.26)$$

Now let's look at what happens to an element of length Δz shown magnified in Fig. 38-10(b). There is a torque $\tau(z)$ at end 1 of the little hunk of rod, and a different torque $\tau(z + \Delta z)$ at end 2. If Δz is small enough, we can use a Taylor expansion and write

$$\tau(z + \Delta z) = \tau(z) + \left(\frac{\partial\tau}{\partial z}\right) \Delta z. \quad (38.27)$$

The net torque $\Delta\tau$ acting on the little piece of rod between z and $z + \Delta z$ is clearly the difference between $\tau(z)$ and $\tau(z + \Delta z)$, or $\Delta\tau = (\partial\tau/\partial z) \Delta z$. Differentiating Eq. (38.26), we get

$$\Delta\tau = \mu \frac{\pi a^4}{2} \frac{\partial^2\phi}{\partial z^2} \Delta z. \quad (38.28)$$

The effect of this net torque is to give an angular acceleration to the little slice of the rod. The mass of the slice is

$$\Delta M = (\pi a^2 \Delta z) \rho,$$

where ρ is the density of the material. We worked out in Chapter 19, Vol. I, that the moment of inertia of a circular cylinder is $mr^2/2$; calling the moment of inertia of our piece ΔI , we have

$$\Delta I = \frac{\pi}{2} \rho a^4 \Delta z. \quad (38.29)$$

Newton's law says the torque is equal to the moment of inertia times the angular acceleration, or

$$\Delta\tau = \Delta I \frac{\partial^2\phi}{\partial t^2}. \quad (38.30)$$

Pulling everything together, we get

$$\mu \frac{\pi a^4}{2} \frac{\partial^2 \phi}{\partial z^2} \Delta z = \frac{\pi}{2} \rho a^4 \Delta z \frac{\partial^2 \phi}{\partial t^2},$$

or

$$\frac{\partial^2 \phi}{\partial z^2} - \frac{\rho}{\mu} \frac{\partial^2 \phi}{\partial t^2} = 0. \quad (38.31)$$

You will recognize this as the one-dimensional wave equation. We have found that waves of torsion will propagate down the rod with the speed

$$C_{\text{shear}} = \sqrt{\frac{\mu}{\rho}}. \quad (38.32)$$

The *denser* the rod—for the same stiffness—the *slower* the waves; and the *stiffer* the rod, the quicker the waves work their way down. The speed does *not* depend upon the diameter of the rod.

Torsional waves are a special example of *shear waves*. In general, shear waves are those in which the strains do not change the *volume* of any part of the material. In torsional waves, we have a particular distribution of such shear stresses—namely, distributed on a circle. But for any arrangement of shear stresses, waves will propagate with the same speed—the one given in Eq. (38.32). For example, the seismologists find such shear waves travelling in the interior of the earth.

We can have another kind of a wave in the elastic world inside a solid material. If you push something, you can start “longitudinal” waves—also called “compressional” waves. They are like the sound waves in air or in water—the displacements are in the same direction as the wave propagation. (At the surfaces of an elastic body there can also be other types of waves—called “Rayleigh waves” or “Love waves.” In them, the strains are neither purely longitudinal nor purely transverse. We will not have time to study them.)

While we’re on the subject of waves, what is the velocity of the pure compressional waves in a *large* solid body like the earth? We say “large” because the speed of sound in a thick body is different from what it is, for instance, along a thin rod. By a “thick” body we mean one in which the transverse dimensions are much larger than the wavelength of the sound. Then, when we push on the object, it cannot expand sideways—it can only compress in one dimension. Fortunately, we have already worked out the special case of the compression of a constrained elastic material. We have also worked out in Chapter 47, Vol. I, the speed of sound waves in a gas. Following the same arguments you can see that the speed of sound in a solid is equal to $\sqrt{Y'/\rho}$, where Y' is the “longitudinal modulus”—or pressure divided by the relative change in length—for the constrained case. This is just the ratio of $\Delta l/l$ to F/A we got in Eq. (38.20). So the speed of the longitudinal waves is given by

$$C_{\text{long}}^2 = \frac{Y'}{\rho} = \frac{1 - \sigma}{(1 + \sigma)(1 - 2\sigma)} \frac{Y}{\rho}. \quad (38.33)$$

So long as σ is between zero and $1/2$, the shear modulus μ is less than Young’s modulus Y , and also Y' is greater than Y , so

$$\mu < Y < Y'.$$

This means that longitudinal waves travel faster than shear waves. One of the most precise ways of measuring the elastic constants of a substance is by measuring the density of the material and the speeds of the two kinds of waves. From this information one can get both Y and σ . It is, incidentally, by measuring the difference in the arrival times of the two kinds of waves from an earthquake that a seismologist can estimate—even from the signals at only one station—the distance to the quake.

38-4 The bent beam

We want now to look at another practical matter—the *bending* of a rod or a beam. What are the forces when we bend a bar of some arbitrary cross section? We will work it out thinking of a bar with a circular cross section, but our answer will be good for any shape. To save time, however, we will cut some corners, so our theory we will work out is only approximate. Our results will be correct only when the radius of the bend is much larger than the thickness of the beam.

Suppose you grab the two ends of a straight bar and bend it into some curve like the one shown in Fig. 38-11. What goes on inside the bar? Well, if it is curved, that means that the material on the inside of the curve is compressed and the material on the outside is stretched. There is some surface which goes along more or less parallel to the axis of the bar that is neither stretched nor compressed. This is called the *neutral surface*. You would expect this surface to be near the “middle” of the cross section. It can be shown (but we won’t do it here) that, for small bending of simple beams, the neutral surface goes through the “center of gravity” of the cross section. This is true only for “pure” bending—if you are not stretching or compressing the beam at the same time.

For pure bending, then, a thin transverse slice of the bar is distorted as shown in Fig. 38-12(a). The material below the neutral surface has a compressional strain which is *proportional to the distance* from the neutral surface; and the material above is stretched, also in proportion to its distance from the neutral surface. So the longitudinal *stretch* Δl is proportional to the height y . The constant of proportionality is just l over the radius of curvature of the bar—see Fig. 38-12:

$$\frac{\Delta l}{l} = \frac{y}{R}.$$

So the force per unit area—the stress—in a small strip at y is also proportional to the distance from the neutral surface

$$\frac{\Delta F}{\Delta A} = Y \frac{y}{R}. \quad (38.34)$$

Now let’s look at the *forces* that would produce such a strain. The forces acting on the little segment drawn in Fig. 38-12 are shown in the figure. If we think of any transverse cut, the forces acting across it are one way above the neutral surface and the other way below. They come in pairs to make a “bending moment” $\mathfrak{M}$ —by which we mean the torque about the neutral line. We can compute the total moment by integrating the force times the distance from the neutral surface for one of the faces of the segment of Fig. 38-12:

$$\mathfrak{M} = \int_{\text{cross sect}} y dF. \quad (38.35)$$

From Eq. (38.34), $dF = Yy/R dA$, so

$$\mathfrak{M} = \frac{Y}{R} \int y^2 dA.$$

The integral of $y^2 dA$ is what we can call the “moment of inertia” of the geometric cross section about a horizontal axis through its “center of mass”;* we will call it I :

$$\mathfrak{M} = \frac{YI}{R} \quad (38.36)$$

$$I = \int y^2 dA. \quad (38.37)$$

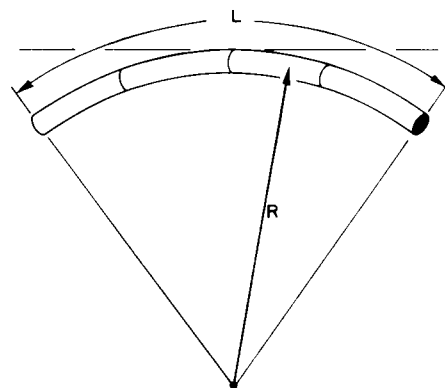


Fig. 38-11. A bent beam.

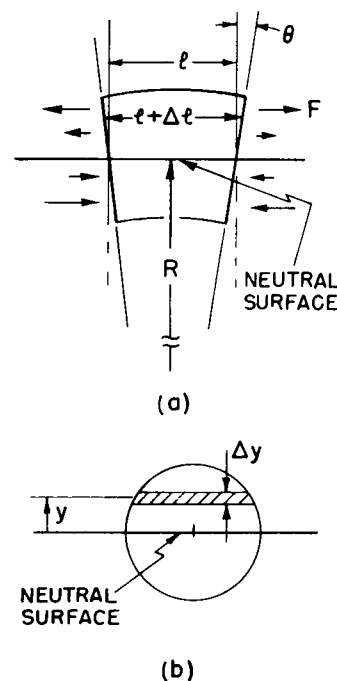


Fig. 38-12. (a) Small segment of a bent beam. (b) Cross section of the beam.

* It is, of course, really the moment of inertia of a slice with unit mass per unit area.

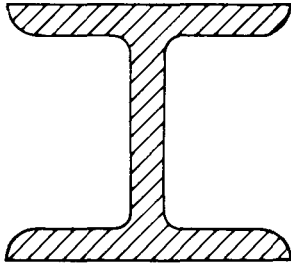


Fig. 38-13. An "I" beam.

Equation (38.36), then, gives us the relation between the bending moment $\mathfrak{M}$ and the curvature $1/R$ of the beam. The "stiffness" of the beam is proportional to Y and to the moment of inertia I . In other words, if you want the stiffest possible beam with a given amount of, say, aluminum, you want to put as much of it as possible as far as you can from the neutral surface, to make a large moment of inertia. You can't carry this to an extreme, however, because then the thing will not curve as we have supposed—it will buckle or twist and become weaker again. But now you see why structural beams are made in the form of an I or an H—as shown in Fig. 38-13.

As an example of the use of our beam equation (38.36), let's work out the deflection of a cantilevered beam with a concentrated force W acting at the free end, as sketched in Fig. 38-14. (By "cantilevered" we simply mean that the beam is supported in such a way that both the position *and* the slope are fixed at one end—it is stuck into a cement wall.) What is the shape of the beam? Let's call the deflection at the distance x from the fixed end z ; we want to know $z(x)$. We'll work it out only for small deflections. We will also assume that the beam is long in comparison with its cross section. Now, as you know from your mathematics courses, the curvature $1/R$ of any curve $z(x)$ is given by

$$\frac{1}{R} = \frac{d^2z/dx^2}{[1 + (dz/dx)^2]^{3/2}}. \quad (38.38)$$

Since we are interested only in small slopes—this is usually the case in engineering structures—we neglect $(dz/dx)^2$ in comparison with 1, and take

$$\frac{1}{R} = \frac{d^2z}{dx^2}. \quad (38.39)$$

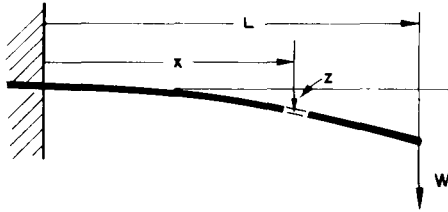


Fig. 38-14. A cantilevered beam with a weight at one end.

We also need to know the bending moment $\mathfrak{M}$. It is a function of x because it is equal to the torque about the neutral axis of any cross section. Let's neglect the weight of the beam and take only the downward force W at the end of the beam. (You can put in the beam weight yourself if you want.) Then the bending moment at x is

$$\mathfrak{M}(x) = W(L - x),$$

because that is the torque about the point at x , exerted by the weight W —the torque which the beam must support of x . We get

$$W(L - x) = \frac{YI}{R} = YI \frac{d^2z}{dx^2}$$

or

$$\frac{d^2z}{dx^2} = \frac{W}{YI} (L - x). \quad (38.40)$$

This one we can integrate without any tricks; we get

$$z = \frac{W}{YI} \left(\frac{Lx^2}{2} - \frac{x^3}{6} \right), \quad (38.41)$$

using our assumptions that $z(0) = 0$ and that dz/dx is also zero at $x = 0$. That is the shape of the beam. The displacement of the end is

$$z(L) = \frac{W}{YI} \frac{L^3}{3}; \quad (38.42)$$

the displacement of the end of a beam increases as the cube of the length.

In deriving our approximate beam theory, we have assumed that the cross section of the beam did not change when the beam was bent. When the thickness of the beam is small compared to the radius of curvature, the cross section changes very little and our result is O.K. In general, however, this effect cannot be neglected, as you can easily demonstrate for yourselves by bending a soft-rubber eraser in your fingers. If the cross section was originally rectangular, you will find that when

it is bent it bulges at the bottom (see Fig. 38-15). This happens because when we compress the bottom, the material expands sideways—as described by Poisson's ratio. Rubber is easy to bend or stretch, but it is somewhat like a liquid in that it's hard to change the *volume*—as shows up nicely when you bend the eraser. For an incompressible material, Poisson's ratio would be exactly 1/2—for rubber it is nearly that.

38-5 Buckling

We want now to use our beam theory to understand the theory of the “buckling” of beams, or columns, or rods. Consider the situation sketched in Fig. 38-16 in which a rod that would normally be straight is held in its bent shape by two opposite forces that push on the ends of the rod. We would like to calculate the shape of the rod and the *magnitude of the forces* on the ends.

Let the deflection of the rod from the straight line between the ends be $y(x)$, where x is the distance from one end. The bending moment $\mathfrak{M}$ at the point P in the figure is equal to the force F multiplied by the moment arm, which is the perpendicular distance y ,

$$\mathfrak{M}(x) = Fy. \quad (38.43)$$

Using the beam equation (38.36), we have

$$\frac{YI}{R} = Fy. \quad (38.44)$$

For small deflections, we can take $1/R = -d^2y/dx^2$ (the minus sign because the curvature is downward). We get

$$\frac{d^2y}{dx^2} = -\frac{F}{YI} y, \quad (38.45)$$

which is the differential equation of a sine wave. So for *small* deflections, the curve of such a bent beam is a sine curve. The “wavelength” λ of the sine wave is twice the distance L between the ends. If the bending is small, this is just twice the unbent length of the rod. So the curve is

$$y = K \sin \pi x/L.$$

Taking the second derivative, we get

$$\frac{d^2y}{dx^2} = -\frac{\pi^2}{L^2} y.$$

Comparing this to Eq. (38.45), we see that the force is

$$F = \pi^2 \frac{YI}{L^2}. \quad (38.46)$$

For small bendings the force is *independent of the bending displacement y* !

We have, then, the following thing physically. If the force is less than the F given in Eq. (38.46), there will be no bending at all. But if it is slightly *greater* than this force, the material will suddenly bend a large amount—that is, for forces above the critical force $\pi^2 YI/L^2$ (often called the “Euler force”) the beam will “buckle.” If the loading on the second floor of a building exceeds the Euler force for the supporting columns, the building will collapse. Another place where the buckling force is most important is in space rockets. On one hand, the rocket must be able to hold its own weight on the launching pad and endure the stresses during acceleration, on the other hand, it is important to keep the weight of the structure to a minimum, so that the payload and fuel capacity may be made as large as possible.

Actually a beam will not necessarily collapse completely when the force exceeds the Euler force. When the displacements get large, the force is larger than

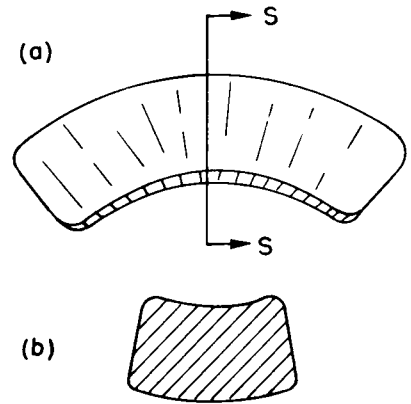


Fig. 38-15 (a) A bent eraser; (b) cross section.

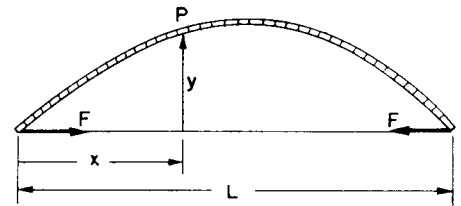


Fig. 38-16. A buckled beam.

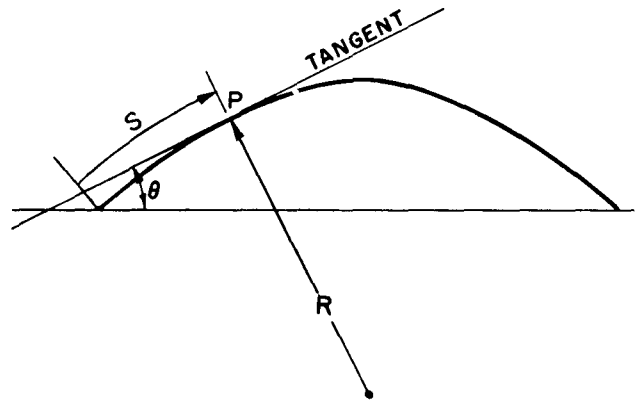


Fig. 38-17. The coordinates S and θ for the curve of a bent beam.

what we have found because of the terms in $1/R$ in Eq. (38.38) that we have neglected. To find the forces for a large bending of the beam, we have to go back to the exact equation, Eq. (38.44), which we had before we used the approximate relation between R and y . Equation (38.44) has a rather simple geometrical property.* It's a little complicated to work out, but rather interesting. Instead of describing the curve in terms of x and y , we can use two new variables: S , the distance along the curve, and θ the slope of the tangent to the curve. See Fig. 38-17. The curvature is the rate of change of angle with distance:

$$\frac{1}{R} = \frac{d\theta}{dS}.$$

We can, therefore write the exact equation (38.44) as

$$\frac{d\theta}{dS} = -\frac{F}{YI} y.$$

If we take the derivative of this equation with respect to S and replace dy/dS by $\sin \theta$, we get

$$\frac{d^2\theta}{dS^2} = -\frac{F}{YI} \sin \theta. \quad (38.47)$$

[If θ is small, we get back Eq. (38.45). Everything is O.K.]

Now it may or may not delight you to know that Eq. (38.47) is exactly the same one you get for the large amplitude oscillations of a pendulum—with F/YI replaced by another constant, of course. We learned way back in Chapter 9, Vol. I, how to find the solution of such an equation by a numerical calculation.† The answers you get are some fascinating curves—known as the curves of the “Elastica.” Figure 38-18 shows three curves for different values of F/YI .

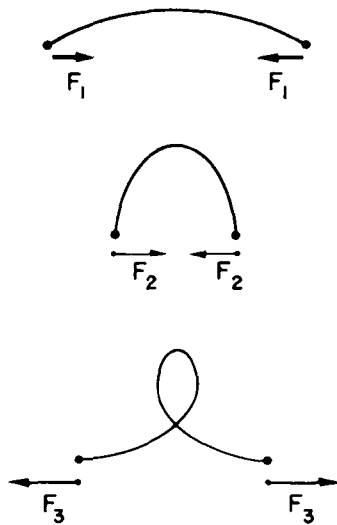


Fig. 38-18. Curves of a bent rod

* The same equation appears, incidentally, in other physical situations—for example, the meniscus at the surface of a liquid contained between parallel planes—and the same geometrical solution can be used.

† The solutions can also be expressed in terms of some functions, called the “Jacobian elliptic functions,” that someone else has already computed.

Elastic Materials

39-1 The tensor of strain

In the last chapter we talked about the distortions of particular elastic objects. In this chapter we want to look at what can happen *in general* inside an elastic material. We would like to be able to describe the conditions of stress and strain inside some big glob of jello which is twisted and squashed in some complicated way. To do this, we need to be able to describe the *local strain* at every point in an elastic body; we can do it by giving a set of six numbers—which are the components of a symmetric tensor—for each point. Earlier, we spoke of the stress tensor (Chapter 31); now we need the tensor of strain.

Imagine that we start with the material initially unstrained and watch the motion of a small speck of “dirt” embedded in the material when the strain is applied. A speck that was at the point P located at $\mathbf{r} = (x, y, z)$ moves to a new position P' at $\mathbf{r}' = (x', y', z')$ as shown in Fig. 39-1. We will call $\mathbf{u}$ the vector displacements from P to P' . Then

$$\mathbf{u} = \mathbf{r}' - \mathbf{r}. \quad (39.1)$$

The displacement $\mathbf{u}$ depends, of course, on which point P we start with, so $\mathbf{u}$ is a vector function of $\mathbf{r}$ —or, if you prefer, of (x, y, z) .

Let's look first at a simple situation in which the strain is constant over the material—so we have what is called a *homogeneous strain*. Suppose, for instance, that we have a block of material and we stretch it uniformly. We just change its dimensions uniformly in one direction—say, in the x -direction, as shown in Fig. 39-2. The motion u_x of a speck at x is proportional to x . In fact,

$$\frac{u_x}{x} = \frac{\Delta l}{l}.$$

We will write u_x this way:

$$u_x = e_{xx}x.$$

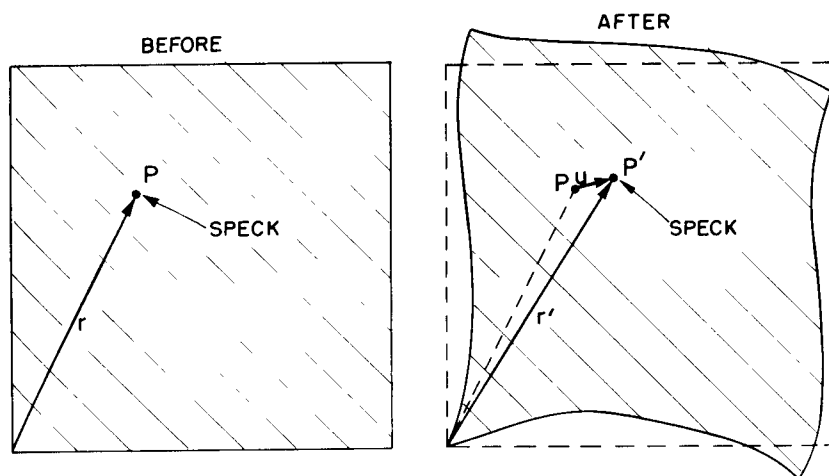


Fig. 39-1. A speck of the material at the point P in an unstrained block moves to P' where the block is strained.

39-1 The tensor of strain

39-2 The tensor of elasticity

39-3 The motions in an elastic body

39-4 Nonelastic behavior

39-5 Calculating the elastic constants

Reference: C. Kittel, *Introduction to Solid State Physics*, John Wiley and Sons, Inc., New York, 2nd ed., 1956.

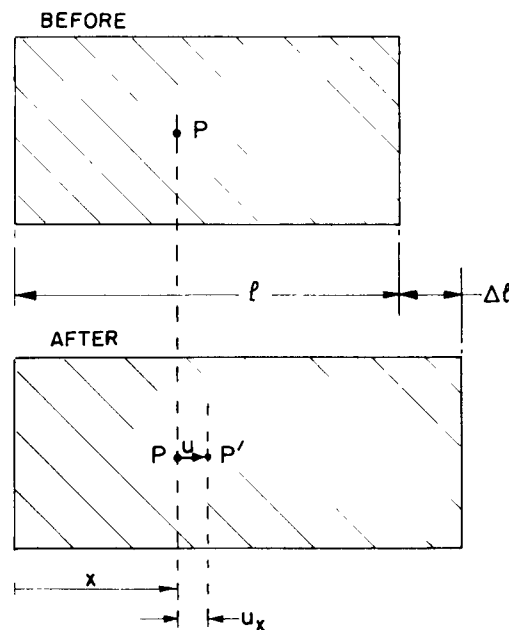


Fig. 39-2. A homogeneous stretch-type strain.

The proportionality constant e_{xx} is, of course, the same thing as $\Delta l/l$. (You will see shortly why we use a double subscript.)

If the strain is not uniform, the relation between u_x and x will vary from place to place in the material. For the general situation, we define the e_{xx} by a kind of local $\Delta l/l$, namely by

$$e_{xx} = \partial u_x / \partial x. \quad (39.2)$$

This number—which is now a function of x , y , and z —describes the amount of stretching in the x -direction throughout the hunk of jello. There may, of course, also be stretching in the y - and z -directions. We describe them by the numbers

$$e_{yy} = \frac{\partial u_y}{\partial y}, \quad e_{zz} = \frac{\partial u_z}{\partial z}. \quad (39.3)$$

We need to be able to describe also the shear-type strains. Suppose we imagine a little cube marked out in the initially undisturbed jello. When the jello is pushed out of shape, this cube may get changed into a parallelogram, as sketched in Fig. 39-3.* In this kind of a strain, the x -motion of each particle is proportional to its y -coordinate,

$$u_x = \frac{\theta}{2} y. \quad (39.4)$$

And there is also a y -motion proportional to x ,

$$u_y = \frac{\theta}{2} x. \quad (39.5)$$

So we can describe such a shear-type strain by writing

$$u_x = e_{xy}y, \quad u_y = e_{yx}x$$

with

$$e_{xy} = e_{yx} = \frac{\theta}{2}.$$

Now you might think that when the strains are not homogeneous we could describe the generalized shear strains by defining the quantities e_{xy} and e_{yx} by

$$e_{xy} = \frac{\partial u_x}{\partial y}, \quad e_{yx} = \frac{\partial u_y}{\partial x}. \quad (39.6)$$

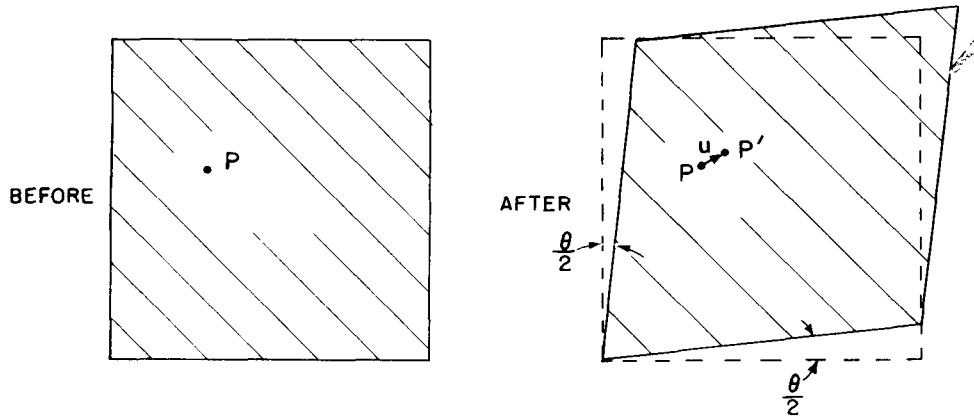


Fig. 39-3. A homogeneous shear strain.

But there is one difficulty. Suppose that the displacements u_x and u_y were given by

$$u_x = \frac{\theta}{2} y, \quad u_y = -\frac{\theta}{2} x.$$

* We choose for the moment to split the total shear angle θ into two equal parts and make the strain symmetric with respect to x and y .

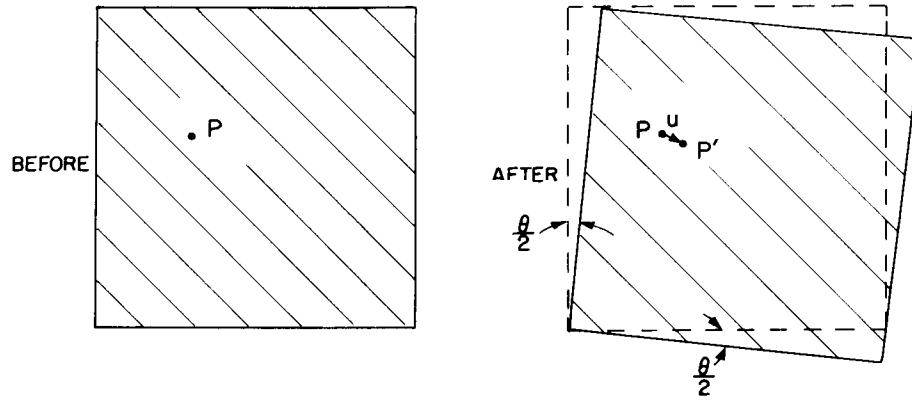


Fig. 39-4. A homogeneous rotation—there is no strain.

They are like Eqs. (39.4) and (39.5) except that the sign of u_y is reversed. With these displacements a little cube in the jello simply gets shifted by the angle $\theta/2$, as shown in Fig. 39-4. There is no strain at all—just a rotation in space. There is no distortion of the material; the *relative* positions of all the atoms are not changed at all. We must somehow make our definitions so that pure rotations are not included in our definitions of a shear strain. The key point is that if $\partial u_y/\partial x$ and $\partial u_x/\partial y$ are equal and opposite, there is no strain; so we can fix things up by *defining*

$$e_{xy} = e_{yx} = \frac{1}{2}(\partial u_y/\partial x + \partial u_x/\partial y).$$

For a pure rotation they are both zero, but for a pure shear we get that e_{xy} is equal to e_{yx} , as we would like.

In the most general distortion—which may include stretching or compression as well as shear—we *define* the state of strain by giving the nine numbers

$$\begin{aligned} e_{xx} &= \frac{\partial u_x}{\partial x}, \\ e_{yy} &= \frac{\partial u_y}{\partial y}, \\ &\vdots \\ e_{xy} &= \frac{1}{2}(\partial u_y/\partial x + \partial u_x/\partial y), \\ &\vdots \end{aligned} \quad (39.7)$$

These are the terms of a *tensor of strain*. Because it is a *symmetric tensor*—our definitions make $e_{xy} = e_{yx}$, always—there are really only six different numbers. You remember (see Chapter 31) that the general characteristic of a tensor is that the terms transform like the products of the components of two vectors. (If $\mathbf{A}$ and $\mathbf{B}$ are vectors, $C_{ij} = A_i B_j$ is a tensor.) Each term of e_{ij} is a product (or the sum of such products) of the components of the vector $\mathbf{u} = (u_x, u_y, u_z)$, and of the operator $\nabla = (\partial/\partial x, \partial/\partial y, \partial/\partial z)$, which we know transforms like a vector. Let's let x_1, x_2 , and x_3 stand for x, y , and z and u_1, u_2 , and u_3 stand for u_x, u_y , and u_z ; then we can write the general term e_{ij} of the strain tensor as

$$e_{ij} = \frac{1}{2}(\partial u_j/\partial x_i + \partial u_i/\partial x_j), \quad (39.8)$$

where i and j can be 1, 2, or 3.

When we have a homogeneous strain—which may include both stretching and shear—all of the e_{ij} are constants, and we can write

$$u_x = e_{xx}x + e_{xy}y + e_{xz}z. \quad (39.9)$$

(We choose our origin of x, y, z at the point where $\mathbf{u}$ is zero.) In this case, the strain tensor e_{ij} gives the relationship between two vectors: the coordinate vector $\mathbf{r} = (x, y, z)$ and the displacement vector $\mathbf{u} = (u_x, u_y, u_z)$.

When the strains are not homogeneous, any piece of the jello may also get somewhat twisted—there will be a local rotation. If the distortions are all small, we would have

$$\Delta u_i = \sum_j (e_{ij} - \omega_{ij}) \Delta x_j, \quad (39.10)$$

where ω_{ij} is an *antisymmetric* tensor,

$$\omega_{ij} = \frac{1}{2}(\partial u_j / \partial x_i - \partial u_i / \partial x_j), \quad (39.11)$$

which describes the rotation. We will, however, not worry any more about rotations, but only about the strains described by the symmetric tensor e_{ij} .

39-2 The tensor of elasticity

Now that we have described the strains, we want to relate them to the internal forces—the stresses in the material. For each small piece of the material, we assume Hooke's law holds and write that the stresses are proportional to the strains. In Chapter 31 we defined the stress tensor S_{ij} as the i th component of the force across a unit area perpendicular to the j -axis. Hooke's law says that each component of S_{ij} is linearly related to *each* of the components of strain. Since S and e each have nine components, there are $9 \times 9 = 81$ possible coefficients which describe the elastic properties of the material. They are constants if the material itself is homogeneous. We write these coefficients as C_{ijkl} and define them by the equation

$$S_{ij} = \sum_{k,l} C_{ijkl} e_{kl}, \quad (39.12)$$

where i, j, k, l all take on the values 1, 2, or 3. Since the coefficients C_{ijkl} relate one tensor to another, they also form a tensor—a tensor of the *fourth rank*. We can call it the *tensor of elasticity*.

Suppose that all the C 's are known and that you put a complicated force on an object of some peculiar shape. There will be all kinds of distortion, and the thing will settle down with some twisted shape. What are the displacements? You can see that it is a complicated problem. If you knew the strains, you could find the stresses from Eq. (39.12)—or vice versa. But the stresses and strains you end up with at any point depend on what happens in all the rest of the material.

The easiest way to get at the problem is by thinking of the energy. When there is a force F proportional to a displacement x , say $F = kx$, the work required for any displacement x is $kx^2/2$. In a similar way, the work w that goes into *each unit volume* of a distorted material turns out to be

$$w = \frac{1}{2} \sum_{ijkl} C_{ijkl} e_{ij} e_{kl}. \quad (39.13)$$

The total work W done in distorting the body is the integral of w over its volume:

$$W = \int \frac{1}{2} \sum_{ijkl} C_{ijkl} e_{ij} e_{kl} d \text{Vol}. \quad (39.14)$$

This is then the potential energy stored in the internal stresses of the material. Now when a body is in equilibrium, this internal energy must be *at a minimum*. So the problem of finding the strains in a body can be solved by finding the set of displacements u throughout the body which will make W a minimum. In Chapter 19 we gave some of the general ideas of the calculus of variations that are used in tackling minimization problems like this. We cannot go into the problem in any more detail here.

What we are mainly interested in now is what we can say about the general properties of the tensor of elasticity. First, it is clear that there are *not* really 81 *different* terms in C_{ijkl} . Since both S_{ij} and e_{ij} are symmetric tensors, each with only six different terms, there can be at most 36 different terms in C_{ijkl} . There are, however, usually many fewer than this.

Let's look at the special case of a cubic crystal. In it, the energy density w starts out like this:

$$w = \frac{1}{2} \{ C_{xxxx} e_{xx}^2 + C_{xxxy} e_{xx} e_{xy} + C_{xxxz} e_{xx} e_{xz} + C_{xxyx} e_{xx} e_{xy} + C_{xyyy} e_{xx} e_{yy} + \dots \text{etc} \dots + C_{yyyy} e_{yy}^2 + \dots \text{etc} \dots \text{etc} \dots \}, \quad (39.15)$$

with 81 terms in all! Now a cubic crystal has certain symmetries. In particular, if the crystal is rotated 90° , it has the same physical properties. It has the same stiffness for stretching in the y -direction as for stretching in the x -direction. Therefore, if we change our definition of the coordinate directions x and y in Eq. (39.15), the energy wouldn't change. It must be that for a cubic crystal

$$C_{xxxx} = C_{yyyy} = C_{zzzz}. \quad (39.16)$$

Next we can show that the terms like C_{xxxy} must be zero. A cubic crystal has the property that it is symmetric under a *reflection* about any plane perpendicular to one of the axes. If we replace y by $-y$, nothing is different. But changing y to $-y$ changes e_{xy} to $-e_{xy}$ —a displacement which was toward $+y$ is now toward $-y$. If the energy is not to change, C_{xxxy} must go into $-C_{xxxy}$ when we make a reflection. But a reflected crystal is the same as before, so C_{xxxy} must be the *same* as $-C_{xxxy}$. This can happen only if both are zero.

You say, "But the same argument will make $C_{yyyy} = 0$!" No, because there are *four* y 's. The sign changes once for each y , and four minuses make a plus. If there are *two* or *four* y 's, the term does not have to be zero. It is zero only when there is *one*, or *three*. So, for a cubic crystal, any nonzero term of C will have only an *even number* of identical subscripts. (The arguments we have made for y obviously hold also for x and z .) We might then have terms like C_{xxyy} , C_{xyxy} , C_{yyxx} , and so on. We have already shown, however, that if we change all x 's to y 's and *vice versa* (or all z 's and x 's, and so on) we must get—for a cubic crystal—the same number. This means that there are *only three different* nonzero possibilities:

$$\begin{aligned} C_{xxxx} (= C_{yyyy} = C_{zzzz}), \\ C_{xxyy} (= C_{yyxx} = C_{xxzz}, \text{ etc.}), \\ C_{xyxy} (= C_{yxxy} = C_{zzzz}, \text{ etc.}). \end{aligned} \quad (39.17)$$

For a cubic crystal, then, the energy density will look like this:

$$w = \frac{1}{2} \{ C_{xxxx} (e_{xx}^2 + e_{yy}^2 + e_{zz}^2) + 2C_{xxyy} (e_{xx} e_{yy} + e_{yy} e_{zz} + e_{zz} e_{xx}) + 4C_{xyxy} (e_{xy}^2 + e_{yz}^2 + e_{zx}^2) \}. \quad (39.18)$$

For an isotropic—that is, noncrystalline—material, the symmetry is still higher. The C 's must be the same for *any* choice of the coordinate system. Then it turns out that there is another relation among the C 's, namely, that

$$C_{xxxx} = C_{xxyy} + C_{xyxy}. \quad (39.19)$$

We can see that this is so by the following general argument. The stress tensor S_{ij} has to be related to e_{ij} in a way that doesn't depend at all on the coordinate directions—it must be related only by *scalar* quantities. "That's easy," you say. "The only way to obtain S_{ij} from e_{ij} is by multiplication by a scalar constant. It's just Hooke's law. It must be that $S_{ij} = (\text{const})e_{ij}$." But that's not quite right; there could also be the *unit tensor* δ_{ij} multiplied by some scalar, linearly related to e_{ij} . The only invariant you can make that is linear in the e 's is $\sum e_{ii}$ (It transforms like $x^2 + y^2 + z^2$, which is a scalar.) So the most general form for the equation relating S_{ij} to e_{ij} —for isotropic materials—is

$$S_{ij} = 2\mu e_{ij} + \lambda \left(\sum_k e_{kk} \right) \delta_{ij}. \quad (39.20)$$

(The first constant is usually written as *two* times μ ; then the coefficient μ is equal

to the shear modulus we defined in the last chapter.) The constants μ and λ are called the Lamé elastic constants. Comparing Eq. (39.20) with Eq. (39.12), you see that

$$\begin{aligned} C_{xxyy} &= \lambda, \\ C_{xyxy} &= 2\mu, \\ C_{xxxx} &= 2\mu + \lambda. \end{aligned} \quad (39.21)$$

So we have proved that Eq. (39.19) is indeed true. You also see that the elastic properties of an isotropic material are completely given by two constants, as we said in the last chapter.

The C 's can be put in terms of any two of the elastic constants we have used earlier—for instance, in terms of Young's modulus Y and Poisson's ratio σ . We will leave it for you to show that

$$\begin{aligned} C_{xxxx} &= \frac{Y}{1 + \sigma} \left(1 + \frac{\sigma}{1 - 2\sigma} \right), \\ C_{xxyy} &= \frac{Y}{1 + \sigma} \left(\frac{\sigma}{1 - 2\sigma} \right), \\ C_{xyxy} &= \frac{Y}{(1 + \sigma)}. \end{aligned} \quad (39.22)$$

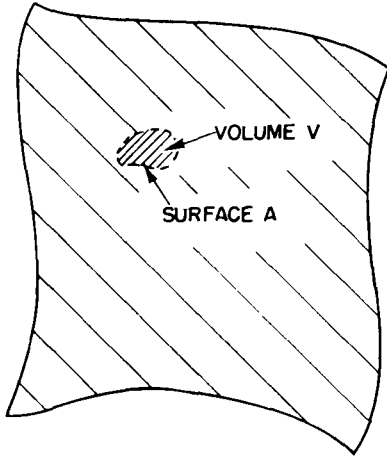


Fig. 39-5. A small volume element V bounded by the surface A .

39-3 The motions in an elastic body

We have pointed out that for an elastic body *in equilibrium* the internal stresses adjust themselves to make the energy a minimum. Now we take a look at what happens when the internal forces are *not* in equilibrium. Let's say we have a small piece of the material inside some surface A . See Fig. 39-5. If the piece is in equilibrium, the total force F acting on it must be zero. We can think of this force as being made up of two parts. There could be one part due to "external" forces like gravity, which act from a distance on the matter in the piece to produce a *force per unit volume* f_{ext} . The total external force F_{ext} is the integral of f_{ext} over the volume of the piece:

$$F_{\text{ext}} = \int f_{\text{ext}} dV. \quad (39.23)$$

In equilibrium, this force would be balanced by the total force F_{int} from the neighboring material which acts across the surface A . When the piece is *not* in equilibrium—if it is moving—the sum of the internal and external forces is equal to the mass times the acceleration. We would have

$$F_{\text{ext}} + F_{\text{int}} = \int \rho \mathbf{r} dV, \quad (39.24)$$

where ρ is the density of the material, and $\mathbf{r}$ is its acceleration. We can now combine Eqs. (39.23) and (39.24), writing

$$F_{\text{int}} = \int_v (-f_{\text{ext}} + \rho \mathbf{r}) dV. \quad (39.25)$$

We will simplify our writing by defining

$$\mathbf{f} = -f_{\text{ext}} + \rho \mathbf{r}. \quad (39.26)$$

Then Eq. (39.25) is written

$$F_{\text{int}} = \int_v \mathbf{f} dV. \quad (39.27)$$

What we have called F_{int} is related to the stresses in the material. The stress tensor S_{ij} was defined (Chapter 31) so that the x -component of the force dF across a surface element da , whose unit normal is $\mathbf{n}$, is given by

$$dF_x = (S_{xx}n_x + S_{xy}n_y + S_{xz}n_z) da. \quad (39.28)$$

The x -component of $\mathbf{F}_{\text{int}}$ on our little piece is then the integral of dF_x over the surface. Substituting this into the x -component of Eq. (39.27), we get

$$\int_A (S_{xx}n_x + S_{xy}n_y + S_{xz}n_z) da = \int_v f_x dV. \quad (39.29)$$

We have a surface integral related to a volume integral—and that reminds us of something we learned in electricity. Note that if you ignore the first subscript x on each of the S 's in the left-hand side of Eq. (39.29), it looks just like the integral of a quantity " $\mathbf{S}$ " $\cdot \mathbf{n}$ —that is, the normal component of a vector—over the surface. It would be the flux of " $\mathbf{S}$ " out of the volume. And this could be written, using Gauss law, as the volume integral of the divergence of " $\mathbf{S}$ ". It is, in fact, true whether the x -subscript is there or not—it is just a mathematical theorem you get by integrating by parts. In other words, we can change Eq. (39.29) into

$$\int_v \left(\frac{\partial S_{xx}}{\partial x} + \frac{\partial S_{xy}}{\partial y} + \frac{\partial S_{xz}}{\partial z} \right) dV = \int_v f_x dV \quad (39.30)$$

Now we can leave off the volume integrals and write the differential equation for the general component of $\mathbf{f}$ as

$$f_i = \sum_j \frac{\partial S_{ij}}{\partial x_j}. \quad (39.31)$$

This tells us how the force per unit volume is related to the stress tensor S_{ij} .

The theory of the motions inside a solid works this way. If we start out knowing the initial displacements—given by, say, $\mathbf{u}$ —we can work out the strains e_{ij} . From the strains we can get the stresses from Eq. (39.12). From the stresses we can get the force density $\mathbf{f}$ in Eq. (39.31). Knowing $\mathbf{f}$, we can get, from Eq. (39.26), the acceleration $\mathbf{r}$ of the material, which tells us how the displacements will be changing. Putting everything together, we get the horrible equation of motion for an elastic solid. We will just write down the results that come out for an isotropic material. If you use (39.20) for S_{ij} , and write the e_{ij} as $\frac{1}{2}\partial u_i/\partial x_j + \partial u_j/\partial x_i$, you end up with the vector equation

$$\mathbf{f} = (\lambda + \mu) \nabla(\nabla \cdot \mathbf{u}) + \mu \nabla^2 \mathbf{u}. \quad (39.32)$$

You can, in fact, see that the equation relating $\mathbf{f}$ and $\mathbf{u}$ *must* have this form. The force must depend on the second derivatives of the displacements $\mathbf{u}$. What second derivatives of $\mathbf{u}$ are there that are vectors? One is $\nabla(\nabla \cdot \mathbf{u})$; that's a true vector. The only other one is $\nabla^2 \mathbf{u}$. So the most general form is

$$\mathbf{f} = a \nabla(\nabla \cdot \mathbf{u}) + b \nabla^2 \mathbf{u},$$

which is just (39.32) with a different definition of the constants. You may be wondering why we don't have a third term using $\nabla \times \nabla \times \mathbf{u}$, which is also a vector. But remember that $\nabla \times \nabla \times \mathbf{u}$ is the same thing as $\nabla^2 \mathbf{u} - \nabla(\nabla \cdot \mathbf{u})$, so it is a linear combination of the two terms we have. Adding it would add nothing new. We have proved once more that isotropic material has only two elastic constants.

For the equation of motion of the material, we can set (39.32) equal to $\rho \partial^2 \mathbf{u}/\partial t^2$ —neglecting for now any body forces like gravity—and get

$$\rho \frac{\partial^2 \mathbf{u}}{\partial t^2} = (\lambda + \mu) \nabla(\nabla \cdot \mathbf{u}) + \mu \nabla^2 \mathbf{u}. \quad (39.33)$$

It looks something like the wave equation we had in electromagnetism, except that there is an additional complicating term. For materials whose elastic properties are everywhere the same we can see what the general solutions look like in the following way. You will remember that any vector field can be written as the sum of two vectors: one whose divergence is zero, and the other whose curl is zero. In

other words, we can put

$$\mathbf{u} = \mathbf{u}_1 + \mathbf{u}_2, \quad (39.34)$$

where

$$\nabla \cdot \mathbf{u}_1 = 0, \quad \nabla \times \mathbf{u}_2 = 0. \quad (39.35)$$

Substituting $\mathbf{u}_1 + \mathbf{u}_2$ for $\mathbf{u}$ in (39.33), we get

$$\rho \partial^2 / \partial t^2 [\mathbf{u}_1 + \mathbf{u}_2] = (\lambda + \mu) \nabla (\nabla \cdot \mathbf{u}_2) + \mu \nabla^2 (\mathbf{u}_1 + \mathbf{u}_2). \quad (39.36)$$

We can eliminate $\mathbf{u}_1$ by taking the divergence of this equation,

$$\rho \partial^2 / \partial t^2 (\nabla \cdot \mathbf{u}_2) = (\lambda + \mu) \nabla^2 (\nabla \cdot \mathbf{u}_2) + \mu \nabla \cdot \nabla^2 \mathbf{u}_2.$$

Since the operators (∇^2) and $(\nabla \cdot)$ can be interchanged, we can factor out the divergence to get

$$\nabla \cdot \{ \rho \partial^2 \mathbf{u}_2 / \partial t^2 - (\lambda + 2\mu) \nabla^2 \mathbf{u}_2 \} = 0. \quad (39.37)$$

Since $\nabla \times \mathbf{u}_2$ is zero by definition, the curl of the bracket $\{ \}$ is also zero; so the bracket itself is identically zero, and

$$\rho \partial^2 \mathbf{u}_2 / \partial t^2 = (\lambda + 2\mu) \nabla^2 \mathbf{u}_2. \quad (39.38)$$

This is the vector wave equation for waves which move at the speed $C_2 = \sqrt{(\lambda + 2\mu)/\rho}$. Since the curl of $\mathbf{u}_2$ is zero, there is no shearing associated with this wave; this wave is just the compressional—sound-type—wave we discussed in the last chapter, and the velocity is just what we found for C_{long} .

In a similar way—by taking the curl of Eq. (39.36)—we can show that $\mathbf{u}_1$ satisfies the equation

$$\rho \partial^2 \mathbf{u}_1 / \partial t^2 = \mu \nabla^2 \mathbf{u}_1. \quad (39.39)$$

This is again a vector wave equation for waves with the speed $C_2 = \sqrt{\mu/\rho}$. Since $\nabla \cdot \mathbf{u}_1$ is zero, $\mathbf{u}_1$ produces no changes in density; the vector $\mathbf{u}_1$ corresponds to the transverse, or shear-type, wave we saw in the last chapter, and $C_2 = C_{\text{shear}}$.

If we wished to know the static stresses in an isotropic material, we could, in principle, find them by solving Eq. (39.32) with f equal to zero—or equal to the static body forces from gravity such as ρg —under certain conditions which are related to the forces acting on the surfaces of our large block of material. This is somewhat more difficult to do than the corresponding problems in electromagnetism. It is more difficult, first, because the equations are a little more difficult to handle, and second, because the shape of the elastic bodies we are likely to be interested in are usually much more complicated. In electromagnetism, we are often interested in solving Maxwell's equations around relatively simple geometric shapes such as cylinders, spheres, and so on, since these are convenient shapes for electrical devices. In elasticity, the objects we would like to analyze may have quite complicated shapes—like a crane hook, or an automobile crankshaft, or the rotor of a gas turbine. Such problems can sometimes be worked out approximately by numerical methods, using the minimum energy principle we mentioned earlier. Another way is to use a model of the object and measure the internal strains experimentally, using polarized light.

It works this way: When a transparent isotropic material—for example, a clear plastic like lucite—is put under stress, it becomes birefringent. If you put polarized light through it, the plane of polarization will be rotated by an amount related to the stress: by measuring the rotation, you can measure the stress. Figure 39-6 shows how such a setup might look. Figure 39-7 is a photograph of a photoelastic model of a complicated shape under stress.

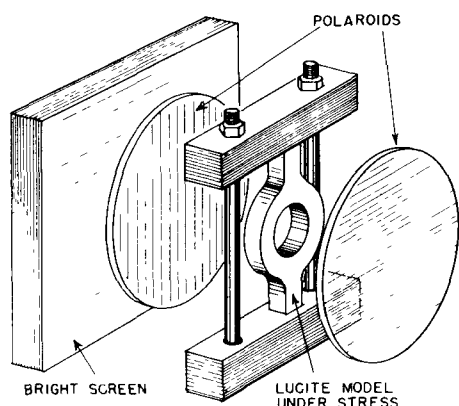


Fig. 39-6. Measuring internal stresses with polarized light.

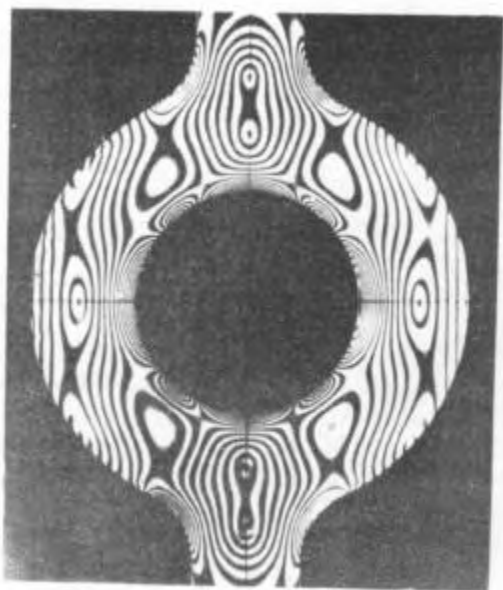


Fig. 39-7. A stressed plastic model as seen between crossed polaroids. [From F. W. Sears, *Optics*, Addison-Wesley Publishing Co., Reading, Mass., 1949.]

39-4 Nonelastic behavior

In all that has been said so far, we have assumed that stress is proportional to strain; in general, that is *not* true. Figure 39-8 shows a typical stress-strain curve for a ductile material. For small strains, the stress is proportional to the

strain. Eventually, however, after a certain point, the relationship between stress and strain begins to deviate from a straight line. For many materials—the ones we would call “brittle”—the object breaks for strains only a little above the point where the curve starts to bend over. In general, there are other complications in the stress-strain relationship. For example, if you strain an object, the stresses may be high at first, but decrease slowly with time. Also if you go to high stresses, but still not to the “breaking” point, when you lower the strain the stress will return along a different curve. There is a small hysteresis effect (like the one we saw between *B* and *H* in magnetic materials).

The stress at which a material will break varies widely from one material to another. Some materials will break when the maximum *tensile* stress reaches a certain value. Other materials will fail when the maximum *shear* stress reaches a certain value. Chalk is an example of a material which is much weaker in tension than in shear. If you pull on the ends of a piece of blackboard chalk, the chalk will break perpendicular to the direction of the applied stress, as shown in Fig. 39-9(a). It breaks perpendicular to the applied force because it is only a bunch of particles packed together which are easily pulled apart. The material is, however, much harder to shear, because the particles get in each other's way. Now you will remember that when we had a rod in torsion there was a shear all around it. Also, we showed that a shear was equivalent to a combination of a tension and compression at 45° . For these reasons, if you *twist* a piece of blackboard chalk, it will break along a complicated surface which starts out at 45° to the axis. A photograph of a piece of chalk broken in this way is shown in Fig. 39-9(b). The chalk breaks where the material is in maximum tension.

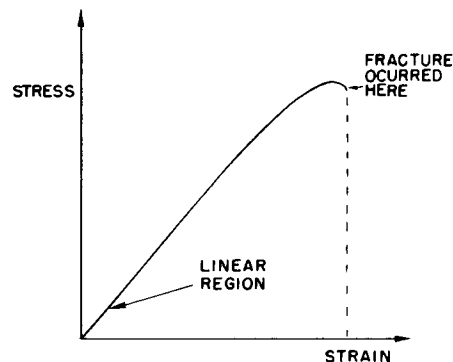


Fig. 39-8. A typical stress-strain relation for large strains.

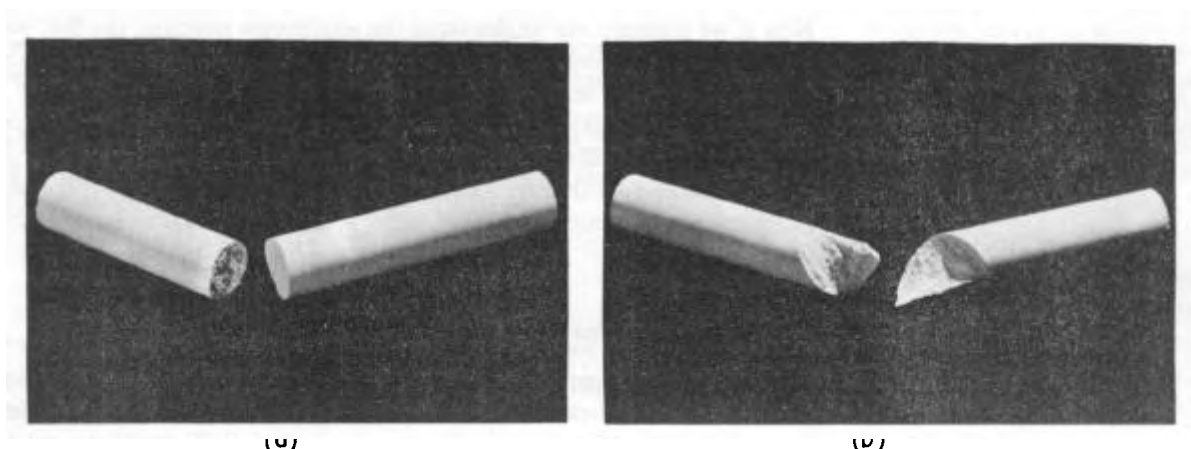


Fig. 39-9. (a) A piece of chalk broken by pulling on the ends; (b) a piece broken by twisting.

Other materials behave in strange and complicated ways. The more complicated the materials are, the more interesting their behavior. If we take a sheet of “Saran-Wrap” and crumple it up into a ball and throw it on the table, it slowly unfolds itself and returns toward its original flat form. At first sight, we might be tempted to think that it is inertia which prevents it from returning to its original form. However, a simple calculation shows that the inertia is several orders of magnitude too small to account for the effect. There appear to be two important competing effects: “something” inside the material “remembers” the shape it had initially and “tries” to get back there, but something else “prefers” the new shape and “resists” the return to the old shape.

We will not attempt to describe the mechanism at play in the Saran plastic, but you can get an idea of how such an effect might come about from the following *model*. Suppose you imagine a material made of long, flexible, but strong, fibers mixed together with some hollow cells filled with a viscous liquid. Imagine also that there are narrow pathways from one cell to the next so the liquid can leak slowly from a cell to its neighbor. When we crumple a sheet of this stuff, we distort the long fibers, squeezing the liquid out of the cells in one place and forcing it into other cells which are being stretched. When we let go, the long fibers try to

return to their original shape. But to do this, they have to force the liquid back to its original location—which will happen relatively slowly because of the viscosity. The forces we apply in crumpling the sheet are much larger than the forces exerted by the fibers. We can crumple the sheet quickly, but it will return more slowly. It is undoubtedly a combination of large stiff molecules and smaller, movable ones in the Saran-Wrap that is responsible for its behavior. This idea also fits with the fact that the material returns more quickly to its original shape when it's warmed up than when it's cold—the heat increases the mobility (decreases the viscosity) of the smaller molecules.

Although we have been discussing how Hooke's law breaks down, the remarkable thing is perhaps not that Hooke's law breaks down for large strains but that it should be so generally true. We can get some idea of why this might be by looking at the strain energy in a material. To say that the stress is proportional to the strain is the same thing as saying that the strain energy varies as the square of the strain. Suppose we have a rod and we twist it through a small angle θ . If Hooke's law holds, the strain energy should be proportional to the square of θ . Suppose we were to assume that the energy were some arbitrary function of the angle; we could write it as a Taylor expansion about zero angle

$$U(\theta) = U(0) + U'(0)\theta + \frac{1}{2}U''(0)\theta^2 + \frac{1}{6}U'''(0)\theta^3 \dots \quad (39.40)$$

The torque τ is the derivative of U with respect to angle; we would have

$$\tau(\theta) = U'(0) + U''(0)\theta + \frac{1}{2}U'''(0)\theta^2 + \dots \quad (39.41)$$

Now if we measure our angles from the *equilibrium* position, the first term is zero. So the first remaining term is proportional to θ ; and for small enough angles, it will dominate the term in θ^2 . [Actually, materials are sufficiently symmetric internally so that $\tau(\theta) = -\tau(-\theta)$; the term in θ^2 will be zero, and the departures from linearity would come only from the θ^3 term. There is, however, no reason why this should be true for compressions and tensions.] The thing we have not explained is why materials usually break soon after the higher-order terms become significant.

39-5 Calculating the elastic constants

As our last topic on elasticity we would like to show how one could try to calculate the elastic constants of a material, starting with some knowledge of the properties of the atoms which make up the material. We will take only the simple case of an *ionic* cubic crystal like sodium chloride. When a crystal is strained, its volume or its shape is changed. Such changes result in an increase in the potential energy of the crystal. To calculate the change in strain energy, we have to know where each atom goes. In complicated crystals, the atoms will rearrange themselves in the lattice in very complicated ways to make the total energy as small as possible. This makes the computation of the strain energy rather difficult. In the case of a simple cubic crystal, however, it is easy to see what will happen. The distortions inside the crystal will be geometrically similar to the distortions of the outside boundaries of the crystal.

We can calculate the elastic constants for a cubic crystal in the following way. First, we assume some force law between each pair of atoms in the crystal. Then, we calculate the change in the internal energy of the crystal when it is distorted from its equilibrium shape. This gives us a relation between the energy and the strains which is quadratic in all the strains. Comparing the energy obtained this way with Eq. (39.13), we can identify the coefficient of each term with the elastic constants C_{ijkl} .

For our example we will assume a simple force law: that the force between neighboring atoms is a *central* force, by which we mean that it acts along the line between the two atoms. We would expect the forces in ionic crystals to be like this, since they are just primarily Coulomb forces. (The forces of covalent bonds are usually more complicated, since they can exert a sideways push on a nearby

atom; we will leave out this complication.) We are also going to include only the forces between each atom and its *nearest* and *next-nearest* neighbors. In other words, we will make an approximation which neglects all forces beyond the next-nearest neighbor. The forces we will include are shown for the xy -plane in Fig. 39-10(a). The corresponding forces in the yz - and zx -planes also have to be included.

Since we are only interested in the elastic coefficients which apply to small strains, and therefore only want the terms in the energy which vary quadratically with the strains, we can imagine that the force between each atom pair varies linearly with the displacements. We can then imagine that each pair of atoms is joined by a linear spring, as drawn in Fig. 39-10(b). All of the springs between a sodium atom and a chlorine atom should have the same spring constant, say k_1 . The springs between two sodiums and between two chlorines could have different constants, but we will make our discussion simpler by taking them equal; we call them k_2 . (We could come back later and make them different after we have seen how the calculations go.)

Now we assume that the crystal is distorted by a homogeneous strain described by the strain tensor e_{ij} . In general, it will have components involving x , y , and z ; but we will consider now only a strain with the three components e_{xx} , e_{xy} , and e_{yy} so that it will be easy to visualize. If we pick one atom as our origin, the displacement of every other atom is given by equations like Eq. (39.9):

$$\begin{aligned} u_x &= e_{xx}x + e_{xy}y, \\ u_y &= e_{xy}x + e_{yy}y. \end{aligned} \quad (39.42)$$

Suppose we call the atom at $x = y = 0$ "atom 1" and number its neighbors in the xy -plane as shown in Fig. 39-11. Calling the lattice constant a , we get the x and y displacements u_x and u_y listed in Table 39-1.

Now we can calculate the energy stored in the springs, which is $k^2/2$ times the square of the extension for each spring. For example, the energy in the horizontal spring between atom 1 and atom 2 is

$$\frac{k_1(e_{xx}a)^2}{2}. \quad (39.43)$$

Note that to first order, the y -displacement of atom 2 does not change the length of the spring between atom 1 and atom 2. To get the strain energy in a diagonal spring, such as that to atom 3, however, we need to calculate the change in length due to both the horizontal and vertical displacements. For small displacements from the

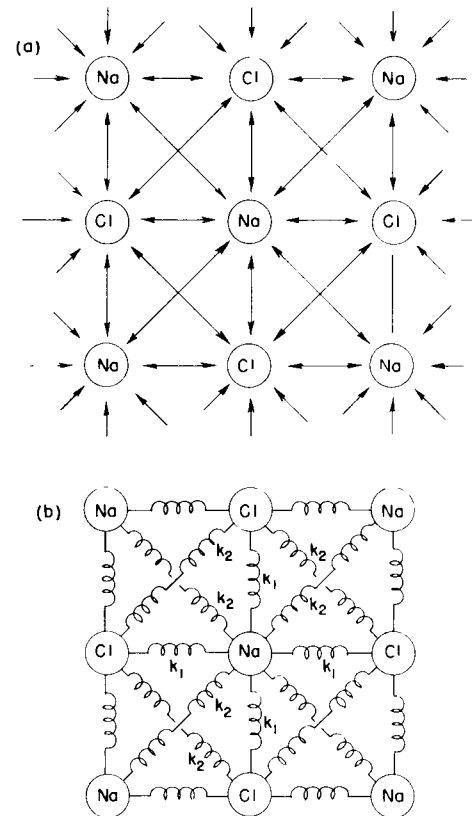


Fig. 39-10. (a) The interatomic forces we are taking into account; (b) a model in which the atoms are connected by springs.

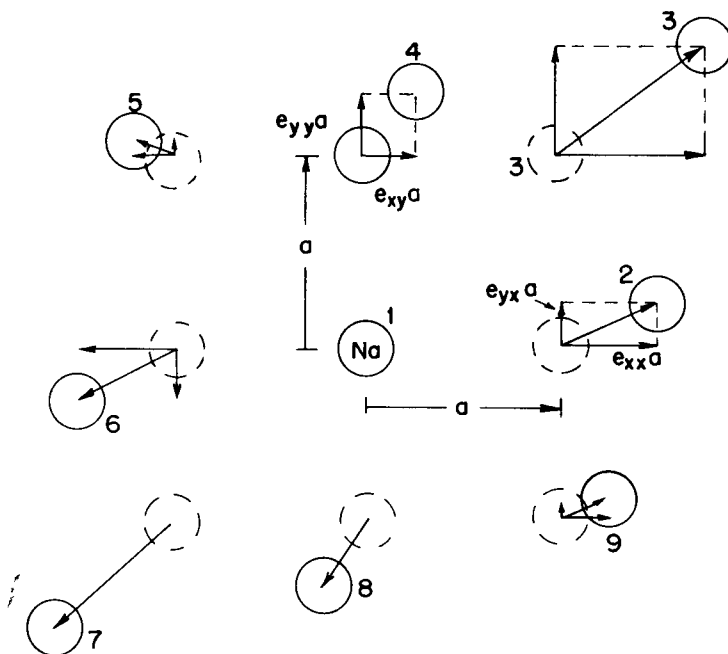


Fig. 39-11. The displacements of the nearest and next-nearest neighbors of atom 1 (exaggerated).

Table 39-1

Atom	Location x, y	u_x	u_y	k
1	0, a	0	0	—
2	$a, 0$	$e_{xx}a$	$e_{yx}a$	k_1
3	a, a	$(e_{xx} + e_{xy})a$	$(e_{yx} + e_{yy})a$	k_2
4	0, a	$e_{xy}a$	$e_{yy}a$	k_1
5	$-a, a$	$(-e_{xx} + e_{xy})a$	$(-e_{yx} + e_{yy})a$	k_2
6	$-a, 0$	$-e_{xx}a$	$-e_{yx}a$	k_1
7	$-a, -a$	$-(e_{xx} + e_{xy})a$	$-(e_{yx} + e_{yy})a$	k_2
8	0, $-a$	$-e_{xy}a$	$-e_{yy}a$	k_1
9	$a, -a$	$(e_{xx} - e_{xy})a$	$(e_{yx} - e_{yy})a$	k_2

original cube, we can write the change in the distance to atom 3 as the sum of the components of u_x and u_y in the diagonal direction, namely as

$$\frac{1}{\sqrt{2}} (u_x + u_y).$$

Using the values of u_x and u_y from the table, we get the energy

$$\frac{k_2}{2} \left(\frac{u_x + u_y}{\sqrt{2}} \right)^2 = \frac{k_2 a^2}{4} (e_{xx} + e_{yx} + e_{xy} + e_{yy})^2. \quad (39.44)$$

For the total energy for all the springs in the xy -plane, we need the sum of eight terms like (39.43) and (39.44). Calling this energy U_0 , we get

$$\begin{aligned} U_0 = \frac{a^2}{2} \left\{ k_1 e_{xx}^2 + \frac{k_2}{2} (e_{xx} + e_{yx} + e_{xy} + e_{yy})^2 \right. \\ + k_1 e_{yy}^2 + \frac{k_2}{2} (e_{xx} - e_{yx} - e_{xy} + e_{yy})^2 \\ + k_1 e_{xx}^2 + \frac{k_2}{2} (e_{xx} + e_{yx} + e_{xy} + e_{yy})^2 \\ \left. + k_1 e_{yy}^2 + \frac{k_2}{2} (e_{xx} - e_{yx} - e_{xy} + e_{yy})^2 \right\}. \quad (39.45) \end{aligned}$$

To get the total energy of all the springs connected to atom 1, we must make one addition to the energy in Eq. (39.45). Even though we have only x - and y -components of the strain, there are still some energies associated with the next-nearest neighbors off the xy -plane. This additional energy is

$$k_2 (e_{xx}^2 a^2 + e_{yy}^2 a^2). \quad (39.46)$$

The elastic constants are related to the energy density w by Eq. (39.13). The energy we have calculated is the energy associated with one atom, or rather, it is *twice* the energy per atom, since one-half of the energy of each spring should be assigned to each of the two atoms it joins. Since there are $1/a^3$ atoms per unit volume, w and U_0 are related by

$$w = \frac{U_0}{2a^3}.$$

To find the elastic constants C_{ijkl} , we need only to expand out the squares in Eq. (39.45)—adding the terms of (39.46)—and compare the coefficients of $e_{ij}e_{kl}$ with the corresponding coefficient in Eq. (39.13). For example, collecting the terms

in e_{xx}^2 and in e_{yy}^2 , we get the factor

$$(k_1 + 2k_2)a^2,$$

so

$$C_{xxxx} = C_{yyyy} = \frac{k_1 + 2k_2}{a}.$$

For the remaining terms, there is a slight complication. Since we cannot distinguish the product of two terms like $e_{xx}e_{yy}$ from $e_{yy}e_{xx}$, the coefficient of such terms in our energy is equal to the sum of two terms in Eq. (39.13). The coefficient of $e_{xx}e_{yy}$ in Eq. (39.45) is $2k_2$, so we have that

$$(C_{xxyy} + C_{yyxx}) = \frac{2k_2}{a}.$$

But because of the symmetry in our crystal, $C_{xxyy} = C_{yyxx}$, so we have that

$$C_{xxyy} = C_{yyxx} = \frac{k_2}{a}.$$

By a similar process, we can also get

$$C_{xyxy} = C_{yxyx} = \frac{k_2}{a}.$$

Finally, you will notice that any term which involves either x or y only once is zero—as we concluded earlier from symmetry arguments. Summarizing our results:

$$\begin{aligned} C_{xxxx} &= C_{yyyy} = \frac{k_1 + 2k_2}{a}, \\ C_{xxyy} &= C_{yyxx} = \frac{k_2}{a}, \\ C_{xyxy} &= C_{yxyx} = \frac{k_2}{a}, \\ C_{xxxy} &= C_{yyxx} = C_{xyyx} = C_{yxxy} = \frac{k_2}{a}, \\ C_{xxxxy} &= C_{xyyyy} = \text{etc.} = 0. \end{aligned} \quad (39.47)$$

We have been able to relate the bulk elastic constants to the atomic properties which appear in the constants k_1 and k_2 . In our particular case, $C_{xxyy} = C_{yyxx}$. It turns out—as you can perhaps see from the way the calculations went—that these terms are *always* equal for a cubic crystal, no matter how many force terms are taken into account, *provided* only that the forces act along the line joining each pair of atoms—that is, so long as the forces between atoms are like springs and don't have a sideways part such as you might get from a cantilevered beam (and you do get in covalent bonds).

We can check this conclusion with the experimental measurements of the elastic constants. In Table 39-2 we give the observed values of the three elastic coefficients for several cubic crystals.* You will notice that C_{xxyy} and C_{xyxy} are, in general, not equal. The reason is that in metals like sodium and potassium the interatomic forces are not along the line joining the atoms, as we assumed in our model. Diamond does not obey the law either, because the forces in diamond are covalent forces and have some directional properties—the bonds would prefer to be at the tetrahedral angle. The ionic crystals like lithium fluoride, sodium chloride, and so on, do have nearly all the physical properties assumed in our model, and the table shows that the constants C_{xxyy} and C_{xyxy} are almost equal. It is not clear why silver chloride should not satisfy the condition that $C_{xxyy} = C_{xyxy}$.

Table 39-2*
Elastic Moduli of Cubic Crystals
in 10^{12} dynes·cm²

	C_{xxxx}	C_{xxyy}	C_{xyxy}
Na	0.055	0.042	0.049
K	0.046	0.037	0.026
Fe	2.37	1.41	1.16
Diamond	10.76	1.25	5.76
Al	1.08	0.62	0.28
LiF	1.19	0.54	0.53
NaCl	0.486	0.127	0.128
KCl	0.40	0.062	0.062
NaBr	0.33	0.13	0.13
KI	0.27	0.043	0.042
AgCl	0.60	0.36	0.062

* From C. Kittel, *Introduction to Solid State Physics*, John Wiley and Sons, Inc., New York, 2nd. ed., 1956, p. 93.

* In the literature you will often find that a different notation is used. For instance, people usually write $C_{xxxx} = C_{11}$, $C_{xxyy} = C_{12}$, and $C_{xyxy} = C_{44}$.

The Flow of Dry Water

40-1 Hydrostatics

The subject of the flow of fluids, and particularly of water, fascinates everybody. We can all remember, as children, playing in the bathtub or in mud puddles with the strange stuff. As we get older, we watch streams, waterfalls, and whirlpools, and we are fascinated by this substance which seems almost alive relative to solids. The behavior of fluids is in many ways very unexpected and interesting—it is the subject of this chapter and the next. The efforts of a child trying to dam a small stream flowing in the street and his surprise at the strange way the water works its way out has its analog in our attempts over the years to understand the flow of fluids. We have tried to dam the water up—in our understanding—by getting the laws and the equations that describe the flow. We will describe these attempts in this chapter. In the next chapter, we will describe the unique way in which water has broken through the dam and escaped our attempts to understand it.

We suppose that the elementary properties of water are already known to you. The main property that distinguishes a fluid from a solid is that a fluid cannot *maintain* a shear stress for any length of time. If a shear is applied to a fluid, it will move under the shear. Thicker liquids like honey move less easily than fluids like air or water. The measure of the ease with which a fluid yields is its viscosity. In this chapter we will consider only situations in which the viscous effects can be ignored. The effects of viscosity will be taken up in the next chapter.

We begin by considering *hydrostatics*, the theory of liquids at rest. When liquids are at rest, there are no shear forces (even for viscous liquids). The law of hydrostatics, therefore, is that the stresses are always normal to any surface inside the fluid. The normal force per unit area is called the *pressure*. From the fact that there is no shear in a static fluid it follows that the pressure stress is the same in all directions (Fig. 40-1). We will let you entertain yourself by proving that if there is no shear on any plane in a fluid, the pressure must be the same in any direction.

The pressure in a fluid may vary from place to place. For example, in a static fluid at the earth's surface the pressure will vary with height because of the weight of the fluid. If the density ρ of the fluid is considered constant, and if the pressure at some arbitrary zero level is called p_0 (Fig. 40-2), then the pressure at a height h above this point is $p = p_0 - \rho gh$, where g is the gravitational force per unit mass. The combination

$$p + \rho gh$$

is, therefore, a constant in the static fluid. This relation is familiar to you, but we will now derive a more general result of which it is a special case.

If we take a small cube of water, what is the net force on it from the pressure? Since the pressure at any place is the same in all directions, there can be a net force per unit volume only because the pressure varies from one point to another. Suppose that the pressure is varying in the x -direction—and we take the coordinate directions parallel to the cube edges. The pressure on the face at x gives the force $p \Delta y \Delta z$ (Fig. 40-3), and the pressure on the face at $x + \Delta x$ gives the force $-[p + (\partial p / \partial x) \Delta x] \Delta y \Delta z$, so that the resultant force is $-(\partial p / \partial x) \Delta x \Delta y \Delta z$. If we take the remaining pairs of faces of the cube, we easily see that the pressure force per unit volume is $-\nabla p$. If there are other forces in addition—such as gravity—then the pressure must balance them to give equilibrium.

40-1 Hydrostatics

40-2 The equations of motion

40-3 Steady flow—Bernoulli's theorem

40-4 Circulation

40-5 Vortex lines

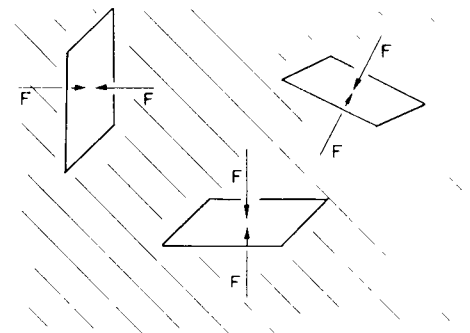


Fig. 40-1. In a static fluid the force per unit area across any surface is normal to the surface and is the same for all orientations of the surface.

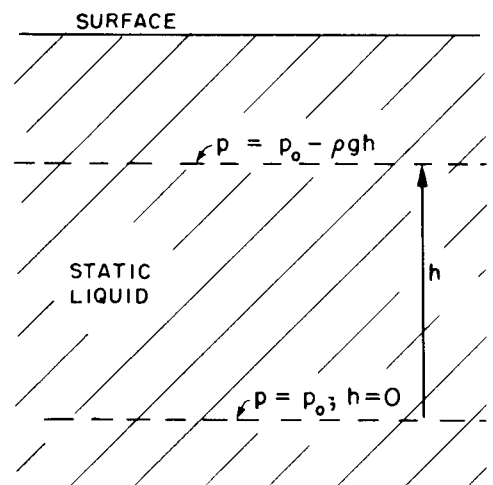


Fig. 40-2. The pressure in a static liquid.

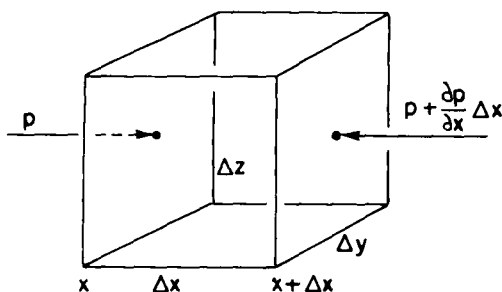


Fig. 40-3. The net pressure force on a cube is $-\nabla p$ per unit volume.

Let's take a circumstance in which such an additional force can be described by a potential energy, as would be true in the case of gravitation; we will let ϕ stand for the potential energy per unit mass. (For gravity, for instance, ϕ is just gz .) The force per unit mass is given in terms of the potential by $-\nabla\phi$, and if ρ is the density of the fluid, the force per unit volume is $-\rho\nabla\phi$. For equilibrium this force per unit volume added to the pressure force per unit volume must give zero:

$$-\nabla p - \rho\nabla\phi = 0. \quad (40.1)$$

Equation (40.1) is the equation of hydrostatics. In *general*, it *has no solution*. If the density varies in space in an arbitrary way, there is no way for the forces to be in balance, and the fluid cannot be in static equilibrium. Convection currents will start up. We can see this from the equation since the pressure term is a pure gradient, whereas for variable ρ the other term is not. Only when ρ is a constant is the potential term a pure gradient. Then the equation has a solution

$$p + \rho\phi = \text{const.}$$

Another possibility which allows hydrostatic equilibrium is for ρ to be a function only of p . However, we will leave the subject of hydrostatics because it is not nearly so interesting as the situation when fluids are in motion.

40-2 The equations of motion

First, we will discuss fluid motions in a purely abstract, theoretical way and then consider special examples. To describe the motion of a fluid, we must give its properties at every point. For example, at different places, the water (let us call the fluid "water") is moving with different *velocities*. To specify the character of the flow, therefore, we must give the three components of velocity at every point and for any time. If we can find the equations that determine the velocity, then we would know how the liquid moves at all times. The velocity, however, is not the only property that the fluid has which varies from point to point. We have just discussed the variation of the *pressure* from point to point. And there are still other variables. There may also be a variation of *density* from point to point. In addition, the fluid may be a conductor and carry an electric *current* whose density j varies from point to point in magnitude and direction. There may be a *temperature* which varies from point to point, or a *magnetic field*, and so on. So the number of fields needed to describe the complete situation will depend on how complicated the problem is. There are interesting phenomena when currents and magnetism play a dominant part in determining the behavior of the fluid; the subject is called *magnetohydrodynamics*, and great attention is being paid to it at the present time. However, we are not going to consider these more complicated situations because there are already interesting phenomena at a lower level of complexity, and even the more elementary level will be complicated enough.

We will take the situation where there is no magnetic field and no conductivity, and we will not worry about the temperature because we will suppose that the density and pressure determine in a unique manner the temperature at any point. As a matter of fact, we will reduce the complexity of our work by making the assumption that the density is a constant—we imagine that the fluid is essentially incompressible. Putting it another way, we are supposing that the variations of pressure are so small that the changes in density produced thereby are negligible. If that is not the case, we would encounter phenomena additional to the ones we will be discussing here—for example, the propagation of sound or of shock waves. We have already discussed the propagation of sound and shocks to some extent, so we will now isolate our consideration of hydrodynamics from these other phenomena by making the approximation that the density ρ is a constant. It is easy to determine when the approximation of constant ρ is a good one. We can say that if the velocities of flow are much less than the speed of a sound wave in the fluid, we do not have to worry about variations in density. The escape that water makes in our attempts to understand it is not related to the approximation of

constant density. The complications that do permit the escape will be discussed in the next chapter.

In the general theory of fluids one must begin with an *equation of state* for the fluid which connects the pressure to the density. In our approximation this equation of state is simply

$$\rho = \text{const.}$$

This then is the first relation for our variables. The next relation expresses the conservation of matter—if matter flows away from a point, there must be a decrease in the amount left behind. If the fluid velocity is $\mathbf{v}$, then the mass which flows in a unit time across a unit area of surface is the component of $\rho\mathbf{v}$ normal to the surface. We have had a similar relation in electricity. We also know from electricity that the divergence of such a quantity gives the rate of decrease of the density per unit time. In the same way, the equation

$$\nabla \cdot (\rho\mathbf{v}) = -\frac{\partial\rho}{\partial t} \quad (40.2)$$

expresses the conservation of mass for a fluid; it is the hydrodynamic *equation of continuity*. In our approximation, which is the incompressible fluid approximation, ρ is a constant, and the equation of continuity is simply

$$\nabla \cdot \mathbf{v} = 0. \quad (40.3)$$

The fluid velocity $\mathbf{v}$ —like the magnetic field $\mathbf{B}$ —has zero divergence. (The hydrodynamic equations are often closely analogous to the electrodynamic equations; that's why we studied electrodynamics first. Some people argue the other way; they think that one should study hydrodynamics first so that it will be easier to understand electricity afterwards. But electrodynamics is really much easier than hydrodynamics.)

We will get our next equation from Newton's law which tells us how the velocity changes because of the forces. The mass of an element of volume of the fluid times its acceleration must be equal to the force on the element. Taking an element of unit volume, and writing the force per unit volume as $\mathbf{f}$, we have

$$\rho \times (\text{acceleration}) = \mathbf{f}.$$

We will write the force density as the sum of three terms. We have already considered the pressure force per unit volume, $-\nabla p$. Then there are the “external” forces which act at a distance—like gravity or electricity. When they are conservative forces with a potential per unit mass, ϕ , they give a force density $-\rho\nabla\phi$. (If the external forces are not conservative, we would have to write $\mathbf{f}_{\text{ext}}$ for the external force per unit volume.) Then there is another “internal” force per unit volume, which is due to the fact that in a *flowing* fluid there can also be a shearing stress. This is called the viscous force, which we will write $\mathbf{f}_{\text{visc}}$. Our equation of motion is

$$\rho \times (\text{acceleration}) = -\nabla p - \rho \nabla\phi + \mathbf{f}_{\text{visc}}. \quad (40.4)$$

For this chapter we are going to suppose that the liquid is “thin” in the sense that the viscosity is unimportant, so we will omit $\mathbf{f}_{\text{visc}}$. When we drop the viscosity term, we will be making an approximation which describes some ideal stuff rather than real water. John von Neumann was well aware of the tremendous difference between what happens when you don't have the viscous terms and when you do, and he was also aware that, during most of the development of hydrodynamics until about 1900, almost the main interest was in solving beautiful *mathematical* problems with this approximation which had almost nothing to do with real fluids. He characterized the theorist who made such analyses as a man who studied “dry water.” Such analyses leave out an *essential* property of the fluid. It is because we are leaving this property out of our calculations in this chapter that we have given it the title “The Flow of Dry Water.” We are postponing a discussion of *real* water to the next chapter.

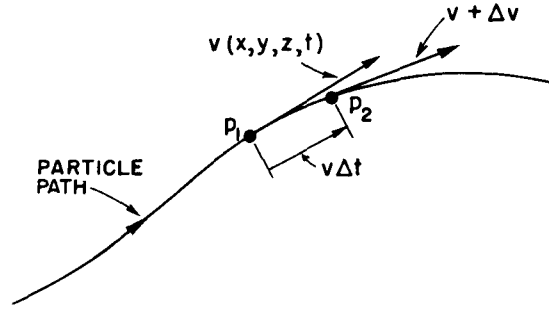


Fig. 40-4. The acceleration of a fluid particle.

If we leave out f_{visc} , we have in Eq. (40.4) everything we need except an expression for the acceleration. You might think that the formula for the acceleration of a fluid particle would be very simple, for it seems obvious that if v is the velocity of a fluid particle at some place in the fluid, the acceleration would just be $\partial v / \partial t$. *It is not*—and for a rather subtle reason. The derivative $\partial v / \partial t$, is the rate at which the velocity $v(x, y, z, t)$ changes at a *fixed point* in space. What we need is how fast the velocity changes for a *particular piece* of fluid. Imagine that we mark one of the drops of water with a colored speck so we can watch it. In a small interval of time Δt , this drop will move to a different location. If the drop is moving along some path as sketched in Fig. 40-4, it might in Δt move from P_1 to P_2 . In fact, it will move in the x -direction by an amount $v_x \Delta t$, in the y -direction by the amount $v_y \Delta t$, and in the z -direction by the amount $v_z \Delta t$. We see that, if $v(x, y, z, t)$ is the velocity of the fluid particle which is at (x, y, z) at the time t , then the velocity of the *same* particle at the time $t + \Delta t$ is given by $v(x + \Delta x, y + \Delta y, z + \Delta z, t + \Delta t)$ —with

$$\Delta x = v_x \Delta t, \quad \Delta y = v_y \Delta t, \quad \text{and} \quad \Delta z = v_z \Delta t.$$

From the definition of the partial derivatives—recall Eq. (2.7)—we have, to first order, that

$$\begin{aligned} v(x + v_x \Delta t, y + v_y \Delta t, z + v_z \Delta t, t + \Delta t) \\ = v(x, y, z, t) + \frac{\partial v}{\partial x} v_x \Delta t + \frac{\partial v}{\partial y} v_y \Delta t + \frac{\partial v}{\partial z} v_z \Delta t + \frac{\partial v}{\partial t} \Delta t. \end{aligned}$$

The acceleration $\Delta v / \Delta t$ is

$$v_x \frac{\partial v}{\partial x} + v_y \frac{\partial v}{\partial y} + v_z \frac{\partial v}{\partial z} + \frac{\partial v}{\partial t}.$$

We can write this symbolically—treating ∇ as a vector—as

$$(v \cdot \nabla)v + \frac{\partial v}{\partial t}. \quad (40.5)$$

Note that there can be an acceleration even though $\partial v / \partial t = 0$ so that velocity *at a given point* is not changing. As an example, water flowing in a circle at a constant speed is accelerating even though the velocity at a given point is not changing. The reason is, of course, that the velocity of a particular piece of water which is initially at one point on the circle has a different direction a moment later; there is a centripetal acceleration.

The rest of our theory is just mathematical—finding solutions of the equation of motion we get by putting the acceleration (40.5) into Eq. (40.4). We get

$$\frac{\partial v}{\partial t} + (v \cdot \nabla)v = -\frac{\nabla p}{\rho} - \nabla \phi, \quad (40.6)$$

where viscosity has been omitted. We can rearrange this equation by using the following identity from vector analysis:

$$(v \cdot \nabla)v = (\nabla \times v) \times v + \frac{1}{2} \nabla(v \cdot v).$$

If we now *define* a new vector field $\mathbf{\Omega}$, as the curl of $\mathbf{v}$,

$$\mathbf{\Omega} = \nabla \times \mathbf{v}, \quad (40.7)$$

the vector identity can be written as

$$(\mathbf{v} \cdot \nabla)\mathbf{v} = \mathbf{\Omega} \times \mathbf{v} + \frac{1}{2} \nabla v^2,$$

and our equation of motion (40.6) becomes

$$\frac{\partial \mathbf{v}}{\partial t} + \mathbf{\Omega} \times \mathbf{v} + \frac{1}{2} \nabla v^2 = -\frac{\nabla p}{\rho} - \nabla \phi. \quad (40.8)$$

You can verify that Eqs. (40.6) and (40.8) are equivalent by checking that the components of the two sides of the equation are equal—and making use of (40.7).

The vector field $\mathbf{\Omega}$ is called the *vorticity*. If the vorticity is zero everywhere, we say that the flow is *irrotational*. We have already defined in Section 3–5 a thing called the *circulation* of a vector field. The circulation around any closed loop in a fluid is the line integral of the fluid velocity, at a given instant of time, around that loop:

$$(\text{Circulation}) = \oint \mathbf{v} \cdot d\mathbf{s}.$$

The circulation *per unit area* for an infinitesimal loop is then—using Stokes' theorem—equal to $\nabla \times \mathbf{v}$. So the vorticity $\mathbf{\Omega}$ is the circulation around a unit area (perpendicular to the direction of $\mathbf{\Omega}$). It also follows that if you put a little piece of dirt—not an infinitesimal point—at any place in the liquid it will rotate with the angular velocity $\mathbf{\Omega}/2$. Try to see if you can prove that. You can also check it out that for a bucket of water on a turntable, $\mathbf{\Omega}$ is equal to twice the local angular velocity of the water.

If we are interested only in the velocity field, we can eliminate the pressure from our equations. Taking the curl of both sides of Eq. (40.8), remembering that ρ is a constant and that the curl of any gradient is zero, and using Eq. (40.3), we get

$$\frac{\partial \mathbf{\Omega}}{\partial t} + \nabla \times (\mathbf{\Omega} \times \mathbf{v}) = 0. \quad (40.9)$$

This equation, together with the equations

$$\mathbf{\Omega} = \nabla \times \mathbf{v} \quad (40.10)$$

and

$$\nabla \cdot \mathbf{v} = 0, \quad (40.11)$$

describes completely the velocity field $\mathbf{v}$. Mathematically speaking, if we know $\mathbf{\Omega}$ at some time, then we know the curl of the velocity vector, and we also know that its divergence is zero, so given the physical situation we have all we need to determine $\mathbf{v}$ everywhere. (It is just like the situation in magnetism where we had $\nabla \cdot \mathbf{B} = 0$ and $\nabla \times \mathbf{B} = \mathbf{j}/\epsilon_0 c^2$.) Thus, a given $\mathbf{\Omega}$ determines $\mathbf{v}$ just as a given $\mathbf{j}$ determines $\mathbf{B}$. Then, knowing $\mathbf{v}$, Eq. (40.9) tells us the rate of change of $\mathbf{\Omega}$ from which we can get the new $\mathbf{\Omega}$ for the next instant. Using Eq. (40.10), again we find the new $\mathbf{v}$, and so on. You see how these equations contain all the machinery for calculating the flow. Note, however, that this procedure gives the velocity field only; we have lost all information about the pressure.

We point out one special consequence of our equation. If $\mathbf{\Omega} = 0$ everywhere at any time t , $\partial \mathbf{\Omega} / \partial t$ also vanishes, so that $\mathbf{\Omega}$ is still zero everywhere at $t + \Delta t$. We have a solution to the equation; the flow is permanently irrotational. If a flow was started with zero rotation, it would always have zero rotation. The equations to be solved then are

$$\nabla \cdot \mathbf{v} = 0, \quad \nabla \times \mathbf{v} = 0.$$

They are just like the equations for the electrostatic or magnetostatic fields in free space. We will come back to them and look at some special problems later.

40-3 Steady flow—Bernoulli's theorem

Now we want to return to the equation of motion, Eq. (40.8), but limit ourselves to situations in which the flow is “steady.” By steady flow we mean that at any one place in the fluid the velocity never changes. The fluid at any point is always replaced by new fluid moving in exactly the same way. The velocity picture always looks the same— $\mathbf{v}$ is a static vector field. In the same way that we drew “field lines” in magnetostatics, we can now draw lines which are always tangent to the fluid velocity as shown in Fig. 40-5. These lines are called *streamlines*. For steady flow, they are evidently the actual paths of fluid particles. (In unsteady flow the streamline pattern changes in time, and the streamline pattern at any instant does not represent the path of a fluid particle.)

A steady flow does not mean that nothing is happening—atoms in the fluid are moving and changing their velocities. It only means that $\partial \mathbf{v} / \partial t = 0$. Then if we take the dot product of $\mathbf{v}$ into the equation of motion, the term $\mathbf{v} \cdot (\boldsymbol{\Omega} \times \mathbf{v})$ drops out, and we are left with

$$\mathbf{v} \cdot \nabla \left\{ \frac{p}{\rho} + \phi + \frac{1}{2} v^2 \right\} = 0. \quad (40.12)$$

This equation says that *for a small displacement in the direction of the fluid velocity* the quantity inside the brackets doesn't change. Now in steady flow all displacements are along streamlines, so Eq. (40.12) tells us that *for all the points along a streamline*, we can write

$$\frac{p}{\rho} + \frac{1}{2} v^2 + \phi = \text{const (streamline)}. \quad (40.13)$$

This is *Bernoulli's theorem*. The constant may in general be different for different streamlines; all we know is that the left-hand side of Eq. (40.13) is the same all along a *given streamline*. Incidentally, we may notice that for steady *irrotational* motion for which $\boldsymbol{\Omega} = 0$, the equation of motion (40.8) gives us the relation

$$\nabla \left\{ \frac{p}{\rho} + \frac{1}{2} v^2 + \phi \right\} = 0,$$

so that

$$\frac{p}{\rho} + \frac{1}{2} v^2 + \phi = \text{const (everywhere)}. \quad (40.14)$$

It's just like Eq. (40.13) *except* that *now* the constant has the *same value throughout the fluid*.

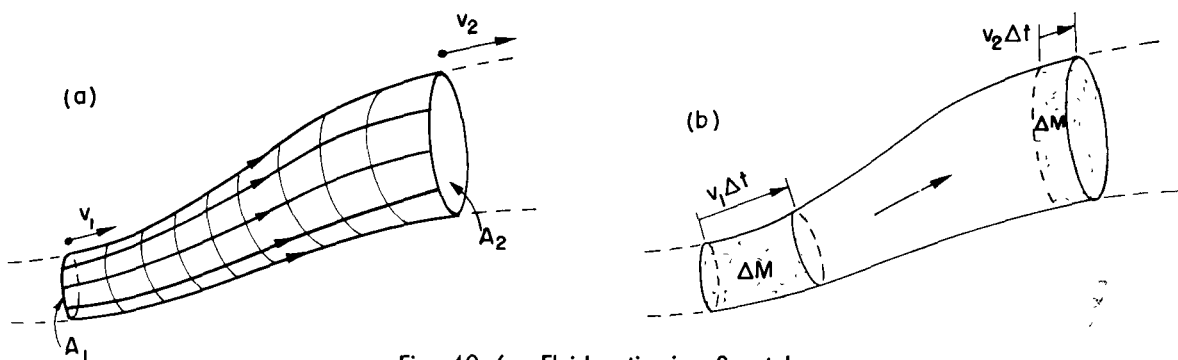


Fig. 40-6. Fluid motion in a flow tube.

The theorem of Bernoulli is in fact nothing more than a statement of the conservation of energy. A conservation theorem such as this gives us a lot of information about a flow without our actually having to solve the detailed equations. Bernoulli's theorem is so important and so simple that we would like to show you how it can be derived in a way that is different from the formal calculations we have just used. Imagine a bundle of adjacent streamlines which form a stream tube as sketched in Fig. 40-6. Since the walls of the tube consist of streamlines, no fluid flows out through the wall. Let's call the area at one end of the stream

tube A_1 , the fluid velocity there v_1 , the density of the fluid ρ_1 , and the potential energy ϕ_1 . At the other end of the tube, we have the corresponding quantities A_2 , v_2 , ρ_2 , and ϕ_2 . Now after a short interval of time Δt , the fluid at A_1 has moved a distance $v_1 \Delta t$, and the fluid at A_2 has moved a distance $v_2 \Delta t$ [Fig. 40-6(b)]. The conservation of mass requires that the mass which enters through A_1 must be equal to the mass which leaves through A_2 . These masses at these two ends must be the same:

$$\Delta M = \rho_1 A_1 v_1 \Delta t = \rho_2 A_2 v_2 \Delta t.$$

So we have the equality

$$\rho_1 A_1 v_1 = \rho_2 A_2 v_2. \quad (40.15)$$

This equation tells us that the velocity varies inversely with the area of the stream tube if ρ is constant.

Now we calculate the work done by the fluid pressure. The work done on the fluid entering at A_1 is $p_1 A_1 v_1 \Delta t$, and the work given up at A_2 is $p_2 A_2 v_2 \Delta t$. The net work on the fluid between A_1 and A_2 is, therefore,

$$p_1 A_1 v_1 \Delta t - p_2 A_2 v_2 \Delta t,$$

which must equal the increase in the energy of a mass ΔM of fluid in going from A_1 to A_2 . In other words,

$$p_1 A_1 v_1 \Delta t - p_2 A_2 v_2 \Delta t = \Delta M (E_2 - E_1), \quad (40.16)$$

where E_1 is the energy per unit mass of fluid at A_1 , and E_2 is the energy per unit mass at A_2 . The energy per unit mass of the fluid can be written as

$$E = \frac{1}{2}v^2 + \phi + U,$$

where $\frac{1}{2}v^2$ is the kinetic energy per unit mass, ϕ is the potential energy per unit mass, and U is an additional term which represents the internal energy per unit mass of fluid. The internal energy might correspond, for example, to the thermal energy in a compressible fluid, or to chemical energy. All these quantities can vary from point to point. Using this form for the energies in (40.16), we have

$$\frac{p_1 A_1 v_1 \Delta t}{\Delta M} - \frac{p_2 A_2 v_2 \Delta t}{\Delta M} = \frac{1}{2} v_2^2 + \phi_2 + U_2 - \frac{1}{2} v_1^2 - \phi_1 - U_1$$

But we have seen that $\Delta M = \rho A v \Delta t$, so we get

$$\frac{p_1}{\rho_1} + \frac{1}{2} v_1^2 + \phi_1 + U_1 = \frac{p_2}{\rho_2} + \frac{1}{2} v_2^2 + \phi_2 + U_2, \quad (40.17)$$

which is the Bernoulli result with an additional term for the internal energy. If the fluid is incompressible, the internal energy term is the same on both sides, and we get again that Eq. (40.14) holds along any streamline.

We consider now some simple examples in which the Bernoulli integral gives us a description of the flow. Suppose we have water flowing out of a hole near the bottom of a tank, as drawn in Fig. 40-7. We take a situation in which the flow speed v_{out} at the hole is much larger than the flow speed near the top of the tank; in other words, we imagine that the diameter of the tank is so large that we can neglect the drop in the liquid level. (We could make a more accurate calculation if we wished.) At the top of the tank the pressure is p_0 , the atmospheric pressure, and the pressure at the sides of the jet is also p_0 . Now we write our Bernoulli equation for a streamline, such as the one shown in the figure. At the top of the tank, we take v equal to zero and we also take the gravity potential ϕ to be zero. At the speed v_{out} , and $\phi = -gh$, so that

$$p_0 = p_0 + \frac{1}{2}\rho v_{\text{out}}^2 - \rho gh,$$

or

$$v_{\text{out}} = \sqrt{2gh}. \quad (40.18)$$

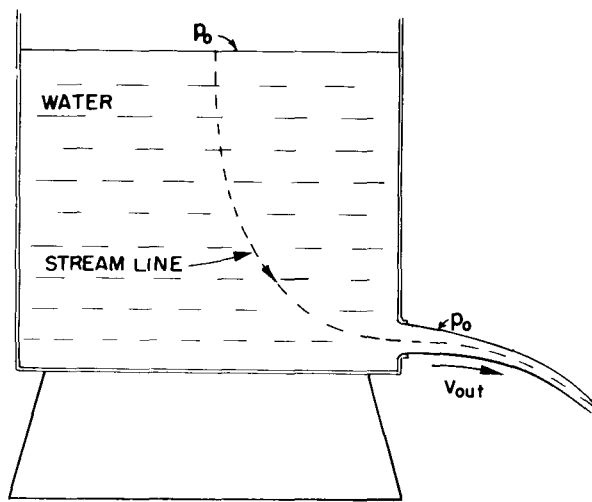


Fig. 40-7. Flow from a tank.

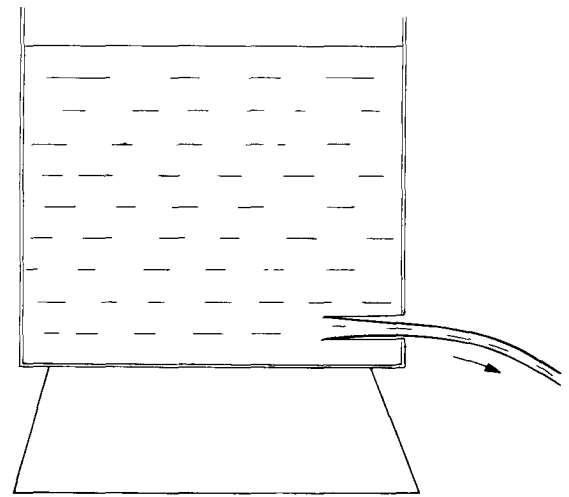


Fig. 40-8. With a re-entrant discharge tube, the stream contracts to one-half the area of the opening.

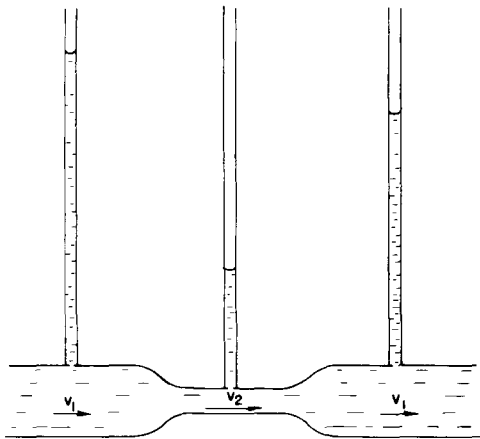


Fig. 40-9. The pressure is lowest where the velocity is highest.

This velocity is just what we would get for something which falls the distance h . It is not too surprising, since the water at the exit gains kinetic energy at the expense of the potential energy of the water at the top. Do not get the idea, however, that you can figure out the rate that the fluid flows out of the tank by multiplying this velocity by the area of the hole. The fluid velocities as the jet leaves the hole are not all parallel to each other but have components inward toward the center of the stream—the jet is converging. After the jet has gone a little way, the contraction stops and the velocities do become parallel. So the total flow is the velocity times the area *at that point*. In fact, if we have a discharge opening which is just a round hole with a sharp edge, the jet contracts to 62 percent of the area of the hole. The reduced effective area of the discharge varies for different shapes of discharge tubes, and experimental contractions are available as tables of *efflux coefficients*.

If the discharge tube is re-entrant, as shown in Fig. 40-8, it is possible to prove in a most beautiful way that the efflux coefficient is exactly 50 percent. We will give just a hint of how the proof goes. We have used the conservation of energy to get the velocity, Eq. (40.18), but there is also momentum conservation to consider. Since there is an outflow of momentum in the discharge jet, there must be a force applied over the cross section of the discharge tube. Where does the force come from? The force must come from the pressure on the walls. As long as the efflux hole is small and away from the walls, the fluid velocity near the walls of the tank will be very small. Therefore, the pressure on every face is almost exactly the same as the static pressure in a fluid at rest—from Eq. (30.14). Then the static pressure at any point on the side of the tank must be matched by an equal pressure at the point on the opposite wall, *except* at the points on the wall opposite the charge tube. If we calculate the momentum poured out through the jet by this pressure, we can show that the efflux coefficient is $1/2$. We cannot use this method for a discharge hole like that shown in Fig. 40-7, however, because the velocity increase along the wall right near the discharge area gives a pressure fall which we are not able to calculate.

Let's look at another example—a horizontal pipe with changing cross section, as shown in Fig. 40-9, with water flowing in one end and out the other. The conservation of energy, namely Bernoulli's formula, says that the pressure is lower in the constricted area where the velocity is higher. We can easily demonstrate this effect by measuring the pressure at different cross sections with small vertical columns of water attached to the flow tube through holes small enough so that they do not disturb the flow. The pressure is then measured by the height of water in these vertical columns. The pressure is found to be less at the constriction than it is on either side. If the area beyond the constriction comes back to the same value it had before the constriction, the pressure rises again.

Bernoulli's formula would predict that the pressure downstream of the constriction should be the same as it was upstream, but actually it is noticeably less. The reason that our prediction is wrong is that we have neglected the frictional, viscous forces which cause a pressure drop along the tube. Despite this pressure drop the pressure is definitely lower at the constriction (because of the increased speed) than it is on either side of it—as predicted by Bernoulli. The speed v_2 must certainly exceed v_1 to get the same amount of water through the narrower tube. So the water accelerates in going from the wide to the narrow part. The force that gives this acceleration comes from the drop in pressure.

We can check our results with another simple demonstration. Suppose we have on a tank a discharge tube which throws a jet of water upward as shown in Fig. 40-10. If the efflux velocity were exactly $\sqrt{2gh}$, the discharge water should rise to a level even with the surface of the water in the tank. Experimentally, it falls somewhat short. Our prediction is roughly right, but again viscous friction which has not been included in our energy conservation formula has resulted in a loss of energy.

Have you ever held two pieces of paper close together and tried to blow them apart? Try it! They come *together*. The reason, of course, is that the air has a higher speed going through the constricted space between the sheets than it does when it gets outside. The pressure between the sheets is *lower* than atmospheric pressure, so they come together rather than separating.

40-4 Circulation

We saw at the beginning of the last section that if we have an incompressible fluid with no circulation, the flow satisfies the following two equations:

$$\nabla \cdot \mathbf{v} = 0, \quad \nabla \times \mathbf{v} = 0. \quad (40.19)$$

They are the same as the equations of electrostatics or magnetostatics in empty space. The divergence of the electric field is zero when there are no charges, and the curl of the electrostatic field is always zero. The curl of the magnetic field is zero if there are no currents, and the divergence of the magnetic field is always zero. Therefore, Eqs. (40.19) have the same solutions as the equations for $\mathbf{E}$ in electrostatics or for $\mathbf{B}$ in magnetostatics. As a matter of fact, we have already solved the problem of the flow of a fluid past a sphere, as an electrostatic analogy, in Section 12-5. The electrostatic analog is a uniform electric field plus a dipole field. The dipole field is so adjusted that the flow velocity normal to the surface of the sphere is zero. The same problem for the flow past a cylinder can be worked out in a similar way by using a suitable line dipole with a uniform flow field. This solution holds for a situation in which the fluid velocity at large distances is constant—both in magnitude and direction. The solution is sketched in Fig. 40-11(a).

There is another solution for the flow around a cylinder when the conditions are such that the fluid at large distances moves in circles around the cylinder. The flow is, then, circular everywhere, as in Fig. 40-11(b). Such a flow has a circulation around the cylinder, although $\nabla \times \mathbf{v}$ is still zero *in the fluid*. How can there be circulation without a curl? We have a circulation around the cylinder because the line integral of $\mathbf{v}$ around any loop *enclosing* the cylinder is not zero. At the same time, the line integral of $\mathbf{v}$ around any closed path which does *not* include the cylinder is zero. We saw the same thing when we found the magnetic field around a wire. The curl of $\mathbf{B}$ was zero outside of the wire, although a line integral of $\mathbf{B}$ around a path which encloses the wire did not vanish. The velocity field in an irrotational circulation around a cylinder is precisely the same as the magnetic field around a wire. For a circular path with its center at the center of the cylinder, the line integral of the velocity is

$$\oint \mathbf{v} \cdot d\mathbf{s} = 2\pi r v.$$

For irrotational flow the integral must be independent of r . Let's call the constant

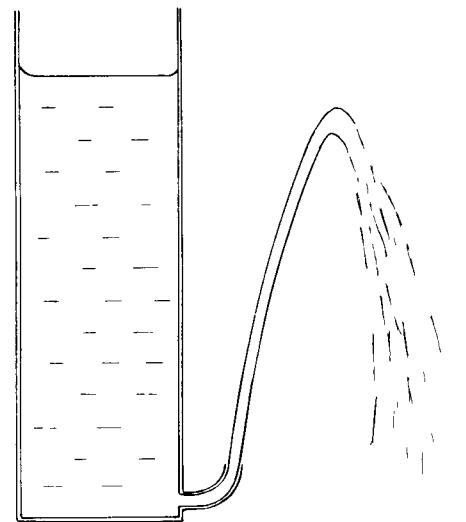


Fig. 40-10. Proof that v is not equal to $\sqrt{2gh}$.

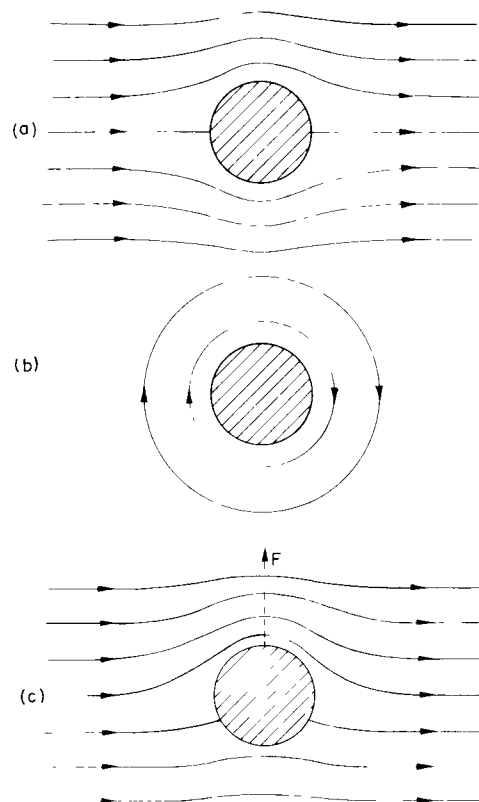


Fig. 40-11. (a) Ideal fluid flow past a cylinder. (b) Circulation around a cylinder. (c) The superposition of (a) and (b).

value C , then we have that

$$v = \frac{C}{2\pi r}, \quad (40.20)$$

where v is the tangential velocity, and r is the distance from the axis.

There is a nice demonstration of a fluid circulating around a hole. You take a transparent cylindrical tank with a drain hole in the center of the bottom. You fill it with water, stir up some circulation with a stick, and pull the drain plug. You get the pretty effect shown in Fig. 40-12. (You've seen a similar thing many times in the bathtub!) Although you put in some ω at beginning, it soon dies down because of viscosity and the flow becomes irrotational—although still with some circulation around the hole.

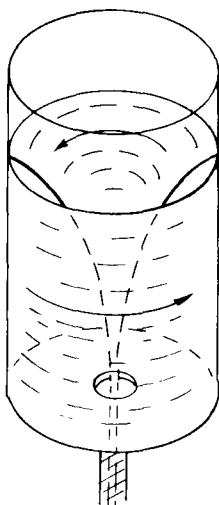


Fig. 40-12. Water with circulation draining from a tank.

From the theory, we can calculate the shape of the inner surface of the water. As a particle of the water moves inward it picks up speed. From Eq. (40.20) the tangential velocity goes as $1/r$ —it's just from the conservation of angular momentum, like the skater pulling in her arms. Also the radial velocity goes as $1/r$. Ignoring the tangential motion, we have water going radially inward toward a hole; from $\nabla \cdot \mathbf{v} = 0$, it follows that the radial velocity is proportional to $1/r$. So the total velocity also increases as $1/r$, and the water goes in along Archimedean spirals. The air-water surface is all at atmospheric pressure, so it must have—from Eq. (40.14)—the property that

$$gz + \frac{1}{2}mv^2 = \text{const.}$$

But v is proportional to $1/r$, so the shape of the surface is

$$(z - z_0) = \frac{k}{r^2}.$$

An interesting point—which is *not true in general* but is true for incompressible, irrotational flow—is that if we have one solution and a second solution, then the sum is also a solution. This is true because the equations in (40.19) are linear. The complete equations of hydrodynamics, Eqs. (40.8), (40.9), and (40.10), are not linear, which makes a vast difference. For the irrotational flow about the cylinder, however, we can superpose the flow of Fig. 40-11(a) on the flow of Fig. 40-11(b) and get the new flow pattern shown in Fig. 40-11(c). This flow is of special interest. The flow velocity is higher on the upper side of the cylinder than on the lower side. The pressures are therefore *lower* on the *upper* side than on the lower side. So when we have a combination of a circulation around a cylinder *and* a net horizontal flow, there is a net *vertical force* on the cylinder—it is called a *lift force*. Of course, if there is no circulation, there is no net force on any body according to our theory of “dry” water.

40-5 Vortex lines

We have already written down the general equations for the flow of an incompressible fluid when there may be vorticity. They are

$$\text{I. } \nabla \cdot \mathbf{v} = 0,$$

$$\text{II. } \boldsymbol{\Omega} = \nabla \times \mathbf{v},$$

$$\text{III. } \frac{\partial \boldsymbol{\Omega}}{\partial t} + \nabla \times (\boldsymbol{\Omega} \times \mathbf{v}) = 0.$$

The physical content of these equations has been described in words by Helmholtz in terms of three theorems. First, imagine that in the fluid we were to draw *vortex lines* rather than streamlines. By vortex lines we mean field lines that have the direction of $\boldsymbol{\Omega}$ and have a density in any region proportional to the magnitude of $\boldsymbol{\Omega}$. From II the divergence of $\boldsymbol{\Omega}$ is *always* zero (remember—Section 3-7—that the divergence of a curl is always zero). So vortex lines are like lines of $\mathbf{B}$ —they never start or stop, and will tend to go in closed loops. Now Helmholtz described III

in words by the following statement: the vortex lines *move with the fluid*. This means that if you were to mark the fluid particles along some vortex lines—by coloring them with ink, for example—then as the fluid moves and carries those particles along, they will always mark the new positions of the vortex lines. In whatever way the atoms of the liquid move, the vortex lines move with them. That is one way to describe the laws.

It also suggests a method for solving any problems. Given the initial flow pattern—say $\mathbf{v}$ everywhere—then you can calculate $\boldsymbol{\Omega}$. From the $\mathbf{v}$ you can also tell where the vortex lines are going to be a little later—they move with the speed $\mathbf{v}$. With the new $\boldsymbol{\Omega}$ you can use I and II to find the new $\mathbf{v}$. (That's just like the problem of finding $\mathbf{B}$, given the currents.) If we are given the flow pattern at one instant we can in principle calculate it for all subsequent times. We have the general solution for nonviscous flow.

We would like to show how Helmholtz's statement—and, therefore, III—can be at least partly understood. It is really just the law of conservation of angular momentum applied to the fluid. Suppose we imagine a small cylinder of the liquid whose axis is parallel to the vortex lines, as in Fig. 40-13(a). At some time later, this *same* piece of fluid will be somewhere else. Generally it will occupy a cylinder with a different diameter and be in a different place. It may also have a different orientation, say as in Fig. 40-13(b). If the diameter has changed, however, the length will have increased to keep the volume constant (since we are assuming an incompressible fluid). Also, since the vortex lines are stuck with the material, their density will go up as the cross-sectional area goes down. The product of the vorticity $\boldsymbol{\Omega}$ and area A of the cylinder will remain constant, so according to Helmholtz, we should have

$$\Omega_2 A_2 = \Omega_1 A_1. \quad (40.21)$$

Now notice that with zero viscosity all the forces on the surface of the cylindrical volume (or *any* volume, for that matter) are perpendicular to the surface. The pressure forces can cause the volume to be moved from place to place, or can cause it to change shape; but with no *tangential* forces the magnitude of the *angular momentum of the material inside* cannot change. The angular momentum of the liquid in the little cylinder is its moment of inertia I times the angular velocity of the liquid, which is proportional to the vorticity $\boldsymbol{\Omega}$. For a cylinder, the moment of inertia is proportional to mr^2 . So from the conservation of angular momentum, we would conclude that

$$(M_1 R_1^2) \Omega_1 = (M_2 R_2^2) \Omega_2.$$

But the mass is the same, $M_1 = M_2$, and the areas are proportional to R^2 , so we get again just Eq. (40.21). Helmholtz's statement—which is equivalent to III—is just a consequence of the fact that in the absence of viscosity the angular momentum of an element of the fluid cannot change.

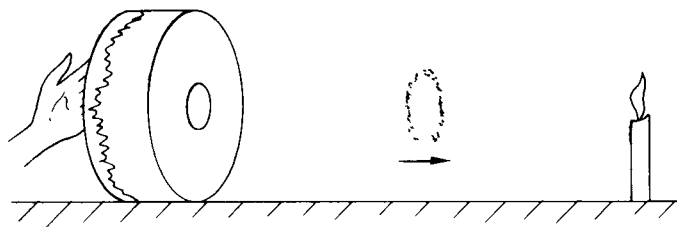


Fig. 40-14. Making a travelling vortex ring.

There is a nice demonstration of a moving vortex which is made with the simple apparatus of Fig 40-14. It is a “drum” two feet in diameter and two feet long made by stretching a thick rubber sheet over the open end of a cylindrical “box.” The “bottom”—the drum is tipped on its side—is solid except for a 3-inch diameter hole. If you give a sharp blow on the rubber diaphragm with your hand, a vortex ring is projected out of the hole. Although the vortex is invisible, you can tell it's there because it will blow out a candle 10 to 20 feet away. By the delay in

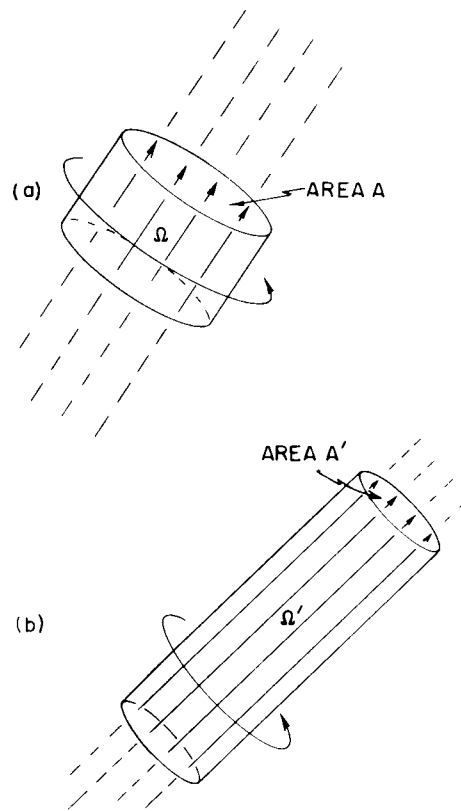


Fig. 40-13. (a) A group of vortex lines at t ; (b) the same lines at a later time t' .

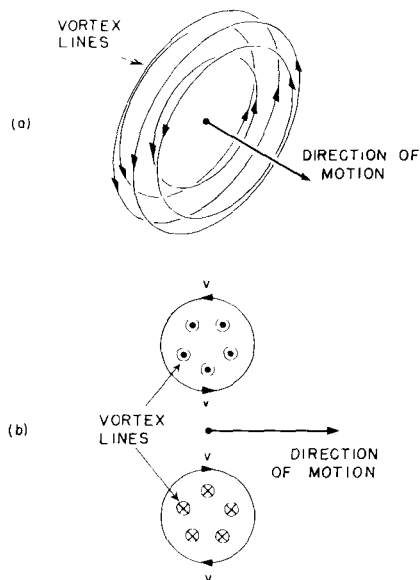


Fig. 40-15. A moving vortex ring (a smoke ring). (a) The vortex lines. (b) A cross section of the ring.

the effect, you can tell that “something” is travelling at a finite speed. You can see better what is going on if you first blow some smoke into the box. Then you see the vortex as a beautiful round “smoke ring.”

The smoke ring is a torus-shaped bundle of vortex lines, as shown in Fig 40-15(a). Since $\Omega = \nabla \times \mathbf{v}$, these vortex lines represent also a circulation of $\mathbf{v}$ as shown in part (b) of the figure. We can understand the forward motion of the ring in the following way: The circulating velocity around the *bottom* of the ring extends up to the top of the ring, having there a forward motion. Since the lines of Ω move with the fluid, they also move ahead with the velocity $\mathbf{v}$. (Of course, the circulation of $\mathbf{v}$ around the top part of the ring is responsible for the forward motion of the vortex lines at the bottom.)

We must now mention a serious difficulty. We have already noted that Eq. (40 9) says that, if Ω is initially zero, it will always be zero. This result is a great failure of the theory of “dry” water, because it means that once Ω is zero it is *always* zero—it is impossible to *produce* any vorticity under any circumstance. Yet, in our simple demonstration with the drum, we can generate a vortex ring starting with air which was initially at rest. (Certainly, $\mathbf{v} = 0$, $\Omega = 0$ everywhere in the box before we hit it.) Also, we all know that we can start some vorticity in a lake with a paddle. Clearly, we must go to a theory of “wet” water to get a complete understanding of the behavior of a fluid.

Another feature of the dry water theory which is incorrect is the supposition we make regarding the flow at the boundary between it and the surface of a solid. When we discussed the flow past a cylinder—as in Fig. 40-11, for example—we permitted the fluid to slide along the surface of the solid. In our theory, the velocity at a solid surface could have any value depending on how it got started, and we did not consider any “friction” between the fluid and the solid. It is an experimental fact, however, that the velocity of a real fluid always goes to zero at the surface of a solid object. Therefore, our solution for the cylinder, with or without circulation, is wrong—as is our result regarding the generation of vorticity. We will tell you about the more correct theories in the next chapter.

The Flow of Wet Water

41-1 Viscosity

In the last chapter we discussed the behavior of water, disregarding the phenomenon of viscosity. Now we would like to discuss the phenomena of the flow of fluids, *including* the effects of viscosity. We want to look at the *real behavior* of fluids. We will describe qualitatively the actual behavior of the fluids under various different circumstances so that you will get some feel for the subject. Although you will see some complicated equations and hear about some complicated things, it is not our purpose that you should learn all these things. This is, in a sense, a “cultural” chapter which will give you some idea of the way the world is. There is only one item which is worth learning, and that is the simple definition of viscosity which we will come to in a moment. The rest is only for your entertainment.

In the last chapter we found that the laws of motion of a fluid are contained in the equation

$$\frac{\partial \mathbf{v}}{\partial t} + (\mathbf{v} \cdot \nabla) \mathbf{v} = -\frac{\nabla p}{\rho} - \nabla \phi + \frac{\mathbf{f}_{\text{visc}}}{\rho}. \quad (41.1)$$

In our “dry” water approximation we left out the last term, so we were neglecting all viscous effects. Also, we sometimes made an additional approximation by considering the fluid as incompressible; then we had the additional equation

$$\nabla \cdot \mathbf{v} = 0.$$

This last approximation is often quite good—particularly when flow speeds are much slower than the speed of sound. But in real fluids it is almost never true that we can neglect the internal friction that we call viscosity; most of the interesting things that happen come from it in one way or another. For example, we saw that in “dry” water the circulation never changes—if there is none to start out with, there will never be any. Yet, circulation in fluids is an everyday occurrence. We must fix up our theory.

We begin with an important experimental fact. When we worked out the flow of “dry” water around or past a cylinder—the so-called “potential flow”—we had no reason not to permit the water to have a velocity tangent to the surface; only the normal component had to be zero. We took no account of the possibility that there might be a shear force between the liquid and the solid. It turns out—although it is not at all self-evident—that in all circumstances where it has been experimentally checked, the *velocity of a fluid is exactly zero at the surface of a solid*. You have noticed, no doubt, that the blade of a fan will collect a thin layer of dust—and that it is still there after the fan has been churning up the air. You can see the same effect even on the great fan of a wind tunnel. Why isn’t the dust blown off by the air? In spite of the fact that the fan blade is moving at high speed through the air, the speed of the air relative to the fan blade goes to zero right at the surface. So the very smallest dust particles are not disturbed.* We must modify the theory to agree with the experimental fact that in all ordinary fluids, the molecules next to a solid surface have zero velocity (relative to the surface).†

* You can blow *large* dust particles from a table top, but *not* the very finest ones. The large ones stick up into the breeze.

† You can imagine circumstances when it is not true: glass is theoretically a “liquid,” but it can certainly be made to slide along a steel surface. So our assertion must break down somewhere.

41-1 Viscosity

41-2 Viscous flow

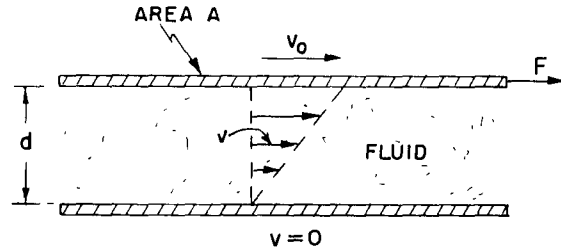
41-3 The Reynolds number

41-4 Flow past a circular cylinder

41-5 The limit of zero viscosity

41-6 Couette flow

Fig. 41-1. Viscous drag between two parallel plates.



We originally characterized a liquid by the fact that if you put a shearing stress on it—no matter how small—it would give way. It flows. In static situations, there are no shear stresses. But before equilibrium is reached—as long as you still push on it—there can be shear forces. *Viscosity* describes these shear forces which exist in a moving fluid. To get a measure of the shear forces during the motion of a fluid, we consider the following kind of experiment. Suppose that we have two solid plane surfaces with water between them, as in Fig. 41-1, and we keep one stationary while moving the other parallel to it at the slow speed v_0 . If you measure the force required to keep the upper plate moving, you find that it is proportional to the area of the plates and to v_0/d , where d is the distance between the plates. So the shear stress F/A is proportional to v_0/d :

$$\frac{F}{A} = \eta \frac{v_0}{d}.$$

The constant of proportionality η is called the *coefficient of viscosity*.

If we have a more complicated situation, we can always consider a little, flat, rectangular cell in the water with its faces parallel to the flow, as in Fig. 41-2. The shear force across this cell is given by

$$\frac{\Delta F}{\Delta A} = \eta \frac{\Delta v_x}{\Delta y} = \eta \frac{\partial v_x}{\partial y}. \quad (41.2)$$

Now, $\partial v_x / \partial y$ is the *rate of change* of the shear strain we defined in Chapter 38, so for a liquid, the shear stress is proportional to the *rate of change* of the shear strain.

In the general case we write

$$S_{xy} = \eta \left(\frac{\partial v_y}{\partial x} + \frac{\partial v_x}{\partial y} \right). \quad (41.3)$$

If there is a uniform rotation of the fluid, $\partial v_x / \partial y$ is the negative of $\partial v_y / \partial x$ and S_{xy} is zero—as it should be since there are no stresses in a uniformly rotating fluid. (We did a similar thing in defining e_{xy} in Chapter 39.) There are, of course, the corresponding expressions for S_{yz} and S_{zx} .

As an example of the application of these ideas, we consider the motion of a fluid between two coaxial cylinders. Let the inner one have the radius a and the peripheral velocity v_a , and let the outer one have radius b and velocity v_b . See Fig. 41-3. We might ask, what is the velocity distribution between the cylinders? To answer this question, we begin by finding a formula for the viscous shear in the fluid at a distance r from the axis. From the symmetry of the problem, we can assume that the flow is always tangential and that its magnitude depends only on r ; $v = v(r)$. If we watch a speck in the water at the radius r , its coordinates as a function of time are

$$x = r \cos \omega t, \quad y = r \sin \omega t,$$

where $\omega = v/r$. Then the x - and y -components of velocity are

$$v_x = -r\omega \sin \omega t = -\omega y \quad \text{and} \quad v_y = r\omega \cos \omega t = \omega x. \quad (41.4)$$

From Eq. (41.3), we have

$$S_{xy} = \eta \left[\frac{\partial}{\partial x} (x\omega) - \frac{\partial}{\partial y} (y\omega) \right] = \eta \left[x \frac{\partial \omega}{\partial x} - y \frac{\partial \omega}{\partial y} \right]. \quad (41.5)$$

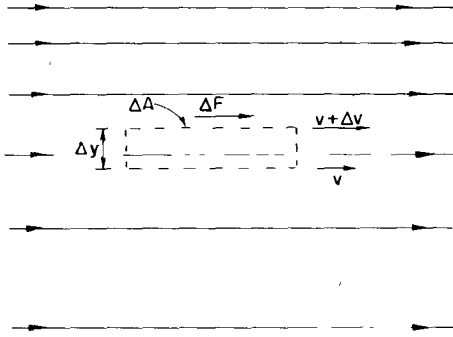


Fig. 41-2. The shear stress in a viscous fluid.

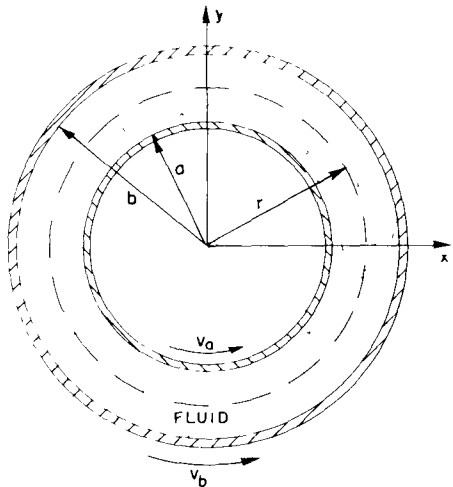


Fig. 41-3. The flow in a fluid between two concentric cylinders rotating at different angular velocities.

For a point at $y = 0$, $\partial\omega/\partial y = 0$, and $x \partial\omega/\partial x$ is the same as $r d\omega/dr$. So at that point

$$(S_{xy})_{y=0} = \eta r \frac{d\omega}{dr}. \quad (41.6)$$

(It is reasonable that S should depend on $\partial\omega/\partial r$: when there is no change in ω with r , the liquid is in uniform rotation and there are no stresses.)

The stress we have calculated is the tangential shear which is the same all around the cylinder. We can get the *torque* acting *across a cylindrical surface* at the radius r by multiplying the shear stress by the moment arm r and the area $2\pi r l$. We get

$$\tau = 2\pi r^2 l (S_{xy})_{y=0} = 2\pi \eta l r^3 \frac{d\omega}{dr}. \quad (41.7)$$

Since the motion of the water is steady—there is no angular acceleration—the net torque on the cylindrical shell of water between r and $r + dr$ must be zero; that is, the torque at r must be balanced by an equal and opposite torque at $r + dr$, so that τ must be independent of r . In other words, $r^3 d\omega/dr$ is equal to some constant, say A , and

$$\frac{d\omega}{dr} = \frac{A}{r^3}. \quad (41.8)$$

Integrating, we find that ω varies with r as

$$\omega = -\frac{A}{2r^2} + B. \quad (41.9)$$

The constants A and B are to be determined to fit the conditions that $\omega = \omega_a$ at $r = a$, and $\omega = \omega_b$ at $r = b$. We get that

$$\begin{aligned} A &= \frac{2a^2b^2}{b^2 - a^2} (\omega_b - \omega_a), \\ B &= \frac{b^2\omega_b - a^2\omega_a}{b^2 - a^2}. \end{aligned} \quad (41.10)$$

So we know ω as a function of r , and from it $v = \omega r$.

If we want the torque, we can get it from Eqs. (41.7) and (41.8):

$$\tau = 2\pi \eta l A$$

or

$$\tau = \frac{4\pi \eta l a^2 b^2}{b^2 - a^2} (\omega_b - \omega_a). \quad (41.11)$$

It is proportional to the relative angular velocities of the two cylinders. One standard apparatus for measuring the coefficients of viscosity is built this way. One cylinder—say the outer one—is on pivots but is held stationary by a spring balance which measures the torque on it, while the inner one is rotated at a constant angular velocity. The coefficient of viscosity is then determined from Eq. (41.11).

From its definition, you see that the units of η are newton-sec/m². For water at 20°C,

$$\eta = 10^3 \text{ newton-sec/m}^2.$$

It is usually more convenient to use the *specific viscosity*, which is η divided by the density ρ . The values for water and air are then comparable:

$$\begin{aligned} \text{water at } 20^\circ\text{C}, \quad \eta/\rho &= 10^{-6} \text{ m}^2/\text{sec}, \\ \text{air at } 20^\circ\text{C}, \quad \eta/\rho &= 15 \times 10^{-6} \text{ m}^2/\text{sec}. \end{aligned} \quad (41.12)$$

Viscosities usually depend strongly on temperature. For instance, for water just above the freezing point, η/ρ is 1.8 times larger than it is at 20°C.

41-2 Viscous flow

We now go to a general theory of viscous flow—at least in the most general form known to man. We already understand that the shear stress components are proportional to the spatial derivatives of the various velocity components such as $\partial v_x/\partial y$ or $\partial v_y/\partial x$. However, in the general case of a *compressible* fluid there is another term in the stress which depends on other derivatives of the velocity. The general expression is

$$S_{ij} = \eta \left(\frac{\partial v_i}{\partial x_j} + \frac{\partial v_j}{\partial x_i} \right) + \eta' \delta_{ij} (\nabla \cdot \mathbf{v}), \quad (41.13)$$

where x_i is any one of the rectangular coordinates x , y , or z , and v_i is any one of the rectangular coordinates of the velocity. (The symbol δ_{ij} is the Kronecker delta which is 1 when $i = j$ and 0 for $i \neq j$.) The additional term adds $\eta' \nabla \cdot \mathbf{v}$ to all the diagonal elements S_{ii} of the stress tensor. If the liquid is incompressible $\nabla \cdot \mathbf{v} = 0$, and this extra term doesn't appear. So it has to do with internal forces during compression. So two constants are required to describe the liquid, just as we had two constants to describe a homogeneous elastic solid. The coefficient η is the “ordinary” coefficient of viscosity which we have already encountered. It is also called the *first coefficient of viscosity* or the “shear viscosity coefficient,” and the new coefficient η' is called the *second coefficient of viscosity*.

Now we want to determine the viscous force per unit volume, f_{visc} , so we can put it into Eq. (41.1) to get the equation of motion for a real fluid. The force on a small cubical volume element of a fluid is the resultant of the forces on all the six faces. Taking them two at a time, we will get differences that depend on the derivatives of the stresses, and, therefore, on the second derivatives of the velocity. This is nice because it will get us back to a vector equation. The component of the viscous force per unit volume in the direction of the rectangular coordinate x_i is

$$\begin{aligned} (f_{\text{visc}})_i &= \sum_{j=1}^3 \frac{\partial S_{ij}}{\partial x_j} \\ &= \eta \sum_{j=1}^3 \frac{\partial}{\partial x_j} \left\{ \eta \left(\frac{\partial v_i}{\partial x_j} + \frac{\partial v_j}{\partial x_i} \right) \right\} + \frac{\partial}{\partial x_i} (\eta' \nabla \cdot \mathbf{v}) \end{aligned} \quad (41.14)$$

Usually, the variation of the viscosity coefficients with position is not significant and can be neglected. Then, the viscous force per unit volume contains only second derivatives of the velocity. We saw in Chapter 39 that the most general form of second derivatives that can occur in a vector equation is the sum of a term in the Laplacian ($\nabla \cdot \nabla \mathbf{v} = \nabla^2 \mathbf{v}$), and a term in the gradient of the divergence ($\nabla(\nabla \cdot \mathbf{v})$). Equation (41.14) is just such a sum with the coefficients η and $(\eta + \eta')$. We get

$$f_{\text{visc}} = \eta \nabla^2 \mathbf{v} + (\eta + \eta') \nabla(\nabla \cdot \mathbf{v}). \quad (41.15)$$

In the incompressible case, $\nabla \cdot \mathbf{v} = 0$, and the viscous force per unit volume is just $\eta \nabla^2 \mathbf{v}$. That is all that many people use; however, if you should want to calculate the absorption of sound in a fluid, you would need the second term.

We can now complete our general equation of motion for a real fluid. Substituting Eq. (41.15) into Eq. (41.1), we get

$$\rho \left\{ \frac{\partial \mathbf{v}}{\partial t} + (\mathbf{v} \cdot \nabla) \mathbf{v} \right\} = -\nabla p - \rho \nabla \phi + \eta \nabla^2 \mathbf{v} + (\eta + \eta') \nabla(\nabla \cdot \mathbf{v})$$

It's complicated. But that's the way nature is.

If we introduce the vorticity $\boldsymbol{\Omega} = \nabla \times \mathbf{v}$, as we did before, we can write our equation as

$$\begin{aligned} \rho \left\{ \frac{\partial \mathbf{v}}{\partial t} + \boldsymbol{\Omega} \times \mathbf{v} + \frac{1}{2} \nabla v^2 \right\} &= -\nabla p - \rho \nabla \phi + \eta \nabla^2 \mathbf{v} \\ &\quad + (\eta + \eta') \nabla(\nabla \cdot \mathbf{v}). \end{aligned} \quad (41.16)$$

We are supposing again that the only body forces acting are conservative forces like gravity. To see what the new term means, let's look at the incompressible fluid case. Then, if we take the curl of Eq. (41.16), we get

$$\frac{\partial \mathbf{\Omega}}{\partial t} + \nabla \times (\mathbf{\Omega} \times \mathbf{v}) = \frac{\eta}{\rho} \nabla^2 \mathbf{\Omega}. \quad (41.17)$$

This is like Eq. (40.9) except for the new term on the right-hand side. When the right-hand side was zero, we had the Helmholtz theorem that the vorticity stays with the fluid. Now, we have the rather complicated nonzero term on the right-hand side which, however, has straightforward physical consequences. If we disregard for the moment the term $\nabla \times (\mathbf{\Omega} \times \mathbf{v})$, we have a *diffusion equation*. The new term means that the vorticity $\mathbf{\Omega}$ *diffuses* through the fluid. If there is a large gradient in the vorticity, it will spread out into the neighboring fluid.

This is the term that causes the smoke ring to get thicker as it goes along. Also, it shows up nicely if you send a "clean" vortex (a "smokeless" ring made by the apparatus described in the last chapter) through a cloud of smoke. When it comes out of the cloud, it will have picked up some smoke, and you will see a hollow shell of a smoke ring. Some of the $\mathbf{\Omega}$ diffuses outward into the smoke, while still maintaining its forward motion with the vortex.

41-3 The Reynolds number

We will now describe the changes which are made in the character of fluid flow as a consequence of the new viscosity term. We will look at two problems in some detail. The first of these is the flow of a fluid past a cylinder—a flow which we tried to calculate in the previous chapter using the theory for nonviscous flow. It turns out that the viscous equations can be solved by man today only for a few special cases. So some of what we will tell you is based on experimental measurements—assuming that the experimental model satisfies Eq. (41.17).

The mathematical problem is this: We would like the solution for the flow of an incompressible, viscous fluid past a long cylinder of diameter D . The flow should be given by Eq. (41.17) and by

$$\mathbf{\Omega} = \nabla \times \mathbf{v} \quad (41.18)$$

with the conditions that the velocity at large distances is some constant velocity, say V (parallel to the x -axis), and at the surface of the cylinder is zero. That is,

$$v_x = v_y = v_z = 0 \quad (41.19)$$

for

$$x^2 + y^2 = \frac{D^2}{4}.$$

That specifies completely the mathematical problem.

If you look at the equations, you see that there are four different parameters to the problem: η , ρ , D , and V . You might think that we would have to give a whole series of cases for different V 's, different D 's, and so on. However, that is not the case. All the different possible solutions correspond to different values of *one parameter*. This is the most important general thing we can say about viscous flow. To see why this is so, notice first that the viscosity and density appear only in the ratio η/ρ —the *specific viscosity*. That reduces the number of independent parameters to three. Now suppose we measure all distances in the only length that appears in the problem, the diameter D of the cylinder; that is, we substitute for x , y , z , the new variables x' , y' , z' with

$$x = x'D, \quad y = y'D, \quad z = z'D.$$

Then D disappears from (41.19). In the same way, if we measure all velocities in terms of V —that is, we set $v = v'V$ —we get rid of the V , and v' is just equal to 1 at large distances. Since we have fixed our units of length and velocity, our unit

of time is now D/V ; so we should set

$$t = t' \frac{D}{V}. \quad (41.20)$$

With our new variables, the derivatives in Eq. (41.18) get changed from $\partial/\partial x$ to $(1/D) \partial/\partial x'$, and so on; so Eq. (41.18) becomes

$$\mathbf{\Omega} = \nabla \times \mathbf{v} = \frac{V}{D} \nabla' \times \mathbf{v}' = \frac{V}{D} \mathbf{\Omega}'. \quad (41.21)$$

Our main equation (41.17) then reads

$$\frac{\partial \mathbf{\Omega}'}{\partial t'} + \nabla' \times (\mathbf{\Omega}' \times \mathbf{v}') = \frac{\eta}{\rho V D} \nabla'^2 \mathbf{\Omega}'.$$

All the constants condense into one factor which we write, following tradition, as $1/\Re$:

$$\Re = \frac{\rho}{\eta} V D. \quad (41.22)$$

If we just remember that all of our equations are to be written with all quantities in the new units, we can omit all the primes. Our equations for the flow are then

$$\frac{\partial \mathbf{\Omega}}{\partial t} + \nabla \times (\mathbf{\Omega} \times \mathbf{v}) = \frac{1}{\Re} \nabla^2 \mathbf{\Omega} \quad (41.23)$$

and

$$\mathbf{\Omega} = \nabla \times \mathbf{v}$$

with the conditions

$$\mathbf{v} = 0$$

for

$$x^2 + y^2 = 1/4 \quad (41.24)$$

and

$$v_x = 1, \quad v_y = v_z = 0$$

for

$$x^2 + y^2 + z^2 \gg 1.$$

What this all means physically is very interesting. It means, for example, that if we solve the problem of the flow for one velocity V_1 and a certain cylinder diameter D_1 , and then ask about the flow for a different diameter D_2 and a different fluid, the flow will be the same for the velocity V_2 which gives the same Reynolds number—that is, when

$$\Re_1 = \frac{\rho_1}{\eta_1} V_1 D_1 = \Re_2 = \frac{\rho_2}{\eta_2} V_2 D_2. \quad (41.25)$$

For any two situations which have the same Reynolds number, the flows will “look” the same—in terms of the appropriate scaled x' , y' , z' , and t' . This is an important proposition because it means that we can determine what the behavior of the flow of air past an airplane wing will be without having to build an airplane and try it. We can, instead, make a model and make measurements using a velocity that gives the same Reynolds number. This is the principle which allows us to apply the results of “wind-tunnel” measurements on small-scale airplanes, or “model-basin” results on scale model boats, to the full-scale objects. Remember, however, that we can only do this provided the compressibility of the fluid can be neglected. Otherwise, a new quantity enters—the speed of sound. And different situations will really correspond to each other only if the ratio of V to the sound speed is also the same. This latter ratio is called the *Mach number*. So, for velocities near the speed of sound or above, the flows are the same in two situations if *both* the *Mach number* and the *Reynolds number* are the same for both situations.

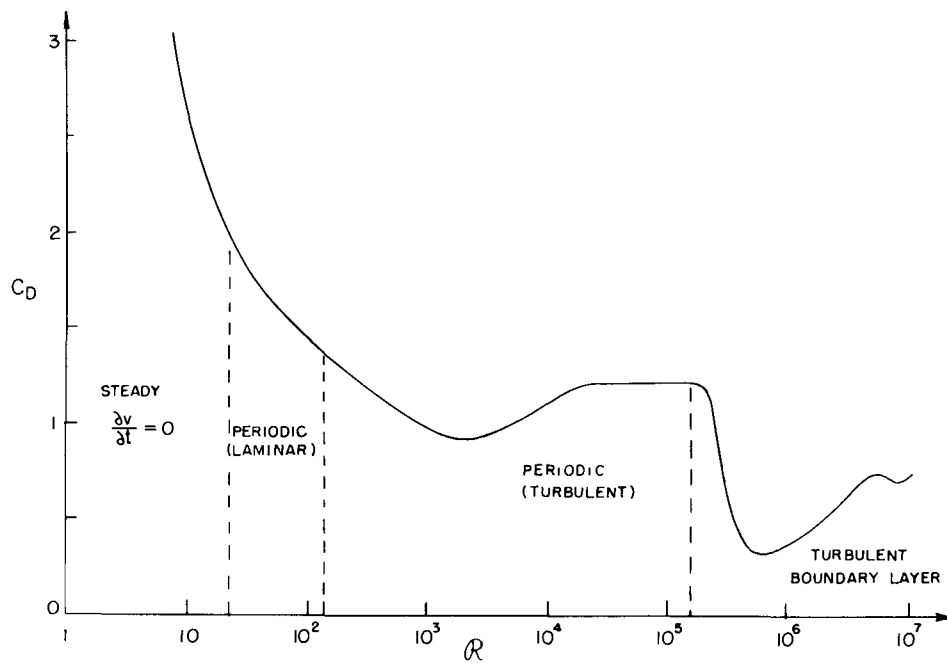


Fig. 41-4. The drag coefficient C_D of a circular cylinder as a function of the Reynolds number.

41-4 Flow past a circular cylinder

Let's go back to the problem of low-speed (nearly incompressible) flow over the cylinder. We will give a qualitative description of the flow of a real fluid. There are many things we might want to know about such a flow—for instance, what is the drag force on the cylinder? The drag force on a cylinder is plotted in Fig. 41-4 as a function of R —which is proportional to the air speed V if everything else is held fixed. What is actually plotted is the so-called *drag coefficient* C_D , which is a dimensionless number equal to the force divided by $\frac{1}{2}\rho V^2 D l$, where D is the diameter, l is the length of the cylinder, and ρ is the density of the liquid:

$$C_D = \frac{F}{\frac{1}{2}\rho V^2 D l}.$$

The coefficient of drag varies in a rather complicated way, giving us a pre-hint that something rather interesting and complicated is happening in the flow. We will now describe the nature of flow for the different ranges of the Reynolds number. First, when the Reynolds number is very small, the flow is quite steady; that is, the velocity is constant at any place, and the flow goes around the cylinder. The actual distribution of the flow lines is, however, not like it is in potential flow. They are solutions of a somewhat different equation. When the velocity is very low or, what is equivalent, when the viscosity is very high so the stuff is like honey, then the inertial terms are negligible and the flow is described by the equation

$$\nabla^2 \Omega = 0.$$

This equation was first solved by Stokes. He also solved the same problem for a sphere. If you have a small sphere moving under such conditions of low Reynolds number, the force needed to drag it is equal to $6\pi\eta aV$, where a is the radius of the sphere and V is its velocity. This is a very useful formula because it tells the speed at which tiny grains of dirt (or other particles which can be approximated as spheres) move through a fluid under a given force—as, for instance, in a centrifuge, or in sedimentation, or diffusion. In the low Reynolds number region—for R less than 1—the lines of v around a *cylinder* are as drawn in Fig. 41-5.

If we now increase the fluid speed to get a Reynolds number somewhat greater than 1, we find that the flow is different. There is a circulation behind the sphere, as shown in Fig. 41-6(b). It is still an open question as to whether there is always

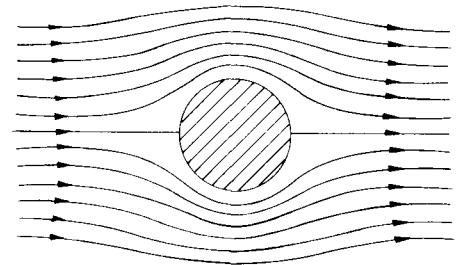


Fig. 41-5. Viscous flow (low velocities) around a circular cylinder.

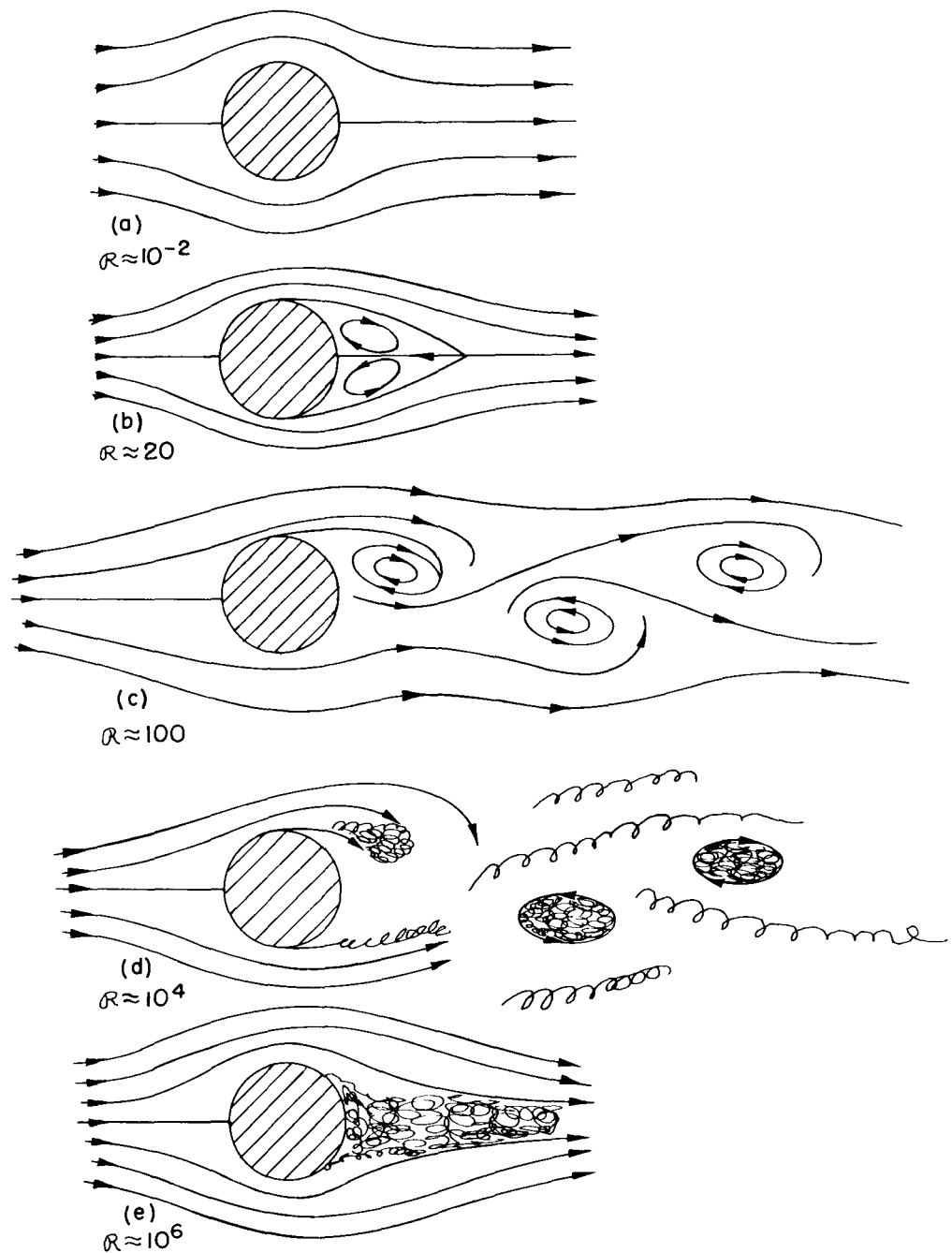


Fig. 41-6. Flow past a cylinder for various Reynolds numbers.

a circulation there even at the smallest Reynolds number or whether things suddenly change at a certain Reynolds number. It used to be thought that the circulation grew continuously. But it is now thought that it appears suddenly, and it is certain that the circulation increases with R . In any case, there is a different character to the flow for R in the region from about 10 to 30. There is a pair of vortices behind the cylinder.

The flow changes again by the time we get to a number of 40 or so. There is suddenly a complete change in the character of the motion. What happens is that one of the vortices behind the cylinder gets so long that it breaks off and travels downstream with the fluid. Then the fluid curls around behind the cylinder and makes a new vortex. The vortices peel off alternately on each side, so an instantaneous view of the flow looks roughly as sketched in Fig. 41-6(c). The stream of

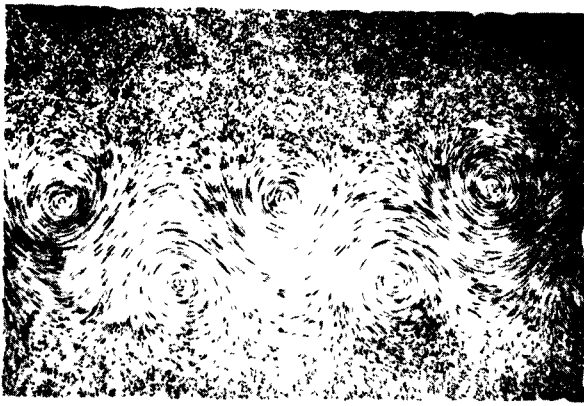


Fig. 41-7. Photograph by Ludwig Prandtl of the "vortex street" in the flow behind a cylinder.

vortices is called a "Kármán vortex street." They always appear for $\mathcal{R} > 40$. We show a photograph of such a flow in Fig. 41-7.

The difference between the two flows in Fig. 41-6(c) and 41-6(b) or 41-6(a) is almost a complete difference in regime. In Fig. 41-6(a) or (b), the velocity is constant, whereas in Fig. 41-6(c), the velocity at any point varies with time. There is no steady solution above $\mathcal{R} = 40$ —which we have marked on Fig. 41-4 by a dashed line. For these higher Reynolds numbers, the flow varies with time but in a *regular*, cyclic fashion.

We can get a physical idea of how these vortices are produced. We know that the fluid velocity must be zero at the surface of the cylinder and that it also increases rapidly away from that surface. Vorticity is created by this large local variation in fluid velocity. Now when the main stream velocity is low enough, there is sufficient time for this vorticity to diffuse out of the thin region near the solid surface where it is produced and to grow into a large region of vorticity. This physical picture should help to prepare us for the next change in the nature of the flow as the main stream velocity, or $\mathcal{R}$, is increased still more.

As the velocity gets higher and higher, there is less and less time for the vorticity to diffuse into a larger region of fluid. By the time we reach a Reynolds number of several hundred, the vorticity begins to fill in a thin band, as shown in Fig. 41-6(d). In this layer the flow is chaotic and irregular. The region is called the *boundary layer* and this irregular flow region works its way farther and farther upstream as $\mathcal{R}$ is increased. In the turbulent region, the velocities are very irregular and "noisy"; also the flow is no longer two-dimensional but twists and turns in all three dimensions. There is still a regular alternating motion superimposed on the turbulent one.

As the Reynolds number is increased further, the turbulent region works its way forward until it reaches the point where the flow lines leave the cylinder—for flows somewhat above $\mathcal{R} = 10^5$. The flow is as shown in Fig. 41-6(e), and we have what is called a "turbulent boundary layer." Also, there is a drastic change in the drag force; it drops by a large factor, as shown in Fig. 41-4. In this speed region, the drag force actually *decreases* with increasing speed. There seems to be little evidence of periodicity.

What happens for still larger Reynolds numbers? As we increase the speed further, the wake increases in size again and the drag increases. The latest experiments—which go up to $\mathcal{R} = 10^7$ or so—indicate that a new periodicity appears in the wake, either because the whole wake is oscillating back and forth in a gross motion or because some new kind of vortex is occurring together with an irregular noisy motion. The details are as yet not entirely clear, and are still being studied experimentally.

41-5 The limit of zero viscosity

We would like to point out that none of the flows we have described are anything like the potential flow solution we found in the preceding chapter. This is, at first sight, quite surprising. After all, $\mathcal{R}$ is proportional to $1/\eta$. So η going to zero is equivalent to $\mathcal{R}$ going to infinity. And if we take the limit of large $\mathcal{R}$ in

Eq. (41.23), we get rid of the right-hand side and get just the equations of the last chapter. Yet, you would find it hard to believe that the highly turbulent flow at $\mathcal{R} = 10^7$ was approaching the smooth flow computed from the equations of “dry” water. How can it be that as we approach $\mathcal{R} = \infty$, the flow described by Eq. (41.23) gives a completely different solution from the one we obtained taking $\eta = 0$ to start out with? The answer is very interesting. Note that the right-hand term of Eq. (41.23) has $1/\mathcal{R}$ times a *second derivative*. It is a higher derivative than any other derivative in the equation. What happens is that although the coefficient $1/\mathcal{R}$ is small, there are very rapid variations of Ω in the space near the surface. These rapid variations compensate for the small coefficient, and the product *does not go to zero* with increasing $\mathcal{R}$. The solutions do not approach the limiting case as the coefficient of $\nabla^2 \Omega$ goes to zero.

You may be wondering, “What is the fine-grain turbulence and how does it maintain itself? How can the vorticity which is made somewhere at the edge of the cylinder generate so much noise in the background?” The answer is again interesting. Vorticity has a tendency to amplify itself. If we forget for a moment about the diffusion of vorticity which causes a loss, the laws of flow say (as we have seen) that the vortex lines are carried along with the fluid, at the velocity v . We can imagine a certain number of lines of Ω which are being distorted and twisted by the complicated flow pattern of v . This pulls the lines closer together and mixes them all up. Lines that were simple before will get knotted and pulled close together. They will be longer and tighter together. The strength of the vorticity will increase and its irregularities—the pluses and minuses—will, in general, increase. So the magnitude of vorticity in three dimensions increases as we twist the fluid about.

You might well ask, “When is the potential flow a satisfactory theory at all?” In the first place, it is satisfactory outside the turbulent region where the vorticity has not entered appreciably by diffusion. By making special streamlined bodies, we can keep the turbulent region as small as possible; the flow around airplane wings—which are carefully designed—is almost entirely true potential flow.

41-6 Couette flow

It is possible to demonstrate that the complex and shifting character of the flow past a cylinder is not special but that the great variety of flow possibilities occurs generally. We have worked out in Section 1 a solution for the viscous flow between two cylinders, and we can compare the results with what actually happens. If we take two concentric cylinders with an oil in the space between them and put a fine aluminum powder as a suspension in the oil, the flow is easy to see. Now if we turn the outer cylinder slowly, nothing unexpected happens; see Fig. 41-8(a). Alternatively, if we turn the inner cylinder slowly, nothing very striking occurs. However, if we turn the inner cylinder at a higher rate, we get a surprise. The fluid breaks into horizontal bands, as indicated in Fig. 41-8(b). When the outer cylinder rotates at a similar rate with the inner one at rest, no such effect occurs. How can it be that there is a difference between rotating the inner or the out cylinder? After all, the flow pattern we derived in Section 1 depended only on $\omega_b - \omega_a$. We can get the answer by looking at the cross sections shown in Fig. 41-9. When the inner layers of the fluid are moving more rapidly than the outer ones, they tend to move *outward*—the centrifugal force is larger than the pressure holding them in place. A whole layer cannot move out uniformly because the outer layers are in the way. It must break into cells and circulate, as shown in Fig. 41-9(b). It is like the convection currents in a room which has hot air at the bottom. When the inner cylinder is at rest and the outer cylinder has a high velocity, the centrifugal forces build up a pressure gradient which keeps everything in equilibrium—see Fig. 41-9(c) (as in a room with hot air at the top).

Now let's speed up the inner cylinder. At first, the number of bands increases. Then suddenly you see the bands become wavy, as in Fig. 41-8(c), and the waves travel around the cylinder. The speed of these waves is easily measured. For high rotation speeds they approach $1/3$ the speed of the inner cylinder. And no one

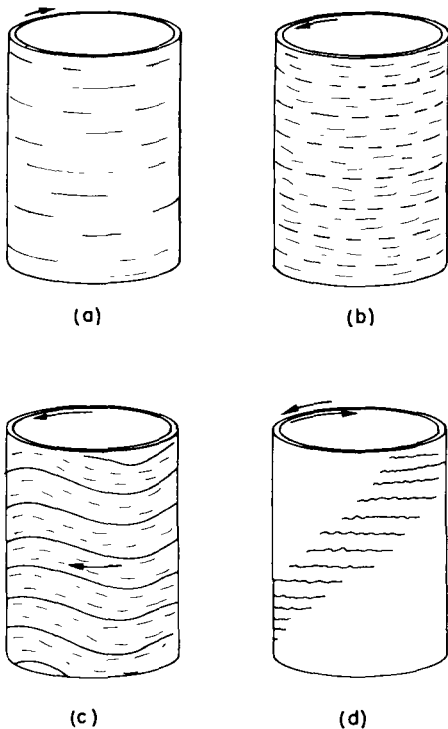


Fig. 41-8. Liquid flow patterns between two transparent rotating cylinders.

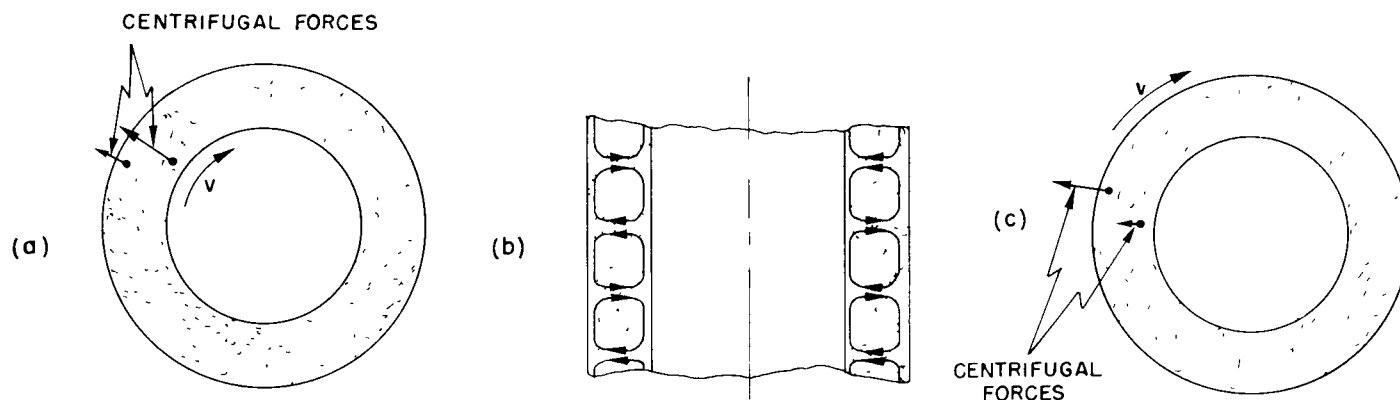


Fig. 41-9. Why the flow breaks up into bands.

knows why! There's a challenge. A simple number like $1/3$, and no explanation. In fact, the whole mechanism of the wave formation is not very well understood, yet it is steady laminar flow.

If we now start rotating the outer cylinder also—but in the opposite direction—the flow pattern starts to break up. We get wavy regions alternating with apparently quiet regions, as sketched in Fig. 41-8(d), making a spiral pattern. In these “quiet” regions, however, we can see that the flow is really quite irregular; it is, in fact completely turbulent. The wavy regions also begin to show irregular turbulent flow. If the cylinders are rotated still more rapidly, the whole flow becomes chaotically turbulent.

In this simple experiment we see many interesting regimes of flow which are quite different, and yet which are all contained in our simple equation for various values of the one parameter $\mathcal{R}$. With our rotating cylinders, we can see many of the effects which occur in the flow past a cylinder: first, there is a steady flow, second, a flow sets in which varies in time but in a regular, smooth way; finally, the flow becomes completely irregular. You have all seen the same effects in the column of smoke rising from a cigarette in quiet air. There is a smooth steady column followed by a series of twistings as the stream of smoke begins to break up, ending finally in an irregular churning cloud of smoke.

The main lesson to be learned from all of this is that a tremendous variety of behavior is hidden in the simple set of equations in (41.23). All the solutions are for the same equations, only with different values of $\mathcal{R}$. We have no reason to think that there are any terms missing from these equations. The only difficulty is that we do not have the mathematical power today to analyze them except for very small Reynolds numbers—that is, in the completely viscous case. That we have written an equation does not remove from the flow of fluids its charm or mystery or its surprise.

If such variety is possible in a simple equation with only one parameter, how much more is possible with more complex equations! Perhaps the fundamental equation that describes the swirling nebulae and the condensing, revolving, and exploding stars and galaxies is just a simple equation for the hydrodynamic behavior of nearly pure hydrogen gas. Often, people in some unjustified fear of physics say you can't write an equation for life. Well, perhaps we can. As a matter of fact, we very possibly already have the equation to a sufficient approximation when we write the equation of quantum mechanics:

$$H\psi = -\frac{\hbar}{i} \frac{\partial \psi}{\partial t}.$$

We have just seen that the complexities of things can so easily and dramatically escape the simplicity of the equations which describe them. Unaware of the scope of simple equations, man has often concluded that nothing short of God, not mere equations, is required to explain the complexities of the world.

We have written the equations of water flow. From experiment, we find a set of concepts and approximations to use to discuss the solution—vortex streets, turbulent wakes, boundary layers. When we have similar equations in a less familiar situation, and one for which we cannot yet experiment, we try to solve the equations in a primitive, halting, and confused way to try to determine what new qualitative features may come out, or what new qualitative forms are a consequence of the equations. Our equations for the sun, for example, as a ball of hydrogen gas, describe a sun without sunspots, without the rice-grain structure of the surface, without prominences, without coronas. Yet, all of these are really in the equations; we just haven't found the way to get them out.

There are those who are going to be disappointed when no life is found on other planets. Not I—I want to be reminded and delighted and surprised once again, through interplanetary exploration, with the infinite variety and novelty of phenomena that can be generated from such simple principles. The test of science is its ability to predict. Had you never visited the earth, could you predict the thunderstorms, the volcanos, the ocean waves, the auroras, and the colorful sunset? A salutary lesson it will be when we learn of all that goes on on each of those dead planets—those eight or ten balls, each agglomerated from the same dust cloud and each obeying exactly the same laws of physics.

The next great era of awakening of human intellect may well produce a method of understanding the *qualitative* content of equations. Today we cannot. Today we cannot see that the water flow equations contain such things as the barber pole structure of turbulence that one sees between rotating cylinders. Today we cannot see whether Schrödinger's equation contains frogs, musical composers, or morality—or whether it does not. We cannot say whether something beyond it like God is needed, or not. And so we can all hold strong opinions either way.

Index

- Aberration, I-27-7, I-34-10
Absolute zero, I-1-5
Absorption, I-31-8 ff
Absorption coefficient, II-32-8
Acceleration, I-8-8 ff
 components of, I-9-3
 of gravity, I-9-4
Accelerator guide field, II-29-4 ff
Activation energy, I-42-7
Active circuit element, II-22-5
Adams, J. C., I-7-5
Adiabatic compression, I-39-5
Adiabatic demagnetization, II-35-9 f
Adiabatic expansion, I-44-5
Affective future, I-17-4
Aharanov, II-15-12
Air trough, I-10-5
Algebra, I-22-1 ff
Alternating-current circuits, II-22-1 ff
Alternating-current generator,
 II-17-6 ff
Alnico V, II-37-10
Amber, II-1-10
Ammeter, II-16-1
Ampere, A., II-13-3
Ampere's law, II-13-4
Amperian current, II-36-2
Amplitudes of oscillation, I-21-3
Amplitude modulation, I-48-3
Analog computer, I-25-8
Anderson, C. D., I-52-10
Angle, of incidence, I-26-3
 of precession, II-34-4
 of reflection, I-26-3
Angstrom (unit), I-1-3
Angular frequency, I-21-3, I-29-2
Angular momentum, I-7-7, I-18-5 f,
 I-20-1
 conservation of, I-4-7, I-18-6 ff,
 I-20-5
 of rigid body, I-20-8
Anomalous refraction, I-33-9 f
Antiferromagnetic material, II-37-11
Antimatter, I-52-10 f
Antiparticle, I-2-8
Aristotle, I-5-1
Atom, I-1-2
 metastable, I-42-10
 Rutherford-Bohr model, II-5-3
 stability of, II-5-3
 Thompson model, II-5-3
Atomic clock, I-5-5
Atomic currents, II-13-5 f
Atomic hypothesis, I-1-2
Atomic orbits, II-1-8
Atomic particles, I-2-9 f
Atomic polarizability, II-32-2
Atomic processes, I-1-5 f
Attenuation, I-31-8
Avogadro, A., I-39-2
Avogadro's number, I-41-10
Axial vector, I-52-6 f
Barkhausen effect, II-37-9
Battery, II-22-6
Becquerel, A. H., I-28-3
Bell, A. G., II-16-3
Benzene molecule, III-49-10 ff
Bernoulli's theorem, II-40-6 ff
Bessel function, II-23-6
Betatron, II-17-5
Biot-Savart law, II-14-10
Birefringence, I-33-3 ff
Blackbody radiation, I-41-5 f
Boehm, I-52-10
Bohm, II-7-7, II-15-12
Bohr, N., I-42-9, II-5-3
Bohr magneton, II-34-12
Bohr radius, I-38-6
Boltzmann, L., I-41-2
Boltzmann's law, I-40-2 f
Bopp, II-28-8
Born, M., I-37-1, I-38-9, II-28-7
Boundary layer, II-41-9
Boundary-value problems, II-7-1
Boyle's law, I-40-8
"Boys" camera, II-9-10
Bragg, L., II-30-9
Bragg-Nye crystal model, II-30-9 ff
Breaking-drop theory, II-9-9
Bremsstrahlung, I-34-6 f
Brewster's angle, I-33-6
Briggs, H., I-22-6
Brown, R., I-41-1
Brownian motion, I-1-8, I-6-5,
 I-41-1 ff
Brush discharge, II-9-9
Bulk modulus, II-38-3
Calculus, differential, I-8-4, II-2-1 ff
 integral, II-3-1 ff
 of variations, II-19-3
Cantilever beam, II-38-10
Capacitance, I-23-5
 mutual, II-22-17
Capacitor, I-14-9, I-23-5, II-22-3 ff,
 II-23-2 ff
 parallel-plate, I-14-9, II-6-11 ff,
 II-8-3
Capacity, II-6-12
 of a condenser, II-8-2
Capillary action, I-51-8
Carnot, S., I-4-2, I-44-2 ff
Carnot cycle, I-44-5 f, I-45-2
Carrier signal, I-48-3
Catalyst, I-42-8
Cavendish, H., I-7-9
Cavendish's experiment, I-7-9
Cavity resonator, II-23-1 ff
Center of mass, I-18-1 f, I-19-1 ff
Centrifugal force, I-7-5, I-12-11
Čerenkov, P. A., I-51-2
Čerenkov radiation, I-51-2
Charge, conservation of, I-4-7,
 II-13-1 f
 on electron, I-12-7
 line of, II-5-3 f
 motion of, II-29-1 ff
 sheet of, II-5-4
 sphere of, II-5-4 f
Charge density, II-5-4
Charge separation, II-9-7 ff
Charged conductor, II-8-2 ff
Chemical energy, I-4-2
Chemical kinetics, I-42-7 f
Chemical reaction, I-1-6 ff
Chromaticity, I-35-6 f
Circuits, alternating-current, II-22-1 ff
 equivalent, II-22-10 f
Circuit elements, II-23-1 f
 active, II-22-5
 passive, II-22-5
Circular motion, I-21-4
Circulation, II-1-5, II-3-8 ff
Classical electron radius, II-28-3
Clausius, R., I-44-2, I-44-3
Clausius-Clapeyron equation, I-45-6 ff
Clausius-Mossotti equation, II-11-6 f,
 II-32-7
Cleavage plane, II-30-1
Coaxial line, II-24-1
Coefficient, absorption, II-32-8
 of coupling, II-17-14
 of friction, I-12-4
 gravitational, I-7-9
 of viscosity, II-41-2
Collision, I-16-6
 elastic, I-10-7
Colloidal particles, II-7-8 ff
Color vision, I-35-1 ff
 physiochemistry of, I-35-9 f
Complex impedance, I-23-7
Complex numbers, I-22-7 ff, I-23-1 ff
Complex variable, II-7-2 ff
Compound eye, I-36-6 ff
Compression, adiabatic, I-39-5
 isothermal, I-44-5
Condenser, parallel-plate, I-14-9,
 II-6-11 ff, II-8-3
Conductivity, II-32-10
 thermal, II-2-8, II-12-2
Conductor, II-1-2
Cones, I-35-1
Conservation, of angular momentum,
 I-4-7, I-18-6 ff, I-20-5
 of charge, I-4-7, II-13-1 f
 of energy, I-3-2, I-4-1 ff, II-27-1 f
 of linear momentum, I-4-7,
 I-10-1 ff

Contraction hypothesis, I-15-3
 Copernicus, I-7-1
 Coriolis force, I-19-8 f
 Cornea, I-35-1
 Cosmic rays, II-9-2
 Couette flow, II-41-10 ff
 Coulomb's law, I-28-2, II-4-2 ff,
 II-5-6
 Coupling, coefficient of, II-17-14
 Covalent bond, II-30-2
 Cross product, II-2-8, II-31-8
 Cross section for scattering, I-32-7
 Crystal, II-30-1 ff
 geometry of, II-30-1 f
 Crystal diffraction, I-38-4 f
 Crystal lattice, II-30-3 f
 Cubic cell, II-30-7
 Curie law, II-11-5
 Curie temperature, II-36-13
 Curie-Weiss law, II-11-9
 Curl operator, II-2-8, II-3-1
 Current, Amperian, II-36-2
 atomic, II-13-5 f
 eddy, II-16-6
 electric, II-13-1 f
 induced, II-16-1 ff
 Current density, II-13-1
 Cutoff frequency, II-22-14

 D'Alembertian, II-25-8
 Debye length, II-7-9
 Dedekind, R , I-22-4
 Degrees of freedom, I-25-2, I-39-12
 Demagnetization, adiabatic, II-35-9 f
 Density, I-1-4
 Derivative, I-8-5 ff
 partial, I-14-9
 Diamagnetism, II-34-1 ff
 Dicke, R H , I-7-11
 Dielectric, II-10-1 ff, II-11-1 ff
 Dielectric constant, II-10-1 f
 Differential calculus, I-8-4, II-2-1 f
 Diffraction, I-30-1 ff
 by screen, I-31-10 f
 Diffraction grating, I-29-5, I-30-3 ff
 Diffusion, I-43-1 ff
 of neutrons, II-12-6 ff
 Dipole, II-21-5 ff
 electric, II-6-2 ff
 magnetic, II-14-7 f
 Dipole moment, I-12-6, II-6-7
 Dipole potential, II-6-4 ff
 Dipole radiator, I-28-5 f, I-29-3 ff
 Dirac, P , I-52-10, II-2-1, II-28-7
 Dirac equation, I-20-6
 Dislocation, II-30-8, II-30-9
 Dispersion, I-31-6 ff
 Distance, I-5-5 ff
 Distance measurement, color brightness,
 I-5-6
 triangulation, I-5-6
 Divergence, II-25-7
 Divergence operator, II-2-7, II-3-1
 Domain, II-37-6
 Doppler effect, I-17-8, I-23-9,
 I-34-7 f, I-38-6
 Dot product, II-2-4, II-25-3
 Double stars, I-7-6
 Drag coefficient, II-41-7
 "Dry" water, II-40-1 ff

 INDEX 2

Dynamics, I-7-2 f, I-9-1 ff
 relativistic, I-15-9 f

 Eddy current, II-16-6
 Efficiency of ideal engine, I-44-7 f
 Einstein, A., I-2-6, I-7-11, I-12-12,
 I-15-1, I-16-1, I-41-8, I-42-8,
 I-42-9
 Elastic collision, I-10-7
 Elastic constants, II-39-6, II-39-10 f
 Elastic energy, I-4-2, I-4-6
 Elastic materials, II-39-1 ff
 Elastica, II-38-12
 Elasticity, II-38-1 ff
 Elasticity tensor, II-39-4 ff
 Electret, II-11-8
 Electric charge density, II-2-8, II-4-3
 Electric current, II-13-1 f
 in the atmosphere, II-9-2 f
 Electric current density, II-2-8
 Electric dipole, II-6-2 ff
 Electric field, I-2-4, I-12-7 f, II-1-2,
 II-1-3, II-6-1 ff, II-7-1 ff
 relativity of, II-13-6 ff
 Electric flux, II-1-4
 Electric potential, II-4-4
 Electric susceptibility, II-10-4
 Electrical energy, I-4-2, II-15-3 ff
 Electrical forces, II-1-1 ff, II-13-1
 Electrodynamics, II-1-3
 relativistic notation, II-25-1 ff
 Electromagnet, II-36-9 ff
 Electromagnetic energy, I-29-2
 Electromagnetic field, I-2-2, I-2-5,
 I-10-9
 Electromagnetic mass, II-28-3 f
 Electromagnetic radiation, I-26-1,
 I-28-1 ff
 Electromagnetic waves, II-21-1 f
 cosmic rays, I-2-5
 gamma rays, I-2-5
 infrared, I-2-5, I-23-8, I-26-1
 light, I-2-5
 ultraviolet, I-2-5, I-26-1
 x-rays, I-2-5, I-26-1
 Electromagnetism, II-1-1 ff
 laws of, II-1-5 ff
 Electromotive force, II-16-2
 Electron, I-2-4, I-37-1, I-37-4 ff
 charge on, I-12-7
 radius of, classical, I-32-4
 Electron cloud, I-6-11
 Electron microscope, II-29-3 f
 Electron-ray tube, I-12-9
 Electron volt (unit), I-34-4
 Electronic polarization, II-11-1 ff
 Electrostatic energy, II-8-1 ff
 of charges, II-8-1 f
 of ionic crystal, II-8-4 ff
 in nuclei, II-8-6 ff
 of a point charge, II-8-12
 Electrostatic equations, II-10-6 f
 Electrostatic field, II-5-1 ff, II-7-1 f
 energy in, II-8-9 ff
 of a grid, II-7-10 f
 Electrostatic lens, II-29-2 f
 Electrostatic potential, equations of,
 II-6-1
 Electrostatics, II-4-1 ff, II-5-1
 Ellipse, I-7-1

Emissivity, II-6-14
 Energy, II-22-11 f
 chemical, I-4-2
 of a condenser, II-8-2 ff
 conservation of, I-3-2, I-4-1 ff,
 II-27-1 f
 elastic, I-4-2, I-4-6
 electrical, I-4-2, II-15-3 ff
 electromagnetic, I-29-2
 electrostatic, II-8-1 ff
 in electrostatic field, II-8-9 ff
 gravitational, I-4-2 ff
 heat, I-4-2, I-4-6, I-10-7, I-10-8
 kinetic, I-1-7, I-4-2, I-4-5 f,
 I-39-4
 magnetic, II-17-12 ff
 mass, I-4-2, I-4-7
 mechanical, II-15-3 ff
 nuclear, I-4-2
 potential, I-4-4, I-13-1 ff, I-14-1 ff
 radiant, I-4-2
 relativistic, I-16-1 ff
 Energy density, II-27-2
 Energy flux, II-27-2
 Energy levels, I-38-7 f
 Energy theorem, I-50-7 f
 Enthalpy, I-45-5
 Entropy, I-44-10 ff, I-46-7 ff
 Eotvös, L., I-7-11
 Equilibrium, I-1-6
 Equipotential surfaces, II-4-11 f
 Equivalent circuits, II-22-10 f
 Euclid, I-5-6
 Euclidean geometry, I-12-3
 Euler force, II-38-11
 Evaporation, I-1-5 f
 of a liquid, I-40-3 f, I-42-1 ff
 Exchange force, II-37-2
 Excited state, II-8-7
 Expansion, adiabatic, I-44-5
 isothermal, I-44-5
 Exponential atmosphere, I-40-1 f
 Eye, compound, I-36-6 ff
 human, I-35-1 f, I-36-3 ff

 Farad (unit), I-25-7, II-6-13
 Faraday, M , II-10-1
 Faraday's law of induction, II-17-2
 Fermat, P , I-26-3
 Fermi (unit), I-5-10
 Fermi, E , I-5-10
 Ferrite, II-37-12
 Ferroelectricity, II-11-8 ff
 Ferromagnetic insulators, II-37-12
 Ferromagnetism, II-34-1 f, II-36-1 ff,
 II-37-1 ff
 Feynman, R., II-28-8
 Fields, I-2-2, I-2-4, I-2-5, I-10-9,
 I-12-7 ff, I-13-8 f, I-14-7 ff
 in a cavity, II-5-8 f
 of a charged conductor, II-6-8
 of a conductor, II-5-7 f
 electric, I-2-4, I-12-7 f, II-1-2,
 II-1-3, II-6-1 ff, II-7-1 ff
 electrostatic, II-5-1 ff, II-7-1 f
 magnetic, II-1-2, II-1-3, II-13-1,
 II-14-1 ff
 magnetizing, II-36-7
 scalar, II-2-2 ff
 superposition of, I-12-9

- two-dimensional, II-7-2 ff
- vector, II-1-4 f, II-2-1 ff
- Field energy, II-27-1 ff
 - of a point charge, II-28-1 f
- Field index, II-29-5
- Field-ion microscope, II-6-14
- Field lines, II-4-11
- Field momentum, II-27-9 ff
 - of a moving charge, II-28-2 f
- Field strength, II-1-4
- Filter, II-22-14 ff
- Flow, fluid, II-12-8 ff
 - irrotationa^l, II-40-5
 - viscous, II-41-4 f
- Fluid flow, II-12-8 ff
- Flux, II-4-7 ff
 - electric, II-1-4
 - of a vector field, II-3-2 ff
- Flux rule, II-17-1 ff
- Focal length, I-27-1 ff
- Focus, I-26-5
- Force, centrifugal, I-7-5, I-12-11
 - components of, I-9-3
 - conservative, I-14-3 ff
 - Coriolis, I-19-8 f
 - electrical, I-2-3 ff, II-1-1 ff, I-13-1
 - electromotive, II-16-2
 - gravitational, I-2-3
 - Lorentz, II-13-1, II-15-14
 - magnetic, II-1-2, II-13-1
 - molecular, I-1-3, I-12-6 f
 - moment of, I-18-5
 - nonconservative, I-14-6 f
 - nuclear, I-12-12
 - pseudo, I-12-10 ff
- Fourier, J., I-50-2 f
- Fourier analysis, I-50-2 ff
- Fourier theorem, II-7-11
- Fourier transform, I-25-4
- Four-vectors, I-15-8 f, I-17-5 ff, II-25-1 ff
- Fovea, I-35-1
- Frank, I., I-51-2
- Franklin, B., II-5-6
- Frequency, angular, I-21-3, I-29-2
 - of oscillation, I-2-5
 - plasma, II-7-6, II-32-12
- Fresnel's reflection formulas, I-33-8
- Friction, I-10-5, I-12-3 ff
 - coefficient of, I-12-4

- Galileo, I-5-1, I-7-2, I-9-1, I-52-3
- Galilean relativity, I-10-3
- Galilean transformation, I-12-11
- Galvanometer, II-1-8, II-16-1
- Garnet, II-37-12
- Gauss (unit), I-34-4
- Gauss, K., II-16-2
- Gauss' law, II-4-9 f, II-5-1 ff
- Gauss' theorem, II-3-5
- Gaussian surface, II-10-1
- Geiger, II-5-3
- Gell-Mann, M., I-2-9
- Generator, alternating-current, II-17-6 ff
 - electric, II-16-1 ff, II-22-5 ff
 - van de Graaff, II-5-9, II-8-7
- Geometrical optics, I-26-1, I-27-1 f
- Gerlach, II-35-3
- Gradient operator, II-2-4, II-3-1

- Gravitation, I-2-3, I-7-1 ff, I-12-2
- Gravitational acceleration, I-9-4
- Gravitational coefficient, I-7-9
- Gravitational energy, I-4-2 ff
- Gravitational field, I-12-8 ff, I-13-8 f
- Gravity, I-13-3 ff
 - acceleration of, I-9-4
- Green's function, I-25-4
- Ground state, II-8-7
- Gyroscope, I-20-5 ff

- Hamilton's first principal function, II-19-8
- Harmonic motion, I-21-4, I-23-1 ff
- Harmonic oscillator, I-10-1, I-21-1 ff
 - forced, I-21-5 f, I-23-3 ff
- Harmonics, I-50-1 ff
- Heat, I-1-3, I-13-3
- Heat conduction, II-3-6 ff
- Heat diffusion equation, II-3-8
- Heat energy, I-4-2, I-4-6, I-10-7, I-10-8
- Heat engines, I-44-1 ff
- Heat flow, II-2-8 f, II-12-2 ff
- Heisenberg, W., I-6-10, I-37-1, I-37-9, I-37-11, I-37-12, I-38-9
- Helmholtz, H., I-35-7, II-40-10
- Henry (unit), I-25-7
- Hess, II-9-2
- Hexagonal cell, II-30-7
- High-voltage breakdown, II-6-13 f
- Hooke's law, I-12-6, II-38-1 f
- Huygens, C., I-15-2, I-26-2
- Hydrodynamics, II-40-2 ff
- Hydrostatics, II-40-1 ff
- Hypocycloid, I-34-3
- Hysteresis curve, II-37-5 ff
- Hysteresis loop, II-36-8

- Ideal gas law, I-39-10 ff
- Illumination, II-12-10 ff
- Image charge, II-6-9
- Impedance, I-25-8 f, II-22-1 ff
 - complex, I-23-7
- Incidence, angle of, I-26-3
- Inclined plane, I-4-4
- Index of refraction, I-31-1 ff
- Induced currents, II-16-1 ff
- Inductance, I-23-6, II-16-4 f, II-17-12 ff, II-22-2 f
 - mutual, II-17-9 ff, II-22-16
 - self-, II-16-4, II-17-11 f
- Induction, laws of, II-17-1 ff
- Inductor, I-23-6
- Inertia, I-2-3, I-7-11
 - moment of, I-18-7, I-19-5 ff
 - principle of, I-9-1
- Infeld, II-28-7
- Infrared radiation, I-23-8, I-26-1
- Integral, I-8-7 f
- Integral calculus, II-3-1 ff
- Insulator, II-1-2, II-10-1
- Interference, I-28-6, I-29-1 ff
- Interfering waves, I-37-4
- Interferometer, I-15-5
- Internal reflection, II-33-12
- Ion, I-1-6
- Ionic bond, II-30-2
- Ionic conductivity, I-43-6 f
- Ionic polarizability, II-11-8

- Ionization energy, I-42-5
- Ionosphere, II-7-5, II-9-3
- Irrotational flow, II-40-5
- Isotherm, II-2-3
- Isothermal atmosphere, I-40-2
- Isothermal compression, I-44-5
- Isothermal expansion, I-44-5
- Isothermal surfaces, II-2-3
- Isotopes, I-3-4 ff

- Jeans, J., I-40-9, I-41-6 f, II-2-6
- Johnson noise, I-41-2, I-41-8
- Joule (unit), I-13-3
- Joule heating, I-24-2

- Kármán vortex street, II-41-9
- Kepler, J., I-7-1
- Kepler's laws, I-7-1 f, I-9-1, I-18-6
- Kerr cell, I-33-5
- Kilocalorie (unit), II-8-5
- Kinetic energy, I-1-7, I-4-2, I-4-5 f, I-39-4
 - rotational, I-19-7 ff
- Kinetic theory, I-42-1 ff
 - of gases, I-39-1 ff
- Kirchhoff's laws, I-25-9, II-22-7 ff
- Kronecker delta, II-31-6

- Lamb, II-5-6
- Lamé elastic constants, II-39-6
- Landé g-factor, II-34-4
- Laplace, P., I-47-7
- Laplace equation, II-6-1, II-7-1
- Laplacian operator, II-2-10
- Larmor frequency, II-34-7
- Larmor's theorem, II-34-6 f
- Laser, I-32-6, I-42-10
- Laughton, II-5-6
- Laws, of electromagnetism, II-1-5 ff
 - of induction, II-17-1 ff
- Least action, principle of, II-19-1 ff
- Least time, principle of, I-26-3 ff, I-26-8
- Leibnitz, G. W., I-8-4
- Lens formula, I-27-6
- Lenz's rule, II-16-4, II-34-2
- Leverrier, U., I-7-5
- Liénard-Wiechert potentials, II-21-11
- Light, II-21-1 f
 - momentum of, I-34-10 f
 - polarized, I-32-9
 - scattering of, I-32-5 ff
 - speed of, I-15-1, II-18-8 f
- Light waves, I-48-1
- Lightning, II-9-10 f
- Line of charge, II-5-3 f
- Line integral, II-3-1
- Linear momentum, conservation of, I-4-7, I-10-1 ff
- Linear systems, I-25-1 ff
- Lodestone, II-1-10
- Logarithms, I-22-4
- Lorentz, H. A., I-15-3
- Lorentz condition, II-25-9
- Lorentz contraction, I-15-7
- Lorentz force, II-13-1, II-15-14
- Lorentz formula, II-21-12 f
- Lorentz gauge, II-18-11
- Lorentz transformation, I-15-3, I-17-1, I-34-8, -52-2, II-25-1
 - of fields, II-26-1 ff

- McCullough, II-1-9
 Mach number, II-41-6
 Magnetic dipole, II-14-7 f
 Magnetic dipole moment, II-14-8
 Magnetic energy, II-17-12 ff
 Magnetic field I-12-9 f, II-1-2, II-1-3, II-13-1, II-14-1 ff
 relativity of, II-13-6 ff
 of steady currents, II-13-3 f
 Magnetic force, II-1-2, II-13-1
 on a current, II-13-2 f
 Magnetic induction, I-12-10
 Magnetic lens, II-29-3
 Magnetic materials, II-37-1 ff
 Magnetic moments, II-34-3 f
 Magnetic resonance, II-35-1 ff
 Magnetic susceptibility, II-35-7
 Magnetism, I-2-4, II-34-1 ff
 Magnetization currents, II-36-1 ff
 Magnetizing fields, II-36-7
 Magnetostatics, II-4-1, II-13-1 ff
 Magnetostriction, II-37-6
 Magnification, I-27-5
 Marsden, II-5-3
 Maser, I-42-10
 Mass, I-9-1, I-15-1
 center of, I-18-1 f, I-19-1 ff
 electromagnetic, II-28-3 f
 relativistic, I-16-6 ff
 Mass energy, I-4-2, I-4-7
 Mass-energy equivalence, I-15-10 f
 Maxwell, J. C., I-6-1, I-6-9, I-28-1, I-40-8, I-41-7, I-46-5, II-1-8, II-1-11, II-5-6, II-18-1 ff
 Maxwell's equations, I-15-2, I-25-3, I-47-7, II-2-1, II-2-8, II-4-1, II-6-1, II-18-1 ff, II-32-3 ff
 currents and charges, II-21-1 ff
 free space, II-20-1 ff
 Mayer, J. R., I-3-2
 Mean free path, I-43-3 f
 Mean square distance, I-6-5, I-41-9
 Mechanical energy, II-15-3 ff
 Mendeléev, I-2-9
 Metastable atom, I-42-10
 Meter (unit), I-5-10
 Mev (unit), I-2-9
 Michelson-Morley experiment, I-15-3 ff
 Miller, W. C., I-35-2
 Minkowski, I-17-8
 Minkowski space, II-31-12
 Modes, I-49-1 ff
 Mossbauer, R., I-23-9
 Mole (unit), I-39-10
 Molecular attraction, I-1-3, I-12-6 f
 Molecular crystal, II-30-2
 Molecular diffusion, I-43-7 ff
 Molecular dipole, II-11-1
 Molecular motion, I-41-1
 Molecule, I-1-3
 Moment, dipole, I-12-6
 of force, I-18-5
 of inertia, I-18-7, I-19-5 ff
 Momentum, I-9-1 f, I-38-2 ff
 angular, I-7-7, I-18-5 ff, I-20-1, I-20-5
 of light, I-34-10 f
 linear, I-4-7, I-10-1 ff
 relativistic, I-10-8 f, I-16-1 ff
 Momentum spectrometer, II-29-1
 Momentum spectrum, II-29-2
 Monatomic gas, I-39-5
 Monoclinic cell, II-30-7
 Motion, I-5-1, I-8-1 ff
 of charge, II-29-1 ff
 circular, I-21-4
 constrained, I-14-3
 harmonic, I-21-4, I-23-1 ff
 parabolic, I-8-10
 planetary, I-7-1 ff, I-9-6 f, I-13-5
 Motors, electric, II-16-1 ff
 Moving charge, field momentum of, II-28-2 f
 Music, I-50-1
 Mutual inductance, II-17-9 ff, II-22-16

 Nernst heat theorem, I-44-11
 Neuman, J. von, II-12-9
 Neutrons, I-2-4
 diffusion of, II-12-6 ff
 Neutron diffusion equation, II-12-7
 Newton, I., I-8-4, I-15-1, I-37-1, II-4-10
 Newton-meter (unit), I-13-3
 Newton's laws, I-2-6, I-7-3 ff, I-7-11, I-9-1 ff, I-10-1 ff, I-11-7 f, I-12-1, I-39-2, I-41-1, I-46-1, II-7-5
 Nishijima, I-2-9
 Nodes, I-49-2
 Noise, I-50-1
 Nonpolar molecule, II-11-1
 Nuclear cross section, I-5-9
 Nuclear energy, I-4-2
 Nuclear forces, I-12-12
 Nuclear *g*-factor, II-34-4
 Nuclear interactions, II-8-7
 Nuclear magnetic resonance, II-35-10 ff
 Nucleus, I-2-4, I-2-8 ff
 Numerical analysis, I-9-6
 Nutation, I-20-7
 Nye, J. F., II-30-9

 Oersted (unit), II-36-6
 Ohm (unit), I-25-7
 Ohm's law, I-25-7, I-43-7
 Operator, curl, II-2-8, II-3-1
 divergence, II-2-7, II-3-1
 gradient, II-2-4, II-3-1
 Laplacian, II-2-10
 vector, II-2-6
 Optic axis, I-33-3
 Optic nerve, I-35-2
 Optics, I-26-1 ff
 geometrical, I-26-1, I-27-1 ff
 Orbital motion, II-34-3
 Orientation polarization, II-11-3 ff
 Oriented magnetic moment, II-35-4
 Orthorhombic cell, II-30-7
 Oscillation, amplitude, of, I-21-3
 damped, I-24-3 f
 frequency of, I-2-5
 period of, I-21-3
 periodic, I-9-4
 phase of, I-21-3

 Oscillator, I-5-2
 harmonic, I-10-1, I-21-1, I-21-5 f, I-23-3 ff

 Pappus, theorem of, I-19-4
 Parabolic antenna, I-30-6 f
 Parabolic motion, I-8-10
 Parallel-axis theorem, I-19-6
 Parallel-plate capacitor, I-14-9, II-6-11 ff, II-8-3
 Paramagnetism, II-34-1 ff, II-35-1 ff
 Paraxial rays, I-27-2
 Partial derivative, I-14-9
 Particles, "strange", II-8-7
 Permalloy, II-37-11
 Permeability, II-36-9
 Pascal's triangle, I-6-4
 Passive circuit element, II-22-5
 Pendulum, I-49-6 f
 Pendulum clock, I-5-2
 Period of oscillation, I-21-3
 Periodic time, I-5-1 f
 Perpetual motion, I-46-2
 Phase of oscillation, I-21-3
 Phase shift, I-21-3
 Phase velocity, I-48-6
 Photon, I-2-7, I-26-1, I-37-8
 Physicochemistry of color vision, I-35-9 f
 Piezoelectricity, II-11-8
 Pines, II-7-7
 Planck, M., I-41-6, I-42-8, I-42-9
 Planck's constant, I-5-10, I-6-10, I-17-8, I-37-11
 Plane lattice, II-30-5
 Plane waves, II-21-1 ff
 Planetary motion, I-7-1 ff, I-9-6 f, I-13-5
 Plasma frequency, II-7-6, II-32-12
 Plasma oscillations, II-7-5 ff
 Plimpton, II-5-6
 Poincaré, H., I-15-3, I-15-5, I-16-1
 Poincaré stress, II-28-4
 Point charge, electrostatic energy of, II-8-12
 field energy of, II-28-1 f
 Poisson's ratio, II-38-2
 Polar molecule, II-11-1, II-11-3 ff
 Polarization, I-33-1 ff, II-32-1 ff
 Polarization charges, II-10-3 ff
 Polarization vector, II-10-2 f
 Polarized light, I-32-9
 Potential energy, I-4-4, I-13-1 ff, I-14-1 ff
 Potential gradient of the atmosphere, II-9-2 f
 Power, I-13-2
 Poynting, J., II-27-3
 Precession, angle of, II-34-4
 of atomic magnets, II-34-4 f
 Pressure, I-1-3
 Priestly, J., II-5-6
 Principle of least action, II-19-1 ff
 Principle of superposition, II-1-3, II-4-2
 Probability, I-6-1 ff
 Probability density, I-6-8 f
 Probability distribution, I-6-7 ff
 Propagation factor, II-22-14
 Proton, I-2-4

Proton spin, II-8-7
 Pseudo force, I-12-10 ff
 Ptolemy, I-26-2
 Purkinje effect, I-35-2
 Pyroelectricity, II-11-8

 Quadrupole lens, II-7-4, II-29-6
 Quadrupole potential, II-6-8
 Quantized magnetic states, II-35-1 ff
 Quantum electrodynamics, I-2-7, I-28-3
 Quantum mechanics, I-2-2, I-2-6 ff, I-6-10, I-10-9, I-37-1 ff, I-38-1 ff

 Rabi, I I., II-35-4
 Rabi molecular-beam method, II-35-4 ff
 Radiant energy, I-4-2
 Radiation, infrared, I-23-8, I-26-1 relativistic effects, I-35-1 ff synchrotron, I-34-3 ff, I-34-6 ultraviolet, I-26-1
 Radiation damping, I-32-3 f
 Radiation resistance, I-32-1 ff
 Radioactive clock I-5-3 ff
 Radius of electron, I-32-4
 Ramsey, N., I-5-5
 Random walk, I-6-5 ff, I-41-8 ff
 Ratchet and pawl machine, I-46-1 ff
 Rayleigh's criterion, I-30-6
 Rayleigh's law, I-41-6
 Rayleigh waves, II-38-8
 Reactance, II-22-11
 Reciprocity principle, I-30-7
 Rectification, I-50-9
 Rectifier, II-22-15
 Reflected waves, II-33-7 ff
 Reflection, I-26-2 f angle of, I-26-3 internal, II-33-12 of light, II-33-1 ff
 Refraction, I-26-2 f anomalous, I-33-9 f index of, I-31-1 ff of light, II-33-1 ff
 Refractive index, II-32-1 ff
 Relative permeability, II-36-9
 Relativistic dynamics, I-15-9 f
 Relativistic energy, I-16-1 ff
 Relativistic mass, I-16-1 ff
 Relativistic momentum, I-10-8 f, I-16-1 ff
 Relativity, of electric field, II-13-6 ff Galilean, I-10-3 of magnetic field, II-13-6 ff special theory of, I-15-1 ff theory of, I-7-11, I-17-1
 Resistance, I-23-5
 Resistor, I-23-5, II-22-4
 Resonant cavity, II-23-6 ff
 Resonant circuits, II-23-10 f
 Resonant mode, II-23-10
 Resonator, cavity, II-23-1 ff
 Resolving power, I-27-7 f, I-30-5 f
 Resonance, I-23-1 ff electrical, I-23-5 ff in nature, I-23-7 ff
 Resonance interaction, I-2-9
 Retarded time, I-28-2
 Retherford, II-5-6
 Retina, I-35-1
 Reynolds' number, II-41-5 f
 Rigid body, I-18-1 angular momentum of, I-20-8 rotation of, I-18-2 ff
 Ritz combination principle, I-38-8
 Rods, I-35-1, I-36-6
 Roemer, O., I-7-5
 Root-mean-square distance, I-6-6
 Rotation, of axes, I-11-3 f plane, I-18-1 of a rigid body, I-18-2 ff in space, I-20-1 ff in two dimensions, I-18-1 ff
 Rushton, I-35-9
 Rutherford, II-5-3
 Rutherford-Bohr atomic model, II-5-3
 Rydberg (unit), I-38-6

 Scalar, I-11-5
 Scalar field, II-2-2 ff
 Scalar product, II-25-3 ff
 Scattering of light, I-32-5 ff
 Schrodinger, E., I-35-6, I-37-1, I-38-9
 Schrödinger equation, II-15-12
 Scientific method, I-2-1 f
 Screw dislocation, II-30-9
 Screw jack, I-4-5
 Second (unit), I-5-5
 Seismograph, I-51-5
 Self-inductance, II-16-4, II-17-11 f
 Shannon, C , I-44-2
 Shear modulus, II-38-5
 Shear wave, I-51-4, II-38-8
 Sheet of charge, II-5-4
 Side bands, I-48-4 f
 Simultaneity, I-15-7 f
 Sinusoidal waves, I-29-2 f
 Skin depth, II-32-11
 Slip dislocation, II-30-9
 Smoluchowski, I-41-8
 Smooth muscle, I-14-2
 Snell, W., I-26-3
 Snell's law, I-26-3, I-31-2, II-33-1
 Solenoid, II-13-5
 Solid-state physics, II-8-6
 Sound, I-2-3, I-47-1 ff, I-50-1 speed of, I-47-7 f
 Space, I-8-2
 Space-time, I-2-6, I-17-1 ff, II-26-12
 Special theory of relativity, I-15-1 ff
 Specific heat, I-40-7 f, I-45-2, II-37-4
 Speed, I-8-2 ff, I-9-2 of light, I-15-1, II-18-8 f of sound, I-47-7 f
 Sphere of charge, II-5-4 f
 Spherical waves, II-20-12 ff, II-21-2 ff
 Spinel, II-37-12
 Spin orbit, II-8-7
 Spontaneous emission, I-42-9
 Standard deviation, I-6-9
 Statics, II-4-1 f
 Statistical fluctuations, I-6-3 ff
 Statistical mechanics, I-3-1, I-40-1 ff
 Steady flow, II-40-6 ff
 Step leader, II-9-10
 Stern, II-35-3
 Stern-Gerlach experiment, II-35-3 ff
 Stevinus, S., I-4-5
 Stokes' theorem, II-3-10
 Strain, II-38-2
 Strain tensor, II-31-11, II-39-1 ff
 "Strange" particles, II-8-7
 "Strangeness" number, I-2-9
 Streamlines, II-40-6
 Stress, II-38-2
 Stress tensor, II-31-9 ff
 Striated muscle, I-14-2
 Supermalloy, II-36-9
 Superposition, II-13-11 f of fields, I-12-9 principle of, I-25-2 ff, II-1-3, II-4-2
 Surface, equipotential, II-4-11 f gaussian, II-10-1 isothermal, II-2-3
 Surface tension, II-12-5
 Symmetry, I-1-4, I-11-1 ff of physical laws, I-16-3, I-52-1 ff
 Synchrotron, I-2-5, I-15-9, I-34-3 ff, I-34-6, II-17-5

 Tamm, I., I-51-2
 Taylor expansion, II-6-7
 Temperature, I-39-6 ff
 Tensor, II-26-7, II-31-1 ff
 Tensor field, II-31-11
 Tetragonal cell, II-30-7
 Thermal conductivity, II-2-8, II-12-2 of a gas, I-43-9 f
 Thermal equilibrium, I-41-3 ff
 Thermal ionization, I-42-5 ff
 Thermodynamics, I-39-2, I-45-1 ff, II-37-4 f laws of, I-44-1 ff
 Thompson, II-5-3
 Thompson atomic model, II-5-3
 Thompson scattering cross section, I-32-8
 Three-body problem, I-10-1
 Three-dimensional waves, II-20-8 f
 Thunderstorms, II-9-5 ff
 Tides, I-7-4 f
 Time, I-2-3, I-5-1 ff, I-8-1, I-8-2 retarded, I-28-2 standard of, I-5-5 transformation of, I-15-5 ff
 Torque, I-18-4, I-20-1 ff
 Torsion bar, II-38-5 ff
 Total internal reflection, II-33-12 f
 Transformation, Fourier, I-25-4 Galilean, I-12-11 linear, I-11-6 Lorentz, I-15-3, I-17-1, I-34-8, I-52-2, II-25-1, II-26-1 ff of time, I-15-5 ff of velocity, I-16-4 ff
 Transformer, II-16-4 f
 Transient, I-24-1 ff electrical, I-24-5 f
 Transient response, I-21-6
 Translation of axes, I-11-1 ff
 Transmission line, II-24-1 ff
 Transmitted waves, II-33-7 ff
 Travelling field, II-18-5 ff
 Triclinic lattice, II-30-7
 Trigonal lattice, II-30-7
 Twin paradox, I-16-3 f

- Two-dimensional field, II-7-2 ff
- Tycho Brahe, I-7-1
- Ultraviolet radiation, I-26-1
- Uncertainty principle, I-2-6, I-6-10 f, I-37-9, I-37-11, I-38-8 f, II-5-3
- Unit cell, I-38-5
- Unit vector, I-11-10, II-2-3
- Unworldliness, II-25-10
- van de Graaff generator, II-5-9, II-8-7
- Vector, I-11-5 ff
- Vector algebra, I-11-6 f
- Vector analysis, I-11-5, I-52-2
- Vector field, II-1-4 f, II-2-1 ff
 - flux of, II-3-2 ff
- Vector integrals, II-3-1 f
- Vector operator, II-2-6
- Vector potential, II-4-1 ff, II-15-1 ff
- Vector product, I-20-4
- Velocity, I-8-3, I-9-2 f
 - components of, I-9-3
 - transformation of, I-16-4 ff
- Velocity potential, II-12-9
- Vinci, Leonardo da, I-36-2
- Virtual work, principle of, I-4-5
- Viscosity, II-41-1 ff
 - coefficient of, II-41-2
- Viscous flow, II-41-4 f
- Vision, I-36-1 ff
 - binocular, I-36-4
 - color, I-35-1 ff
- Visual cortex, I-36-4
- Visual purple, I-35-9
- Voltmeter, II-16-1
- Volume strain, II-38-3
- Volume stress, II-38-3
- von Neumann, J., II-40-3
- Vortex lines, II-40-10 ff
- Vorticity, II-40-5
- Wall energy, II-37-6
- Wapstra, I-52-10
- Watt (unit), I-13-3
- Wave, I-51-1 ff, II-20-1 ff
 - electromagnetic, II-21-1 f
 - light, I-48-1
 - plane, II-20-1 ff
 - reflected, II-33-7 ff
 - shear, I-51-4, II-38-8
 - sinusoidal, I-29-2 f
 - spherical, II-20-12 ff, II-21-2 ff
 - three-dimensional, II-20-8 f
 - transmitted, II-33-7 ff
- Wave equation, I-47-1 ff, II-18-9 ff
- Wavefront, I-47-3
- Waveguides, II-24-1 ff
- Wavelength, I-19-3, I-26-1
- Wave number, I-29-2
- Weber, II-16-2
- Weber (unit), II-13-1
- "Wet" water, II-41-1 ff
- Weyl, H., I-11-1
- Wheeler, II-28-8
- Wilson, C. T. R., II-9-9
- Work, I-13-1 ff, I-14-1 ff
- X-rays, I-2-5, I-26-1
- X-ray diffraction, II-30-1
- Young, I-35-7
- Young's modulus, II-38-2
- Yukawa, H., I-2-8, II-28-13
- Yukawa potential, II-28-13
- Yustova, I-35-8
- Zeno, I-8-3
- Zero, absolute, I-1-5
- Zero curl, II-3-10 f, II-4-1
- Zero divergence, II-3-10 f, II-4-1
- Zero mass, I-2-10

Feynman's Preface



These are the lectures in physics that I gave last year and the year before to the freshman and sophomore classes at Caltech. The lectures are, of course, not verbatim—they have been edited, sometimes extensively and sometimes less so. The lectures form only part of the complete course. The whole group of 180 students gathered in a big lecture room twice a week to hear these lectures and then they broke up into small groups of 15 to 20 students in recitation sections under the guidance of a teaching assistant. In addition, there was a laboratory session once a week.

The special problem we tried to get at with these lectures was to maintain the interest of the very enthusiastic and rather smart students coming out of the high schools and into Caltech. They have heard a lot about how interesting and exciting physics is—the theory of relativity, quantum mechanics, and other modern ideas. By the end of two years of our previous course, many would be very discouraged because there were really very few grand, new, modern ideas presented to them. They were made to study inclined planes, electrostatics, and so forth, and after two years it was quite stultifying. The problem was whether or not we could make a course which would save the more advanced and excited student by maintaining his enthusiasm.

The lectures here are not in any way meant to be a survey course, but are very serious. I thought to address them to the most intelligent in the class and to make sure, if possible, that even the most intelligent student was unable to completely encompass everything that was in the lectures—by putting in suggestions of applications of the ideas and concepts in various directions outside the main line of attack. For this reason, though, I tried very hard to make all the statements as accurate as possible, to point out in every case where the equations and ideas fitted into the body of physics, and how—when they learned more—things would be modified. I also felt that for such students it is important to indicate what it is that they should—if they are sufficiently clever—be able to understand by deduction from what has been said before, and what is being put in as something new. When new ideas came in, I would try either to deduce them if they were deducible, or to explain that it *was* a new idea which hadn't any basis in terms of things they had already learned and which was not supposed to be provable—but was just added in.

At the start of these lectures, I assumed that the students knew something when they came out of high school—such things as geometrical optics, simple chemistry ideas, and so on. I also didn't see that there was any reason to make the lectures

Foreword

A great triumph of twentieth-century physics, the theory of quantum mechanics, is now nearly 40 years old, yet we have generally been giving our students their introductory course in physics (for many students, their last) with hardly more than a casual allusion to this central part of our knowledge of the physical world. We should do better by them. These lectures are an attempt to present them with the basic and essential ideas of the quantum mechanics in a way that would, hopefully, be comprehensible. The approach you will find here is novel, particularly at the level of a sophomore course, and was considered very much an experiment. After seeing how easily some of the students take to it, however, I believe that the experiment was a success. There is, of course, room for improvement, and it will come with more experience in the classroom. What you will find here is a record of that first experiment.

In the two-year sequence of the Feynman Lectures on Physics which were given from September 1961 through May 1963 for the introductory physics course at Caltech, the concepts of quantum physics were brought in whenever they were necessary for an understanding of the phenomena being described. In addition, the last twelve lectures of the second year were given over to a more coherent introduction to some of the concepts of quantum mechanics. It became clear as the lectures drew to a close, however, that not enough time had been left for the quantum mechanics. As the material was prepared, it was continually discovered that other important and interesting topics could be treated with the elementary tools that had been developed. There was also a fear that the too brief treatment of the Schrödinger wave function which had been included in the twelfth lecture would not provide a sufficient bridge to the more conventional treatments of many books the students might hope to read. It was therefore decided to extend the series with seven additional lectures; they were given to the sophomore class in May of 1964. These lectures rounded out and extended somewhat the material developed in the earlier lectures.

In this volume we have put together the lectures from both years with some adjustment of the sequence. In addition, two lectures originally given to the freshman class as an introduction to quantum physics have been lifted bodily from Volume I (where they were Chapters 37 and 38) and placed as the first two chapters here—to make this volume a self-contained unit, relatively independent of the first two. A few ideas about the quantization of angular momentum (including a discussion of the Stern-Gerlach experiment) had been introduced in Chapters 34 and 35 of Volume II, and familiarity with them is assumed; for the convenience of those who will not have that volume at hand, those two chapters are reproduced here as an Appendix.

This set of lectures tries to elucidate from the beginning those features of the quantum mechanics which are most basic and most general. The first lectures tackle head on the ideas of a probability amplitude, the interference of amplitudes, the abstract notion of a state, and the superposition and resolution of states—and the Dirac notation is used from the start. In each instance the ideas are introduced together with a detailed discussion of some specific examples—to try to make the physical ideas as real as possible. The time dependence of states including states of definite energy comes next, and the ideas are applied at once to the study of two-state systems. A detailed discussion of the ammonia maser provides the frame-

work for the introduction to radiation absorption and induced transitions. The lectures then go on to consider more complex systems, leading to a discussion of the propagation of electrons in a crystal, and to a rather complete treatment of the quantum mechanics of angular momentum. Our introduction to quantum mechanics ends in Chapter 20 with a discussion of the Schrödinger wave function, its differential equation, and the solution for the hydrogen atom.

The last chapter of this volume is not intended to be a part of the “course.” It is a “seminar” on superconductivity and was given in the spirit of some of the entertainment lectures of the first two volumes, with the intent of opening to the students a broader view of the relation of what they were learning to the general culture of physics. Feynman’s “epilogue” serves as the period to the three-volume series.

As explained in the Foreword to Volume I, these lectures were but one aspect of a program for the development of a new introductory course carried out at the California Institute of Technology under the supervision of the Physics Course Revision Committee (Robert Leighton, Victor Neher, and Matthew Sands). The program was made possible by a grant from the Ford Foundation. Many people helped with the technical details of the preparation of this volume: Marylou Clayton, Julie Curcio, James Hartle, Tom Harvey, Martin Israel, Patricia Preuss, Fanny Warren, and Barbara Zimmerman. Professors Gerry Neugebauer and Charles Wilts contributed greatly to the accuracy and clarity of the material by reviewing carefully much of the manuscript.

But the story of quantum mechanics you will find here is Richard Feynman’s. Our labors will have been well spent if we have been able to bring to others even some of the intellectual excitement we experienced as we saw the ideas unfold in his real-life Lectures on Physics.

December, 1964

MATTHEW SANDS

Contents

CHAPTER 1. QUANTUM BEHAVIOR

- 1-1 Atomic mechanics 1-1
- 1-2 An experiment with bullets 1-1
- 1-3 An experiment with waves 1-3
- 1-4 An experiment with electrons 1-4
- 1-5 The interference of electron waves 1-5
- 1-6 Watching the electrons 1-6
- 1-7 First principles of quantum mechanics 1-9
- 1-8 The uncertainty principle 1-11

CHAPTER 2. THE RELATION OF WAVE AND PARTICLE VIEWPOINTS

- 2-1 Probability wave amplitudes 2-1
- 2-2 Measurement of position and momentum 2-2
- 2-3 Crystal diffraction 2-4
- 2-4 The size of an atom 2-5
- 2-5 Energy levels 2-7
- 2-6 Philosophical implications 2-8

CHAPTER 3. PROBABILITY AMPLITUDES

- 3-1 The laws of combining amplitudes 3-1
- 3-2 The two-slit interference pattern 3-5
- 3-3 Scattering from a crystal 3-7
- 3-4 Identical particles 3-9

CHAPTER 4. IDENTICAL PARTICLES

- 4-1 Bose particles and Fermi particles 4-1
- 4-2 States with two Bose particles 4-3
- 4-3 States with n Bose particles 4-6
- 4-4 Emission and absorption of photons 4-7
- 4-5 The blackbody spectrum 4-8
- 4-6 Liquid helium 4-12
- 4-7 The exclusion principle 4-12

CHAPTER 5. SPIN ONE

- 5-1 Filtering atoms with a Stern-Gerlach apparatus 5-1
- 5-2 Experiments with filtered atoms 5-5
- 5-3 Stern-Gerlach filters in series 5-6
- 5-4 Base states 5-8
- 5-5 Interfering amplitudes 5-10
- 5-6 The machinery of quantum mechanics 5-12
- 5-7 Transforming to a different base 5-15
- 5-8 Other situations 5-16

CHAPTER 6. SPIN ONE-HALF

- 6-1 Transforming amplitudes 6-1
- 6-2 Transforming to a rotated coordinate system 6-3
- 6-3 Rotations about the z -axis 6-6
- 6-4 Rotations of 180° and 90° about y 6-9
- 6-5 Rotations about x 6-11
- 6-6 Arbitrary rotations 6-12

CHAPTER 7. THE DEPENDENCE OF AMPLITUDES ON TIME

- 7-1 Atoms at rest; stationary states 7-1
- 7-2 Uniform motion 7-3
- 7-3 Potential energy; energy conservation 7-6
- 7-4 Forces; the classical limit 7-9
- 7-5 The "precession" of a spin one-half particle 7-10

CHAPTER 8. THE HAMILTONIAN MATRIX

- 8-1 Amplitudes and vectors 8-1
- 8-2 Resolving state vectors 8-3
- 8-3 What are the base states of the world? 8-5
- 8-4 How states change with time 8-7
- 8-5 The Hamiltonian matrix 8-10
- 8-6 The ammonia molecule 8-11

CHAPTER 9. THE AMMONIA MASER

- 9-1 The states of an ammonia molecule 9-1
- 9-2 The molecule in a static electric field 9-5
- 9-3 Transitions in a time-dependent field 9-9
- 9-4 Transitions at resonance 9-11
- 9-5 Transitions off resonance 9-13
- 9-6 The absorption of light 9-14

CHAPTER 10. OTHER TWO-STATE SYSTEMS

- 10-1 The hydrogen molecular ion 10-1
- 10-2 Nuclear forces 10-6
- 10-3 The hydrogen molecule 10-8
- 10-4 The benzene molecule 10-10
- 10-5 Dyes 10-12
- 10-6 The Hamiltonian of a spin one-half particle in a magnetic field 10-12
- 10-7 The spinning electron in a magnetic field 10-15

CHAPTER 11. MORE TWO-STATE SYSTEMS

- 11-1 The Pauli spin matrices 11-1
- 11-2 The spin matrices as operators 11-5
- 11-3 The solution of the two-state equations 11-8
- 11-4 The polarization states of the photon 11-9
- 11-5 The neutral K-meson 11-12
- 11-6 Generalization to N -state systems 11-20

CHAPTER 12. THE HYPERFINE SPLITTING IN HYDROGEN

- 12-1 Base states for a system with two spin one-half particles 12-1
- 12-2 The Hamiltonian for the ground state of hydrogen 12-3
- 12-3 The energy levels 12-7
- 12-4 The Zeeman splitting 12-9
- 12-5 The states in a magnetic field 12-12
- 12-6 The projection matrix for spin one 12-14

CHAPTER 13. PROPAGATION IN A CRYSTAL LATTICE

- 13-1 States for an electron in a one-dimensional lattice 13-1
- 13-2 States of definite energy 13-3
- 13-3 Time-dependent states 13-6
- 13-4 An electron in a three-dimensional lattice 13-7
- 13-5 Other states in a lattice 13-8
- 13-6 Scattering by imperfections in the lattice 13-10
- 13-7 Trapping by a lattice imperfection 13-12
- 13-8 Scattering amplitudes and bound states 13-13

CHAPTER 14. SEMICONDUCTORS

- 14-1 Electrons and holes in semiconductors 14-1
- 14-2 Impure semiconductors 14-4
- 14-3 The Hall effect 14-7
- 14-4 Semiconductor junctions 14-8
- 14-5 Rectification at a semiconductor junction 14-10
- 14-6 The transistor 14-11

CHAPTER 15. THE INDEPENDENT PARTICLE APPROXIMATION

- 15-1 Spin waves 15-1
- 15-2 Two spin waves 15-4
- 15-3 Independent particles 15-6
- 15-4 The benzene molecule 15-7
- 15-5 More organic chemistry 15-10
- 15-6 Other uses of the approximation 15-12

CHAPTER 16. THE DEPENDENCE OF AMPLITUDES ON POSITION

- 16-1 Amplitudes on a line 16-1
- 16-2 The wave function 16-5
- 16-3 States of definite momentum 16-7
- 16-4 Normalization of the states in x 16-9
- 16-5 The Schrödinger equation 16-11
- 16-6 Quantized energy levels 16-14

CHAPTER 17. SYMMETRY AND CONSERVATION LAWS

- 17-1 Symmetry 17-1
- 17-2 Symmetry and conservation 17-3
- 17-3 The conservation laws 17-7
- 17-4 Polarized light 17-9
- 17-5 The disintegration of the Λ^0 17-11
- 17-6 Summary of the rotation matrices 17-15

CHAPTER 18. ANGULAR MOMENTUM

- 18-1 Electric dipole radiation 18-1
- 18-2 Light scattering 18-3
- 18-3 The annihilation of positronium 18-5
- 18-4 Rotation matrix for any spin 18-9
- 18-5 Measuring a nuclear spin 18-13
- 18-6 Composition of angular momentum 18-14
- Added Note 1: Derivation of the rotation matrix 18-19
- Added Note 2: Conservation of parity in photon emission 18-22

CHAPTER 19. THE HYDROGEN ATOM AND THE PERIODIC TABLE

- 19-1 Schrödinger's equation for the hydrogen atom 19-1
- 19-2 Spherically symmetric solutions 19-2
- 19-3 States with an angular dependence 19-6
- 19-4 The general solution for hydrogen 19-10
- 19-5 The hydrogen wave functions 19-12
- 19-6 The periodic table 19-13

CHAPTER 20. OPERATORS

- 20-1 Operations and operators 20-1
- 20-2 Average energies 20-3
- 20-3 The average energy of an atom 20-6
- 20-4 The position operator 20-8
- 20-5 The momentum operator 20-9
- 20-6 Angular momentum 20-14
- 20-7 The change of averages with time 20-15

CHAPTER 21. THE SCHRÖDINGER EQUATION IN A CLASSICAL CONTEXT: A SEMINAR ON SUPERCONDUCTIVITY

- 21-1 Schrödinger's equation in a magnetic field 21-1
- 21-2 The equation of continuity for probabilities 21-3
- 21-3 Two kinds of momentum 21-4
- 21-4 The meaning of the wave function 21-6
- 21-5 Superconductivity 21-7
- 21-6 The Meissner effect 21-8
- 21-7 Flux quantization 21-10
- 21-8 The dynamics of superconductivity 21-12
- 21-9 The Josephson junction 21-14

FEYNMAN'S EPILOGUE

APPENDIX

INDEX

Quantum Behavior

1-1 Atomic mechanics

“Quantum mechanics” is the description of the behavior of matter and light in all its details and, in particular, of the happenings on an atomic scale. Things on a very small scale behave like nothing that you have any direct experience about. They do not behave like waves, they do not behave like particles, they do not behave like clouds, or billiard balls, or weights on springs, or like anything that you have ever seen.

Newton thought that light was made up of particles, but then it was discovered that it behaves like a wave. Later, however (in the beginning of the twentieth century), it was found that light did indeed sometimes behave like a particle. Historically, the electron, for example, was thought to behave like a particle, and then it was found that in many respects it behaved like a wave. So it really behaves like neither. Now we have given up. We say: “It is like *neither*.”

There is one lucky break, however—electrons behave just like light. The quantum behavior of atomic objects (electrons, protons, neutrons, photons, and so on) is the same for all, they are all “particle waves,” or whatever you want to call them. So what we learn about the properties of electrons (which we shall use for our examples) will apply also to all “particles,” including photons of light.

The gradual accumulation of information about atomic and small-scale behavior during the first quarter of this century, which gave some indications about how small things do behave, produced an increasing confusion which was finally resolved in 1926 and 1927 by Schrödinger, Heisenberg, and Born. They finally obtained a consistent description of the behavior of matter on a small scale. We take up the main features of that description in this chapter.

Because atomic behavior is so unlike ordinary experience, it is very difficult to get used to, and it appears peculiar and mysterious to everyone—both to the novice and to the experienced physicist. Even the experts do not understand it the way they would like to, and it is perfectly reasonable that they should not, because all of direct, human experience and of human intuition applies to large objects. We know how large objects will act, but things on a small scale just do not act that way. So we have to learn about them in a sort of abstract or imaginative fashion and not by connection with our direct experience.

In this chapter we shall tackle immediately the basic element of the mysterious behavior in its most strange form. We choose to examine a phenomenon which is impossible, *absolutely* impossible, to explain in any classical way, and which has in it the heart of quantum mechanics. In reality, it contains the *only* mystery. We cannot make the mystery go away by “explaining” how it works. We will just *tell* you how it works. In telling you how it works we will have told you about the basic peculiarities of all quantum mechanics.

1-2 An experiment with bullets

To try to understand the quantum behavior of electrons, we shall compare and contrast their behavior, in a particular experimental setup, with the more familiar behavior of particles like bullets, and with the behavior of waves like water waves. We consider first the behavior of bullets in the experimental setup shown diagrammatically in Fig. 1-1. We have a machine gun that shoots a stream of bullets. It is not a very good gun, in that it sprays the bullets (randomly) over a fairly large angular spread, as indicated in the figure. In front of the gun we have

1-1 Atomic mechanics

1-2 An experiment with bullets

1-3 An experiment with waves

1-4 An experiment with electrons

1-5 The interference of electron waves

1-6 Watching the electrons

1-7 First principles of quantum mechanics

1-8 The uncertainty principle

Note: This chapter is almost exactly the same as Chapter 37 of Volume I.

a wall (made of armor plate) that has in it two holes just about big enough to let a bullet through. Beyond the wall is a backstop (say a thick wall of wood) which will “absorb” the bullets when they hit it. In front of the wall we have an object which we shall call a “detector” of bullets. It might be a box containing sand. Any bullet that enters the detector will be stopped and accumulated. When we wish, we can empty the box and count the number of bullets that have been caught. The detector can be moved back and forth (in what we will call the x -direction). With this apparatus, we can find out experimentally the answer to the question: “What is the probability that a bullet which passes through the holes in the wall will arrive at the backstop at the distance x from the center?” First, you should realize that we should talk about probability, because we cannot say definitely where any particular bullet will go. A bullet which happens to hit one of the holes may bounce off the edges of the hole, and may end up anywhere at all. By “probability” we mean the chance that the bullet will arrive at the detector, which we can measure by counting the number which arrive at the detector in a certain time and then taking the ratio of this number to the *total* number that hit the backstop during that time. Or, if we assume that the gun always shoots at the same rate during the measurements, the probability we want is just proportional to the number that reach the detector in some standard time interval.

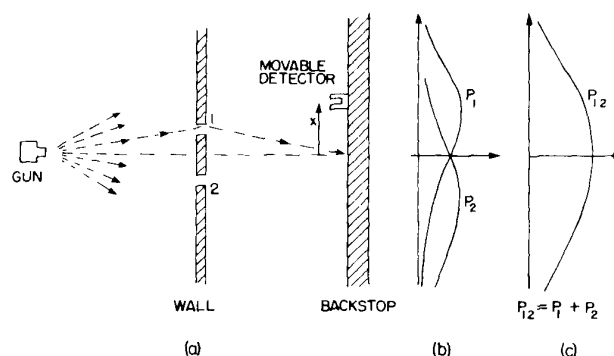


Fig. 1-1. Interference experiment with bullets.

For our present purposes we would like to imagine a somewhat idealized experiment in which the bullets are not real bullets, but are *indestructible* bullets—they cannot break in half. In our experiment we find that bullets always arrive in lumps, and when we find something in the detector, it is always one whole bullet. If the rate at which the machine gun fires is made very low, we find that at any given moment either nothing arrives, or one and only one—exactly one—bullet arrives at the backstop. Also, the size of the lump certainly does not depend on the rate of firing of the gun. We shall say: “Bullets *always* arrive in identical lumps.” What we measure with our detector is the probability of arrival of a lump. And we measure the probability as a function of x . The result of such measurements with this apparatus (we have not yet done the experiment, so we are really imagining the result) are plotted in the graph drawn in part (c) of Fig. 1-1. In the graph we plot the probability to the right and x vertically, so that the x -scale fits the diagram of the apparatus. We call the probability P_{12} because the bullets may have come either through hole 1 or through hole 2. You will not be surprised that P_{12} is large near the middle of the graph but gets small if x is very large. You may wonder, however, why P_{12} has its maximum value at $x = 0$. We can understand this fact if we do our experiment again after covering up hole 2, and once more while covering up hole 1. When hole 2 is covered, bullets can pass only through hole 1, and we get the curve marked P_1 in part (b) of the figure. As you would expect, the maximum of P_1 occurs at the value of x which is on a straight line with the gun and hole 1. When hole 1 is closed, we get the symmetric curve P_2 drawn in the figure. P_2 is the probability distribution for bullets that pass through hole 2. Comparing parts (b) and (c) of Fig. 1-1, we find the important result that

$$P_{12} = P_1 + P_2. \quad (1.1)$$

The probabilities just add together. The effect with both holes open is the sum of the effects with each hole open alone. We shall call this result an observation of “no interference,” for a reason that you will see later. So much for bullets. They come in lumps, and their probability of arrival shows no interference.

1-3 An experiment with waves

Now we wish to consider an experiment with water waves. The apparatus is shown diagrammatically in Fig. 1-2. We have a shallow trough of water. A small object labeled the “wave source” is jigged up and down by a motor and makes circular waves. To the right of the source we have again a wall with two holes, and beyond that is a second wall, which, to keep things simple, is an “absorber,” so that there is no reflection of the waves that arrive there. This can be done by building a gradual sand “beach.” In front of the beach we place a detector which can be moved back and forth in the x -direction, as before. The detector is now a device which measures the “intensity” of the wave motion. You can imagine a gadget which measures the height of the wave motion, but whose scale is calibrated in proportion to the *square* of the actual height, so that the reading is proportional to the intensity of the wave. Our detector reads, then, in proportion to the *energy* being carried by the wave—or rather, the rate at which energy is carried to the detector.

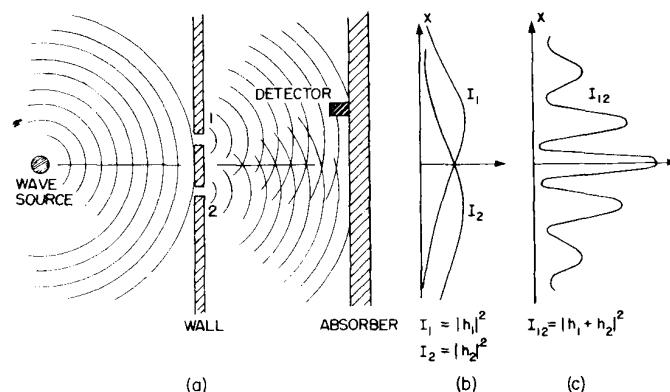


Fig. 1-2. Interference experiment with water waves.

With our wave apparatus, the first thing to notice is that the intensity can have *any* size. If the source just moves a very small amount, then there is just a little bit of wave motion at the detector. When there is more motion at the source, there is more intensity at the detector. The intensity of the wave can have any value at all. We would *not* say that there was any “lumpiness” in the wave intensity.

Now let us measure the wave intensity for various values of x (keeping the wave source operating always in the same way). We get the interesting-looking curve marked I_{12} in part (c) of the figure.

We have already worked out how such patterns can come about when we studied the interference of electric waves in Volume I. In this case we would observe that the original wave is diffracted at the holes, and new circular waves spread out from each hole. If we cover one hole at a time and measure the intensity distribution at the absorber we find the rather simple intensity curves shown in part (b) of the figure. I_1 is the intensity of the wave from hole 1 (which we find by measuring when hole 2 is blocked off) and I_2 is the intensity of the wave from hole 2 (seen when hole 1 is blocked).

The intensity I_{12} observed when both holes are open is certainly *not* the sum of I_1 and I_2 . We say that there is “interference” of the two waves. At some places (where the curve I_{12} has its maxima) the waves are “in phase” and the wave peaks add together to give a large amplitude and, therefore, a large intensity. We say that the two waves are “interfering constructively” at such places. There will be such constructive interference wherever the distance from the detector to one hole is a whole number of wavelengths larger (or shorter) than the distance from the detector to the other hole.

At those places where the two waves arrive at the detector with a phase difference of π (where they are “out of phase”) the resulting wave motion at the detector will be the difference of the two amplitudes. The waves “interfere destructively,” and we get a low value for the wave intensity. We expect such low values wherever the distance between hole 1 and the detector is different from the distance between hole 2 and the detector by an odd number of half-wavelengths. The low values of I_{12} in Fig. 1-2 correspond to the places where the two waves interfere destructively.

You will remember that the quantitative relationship between I_1 , I_2 , and I_{12} can be expressed in the following way: The instantaneous height of the water wave at the detector for the wave from hole 1 can be written as (the real part of) $h_1 e^{i\omega t}$, where the “amplitude” h_1 is, in general, a complex number. The intensity is proportional to the mean squared height or, when we use the complex numbers, to the absolute value squared $|h_1|^2$. Similarly, for hole 2 the height is $h_2 e^{i\omega t}$ and the intensity is proportional to $|h_2|^2$. When both holes are open, the wave heights add to give the height $(h_1 + h_2)e^{i\omega t}$ and the intensity $|h_1 + h_2|^2$. Omitting the constant of proportionality for our present purposes, the proper relations for *interfering waves* are

$$I_1 = |h_1|^2, \quad I_2 = |h_2|^2, \quad I_{12} = |h_1 + h_2|^2. \quad (1.2)$$

You will notice that the result is quite different from that obtained with bullets (Eq. 1-1). If we expand $|h_1 + h_2|^2$ we see that

$$|h_1 + h_2|^2 = |h_1|^2 + |h_2|^2 + 2|h_1||h_2|\cos\delta, \quad (1.3)$$

where δ is the phase difference between h_1 and h_2 . In terms of the intensities, we could write

$$I_{12} = I_1 + I_2 + 2\sqrt{I_1 I_2} \cos\delta. \quad (1.4)$$

The last term in (1.4) is the “interference term.” So much for water waves. The intensity can have any value, and it shows interference.

1-4 An experiment with electrons

Now we imagine a similar experiment with electrons. It is shown diagrammatically in Fig. 1-3. We make an electron gun which consists of a tungsten wire heated by an electric current and surrounded by a metal box with a hole in it. If the wire is at a negative voltage with respect to the box, electrons emitted by the wire will be accelerated toward the walls and some will pass through the hole. All the electrons which come out of the gun will have (nearly) the same energy. In front of the gun is again a wall (just a thin metal plate) with two holes in it. Beyond the wall is another plate which will serve as a “backstop.” In front of the backstop we place a movable detector. The detector might be a geiger counter or, perhaps better, an electron multiplier, which is connected to a loudspeaker.

We should say right away that you should not try to set up this experiment (as you could have done with the two we have already described). This experiment

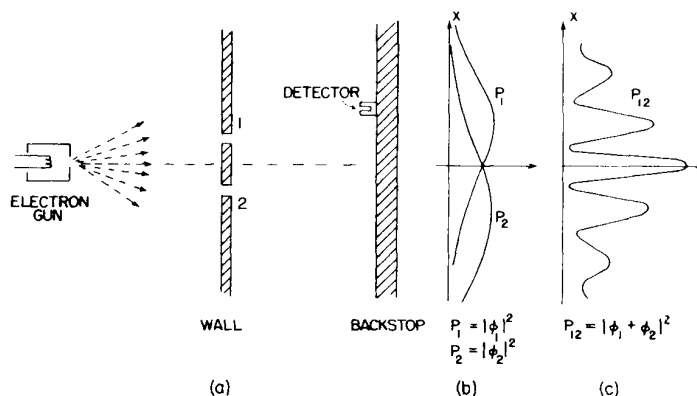


Fig. 1-3. Interference experiment with electrons.

has never been done in just this way. The trouble is that the apparatus would have to be made on an impossibly small scale to show the effects we are interested in. We are doing a “thought experiment,” which we have chosen because it is easy to think about. We know the results that *would* be obtained because there *are* many experiments that have been done, in which the scale and the proportions have been chosen to show the effects we shall describe.

The first thing we notice with our electron experiment is that we hear sharp “clicks” from the detector (that is, from the loudspeaker). And all “clicks” are the same. There are *no* “half-clicks.”

We would also notice that the “clicks” come very erratically. Something like: click click-click . . . click click click-click click . . . , etc., just as you have, no doubt, heard a geiger counter operating. If we count the clicks which arrive in a sufficiently long time—say for many minutes—and then count again for another equal period, we find that the two numbers are very nearly the same. So we can speak of the *average rate* at which the clicks are heard (so-and-so-many clicks per minute on the average).

As we move the detector around, the *rate* at which the clicks appear is faster or slower, but the size (loudness) of each click is always the same. If we lower the temperature of the wire in the gun, the rate of clicking slows down, but still each click sounds the same. We would notice also that if we put two separate detectors at the backstop, one *or* the other would click, but never both at once. (Except that once in a while, if there were two clicks very close together in time, our ear might not sense the separation.) We conclude, therefore, that whatever arrives at the backstop arrives in “lumps.” All the “lumps” are the same size: only whole “lumps” arrive, and they arrive one at a time at the backstop. We shall say: “Electrons always arrive in identical lumps.”

Just as for our experiment with bullets, we can now proceed to find experimentally the answer to the question: “What is the relative probability that an electron ‘lump’ will arrive at the backstop at various distances x from the center?” As before, we obtain the relative probability by observing the rate of clicks, holding the operation of the gun constant. The probability that lumps will arrive at a particular x is proportional to the average rate of clicks at that x .

The result of our experiment is the interesting curve marked P_{12} in part (c) of Fig. 1-3. Yes! That is the way electrons go.

1-5 The interference of electron waves

Now let us try to analyze the curve of Fig. 1-3 to see whether we can understand the behavior of the electrons. The first thing we would say is that since they come in lumps, each lump, which we may as well call an electron, has come either through hole 1 or through hole 2. Let us write this in the form of a “Proposition”:

Proposition A: Each electron *either* goes through hole 1 *or* it goes through hole 2.

Assuming Proposition A, all electrons that arrive at the backstop can be divided into two classes: (1) those that come through hole 1, and (2) those that come through hole 2. So our observed curve must be the sum of the effects of the electrons which come through hole 1 and the electrons which come through hole 2. Let us check this idea by experiment. First, we will make a measurement for those electrons that come through hole 1. We block off hole 2 and make our counts of the clicks from the detector. From the clicking rate, we get P_1 . The result of the measurement is shown by the curve marked P_1 in part (b) of Fig. 1-3. The result seems quite reasonable. In a similar way, we measure P_2 , the probability distribution for the electrons that come through hole 2. The result of this measurement is also drawn in the figure.

The result P_{12} obtained with *both* holes open is clearly not the sum of P_1 and P_2 , the probabilities for each hole alone. In analogy with our water-wave experi-

ment, we say: "There is interference."

$$\text{For electrons: } P_{12} \neq P_1 + P_2. \quad (1.5)$$

How can such an interference come about? Perhaps we should say: "Well, that means, presumably, that it is *not true* that the lumps go either through hole 1 or hole 2, because if they did, the probabilities should add. Perhaps they go in a more complicated way. They split in half and . . ." But no! They cannot, they always arrive in lumps . . . "Well, perhaps some of them go through 1, and then they go around through 2, and then around a few more times, or by some other complicated path . . . then by closing hole 2, we changed the chance that an electron that *started out* through hole 1 would finally get to the backstop . . ." But notice! There are some points at which very few electrons arrive when *both* holes are open, but which receive many electrons if we close one hole, so *closing* one hole *increased* the number from the other. Notice, however, that at the center of the pattern, P_{12} is more than twice as large as $P_1 + P_2$. It is as though closing one hole *decreased* the number of electrons which come through the other hole. It seems hard to explain *both* effects by proposing that the electrons travel in complicated paths.

It is all quite mysterious. And the more you look at it the more mysterious it seems. Many ideas have been concocted to try to explain the curve for P_{12} in terms of individual electrons going around in complicated ways through the holes. None of them has succeeded. None of them can get the right curve for P_{12} in terms of P_1 and P_2 .

Yet, surprisingly enough, the *mathematics* for relating P_1 and P_2 to P_{12} is extremely simple. For P_{12} is just like the curve I_{12} of Fig. 1-2, and *that* was simple. What is going on at the backstop can be described by two complex numbers that we can call ϕ_1 and ϕ_2 (they are functions of x , of course). The absolute square of ϕ_1 gives the effect with only hole 1 open. That is, $P_1 = |\phi_1|^2$. The effect with only hole 2 open is given by ϕ_2 in the same way. That is, $P_2 = |\phi_2|^2$. And the combined effect of the two holes is just $P_{12} = |\phi_1 + \phi_2|^2$. The *mathematics* is the same as that we had for the water waves! (It is hard to see how one could get such a simple result from a complicated game of electrons going back and forth through the plate on some strange trajectory.)

We conclude the following: The electrons arrive in lumps, like particles, and the probability of arrival of these lumps is distributed like the distribution of intensity of a wave. It is in this sense that an electron behaves "sometimes like a particle and sometimes like a wave."

Incidentally, when we were dealing with classical waves we defined the intensity as the mean over time of the square of the wave amplitude, and we used complex numbers as a mathematical trick to simplify the analysis. But in quantum mechanics it turns out that the amplitudes *must* be represented by complex numbers. The real parts alone will not do. That is a technical point, for the moment, because the formulas look just the same.

Since the probability of arrival through both holes is given so simply, although it is not equal to $(P_1 + P_2)$, that is really all there is to say. But there are a large number of subtleties involved in the fact that nature does work this way. We would like to illustrate some of these subtleties for you now. First, since the number that arrives at a particular point is *not* equal to the number that arrives through 1 plus the number that arrives through 2, as we would have concluded from Proposition A, undoubtedly we should conclude that *Proposition A is false*. It is *not* true that the electrons go *either* through hole 1 or hole 2. But that conclusion can be tested by another experiment.

1-6 Watching the electrons

We shall now try the following experiment. To our electron apparatus we add a very strong light source, placed behind the wall and between the two holes, as shown in Fig. 1-4. We know that electric charges scatter light. So when an

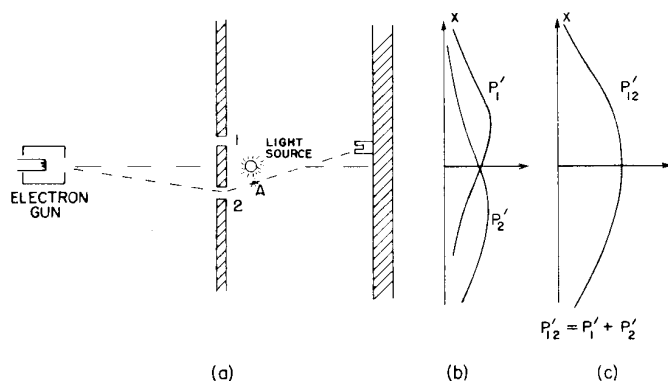


Fig. 1-4. A different electron experiment.

electron passes, however it does pass, on its way to the detector, it will scatter some light to our eye, and we can *see* where the electron goes. If, for instance, an electron were to take the path via hole 2 that is sketched in Fig. 1-4, we should see a flash of light coming from the vicinity of the place marked *A* in the figure. If an electron passes through hole 1, we would expect to see a flash from the vicinity of the upper hole. If it should happen that we get light from both places at the same time, because the electron divides in half . . . Let us just do the experiment!

Here is what we see: *every* time that we hear a “click” from our electron detector (at the backstop), we *also see* a flash of light *either* near hole 1 *or* near hole 2, but *never* both at once! And we observe the same result no matter where we put the detector. From this observation we conclude that when we look at the electrons we find that the electrons go either through one hole or the other. Experimentally, Proposition A is necessarily true.

What, then, is wrong with our argument *against* Proposition A? Why *isn't* P_{12} just equal to $P_1 + P_2$? Back to experiment! Let us keep track of the electrons and find out what they are doing. For each position (*x*-location) of the detector we will count the electrons that arrive and *also* keep track of which hole they went through, by watching for the flashes. We can keep track of things this way: whenever we hear a “click” we will put a count in Column 1 if we see the flash near hole 1, and if we see the flash near hole 2, we will record a count in Column 2. Every electron which arrives is recorded in one of two classes: those which come through 1 and those which come through 2. From the number recorded in Column 1 we get the probability P'_1 that an electron will arrive at the detector via hole 1; and from the number recorded in Column 2 we get P'_2 , the probability that an electron will arrive at the detector via hole 2. If we now repeat such a measurement for many values of *x*, we get the curves for P'_1 and P'_2 shown in part (b) of Fig. 1-4.

Well, that is not too surprising! We get for P'_1 something quite similar to what we got before for P_1 by blocking off hole 2; and P'_2 is similar to what we got by blocking hole 1. So there is *not* any complicated business like going through both holes. When we watch them, the electrons come through just as we would expect them to come through. Whether the holes are closed or open, those which we see come through hole 1 are distributed in the same way whether hole 2 is open or closed.

But wait! What do we have *now* for the *total* probability, the probability that an electron will arrive at the detector by any route? We already have that information. We just pretend that we never looked at the light flashes, and we lump together the detector clicks which we have separated into the two columns. We *must* just *add* the numbers. For the probability that an electron will arrive at the backstop by passing through *either* hole, we do find $P'_{12} = P'_1 + P'_2$. That is, although we succeeded in watching which hole our electrons come through, we no longer get the old interference curve P_{12} , but a new one, P'_{12} , showing no interference! If we turn out the light P_{12} is restored.

We must conclude that *when we look at the electrons* the distribution of them on the screen is different than when we do not look. Perhaps it is turning on our light source that disturbs things? It must be that the electrons are very delicate, and the light, when it scatters off the electrons, gives them a jolt that changes their

motion. We know that the electric field of the light acting on a charge will exert a force on it. So perhaps we *should* expect the motion to be changed. Anyway, the light exerts a big influence on the electrons. By trying to “watch” the electrons we have changed their motions. That is, the jolt given to the electron when the photon is scattered by it is such as to change the electron’s motion enough so that if it *might* have gone to where P_{12} was at a maximum it will instead land where P_{12} was a minimum; that is why we no longer see the wavy interference effects.

You may be thinking: “Don’t use such a bright source! Turn the brightness down! The light waves will then be weaker and will not disturb the electrons so much. Surely, by making the light dimmer and dimmer, eventually the wave will be weak enough that it will have a negligible effect.” O.K. Let’s try it. The first thing we observe is that the flashes of light scattered from the electrons as they pass by does *not* get weaker. *It is always the same-sized flash.* The only thing that happens as the light is made dimmer is that sometimes we hear a “click” from the detector but see *no flash at all*. The electron has gone by without being “seen.” What we are observing is that light *also* acts like electrons, we *knew* that it was “wavy,” but now we find that it is also “lumpy.” It always arrives—or is scattered—in lumps that we call “photons.” As we turn down the *intensity* of the light source we do not change the *size* of the photons, only the *rate* at which they are emitted. *That* explains why, when our source is dim, some electrons get by without being seen. There did not happen to be a photon around at the time the electron went through.

This is all a little discouraging. If it is true that whenever we “see” the electron we see the same-sized flash, then those electrons we see are *always* the disturbed ones. Let us try the experiment with a dim light anyway. Now whenever we hear a click in the detector we will keep a count in three columns: in Column (1) those electrons seen by hole 1, in Column (2) those electrons seen by hole 2, and in Column (3) those electrons not seen at all. When we work up our data (computing the probabilities) we find these results: Those “seen by hole 1” have a distribution like P'_1 ; those “seen by hole 2” have a distribution like P'_2 (so that those “seen by either hole 1 or 2” have a distribution like P'_{12}); and those “not seen at all” have a “wavy” distribution just like P_{12} of Fig. 1-3! *If the electrons are not seen, we have interference!*

That is understandable. When we do not see the electron, no photon disturbs it, and when we do see it, a photon has disturbed it. There is always the same amount of disturbance because the light photons all produce the same-sized effects and the effect of the photons being scattered is enough to smear out any interference effect.

Is there not *some* way we can see the electrons without disturbing them? We learned in an earlier chapter that the momentum carried by a “photon” is inversely proportional to its wavelength ($p = h/\lambda$). Certainly the jolt given to the electron when the photon is scattered toward our eye depends on the momentum that photon carries. Aha! If we want to disturb the electrons only slightly we should not have lowered the *intensity* of the light, we should have lowered its *frequency* (the same as increasing its wavelength). Let us use light of a redder color. We could even use infrared light, or radiowaves (like radar), and “see” where the electron went with the help of some equipment that can “see” light of these longer wavelengths. If we use “gentler” light perhaps we can avoid disturbing the electrons so much.

Let us try the experiment with longer waves. We shall keep repeating our experiment, each time with light of a longer wavelength. At first, nothing seems to change. The results are the same. Then a terrible thing happens. You remember that when we discussed the microscope we pointed out that, due to the *wave nature* of the light, there is a limitation on how close two spots can be and still be seen as two separate spots. This distance is of the order of the wavelength of light. So now, when we make the wavelength longer than the distance between our holes, we see a *big* fuzzy flash when the light is scattered by the electrons. We can no longer tell which hole the electron went through! We just know it went somewhere! And it is just with light of this color that we find that the jolts given to the electron

are small enough so that P'_{12} begins to look like P_{12} —that we begin to get some interference effect. And it is only for wavelengths much longer than the separation of the two holes (when we have no chance at all of telling where the electron went) that the disturbance due to the light gets sufficiently small that we again get the curve P_{12} shown in Fig. 1-3.

In our experiment we find that it is impossible to arrange the light in such a way that one can tell which hole the electron went through, and at the same time not disturb the pattern. It was suggested by Heisenberg that the then new laws of nature could only be consistent if there were some basic limitation on our experimental capabilities not previously recognized. He proposed, as a general principle, his *uncertainty principle*, which we can state in terms of our experiment as follows: “It is impossible to design an apparatus to determine which hole the electron passes through, that will not at the same time disturb the electrons enough to destroy the interference pattern.” If an apparatus is capable of determining which hole the electron goes through, it *cannot* be so delicate that it does not disturb the pattern in an essential way. No one has ever found (or even thought of) a way around the uncertainty principle. So we must assume that it describes a basic characteristic of nature.

The complete theory of quantum mechanics which we now use to describe atoms and, in fact, all matter, depends on the correctness of the uncertainty principle. Since quantum mechanics is such a successful theory, our belief in the uncertainty principle is reinforced. But if a way to “beat” the uncertainty principle were ever discovered, quantum mechanics would give inconsistent results and would have to be discarded as a valid theory of nature.

“Well,” you say, “what about Proposition A? Is it true, or is it *not* true, that the electron either goes through hole 1 or it goes through hole 2?” The only answer that can be given is that we have found from experiment that there is a certain special way that we have to think in order that we do not get into inconsistencies. What we must say (to avoid making wrong predictions) is the following. If one looks at the holes or, more accurately, if one has a piece of apparatus which is capable of determining whether the electrons go through hole 1 or hole 2, then one *can* say that it goes either through hole 1 or hole 2. *But*, when one does *not* try to tell which way the electron goes, when there is nothing in the experiment to disturb the electrons, then one may *not* say that an electron goes either through hole 1 or hole 2. If one does say that, and starts to make any deductions from the statement, he will make errors in the analysis. This is the logical tightrope on which we must walk if we wish to describe nature successfully.

If the motion of all matter—as well as electrons—must be described in terms of waves, what about the bullets in our first experiment? Why didn’t we see an interference pattern there? It turns out that for the bullets the wavelengths were so tiny that the interference patterns became very fine. So fine, in fact, that with any detector of finite size one could not distinguish the separate maxima and minima. What we saw was only a kind of average, which is the classical curve. In Fig. 1-5 we have tried to indicate schematically what happens with large-scale objects. Part (a) of the figure shows the probability distribution one might predict for bullets, using quantum mechanics. The rapid wiggles are supposed to represent the interference pattern one gets for waves of very short wavelength. Any physical detector, however, straddles several wiggles of the probability curve, so that the measurements show the smooth curve drawn in part (b) of the figure.

1-7 First principles of quantum mechanics

We will now write a summary of the main conclusions of our experiments. We will, however, put the results in a form which makes them true for a general class of such experiments. We can write our summary more simply if we first define an “ideal experiment” as one in which there are no uncertain external influences, i.e., no jiggling or other things going on that we cannot take into ac-

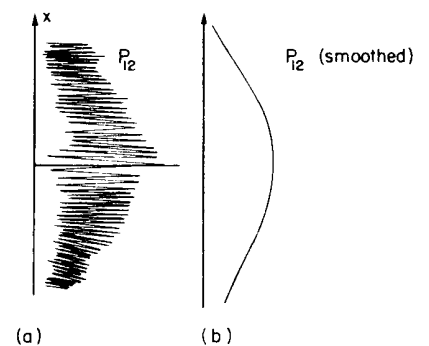


Fig. 1-5. Interference pattern with bullets: (a) actual (schematic), (b) observed.

count. We would be quite precise if we said: "An ideal experiment is one in which all of the initial and final conditions of the experiment are completely specified." What we will call "an event" is, in general, just a specific set of initial and final conditions. (For example: "an electron leaves the gun, arrives at the detector, and nothing else happens.") Now for our summary.

SUMMARY

- (1) The probability of an event in an ideal experiment is given by the square of the absolute value of a complex number ϕ which is called the probability amplitude:

$$\begin{aligned} P &= \text{probability,} \\ \phi &= \text{probability amplitude,} \\ P &= |\phi|^2 \end{aligned} \tag{1.6}$$

- (2) When an event can occur in several alternative ways, the probability amplitude for the event is the sum of the probability amplitudes for each way considered separately. There is interference:

$$\begin{aligned} \phi &= \phi_1 + \phi_2, \\ P &= |\phi_1 + \phi_2|^2 \end{aligned} \tag{1.7}$$

- (3) If an experiment is performed which is capable of determining whether one or another alternative is actually taken, the probability of the event is the sum of the probabilities for each alternative. The interference is lost:

$$P = P_1 + P_2. \tag{1.8}$$

One might still like to ask: "How does it work? What is the machinery behind the law?" No one has found any machinery behind the law. No one can "explain" any more than we have just "explained." No one will give you any deeper representation of the situation. We have no ideas about a more basic mechanism from which these results can be deduced.

We would like to emphasize a very important difference between classical and quantum mechanics. We have been talking about the probability that an electron will arrive in a given circumstance. We have implied that in our experimental arrangement (or even in the best possible one) it would be impossible to predict exactly what would happen. We can only predict the odds! This would mean, if it were true, that physics has given up on the problem of trying to predict exactly what will happen in a definite circumstance. Yes! physics *has* given up. *We do not know how to predict what would happen in a given circumstance*, and we believe now that it is impossible—that the only thing that can be predicted is the probability of different events. It must be recognized that this is a retrenchment in our earlier ideal of understanding nature. It may be a backward step, but no one has seen a way to avoid it.

We make now a few remarks on a suggestion that has sometimes been made to try to avoid the description we have given: "Perhaps the electron has some kind of internal works—some inner variables—that we do not yet know about. Perhaps that is why we cannot predict what will happen. If we could look more closely at the electron, we could be able to tell where it would end up." So far as we know, that is impossible. We would still be in difficulty. Suppose we were to assume that inside the electron there is some kind of machinery that determines where it is going to end up. That machine must *also* determine which hole it is going to go through on its way. But we must not forget that what is inside the electron should not be dependent on what *we* do, and in particular upon whether we open or close one of the holes. So if an electron, before it starts, has already made up its mind (a) which hole it is going to use, and (b) where it is going to land, we should find P_1 for those electrons that have chosen hole 1, P_2 for those that have chosen hole 2, and *necessarily* the sum $P_1 + P_2$ for those that arrive through the two holes. There seems to be no way around this. But we have verified experimentally that that is not the case. And no one has figured a way out of this puzzle. So at the

present time we must limit ourselves to computing probabilities. We say “at the present time,” but we suspect very strongly that it is something that will be with us forever—that it is impossible to beat that puzzle—that this is the way nature really *is*.

1-8 The uncertainty principle

This is the way Heisenberg stated the uncertainty principle originally: If you make the measurement on any object, and you can determine the x -component of its momentum with an uncertainty Δp , you cannot, at the same time, know its x -position more accurately than $\Delta x = h/\Delta p$, where h is a definite fixed number given by nature. It is called “Planck’s constant,” and is approximately 6.63×10^{-34} joule-seconds. The uncertainties in the position and momentum of a particle at any instant must have their product greater than Planck’s constant. This is a special case of the uncertainty principle that was stated above more generally. The more general statement was that one cannot design equipment in any way to determine which of two alternatives is taken, without, at the same time, destroying the pattern of interference.

Let us show for one particular case that the kind of relation given by Heisenberg must be true in order to keep from getting into trouble. We imagine a modification of the experiment of Fig. 1-3, in which the wall with the holes consists of a plate mounted on rollers so that it can move freely up and down (in the x -direction), as shown in Fig. 1-6. By watching the motion of the plate carefully we can try to tell which hole an electron goes through. Imagine what happens when the detector is placed at $x = 0$. We would expect that an electron which passes through hole 1 must be deflected downward by the plate to reach the detector. Since the vertical component of the electron momentum is changed, the plate must recoil with an equal momentum in the opposite direction. The plate will get an upward kick. If the electron goes through the lower hole, the plate should feel a downward kick. It is clear that for every position of the detector, the momentum received by the plate will have a different value for a traversal via hole 1 than for a traversal via hole 2. So! Without disturbing the electrons *at all*, but just by watching the *plate*, we can tell which path the electron used.

Now in order to do this it is necessary to know what the momentum of the screen is, before the electron goes through. So when we measure the momentum after the electron goes by, we can figure out how much the plate’s momentum has changed. But remember, according to the uncertainty principle we cannot at the same time know the position of the plate with an arbitrary accuracy. But if we do not know exactly *where* the plate is, we cannot say precisely where the two holes are. They will be in a different place for every electron that goes through. This means that the center of our interference pattern will have a different location for each electron. The wiggles of the interference pattern will be smeared out. We shall show quantitatively in the next chapter that if we determine the momentum of the plate sufficiently accurately to determine from the recoil measurement which hole was used, then the uncertainty in the x -position of the plate will, according to the uncertainty principle, be enough to shift the pattern observed at the detector up and down in the x -direction about the distance from a maximum to its nearest minimum. Such a random shift is just enough to smear out the pattern so that no interference is observed.

The uncertainty principle “protects” quantum mechanics. Heisenberg recognized that if it were possible to measure the momentum and the position simultaneously with a greater accuracy, the quantum mechanics would collapse. So he proposed that it must be impossible. Then people sat down and tried to figure out ways of doing it, and nobody could figure out a way to measure the position and the momentum of anything—a screen, an electron, a billiard ball, anything—with any greater accuracy. Quantum mechanics maintains its perilous but still correct existence.

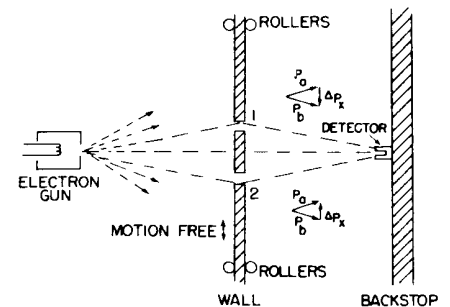


Fig. 1-6. An experiment in which the recoil of the wall is measured.

The Relation of Wave and Particle Viewpoints

2-1 Probability wave amplitudes

In this chapter we shall discuss the relationship of the wave and particle viewpoints. We already know, from the last chapter, that neither the wave viewpoint nor the particle viewpoint is correct. We would always like to present things accurately, or at least precisely enough that they will not have to be changed when we learn more—it may be extended, but it will not be changed! But when we try to talk about the wave picture or the particle picture, both are approximate, and both will change. Therefore what we learn in this chapter will not be accurate in a certain sense; we will deal with some half-intuitive arguments which will be made more precise later. But certain things will be changed a little bit when we interpret them correctly in quantum mechanics. We are doing this so that you can have some qualitative feeling for some quantum phenomena before we get into the mathematical details of quantum mechanics. Furthermore, all our experiences are with waves and with particles, and so it is rather handy to use the wave and particle ideas to get some understanding of what happens in given circumstances before we know the complete mathematics of the quantum-mechanical amplitudes. We shall try to indicate the weakest places as we go along, but most of it is very nearly correct—it is just a matter of interpretation.

First of all, we know that the new way of representing the world in quantum mechanics—the new framework—is to give an amplitude for every event that can occur, and if the event involves the reception of one particle, then we can give the amplitude to find that one particle at different places and at different times. The probability of finding the particle is then proportional to the absolute square of the amplitude. In general, the amplitude to find a particle in different places at different times varies with position and time.

In some special case it can be that the amplitude varies sinusoidally in space and time like $e^{i(\omega t - \mathbf{k} \cdot \mathbf{r})}$, where $\mathbf{r}$ is the vector position from some origin. (Do not forget that these amplitudes are complex numbers, not real numbers.) Such an amplitude varies according to a definite frequency ω and wave number $\mathbf{k}$. Then it turns out that this corresponds to a classical limiting situation where we would have believed that we have a particle whose energy E was known and is related to the frequency by

$$E = \hbar\omega, \quad (2.1)$$

and whose momentum $\mathbf{p}$ is also known and is related to the wave number by

$$\mathbf{p} = \hbar\mathbf{k}. \quad (2.2)$$

(The symbol $\hbar$ represents the number h divided by 2π ; $\hbar = h/2\pi$.)

This means that the idea of a particle is limited. The idea of a particle—its location, its momentum, etc.—which we use so much, is in certain ways unsatisfactory. For instance, if an amplitude to find a particle at different places is given by $e^{i(\omega t - \mathbf{k} \cdot \mathbf{r})}$, whose absolute square is a constant, that would mean that the probability of finding a particle is the same at all points. That means we do not know *where* it is—it can be anywhere—there is a great uncertainty in its location.

On the other hand, if the position of a particle is more or less well known and we can predict it fairly accurately, then the probability of finding it in different places must be confined to a certain region, whose length we call Δx . Outside this region, the probability is zero. Now this probability is the absolute square of an amplitude, and if the absolute square is zero, the amplitude is also zero, so that

2-1 Probability wave amplitudes

2-2 Measurement of position and momentum

2-3 Crystal diffraction

2-4 The size of an atom

2-5 Energy levels

2-6 Philosophical implications

Note: This chapter is almost exactly the same as Chapter 38 of Volume I.

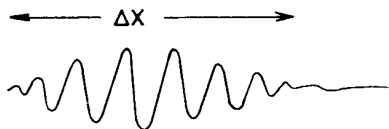


Fig. 2-1. A wave packet of length Δx .

we have a wave train whose length is Δx (Fig. 2-1), and the wavelength (the distance between nodes of the waves in the train) of that wave train is what corresponds to the particle momentum.

Here we encounter a strange thing about waves; a very simple thing which has nothing to do with quantum mechanics strictly. It is something that anybody who works with waves, even if he knows no quantum mechanics, knows: namely, *we cannot define a unique wavelength for a short wave train*. Such a wave train does not *have* a definite wavelength; there is an indefiniteness in the wave number that is related to the finite length of the train, and thus there is an indefiniteness in the momentum.

2-2 Measurement of position and momentum

Let us consider two examples of this idea—to see the reason that there is an uncertainty in the position and/or the momentum, if quantum mechanics is right. We have also seen before that if there were not such a thing—if it were possible to measure the position and the momentum of anything simultaneously—we would have a paradox; it is fortunate that we do not have such a paradox, and the fact that such an uncertainty comes naturally from the wave picture shows that everything is mutually consistent.

Here is one example which shows the relationship between the position and the momentum in a circumstance that is easy to understand. Suppose we have a single slit, and particles are coming from very far away with a certain energy—so that they are all coming essentially horizontally (Fig. 2-2). We are going to concentrate on the vertical components of momentum. All of these particles have a certain horizontal momentum p_0 , say, in a classical sense. So, in the classical sense, the vertical momentum p_y , before the particle goes through the hole, is definitely known. The particle is moving neither up nor down, because it came from a source that is far away—and so the vertical momentum is of course zero. But now let us suppose that it goes through a hole whose width is B . Then after it has come out through the hole, we know the position vertically—the y -position—with considerable accuracy—namely $\pm B/2$.[†] That is, the uncertainty in position, Δy , is of order B . Now we might also want to say, since we known the momentum is absolutely horizontal, that Δp_y is zero; but that is wrong. We *once* knew the momentum was horizontal, but we do not know it any more. Before the particles passed through the hole, we did not know their vertical positions. Now that we have found the vertical position by having the particle come through the hole, we have lost our information on the vertical momentum! Why? According to the wave theory, there is a spreading out, or diffraction, of the waves after they go through the slit, just as for light. Therefore there is a certain probability that particles coming out of the slit are not coming exactly straight. The pattern is spread out by the diffraction effect, and the angle of spread, which we can define as the angle of the first minimum, is a measure of the uncertainty in the final angle.

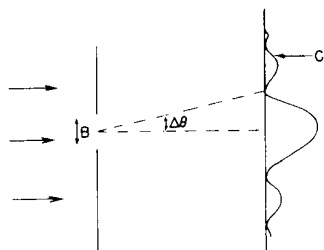


Fig. 2-2. Diffraction of particles passing through a slit.

How does the pattern become spread? To say it is spread means that there is some chance for the particle to be moving up or down, that is, to have a component of momentum up or down. We say *chance* and *particle* because we can detect this diffraction pattern with a particle counter, and when the counter receives the particle, say at C in Fig. 2-2, it receives the *entire* particle, so that, in a classical sense, the particle has a vertical momentum, in order to get from the slit up to C .

To get a rough idea of the spread of the momentum, the vertical momentum p_y has a spread which is equal to $p_0 \Delta\theta$, where p_0 is the horizontal momentum. And how big is $\Delta\theta$ in the spread-out pattern? We know that the first minimum occurs at an angle $\Delta\theta$ such that the waves from one edge of the slit have to travel one wavelength farther than the waves from the other side—we worked that out before (Chapter 30 of Vol. I). Therefore $\Delta\theta$ is λ/B , and so Δp_y in this experiment is $p_0 \lambda/B$. Note that if we make B smaller and make a more accurate measurement

[†] More precisely, the error in our knowledge of y is $\pm B/2$. But we are now only interested in the general idea, so we won't worry about factors of 2.

of the position of the particle, the diffraction pattern gets wider. So the narrower we make the slit, the wider the pattern gets, and the more is the likelihood that we would find that the particle has sidewise momentum. Thus the uncertainty in the vertical momentum is inversely proportional to the uncertainty of y . In fact, we see that the product of the two is equal to $p_0\lambda$. But λ is the wavelength and p_0 is the momentum, and in accordance with quantum mechanics, the wavelength times the momentum is Planck's constant h . So we obtain the rule that the uncertainties in the vertical momentum and in the vertical position have a product of the order h :

$$\Delta y \Delta p_y \approx h. \quad (2.3)$$

We cannot prepare a system in which we know the vertical position of a particle and can predict how it will move vertically with greater certainty than given by (2.3). That is, the uncertainty in the vertical momentum must exceed $h/\Delta y$, where Δy is the uncertainty in our knowledge of the position.

Sometimes people say quantum mechanics is all wrong. When the particle arrived from the left, its vertical momentum was zero. And now that it has gone through the slit, its position is known. Both position and momentum seem to be known with arbitrary accuracy. It is quite true that we can receive a particle, and on reception determine what its position is and what its momentum would have had to have been to have gotten there. That is true, but that is not what the uncertainty relation (2.3) refers to. Equation (2.3) refers to the *predictability* of a situation, not remarks about the *past*. It does no good to say "I knew what the momentum was before it went through the slit, and now I know the position," because now the momentum knowledge is lost. The fact that it went through the slit no longer permits us to predict the vertical momentum. We are talking about a predictive theory, not just measurements after the fact. So we must talk about what we can predict.

Now let us take the thing the other way around. Let us take another example of the same phenomenon, a little more quantitatively. In the previous example we measured the momentum by a classical method. Namely, we considered the direction and the velocity and the angles, etc., so we got the momentum by classical analysis. But since momentum is related to wave number, there exists in nature still another way to measure the momentum of a particle—photon or otherwise—which has no classical analog, because it uses Eq. (2.2). We measure the *wavelengths of the waves*. Let us try to measure momentum in this way.

Suppose we have a grating with a large number of lines (Fig. 2-3), and send a beam of particles at the grating. We have often discussed this problem: if the particles have a definite momentum, then we get a very sharp pattern in a certain direction, because of the interference. And we have also talked about how accurately we can determine that momentum, that is to say, what the resolving power of such a grating is. Rather than derive it again, we refer to Chapter 30 of Volume I, where we found that the relative uncertainty in the wavelength that can be measured with a given grating is $1/Nm$, where N is the number of lines on the grating and m is the order of the diffraction pattern. That is,

$$\Delta\lambda/\lambda = 1/Nm. \quad (2.4)$$

Now formula (2.4) can be rewritten as

$$\Delta\lambda/\lambda^2 = 1/Nm\lambda = 1/L, \quad (2.5)$$

where L is the distance shown in Fig. 2-3. This distance is the difference between the total distance that the particle or wave or whatever it is has to travel if it is reflected from the bottom of the grating, and the distance that it has to travel if it is reflected from the top of the grating. That is, the waves which form the diffraction pattern are waves which come from different parts of the grating. The first ones that arrive come from the bottom end of the grating, from the beginning of the wave train, and the rest of them come from later parts of the wave train, coming from different parts of the grating, until the last one finally arrives, and that involves a point in the wave train a distance L behind the first point. So in order that we

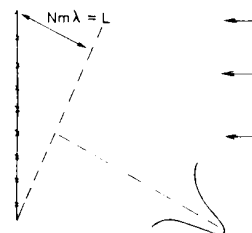


Fig. 2-3. Determination of momentum by using a diffraction grating.

shall have a sharp line in our spectrum corresponding to a definite momentum, with an uncertainty given by (2.4), we have to have a wave train of at least length L . If the wave train is too short, we are not using the entire grating. The waves which form the spectrum are being reflected from only a very short sector of the grating if the wave train is too short, and the grating will not work right—we will find a big angular spread. In order to get a narrower one, we need to use the whole grating, so that at least at some moment the whole wave train is scattering simultaneously from all parts of the grating. Thus the wave train must be of length L in order to have an uncertainty in the wavelength less than that given by (2.5). Incidentally,

$$\Delta\lambda/\lambda^2 = \Delta(1/\lambda) = \Delta k/2\pi. \quad (2.6)$$

Therefore

$$\Delta k = 2\pi/L, \quad (2.7)$$

where L is the length of the wave train.

This means that if we have a wave train whose length is less than L , the uncertainty in the wave number must exceed $2\pi/L$. Or the uncertainty in a wave number times the length of the wave train—we will call that for a moment Δx —exceeds 2π . We call it Δx because that is the uncertainty in the location of the particle. If the wave train exists only in a finite length, then that is where we could find the particle, within an uncertainty Δx . Now this property of waves, that the length of the wave train times the uncertainty of the wave number associated with it is at least 2π , is a property that is known to everyone who studies them. It has nothing to do with quantum mechanics. It is simply that if we have a finite train, we cannot count the waves in it very precisely.

Let us try another way to see the reason for that. Suppose that we have a finite train of length L ; then because of the way it has to decrease at the ends, as in Fig. 2-1, the number of waves in the length L is uncertain by something like ± 1 . But the number of waves in L is $kL/2\pi$. Thus k is uncertain, and we again get the result (2.7), a property merely of waves. The same thing works whether the waves are in space and k is the number of radians per centimeter and L is the length of the train, or the waves are in time and ω is the number of oscillations per second and T is the “length” in time that the wave train comes in. That is, if we have a wave train lasting only for a certain finite time T , then the uncertainty in the frequency is given by

$$\Delta\omega = 2\pi/T. \quad (2.8)$$

We have tried to emphasize that these are properties of waves alone, and they are well known, for example, in the theory of sound.

The point is that in quantum mechanics we interpret the wave number as being a measure of the momentum of a particle, with the rule that $p = \hbar k$, so that relation (2.7) tells us that $\Delta p \approx \hbar/\Delta x$. This, then, is a limitation of the classical idea of momentum. (Naturally, it has to be limited in some ways if we are going to represent particles by waves!) It is nice that we have found a rule that gives us some idea of when there is a failure of classical ideas.

2-3 Crystal diffraction

Next let us consider the reflection of particle waves from a crystal. A crystal is a thick thing which has a whole lot of similar atoms—we will include some complications later—in a nice array. The question is how to set the array so that we get a strong reflected maximum in a given direction for a given beam of, say, light (x-rays), electrons, neutrons, or anything else. In order to obtain a strong reflection, the scattering from all of the atoms must be in phase. There cannot be equal numbers in phase and out of phase, or the waves will cancel out. The way to arrange things is to find the regions of constant phase, as we have already explained; they are planes which make equal angles with the initial and final directions (Fig. 2-4).

If we consider two parallel planes, as in Fig. 2-4, the waves scattered from the two planes will be in phase, provided the difference in distance traveled by a wave

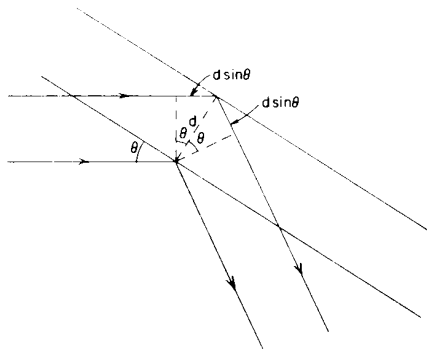


Fig. 2-4. Scattering of waves by crystal planes.

front is an integral number of wavelengths. This difference can be seen to be $2d \sin \theta$, where d is the perpendicular distance between the planes. Thus the condition for coherent reflection is

$$2d \sin \theta = n\lambda \quad (n = 1, 2, \dots). \quad (2.9)$$

If, for example, the crystal is such that the atoms happen to lie on planes obeying condition (2.9) with $n = 1$, then there will be a strong reflection. If, on the other hand, there are other atoms of the same nature (equal in density) halfway between, then the intermediate planes will also scatter equally strongly and will interfere with the others and produce no effect. So d in (2.9) must refer to *adjacent* planes; we cannot take a plane five layers farther back and use this formula!

As a matter of interest, actual crystals are not usually as simple as a single kind of atom repeated in a certain way. Instead, if we make a two-dimensional analog, they are much like wallpaper, in which there is some kind of figure which repeats all over the wallpaper. By "figure" we mean, in the case of atoms, some arrangement—calcium and a carbon and three oxygens, etc., for calcium carbonate, and so on—which may involve a relatively large number of atoms. But whatever it is, the figure is repeated in a pattern. This basic figure is called a *unit cell*.

The basic pattern of repetition defines what we call the *lattice type*; the lattice type can be immediately determined by looking at the reflections and seeing what their symmetry is. In other words, where we find any reflections *at all* determines the lattice type, but in order to determine what is in each of the elements of the lattice one must take into account the *intensity* of the scattering at the various directions. *Which* directions scatter depends on the type of lattice, but *how strongly* each scatters is determined by what is inside each unit cell, and in that way the structure of crystals is worked out.

Two photographs of x-ray diffraction patterns are shown in Figs. 2-5 and 2-6; they illustrate scattering from rock salt and myoglobin, respectively.

Incidentally, an interesting thing happens if the spacings of the nearest planes are less than $\lambda/2$. In this case (2.9) has no solution for n . Thus if λ is bigger than twice the distance between adjacent planes, then there is no side diffraction pattern, and the light—or whatever it is—will go right through the material without bouncing off or getting lost. So in the case of light, where λ is much bigger than the spacing, of course it does go through and there is no pattern of reflection from the planes of the crystal.

This fact also has an interesting consequence in the case of piles which make neutrons (these are obviously particles, for anybody's money!). If we take these neutrons and let them into a long block of graphite, the neutrons diffuse and work their way along (Fig. 2-7). They diffuse because they are bounced by the atoms, but strictly, in the wave theory, they are bounced by the atoms because of diffraction from the crystal planes. It turns out that if we take a very long piece of graphite, the neutrons that come out the far end are all of long wavelength! In fact, if one plots the intensity as a function of wavelength, we get nothing except for wavelengths longer than a certain minimum (Fig. 2-8). In other words, we can get very slow neutrons that way. Only the slowest neutrons come through; they are not diffracted or scattered by the crystal planes of the graphite, but keep going right through like light through glass, and are not scattered out the sides. There are many other demonstrations of the reality of neutron waves and waves of other particles.

2-4 The size of an atom

We now consider another application of the uncertainty relation, Eq. (2.3). It must not be taken too seriously; the idea is right but the analysis is not very accurate. The idea has to do with the determination of the size of atoms, and the fact that, classically, the electrons would radiate light and spiral in until they settle down right on top of the nucleus. But that cannot be right quantum-mechanically because then we would know where each electron was and how fast it was moving.

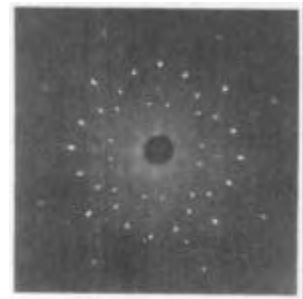


Fig. 2-5. The pattern produced by the diffraction of a beam of x-rays in a crystal of sodium chloride.

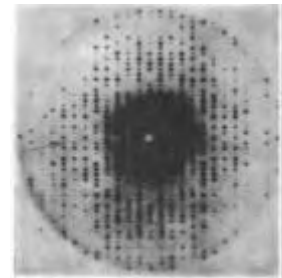


Fig. 2-6. The x-ray diffraction pattern of myoglobin.

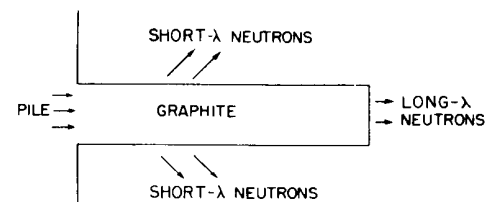


Fig. 2-7. Diffusion of pile neutrons through graphite block.

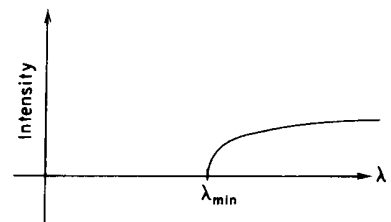


Fig. 2-8. Intensity of neutrons out of graphite rod as function of wavelength.

spectral frequencies was noted before quantum mechanics was discovered, and it is called the *Ritz combination principle*. This is again a mystery from the point of view of classical mechanics. Let us not belabor the point that classical mechanics is a failure in the atomic domain; we seem to have demonstrated that pretty well.

We have already talked about quantum mechanics as being represented by amplitudes which behave like waves, with certain frequencies and wave numbers. Let us observe how it comes about from the point of view of amplitudes that the atom has definite energy states. This is something we cannot understand from what has been said so far, but we are all familiar with the fact that confined waves have definite frequencies. For instance, if sound is confined to an organ pipe, or anything like that, then there is more than one way that the sound can vibrate, but for each such way there is a definite frequency. Thus an object in which the waves are confined has certain resonance frequencies. It is therefore a property of waves in a confined space—a subject which we will discuss in detail with formulas later on—that they exist only at definite frequencies. And since the general relation exists between frequencies of the amplitude and energy, we are not surprised to find definite energies associated with electrons bound in atoms.

2-6 Philosophical implications

Let us consider briefly some philosophical implications of quantum mechanics. As always, there are two aspects of the problem: one is the philosophical implication for physics, and the other is the extrapolation of philosophical matters to other fields. When philosophical ideas associated with science are dragged into another field, they are usually completely distorted. Therefore we shall confine our remarks as much as possible to physics itself.

First of all, the most interesting aspect is the idea of the uncertainty principle; making an observation affects the phenomenon. It has always been known that making observations affects a phenomenon, but the point is that the effect cannot be disregarded or minimized or decreased arbitrarily by rearranging the apparatus. When we look for a certain phenomenon we cannot help but disturb it in a certain minimum way, and *the disturbance is necessary for the consistency of the viewpoint*. The observer was sometimes important in prequantum physics, but only in a trivial sense. The problem has been raised: if a tree falls in a forest and there is nobody there to hear it, does it make a noise? A *real* tree falling in a *real* forest makes a sound, of course, even if nobody is there. Even if no one is present to hear it, there are other traces left. The sound will shake some leaves, and if we were careful enough we might find somewhere that some thorn had rubbed against a leaf and made a tiny scratch that could not be explained unless we assumed the leaf were vibrating. So in a certain sense we would have to admit that there is a sound made. We might ask: was there a *sensation* of sound? No, sensations have to do, presumably, with consciousness. And whether ants are conscious and whether there were ants in the forest, or whether the tree was conscious, we do not know. Let us leave the problem in that form.

Another thing that people have emphasized since quantum mechanics was developed is the idea that we should not speak about those things which we cannot measure. (Actually relativity theory also said this.) Unless a thing can be defined by measurement, it has no place in a theory. And since an accurate value of the momentum of a localized particle cannot be defined by measurement it therefore has no place in the theory. The idea that this is what was the matter with classical theory *is a false position*. It is a careless analysis of the situation. Just because we cannot *measure* position and momentum precisely does not *a priori* mean that we *cannot* talk about them. It only means that we *need* not talk about them. The situation in the sciences is this: A concept or an idea which cannot be measured or cannot be referred directly to experiment may or may not be useful. It need not exist in a theory. In other words, suppose we compare the classical theory of the world with the quantum theory of the world, and suppose that it is true experimentally that we can measure position and momentum only imprecisely. The question is whether the *ideas* of the exact position of a particle and the exact

momentum of a particle are valid or not. The classical theory admits the ideas; the quantum theory does not. This does not in itself mean that classical physics is wrong. When the new quantum mechanics was discovered, the classical people—which included everybody except Heisenberg, Schrödinger, and Born—said: “Look, your theory is not any good because you cannot answer certain questions like: what is the exact position of a particle?, which hole does it go through?, and some others.” Heisenberg’s answer was: “I do not need to answer such questions because you cannot ask such a question experimentally.” It is that we do not *have* to. Consider two theories (a) and (b); (a) contains an idea that cannot be checked directly but which is used in the analysis, and the other, (b), does not contain the idea. If they disagree in their predictions, one could not claim that (b) is false because it cannot explain this idea that is in (a), because that idea is one of the things that cannot be checked directly. It is always good to know which ideas cannot be checked directly, but it is not necessary to remove them all. It is not true that we can pursue science completely by using only those concepts which are directly subject to experiment.

In quantum mechanics itself there is a probability amplitude, there is a potential, and there are many constructs that we cannot measure directly. The basis of a science is its ability to *predict*. To predict means to tell what will happen in an experiment that has never been done. How can we do that? By assuming that we know what is there, independent of the experiment. We must extrapolate the experiments to a region where they have not been done. We must take our concepts and extend them to places where they have not yet been checked. If we do not do that, we have no prediction. So it was perfectly sensible for the classical physicists to go happily along and suppose that the position—which obviously means something for a baseball—meant something also for an electron. It was not stupidity. It was a sensible procedure. Today we say that the law of relativity is supposed to be true at all energies, but someday somebody may come along and say how stupid we were. We do not know where we are “stupid” until we “stick our neck out,” and so the whole idea is to put our neck out. And the only way to find out that we are wrong is to find out *what* our predictions are. It is absolutely necessary to make constructs.

We have already made a few remarks about the indeterminacy of quantum mechanics. That is, that we are unable now to predict what will happen in physics in a given physical circumstance which is arranged as carefully as possible. If we have an atom that is in an excited state and so is going to emit a photon, we cannot say *when* it will emit the photon. It has a certain amplitude to emit the photon at any time, and we can predict only a probability for emission; we cannot predict the future exactly. This has given rise to all kinds of nonsense and questions on the meaning of freedom of will, and of the idea that the world is uncertain.

Of course we must emphasize that classical physics is also indeterminate, in a sense. It is usually thought that this indeterminacy, that we cannot predict the future, is an important quantum-mechanical thing, and this is said to explain the behavior of the mind, feelings of free will, etc. But if the world *were* classical—if the laws of mechanics were classical—it is not quite obvious that the mind would not feel more or less the same. It is true classically that if we knew the position and the velocity of every particle in the world, or in a box of gas, we could predict exactly what would happen. And therefore the classical world is deterministic. Suppose, however, that we have a finite accuracy and do not know *exactly* where just one atom is, say to one part in a billion. Then as it goes along it hits another atom, and because we did not know the position better than to one part in a billion, we find an even larger error in the position after the collision. And that is amplified, of course, in the next collision, so that if we start with only a tiny error it rapidly magnifies to a very great uncertainty. To give an example: if water falls over a dam, it splashes. If we stand nearby, every now and then a drop will land on our nose. This appears to be completely random, yet such a behavior would be predicted by purely classical laws. The exact position of all the drops depends upon the precise wiggles of the water before it goes over the dam. How? The tiniest irregularities are magnified in falling, so that we get complete randomness. Ob-

viously, we cannot really predict the position of the drops unless we know the motion of the water *absolutely exactly*.

Speaking more precisely, given an arbitrary accuracy, no matter how precise, one can find a time long enough that we cannot make predictions valid for that long a time. Now the point is that this length of time is not very large. It is not that the time is millions of years if the accuracy is one part in a billion. The time goes, in fact, only logarithmically with the error, and it turns out that in only a very, very tiny time we lose all our information. If the accuracy is taken to be one part in billions and billions and billions—no matter how many billions we wish, provided we do stop somewhere—then we can find a time less than the time it took to state the accuracy—after which we can no longer predict what is going to happen! It is therefore not fair to say that from the apparent freedom and indeterminacy of the human mind, we should have realized that classical “deterministic” physics could not ever hope to understand it, and to welcome quantum mechanics as a release from a “completely mechanistic” universe. For already in classical mechanics there was indeterminability from a practical point of view.

Probability Amplitudes

3-1 The laws for combining amplitudes

When Schrödinger first discovered the correct laws of quantum mechanics, he wrote an equation which described the amplitude to find a particle in various places. This equation was very similar to the equations that were already known to classical physicists—equations that they had used in describing the motion of air in a sound wave, the transmission of light, and so on. So most of the time at the beginning of quantum mechanics was spent in solving this equation. But at the same time an understanding was being developed, particularly by Born and Dirac, of the basically new physical ideas behind quantum mechanics. As quantum mechanics developed further, it turned out that there were a large number of things which were not directly encompassed in the Schrödinger equation—such as the spin of the electron, and various relativistic phenomena. Traditionally, all courses in quantum mechanics have begun in the same way, retracing the path followed in the historical development of the subject. One first learns a great deal about classical mechanics so that he will be able to understand how to solve the Schrödinger equation. Then he spends a long time working out various solutions. Only after a detailed study of this equation does he get to the “advanced” subject of the electron’s spin.

We had also originally considered that the right way to conclude these lectures on physics was to show how to solve the equations of classical physics in complicated situations—such as the description of sound waves in enclosed regions, modes of electromagnetic radiation in cylindrical cavities, and so on. That was the original plan for this course. However, we have decided to abandon that plan and to give instead an introduction to the quantum mechanics. We have come to the conclusion that what are usually called the advanced parts of quantum mechanics are, in fact, quite simple. The mathematics that is involved is particularly simple, involving simple algebraic operations and no differential equations or at most only very simple ones. The only problem is that we must jump the gap of no longer being able to describe the behavior *in detail* of particles in space. So this is what we are going to try to do: to tell you about what conventionally would be called the “advanced” parts of quantum mechanics. But they are, we assure you, by all odds the simplest parts—in a deep sense of the word—as well as the most basic parts. This is frankly a pedagogical experiment; it has never been done before, as far as we know.

In this subject we have, of course, the difficulty that the quantum mechanical behavior of things is quite strange. Nobody has an everyday experience to lean on to get a rough, intuitive idea of what will happen. So there are two ways of presenting the subject: We could either describe what can happen in a rather rough physical way, telling you more or less what happens without giving the precise laws of everything; or we could, on the other hand, give the precise laws in their abstract form. But, then because of the abstractions, you wouldn’t know what they were all about, physically. The latter method is unsatisfactory because it is completely abstract, and the first way leaves an uncomfortable feeling because one doesn’t know exactly what is true and what is false. We are not sure how to overcome this difficulty. You will notice, in fact, that Chapters 1 and 2 showed this problem. The first chapter was relatively precise; but the second chapter was a rough description of the characteristics of different phenomena. Here, we will try to find a happy medium between the two extremes.

3-1 The laws for combining amplitudes

3-2 The two-slit interference pattern

3-3 Scattering from a crystal

3-4 Identical particles

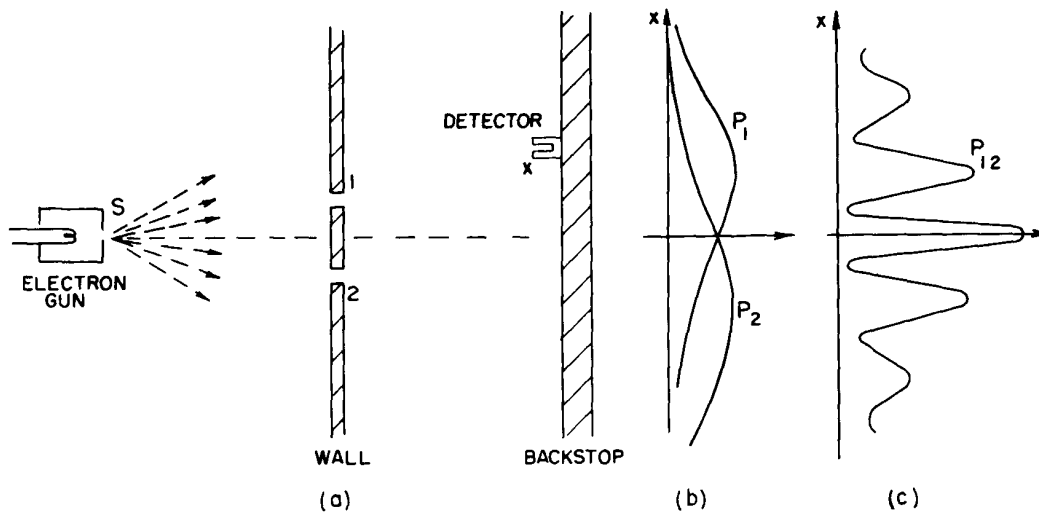


Fig 3-1. Interference experiment with electrons.

We will begin in this chapter by dealing with some general quantum mechanical ideas. Some of the statements will be quite precise, others only partially precise. It will be hard to tell you as we go along which is which, but by the time you have finished the rest of the book, you will understand in looking back which parts hold up and which parts were only explained roughly. The chapters which follow this one will not be so imprecise. In fact, one of the reasons we have tried carefully to be precise in the succeeding chapters is so that we can show you one of the most beautiful things about quantum mechanics—how much can be deduced from so little.

We begin by discussing again the superposition of *probability amplitudes*. As an example we will refer to the experiment described in Chapter 1, and shown again here in Fig. 3-1. There is a source s of particles, say electrons; then there is a wall with two slits in it; after the wall, there is a detector located at some position x . We ask for the probability that a particle will be found at x . Our *first general principle* in quantum mechanics is that the *probability* that a particle will arrive at x , when let out at the source s , can be represented quantitatively by the absolute square of a complex number called a *probability amplitude*—in this case, the “amplitude that a particle from s will arrive at x .” We will use such amplitudes so frequently that we will use a shorthand notation—invented by Dirac and generally used in quantum mechanics—to represent this idea. We write the probability amplitude this way:

$$\langle \text{Particle arrives at } x \mid \text{particle leaves } s \rangle. \quad (3.1)$$

In other words, the two brackets $\langle \rangle$ are a sign equivalent to “the amplitude that”; the expression at the *right* of the vertical line always gives the *starting* condition, and the one at the left, the *final* condition. Sometimes it will also be convenient to abbreviate still more and describe the initial and final conditions by single letters. For example, we may on occasion write the amplitude (3.1) as

$$\langle x \mid s \rangle. \quad (3.2)$$

We want to emphasize that such an amplitude is, of course, just a single number—a *complex* number.

We have already seen in the discussion of Chapter 1 that when there are two ways for the particle to reach the detector, the resulting probability is not the sum of the two probabilities, but must be written as the absolute square of the sum of two amplitudes. We had that the probability that an electron arrives at the detector when both paths are open is

$$P_{12} = |\phi_1 + \phi_2|^2. \quad (3.3)$$

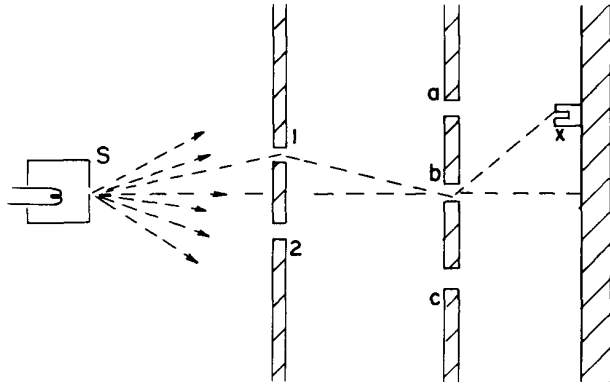


Fig. 3-2. A more complicated interference experiment.

We wish now to put this result in terms of our new notation. First, however, we want to state our *second general principle* of quantum mechanics: When a particle can reach a given state by two possible routes, the total amplitude for the process is the *sum of the amplitudes* for the two routes considered separately. In our new notation we write that

$$\langle x | s \rangle_{\text{both holes open}} = \langle x | s \rangle_{\text{through 1}} + \langle x | s \rangle_{\text{through 2}}. \quad (3.4)$$

Incidentally, we are going to suppose that the holes 1 and 2 are small enough that when we say an electron goes through the hole, we don't have to discuss which part of the hole. We could, of course, split each hole into pieces with a certain amplitude that the electron goes to the top of the hole and the bottom of the hole and so on. We will suppose that the hole is small enough so that we don't have to worry about this detail. That is part of the roughness involved; the matter can be made more precise, but we don't want to do so at this stage.

Now we want to write out in more detail what we can say about the amplitude for the process in which the electron reaches the detector at x by way of hole 1. We can do that by using our *third general principle*: When a particle goes by some particular route the amplitude for that route can be written as the *product* of the *amplitude* to go part way with the *amplitude* to go the rest of the way. For the setup of Fig. 3-1 the amplitude to go from s to x by way of hole 1 is equal to the amplitude to go from s to 1, multiplied by the amplitude to go from 1 to x .

$$\langle x | s \rangle_{\text{via 1}} = \langle x | 1 \rangle \langle 1 | s \rangle. \quad (3.5)$$

Again this result is not completely precise. We should also include a factor for the amplitude that the electron will get through the hole at 1; but in the present case it is a simple hole, and we will take this factor to be unity.

You will note that Eq. (3.5) appears to be written in reverse order. It is to be read from right to left: The electron goes from s to 1 and then from 1 to x . In summary, if events occur in succession—that is, if you can analyze one of the routes of the particle by saying it does this, then it does this, then it does that—the resultant amplitude for that route is calculated by multiplying in succession the amplitude for each of the successive events. Using this law we can rewrite Eq. (3.4) as

$$\langle x | s \rangle_{\text{both}} = \langle x | 1 \rangle \langle 1 | s \rangle + \langle x | 2 \rangle \langle 2 | s \rangle.$$

Now we wish to show that just using these principles we can calculate a much more complicated problem like the one shown in Fig. 3-2. Here we have two walls, one with two holes, 1 and 2, and another which has three holes, a , b , and c . Behind the second wall there is a detector at x , and we want to know the amplitude for a particle to arrive there. Well, one way you can find this is by calculating the superposition, or interference, of the waves that go through; but you can also do it by saying that there are six possible routes and superposing an amplitude for each. The electron can go through hole 1, then through hole a , and then to x ; or it could go through hole 1, then through hole b , and then to x ; and so on. According to our second principle, the amplitudes for alternative routes add, so we should

be able to write the amplitude from s to x as a sum of six separate amplitudes. On the other hand, using the third principle, each of these separate amplitudes can be written as a product of three amplitudes. For example, one of them is the amplitude for s to 1, times the amplitude for 1 to a , times the amplitude for a to x . Using our shorthand notation, we can write the complete amplitude to go from s to x as

$$\langle x | s \rangle = \langle x | a \rangle \langle a | 1 \rangle \langle 1 | s \rangle + \langle x | b \rangle \langle b | 1 \rangle \langle 1 | s \rangle + \cdots + \langle x | c \rangle \langle c | 2 \rangle \langle 2 | s \rangle.$$

We can save writing by using the summation notation

$$\langle x | s \rangle = \sum_{\substack{i=1,2 \\ \alpha=a,b,c}} \langle x | \alpha \rangle \langle \alpha | i \rangle \langle i | s \rangle. \quad (3.6)$$

In order to make any calculations using these methods, it is, naturally, necessary to know the amplitude to get from one place to another. We will give a rough idea of a typical amplitude. It leaves out certain things like the polarization of light or the spin of the electron, but aside from such features it is quite accurate. We give it so that you can solve problems involving various combinations of slits. Suppose a particle with a definite energy is going in empty space from a location $\mathbf{r}_1$ to a location $\mathbf{r}_2$. In other words, it is a free particle with no forces on it. Except for a numerical factor in front, the amplitude to go from $\mathbf{r}_1$ to $\mathbf{r}_2$ is

$$\langle \mathbf{r}_2 | \mathbf{r}_1 \rangle = \frac{e^{i\mathbf{p} \cdot \mathbf{r}_{12} / \hbar}}{r_{12}}, \quad (3.7)$$

where $\mathbf{r}_{12} = \mathbf{r}_2 - \mathbf{r}_1$, and $\mathbf{p}$ is the momentum which is related to the energy E by the relativistic equation

$$p^2 c^2 = E^2 - (m_0 c^2)^2,$$

or the nonrelativistic equation

$$\frac{p^2}{2m} = \text{Kinetic energy}.$$

Equation (3.7) says in effect that the particle has wavelike properties, the amplitude propagating as a wave with a wave number equal to the momentum divided by $\hbar$.

In the most general case, the amplitude and the corresponding probability will also involve the time. For most of these initial discussions we will suppose that the source always emits the particles with a given energy so we will not need to worry about the time. But we could, in the general case, be interested in some other questions. Suppose that a particle is liberated at a certain place P at a certain time, and you would like to know the amplitude for it to arrive at some location, say $\mathbf{r}$, at some later time. This could be represented symbolically as the amplitude $\langle \mathbf{r}, t = t_1 | P, t = 0 \rangle$. Clearly, this will depend upon both $\mathbf{r}$ and t . You will get different results if you put the detector in different places and measure at different times. This function of $\mathbf{r}$ and t , in general, satisfies a differential equation which is a wave equation. For example, in a nonrelativistic case it is the Schrödinger equation. One has then a wave equation analogous to the equation for electromagnetic waves or waves of sound in a gas. However, it must be emphasized that the wave function that satisfies the equation is not like a real wave in space; one cannot picture any kind of reality to this wave as one does for a sound wave.

Although one may be tempted to think in terms of "particle waves" when dealing with one particle, it is not a good idea, for if there are, say, two particles, the amplitude to find one at $\mathbf{r}_1$ and the other at $\mathbf{r}_2$ is not a simple wave in three-dimensional space, but depends on the six space variables $\mathbf{r}_1$ and $\mathbf{r}_2$. If we are, for example, dealing with two (or more) particles, we will need the following additional principle: Provided that the two particles do not interact, the amplitude that one particle will do one thing *and* the other one something else is the product of the two amplitudes that the two particles would do the two things separately. For example, if $\langle a | s_1 \rangle$ is the amplitude for particle 1 to go from s_1 to a , and $\langle b | s_2 \rangle$

is the amplitude for particle 2 to go from s_2 to b , the amplitude that *both* things will happen together is

$$\langle a | s_1 \rangle \langle b | s_2 \rangle.$$

There is one more point to emphasize. Suppose that we didn't know where the particles in Fig. 3-2 come from before arriving at holes 1 and 2 of the first wall. We can still make a prediction of what will happen beyond the wall (for example, the amplitude to arrive at x) provided that we are given two numbers: the amplitude to have arrived at 1 and the amplitude to have arrived at 2. In other words, because of the fact that the amplitude for successive events multiplies, as shown in Eq. (3.6), all you need to know to continue the analysis is two numbers—in this particular case $\langle 1 | s \rangle$ and $\langle 2 | s \rangle$. These two complex numbers are enough to predict all the future. That is what really makes quantum mechanics easy. It turns out that in later chapters we are going to do just such a thing when we specify a starting condition in terms of two (or a few) numbers. Of course, these numbers depend upon where the source is located and possibly other details about the apparatus, but given the two numbers, we do not need to know any more about such details.

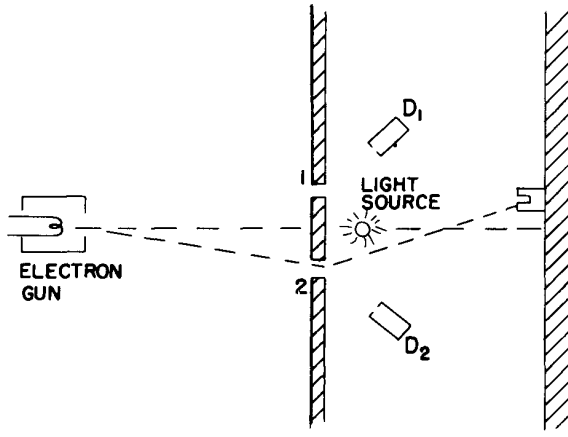


Fig. 3-3. An experiment to determine which hole the electron goes through.

3-2 The two-slit interference pattern

Now we would like to consider a matter which was discussed in some detail in Chapter 1. This time we will do it with the full glory of the amplitude idea to show you how it works out. We take the same experiment shown in Fig. 3-1, but now with the addition of a light source behind the two holes, as shown in Fig. 3-3. In Chapter 1, we discovered the following interesting result. If we looked behind slit 1 and saw a photon scattered from there, then the distribution obtained for the electrons at x in coincidence with these photons was the same as though slit 2 were closed. The total distribution for electrons that had been “seen” at either slit 1 or slit 2 was the sum of the separate distributions and was completely different from the distribution with the light turned off. This was true at least if we used light of short enough wavelength. If the wavelength was made longer so we could not be sure at which hole the scattering had occurred, the distribution became more like the one with the light turned off.

Let's examine what is happening by using our new notation and the principles of combining amplitudes. To simplify the writing, we can again let ϕ_1 stand for the amplitude that the electron will arrive at x by way of hole 1, that is,

$$\phi_1 = \langle x | 1 \rangle \langle 1 | s \rangle.$$

Similarly, we'll let ϕ_2 stand for the amplitude that the electron gets to the detector by way of hole 2:

$$\phi_2 = \langle x | 2 \rangle \langle 2 | s \rangle.$$

These are the amplitudes to go through the two holes and arrive at x if there is no light. Now if there is light, we ask ourselves the question: What is the amplitude for the process in which the electron starts at s and a photon is liberated by the

light source L , ending with the electron at x and a photon seen behind slit 1? Suppose that we observe the photon behind slit 1 by means of a detector D_1 , as shown in Fig. 3-3, and use a similar detector D_2 to count photons scattered behind hole 2. There will be an amplitude for a photon to arrive at D_1 and an electron at x , and also an amplitude for a photon to arrive at D_2 and an electron at x . Let's try to calculate them.

Although we don't have the correct mathematical formula for all the factors that go into this calculation, you will see the spirit of it in the following discussion. First, there is the amplitude $\langle 1 | s \rangle$ that an electron goes from the source to hole 1. Then we can suppose that there is a certain amplitude that while the electron is at hole 1 it scatters a photon into the detector D_1 . Let us represent this amplitude by a . Then there is the amplitude $\langle x | 1 \rangle$ that the electron goes from slit 1 to the electron detector at x . The amplitude that the electron goes from s to x via slit 1 and scatters a photon into D_1 is then

$$\langle x | 1 \rangle a \langle 1 | s \rangle.$$

Or, in our previous notation, it is just $a\phi_1$.

There is also some amplitude that an electron going through slit 2 will scatter a photon into counter D_1 . You say, "That's impossible; how can it scatter into counter D_1 if it is only looking at hole 1?" If the wavelength is long enough, there are diffraction effects, and it is certainly possible. If the apparatus is built well and if we use photons of short wavelength, then the amplitude that a photon will be scattered into detector 1, from an electron at 2 is very small. But to keep the discussion general we want to take into account that there is always some such amplitude, which we will call b . Then the amplitude that an electron goes via slit 2 and scatters a photon into D_1 is

$$\langle x | 2 \rangle b \langle 2 | s \rangle = b\phi_2.$$

The amplitude to find the electron at x and the photon in D_1 is the sum of two terms, one for each possible path for the electron. Each term is in turn made up of two factors: first, that the electron went through a hole, and second, that the photon is scattered by such an electron into detector 1; we have

$$\left\langle \begin{array}{l} \text{electron at } x \\ \text{photon at } D_1 \end{array} \middle| \begin{array}{l} \text{electron from } s \\ \text{photon from } L \end{array} \right\rangle = a\phi_1 + b\phi_2. \quad (3.8)$$

We can get a similar expression when the photon is found in the other detector D_2 . If we assume for simplicity that the system is symmetrical, then a is also the amplitude for a photon in D_2 when an electron passes through hole 2, and b is the amplitude for a photon in D_2 when the electron passes through hole 1. The corresponding total amplitude for a photon at D_2 and an electron at x is

$$\left\langle \begin{array}{l} \text{electron at } x \\ \text{photon at } D_2 \end{array} \middle| \begin{array}{l} \text{electron from } s \\ \text{photon from } L \end{array} \right\rangle = a\phi_2 + b\phi_1. \quad (3.9)$$

Now we are finished. We can easily calculate the probability for various situations. Suppose that we want to know with what probability we get a count in D_1 and an electron at x . That will be the absolute square of the amplitude given in Eq. (3.8), namely, just $|a\phi_1 + b\phi_2|^2$. Let's look more carefully at this expression. First of all, if b is zero—which is the way we would like to design the apparatus—then the answer is simply $|\phi_1|^2$ diminished in total amplitude by the factor $|a|^2$. This is the probability distribution that you would get if there were only one hole—as shown in the graph of Fig. 3-4(a). On the other hand, if the wavelength is very long, the scattering behind hole 2 into D_1 may be just about the same as for hole 1. Although there may be some phases involved in a and b , we can ask about a simple case in which the two phases are equal. If a is practically equal to b , then the total probability becomes $|\phi_1 + \phi_2|^2$ multiplied by $|a|^2$, since the common factor a can be taken out. This, however, is just the probability

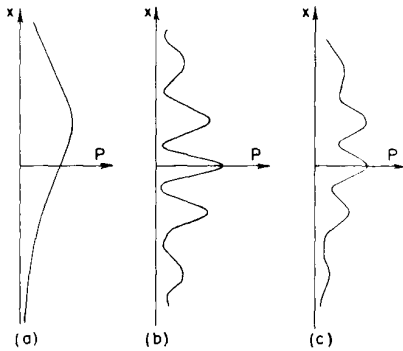


Fig. 3-4. The probability of counting an electron at x in coincidence with a photon at D in the experiment of Fig. 3-3: (a) for $b = 0$; (b) for $b = a$; (c) for $0 < b < a$.

light source L , ending with the electron at x and a photon seen behind slit 1? Suppose that we observe the photon behind slit 1 by means of a detector D_1 , as shown in Fig. 3-3, and use a similar detector D_2 to count photons scattered behind hole 2. There will be an amplitude for a photon to arrive at D_1 and an electron at x , and also an amplitude for a photon to arrive at D_2 and an electron at x . Let's try to calculate them.

Although we don't have the correct mathematical formula for all the factors that go into this calculation, you will see the spirit of it in the following discussion. First, there is the amplitude $\langle 1 | s \rangle$ that an electron goes from the source to hole 1. Then we can suppose that there is a certain amplitude that while the electron is at hole 1 it scatters a photon into the detector D_1 . Let us represent this amplitude by a . Then there is the amplitude $\langle x | 1 \rangle$ that the electron goes from slit 1 to the electron detector at x . The amplitude that the electron goes from s to x via slit 1 and scatters a photon into D_1 is then

$$\langle x | 1 \rangle a \langle 1 | s \rangle.$$

Or, in our previous notation, it is just $a\phi_1$.

There is also some amplitude that an electron going through slit 2 will scatter a photon into counter D_1 . You say, "That's impossible; how can it scatter into counter D_1 if it is only looking at hole 1?" If the wavelength is long enough, there are diffraction effects, and it is certainly possible. If the apparatus is built well and if we use photons of short wavelength, then the amplitude that a photon will be scattered into detector 1, from an electron at 2 is very small. But to keep the discussion general we want to take into account that there is always some such amplitude, which we will call b . Then the amplitude that an electron goes via slit 2 and scatters a photon into D_1 is

$$\langle x | 2 \rangle b \langle 2 | s \rangle = b\phi_2.$$

The amplitude to find the electron at x and the photon in D_1 is the sum of two terms, one for each possible path for the electron. Each term is in turn made up of two factors: first, that the electron went through a hole, and second, that the photon is scattered by such an electron into detector 1; we have

$$\left\langle \begin{array}{l} \text{electron at } x \\ \text{photon at } D_1 \end{array} \middle| \begin{array}{l} \text{electron from } s \\ \text{photon from } L \end{array} \right\rangle = a\phi_1 + b\phi_2. \quad (3.8)$$

We can get a similar expression when the photon is found in the other detector D_2 . If we assume for simplicity that the system is symmetrical, then a is also the amplitude for a photon in D_2 when an electron passes through hole 2, and b is the amplitude for a photon in D_2 when the electron passes through hole 1. The corresponding total amplitude for a photon at D_2 and an electron at x is

$$\left\langle \begin{array}{l} \text{electron at } x \\ \text{photon at } D_2 \end{array} \middle| \begin{array}{l} \text{electron from } s \\ \text{photon from } L \end{array} \right\rangle = a\phi_2 + b\phi_1. \quad (3.9)$$

Now we are finished. We can easily calculate the probability for various situations. Suppose that we want to know with what probability we get a count in D_1 and an electron at x . That will be the absolute square of the amplitude given in Eq. (3.8), namely, just $|a\phi_1 + b\phi_2|^2$. Let's look more carefully at this expression. First of all, if b is zero—which is the way we would like to design the apparatus—then the answer is simply $|\phi_1|^2$ diminished in total amplitude by the factor $|a|^2$. This is the probability distribution that you would get if there were only one hole—as shown in the graph of Fig. 3-4(a). On the other hand, if the wavelength is very long, the scattering behind hole 2 into D_1 may be just about the same as for hole 1. Although there may be some phases involved in a and b , we can ask about a simple case in which the two phases are equal. If a is practically equal to b , then the total probability becomes $|\phi_1 + \phi_2|^2$ multiplied by $|a|^2$, since the common factor a can be taken out. This, however, is just the probability

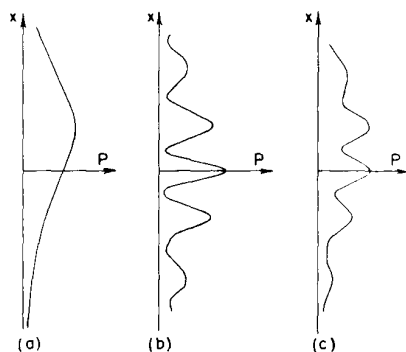


Fig. 3-4. The probability of counting an electron at x in coincidence with a photon at D in the experiment of Fig. 3-3: (a) for $b = 0$; (b) for $b = a$; (c) for $0 < b < a$.

distribution we would have gotten without the photons at all. Therefore, in the case that the wavelength is very long—and the photon detection ineffective—you return to the original distribution curve which shows interference effects, as shown in Fig. 3-4(b). In the case that the detection is partially effective, there is an interference between a lot of ϕ_1 and a little of ϕ_2 , and you will get an intermediate distribution such as is sketched in Fig. 3-4(c). Needless to say, if we look for coincidence counts of photons at D_2 and electrons at x , we will get the same kinds of results. If you remember the discussion in Chapter 1, you will see that these results give a quantitative description of what was described there.

Now we would like to emphasize an important point so that you will avoid a common error. Suppose that you only want the amplitude that the electron arrives at x , *regardless* of whether the photon was counted at D_1 or D_2 . Should you add the amplitudes given in Eqs. (3.8) and (3.9)? No! You must *never add amplitudes for different and distinct final states*. Once the photon is accepted by one of the photon counters, we can always determine which alternative occurred if we want, without any further disturbance to the system. Each alternative has a probability completely independent of the other. To repeat, do not add amplitudes for different *final* conditions, where by “final” we mean at that moment the *probability* is desired—that is, when the experiment is “finished.” You do add the amplitudes for the different *indistinguishable* alternatives inside the experiment, before the complete process is finished. At the end of the process you may say that you “don’t want to look at the photon.” That’s your business, but you still do not add the amplitudes. Nature does not know what you are looking at, and she behaves the way she is going to behave whether you bother to take down the data or not. So here we must not add the amplitudes. We first square the amplitudes for all possible different final events and then sum. The correct result for an electron at x and a photon at either D_1 or D_2 is

$$\begin{aligned} & \left| \langle e \text{ at } x \mid e \text{ from } s \rangle \right|^2 + \left| \langle e \text{ at } x \mid e \text{ from } s \rangle \right|^2 \\ & \left| \langle \text{ph at } D_1 \mid \text{ph from } L \rangle \right|^2 + \left| \langle \text{ph at } D_2 \mid \text{ph from } L \rangle \right|^2 \\ & = |a\phi_1 + b\phi_2|^2 + |a\phi_2 + b\phi_1|^2. \end{aligned} \quad (3.10)$$

3-3 Scattering from a crystal

Our next example is a phenomenon in which we have to analyze the interference of probability amplitudes somewhat carefully. We look at the process of the scattering of neutrons from a crystal. Suppose we have a crystal which has a lot of atoms with nuclei at their centers, arranged in a periodic array, and a neutron beam that comes from far away. We can label the various nuclei in the crystal by an index i , where i runs over the integers 1, 2, 3, . . . N , with N equal to the total number of atoms. The problem is to calculate the probability of getting a neutron into a counter with the arrangement shown in Fig. 3-5. For any particular atom i , the amplitude that the neutron arrives at the counter C is the amplitude that the neutron gets from the source S to nucleus i , multiplied by the amplitude a that it gets scattered there, multiplied by the amplitude that it gets from i to the counter C . Let’s write that down:

$$\langle \text{neutron at } C \mid \text{neutron from } S \rangle_{\text{via } i} = \langle C \mid i \rangle a \langle i \mid S \rangle. \quad (3.11)$$

In writing this equation we have assumed that the scattering amplitude a is the same for all atoms. We have here a large number of apparently indistinguishable routes. They are indistinguishable because a low-energy neutron is scattered from a nucleus without knocking the atom out of its place in the crystal—no “record” is left of the scattering. According to the earlier discussion, the total amplitude for a neutron at C involves a sum of Eq. (3.11) over all the atoms:

$$\langle \text{neutron at } C \mid \text{neutron from } S \rangle = \sum_{i=1}^N \langle C \mid i \rangle a \langle i \mid S \rangle. \quad (3.12)$$

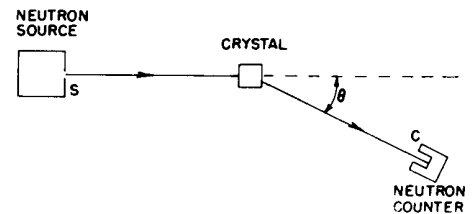


Fig. 3-5. Measuring the scattering of neutrons by a crystal.

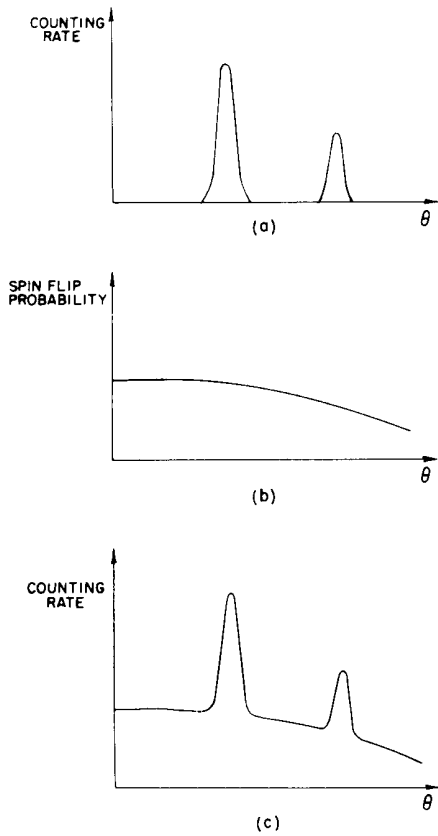


Fig. 3-6. The neutron counting rate as a function of angle: (a) for spin zero nuclei; (b) the probability of scattering with spin flip; (c) the observed counting rate with a spin one-half nucleus.

Because we are adding amplitudes of scattering from atoms with different space positions, the amplitudes will have different phases giving the characteristic interference pattern that we have already analyzed in the case of the scattering of light from a grating.

The neutron intensity as a function of angle in such an experiment is indeed often found to show tremendous variations, with very sharp interference peaks and almost nothing in between—as shown in Fig. 3-6(a). However, for certain kinds of crystals it does not work this way, and there is—along with the interference peaks discussed above—a general background of scattering in all directions. We must try to understand the apparently mysterious reasons for this. Well, we have not considered one important property of the neutron. It has a spin of one-half, and so there are two conditions in which it can be: either spin “up” (say perpendicular to the page in Fig. 3-5) or spin “down.” If the nuclei of the crystal have no spin, the neutron spin doesn’t have any effect. But when the nuclei of the crystal also have a spin, say a spin of one-half, you will observe the background of smeared-out scattering described above. The explanation is as follows.

If the neutron has one direction of spin and the atomic nucleus has the same spin, then no change of spin can occur in the scattering process. If the neutron and atomic nucleus have opposite spin, then scattering can occur by two processes, one in which the spins are unchanged and another in which the spin directions are exchanged. This rule for no net change of the sum of the spins is analogous to our classical law of conservation of angular momentum. We can begin to understand the phenomenon if we assume that all the scattering nuclei are set up with spins in one direction. A neutron with the same spin will scatter with the expected sharp interference distribution. What about one with opposite spin? If it scatters without spin flip, then nothing is changed from the above; but if the two spins flip over in the scattering, we could, in principle, find out which nucleus had done the scattering, since it would be the only one with spin turned over. Well, if we can tell which atom did the scattering, what have the other atoms got to do with it? Nothing, of course. The scattering is exactly the same as that from a single atom.

To include this effect, the mathematical formulation of Eq. (3.12) must be modified since we haven’t described the states completely in that analysis. Let’s start with all neutrons from the source having spin up and all the nuclei of the crystal having spin down. First, we would like the amplitude that at the counter the spin of the neutron is up *and* all spins of the crystal are still down. This is not different from our previous discussion. We will let a be the amplitude to scatter with no flip or spin. The amplitude for scattering from the i th atom is, of course,

$$\langle C_{\text{up}}, \text{crystal all down} | S_{\text{up}}, \text{crystal all down} \rangle = \langle C | i \rangle a \langle i | S \rangle.$$

Since all the atomic spins are still down, the various alternatives (different values of i) cannot be distinguished. There is clearly no way to tell which atom did the scattering. For this process, all the amplitudes interfere.

We have another case, however, where the spin of the detected neutron is down although it started from S with spin up. In the crystal, one of the spins must be changed to the up direction—let’s say that of the k th atom. We will assume that there is the same scattering amplitude with spin flip for every atom, namely b . (In a real crystal there is the disagreeable possibility that the reversed spin moves to some other atom, but let’s take the case of a crystal for which this probability is very low.) The scattering amplitude is then

$$\langle C_{\text{down}}, \text{nucleus } k \text{ up} | S_{\text{up}}, \text{crystal all down} \rangle = \langle C | k \rangle b \langle k | S \rangle. \quad (3.13)$$

If we ask for the probability of finding the neutron spin down and the k th nucleus spin up, it is equal to the absolute square of this amplitude, which is simply $|b|^2$ times $|\langle C | k \rangle \langle k | S \rangle|^2$. The second factor is almost independent of location in the crystal, and all phases have disappeared in taking the absolute square. The

probability of scattering from *any nucleus* in the crystal with spin flip is now

$$|b|^2 \sum_{k=1}^N |\langle C | k \rangle \langle k | S \rangle|^2,$$

which will show a smooth distribution as in Fig. 3-6(b).

You may argue, "I don't care which atom is up." Perhaps you don't, but nature knows; and the probability is, in fact, what we gave above—there isn't any interference. On the other hand, if we ask for the probability that the spin is up at the detector and all the atoms still have spin down, then we must take the absolute square of

$$\sum_{i=1}^N \langle C | i \rangle a \langle i | S \rangle.$$

Since the terms in this sum have phases, they do interfere, and we get a sharp interference pattern. If we do an experiment in which we don't observe the spin of the detected neutron, then both kinds of events can occur; and the separate probabilities add. The total probability (or counting rate) as a function of angle then looks like the graph in Fig. 3-6(c).

Let's review the physics of this experiment. If you could, *in principle*, distinguish the alternative *final* states (even though you do not bother to do so), the total, final probability is obtained by calculating the *probability* for each state (not the amplitude) and then adding them together. If you *cannot* distinguish the final states *even in principle*, then the probability amplitudes must be summed before taking the absolute square to find the actual probability. The thing you should notice particularly is that if you were to try to represent the neutron by a wave alone, you would get the same kind of distribution for the scattering of a down-spinning neutron as for an up-spinning neutron. You would have to say that the "wave" would come from all the different atoms and interfere just as for the up-spinning one with the same wavelength. But we know that is not the way it works. So as we stated earlier, we must be careful not to attribute too much reality to the waves in space. They are useful for certain problems but not for all.

3-4 Identical particles

The next experiment we will describe is one which shows one of the beautiful consequences of quantum mechanics. It again involves a physical situation in which a thing can happen in two *indistinguishable* ways, so that there is an interference of amplitudes—as is *always* true in such circumstances. We are going to discuss the scattering, at relatively low energy, of nuclei on other nuclei. We start by thinking of α -particles (which, as you know, are helium nuclei) bombarding, say, oxygen. To make it easier for us to analyze the reaction, we will look at it in the center-of-mass system, in which the oxygen nucleus and the α -particle have their velocities in opposite directions before the collision and again in exactly opposite directions after the collision. See Fig. 3-7(a). (The magnitudes of the velocities are, of course, different, since the masses are different.) We will also suppose that there is conservation of energy and that the collision energy is low enough that neither particle is broken up or left in an excited state. The reason that the two particles deflect each other is, of course, that each particle carries a positive charge and, classically speaking, there is an electrical repulsion as they go by. The scattering will happen at different angles with different probabilities, and we would like to discuss something about the angle dependence of such scatterings. (It is possible, of course, to calculate this thing classically, and it is one of the most remarkable accidents of quantum mechanics that the answer to this problem comes out the same as it does classically. This is a curious point because it happens for no other force except the inverse square law—so it is indeed an accident.)

The probability of scattering in different directions can be measured by an experiment as shown in Fig. 3-7(a). The counter at position 1 could be designed to detect only α -particles; the counter at position 2 could be designed to detect

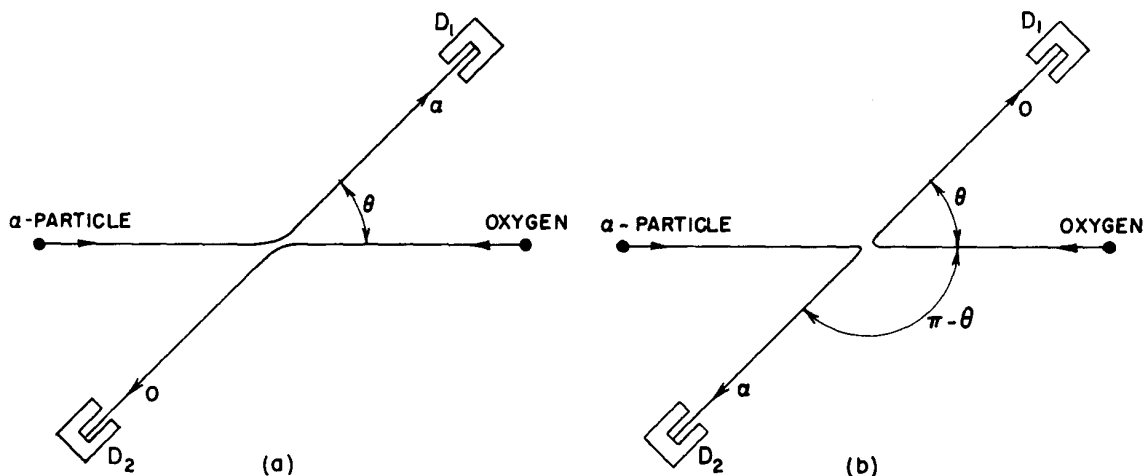


Fig. 3-7. The scattering of α -particles from oxygen nuclei, as seen in the center-of-mass system.

only oxygen—just as a check. (In the laboratory system the detectors would not be opposite; but in the CM system they are.) Our experiment consists in measuring the probability of scattering in various directions. Let's call $f(\theta)$ the amplitude to scatter into the counters when they are at the angle θ ; then $|f(\theta)|^2$ will be our experimentally determined probability.

Now we could set up another experiment in which our counters would respond to *either* the α -particle *or* the oxygen nucleus. Then we have to work out what happens when we do not bother to distinguish which particles are counted. Of course, if we are to get an oxygen in the position θ , there must be an α -particle on the opposite side at the angle $(\pi - \theta)$, as shown in Fig. 3-7(b). So if $f(\theta)$ is the amplitude for α -scattering through the angle θ , then $f(\pi - \theta)$ is the amplitude for oxygen scattering through the angle θ .[†] Thus, the probability for having *some* particle in the detector at position 1 is:

$$\text{Probability of some particle in } D_1 = |f(\theta)|^2 + |f(\pi - \theta)|^2. \quad (3.14)$$

Note that the two states are distinguishable in principle. Even though in this experiment we *do not* distinguish them, *we could*. According to the earlier discussion, then, we must add the probabilities, not the amplitudes.

The result given above is correct for a variety of target nuclei—for α -particles on oxygen, on carbon, on beryllium, on hydrogen. *But it is wrong for α -particles on α -particles.* For the one case in which both particles are exactly the same, the experimental data disagree with the prediction of (3.14). For example, the scattering probability at 90° is exactly twice what the above theory predicts and has nothing to do with the particles being “helium” nuclei. If the target is He^3 , but the projectiles are α -particles (He^4), then there is agreement. Only when the target is He^4 —so its nuclei are identical with the incoming α -particle—does the scattering vary in a peculiar way with angle.

Perhaps you can already see the explanation. There are two ways to get an α -particle into the counter: by scattering the bombarding α -particle at an angle θ , or by scattering it at an angle of $(\pi - \theta)$. How can we tell whether the bombarding particle or the target particle entered the counter? The answer is that we cannot. In the case of α -particles with α -particles there are two alternatives that cannot be distinguished. Here, we must let the probability *amplitudes* interfere by addition,

[†] In general, a scattering direction should, of course, be described by two angles, the polar angle ϕ , as well as the azimuthal angle θ . We would then say that an oxygen nucleus at (θ, ϕ) means that the α -particle is at $(\pi - \theta, \phi + \pi)$. However, for Coulomb scattering (and for many other cases), the scattering amplitude is independent of ϕ . Then the amplitude to get an oxygen at θ is the same as the amplitude to get the α -particle at $(\pi - \theta)$.

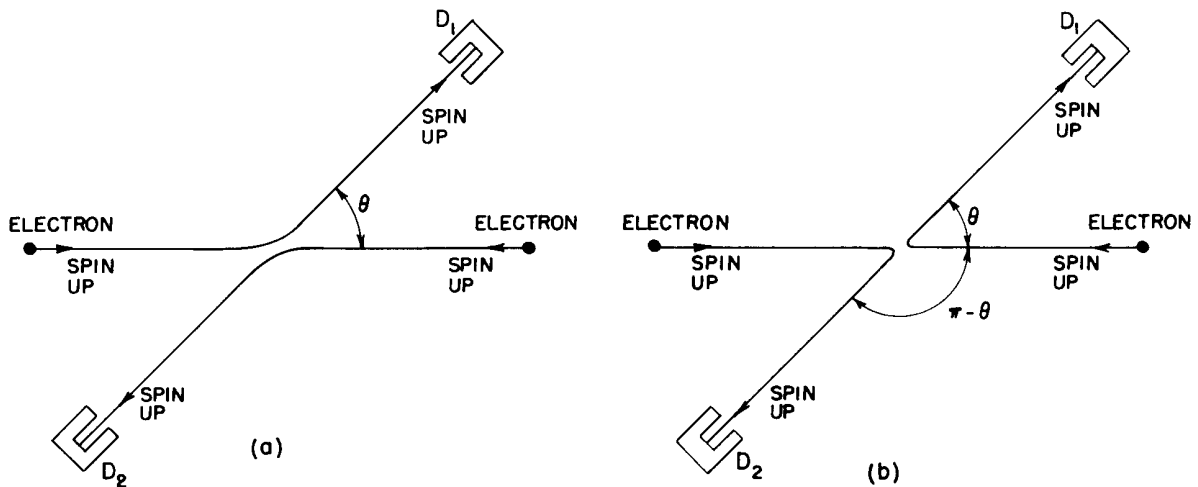


Fig. 3-8. The scattering of electrons on electrons. If the incoming electrons have parallel spins, the processes (a) and (b) are indistinguishable.

and the probability of finding an α -particle in the counter is the square of their sum:

$$\text{Probability of an } \alpha\text{-particle at } D_1 = |f(\theta) + f(\pi - \theta)|^2. \quad (3.15)$$

This is quite a different result than that in Eq. (3.14). We can take an angle of $\pi/2$ as an example, because it is easy to figure out. For $\theta = \pi/2$, we obviously have $f(\theta) = f(\pi - \theta)$, so the probability in Eq. (3.15) becomes $|f(\pi/2) + f(\pi/2)|^2 = 4|f(\pi/2)|^2$.

On the other hand, if they did not interfere, the result of Eq. (3.14) gives only $2|f(\pi/2)|^2$. So there is twice as much scattering at 90° as we might have expected. Of course, at other angles the results will also be different. And so you have the unusual result that when particles are identical, a certain new thing happens that doesn't happen when particles can be distinguished. In the mathematical description you must add the amplitudes for alternative process in which the two particles simply exchange roles and there is an interference.

An even more perplexing thing happens when we do the same kind of experiment by scattering electrons on electrons, or protons on protons. Neither of the above results is then correct! For these particles, we must invoke still a new rule, a most peculiar rule, which is the following: When you have a situation in which the identity of the electron which is arriving at a point is exchanged with another one, the new amplitude interferes with the old one with an *opposite phase*. It is interference all right, but with a minus sign. In the case of α -particles, when you exchange the α -particle entering the detector, the interfering amplitudes interfere with the positive sign. *In the case of electrons, the interfering amplitudes for exchange interfere with a negative sign.* Except for another detail to be discussed below, the proper equation for electrons in an experiment like the one shown in Fig. 3-8 is

$$\text{Probability of e at } D_1 = |f(\theta) - f(\pi - \theta)|^2. \quad (3.16)$$

The above statement must be qualified, because we have not considered the spin of the electron (α -particles have no spin). The electron spin may be considered to be either "up" or "down" with respect to the plane of the scattering. If the energy of the experiment is low enough, the magnetic forces due to the currents will be small and the spin will not be affected. We will assume that this is the case for the present analysis, so that there is no chance that the spins are changed during the collision. Whatever spin the electron has, it carries along with it. Now you see there are many possibilities. The bombarding and target particles can have both spins up, both down, or opposite spins. If both spins are up, as in Fig. 3-8 (or if both spins are down), the same will be true of the recoil particles and the *amplitude* for the process is the *difference* of the amplitudes for the two possibilities

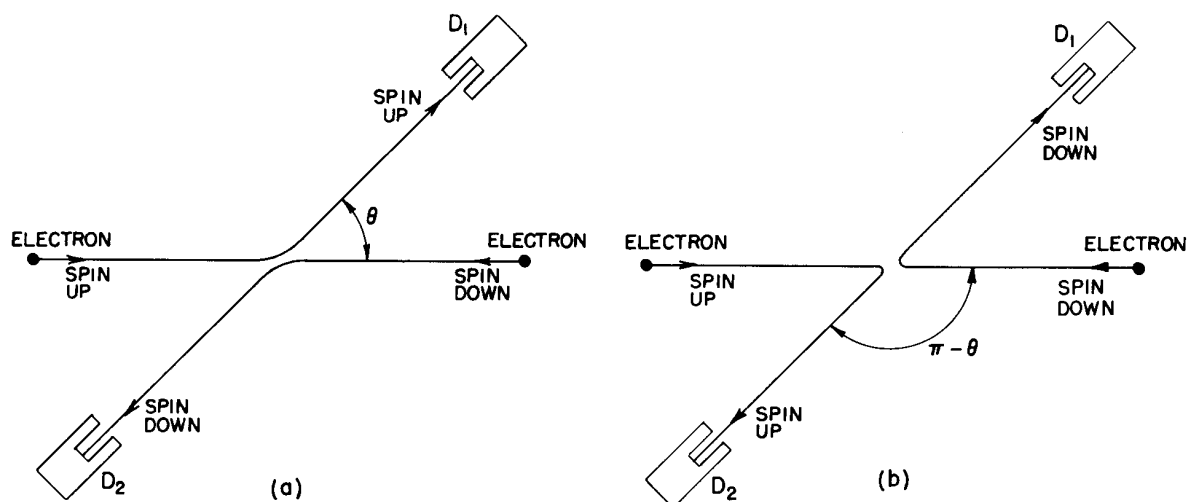


Fig. 3-9. The scattering of electrons with antiparallel spins.

shown in Fig. 3-8(a) and (b). The *probability* of detecting an electron in D_1 is then given by Eq. (3.16).

Suppose, however, the “bombarding” spin is up and the “target” spin is down. The electron entering counter 1 can have spin up or spin down, and by measuring this spin we can tell whether it came from the bombarding beam or from the target. The two possibilities are shown in Fig. 3-9(a) and (b); they are distinguishable in principle, and hence there will be no interference—merely an addition of the two probabilities. The same argument holds if both of the original spins are reversed—that is, if the left-hand spin is down and the right-hand spin is up.

Now if we take our electrons at random—as from a tungsten filament in which the electrons are completely unpolarized—then the odds are fifty-fifty that any particular electron comes out with spin up or spin down. If we don’t bother to measure the spin of the electrons at any point in the experiment, we have what we call an unpolarized experiment. The results for this experiment are best calculated by listing all of the various possibilities as we have done in Table 3-1. A separate *probability* is computed for each distinguishable alternative. The total probability is then the sum of all the separate probabilities. Note that for unpolarized beams the result for $\theta = \pi/2$ is one-half that of the classical result with independent particles. The behavior of identical particles has many interesting consequences; we will discuss them in greater detail in the next chapter.

Table 3-1

Scattering of unpolarized spin one-half particles

Fraction of cases	Spin of particle 1	Spin of particle 2	Spin at D_1	Spin at D_2	Probability
$\frac{1}{4}$	up	up	up	up	$ f(\theta) - f(\pi - \theta) ^2$
$\frac{1}{4}$	down	down	down	down	$ f(\theta) - f(\pi - \theta) ^2$
$\frac{1}{4}$	up	down	up	down	$ f(\theta) ^2$
			down	up	$ f(\pi - \theta) ^2$
$\frac{1}{4}$	down	up	up	down	$ f(\pi - \theta) ^2$
			down	up	$ f(\theta) ^2$
Total probability = $\frac{1}{2} f(\theta) - f(\pi - \theta) ^2 + \frac{1}{2} f(\theta) ^2 + \frac{1}{2} f(\pi - \theta) ^2$					

Identical Particles

4-1 Bose particles and Fermi particles

In the last chapter we began to consider the special rules for the interference that occurs in processes with two *identical* particles. By *identical particles* we mean things like electrons which can in no way be distinguished one from another. If a process involves two particles that are identical, reversing which one arrives at a counter is an alternative which cannot be distinguished and—like all cases of alternatives which cannot be distinguished—interferes with the original, un-exchanged case. The amplitude for an event is then the sum of the two interfering amplitudes; but, interestingly enough, the interference is in some cases with the *same* phase and, in others, with the *opposite* phase.

Suppose we have a collision of two particles a and b in which particle a scatters in the direction 1 and particle b scatters in the direction 2, as sketched in Fig. 4-1(a). Let's call $f(\theta)$ the amplitude for this process; then the probability P_1 of observing such an event is proportional to $|f(\theta)|^2$. Of course, it could also happen that particle b scattered into counter 1 and particle a went into counter 2, as shown in Fig. 4-1(b). Assuming that there are no special directions defined by spins or such, the probability P_2 for this process is just $|f(\pi - \theta)|^2$, because it is just equivalent to the first process with counter 1 moved over to the angle $\pi - \theta$. You might also think that the *amplitude* for the second process is just $f(\pi - \theta)$. But that is not necessarily so, because there could be an arbitrary phase factor. That is, the amplitude could be

$$e^{i\delta} f(\pi - \theta).$$

Such an amplitude still gives a probability P_2 equal to $|f(\pi - \theta)|^2$.

Now let's see what happens if a and b are identical particles. Then the two different processes shown in the two diagrams of Fig. 4-1 cannot be distinguished. There is an amplitude that *either* a or b goes into counter 1, while the other goes into counter 2. This amplitude is the sum of the amplitudes for the two processes shown in Fig. 4-1. If we call the first one $f(\theta)$, then the second one is $e^{i\delta} f(\pi - \theta)$, where now the phase factor is very important because we are going to be adding two amplitudes. Suppose we have to multiply the amplitude by a certain phase factor when we exchange the roles of the two particles. If we exchange them again we should get the same factor again. But we are then back to the first process.

4-1 Bose particles and Fermi particles

4-2 States with two Bose particles

4-3 States with n Bose particles

4-4 Emission and absorption of photons

4-5 The blackbody spectrum

4-6 Liquid helium

4-7 The exclusion principle

Review: Blackbody radiation in:
Chapter 41, Vol. I, *The Brownian Movement*

Chapter 42, Vol. I, *Applications of Kinetic Theory*

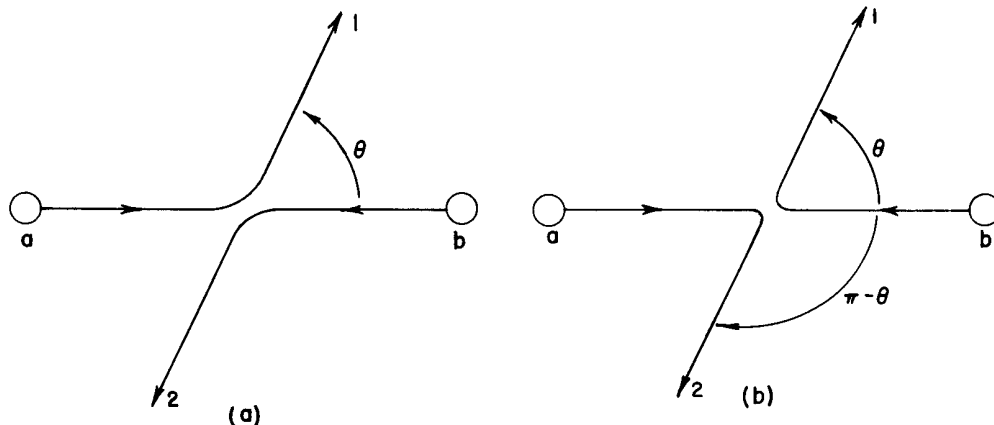


Fig. 4-1. In the scattering of two identical particles, the processes (a) and (b) are indistinguishable.

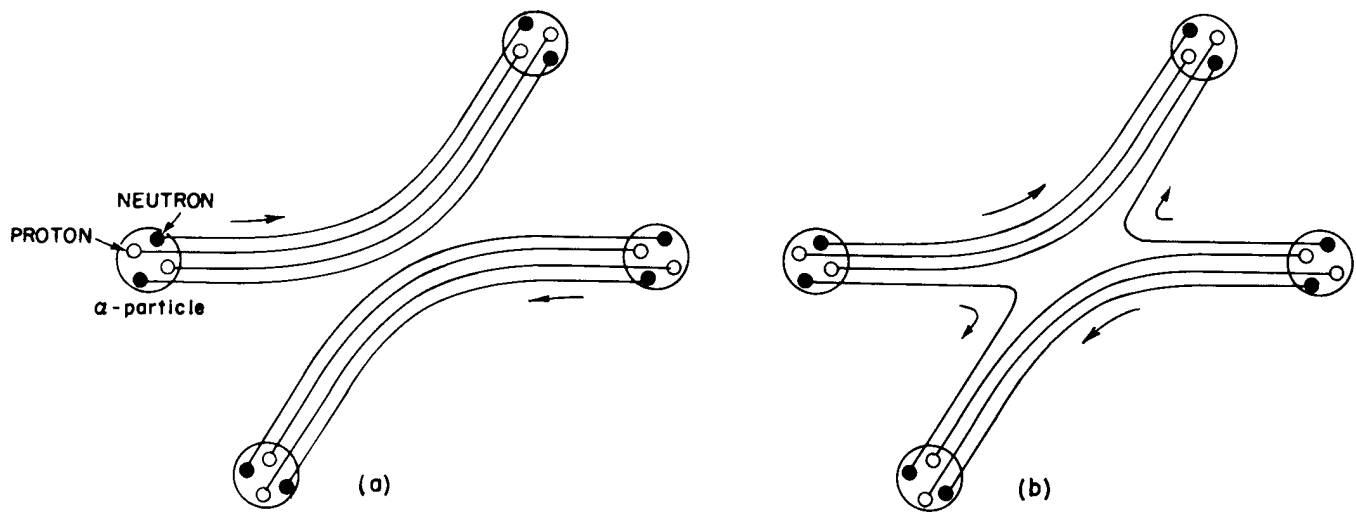


Fig. 4-2. The scattering of two α -particles. In (a) the two particles retain their identity; in (b) a neutron is exchanged during the collision.

The phase factor taken twice must bring us back where we started—its square must be equal to 1. There are only two possibilities: $e^{i\delta}$ is equal to $+1$, or is equal to -1 . Either the exchanged case contributes with the *same* sign, or it contributes with the *opposite* sign. Both cases exist in nature, each for a different class of particles. Particles which interfere with a *positive* sign are called *Bose particles* and those which interfere with a *negative* sign are called *Fermi particles*. The Bose particles are the photon, the mesons, and the graviton. The Fermi particles are the electron, the muon, the neutrinos, the nucleons, and the baryons. We have, then, that the amplitude for the scattering of identical particles is:

Bose particles:

$$(\text{Amplitude direct}) + (\text{Amplitude exchanged}). \quad (4.1)$$

Fermi particles:

$$(\text{Amplitude direct}) - (\text{Amplitude exchanged}). \quad (4.2)$$

For particles with spin—like electrons—there is an additional complication. We must specify not only the location of the particles but the direction of their spins. It is only for identical particles *with identical spin states* that the amplitudes interfere when the particles are exchanged. If you think of the scattering of unpolarized beams—which are a mixture of different spin states—there is some extra arithmetic.

Now an interesting problem arises when there are two or more particles bound tightly together. For example, an α -particle has four particles in it—two neutrons and two protons. When two α -particles scatter, there are several possibilities. It may be that during the scattering there is a certain amplitude that one of the neutrons will leap across from one α -particle to the other, while a neutron from the other α -particle leaps the other way so that the two alphas which come out of the scattering are not the original ones—there has been an exchange of a pair of neutrons. See Fig. 4-2. The amplitude for scattering with an exchange of a pair of neutrons will interfere with the amplitude for scattering with no such exchange, and the interference must be with a minus sign because there has been an exchange of one pair of Fermi particles. On the other hand, if the relative energy of the two α -particles is so low that they stay fairly far apart—say, due to the Coulomb repulsion—and there is never any appreciable probability of exchanging any of the internal particles, we can consider the α -particle as a simple object, and we do not need to worry about its internal details. In such circumstances, there are only two contributions to the scattering amplitude. Either there is no exchange, or all four of the nucleons are exchanged in the scattering. Since the protons and the

neutrons in the α -particle are all Fermi particles, an exchange of any pair reverses the sign of the scattering amplitude. So long as there are no internal changes in the α -particles, interchanging the two α -particles is the same as interchanging four pairs of Fermi particles. There is a change in sign for each pair, so the net result is that the amplitudes combine with a positive sign. The α -particle behaves like a Bose particle.

So the rule is that composite objects, in circumstances in which the composite object can be considered as a single object, behave like Fermi particles or Bose particles, depending on whether they contain an odd number or an even number of Fermi particles.

All the elementary Fermi particles we have mentioned—such as the electron, the proton, the neutron, and so on—have a spin $j = 1/2$. If several such Fermi particles are put together to form a composite object, the resulting spin may be either integral or half-integral. For example, the common isotope of helium, He^4 , which has two neutrons and two protons, has a spin of zero, whereas Li^7 which has three protons and four neutrons, has a spin of $3/2$. We will learn later the rules for compounding angular momentum, and will just mention now that every composite object which has a *half-integral spin* imitates a *Fermi particle*, whereas every composite object with an *integral spin* imitates a *Bose particle*.

This brings up an interesting question: Why is it that particles with half-integral spin are Fermi particles whose amplitudes add with the minus sign, whereas particles with integral spin are Bose particles whose amplitudes add with the positive sign? We apologize for the fact that we cannot give you an elementary explanation. An explanation has been worked out by Pauli from complicated arguments of quantum field theory and relativity. He has shown that the two must necessarily go together, but we have not been able to find a way of reproducing his arguments on an elementary level. It appears to be one of the few places in physics where there is a rule which can be stated very simply, but for which no one has found a simple and easy explanation. The explanation is deep down in relativistic quantum mechanics. This probably means that we do not have a complete understanding of the fundamental principle involved. For the moment, you will just have to take it as one of the rules of the world.

4-2 States with two Bose particles

Now we would like to discuss an interesting consequence of the addition rule for Bose particles. It has to do with their behavior when there are several particles present. We begin by considering a situation in which two Bose particles are scattered from two different scatterers. We won't worry about the details of the scattering mechanism. We are interested only in what happens to the scattered particles. Suppose we have the situation shown in Fig. 4-3. The particle a is scattered into the state 1. By a *state* we mean a given direction and energy, or some other given condition. The particle b is scattered into the state 2. We want to assume that the two states 1 and 2 are nearly the same. (What we really want to find out eventually is the amplitude that the two particles are scattered into identical directions, or states; but it is best if we think first about what happens if the states are almost the same and then work out what happens when they become identical.)

Suppose that we had only particle a ; then it would have a certain amplitude for scattering in direction 1, say $\langle 1 | a \rangle$. And particle b alone would have the amplitude $\langle 2 | b \rangle$ for landing in direction 2. If the two particles are not identical, the amplitude for the two scatterings to occur at the same time is just the product

$$\langle 1 | a \rangle \langle 2 | b \rangle.$$

The probability for such an event is then

$$|\langle 1 | a \rangle \langle 2 | b \rangle|^2,$$

which is also equal to

$$|\langle 1 | a \rangle|^2 |\langle 2 | b \rangle|^2.$$

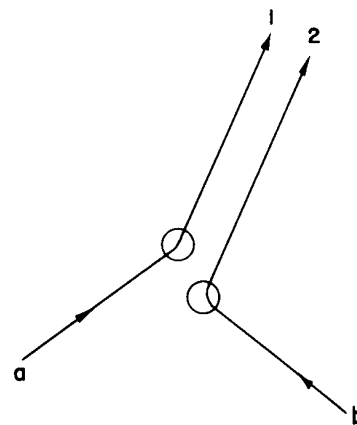


Fig. 4-3. A double scattering into nearby final states.

To save writing for the present arguments, we will sometimes set

$$\langle 1 | a \rangle = a_1, \quad \langle 2 | b \rangle = b_2.$$

Then the probability of the double scattering is

$$|a_1|^2 |b_2|^2.$$

It could also happen that particle b is scattered into direction 1, while particle a goes into direction 2. The amplitude for this process is

$$\langle 2 | a \rangle \langle 1 | b \rangle,$$

and the probability of such an event is

$$|\langle 2 | a \rangle \langle 1 | b \rangle|^2 = |a_2|^2 |b_1|^2.$$

Imagine now that we have a pair of tiny counters that pick up the two scattered particles. The probability P_2 that they will pick up two particles together is just the sum

$$P_2 = |a_1|^2 |b_2|^2 + |a_2|^2 |b_1|^2. \quad (4.3)$$

Now let's suppose that the directions 1 and 2 are very close together. We expect that a should vary smoothly with direction, so a_1 and a_2 must approach each other as 1 and 2 get close together. If they are close enough, the amplitudes a_1 and a_2 will be equal. We can set $a_1 = a_2$ and call them both just a ; similarly, we set $b_1 = b_2 = b$. Then we get that

$$P_2 = 2|a|^2 |b|^2. \quad (4.4)$$

Now suppose, however, that a and b are identical Bose particles. Then the process of a going into 1 and b going into 2 cannot be distinguished from the exchanged process in which a goes into 2 and b goes into 1. In this case the *amplitudes* for the two different processes can interfere. The *total* amplitude to obtain a particle in each of the two counters is

$$\langle 1 | a \rangle \langle 2 | b \rangle + \langle 2 | a \rangle \langle 1 | b \rangle. \quad (4.5)$$

And the probability that we get a pair is the absolute square of this amplitude,

$$P_2 = |a_1 b_2 + a_2 b_1|^2 = 4|a|^2 |b|^2. \quad (4.6)$$

We have the result that it is *twice as likely* to find two *identical* Bose particles scattered into the same state *as you would calculate assuming the particles were different*.

Although we have been considering that the two particles are observed in separate counters, this is not essential—as we can see in the following way. Let's imagine that both the directions 1 and 2 would bring the particles into a *single* small counter which is some distance away. We will let the direction 1 be defined by saying that it heads toward the element of area dS_1 of the counter. Direction 2 heads toward the surface element dS_2 of the counter. (We imagine that the counter presents a surface at right angles to the line from the scatterings.) Now we cannot give a probability that a particle will go into a precise direction or to a *particular* point in space. Such a thing is impossible—the chance for any exact direction is zero. When we want to be so specific, we shall have to define our amplitudes so that they give the probability of arriving *per unit area* of a counter. Suppose that we had only particle a ; it would have a certain amplitude for scattering in direction 1. Let's define $\langle 1 | a \rangle = a_1$ to be the amplitude that a will scatter *into a unit area* of the counter in the direction 1. In other words, the scale of a_1 is chosen—we say it is “normalized” so that the probability that it will scatter *into an element of area* dS_1 is

$$|\langle 1 | a \rangle|^2 dS_1 = |a_1|^2 dS_1. \quad (4.7)$$

If our counter has the total area ΔS , and we let dS_1 range over this area, the total probability that the particle a will be scattered into the counter is

$$\int_{\Delta S} |a_1|^2 dS_1. \quad (4.8)$$

As before, we want to assume that the counter is sufficiently small so that the amplitude a_1 doesn't vary significantly over the surface of the counter; a_1 is then a constant amplitude which we can call a . Then the probability that particle a is scattered somewhere into the counter is

$$p_a = |a|^2 \Delta S. \quad (4.9)$$

In the same way, we will have that the probability that particle b —when it is alone—scatters into some element of area, say dS_2 , is

$$|b_2|^2 dS_2.$$

(We use dS_2 instead of dS_1 because we will later want a and b to go into different directions.) Again we set b_2 equal to the constant amplitude b ; then the probability that particle b is counted in the detector is

$$p_b = |b|^2 \Delta S. \quad (4.10)$$

Now when both particles are present, the probability that a is scattered into dS_1 and b is scattered into dS_2 is

$$|a_1 b_2|^2 dS_1 dS_2 = |a|^2 |b|^2 dS_1 dS_2. \quad (4.11)$$

If we want the probability that *both* a and b get into the counter, we integrate both dS_1 and dS_2 over ΔS and find that

$$P_2 = |a|^2 |b|^2 (\Delta S)^2. \quad (4.12)$$

We notice, incidentally, that this is just equal to $p_a \cdot p_b$, just as you would suppose assuming that the particles a and b act independently of each other.

When the two particles are identical, however, there are two indistinguishable possibilities for each pair of surface elements dS_1 and dS_2 . Particle a going into dS_2 and particle b going into dS_1 is indistinguishable from a into dS_1 and b into dS_2 , so the amplitudes for these processes will interfere. (When we had two *different* particles above—although we did not *in fact* care which particle went where in the counter—we could, *in principle*, have found out; so there was no interference. For identical particles we cannot tell, even in principle.) We must write, then, that the probability that the two particles arrive at dS_1 and dS_2 is

$$|a_1 b_2 + a_2 b_1|^2 dS_1 dS_2. \quad (4.13)$$

Now, however, when we integrate over the area of the counter, we must be careful. If we let dS_1 and dS_2 range over the whole area ΔS , we would count each part of the area *twice* since (4.13) contains everything that can happen with any pair of surface elements dS_1 and dS_2 †. We can still do the integral that way, if we correct for the double counting by dividing the result by 2. We get then that P_2 for identical Bose particles is

$$P_2(\text{Bose}) = \frac{1}{2} \{4|a|^2 |b|^2 (\Delta S)^2\} = 2|a|^2 |b|^2 (\Delta S)^2. \quad (4.14)$$

Again, this is just twice what we got in Eq. (4.12) for distinguishable particles.

If we imagine for a moment that we knew that the b channel had already sent its particle into the particular direction, we can say that the *probability* that a second particle will go into the same direction is twice as great as we would have

† In (4.11) interchanging dS_1 and dS_2 gives a different event, so both surface elements should range over the whole area of the counter. In (4.13) we are treating dS_1 and dS_2 as a *pair* and including everything that can happen. If the integrals include again what happens when dS_1 and dS_2 are reversed, everything is counted twice.

expected if we had calculated it as an independent event. It is a property of Bose particles that if there is already one particle in a condition of some kind, the *probability* of getting a second one in the same condition is twice as great as it would be if the first one were not already there. This fact is often stated in the following way: If there is already one Bose particle in a given state, the amplitude for putting an identical one on top of it is $\sqrt{2}$ greater than if it weren't there. (This is not a proper way of stating the result from the physical point of view we have taken, but if it is used consistently as a rule, it will, of course, give the correct result.)

4-3 States with n Bose particles

Let's extend our result to a situation in which there are n particles present. We imagine the circumstance shown in Fig. 4-4. We have n particles $a, b, c, \dots$, which are scattered and end up in the directions $1, 2, 3, \dots, n$. All n directions are headed toward a small counter a long distance away. As in the last section, we choose to normalize all the amplitudes so that the probability that each particle acting alone would go into an element of surface dS of the counter is

$$|\langle \rangle|^2 dS.$$

First, let's assume that the particles are all distinguishable; then the probability that n particles will be counted together in n different surface elements is

$$|a_1 b_2 c_3 \dots|^2 dS_1 dS_2 dS_3 \dots \quad (4.15)$$

Again we take that the amplitudes don't depend on where dS is located in the counter (assumed small) and call them simply $a, b, c, \dots$. The probability (4.15) becomes

$$|a|^2 |b|^2 |c|^2 \dots dS_1 dS_2 dS_3 \dots \quad (4.16)$$

Integrating each dS over the surface ΔS of the counter, we have that P_n (different), the probability of counting n different particles at once, is

$$P_r \text{ (different)} = |a|^2 |b|^2 |c|^2 \dots (\Delta S)^n \quad (4.17)$$

This is just the product of the probabilities for each particle to enter the counter separately. They all act independently—the probability for one to enter does not depend on how many others are also entering.

Now suppose that all the particles are identical Bose particles. For each set of directions $1, 2, 3, \dots$ there are many indistinguishable possibilities. If there were, for instance, just three particles, we would have the following possibilities:

$a \rightarrow 1$	$a \rightarrow 1$	$a \rightarrow 2$
$b \rightarrow 2$	$b \rightarrow 3$	$b \rightarrow 1$
$c \rightarrow 3$	$c \rightarrow 2$	$c \rightarrow 3$
$a \rightarrow 2$	$a \rightarrow 3$	$a \rightarrow 3$
$b \rightarrow 3$	$b \rightarrow 1$	$b \rightarrow 2$
$c \rightarrow 1$	$c \rightarrow 2$	$c \rightarrow 1$

There are six different combinations. With n particles, there are $n!$ different, but *indistinguishable*, possibilities for which we must add amplitudes. The probability that n particles will be counted in n surface elements is then

$$|a_1 b_2 c_3 \dots + a_1 b_3 c_2 \dots + a_2 b_1 c_3 \dots + a_2 b_3 c_1 \dots + \text{etc.} + \text{etc.}|^2 dS_1 dS_2 dS_3 \dots dS_n. \quad (4.18)$$

Once more we assume that all the directions are so close that we can set $a_1 = a_2 = \dots = a = a_n$, and similarly for $b, c, \dots$; the probability of (4.18) becomes

$$|n! abc \dots|^2 dS_1 dS_2 \dots dS_n. \quad (4.19)$$

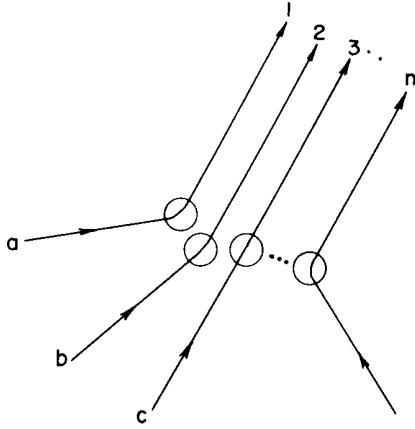


Fig. 4-4. The scattering of n particles into nearby final states.

When we integrate each dS over the area ΔS of the counter, each possible product of surface elements is counted $n!$ times; we correct for this by dividing by $n!$ and get

$$P_n(\text{Bose}) = \frac{1}{n!} |n! abc \dots|^2 (\Delta S)^n$$

or

$$P_n(\text{Bose}) = n! |abc \dots|^2 (\Delta S)^n. \quad (4.20)$$

Comparing this result with Eq. (4.17), we see that the probability of counting n Bose particles together is $n!$ greater than we would calculate assuming that the particles were all distinguishable. We can summarize our result this way:

$$P_n(\text{Bose}) = n! P_n(\text{different}). \quad (4.21)$$

Thus, the probability in the Bose case is larger by $n!$ than you would calculate assuming that the particles acted independently.

We can see better what this means if we ask the following question: What is the probability that a Bose particle will go into a particular state when there are *already* n others present? Let's call the newly added particle w . If we have $(n + 1)$ particles, including w , Eq. (4.20) becomes

$$P_{n+1}(\text{Bose}) = (n + 1)! |abc \dots w|^2 (\Delta S)^{n+1}. \quad (4.22)$$

We can write this as

$$P_{n+1}(\text{Bose}) = \{(n + 1)|w|^2 \Delta S\} n! |abc \dots|^2 \Delta S^n$$

or

$$P_{n+1}(\text{Bose}) = (n + 1)|w|^2 \Delta S P_n(\text{Bose}). \quad (4.23)$$

We can look at this result in the following way: The number $|w|^2 \Delta S$ is the probability for getting particle w into the detector if no other particles were present; $P_n(\text{Bose})$ is the chance that there are already n other Bose particles present. So Eq. (4.23) says that *when* there are n other identical Bose particles present, the probability that *one more* particle will enter the same state is *enhanced* by the factor $(n + 1)$. The probability of getting a boson, where there are already n , is $(n + 1)$ times stronger than it would be if there were none before. The *presence* of the other particles increases the probability of getting one more.

4-4 Emission and absorption of photons

Throughout our discussion we have talked about a process like the scattering of α -particles. But that is not essential; we could have been speaking of the creation of particles, as for instance the emission of light. When the light is emitted, a photon is "created." In such a case, we don't need the incoming lines in Fig. 4-4; we can consider merely that there are n atoms $a, b, c, \dots$ emitting light, as in Fig. 4-5. So our result can also be stated: *The probability that an atom will emit a photon into a particular final state is increased by the factor $(n + 1)$ if there are already n photons in that state.*

People like to summarize this result by saying that the *amplitude* to emit a photon is increased by the factor $\sqrt{n + 1}$ when there are already n photons present. It is, of course, another way of saying the same thing if it is understood to mean that this amplitude is just to be squared to get the probability.

It is generally true in quantum mechanics that the amplitude to get from any condition ϕ to any other condition χ is the complex conjugate of the amplitude to get from χ to ϕ :

$$\langle \chi | \phi \rangle = \langle \phi | \chi \rangle^*. \quad (4.24)$$

We will learn about this law a little later, but for the moment we will just assume it is true. We can use it to find out how photons are scattered or absorbed *out* of a given state. We have that the amplitude that a photon will be added to some state, say i , when there are already n photons present is, say,

$$\langle n + 1 | i \rangle = \sqrt{n + 1} a, \quad (4.25)$$

4-7

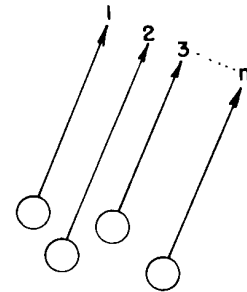


Fig. 4-5. The creation of n photons in nearby states.

where $a = \langle i | a \rangle$ is the amplitude when there are no others present. Using Eq. (4.24), the amplitude to go the other way—from $(n + 1)$ photons to n —is

$$\langle n | n + 1 \rangle = \sqrt{n + 1} a^*. \quad (4.26)$$

This isn't the way people usually say it; they don't like to think of going from $(n + 1)$ to n , but prefer always to start with n photons present. Then they say that the amplitude to absorb a photon when there are n present—in other words, to go from n to $(n + 1)$ —is

$$\langle n + 1 | n \rangle = \sqrt{n} a^*. \quad (4.27)$$

which is, of course, just the same as Eq. (4.26). Then they have trouble trying to remember when to use $\sqrt{n}$ or $\sqrt{n + 1}$. Here's the way to remember: The factor is always the square root of the largest number of photons present, whether it is before or after the reaction. Equations (4.25) and (4.26) show that the law is really symmetric—it only appears unsymmetric if you write it as Eq. (4.27).

There are many physical consequences of these new rules; we want to describe one of them having to do with the emission of light. Suppose we imagine a situation in which photons are contained in a box—you can imagine a box with mirrors for walls. Now say that in the box we have n photons, all of the same state—the same frequency, direction, and polarization—so they can't be distinguished, and that also there is an atom in the box that can emit another photon into the same state. Then the probability that it will emit a photon is

$$(n + 1)|a|^2, \quad (4.28)$$

and the probability that it will absorb a photon is

$$n|a|^2, \quad (4.29)$$

where $|a|^2$ is the probability it would emit if no photons were present. We have already discussed these rules in a somewhat different way in Chapter 42 of Vol. I. Equation (4.29) says that the probability that an atom will *absorb* a photon and make a transition to a higher energy state is proportional to the intensity of the light shining on it. But, as Einstein first pointed out, the rate at which an atom will make a transition *downward* has two parts. There is the probability that it will make a spontaneous transition $|a|^2$, plus the probability of an induced transition $n|a|^2$, which is proportional to the intensity of the light—that is, to the number of photons present. Furthermore, as Einstein said, the coefficients of absorption and of induced emission are equal and are related to the probability of spontaneous emission. What we learn here is that if the light intensity is measured in terms of the number of photons present (instead of as the energy per unit area, and per sec), the coefficients of absorption of induced emission and of spontaneous emission are all equal. This is the content of the relation between the Einstein coefficients A and B of Chapter 42, Vol. I, Eq. (42.18).

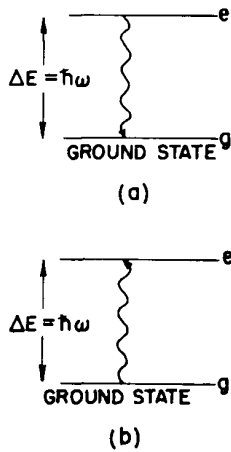


Fig. 4-6. Radiation and absorption of a photon with the frequency ω .

4-5 The blackbody spectrum

We would like to use our rules for Bose particles to discuss once more the spectrum of blackbody radiation (see Chapter 42, Vol. I). We will do it by finding out how many photons there are in a box if the radiation is in thermal equilibrium with some atoms in the box. Suppose that for each light frequency ω , there are a certain number N of atoms which have two energy states separated by the energy $\Delta E = \hbar\omega$. See Fig. 4-6. We'll call the lower-energy state the "ground" state and the upper state the "excited" state. Let N_g and N_e be the average numbers of atoms in the ground and excited states; then in thermal equilibrium at the temperature T , we have from statistical mechanics that

$$\frac{N_e}{N_g} = e^{-\Delta E/kT} = e^{-\hbar\omega/kT}. \quad (4.30)$$

Each atom in the ground state can absorb a photon and go into the excited state, and each atom in the excited state can emit a photon and go to the ground state. In equilibrium, the rates for these two processes must be equal. The rates are proportional to the probability for the event and to the number of atoms present. Let's let $\bar{n}$ be the average number of photons present in a given state with the frequency ω . Then the absorption rate from that state is $N_g \bar{n} |a|^2$, and the emission rate into that state is $N_e (\bar{n} + 1) |a|^2$. Setting the two rates equal, we have that

$$N_g \bar{n} = N_e (\bar{n} + 1). \quad (4.31)$$

Combining this with Eq. (4.30), we have

$$\frac{\bar{n}}{\bar{n} + 1} = e^{-\hbar\omega/kT}.$$

Solving for $\bar{n}$, we have

$$\bar{n} = \frac{1}{e^{\hbar\omega/kT} - 1}, \quad (4.32)$$

which is the mean number of photons in any state with frequency ω , for a cavity in thermal equilibrium. Since each photon has the energy $\hbar\omega$, the energy in the photons of a given state is $\bar{n}\hbar\omega$, or

$$\frac{\hbar\omega}{e^{\hbar\omega/kT} - 1}. \quad (4.33)$$

Incidentally, we once found a similar equation in another context [Chapter 41, Vol. I, Eq. (41.15)]. You remember that for any harmonic oscillator—such as a weight on a spring—the quantum mechanical energy levels are equally spaced with a separation $\hbar\omega$, as drawn in Fig. 4-7. If we call the energy of the n th level $n\hbar\omega$, we find that the mean energy of such an oscillator is also given by Eq. (4.33). Yet this equation was derived here for photons, by counting particles, and it gives the same results. That is one of the marvelous miracles of quantum mechanics. If one begins by considering a kind of state or condition for Bose particles which do not interact with each other (we have assumed that the photons do not interact with each other), and then considers that into this state there can be put either zero, or one, or two, . . . up to any number n of particles, one finds that this system behaves for all quantum mechanical purposes exactly like a harmonic oscillator. By such an oscillator we mean a dynamic system like a weight on a spring or a standing wave in a resonant cavity. And that is why it is possible to represent the electromagnetic field by photon particles. From one point of view, we can analyze the electromagnetic field in a box or cavity in terms of a lot of harmonic oscillators, treating each mode of oscillation according to quantum mechanics as a harmonic oscillator. From a different point of view, we can analyze the same physics in terms of identical Bose particles. And the results of both ways of working *are always in exact agreement*. There is no way to make up your mind whether the electromagnetic field is really to be described as a quantized harmonic oscillator or by giving how many photons there are in each condition. The two views turn out to be mathematically identical. So in the future we can speak either about the number of photons in a particular state in a box or the number of the energy level associated with a particular mode of oscillation of the electromagnetic field. They are two ways of saying the same thing. The same is true of photons in free space. They are equivalent to oscillations of a cavity whose walls have receded to infinity.

We have computed the mean energy in any particular mode in a box at the temperature T ; we need only one more thing to get the blackbody radiation law: We need to know how many modes there are at each energy. (We assume that for every mode there are some atoms in the box—or in the walls—which have energy levels that can radiate into that mode, so that each mode can get into thermal equilibrium.) The blackbody radiation law is usually stated by giving the energy per unit volume carried by the light in a small frequency interval from ω to $\omega + \Delta\omega$. So we need to know how many modes there are in a box with frequencies in the

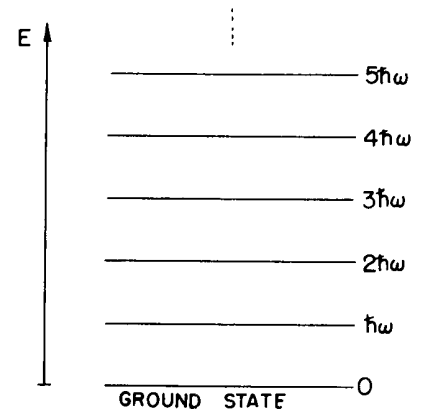


Fig. 4-7. The energy levels of a harmonic oscillator.

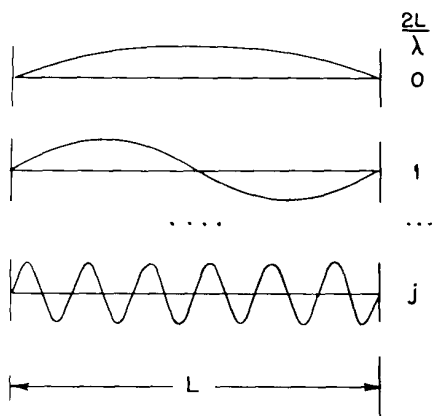


Fig. 4-8. The standing wave modes on a line.

interval $\Delta\omega$. Although this question continually comes up in quantum mechanics, it is purely a classical question about standing waves.

We will get the answer only for a rectangular box. It comes out the same for a box of any shape, but it's very complicated to compute for the arbitrary case. Also, we are only interested in a box whose dimensions are very large compared with a wavelength of the light. Then there are billions and billions of modes; there will be many in any small frequency interval $\Delta\omega$, so we can speak of the "average number" in any $\Delta\omega$ at the frequency ω . Let's start by asking how many modes there are in a one-dimensional case—as for waves on a stretched string. You know that each mode is a sine wave that has to go to zero at both ends; in other words, there must be an integral number of half-wavelengths in the length of the line, as shown in Fig. 4-8. We prefer to use the wave number $k = 2\pi/\lambda$; calling k_j the wave number of the j th mode, we have that

$$k_j = \frac{j\pi}{L}, \quad (4.34)$$

where j is any integer. The separation δk between successive modes is

$$\delta k = k_{j+1} - k_j = \frac{\pi}{L}.$$

We want to assume that kL is so large that in a small interval Δk , there are many modes. Calling $\Delta\mathfrak{N}$ the number of modes in the interval Δk , we have

$$\Delta\mathfrak{N} = \frac{\Delta k}{\delta k} = \frac{L}{\pi} \Delta k. \quad (4.35)$$

Now theoretical physicists working in quantum mechanics usually prefer to say that there are one-half as many modes; they write

$$\Delta\mathfrak{N} = \frac{L}{2\pi} \Delta k. \quad (4.36)$$

We would like to explain why. They usually like to think in terms of travelling waves—some going to the right (with a positive k) and some going to the left (with a negative k). But a "mode" is a *standing* wave which is the sum of two waves, one going in each direction. In other words, they consider each standing wave as containing two distinct photon "states." So if by $\Delta\mathfrak{N}$, one prefers to mean the number of photon states of a given k (where now k ranges over positive and negative values), one should then take $\Delta\mathfrak{N}$ half as big. (All integrals must now go from $k = -\infty$ to $k = +\infty$, and the total number of states up to any given absolute value of k will come out O.K.) Of course, we are not then describing standing waves very well, but we are counting modes in a consistent way.

Now we want to extend the results to three dimensions. A standing wave in a rectangular box must have an integral number of half-waves *along each axis*. The situation for two of the dimensions is shown in Fig. 4-9. Each wave direction and frequency is described by a vector wave number $\mathbf{k}$, whose x , y , and z components must satisfy equations like Eq. (4.34). So we have that

$$k_x = \frac{j_x\pi}{L_x},$$

$$k_y = \frac{j_y\pi}{L_y},$$

$$k_z = \frac{j_z\pi}{L_z}.$$

The number of modes with k_x in an interval Δk_x is, as before,

$$\frac{L_x}{2\pi} \Delta k_x,$$

and similarly for Δk_y and Δk_z . If we call $\Delta\mathfrak{N}(\mathbf{k})$ the number of modes for a vector

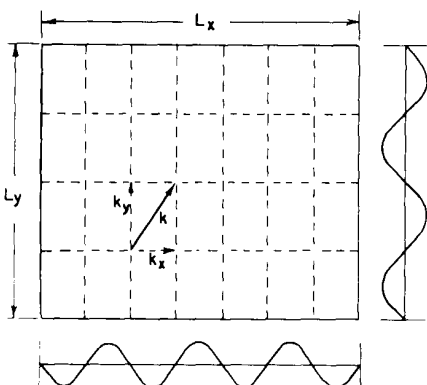


Fig. 4-9. Standing wave modes in two dimensions.

wave number $\mathbf{k}$ whose x -component is between k_x and $k_x + \Delta k_x$, whose y -component is between k_y and $k_y + \Delta k_y$, and whose z -component is between k_z and $k_z + \Delta k_z$, then

$$\Delta \mathcal{N}(\mathbf{k}) = \frac{L_x L_y L_z}{(2\pi)^3} \Delta k_x \Delta k_y \Delta k_z. \quad (4.37)$$

The product $L_x L_y L_z$ is equal to the volume V of the box. So we have the important result that for high frequencies (wavelengths small compared with the dimensions), the number of modes in a cavity is proportional to the volume V of the box and to the “volume in k -space” $\Delta k_x \Delta k_y \Delta k_z$. This result comes up again and again in many problems and should be memorized:

$$d\mathcal{N}(\mathbf{k}) = V \frac{d^3 \mathbf{k}}{(2\pi)^3}. \quad (4.38)$$

Although we have not proved it, the result is independent of the shape of the box.

We will now apply this result to find the number of photon modes for photons with frequencies in the range $\Delta\omega$. We are just interested in the energy in various modes—but not interested in the directions of the waves. We would like to know the number of modes in a given range of frequencies. In a vacuum the magnitude of $\mathbf{k}$ is related to the frequency by

$$|\mathbf{k}| = \frac{\omega}{c}. \quad (4.39)$$

So in a frequency interval $\Delta\omega$, these are all the modes which correspond to $\mathbf{k}$'s with a *magnitude* between k and $k + \Delta k$, independent of the direction. The “volume in k -space” between k and $k + \Delta k$ is a spherical shell of volume

$$4\pi k^2 \Delta k.$$

The number of modes is then

$$\Delta \mathcal{N}(\omega) = \frac{V 4\pi k^2 \Delta k}{(2\pi)^3}. \quad (4.40)$$

However, since we are now interested in frequencies, we should substitute $k = \omega/c$, so we get

$$\Delta \mathcal{N}(\omega) = \frac{V 4\pi \omega^2 \Delta\omega}{(2\pi)^3 c^3}. \quad (4.41)$$

There is one more complication. If we are talking about modes of an electromagnetic wave, for any given wave vector $\mathbf{k}$ there can be either of two polarizations (at right angles to each other). Since these modes are independent, we must—*for light*—double the number of modes. So we have

$$\Delta \mathcal{N}(\omega) = \frac{V \omega^2 \Delta\omega}{\pi^2 c^3} \quad (\text{for light}). \quad (4.42)$$

We have shown, Eq. (4.33), that each mode (or each “state”) has on the average the energy

$$\bar{n} \hbar \omega = \frac{\hbar \omega}{e^{\hbar \omega / kT} - 1}.$$

Multiplying this by the number of modes, we get the energy ΔE in the modes that lie in the interval $\Delta\omega$:

$$\Delta E = \frac{\hbar \omega}{e^{\hbar \omega / kT} - 1} \frac{V \omega^2 \Delta\omega}{\pi^2 c^3}. \quad (4.43)$$

This is the law for the frequency spectrum of blackbody radiation, which we have already found in Chapter 41 of Vol. I. The spectrum is plotted in Fig. 4-10. You see now that the answer depends on the fact that photons are Bose particles, which

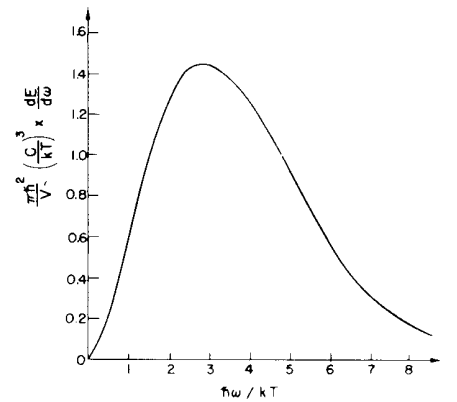


Fig. 4-10. The frequency spectrum of radiation in a cavity in thermal equilibrium, the “blackbody” spectrum.

have a tendency to try to get all into the same state (because the amplitude for doing so is large). You will remember, it was Planck's study of the blackbody spectrum (which was a mystery to classical physics), and his discovery of the formula in Eq. (4.43) that started the whole subject of quantum mechanics.

4-6 Liquid helium

Liquid helium has at low temperatures many odd properties which we cannot unfortunately take the time to describe in detail right now, but many of them arise from the fact that a helium atom is a Bose particle. One of the things is that liquid helium flows without any viscous resistance. It is, in fact, the ideal "dry" water we have been talking about in one of the earlier chapters—provided that the velocities are low enough. The reason is the following. In order for a liquid to have viscosity, there must be internal energy losses; there must be some way for one part of the liquid to have a motion that is different from that of the rest of the liquid. This means that it must be possible to knock some of the atoms into states that are different from the states occupied by other atoms. But at sufficiently low temperatures, when the thermal motions get very small, all the atoms try to get into the same condition. So, if some of them are moving along, then all the atoms try to move together in the same state. There is a kind of rigidity to the motion, and it is hard to break the motion up into irregular patterns of turbulence, as would happen, for example, with independent particles. So in a liquid of Bose particles, there is a strong tendency for all the atoms to go into the same state—which is represented by the $\sqrt{n+1}$ factor we found earlier. (For a bottle of liquid helium n is, of course, a very large number!) This cooperative motion does not happen at high temperatures, because then there is sufficient thermal energy to put the various atoms into various different higher states. But at a sufficiently low temperature there suddenly comes a moment in which all the helium atoms try to go into the same state. The helium becomes a superfluid. Incidentally, this phenomenon only appears with the isotope of helium which has atomic weight 4. For the helium isotope of atomic weight 3, the individual atoms are Fermi particles, and the liquid is a normal fluid. Since superfluidity occurs only with He^4 , it is evidently a quantum mechanical effect—due to the Bose nature of the α -particle.

4-7 The exclusion principle

Fermi particles act in a completely different way. Let's see what happens if we try to put two Fermi particles into the same state. We will go back to our original example and ask for the amplitude that two identical Fermi particles will be scattered into almost exactly the same direction. The amplitude that particle a will go in direction 1 and particle b will go in direction 2 is

$$\langle 1 | a \rangle \langle 2 | b \rangle,$$

whereas the amplitude that the outgoing directions will be interchanged is

$$\langle 2 | a \rangle \langle 1 | b \rangle.$$

Since we have Fermi particles, the amplitude for the process is the difference of these two amplitudes:

$$\langle 1 | a \rangle \langle 2 | b \rangle - \langle 2 | a \rangle \langle 1 | b \rangle. \quad (4.44)$$

Let's say that by "direction 1" we mean that the particle has not only a certain direction but also a given direction of its spin, and that "direction 2" is almost exactly the same as direction 1 and corresponds to the *same* spin direction. Then $\langle 1 | a \rangle$ and $\langle 2 | a \rangle$ are nearly equal. (This would not necessarily be true if the outgoing states 1 and 2 did not have the same spin, because there might be some reason why the amplitude would depend on the spin direction.) Now if we let

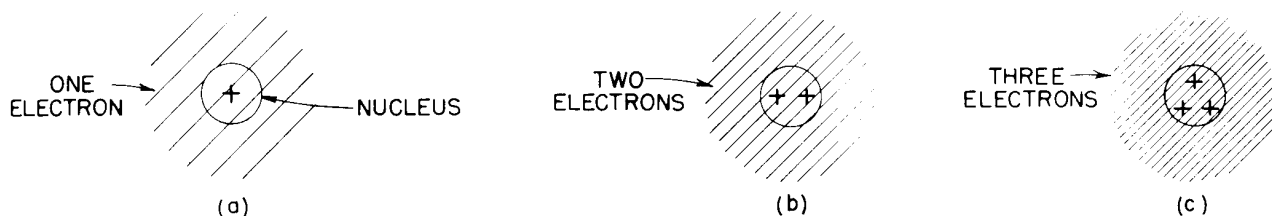


Fig. 4-11. How atoms might look if electrons behaved like Bose particles.

directions 1 and 2 approach each other, the total amplitude in Eq. (4.44) becomes zero. The result for Fermi particles is much simpler than for Bose particles. It just isn't possible at all for two Fermi particles—such as two electrons—to get into exactly the same state. You will never find two electrons in the same position with their two spins in the same direction. It is not possible for two electrons to have the same momentum and the same spin directions. If they are at the same location or with the same state of motion, the only possibility is that they must be spinning opposite to each other.

What are the consequences of this? There are a number of most remarkable effects which are a consequence of the fact that two Fermi particles cannot get into the same state. In fact, almost all the peculiarities of the material world hinge on this wonderful fact. The variety that is represented in the periodic table is basically a consequence of this one rule.

Of course, we cannot say what the world would be like if this one rule were changed, because it is just a part of the whole structure of quantum mechanics, and it is impossible to say what else would change if the rule about Fermi particles were different. Anyway, let's just try to see what would happen if only this one rule were changed. First, we can show that every atom would be more or less the same. Let's start with the hydrogen atom. It would not be noticeably affected. The proton of the nucleus would be surrounded by a spherically symmetric electron cloud, as shown in Fig. 4-11(a). As we have described in Chapter 2, the electron is attracted to the center, but the uncertainty principle requires that there be a balance between the concentration in space and in momentum. The balance means that there must be a certain energy and a certain spread in the electron distribution which determines the characteristic dimension of the hydrogen atom.

Now suppose that we have a nucleus with two units of charge, such as the helium nucleus. This nucleus would attract two electrons, and if they were Bose particles, they would—except for their electric repulsion—both crowd in as close as possible to the nucleus. A helium atom might look as shown in part (b) of the figure. Similarly, a lithium atom which has a triply charged nucleus would have an electron distribution like that shown in part (c) of Fig. 4-11. Every atom would look more or less the same—a little round ball with all the electrons sitting near the nucleus, nothing directional and nothing complicated.

Because electrons are Fermi particles, however, the actual situation is quite different. For the hydrogen atom the situation is essentially unchanged. The only difference is that the electron has a spin which we indicate by the little arrow in Fig. 4-12(a). In the case of a helium atom, however, we cannot put two electrons on top of each other. But wait, that is only true if their spins are the same. Two electrons *can* occupy the same state if their spins are opposite. So the helium atom does not look much different either. It would appear as shown in part (b) of Fig. 4-12. For lithium, however, the situation becomes quite different. Where can we put the third electron? The third electron cannot go on top of the other two because both spin directions are occupied. (You remember that for an electron or any particle with spin $1/2$ there are only two possible directions for the spin.) The third electron can't go near the place occupied by the other two, so it must take up a special condition in a different kind of state farther away from the nucleus in part (c) of the figure. (We are speaking only in a rather rough way here, because in reality all three electrons are identical; since we cannot really distinguish which one is which, our picture is only an approximate one.)

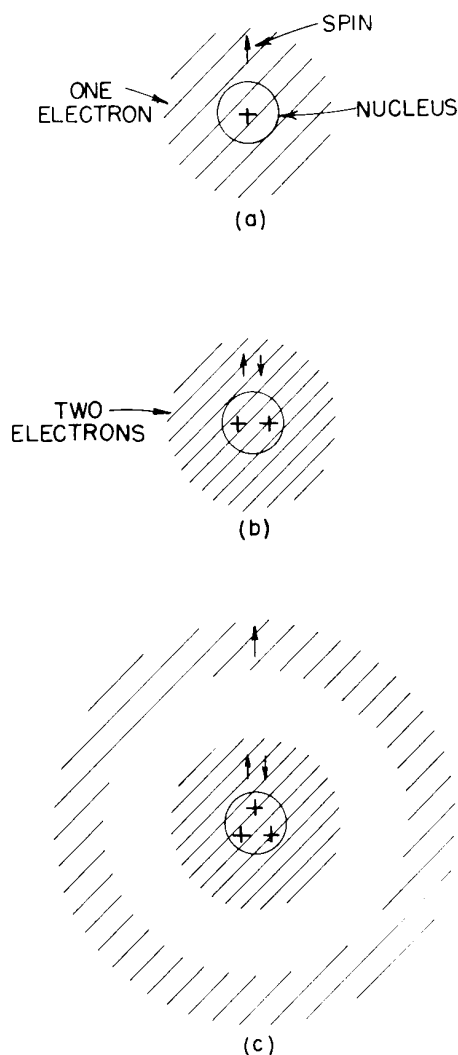


Fig. 4-12. Atomic configurations for real, Fermi-type, spin one-half electrons.

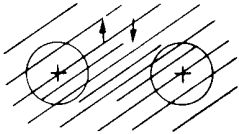


Fig. 4-13. The hydrogen molecule.

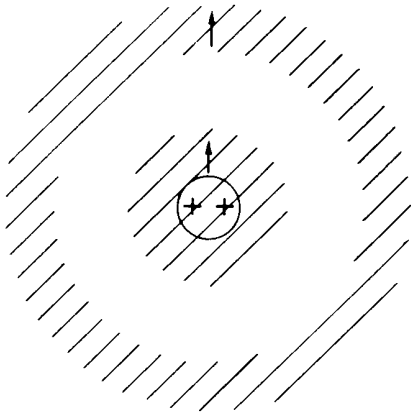


Fig. 4-14. Helium with one electron in a higher energy state.

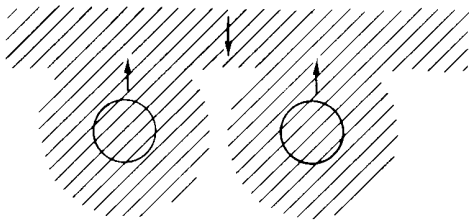


Fig. 4-15. The likely mechanism in a ferromagnetic crystal; the conduction electron is antiparallel to the unpaired inner electrons.

Now we can begin to see why different atoms will have different chemical properties. Because the third electron in lithium is farther out, it is relatively more loosely bound. It is much easier to remove an electron from lithium than from helium. (Experimentally, it takes 25 volts to ionize helium but only 5 volts to ionize lithium.) This accounts for the valence of the lithium atom. The directional properties of the valence have to do with the pattern of the waves of the outer electron, which we will not go into at the moment. But we can already see the importance of the so-called *exclusion principle*—which states that no two electrons can be found in exactly the same state (including spin).

The exclusion principle is also responsible for the stability of matter on a large scale. We explained earlier that the individual atoms in matter did not collapse because of the uncertainty principle; but this does not explain why it is that two hydrogen atoms can't be squeezed together as close as you want—why it is that all the protons don't get close together with one big smear of electrons around them. The answer is, of course, that since no more than two electrons—with opposite spins—can be in roughly the same place, the hydrogen atoms must keep away from each other. So the stability of matter on a large scale is really a consequence of the Fermi particle nature of the electrons.

Of course, if the outer electrons on two atoms have spins in opposite directions, they can get close to each other. This is, in fact, just the way that the chemical bond comes about. It turns out that two atoms together will generally have the lowest energy if there is an electron between them. It is a kind of an electrical attraction for the two positive nuclei toward the electron in the middle. It is possible to put two electrons more or less between the two nuclei so long as their spins are opposite, and the strongest chemical binding comes about this way. There is no stronger binding, because the exclusion principle does not allow there to be more than two electrons in the space between the atoms. We expect the hydrogen molecule to look more or less as shown in Fig. 4-13.

We want to mention one more consequence of the exclusion principle. You remember that if both electrons in the helium atom are to be close to the nucleus, their spins are necessarily opposite. Now suppose that we would like to try to arrange to have both electrons with the same spin—as we might consider doing by putting on a fantastically strong magnetic field that would try to line up the spins in the same direction. But then the two electrons could not occupy the same state in space. One of them would have to take on a different geometrical position, as indicated in Fig. 4-14. The electron which is located farther from the nucleus has less binding energy. The energy of the whole atom is therefore quite a bit higher. In other words, when the two spins are opposite, there is a much stronger total attraction.

So, there is an apparent, enormous force trying to line up spins opposite to each other when two electrons are close together. If two electrons are trying to go in the same place, there is a very strong tendency for the spins to become lined opposite. This apparent force trying to orient the two spins opposite to each other is much more powerful than the tiny force between the two magnetic moments of the electrons. You remember when we were speaking of ferromagnetism there was the mystery of why the electrons in different atoms had a strong tendency to line up parallel. Although there is still no quantitative explanation, it is believed that what happens is that the electrons around the core of one atom interact through the exclusion principle with the outer electrons which have become free to wander throughout the crystal. This interaction causes the spins of the free electrons and the inner electrons to take on opposite directions. But the free electrons and the inner atomic electrons can only be opposite provided all the inner electrons have the same spin direction, as indicated in Fig. 4-15. It seems probable that it is the effect of the exclusion principle acting indirectly through the free electrons that gives rise to the strong aligning forces responsible for ferromagnetism.

We will mention one further example of the influence of the exclusion principle. We have said earlier that the nuclear forces are the same between the neutron and the proton, between the proton and the proton, and between the proton and the neutron. Why is it then that a proton and a neutron can stick together to make a

deuterium nucleus, whereas there is no nucleus with just two protons or with just two neutrons? The deuteron is, as a matter of fact, bound by an energy of about 2.2 million volts, yet, there is no corresponding binding between a pair of protons to make an isotope of helium with the atomic weight 2. Such nuclei do not exist. The combination of two protons does not make a bound state.

The answer is a result of two effects: first, the exclusion principle; and second, the fact that the nuclear forces are somewhat sensitive to the direction of spin. The force between a neutron and a proton is attractive and somewhat stronger when the spins are parallel than when they are opposite. It happens that these forces are just different enough that a deuteron can only be made if the neutron and proton have their spins parallel; when their spins are opposite, the attraction is not quite strong enough to bind them together. Since the spins of the neutron and proton are each one-half and are in the same direction, the deuteron has a spin of one. We know, however, that two protons are not allowed to sit on top of each other if their spins are parallel. If it were not for the exclusion principle, two protons would be bound, but since they cannot exist at the same place and with the same spin directions, the He^2 nucleus does not exist. The protons could come together with their spins opposite, but then there is not enough binding to make a stable nucleus, because the nuclear force for opposite spins is too weak to bind a pair of nucleons. The attractive force between neutrons and protons of opposite spins can be seen by scattering experiments. Similar scattering experiments with two protons with parallel spins show that there is the corresponding attraction. So it is the exclusion principle that helps explain why deuterium can exist when He^2 cannot.

Spin One

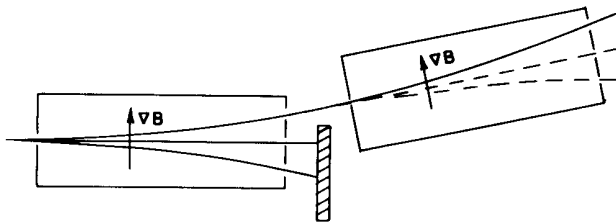
5-1 Filtering atoms with a Stern-Gerlach apparatus

In this chapter we really begin the quantum mechanics proper—in the sense that we are going to describe a quantum mechanical phenomenon in a completely quantum mechanical way. We will make no apologies and no attempt to find connections to classical mechanics. We want to talk about something new in a new language. The particular situation which we are going to describe is the behavior of the so-called quantization of the angular momentum, for a particle of *spin one*. But we won't use words like "angular momentum" or other concepts of classical mechanics until later. We have chosen this particular example because it is relatively simple, although not the simplest possible example. It is sufficiently complicated that it can stand as a prototype which can be generalized for the description of all quantum mechanical phenomena. Thus, although we are dealing with a particular example, all the laws which we mention are immediately generalizable, and we will give the generalizations so that you will see the general characteristics of a quantum mechanical description. We begin with the phenomenon of the splitting of a beam of atoms into three separate beams in a Stern-Gerlach experiment.

You remember that if we have an inhomogeneous magnetic field made by a magnet with a pointed pole tip and we send a beam through the apparatus, the beam of particles may be split into a number of beams—the number depending on the particular kind of atom and its state. We are going to take the case of an atom which gives three beams, and we are going to call that a particle of *spin one*. You can do for yourself the case of five beams, seven beams, two beams, etc.—you just copy everything down and where we have three terms, you will have five terms, seven terms, and so on.

Imagine the apparatus drawn schematically in Fig. 5-1. A beam of atoms (or particles of any kind) is collimated by some slits and passes through a non-uniform field. Let's say that the beam moves in the y -direction and that the magnetic field and its gradient are both in the z -direction. Then, looking from the side, we will see the beam split vertically into three beams, as shown in the figure. Now at the output end of the magnet we could put small counters which count the rate of arrival of particles in any one of the three beams. Or we can block off two of the beams and let the third one go on.

Suppose we block off the lower two beams and let the top-most beam go on and enter a second Stern-Gerlach apparatus of the same kind, as shown in Fig. 5-2. What happens? There are *not* three beams in the second apparatus; there is only the top beam.† This is what you would expect if you think of the second apparatus as simply an extension of the first. Those atoms which are being pushed upward continue to be pushed upward in the second magnet.



† We are assuming that the deflection angles are very small.

5-1 Filtering atoms with a Stern-Gerlach apparatus

5-2 Experiments with filtered atoms

5-3 Stern-Gerlach filters in series

5-4 Base states

5-5 Interfering amplitudes

5-6 The machinery of quantum mechanics

5-7 Transforming to a different base

5-8 Other situations

Review: Chapter 35, Vol. II, *Paramagnetism and Magnetic Resonance*. For your convenience this chapter is reproduced in the Appendix of this volume.

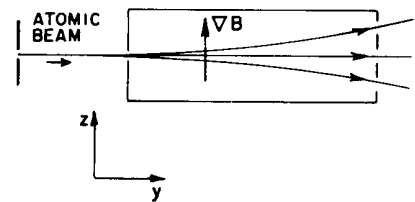


Fig. 5-1. In a Stern-Gerlach experiment, atoms of spin one are split into three beams.

Fig. 5-2. The atoms from one of the beams are sent into a second identical apparatus.

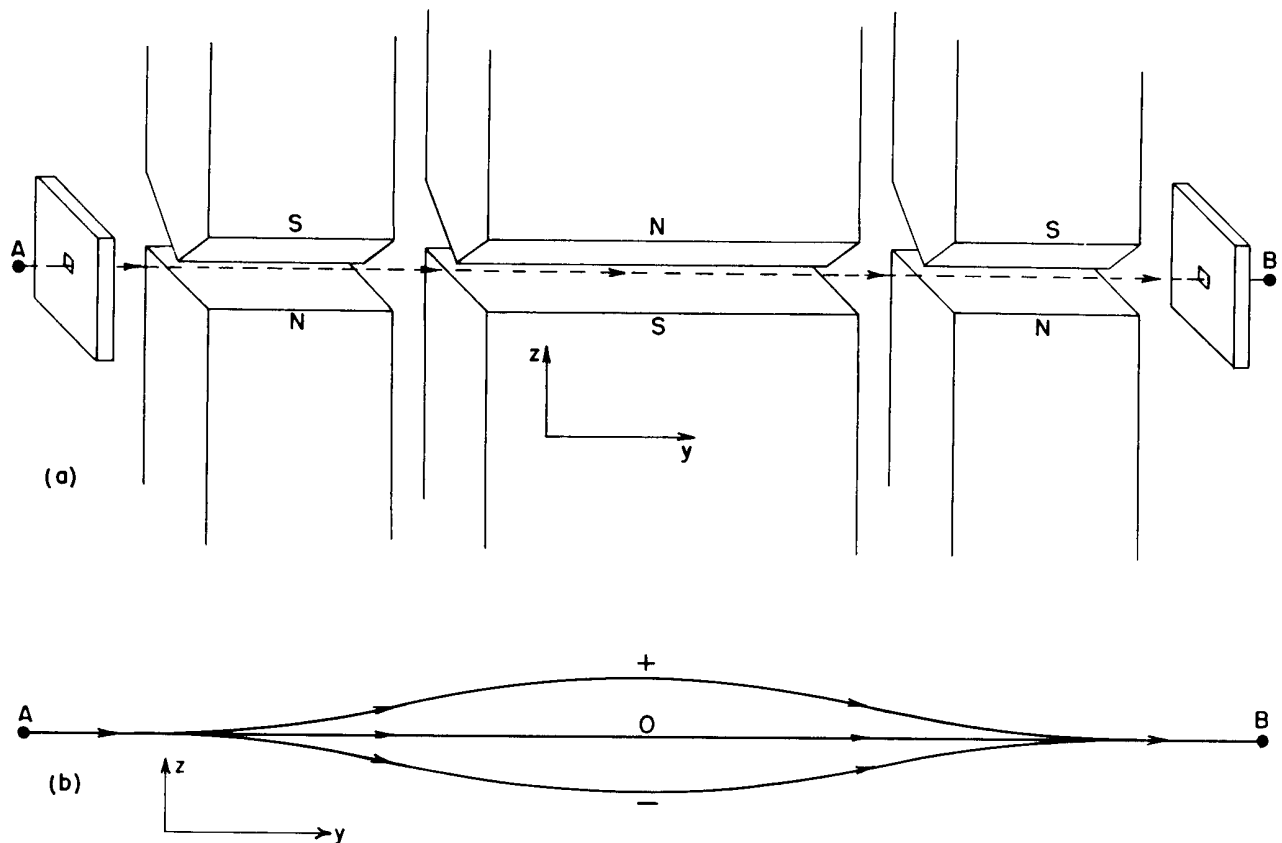


Fig. 5-3. (a) An imagined modification of a Stern-Gerlach apparatus. (b) The paths of spin-one atoms.

You can see then that the first apparatus has produced a beam of “purified” objects—atoms that get bent upward in the particular inhomogeneous field. The atoms, as they enter the original Stern-Gerlach apparatus, are of three “varieties,” and the three kinds take different trajectories. By filtering out all but one of the varieties, we can make a beam whose future behavior in the same kind of apparatus is determined and predictable. We will call this a *filtered* beam, or a *polarized* beam, or a beam in which the atoms all are known to be in a *definite state*.

For the rest of our discussion, it will be more convenient if we consider a somewhat modified apparatus of the Stern-Gerlach type. The apparatus looks more complicated at first, but it will make all the arguments simpler. Anyway, since they are only “thought experiments,” it doesn’t cost anything to complicate the equipment. (Incidentally, no one has ever done all of the experiments we will describe in just this way, but we know what *would* happen from the laws of quantum mechanics, which are, of course, based on other similar experiments. These other experiments are harder to understand at the beginning, so we want to describe some idealized—but possible—experiments.)

Figure 5-3(a) shows a drawing of the “modified Stern-Gerlach apparatus” we would like to use. It consists of a sequence of three high-gradient magnets. The first one (on the left) is just the usual Stern-Gerlach magnet and splits the incoming beam of spin-one particles into three separate beams. The second magnet has the same cross section as the first, but is twice as long *and* the polarity of its magnetic field is opposite the field in magnet 1. The second magnet pushes in the opposite direction on the atomic magnets and bends their paths back toward the axis, as shown in the trajectories drawn in the lower part of the figure. The third magnet is just like the first, and brings the three beams back together again, so that leaves the exit hole along the axis. Finally, we would like to imagine that in front of the hole at *A* there is some mechanism which can get the atoms started from rest and that after the exit hole at *B* there is a decelerating mechanism that brings the atoms back to rest at *B*. That is not essential, but it will mean that in

our analysis we won't have to worry about including any effects of the motion as the atoms come out, and can concentrate on those matters having only to do with the spin. The whole purpose of the "improved" apparatus is just to bring all the particles to the same place, and with zero speed.

Now if we want to do an experiment like the one in Fig. 5-2, we can first make a filtered beam by putting a plate in the middle of the apparatus that blocks two of the beams, as shown in Fig. 5-4. If we now put the polarized atoms through a second identical apparatus, all the atoms will take the upper path, as can be verified by putting similar plates in the way of the various beams of the second S filter and seeing whether particles get through.

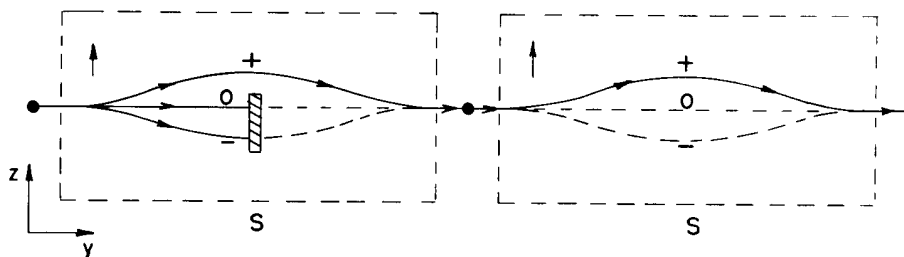


Fig. 5-4. The "improved" Stern-Gerlach apparatus as a filter.

Suppose we call the first apparatus by the name S . (We are going to consider all sorts of combinations, and we will need labels to keep things straight.) We will say that the atoms which take the top path in S are in the "plus state with respect to S "; the ones which take the middle path are in the "zero state with respect to S "; and the ones which take the lowest path are in the "minus state with respect to S ." (In the more usual language we would say that the z -component of the angular momentum was $+\hbar$, 0 , and $-\hbar$, but we are not using that language now.) Now in Fig. 5-4 the second apparatus is oriented just like the first, so the filtered atoms will all go on the upper path. Or if we had blocked off the upper and lower beams in the first apparatus and let only the zero state through, all the filtered atoms would go through the middle path of the second apparatus. And if we had blocked off all but the lowest beam in the first, there would be only a low beam in the second. We can say that in each case our first apparatus has produced a filtered beam in a *pure* state with respect to S (+, 0, or -), and we can test which state is present by putting the atoms through a second, identical apparatus.

We can make our second apparatus so that it transmits only atoms of a particular state—by putting masks inside it as we did for the first one—and then we can test the state of the incoming beam just by seeing whether anything comes out the far end. For instance, if we block off the two lower paths in the second apparatus, 100 percent of the atoms will still come through; but if we block off the upper path, nothing will get through.

To make this kind of discussion easier, we are going to invent a shorthand symbol to represent one of our improved Stern-Gerlach apparatuses. We will let the symbol

$$\left\{ \begin{array}{c} + \\ 0 \\ - \end{array} \right\}_S \quad (5.1)$$

stand for one complete apparatus. (This is *not* a symbol you will ever find used in quantum mechanics; we've just invented it for this chapter. It is simply meant to be a shorthand picture of the apparatus of Fig. 5-3.) Since we are going to want to use several apparatuses at once, and with various orientations, we will identify each with a letter underneath. So the symbol in (5.1) stands for the apparatus S . When we block off one or more of the beams inside, we will show that by some

vertical bars indicating which beam is blocked, like this:

$$\left\{ \begin{array}{c} + \\ 0 \\ - \end{array} \right\} \quad S \quad (5.2)$$

The various possible combinations we will be using are shown in Fig. 5-5.

If we have two filters in succession (as in Fig. 5-4), we will put the two symbols next to each other, like this:

$$\left\{ \begin{array}{c} + \\ 0 \\ - \end{array} \right\} \quad S \quad \left\{ \begin{array}{c} + \\ 0 \\ - \end{array} \right\} \quad S \quad (5.3)$$

For this setup, everything that comes through the first also gets through the second. In fact, even if we block off the “zero” and “minus” channels of the second apparatus, so that we have

$$\left\{ \begin{array}{c} + \\ 0 \\ - \end{array} \right\} \quad S \quad \left\{ \begin{array}{c} + \\ 0 \\ - \end{array} \right\} \quad S \quad (5.4)$$

we still get 100 percent transmission through the second apparatus. On the other hand, if we have

$$\left\{ \begin{array}{c} + \\ 0 \\ - \end{array} \right\} \quad S \quad \left\{ \begin{array}{c} + \\ 0 \\ - \end{array} \right\} \quad S \quad (5.5)$$

nothing at all comes out of the far end. Similarly,

$$\left\{ \begin{array}{c} + \\ 0 \\ - \end{array} \right\} \quad S \quad \left\{ \begin{array}{c} + \\ 0 \\ - \end{array} \right\} \quad S \quad (5.6)$$

would give nothing out. On the other hand,

$$\left\{ \begin{array}{c} + \\ 0 \\ - \end{array} \right\} \quad S \quad \left\{ \begin{array}{c} + \\ 0 \\ - \end{array} \right\} \quad S \quad (5.7)$$

would be just equivalent to

$$\left\{ \begin{array}{c} + \\ 0 \\ - \end{array} \right\} \quad S$$

by itself.

Now we want to describe these experiments quantum mechanically. We will say that an atom is in the $(+S)$ state if it has gone through the apparatus of Fig. 5-5(b), that it is in a $(0S)$ state if it has gone through (c), and in a $(-S)$ state if it has gone through (d).† Then we let $\langle b | a \rangle$ be the *amplitude* that an atom which is in state a will get through an apparatus into the b state. We can say: $\langle b | a \rangle$ is the amplitude for an atom *in* the state a to *get into* the state b . The experiment (5.4) gives us that

$$\langle +S | +S \rangle = 1,$$

† Read: $(+S)$ = “plus- S ”; $(0S)$ = “zero- S ”; $(-S)$ = “minus- S .”

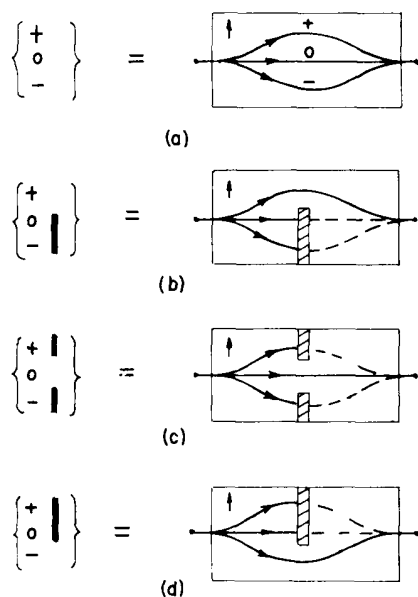


Fig. 5-5. Special shorthand symbols for Stern-Gerlach type filters.

whereas (5.5) gives us

$$\langle -S | +S \rangle = 0.$$

Similarly, the result of (5.6) is

$$\langle +S | -S \rangle = 0,$$

and of (5.7) is

$$\langle -S | -S \rangle = 1.$$

As long as we deal only with “pure” states—that is, we have only one channel open—there are nine such amplitudes, and we can write them in a table:

$$\begin{array}{c} \text{to} \\ \begin{array}{c} +S \\ 0S \\ -S \end{array} \end{array} \begin{array}{c} \text{from} \\ \begin{array}{ccc} +S & 0S & -S \end{array} \\ \left| \begin{array}{ccc} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{array} \right| \end{array} \quad (5.8)$$

This array of nine numbers—called a *matrix*—summarizes the phenomena we’ve been describing.

5-2 Experiments with filtered atoms

Now comes the big question: What happens if the second apparatus is tipped to a different angle, so that its field axis is no longer parallel to the first? It could be not only tipped, but also pointed in a different direction—for instance, it could take the beam off at 90° with respect to the original direction. To take it easy at first, let’s first think about an arrangement in which the second Stern-Gerlach experiment is tilted by some angle α about the y -axis, as shown in Fig. 5-6. We’ll call the second apparatus T . Suppose that we now set up the following experiment:

$$\begin{array}{c} \left\{ \begin{array}{c} + \\ 0 \\ - \end{array} \right\} \\ S \end{array} \quad \begin{array}{c} \left\{ \begin{array}{c} + \\ 0 \\ - \end{array} \right\} \\ T \end{array},$$

or the experiment:

$$\begin{array}{c} \left\{ \begin{array}{c} + \\ 0 \\ - \end{array} \right\} \\ S \end{array} \quad \begin{array}{c} \left\{ \begin{array}{c} + \\ 0 \\ - \end{array} \right\} \\ T \end{array}.$$

What comes out at the far end in these cases?

The answer is this: If the atoms are in a definite state with respect to S , they are *not* in the same state with respect to T —a $(+S)$ state is *not* also a $(+T)$ state. There *is*, however, a certain *amplitude* to find the atom in a $(+T)$ state—or a $(0T)$ state or a $(-T)$ state.

In other words, as careful as we have been to make sure that we have the atoms in a definite condition, the fact of the matter is that if it goes through an apparatus which is tilted at a different angle it has, so to speak, to “reorient”

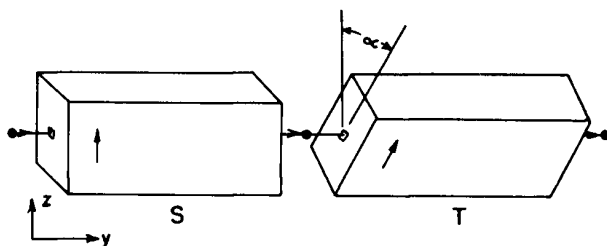


Fig. 5-6. Two Stern-Gerlach type filters in series; the second is tilted at the angle α with respect to the first.

itself—which it does, don't forget, by luck. We can put only one particle through at a time, and then we can only ask the question: What is the probability that it gets through? Some of the atoms that have gone through S will end in a $(+T)$ state, some of them will end in a $(0T)$, and some in a $(-T)$ state—all with different odds. These odds can be calculated by the absolute squares of complex amplitudes; what we want is some mathematical method, or quantum mechanical description, for these amplitudes. What we need to know are various quantities like

$$\langle -T | +S \rangle,$$

by which we mean the amplitude that an atom initially in the $(+S)$ state can get into the $(-T)$ condition (which is *not* zero unless T and S are lined up parallel to each other). There are other amplitudes like

$$\langle +T | 0S \rangle, \quad \text{or} \quad \langle 0T | -S \rangle, \quad \text{etc.}$$

There are, in fact, nine such amplitudes—another matrix—that a theory of particles should tell us how to calculate. Just as $F = ma$ tells us how to calculate what happens to a classical particle in any circumstance, the laws of quantum mechanics permit us to determine the amplitude that a particle will get through a particular apparatus. The central problem, then, is to be able to calculate—for any given tilt angle α , or in fact for any orientation whatever—the nine amplitudes:

$$\begin{array}{lll} \langle +T | +S \rangle, & \langle +T | 0S \rangle, & \langle +T | -S \rangle, \\ \langle 0T | +S \rangle, & \langle 0T | 0S \rangle, & \langle 0T | -S \rangle, \\ \langle -T | +S \rangle, & \langle -T | 0S \rangle, & \langle -T | -S \rangle. \end{array} \quad (5.9)$$

We can already figure out some relations among these amplitudes. First, according to our definitions, the absolute square

$$|\langle +T | +S \rangle|^2$$

is the *probability* that an atom in a $(+S)$ state will enter a $(+T)$ state. We will often find it more convenient to write such squares in the equivalent form

$$\langle +T | +S \rangle \langle +T | +S \rangle^*.$$

In the same notation the number

$$\langle 0T | +S \rangle \langle 0T | +S \rangle^*$$

is the probability that a particle in the $(+S)$ state will enter the $(0T)$ state, and

$$\langle -T | +S \rangle \langle -T | +S \rangle^*$$

is the probability that it will enter the $(-T)$ state. But the way our apparatuses are made, every atom which enters the T apparatus must be found in *some* one of the three states of the T apparatus—there's nowhere else for a given kind of atom to go. So the sum of the three probabilities we've just written must be equal to 100 percent. We have the relation

$$\begin{aligned} \langle +T | +S \rangle \langle +T | +S \rangle^* + \langle 0T | +S \rangle \langle 0T | +S \rangle^* \\ + \langle -T | +S \rangle \langle -T | +S \rangle^* = 1. \end{aligned} \quad (5.10)$$

There are, of course, two other such equations that we get if we start with a $(0S)$ or a $(-S)$ state. But they are all we can easily get, so we'll go on to some other general questions.

5-3 Stern-Gerlach filters in series

Here is an interesting question: Suppose we had atoms filtered into the $(+S)$ state, then we put them through a second filter, say into a $(0T)$ state, and *then* through *another* $+S$ filter. (We'll call the last filter S' just so we can distinguish

it from the first S -filter.) Do the atoms remember that they were once in a $(+S)$ state? In other words, we have the following experiment:

$$\left\{ \begin{array}{c} + \\ 0 \\ - \end{array} \right\}_S \quad \left\{ \begin{array}{c} + \\ 0 \\ - \end{array} \right\}_T \quad \left\{ \begin{array}{c} + \\ 0 \\ - \end{array} \right\}_{S'}. \quad (5.11)$$

We want to know whether all those that get through T also get through S' . *They do not.* Once they have been filtered by T , they *do not remember* in any way that they were in a $(+S)$ state when they entered T . Note that the second S apparatus in (5.11) is oriented exactly the same as the first, so it is still an S -type filter. The states filtered by S' are, of course, still $(+S)$, $(0S)$, and $(-S)$.

The important point is this: *If the T filter passes only one beam*, the fraction that gets through the second S filter depends only on the setup of the T filter, and is completely independent of what precedes it. The fact that the same atoms were once sorted by an S filter has no influence whatever on what they will do once they have been sorted again into a pure beam by a T apparatus. From then on, the probability for getting into different states is the same no matter what happened before they got into the T apparatus.

As an example, let's compare the experiment of (5.11) with the following experiment:

$$\left\{ \begin{array}{c} + \\ 0 \\ - \end{array} \right\}_S \quad \left\{ \begin{array}{c} + \\ 0 \\ - \end{array} \right\}_T \quad \left\{ \begin{array}{c} + \\ 0 \\ - \end{array} \right\}_{S'} \quad (5.12)$$

in which only the first S is changed. Let's say that the angle α (between S and T) is such that in experiment (5.11) one-third of the atoms that get through T also get through S' . In experiment (5.12), although there will, in general, be a different number of atoms coming through T , the *same fraction of these*—one-third—will also get through S' .

We can, in fact, show from what you have learned earlier that the fraction of the atoms that come out of T and get through any particular S' depends only on T and S' , not on anything that happened earlier. Let's compare experiment (5.12) with

$$\left\{ \begin{array}{c} + \\ 0 \\ - \end{array} \right\}_S \quad \left\{ \begin{array}{c} + \\ 0 \\ - \end{array} \right\}_T \quad \left\{ \begin{array}{c} + \\ 0 \\ - \end{array} \right\}_{S'}. \quad (5.13)$$

The amplitude that an atom that comes out of S will also get through both T and S' is, for the experiments of (5.12),

$$\langle +S | 0T \rangle \langle 0T | 0S \rangle.$$

The corresponding probability is

$$|\langle +S | 0T \rangle \langle 0T | 0S \rangle|^2 = |\langle +S | 0T \rangle|^2 |\langle 0T | 0S \rangle|^2.$$

The probability for experiment (5.13) is

$$|\langle 0S | 0T \rangle \langle 0T | 0S \rangle|^2 = |\langle 0S | 0T \rangle|^2 |\langle 0T | 0S \rangle|^2.$$

The ratio is

$$\frac{|\langle 0S | 0T \rangle|^2}{|\langle +S | 0T \rangle|^2}$$

and depends only on T and S' , and not at all on which beam $(+S)$, $(0S)$, or $(-S)$ is selected by S . (The absolute numbers may go up and down together depending on how much gets through T .) We would, of course, find the same result if we compared the probabilities that the atoms would go into the plus or the minus

states with respect to S' , or the ratio of the probabilities to go into the zero or minus states.

In fact, since these ratios depend only on which beam is allowed to pass through T , and not on the selection made by the first S filter, it is clear that we would get the same result even if the last apparatus were not an S filter. If we use for the third apparatus—which we will now call R —one rotated by some arbitrary angle with respect to T , we would find that a ratio such as $|\langle 0 R | 0 T \rangle|^2 / |\langle + R | 0 T \rangle|^2$ was independent of which beam was passed by the first filter S .

5-4 Base states

These results illustrate one of the basic principles of quantum mechanics: Any atomic system can be separated by a filtering process into a certain set of what we will call *base states*, and the future behavior of the atoms in any single given base state depends only on the nature of the base state—it is independent of any previous history.† The base states depend, of course, on the filter used; for instance, the three states $(+T)$, $(0 T)$, and $(-T)$ are one set of base states; the three states $(+S)$, $(0 S)$, and $(-S)$ are another. There are any number of possibilities each as good as any other.

We should be careful to say that we are considering *good* filters which do indeed produce “pure” beams. If, for instance, our Stern-Gerlach apparatus didn’t produce a good separation of the three beams so that we could not separate them cleanly by our masks, then we could not make a complete separation into base states. We can tell if we have pure base states by seeing whether or not the beams can be split again in another filter of the same kind. If we have a pure $(+T)$ state, for instance, all the atoms will go through

$$\left\{ \begin{array}{c} + \\ 0 \\ - \end{array} \right\}_T,$$

and none will go through

$$\left\{ \begin{array}{c} + \\ 0 \\ - \end{array} \right\}_T,$$

or through

$$\left\{ \begin{array}{c} + \\ 0 \\ - \end{array} \right\}_T.$$

Our statement about base states means that it is *possible* to filter to some pure state, so that no further filtering by an identical apparatus is possible.

We must also point out that what we are saying is exactly true only in rather idealized situations. In any real Stern-Gerlach apparatus, we would have to worry about diffraction by the slits that could cause some atoms to go into states corresponding to different angles, or about whether the beams might contain atoms with different excitations of their internal states, and so on. We have idealized the situation so that we are talking only about the states that are split in a magnetic field; we are ignoring things having to do with position, momentum, internal excitations, and the like. In general, one would need to consider also base states which are sorted out with respect to such things also. But to keep the concepts simple, we are considering only our set of three states, which is sufficient for the exact treatment of the idealized situation in which the atoms don’t get torn up in

† We do not intend the word “base state” to imply anything more than what is said here. They are not to be thought of as “basic” in any sense. We are using the word base with the thought of a *basis* for a description, somewhat in the sense that one speaks of “numbers to the base ten.”

going through the apparatus, or otherwise badly treated, and come to rest when they leave the apparatus.

You will note that we always begin our thought experiments by taking a filter with only one channel open, so that we start with some definite base state. We do this because atoms come out of a furnace in various states determined at random by the accidental happenings inside the furnace. (It gives what is called an “unpolarized” beam.) This randomness involves probabilities of the “classical” kind—as in coin tossing—which are different from the quantum mechanical probabilities we are worrying about now. Dealing with an unpolarized beam would get us into additional complications that are better to avoid until after we understand the behavior of polarized beams. So don’t try to consider at this point what happens if the *first* apparatus lets more than one beam through. (We will tell you how you can handle such cases at the end of the chapter.)

Let’s now go back and see what happens when we go from a base state for one filter to a base state for a different filter. Suppose we start again with

$$\left\{ \begin{array}{c} + \\ 0 \\ - \end{array} \right\}_S \quad \left\{ \begin{array}{c} + \\ 0 \\ - \end{array} \right\}_T.$$

The atoms which come out of T are in the base state $(0\ T)$ and have no memory that they were once in the state $(+S)$. Some people would say that in the filtering by T we have “lost the information” about the previous state $(+S)$ because we have “disturbed” the atoms when we separated them into three beams in the apparatus T . But that is not true. The past information is not lost by the *separation* into three beams, but by the *blocking masks* that are put in—as we can see by the following set of experiments.

We start with a $+S$ filter and will call N the number of atoms that come through it. If we follow this by a $0\ T$ filter, the number of atoms that come out is some fraction of the original number, say αN . If we then put another $+S$ filter, only some fraction β of these atoms will get to the far end. We can indicate this in the following way:

$$\left\{ \begin{array}{c} + \\ 0 \\ - \end{array} \right\}_S \xrightarrow{N} \left\{ \begin{array}{c} + \\ 0 \\ - \end{array} \right\}_T \xrightarrow{\alpha N} \left\{ \begin{array}{c} + \\ 0 \\ - \end{array} \right\}_{S'} \xrightarrow{\beta \alpha N}. \quad (5.14)$$

If our third apparatus S' selected a different state, say the $(0\ S)$ state, a different fraction, say γ , would get through.† We would have

$$\left\{ \begin{array}{c} + \\ 0 \\ - \end{array} \right\}_S \xrightarrow{N} \left\{ \begin{array}{c} + \\ 0 \\ - \end{array} \right\}_T \xrightarrow{\alpha N} \left\{ \begin{array}{c} + \\ 0 \\ - \end{array} \right\}_{S'} \xrightarrow{\gamma \alpha N}. \quad (5.15)$$

Now suppose we repeat these two experiments but remove all the masks from T . We would then find the remarkable results as follows:

$$\left\{ \begin{array}{c} + \\ 0 \\ - \end{array} \right\}_S \xrightarrow{N} \left\{ \begin{array}{c} + \\ 0 \\ - \end{array} \right\}_T \xrightarrow{N} \left\{ \begin{array}{c} + \\ 0 \\ - \end{array} \right\}_{S'} \xrightarrow{N}, \quad (5.16)$$

$$\left\{ \begin{array}{c} + \\ 0 \\ - \end{array} \right\}_S \xrightarrow{N} \left\{ \begin{array}{c} + \\ 0 \\ - \end{array} \right\}_T \xrightarrow{N} \left\{ \begin{array}{c} + \\ 0 \\ - \end{array} \right\}_{S'} \xrightarrow{0}. \quad (5.17)$$

† In terms of our earlier notation $\alpha = |\langle 0\ T | +\ S \rangle|^2$, $\beta = |\langle +\ S | 0\ T \rangle|^2$, and $\gamma = |\langle 0\ S | 0\ T \rangle|^2$.

All the atoms get through S' in the first case, but *none* in the second case! This is one of the great laws of quantum mechanics. That nature works this way is not self-evident, but the results we have given correspond for our idealized situation to the quantum mechanical behavior observed in innumerable experiments.

5-5 Interfering amplitudes

How can it be that in going from (5.15) to (5.17)—by *opening more channels*—we let *fewer* atoms through? This is the old, deep mystery of quantum mechanics—the interference of amplitudes. It's the same kind of thing we first saw in the two-slit interference experiment with electrons. We saw that we could get fewer electrons at some places with both slits open than we got with one slit open. It works quantitatively this way. We can write the amplitude that an atom will get through T and S' in the apparatus of (5.17) as the sum of three amplitudes, one for each of the three beams in T ; the sum is equal to zero:

$$\langle 0 S | +T \rangle \langle +T | +S \rangle + \langle 0 S | 0T \rangle \langle 0T | +S \rangle + \langle 0 S | -T \rangle \langle -T | +S \rangle = 0. \quad (5.18)$$

None of the three individual amplitudes is zero—for example, the absolute square of the second amplitude is γ_α , see (5.15)—but the *sum is zero*. We would have also the same answer if S' were set to select the $(-S)$ state. However, in the setup of (5.16), the answer is different. If we call a the amplitude to get through T and S' , in this case we have†

$$a = \langle +S | +T \rangle \langle +T | +S \rangle + \langle +S | 0T \rangle \langle 0T | +S \rangle + \langle +S | -T \rangle \langle -T | +S \rangle = 1. \quad (5.19)$$

In the experiment (5.16) the beam has been split and recombined. Humpty Dumpty has been put back together again. The information about the original $(+S)$ state is retained—it is just as though the T apparatus were not there at all. This is true whatever is put after the “wide-open” T apparatus. We could follow it with an R filter—a filter at some odd angle—or anything we want. The answer will always be the same as if the atoms were taken directly from the first S filter.

So this is the important principle: A T filter—or any filter—with wide-open masks produces no change at all. We should make one additional condition. The wide-open filter must not only transmit all three beams, but it must also *not* produce unequal disturbances on the three beams. For instance, it should not have a strong electric field near one beam and not the others. The reason is that even if this extra disturbance would still let all the atoms through the filter, it could change the *phases* of some of the amplitudes. Then the interference would be changed, and the amplitudes in Eqs. (5.18) and (5.19) would be different. We will always assume that there are no such extra disturbances.

Let's rewrite Eqs. (5.18) and (5.19) in an improved notation. We will let i stand for any one of the three states $(+T)$, $(0T)$, or $(-T)$; then the equations can be written:

$$\sum_{\text{all } i} \langle 0 S | i \rangle \langle i | +S \rangle = 0 \quad (5.20)$$

and

$$\sum_{\text{all } i} \langle +S | i \rangle \langle i | +S \rangle = 1. \quad (5.21)$$

Similarly, for an experiment where S' is replaced by a completely arbitrary filter R , we have

$$\left\{ \begin{array}{c} + \\ 0 \\ - \end{array} \right\}_S \quad \left\{ \begin{array}{c} + \\ 0 \\ - \end{array} \right\}_T \quad \left\{ \begin{array}{c} + \\ 0 \\ - \end{array} \right\}_R. \quad (5.22)$$

† We really cannot conclude from the experiment that $a = 1$, but only that $|a|^2 = 1$, so a might be $e^{i\delta}$, but it can be shown that the choice $\delta = 0$ represents no real loss of generality.

The results will always be the same as if the T apparatus were left out and we had only

$$\left\{ \begin{array}{c} + \\ 0 \\ - \end{array} \right\}_S \quad \left\{ \begin{array}{c} + \\ 0 \\ - \end{array} \right\}_R.$$

Or, expressed mathematically,

$$\sum_{\text{all } i} \langle +R | i \rangle \langle i | +S \rangle = \langle +R | +S \rangle. \quad (5.23)$$

This is our fundamental law, and it is generally true so long as i stands for the three base states of any filter.

You will notice that in the experiment (5.22) there is no special relation of S and R to T . Furthermore, the arguments would be the same no matter what states they selected. To write the equation in a general way, without having to refer to the specific states selected by S and R , let's call ϕ ("phi") the state prepared by the first filter (in our special example, $+S$) and χ ("khi") the state tested by the final filter (in our example, $+R$). Then we can state our fundamental law of Eq. (5.23) in the form

$$\langle \chi | \phi \rangle = \sum_{\text{all } i} \langle \chi | i \rangle \langle i | \phi \rangle, \quad (5.24)$$

where i is to range over the three base states of some particular filter.

We want to emphasize again what we mean by base states. They are like the three states which can be selected by one of our Stern-Gerlach apparatuses. One condition is that if you have a base state, then the future is independent of the past. Another condition is that if you have a complete set of base states, Eq. (5.24) is true for any set of beginning and ending states ϕ and χ . There is, however, *no unique* set of base states. We began by considering base states *with respect to* a particular apparatus T . We could equally well consider a *different set* of base states with respect to an apparatus S , or with respect to R , etc.† We usually speak of the base states "in a certain representation."

Another condition on a set of base states in any particular representation is that they are all completely different. By that we mean that if we have a $(+T)$ state, there is no amplitude for it to go into a $(0T)$ or a $(-T)$ state. If we let i and j stand for any two base states of a particular set, the general rules discussed in connection with (5.8) are that

$$\langle j | i \rangle = 0$$

for all i and j that are not equal. Of course, we know that

$$\langle i | i \rangle = 1.$$

These two equations are usually written as

$$\langle j | i \rangle = \delta_{ji}, \quad (5.25)$$

where δ_{ji} (the "Kronecker delta") is a symbol that is defined to be zero for $i \neq j$, and to be one for $i = j$.

Equation (5.25) is not independent of the other laws we have mentioned. It happens that we are not particularly interested in the mathematical problem of finding the minimum set of independent axioms that will give all the laws as consequences.‡ We are satisfied if we have a set that is complete and not apparently inconsistent. We can, however, show that Eqs. (5.25) and (5.24) are not independent. Suppose we let ϕ in Eq. (5.24) represent one of the base states of the

† In fact, for atomic systems with three or more base states, there exist other kinds of filters—quite different from a Stern-Gerlach apparatus—which can be used to get more choices for the set of base states (each set with the same *number* of states).

‡ Redundant *truth* doesn't bother us!

same set as i , say the j th state; then we have

$$\langle \chi | j \rangle = \sum_i \langle \chi | i \rangle \langle i | j \rangle.$$

But Eq. (5.25) says that $\langle i | j \rangle$ is zero unless $i = j$, so the sum becomes just $\langle \chi | j \rangle$ and we have an identity, which shows that the two laws are not independent.

We can see that there must be another relation among the amplitudes if both Eqs. (5.10) and (5.24) are true. Equation (5.10) is

$$\langle +T | +S \rangle \langle +T | +S \rangle^* + \langle 0T | +S \rangle \langle 0T | +S \rangle^* + \langle -T | +S \rangle \langle -T | +S \rangle^* = 1.$$

If we write Eq. (5.24), letting both ϕ and χ be the state $(+S)$, the left-hand side is $\langle +S | +S \rangle$, which is clearly $= 1$; so we get once more Eq. (5.19),

$$\langle +S | +T \rangle \langle +T | +S \rangle + \langle +S | 0T \rangle \langle 0T | +S \rangle + \langle +S | -T \rangle \langle -T | +S \rangle = 1.$$

These two equations are consistent (for all relative orientations of the T and S apparatuses) only if

$$\begin{aligned}\langle +S | +T \rangle &= \langle +T | +S \rangle^*, \\ \langle +S | 0T \rangle &= \langle 0T | +S \rangle^*, \\ \langle +S | -T \rangle &= \langle -T | +S \rangle^*.\end{aligned}$$

And it follows that for any states ϕ and χ ,

$$\langle \phi | \chi \rangle = \langle \chi | \phi \rangle^*. \quad (5.26)$$

If this were not true, probability wouldn't be "conserved," and particles would get "lost."

Before going on, we want to summarize the three important general laws about amplitudes. They are Eqs. (5.24), (5.25), and (5.26):

$$\begin{aligned}\text{I} \quad \langle j | i \rangle &= \delta_{ji}, \\ \text{II} \quad \langle \chi | \phi \rangle &= \sum_{\text{all } i} \langle \chi | i \rangle \langle i | \phi \rangle, \\ \text{III} \quad \langle \phi | \chi \rangle &= \langle \chi | \phi \rangle^*.\end{aligned} \quad (5.27)$$

In these equations the i and j refer to *all* the base states of some *one* representation, while ϕ and χ represent any possible states of the atom. It is important to note that II is valid only if the sum is carried out over *all* the base states of the system (in our case, three: $+T$, $0T$, $-T$). These laws say nothing about what we should choose for a base for our set of base states. We began by using a T apparatus, which is a Stern-Gerlach experiment with some arbitrary orientation; but any other orientation, say W , would be just as good. We would have a different set of states to use for i and j , but all the laws would still be good—there is no unique set. One of the great games of quantum mechanics is to make use of the fact that things can be calculated in more than one way.

5-6 The machinery of quantum mechanics

We want to show you why these laws are useful. Suppose we have an atom in a given condition (by which we mean that it was prepared in a certain way), and we want to know what will happen to it in some experiment. In other words, we start with our atom in the state ϕ and want to know what are the *odds* that it will go through some apparatus which accepts atoms only in the condition χ . The laws say that we can describe the apparatus completely in terms of three complex numbers $\langle \chi | i \rangle$, the amplitudes for each base state to be in the condition χ ; and that we can tell what will happen if an atom is put into the apparatus if we describe the state of the atom by giving three numbers $\langle i | \phi \rangle$, the amplitudes for the atom in its original condition to be found in each of the three base states. This is an important idea.

Let's consider another illustration. Think of the following problem: We start with an S apparatus; then we have a complicated mess of junk, which we can call A , and then an R apparatus—like this:

$$\left\{ \begin{array}{c} + \\ 0 \\ - \end{array} \right\} \left\{ \begin{array}{c} \\ A \\ \end{array} \right\} \left\{ \begin{array}{c} + \\ 0 \\ - \end{array} \right\}. \quad (5.28)$$

$S \qquad \qquad R$

By A we mean any complicated arrangement of Stern-Gerlach apparatuses with masks or half-masks, oriented at peculiar angles, with odd electric and magnetic fields . . . almost anything you want to put. (It's nice to do thought experiments—you don't have to go to all the trouble of actually *building* the apparatus!) The problem then is: With what amplitude does a particle that enters the section A in a $(+S)$ state come out of A in the $(0R)$ state, so that it will get through the last R filter? There is a regular notation for such an amplitude; it is

$$\langle 0R | A | +S \rangle.$$

As usual, it is to be read from right to left (like Hebrew):

$$\langle \text{finish} | \text{through} | \text{start} \rangle.$$

If by chance A doesn't do anything—but is just an open channel—then we write

$$\langle 0R | 1 | +S \rangle = \langle 0R | +S \rangle; \quad (5.29)$$

the two symbols are equivalent. For a more general problem, we might replace $(+S)$ by a general starting state ϕ and $(0R)$ by a general finishing state χ , and we would want to know the amplitude

$$\langle \chi | A | \phi \rangle.$$

A complete analysis of the apparatus A would have to give the amplitude $\langle \chi | A | \phi \rangle$ for every possible pair of states ϕ and χ —an infinite number of combinations! How then can we give a concise description of the behavior of the apparatus A ? We can do it in the following way. Imagine that the apparatus of (5.28) is modified to be

$$\left\{ \begin{array}{c} + \\ 0 \\ - \end{array} \right\} \left\{ \begin{array}{c} + \\ 0 \\ - \end{array} \right\} \left\{ \begin{array}{c} \\ A \\ \end{array} \right\} \left\{ \begin{array}{c} + \\ 0 \\ - \end{array} \right\} \left\{ \begin{array}{c} + \\ 0 \\ - \end{array} \right\}. \quad (5.30)$$

$S \qquad T \qquad \qquad T \qquad R$

This is really no modification at all since the wide-open T apparatuses don't do anything. But they do suggest how we can analyze the problem. There is a certain set of amplitudes $\langle i | +S \rangle$ that the atoms from S will get into the i state of T . Then there is another set of amplitudes that an i state (with respect to T) entering A will come out as a j state (with respect to T). And finally there is an amplitude that each j state will get through the last filter as a $(0R)$ state. For each possible alternative path, there is an amplitude of the form

$$\langle 0R | j \rangle \langle j | A | i \rangle \langle i | +S \rangle,$$

and the total amplitude is the sum of the terms we can get with all possible combinations of i and j . The amplitude we want is

$$\sum_{ij} \langle 0R | j \rangle \langle j | A | i \rangle \langle i | +S \rangle. \quad (5.31)$$

If $(0R)$ and $(+S)$ are replaced by general states χ and ϕ , we would have the same kind of expression; so we have the general result

$$\langle \chi | A | \phi \rangle = \sum_{ij} \langle \chi | j \rangle \langle j | A | i \rangle \langle i | \phi \rangle. \quad (5.32)$$

This then is the machinery of quantum mechanics for a spin-one particle. Every *state* is described by three numbers which are the amplitudes to be in each of some selected set of base states. Every apparatus is described by nine numbers which are the amplitudes to go from one base state to another in the apparatus. From these numbers anything can be calculated.

$$\begin{array}{rcccl}
& & & \text{from} & \\
& & + & 0 & - \\
\text{to} & + & \langle + | A | + \rangle & \langle + | A | 0 \rangle & \langle + | A | - \rangle \\
& 0 & \langle 0 | A | + \rangle & \langle 0 | A | 0 \rangle & \langle 0 | A | - \rangle \\
& - & \langle - | A | + \rangle & \langle - | A | 0 \rangle & \langle - | A | - \rangle
\end{array} \quad (5.33)$$
$$\left\{ \begin{array}{c} + \\ 0 \\ - \end{array} \right\}_S \quad \left\{ C \right\} \quad \left\{ \begin{array}{c} + \\ 0 \\ - \end{array} \right\}_R. \quad (5.34)$$
$$\left\{ C \right\} = \left\{ A \right\} \cdot \left\{ B \right\} . \quad (5.35)$$
$$\langle \chi | C | \phi \rangle$$
$$\langle \chi | C | \phi \rangle = \sum_k \langle \chi | B | k \rangle \langle k | A | \phi \rangle.$$
$$\langle j | C | i \rangle = \sum_k \langle j | B | k \rangle \langle k | A | i \rangle. \quad (5.36)$$

5-14

5-7 Transforming to a different base

We want to make one final point about the base states used in the calculations. Suppose we have chosen to work with some particular base—say the S base—and another fellow decides to do the same calculations with a different base—say the T base. To keep things straight let's call our base states the (iS) states, where $i = +, 0, -$. Similarly, we can call his base states (jT) . How can we compare our work with his? The final answers for the result of any measurement should come out the same, but in the calculations the various amplitudes and matrices used will be different. How are they related? For instance, if we both start with the same ϕ , we will describe it in terms of the three amplitudes $\langle iS | \phi \rangle$ that ϕ goes into our base states in the S representation, whereas he will describe it by the amplitudes $\langle jT | \phi \rangle$ that the state ϕ goes into the base states in his T representation. How can we check that we are really both describing the same state ϕ ? We can do it with the general rule II in (5.27). Replacing x by any one of his states jT , we have

$$\langle jT | \phi \rangle = \sum_i \langle jT | iS \rangle \langle iS | \phi \rangle. \quad (5.37)$$

To relate the two representations, we need only give the nine complex numbers of the matrix $\langle jT | iS \rangle$. This matrix can then be used to convert all of his equations to our form. It tells us how to *transform* from one set of base states to another. (For this reason $\langle jT | iS \rangle$ is sometimes called “the transformation matrix from representation S to representation T .” Big words!) Big words!

For the case of spin-one particles for which we have only three base states (for higher spins, there are more) the mathematical situation is analogous to what we have seen in vector algebra. Every vector can be represented by giving three numbers—the components along the axes x , y , and z . That is, every vector can be resolved into three “base” vectors which are vectors along the three axes. But suppose someone else chooses to use a different set of axes— x' , y' , and z' . He will be using different numbers to represent any particular vector. His calculations will look different, but the final results will be the same. We have considered this before and know the rules for transforming vectors from one set of axes to another.

You may want to see how the quantum mechanical transformations work by trying some out; so we will give here, without proof, the transformation matrices for converting the spin-one amplitudes in one representation S to another representation T , for various special relative orientations of the S and T filters. (We will show you in a later chapter how to derive these same results.)

First case: The T apparatus has the same y -axis (along which the particles move) as the S apparatus, but is rotated about the common y -axis by the angle α (as in Fig. 5-6). (To be specific, a set of coordinates x' , y' , z' is fixed in the T apparatus, related to the x , y , z coordinates of the S apparatus by: $z' = z \cos \alpha + x \sin \alpha$, $x' = x \cos \alpha - z \sin \alpha$, $y' = y$.) Then the transformation amplitudes are:

$$\begin{aligned} \langle +T | +S \rangle &= \frac{1}{2}(1 + \cos \alpha), \\ \langle 0T | +S \rangle &= -\frac{1}{\sqrt{2}} \sin \alpha, \\ \langle -T | +S \rangle &= \frac{1}{2}(1 - \cos \alpha), \\ \langle +T | 0S \rangle &= +\frac{1}{\sqrt{2}} \sin \alpha, \\ \langle 0T | 0S \rangle &= \cos \alpha, \\ \langle -T | 0S \rangle &= -\frac{1}{\sqrt{2}} \sin \alpha, \\ \langle +T | -S \rangle &= \frac{1}{2}(1 - \cos \alpha), \\ \langle 0T | -S \rangle &= +\frac{1}{\sqrt{2}} \sin \alpha, \\ \langle -T | -S \rangle &= \frac{1}{2}(1 + \cos \alpha). \end{aligned} \quad (5.38)$$